



737-7/8/9/10 Maintenance Training Manual

Boeing Training and Professional Services
FS328

Boeing Proprietary

The information contained herein is proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company.

Copyright © 2020 The Boeing Company. All rights reserved.

EXPORT CONTROLLED - This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN 9E991

ENGINE MANUFACTURER PROPRIETARY INFORMATION

A portion of the information in this training material is the property of Rolls-Royce plc, General Electric or CFM International, Inc. (the "Engine Manufacturer") and has been included with the Engine Manufacturer's permission. It may not be copied, or communicated to a third party, or used for any purpose other than that for which it is supplied without the express written consent of the applicable Engine Manufacturer. In this regard, the Engine Manufacturer training material herein is not an official publication of the Engine Manufacturer and must not be used for operating or maintaining the equipment described. Official Engine Manufacturer publications and manuals must be used for those purposes.

To the extent the Engine Manufacturer has an agreement covering the control and disclosure of information with the company, organization or individual receiving this training material then the Engine Manufacturer portion of the training material must be held in confidence and controlled in accordance with such agreement's terms. While this information is given in good faith based upon the latest information available to the Engine Manufacturer, no warranty or representation is given concerning such information, which must not be taken as establishing any contractual or other commitment binding upon the Engine Manufacturer or any of its subsidiary or associated companies.



737-7/8/9/10 Maintenance Training Manual

This page is intentionally left blank.



737-7/8/9/10 Maintenance Training Manual

Boeing Training and Professional Services
FS328

HIGHLIGHTS

<u>Reason for Change</u>	<u>Page</u>
Added: Minor edits to graphic callouts.Added: Minor edits to graphic callouts.	5
Modified: Inclusion of paragraph for Fairings.Modified: Inclusion of paragraph for Fairings.	13
Modified: Elaboration of dropped tool access panel's function.Modified: Elaboration of dropped tool access panel's function.	13
Modified: Inclusion of paragraph for Thrust ReverserModified: Inclusion of paragraph for Thrust Reverser	13
Added: Minor edits to Fan Cowl Latches discussion.	18
Added: Added Fan Cowl Latch Indication System discussion.	18
Added: Minor edits to Fan Cowl Hold Open Rod discussion.	18
Added: Added illustration of tension latch handle and keeper.	19
Added: Added photos of tension latches and specified locations.Added: Added photos of tension latches and specified locations.	23
Added: Minor edits to one illustration and several callouts.Added: Minor edits to one illustration and several callouts.	55
Added: Enhanced graphic of engine removal equipment.Added: Enhanced graphic of engine removal equipment.	57
Added: Minor edits to graphic.Added: Minor edits to graphic.	67
Modified: Inclusion of Boeing Equivalent Thrust	105
Modified: 'engine center shaft' replaced with 'oil vent tube'.Modified: 'engine center shaft' replaced with 'oil vent tube'.	112
Modified: Addition of description on the locations of the various sensors and ports.Modified: Addition of description on the locations of the various sensors and ports.	114
Modified: Inclusion of description on the locations of the major modules.Modified: Inclusion of description on the locations of the major modules.	116
Modified: Replacing 'Fan Composit Case' with 'Fan Case' to align with ASNModified: Replacing 'Fan Composit Case' with 'Fan Case' to align with ASN	118
Modified: Addition of IGB and No.3 Bearing Support into the list of major modulesModified: Addition of IGB and No.3 Bearing Support into the list of major modules	118



Modified: Inclusion of paragraph for Fan Case	118
Modified: Inclusion of paragraph for Fan Case	118
Added: 'Platform Front Shroud' added as a random list item.	120
Added: Paragraph introduced for Front Spinner Cone	120
Modified: Paragraph edited to remove front spinner cone.	120
Modified: Location for the booster added.	120
Modified: Inclusion of 'description'.	122
Modified: Inclusion of sentence to explain that installation is at a shop level.	122
Modified: Changed 2 engines are stopped to all EEC's are OFF.	178
Re-edited for clarity.	184
Deleted: Removed references to alternate #1 bearing accelerometer.	186
Deleted: Removed references to alternate #1 bearing accelerometer.	186
Deleted: Removed references to alternate #1 bearing accelerometer.	186
Modified: Edited to provide clarity on "Cockpit Units".	189
Modified: Edited the illustrated EPCS vibration values to correspond with the EI.	191
Modified: Modified section due to removal of fuel manifold insulation	238
Modified: Modified section due to removal of fuel manifold insulation per GE comunique	239
Modified: Inclusion of EEC ON and OFF	306
Modified: Reference to ATA 71	306
Modified: New Art Introduced	307
Deleted: The EEC cooling fans have been removed from the engine.	316
Deleted: The EEC cooling fans have been removed from the engine.	317
Deleted: The EEC cooling blowers have been removed from the engine.	325
Modified: The word 'vane' is deleted. Changed to 8 cascade segments for both I/B and O/B T/R sleeves.	480
Modified: The word 'vane' is deleted. Changed to 8 cascade segments for both I/B and O/B T/R sleeves.	480
Modified: Changed to show 16 cascade segments.	481
Modified: Changed to show 16 cascade segments.	481
Description of function corrected.	538
Description of function corrected.	538

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
POWER PLANT -- INTRODUCTION	.71-00-00 . . .	2
GENERAL DESCRIPTION		4
ENGINE HAZARDS		6
NOISE HAZARD		8
ENGINE COWLING -- INTRODUCTION	.71-11-00 . .	10
GENERAL DESCRIPTION		13
OPENING INTERFERENCE		16
FAN COWL		18
FAN COWL HOR AND SDOS		20
THRUST REVERSER -- TENSION LATCHES	.78-31-00 . .	22
THRUST REVERSER OPENING ACTUATOR		25
POWER PLANT -- MOUNTS AND DRAINS - INTRODUCTION (cont'd on next page).	.71-00-00 . .	28
ENGINE MOUNTS		30
ENGINE DRAINS		32
TRAINING INFORMATION POINTS -- INTRODUCTION		34
LEAP-1B REMOVAL - GENERAL DESCRIPTION		36
LEAP-1B REMOVAL - PREPARATION 1		38
LEAP-1B REMOVAL - PREPARATION 2		40
LEAP-1B REMOVAL - FAN COWL AND THRUST REVERSER		42
LEAP-1B REMOVAL - INSTALL T/R HOLD OPEN EQUIPMENT		44
LEAP-1B REMOVAL - DISCONNECT T/R OPENING ACTUATOR		46
LEAP-1B REMOVAL - FAN CASE DISCONNECT LH		48
LEAP-1B REMOVAL - FAN CASE DISCONNECT RH		50
LEAP-1B REMOVAL - ENGINE CORE		52
LEAP-1B REMOVAL - INSTALL FORWARD BOOTSTRAP		54
LEAP-1B REMOVAL - ENGINE HANDLING FWD ADAPTERS AND TOOL		56
LEAP-1B REMOVAL - INSTALL AFT BOOTSTRAP		58
LEAP-1B REMOVAL - AFT BOOTSTRAP TO ENGINE		60

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
POWER PLANT -- MOUNTS AND DRAINS - INTRODUCTION	.71-00-00 . .	28
LEAP-1B REMOVAL - PREPARE TO REMOVE POWER PLANT		62
LEAP-1B REMOVAL - DISCONNECT ENGINE MOUNTS		64
LEAP-1B REMOVAL - LOWER THE POWER PLANT		66
LEAP-1B INSTALLATION - GENERAL		68
LEAP-1B INSTALLATION - GENERAL DESCRIPTION		70
LEAP-1B INSTALLATION - PREPARATION		72
LEAP-1B INSTALLATION - INSTALL PREPARATION 2		74
LEAP-1B INSTALLATION - DISCONNECT ENG ADAPTERS		76
LEAP-1B INSTALLATION - LIFT THE ENGINE		78
LEAP-1B INSTALLATION - ATTACH THE ENGINE TO STRUT		80
LEAP-1B INSTALLATION - REMOVE BOOTSTRAP EQUIPMENT		82
LEAP-1B INSTALLATION - LEFT FAN CASE CONNECTIONS		84
LEAP-1B INSTALLATION - RIGHT FAN CASE CONNECTIONS		86
LEAP-1B INSTALLATION - CORE CONNECTIONS		88
LEAP-1B INSTALLATION - CONNECT T/R OPENING ACTUATOR		90
LEAP-1B INSTALLATION - REMOVE HOLD-OPEN EQUIPMENT		92
LEAP-1B INSTALLATION - PREPARE FOR SERVICE		94
LEAP-1B INSTALLATION - CLOSE CIRCUIT BREAKERS		96
LEAP-1B INSTALLATION - ENGINE CONFIGURATION		98
LEAP-1B - PRESERVATION/DEPRESERVATION		100
ENGINE -- INTRODUCTION (cont'd on next page).	.72-00-00 . .	102
NAMEPLATE DATA		105
GENERAL DESCRIPTION		109
MAIN ENGINE BEARINGS		112
AERODYNAMIC STATIONS		114
MAJOR MODULES		116
FAN MAJOR MODULE		118
FAN AND BOOSTER		120
COMPONENT DESCRIPTION		122

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
ENGINE -- INTRODUCTION	.72-00-00	102
FAN BLADES AND SPINNER		124
CORE MAJOR MODULE		126
HIGH PRESSURE COMPRESSOR (HPC)		128
COMBUSTION SECTION		130
HIGH PRESSURE TURBINE (HPT)		132
LOW PRESSURE TURBINE (LPT) MAJOR MODULE		134
LPT SHAFT MODULE		136
LPT ROTOR AND STATOR MODULE		138
TURBINE CENTER FRAME (TCF)		140
TURBINE REAR VANE MODULE		142
ACCESSORY DRIVE		144
ACCESSORY GEARBOX (AGB)		146
ACCESSORIES		148
BORESCOPE INSPECTION PORTS - LOCATION		150
BORESCOPE INSPECTION PORTS - GENERAL DESCRIPTION		152
ENGINE INDICATING -- INTRODUCTION	.77-00-00	154
GENERAL DESCRIPTION		157
ENGINE TACHOMETER SYSTEM -- GENERAL DESCRIPTION	.77-11-00	160
N1 SPEED SENSOR		162
N2 SPEED SENSOR		164
FUNCTIONAL DESCRIPTION		166
EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- GENERAL DESCRIPTION	.77-21-00	170
EGT SENSORS AND HARNESSSES		172
FUNCTIONAL DESCRIPTION		175
ENGINE INDICATING -- TRAINING INFORMATION POINTS - ENGINE EXCEEDANCES (cont'd on next page)	.77-00-00	178
TRAINING INFORMATION POINTS - PERFORMANCE MAINTENANCE PAGE		180

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
ENGINE INDICATING -- TRAINING INFORMATION POINTS - ENGINE EXCEEDANCES	.77-00-00	178
TRAINING INFORMATION POINTS - EPCS MAINTENANCE PAGES		182
AIRBORNE VIBRATION MONITORING (AVM) SYSTEM -- GENERAL DESCRIPTION	.77-31-00	184
COMPONENT LOCATION		186
VIBRATION INDICATIONS		189
FUNCTIONAL DESCRIPTION		193
OPERATION		196
ENGINE CONTROLS -- INTRODUCTION	.76-00-00	200
GENERAL DESCRIPTION		202
COMPONENT LOCATION		204
INTERFACES		206
THRUST LEVER		208
THRUST LEVER ANGLE RESOLVER		210
START LEVER		212
START LEVER - FUNCTIONAL DESCRIPTION		214
REVERSE THRUST INTERLOCK SOLENOIDS		216
ENGINE FUEL AND CONTROL -- INTRODUCTION	.73-00-00	219
GENERAL DESCRIPTION		223
ENGINE FUEL AND CONTROL -- DISTRIBUTION - INTRODUCTION	.73-11-00	226
DISTRIBUTION - GENERAL DESCRIPTION		228
DISTRIBUTION - COMPONENT LOCATION		230
DISTRIBUTION - MAIN FUEL PUMP		232
DISTRIBUTION - MAIN FUEL/OIL HEAT EXCHANGER		234
DISTRIBUTION - MAIN FUEL FILTER		236
DISTRIBUTION - FUEL MANIFOLDS		238
DISTRIBUTION - FUEL NOZZLES		240
DISTRIBUTION - FUNCTIONAL DESCRIPTION		243

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
ENGINE FUEL AND CONTROL -- ENGINE CONTROL - INTRODUCTION	.73-21-00 . .	246
ENGINE CONTROL - GENERAL DESCRIPTION		248
ENGINE CONTROL - COMPONENT LOCATION - LEFT SIDE		250
ENGINE CONTROL - COMPONENT LOCATION - RIGHT SIDE		252
ENGINE CONTROL - ANALOG INTERFACES		254
ENGINE CONTROL - DIGITAL INTERFACES		258
ENGINE CONTROL - ELECTRONIC ENGINE CONTROL		260
ENGINE CONTROL - PRESSURE SUBSYSTEM		264
ENGINE CONTROL - DATA ENTRY PLUG		266
ENGINE CONTROL - ENGINE CONTROL LIGHT AND EEC SWITCHES		268
ENGINE CONTROL - FUEL METERING UNIT		270
ENGINE CONTROL - SPLIT CONTROL UNIT		272
ENGINE CONTROL - T12 SENSOR		274
ENGINE CONTROL - T25 SENSOR		276
ENGINE CONTROL - T3 SENSOR		278
ENGINE CONTROL - PERMANENT MAGNET ALTERNATOR		280
ENGINE CONTROL - EECS ENGINE 1 POWER SUPPLY- FUNCTIONAL DESCRIPTION		282
ENGINE CONTROL - EECS ENGINE 2 POWER SUPPLY - FUNCTIONAL DESCRIPTION		284
ENGINE CONTROL - IDLE CONTROL - FUNCTIONAL DESCRIPTION		286
ENGINE CONTROL - THRUST CONTROL - FUNCTIONAL DESCRIPTION		289
ENGINE CONTROL - EOS AND TCMA		292
ENGINE CONTROL - SCU - FUNCTIONAL DESCRIPTION		295
ENGINE CONTROL - FUNCTIONAL DESCRIPTION		298
ENGINE CONTROL - TRAINING INFORMATION POINT - EEC BITE - INDICATIONS		300
OMF - OPERATIONAL TEST		302
OMF - SYSTEM TESTS		304
OMF - OTHER FUNCTIONS		306
ENGINE FUEL AND CONTROL -- FUEL INDICATING - INTRODUCTION (cont'd on next page).	.73-30-00 . .	308
FUEL INDICATING - GENERAL DESCRIPTION		310
FUEL INDICATING - FUEL FILTER DIFFERENTIAL PRESSURE SENSOR		312

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
ENGINE FUEL AND CONTROL -- FUEL INDICATING - INTRODUCTION	.73-30-00 . .	308
FUEL INDICATING - FUEL FLOW TRANSMITTER		314
ENGINE AIR -- INTRODUCTION	.75-00-00 . .	316
GENERAL DESCRIPTION		318
ENGINE AIR -- LEAP-1B ENGINE - COMPONENT LOCATION - LEFT SIDE	.75-10-00 . .	320
LEAP-1B ENGINE - COMPONENT LOCATION - RIGHT SIDE		322
ENGINE AIR -- TURBINE CLEARANCE CONTROL - INTRODUCTION	.75-20-00 . .	324
ENGINE AIR -- HPTACC - COMPONENT LOCATION	.75-21-00 . .	326
HPTACC - VALVE ASSEMBLY		328
HPTACC - FUNCTIONAL DESCRIPTION		330
ENGINE AIR -- LPTACC - COMPONENT LOCATION	.75-22-00 . .	332
LPTACC - VALVE ASSEMBLY		334
LPTACC - FUNCTIONAL DESCRIPTION		336
ENGINE AIR -- MTC - COMPONENT LOCATION	.75-23-00 . .	338
MTC - VALVE/ACTUATOR ASSEMBLY		340
MTC - FUNCTIONAL DESCRIPTION		342
ENGINE AIR -- COMPRESSOR CONTROL - INTRODUCTION	.75-30-00 . .	344
ENGINE AIR -- VSV SYSTEM - COMPONENT LOCATION	.75-31-00 . .	346
VSV SYSTEM - VSV ACTUATORS		348
VSV SYSTEM - FUNCTIONAL DESCRIPTION		350
ENGINE AIR -- VBV SYSTEM - COMPONENT LOCATION (cont'd on next page)	.75-32-00 . .	352
VBV SYSTEM - VBV ACTUATORS		354

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
ENGINE AIR -- VBVS SYSTEM - COMPONENT LOCATION	.75-32-00 . .	352
VBVS SYSTEM - DOORS		356
VBVS SYSTEM - FUNCTIONAL DESCRIPTION		358
ENGINE AIR -- TBV - COMPONENT LOCATION	.75-23-00 . .	360
TBV - VALVE ASSEMBLY		362
TBV - FUNCTIONAL DESCRIPTION		364
ENGINE AIR -- START BLEED/BOOSTER ANTI-ICE VALVE - COMPONENT LOCATION	.75-33-00 . .	366
START BLEED/BOOSTER ANTI-ICE VALVE - VALVE ASSEMBLY		368
START BLEED/BOOSTER ANTI-ICE VALVE - FUNCTIONAL DESCRIPTION		370
ENGINE AIR -- FUNCTIONAL DESCRIPTION	.75-00-00 . .	373
ENGINE OIL -- INTRODUCTION	.79-00-00 . .	376
GENERAL DESCRIPTION		378
ENGINE OIL -- STORAGE - OIL TANK	.79-10-00 . .	380
STORAGE - SERVICING		382
ENGINE OIL -- DISTRIBUTION - GENERAL DESCRIPTION (cont'd on next page)	.79-20-00 . .	384
DISTRIBUTION - COMPONENT LOCATION		386
DISTRIBUTION - LUBRICATION UNIT		388
DISTRIBUTION - OIL SUPPLY FILTER		390
DISTRIBUTION - ENGINE SURFACE AIR COOLING OIL COOLERS		392
DISTRIBUTION - MAIN OIL/FUEL HEAT EXCHANGER		394
DISTRIBUTION - SCAVENGE SCREEN PLUGS		396
DISTRIBUTION - AIR/OIL/DEBRIS SEPARATOR		398
DISTRIBUTION -ODMS SENSOR		400
DISTRIBUTION - ODMS UNIT		402
DISTRIBUTION - OIL EDUCTOR VALVE		404

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
ENGINE OIL -- DISTRIBUTION - GENERAL DESCRIPTION	.79-20-00 . .	384
DISTRIBUTION - FUNCTIONAL DESCRIPTION		407
ENGINE OIL -- INDICATING - GENERAL DESCRIPTION	.79-30-00 . .	410
INDICATING - COMPONENT LOCATION		412
INDICATING - OIL QUANTITY INDICATING SYSTEM		414
INDICATING - OIL PRESSURE INDICATING SYSTEM		416
INDICATING - OIL TEMPERATURE INDICATING SYSTEM		418
INDICATING - OIL FILTER BYPASS WARNING SYSTEM		420
IGNITION -- INTRODUCTION	.74-00-00 . .	422
GENERAL DESCRIPTION		424
FLIGHT COMPARTMENT - COMPONENT LOCATION		426
ENGINE - COMPONENT LOCATION		428
IGNITION EXCITER		430
DISTRIBUTION - IGNITION LEADS AND SPARK IGNITERS		432
FUNCTIONAL DESCRIPTION		434
OPERATION		436
ENGINE STARTING -- INTRODUCTION	.80-00-00 . .	438
GENERAL DESCRIPTION		440
COMPONENT LOCATIONS - ENGINE		442
COMPONENT LOCATIONS - FLIGHT COMPARTMENT AND EE COMPARTMENT		444
STARTER AIR VALVE (SAV)		446
STARTER		448
FUNCTIONAL DESCRIPTION		451
BOWED ROTOR		454
BOWED ROTOR PROTECTION		456
FUNCTIONAL DESCRIPTION - START PROFILE		458
OPERATION		462

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
ENGINE EXHAUST SYSTEM -- INTRODUCTION	.78-00-00	. . 464
GENERAL DESCRIPTION		. . 466
TURBINE EXHAUST SYSTEM -- INTRODUCTION	.78-11-00	. . 468
GENERAL DESCRIPTION		. . 470
EXHAUST NOZZLE		. . 472
EXHAUST PLUG		. . 474
THRUST REVERSER SYSTEM -- INTRODUCTION	.78-30-00	. . 476
THRUST REVERSER -- INTRODUCTION	.78-31-00	. . 478
GENERAL DESCRIPTION		. . 480
COMPONENT LOCATION		. . 482
HYDRAULIC ACTUATORS AND SYNC SHAFTS - GENERAL DESCRIPTION		. . 484
HYDRAULIC ACTUATORS - GENERAL DESCRIPTION		. . 487
CASCADE SEGMENTS		. . 490
ACCESS PANELS		. . 492
DEACTIVATION FOR FLIGHT DISPATCH - TRAINING INFORMATION POINTS		. . 494
LOCKING HYDRAULIC ACTUATORS -- FUNCTIONAL DESCRIPTION		. . 497
THRUST REVERSER CONTROL SYSTEM -- INTRODUCTION	.78-34-00	. . 500
GENERAL DESCRIPTION		. . 503
COMPONENT LOCATIONS		. . 506
SYNC LOCK SWITCHES		. . 508
SYNC LOCK		. . 510
CONTROL SWITCH		. . 512
CONTROL VALVE MODULE		. . 514
THRUST REVERSER ACTUATOR LOCK PROXIMITY SENSORS		. . 516
FUNCTIONAL DESCRIPTION - DEPLOY AND STOW CONTROL		. . 518
FUNCTIONAL DESCRIPTION - DEPLOY HYDRAULIC FLOW		. . 523
FUNCTIONAL DESCRIPTION - STOW HYDRAULIC FLOW		. . 527

TABLE OF CONTENTS

<u>Subject</u>	<u>Chapter Section Subject</u>	<u>Page</u>
THRUST REVERSER INDICATING SYSTEM -- INTRODUCTION	78-36-00 . .	530
GENERAL DESCRIPTION		532
COMPONENT LOCATION - FLIGHT COMPARTMENT		534
LINEAR VARIABLE DIFFERENTIAL TRANSFORMER		536
REVERSER AIR/GRD LIGHT - FUNCTIONAL DESCRIPTION		538
REVERSER COMMAND LIGHT - FUNCTIONAL DESCRIPTION		540
REVERSER LIMITED LIGHT - FUNCTIONAL DESCRIPTION		542



POWER PLANT -- INTRODUCTION

THIS PAGE INTENTIONALLY LEFT BLANK

POWER PLANT -- INTRODUCTION

General

Two LEAP-1B engines supply thrust for the airplane. The engines also supply power for these systems:

- Electric
- Hydraulic
- Pneumatic.

The LEAP-1B is a high-bypass ratio, dual-rotor, turbofan engine.

LEAP-1B

Information about the LEAP-1B is in this chapter order:

- Power plant (71)
- Engine (72)
- Engine indicating (77)
- Engine controls (76)
- Engine fuel and control (73)
- Engine air (75)
- Engine Oil (79)
- Ignition (74)
- Starting (80)
- Exhaust (78).

Acronyms and Abbreviations

C	- celsius
cm	- centimeters
daN	- deca newton
EEC	- electronic engine control
FMU	- fuel metering unit
ft	- feet
HPTACC	- high pressure turbine active clearance control

LPTACC	- low pressure turbine active clearance control
IDG	- integrated drive generator
in	- inches
kg	- kilograms
lbs	- pounds
m	- meters
RPM	- revolutions per minute
TBV	- transient bleed valve
VBV	- variable bleed valve
VSV	- variable stator vanes

DMC-B737T-A-71-00-00-04A-018A-T_004-00

POWER PLANT
(71)

ENGINE
(72)

ENGINE
INDICATING
(77)

ENGINE
CONTROLS
(76)

ENGINE FUEL
AND CONTROL
(73)



ENGINE AIR
(75)

ENGINE OIL
(79)

IGNITION
(74)

STARTING
(80)

EXHAUST
(78)

DMC-B737T-A-71-00-00-04A-018A-T_004-00
ICN-B737T-A-710000-A-81205-00050-A-005-01

POWER PLANT -- INTRODUCTION

71-00-00

02-Dec-2020

POWER PLANT -- GENERAL DESCRIPTION

Engine Left Side

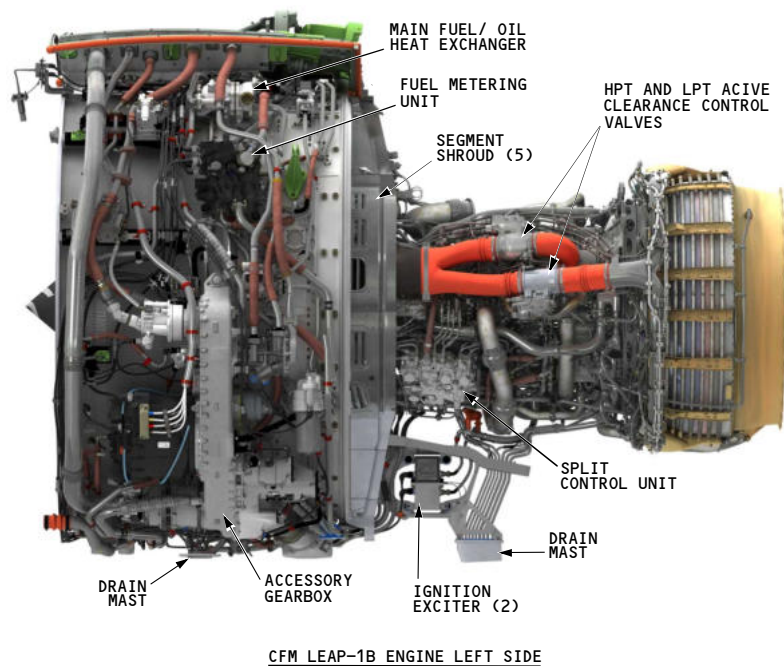
These major components are on the left side of the engine:

- Accessory gearbox
- Ignition system
- Fuel metering unit (FMU)
- Split control unit (SCU)
- High pressure and low pressure turbine active clearance control systems (HPTACC / LPTACC).

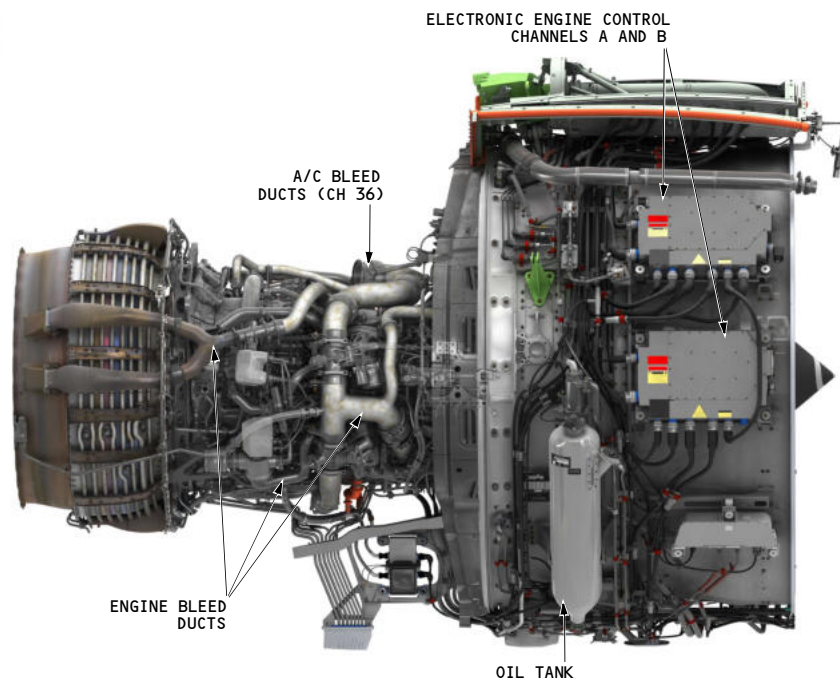
Engine Right Side

These major components are on the right side of the engine:

- Electronic engine control (EEC) channels A and B
- Oil tank
- Bleed ducts and valves.



CFM LEAP-1B ENGINE LEFT SIDE



CFM LEAP-1B ENGINE RIGHT SIDE

POWER PLANT -- GENERAL DESCRIPTION

71-00-00

DMC-B737T-A-71-00-00-11A-040A-T_004-00
ICN-B737T-A-710000-A-81205-00166-A-003-01

02-Dec-2020

POWER PLANT -- ENGINE HAZARDS

General

It is dangerous to work around an operating engine.

These are the hazards around an engine in operation:

- Inlet suction
- Exhaust heat
- Exhaust velocity
- Engine noise.

CAUTION: PERFORM A FOD WALK-AROUND IN FRONT OF AND AROUND THE ENGINE INGESTION AREA PRIOR TO ENGINE START.

When you see an anti-collision light on, an engine is in operation or will start.

Inlet Suction

Engine inlet suction can pull people and large objects into the engine. At idle power, the inlet hazard area is a 10.6 ft (3.2 m) radius around and 4.5 ft (1.4 m) aft of the inlet.

WARNING: IF THE WIND IS OVER 25 KNOTS, INCREASE THE INLET HAZARD AREA BY 20 PERCENT.

Exhaust Heat

The engine exhaust is very hot for long distances behind the engine. This can injure personnel and cause damage to equipment.

Exhaust Velocity

Exhaust velocity is very high for long distances behind the engine. This can injure personnel and cause damage to equipment.

Engine Noise

Engine noise can cause temporary and permanent loss of hearing. Wear ear protection when near an engine in operation.

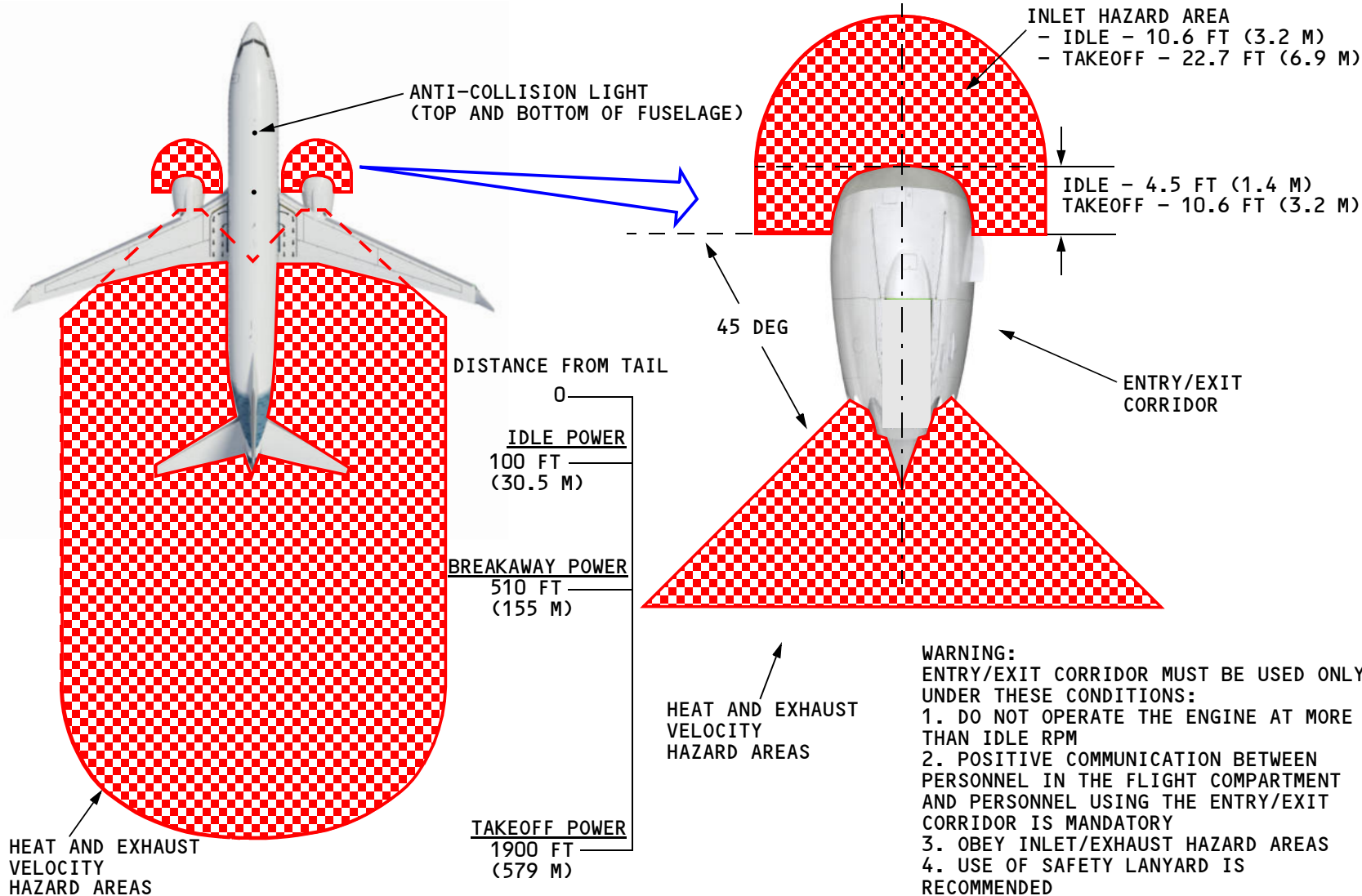
Engine Entry/Exit Corridor

The entry/exit corridor is only used to approach or leave an engine that is operating at or less than idle speed. You must also have positive communication with the flight compartment.

The engine entry/exit corridor begins at the red vertical line on the fan cowl (5 ft/1.5 m aft of the engine inlet) and continues to the back of the thrust reverser (T/R). The engine entry/exit corridor extends from the back of the T/R away from the engine at a 45° angle.

For additional safety, wear a safety harness when the engine is in operation.

WARNING: ENTER THE ENTRY/EXIT CORRIDOR ONLY WHEN THE ENGINE IS AT IDLE AND YOU HAVE POSITIVE COMMUNICATION WITH THE FLIGHT COMPARTMENT. THERE IS NO SAFE AREA NEAR AN ENGINE THAT IS OPERATING ABOVE IDLE.



DMC-B737T-A-71-00-00-06A-034B-T_003-00
ICN-B737T-A-710000-A-81205-00051-A-003-01

POWER PLANT -- ENGINE HAZARDS

02-Dec-2020

71-00-00



POWER PLANT -- NOISE HAZARD

Safety

An operating engine makes noise due to high velocity air that goes into and out of engine.

The noise comes from these areas:

- Air inlet
- Compressor
- Turbine
- Exhaust.

Noise also comes from the operating accessories.

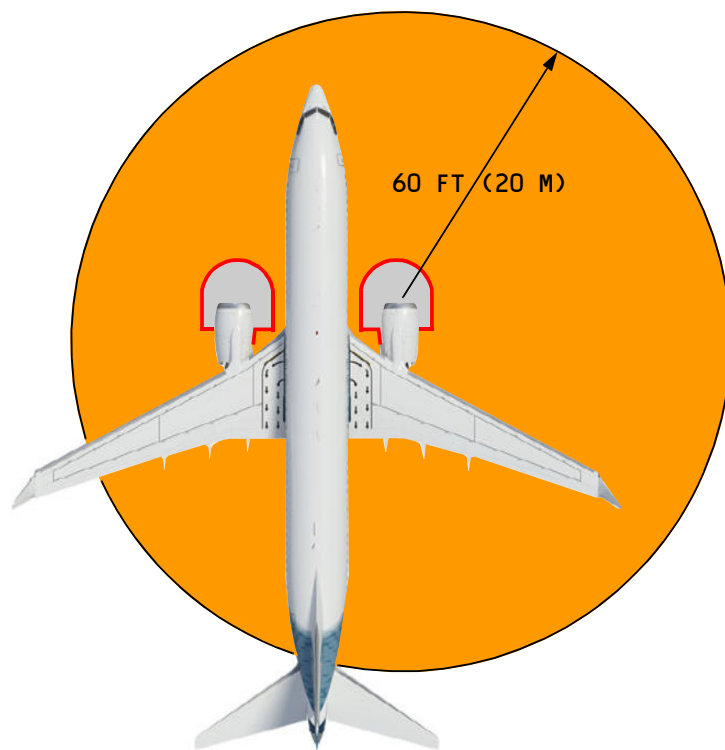
The noise level increases as engine power settings increase.

Ear protection must be used at all times in the vicinity of the operating engine. Obey the noise exposure limits to prevent ear damage.

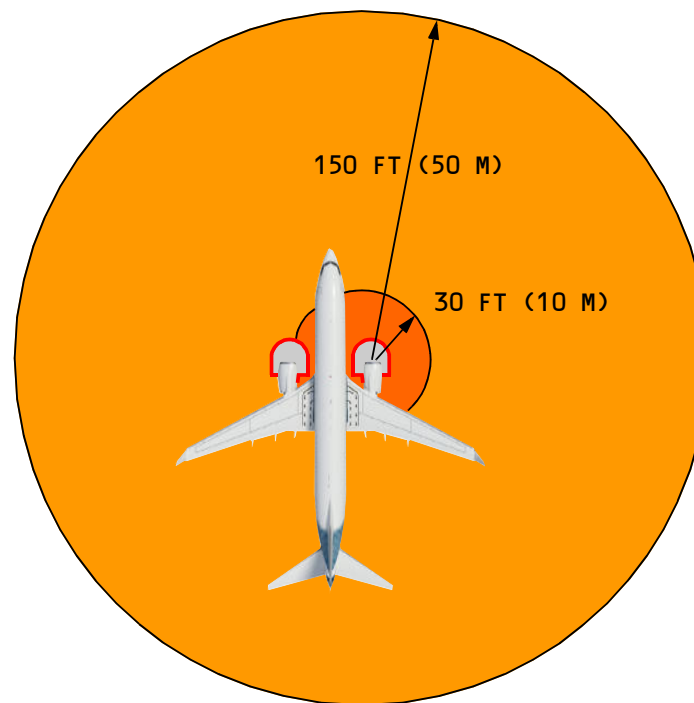
DMC-B737T-A-71-00-00-02A-012A-T_002-00

02-Dec-2020

71-00-00



IDLE POWER



TAKEOFF POWER

-  **WARNING:** EAR PROTECTION MANDATORY WITHIN THIS AREA
-  **WARNING:** PROLONGED EXPOSURE WITHIN THIS AREA OF MORE THAN SIX MINUTES, EVEN WITH EAR PROTECTION CAN CAUSE EAR DAMAGE.

POWER PLANT -- NOISE HAZARD

71-00-00

DMC-B737T-A-71-00-00-02A-012A-T_002-00
ICN-B737T-A-710000-A-81205-00052-A-002-01

02-Dec-2020

ENGINE COWLING -- INTRODUCTION

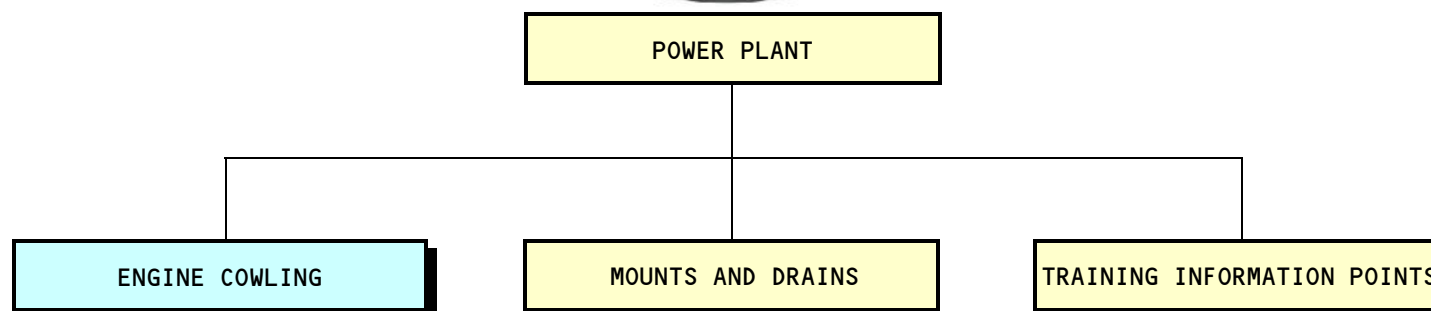
General

The engine cowl contains and protects the engine components. The engine cowl also controls the airflow through and around the engine. Some of the engine cowl has panels for maintenance access.

The engine cowl has three parts:

- Inlet cowl
- Fan cowl
- Thrust reverser cowl.

This chapter also gives information for opening and closing the thrust reverser cowl. For more information about the thrust reverser cowl, see chapter 78.



ENGINE COWLING -- INTRODUCTION

71-11-00

DMC-B737T-A-71-11-00-02A-018A-T_001-00
ICN-B737T-A-71-1100-A-81205-00009-A-001-01

02-Dec-2020



ENGINE COWLING -- GENERAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE COWLING -- GENERAL DESCRIPTION

General

The engine cowling gives an aerodynamically smooth surface into and over the engine. It also gives a protective area for engine components and accessories.

These are the parts of the engine cowling:

- Inlet cowl
- Fan cowl
- Thrust reverser.

Fairings

The thumbnail, mid-strut and over-wing fairings are above the engine cowls. The fairings give an aerodynamically smooth surface over the engine strut. The thumbnail and mid-strut fairings each have a pressure relief door.

Inlet Cowl

The inlet cowl directs air into the engine. The inlet cowl attaches to the engine fan case.

The T12 access panel is on the inlet cowl. The T12 access panel permits access to the T12 sensor.

A dropped tool access panel is at the bottom of the inlet cowl outer barrel. Tools that have fallen, during replacement of the T12 sensor, are retrieved through this access panel.

Fan Cowls

The fan cowls give an aerodynamically smooth surface over the fan case. The fan cowls attach to the fan cowl support beam. The fan cowls open for maintenance.

These items are on the fan cowls:

- IDG access door (left)
- Chine (inboard only)
- Starter valve access port (left)
- Oil tank access door (right).

IDG Access Door

The IDG access door permits access to the IDG for servicing. It is on the left fan cowl.

Chine

The chine is a vortex control device that smooths airflow around the wing. It is only on the inboard fan cowl.

Starter Valve Access Port

The starter valve access port permits access to the starter air valve (SAV) for manual operation. It is on the left fan cowl.

Oil Tank Access Door

The oil tank access door permits access to the oil tank for servicing. It is on the right fan cowl. The oil tank access door is also a pressure relief door for the fan compartment.

Thrust Reverser

The T/R system has two thrust reversers. T/R 1 is the thrust reverser for engine 1 (left). T/R 2 is the thrust reverser for engine 2 (right). Each T/R has a left and right half. Each half has a translating sleeve which moves aft (deploy position) for reverse thrust.

These are features on the thrust reverser halves:

ENGINE COWLING -- GENERAL DESCRIPTION

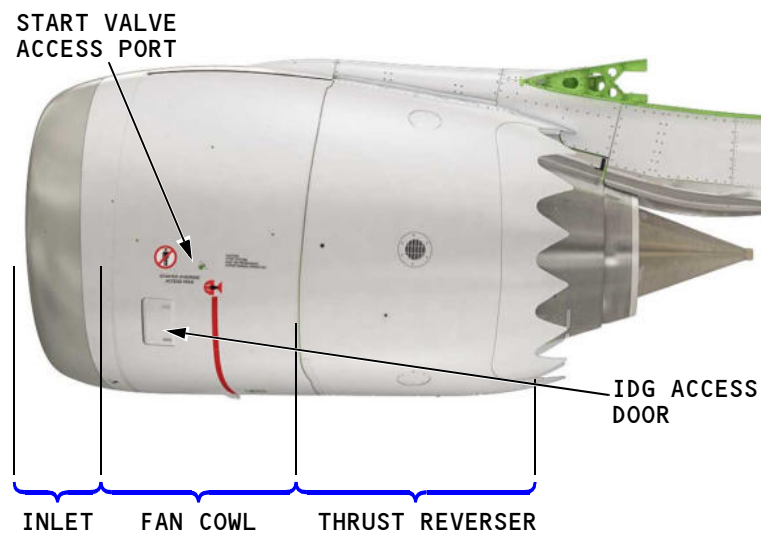
- Kruger flap deflectors (only on inboard translating sleeves)
- Chevron cold exhaust edges (reduces exhaust noise)
- Deactivation fitting (one on each translating sleeve)
- Access doors (one for each sleeve actuating rod end).

More information about the thrust reverser halves is discussed in ATA
78 - Engine Exhaust.

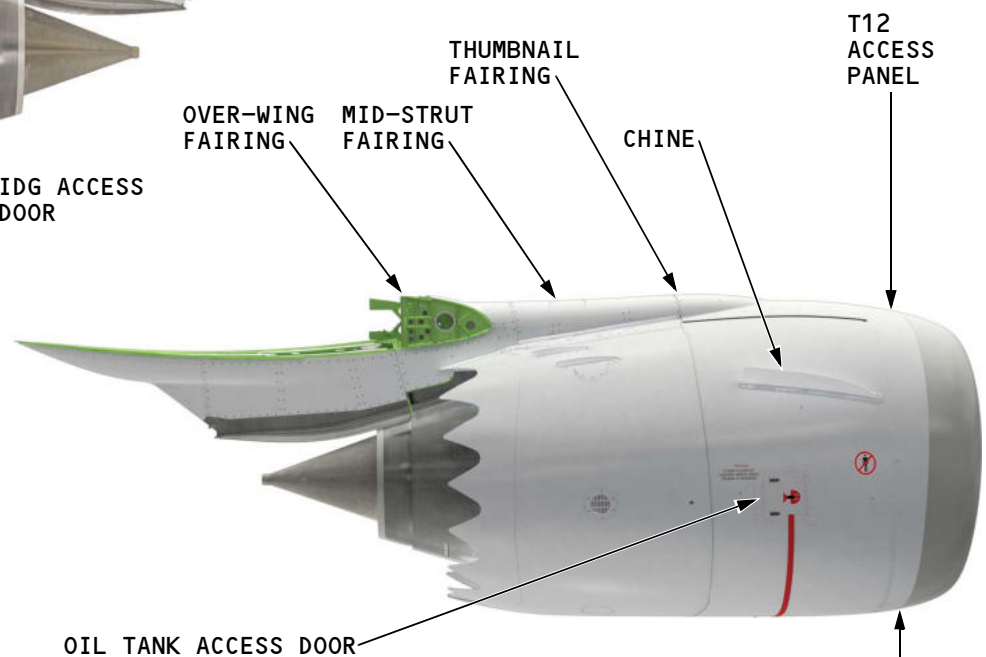
DMC-B737T-A-71-11-00-02A-040A-T_007-01

02-Dec-2020

71-11-00



ENGINE #1 - LEFT VIEW



ENGINE #1 - RIGHT VIEW

ENGINE COWLING -- GENERAL DESCRIPTION

71-11-00

DMC-B737T-A-71-11-00-02A-040A-T_007-01
ICN-B737T-A-71-1100-A-81205-00006-A-009-01

02-Dec-2020

ENGINE COWLING -- OPENING INTERFERENCE

General

There are different conditions where the leading edge (LE) flaps and/or slats can hit the engine fan cowl, thrust reversers, or deployed thrust reverser translating sleeves. This interference condition can cause damage to equipment.

To prevent equipment damage, you must retract and deactivate the leading edge flaps/slats before you open a fan cowl or thrust reverser cowl.

LE Flaps and Slats Positions

The LE edge flaps are inboard of the engine strut, and have these two positions:

- Retract - when the flap lever is in the 0 detent
- Extend - when the flap lever is at any other detent.

The LE slats are outboard of the engine strut, and have these three positions:

- Retract - when the flap lever is in the 0 detent
- Extend - when the flap lever is in the 1, 2, 5, 10, 15 or 25 detents
- Full extend - when the flap lever is in the 30 or 40 detents.

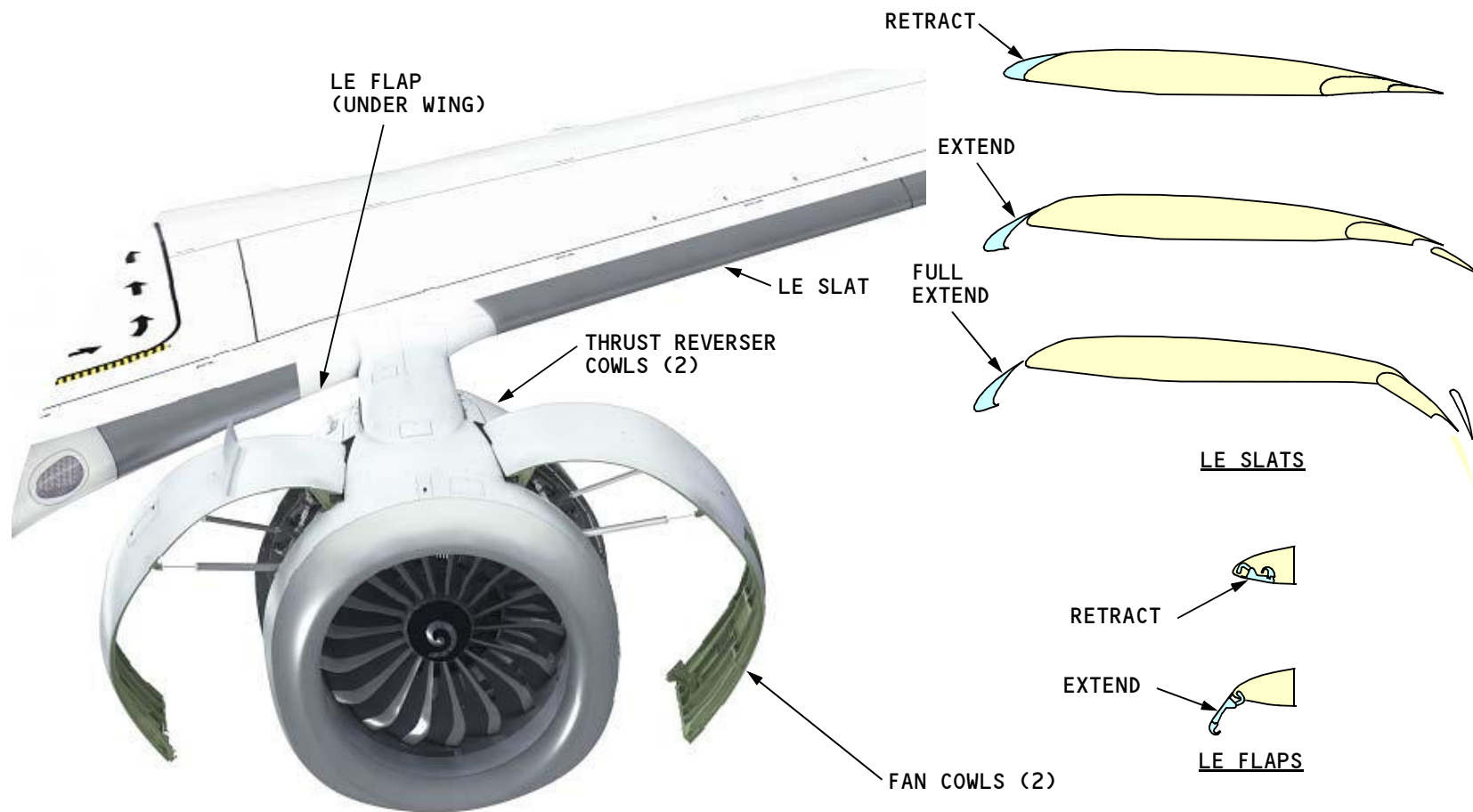
Cowling Opening

CAUTION: BEFORE YOU EXTEND THE LE FLAPS AND SLATS, YOU MUST MAKE SURE THE INBOARD FAN COWLS ARE CLOSED OR REMOVED. YOU MUST ALSO MAKE SURE THE THRUST REVERSER COWLS ARE CLOSED. THERE IS NOT SUFFICIENT CLEARANCE FOR THE LE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL AND THE THRUST REVERSER COWL ARE IN

THE OPEN (MAINTENANCE) POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

The table shows the conditions where the LE flaps or slats have an interference with either the engine fan cowl, thrust reversers, or thrust reverser translating sleeves.

Item	Condition	Result
Inbd fan cowl	Open	LE flaps hit at extend
Outbd fan cowl	Closed or Open	No contact with LE slats
Inbd T/R	Deployed	LE flaps slide on deflector during LE retraction and extension
Outbd T/R	Deployed	No contact with LE slats
Inbd T/R cowl	Raised	LE flaps hit at extend
Outbd T/R cowl	Raised	LE slats hit at extend



DMC-B737T-A-71-11-00-05A-034B-T_005-00
ICN-B737T-A-711100-A-81205-00008-A-005-01

ENGINE COWLING -- OPENING INTERFERENCE

71-11-00

02-Dec-2020

ENGINE COWLING -- FAN COWL

General

There are two fan cowls for each engine. Each fan cowl attaches to the strut with three hinges. The left and right fan cowls are not interchangeable. The inboard fan cowl has a vortex control device (chine).

The fan cowl skin, frames and intercostals are aluminum. The upper and lower longerons are titanium. A fiberglass sheet is bonded to the inner surface of the fan cowl skin for fire protection.

The outboard fan cowl weighs 143 lbs (64 kg). The inboard fan cowl weighs 152 lbs (69 kg).

Fan Cowl Latches

Three fan cowl latches secure the left and right fan cowls together. The latches are at 6:00 of the fan cowls. Push the latch trigger to release a fan cowl latch. Rotate the latch to unhook the latch from the keeper pin at the other side of the fan cowl.

Fan Cowl Latch Indication System

This system gives a visual alert when the fan cowl latches are not completely latched and closed. To unlatch the fan cowls, you must first open the oil tank access door to pull the latch indication system (LIS) handle outwards of the door frame. This will move the blade away from covering the center latch handle. The LIS handle can only be moved back completely into the oil tank access door frame when the center tension latch handle is completely stowed.

Fan Cowl Hinges

Each fan cowl attaches to the strut with three hinges.

Each fan cowl hinge has these components:

- Fan cowl clevis
- Quick release pin
- Strut lug.

The fan cowl clevis is on the fan cowl. The strut lug is on the strut. The quick release pins make it easy to remove a fan cowl.

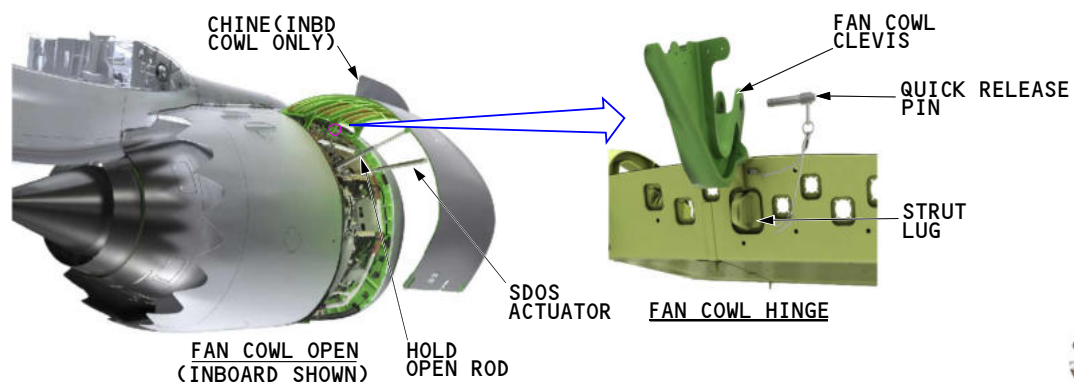
Fan Cowl Hold Open Rod

There is a hold open rod (HOR) at the aft side of each fan cowl. The HOR attaches at the fan cowl and the engine fan case engine build up (EBU) bracket. The extending HOR automatically locks when the fan cowl is moved to full open at 61 degrees.

Spring Door Opening System

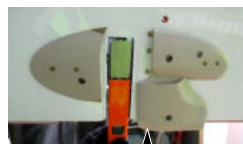
There is a spring door opening system (SDOS) actuator at the forward side of each fan cowl. The SDOS actuator attaches at the fan cowl and the engine fan case. The extending SDOS actuator automatically locks when the fan cowl is moved to full open at 61 degrees.

SDOS reduces the force to lift the fan cowl open to about 35 pounds (~16 kg). The SDOS actuator and the HOR together lock the fan cowl open for safety.

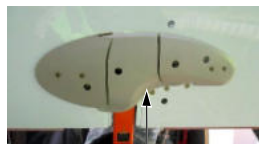


LATCH INDICATION SYSTEM HANDLE (OPEN POSITION)

OIL TANK ACCESS DOOR

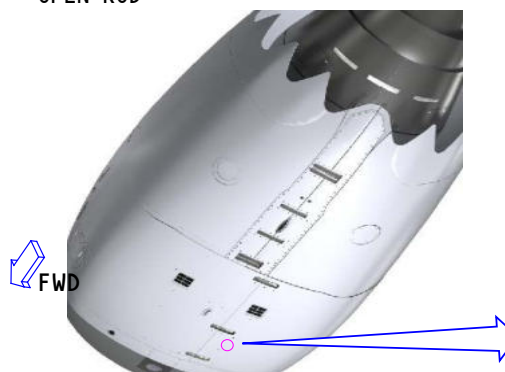


BLADE OPEN



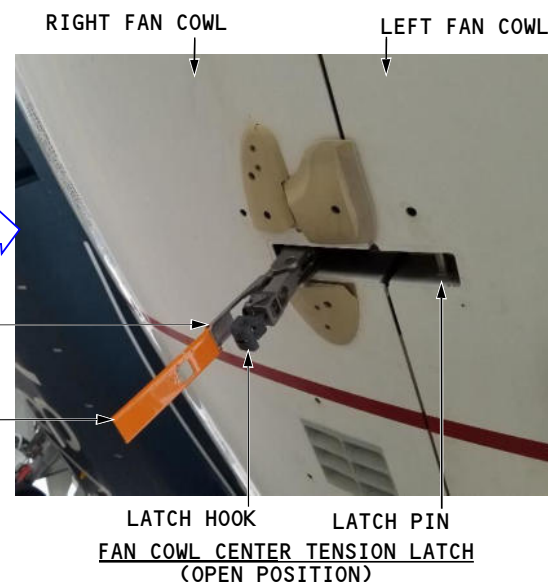
BLADE CLOSE

FAN COWL LATCH INDICATION SYSTEM



LATCH HANDLE

LATCH EXTENDED HANDLE



ENGINE COWLING -- FAN COWL

71-11-00

ENGINE COWLING -- FAN COWL HOR AND SDOS

General

Each fan cowl has a hold open rod (HOR) assembly and a spring door system (SDOS) rod assembly.

One end of the SDOS and the HOR connects to the engine fan case. The other end connects to the fan cowl.

The SDOS reduces the force that you use to open the fan cowl. As the fan cowl opens, the SDOS and HOR assemblies hold the fan cowl in a safe locked-open position for maintenance. This locking feature gives safety while you do maintenance under the fan cowl.

Operation

To open the fan cowl, unlatch the 3 fan cowl latches and pull at the bottom center of the fan cowl. As the fan cowl opens, the SDOS gives an assisting force. With this assistance, you will use about 35 pounds of force to open the fan cowl.

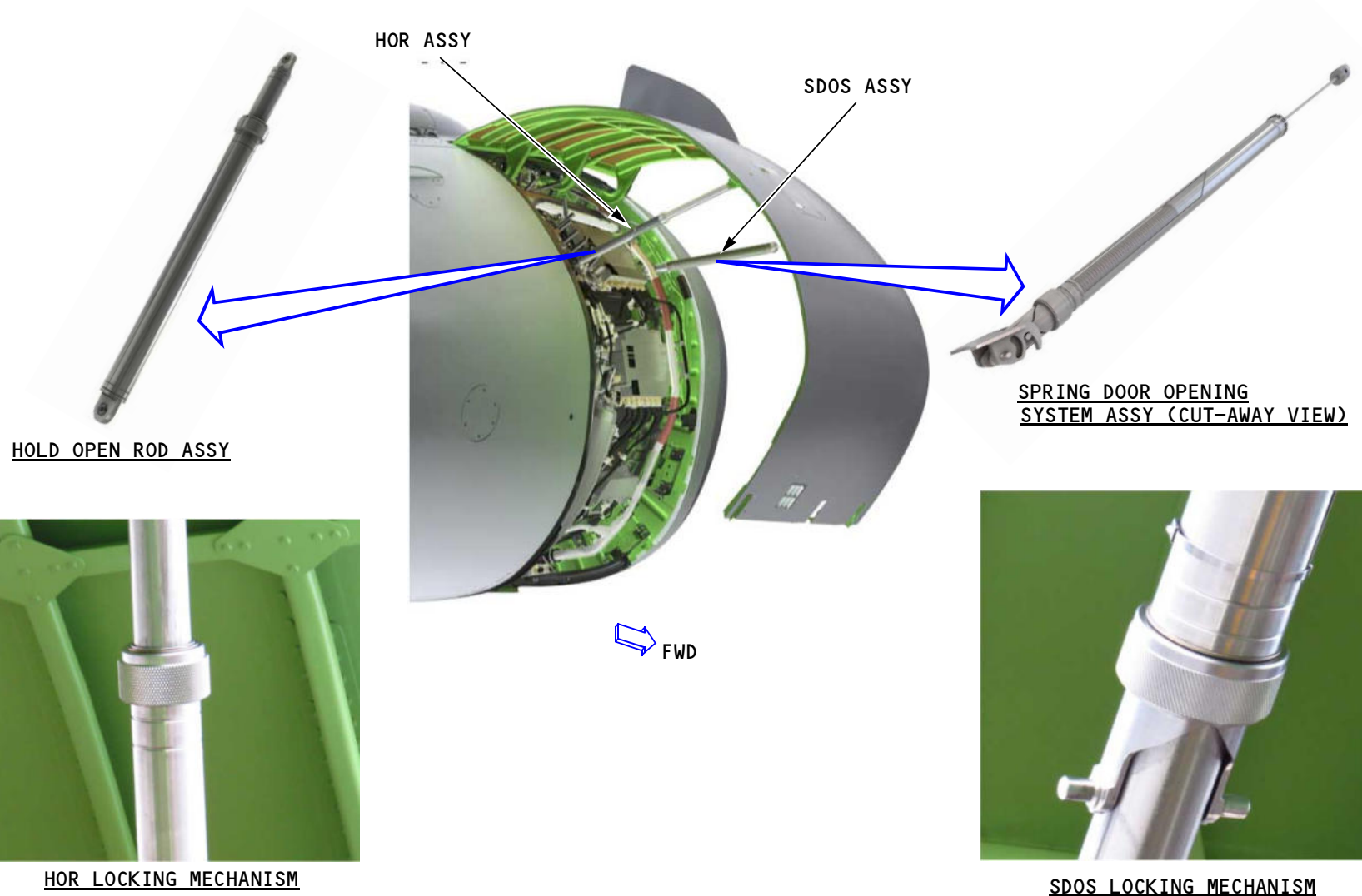
The SDOS and HOR locking mechanisms of the fan cowls automatically engage at these angles:

- Inboard fan cowl: 36 degrees
- Outboard fan cowl: 55 degrees.

To close the fan cowl, release the HOR lock mechanism to unlock. The fan cowl will sag on the HOR after it unlocks. Next, while supporting the fan cowl by hand, pull on the locking mechanism of the SDOS to unlock.

With the SDOS unlocked, the fan cowl will slowly close. From the full open position, the fan cowl closes in about 11 seconds.

DMC-B737T-A-71-11-00-07A-034B-T_004-00



DMC-B737T-A-71-11-00-07A-034B-T_004-00
ICN-B737T-A-711100-A-81205-00011-A-004-01

ENGINE COWLING -- FAN COWL HOR AND SDOS

71-11-00

02-Dec-2020

THRUST REVERSER -- TENSION LATCHES

Purpose

The tension latches hold the T/R halves together.

General

There are nine tension latches for each T/R. Numbers identify each latch.

You must follow the correct maintenance manual procedures to open and close the thrust reverser latches. You must operate the latches in the correct sequence to prevent damage to the thrust reversers or to the engine. The latch sequences to open and close the thrust reversers are different.

Location

All thrust reverser latches are on the bottom of the engine. Latches number one through seven are in a line parallel to the engine centerline. You identify them in a sequence, with latch number one most forward.

Latches number eight and nine are perpendicular to the engine centerline on the rear of the engine. They are not on the engine's centerline. When you look from the rear of the engine, latch number eight is to the left of latch number seven. Latch number nine is to the right of latch number seven.

Operation

The handles and mechanisms for latches one through six are on the left thrust reverser. The keeper pins for latches one through six are on the right thrust reverser.

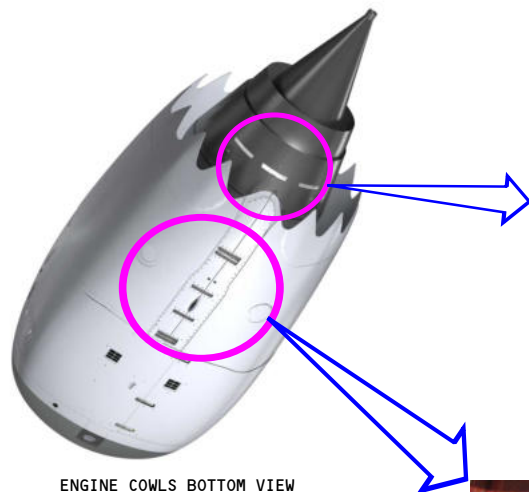
Tension latches numbers 3, 4, 5, and 6 have progressive hooks and primary hooks. The primary hooks are the hooks nearest to the latch body. Progressive hooks help you pull the thrust reverser halves together while you close the thrust reversers. They make the latches do the function of levers.

You connect the first progressive hook on latch 4 first and close the latch. Then you connect the first progressive hook on latch 3 and close the latch. Then you connect the next progressive hook on latches 4 and 3 in the same sequence.

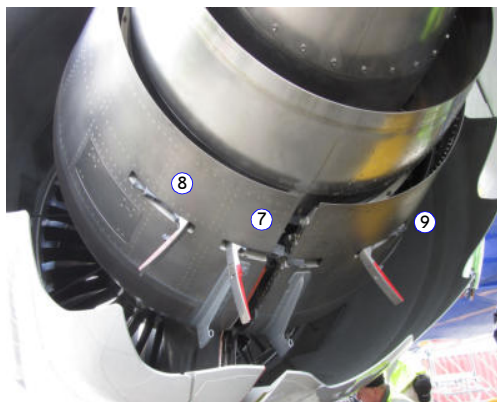
Note: While you use one latch's progressive hooks, the handle of a different latch could fall. As an example: when you engage a progressive hook on latch 3, latch 4's handle could fall. Then when you engage the next progressive hook on latch 4, latch 3's handle could fall. This is because the progressive hooks gradually decrease the distance between the thrust reversers. This decreases the tension on other latches until you engage their next progressive hook.

Continue this procedure until latches 3 and 4 use the last progressive hook before the primary hook. Engage latch 5's progressive hook. At this time you close all of the latches to close the thrust reverser halves. Engage the primary hooks of latches 3, 4, 5, and 6 in that sequence. Then engage latches 1, 2, 7, 8, and 9 in that sequence.

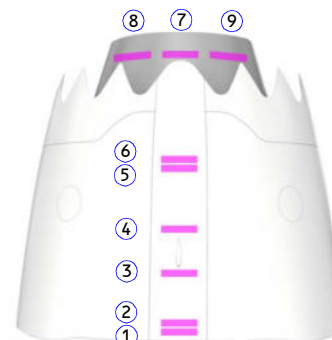
WARNING: BEFORE YOU START TO OPEN OR CLOSE THE THRUST REVERSERS, READ THE MAINTENANCE MANUAL TASK CAREFULLY. DO EACH SAFETY PROCEDURE FOR THESE TASKS IN THE AIRCRAFT MAINTENANCE MANUAL. IF YOU DO NOT DO THESE PROCEDURES, EQUIPMENT DAMAGE, INJURY, AND DEATH ARE POSSIBLE.



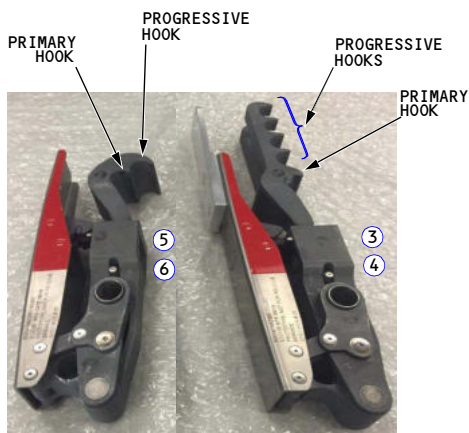
ENGINE COWLS BOTTOM VIEW



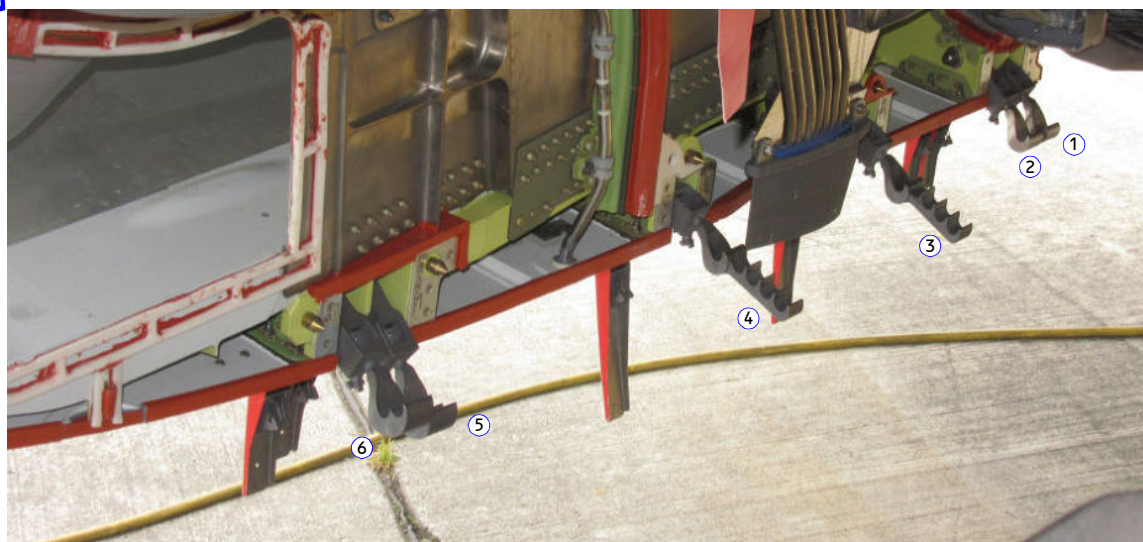
THRUST REVERSER AFT TENSION LATCHES 7, 8, AND 9



TR COWL TENSION LATCH NUMBERING



TENSION LATCHES WITH PROGRESSIVE HOOK



LEFT THRUST REVERSER HALF WITH TENSION LATCH HANDLES AND HOOKS

THRUST REVERSER -- TENSION LATCHES

78-31-00



THRUST REVERSER -- THRUST REVERSER OPENING ACTUATOR

THIS PAGE INTENTIONALLY LEFT BLANK

THRUST REVERSER -- THRUST REVERSER OPENING ACTUATOR

Purpose

Each T/R assembly opens for maintenance. When you open a T/R assembly for maintenance, it moves around the 3 hinges that connect it to the pylon. You use the T/R manual door opening system (MDOS) actuator to pump the cowl to the open position.

Location

Each T/R opening actuator is on the forward face of its T/R cowl. The upper end of the T/R opening actuator attaches to the T/R cowl. The lower end attaches to the engine fan frame extension ring. Open the fan cowls to get access to the T/R opening actuators.

Physical Description

The T/R opening actuator has these components:

- Hydraulic piston housing
- Rod
- Lock collar mechanism
- Inlet fitting
- Internal snubber assembly.

There are two different MDOS actuators on each engine. The actuator on the outboard T/R assembly is larger than the actuator on the inboard T/R assembly. The outboard MDOS actuator lets the outboard T/R assembly do these:

- Swing through an arc of 56 degrees
- Lock in position at 55 degrees.

The MDOS actuator on the inboard T/R assembly is smaller than the actuator on the outboard T/R assembly. The inboard MDOS actuator lets the inboard T/R assembly do these:

- Swing through an arc of 37 degrees
- Lock in position at 36 degrees.

Functional Description

Fluid from the hand pump causes the T/R opening actuator rod to extend and open the T/R half. As the actuator gets near the full extend position, the lock collar goes into the lock position.

When the T/R half is moved to the close position, fluid goes from the opening actuator back to the hand pump.

Operation

There are two procedures to open the T/Rs. The pump procedure and the manual procedure.

A hand pump is usually used to operate the T/R opening actuator. As the rod extends, the T/R cowl opens and the lock collar moves into the lock position. When the MDOS actuator is locked, you see the word "LOCKED" inboard of the locking collar on the actuator. This indication is engraved on the actuator rod. Refer to the pump procedure in the airplane maintenance manual (AMM).

The AMM also has the manual open and close procedure. For the manual procedure, lift and move the cowl up until the actuator lock goes into the lock position.

Training Information Point

The leading edge flaps can extend and damage the T/R cowls (halves) when they are in the open (maintenance) position. Use the AMM procedures to prevent the leading edge flap operation before you open the T/Rs.

THRUST REVERSER -- THRUST REVERSER OPENING ACTUATOR

Use the hand pump hose safety brackets when you open or close the thrust reversers. This will help prevent damage to the thrust reverser MDOS actuator and the hose fitting.

Put an actuator safety lock around the T/R opening actuator rod after the T/R cowl is open. The actuator safety lock is a safety device and is a backup if the lock collar has a failure. Two quick release pins hold the actuator safety lock around the T/R opening actuator rod.

WARNING: DO NOT OPERATE THE OPENING SYSTEM FOR THE THRUST REVERSER IF THE WIND VELOCITY IS MORE THAN 40 KNOTS. THE OPENING SYSTEM FOR THE THRUST REVERSER CAN HAVE A FAILURE IN LARGE WINDS WHICH CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

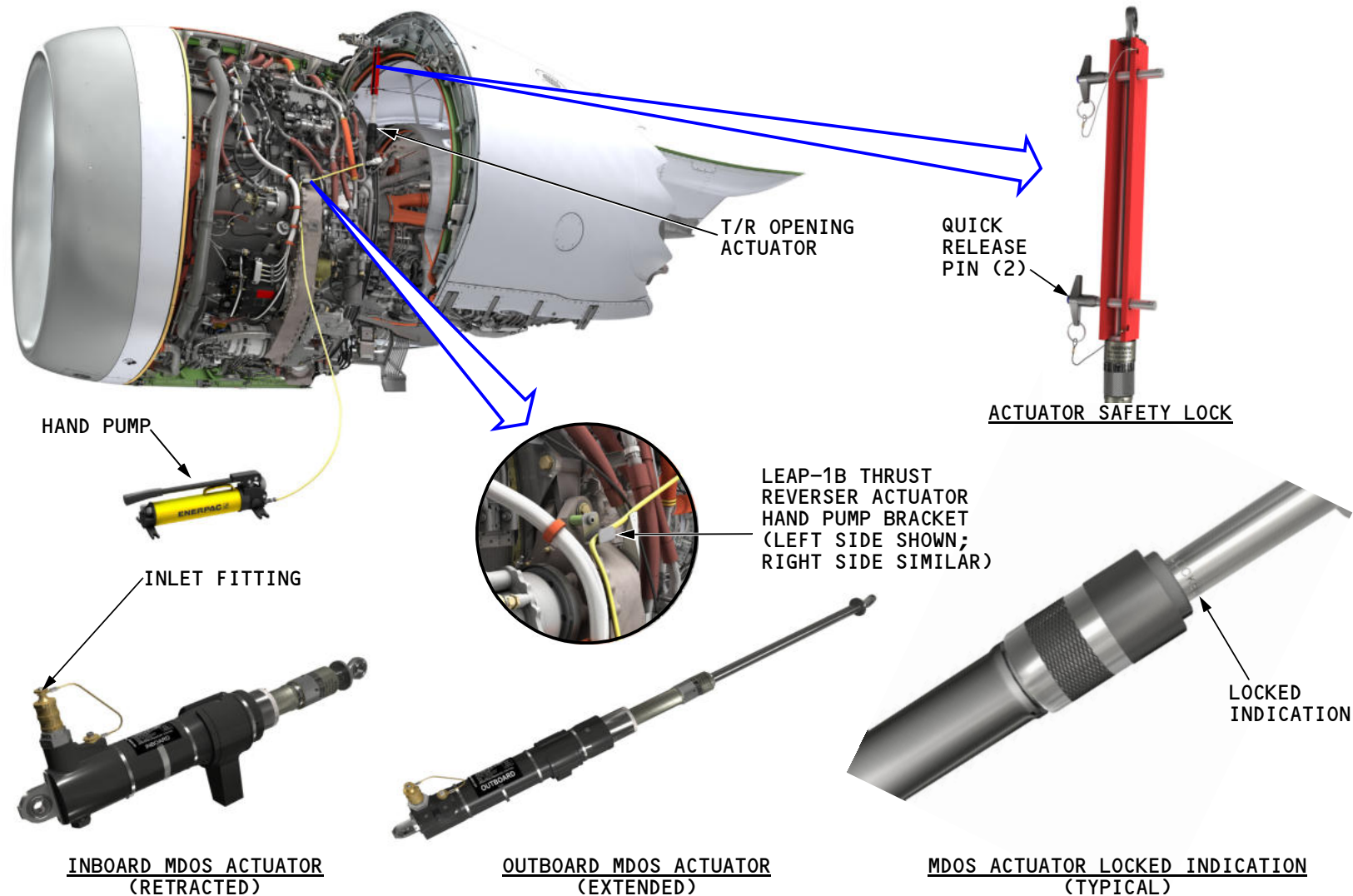
CAUTION: MAKE SURE THE ADJACENT FAN COWL PANEL IS FULLY OPEN BEFORE YOU OPEN THE REVERSER HALF. IF THE ADJACENT FAN COWL PANEL IS NOT FULLY OPEN, YOU CAN DAMAGE IT WHEN YOU OPEN THE REVERSER HALF.

CAUTION: DO NOT CAUSE THE THRUST REVERSER SLEEVE TO MOVE TO THE DEPLOYED POSITION WHEN THE THRUST REVERSER IS OPEN. THIS CAN CAUSE DAMAGE TO THE TRANSLATING SLEEVE AND THE LEADING EDGE FLAP.

DMC-B737T-A-78-31-00-06A-034B-T_004-00

02-Dec-2020

78-31-00



DMC-B737T-A-78-31-00-06A-034B-T_004-00
ICN-B737T-A-783100-A-81205-00017-A-003-01

THRUST REVERSER -- THRUST REVERSER OPENING ACTUATOR

78-31-00

02-Dec-2020



POWER PLANT -- MOUNTS AND DRAINS - INTRODUCTION

General

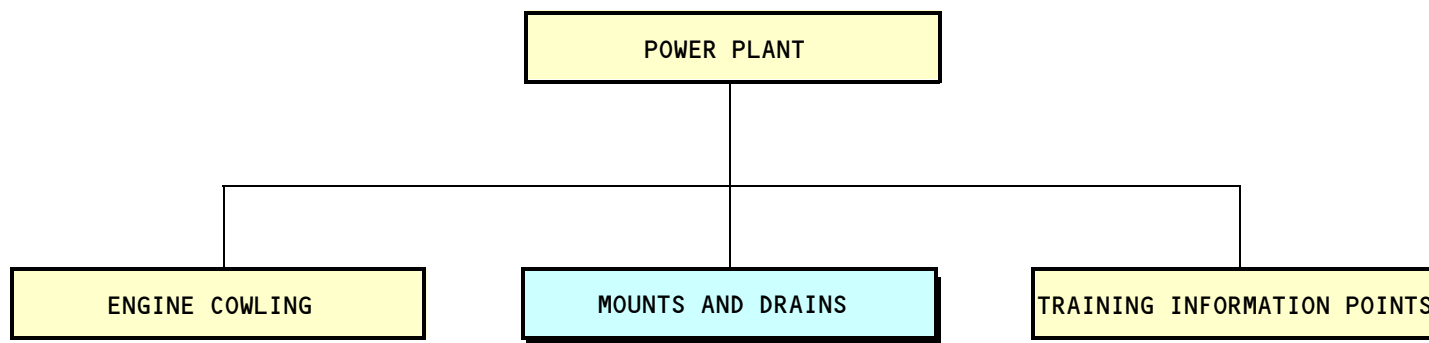
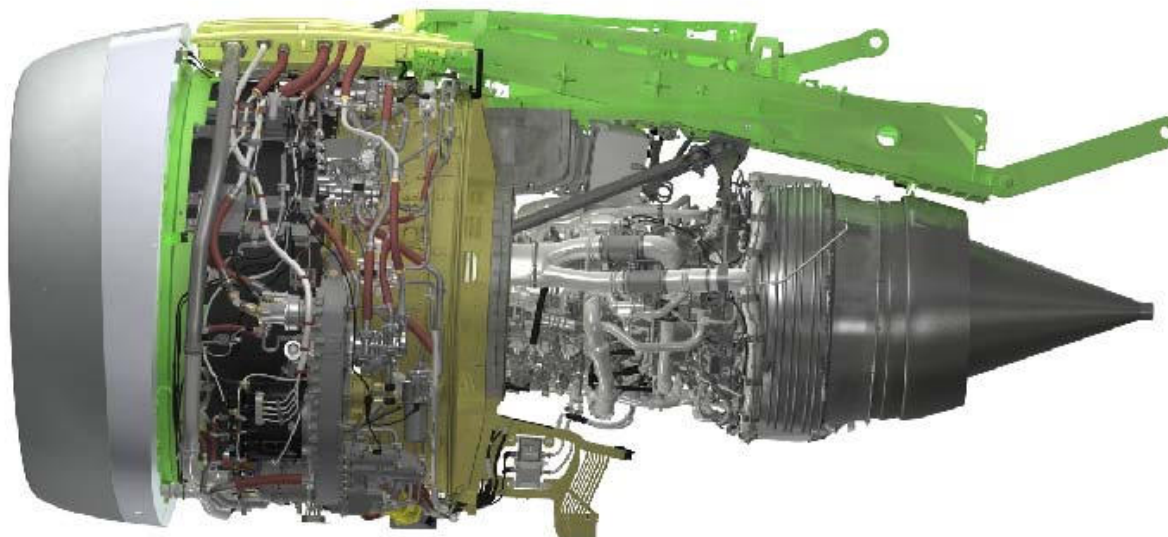
Engine mounts give the engine attachment to the airplane.

Engine drains let fluid from leaks or spillage go overboard through the bottom of the engine cowl.

DMC-B737T-A-71-00-00-05A-018A-T_001-00

02-Dec-2020

71-00-00



POWER PLANT -- MOUNTS AND DRAINS - INTRODUCTION

71-00-00

DMC-B737T-A-71-00-00-05A-018A-T_001-00
ICN-B737T-A-710000-A-81205-00056-A-002-01

02-Dec-2020



POWER PLANT -- ENGINE MOUNTS

General

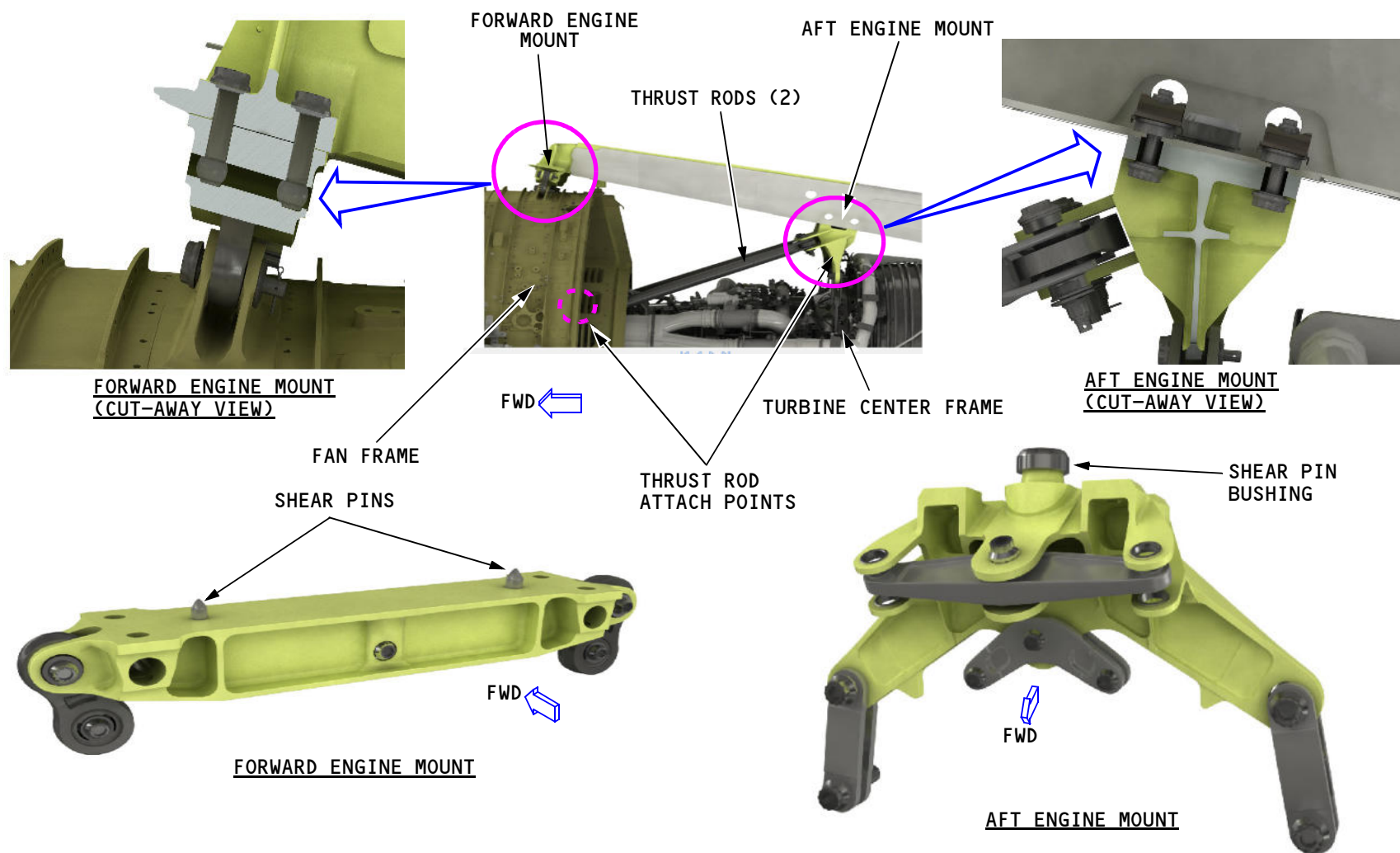
There is a forward and an aft engine mount. Each engine mount attaches the engine to the strut. The forward engine mount attaches to the fan frame. The aft engine mount attaches to the turbine center frame (TCF).

Two thrust rods attach the aft engine mount to the fan frame. The thrust rods transmit engine thrust to the engine strut through the rear engine mount.

DMC-B737T-A-71-00-00-07A-034B-T_001-00

02-Dec-2020

71-00-00



DMC-B737T-A-71-00-00-07A-034B-T_001-00
ICN-B737T-A-710000-A-81205-00053-A-001-01

POWER PLANT -- ENGINE MOUNTS

71-00-00

POWER PLANT -- ENGINE DRAINS

General

Engine drains prevent fluid contact with hot engine areas. You use engine drains to detect component failures. Engine drains direct these fluids overboard:

- Oil
- Fuel
- Hydraulic fluid
- Water
- Vapor.

Two independent engine drains direct fluids overboard from each area. There is an engine drain in the fan compartment and one in the core compartment.

Fan Compartment Drains

These components drain fluids through the fan drain mast:

- Oil tank scupper
- Fuel pump
- Hand cranking pad
- Hydraulic pump
- Integrated Drive Generator (IDG)
- Support strut.

There is a dedicated drain line for each of the components. The fan drain mast is marked to identify each drain line. A label on the oil tank also shows a schematic of these drain lines.

Core Compartment Drains

These components drain fluids from the core compartment through the core drain mast:

- A-Sump 1R
- A-Sump 2R
- B-Sump
- C-Sump
- Modulated turbine cooling (MTC) actuators
- Variable stator vane (VSV) actuators
- Variable bleed valve (VBV) actuators
- High pressure turbine active clearance control (HPTACC) valve
- Low pressure turbine active clearance control (LPTACC) valve
- Booster anti-ice/start bleed valve (BAI/SBV)
- Transient bleed valve (TBV)
- Auxiliary drain.

The oil sump drain lines are dedicated drain lines. The fuel component drain lines are joined by group. There is a drain line for components on the left and a drain line for components on the right. Coupling joints and witness tees assist in fault detection for these engine fuel drain lines. The core drain mast has markings to identify each drain line.

Leakage Limits

Leaks from drain lines can occur on an engine that is not operating or during engine start. These leaks are normal, and may stop after the engine operates at idle power for 5 minutes.

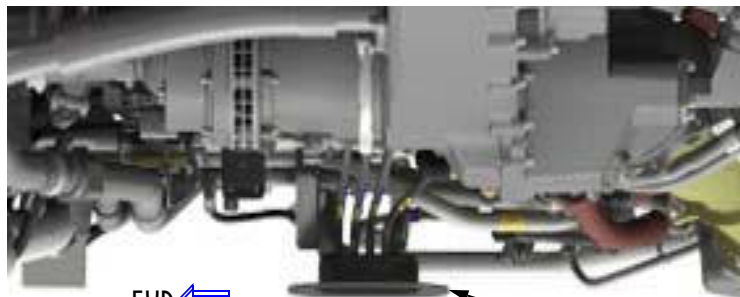
Oil leaks are classified as:

- less than the threshold limit, no maintenance required
- more than the threshold limit but less than the serviceable limit, corrective action required within 25 flight cycles
- more than the serviceable limit, corrective action is required.

The aircraft maintenance manual (AMM) Chapter 71 contains leakage limit specifications for each drain line.

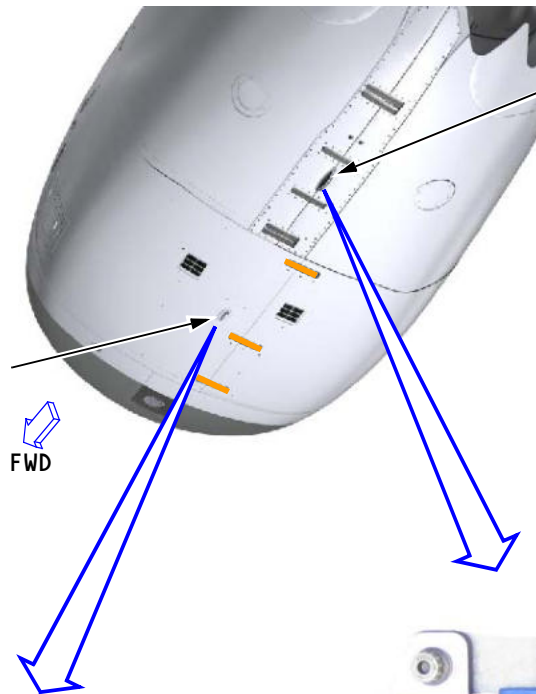


DRAIN PLACARD (ON OIL TANK)



FWD

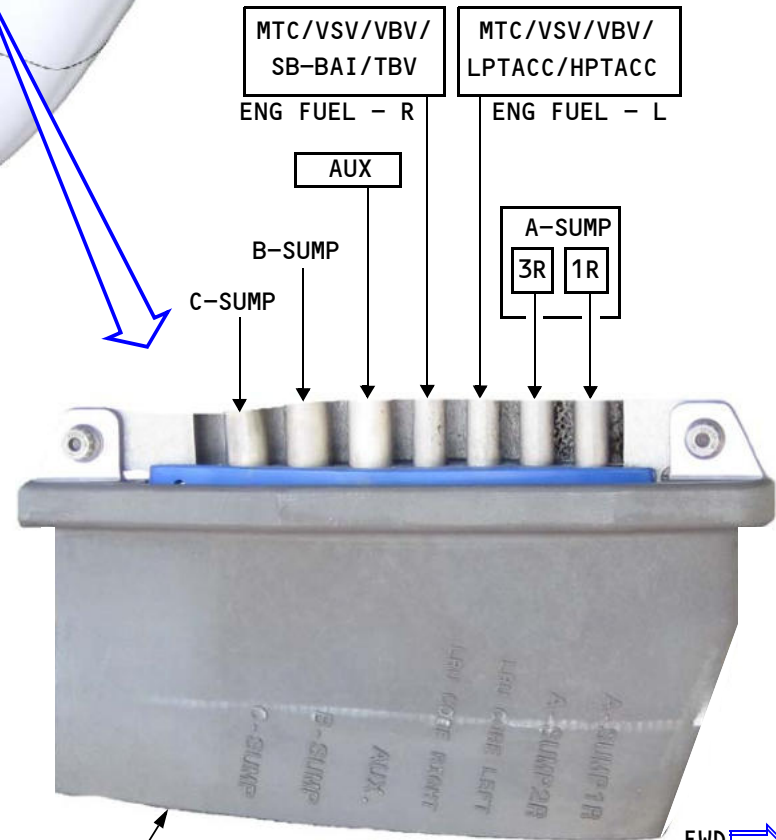
FAN COMPARTMENT DRAINS



FAN DRAIN
MAST

FWD

CORE DRAIN MAST



CORE DRAIN MAST

CORE COMPARTMENT DRAINS

FWD

POWER PLANT -- ENGINE DRAINS

71-00-00

POWER PLANT -- TRAINING INFORMATION POINTS - - INTRODUCTION

General

This section gives procedures to remove and install the power plant.

It also gives information about the procedures to preserve and depreserve the power plant on the airplane.

Power Plant Removal and Installation

The procedure to remove and install the power plant has these general sections:

- Preparation
- Remove the power plant
- Install the power plant.

This is a summary of procedures in the Aircraft Maintenance Manual (AMM). For detailed procedures, refer to the AMM.

Preservation and Depreservation

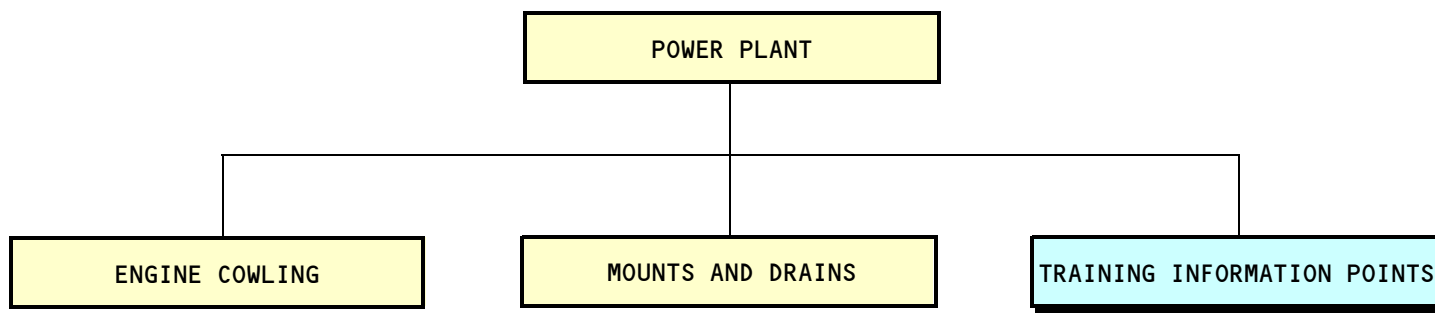
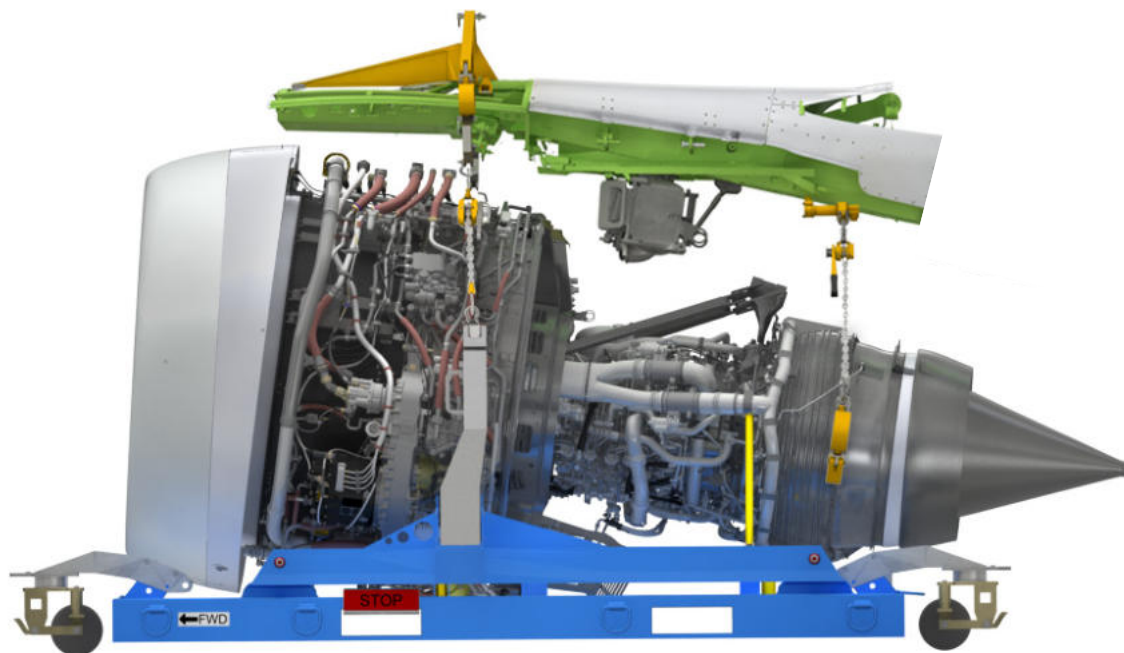
Preservation procedures for the power plant give protection from environmental damage when the airplane is not in use for an extended period of time.

Depreservation of the power plant returns power plant systems to normal operation after the airplane has not been used for an extended period of time.

DMC-B737T-A-71-00-00-02A-300A-T_001-00

02-Dec-2020

71-00-00



POWER PLANT -- TRAINING INFORMATION POINTS -- INTRODUCTION

71-00-00

DMC-B737T-A-71-00-00-02A-300A-T_001-00
ICN-B737T-A-710000-A-81205-00057-A-001-01

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - GENERAL DESCRIPTION

General

The power plant removal procedure is in the Aircraft Maintenance Manual (AMM).

Before you remove the power plant, you must prepare the airplane and the ground support equipment

Power Plant Removal Preparation

These are the general procedures to prepare for power plant removal:

- If you replace the engine with a different engine, record engine related faults and vibration history
- Ground airplane
- Make airplane level
- Make sure that there is enough clearance for engine stand under the engine
- Open circuit breakers
- Remove fan cowls
- Open thrust reverser.

It is an option to remove the inlet cowl when the power plant is on the wing or on the engine stand.

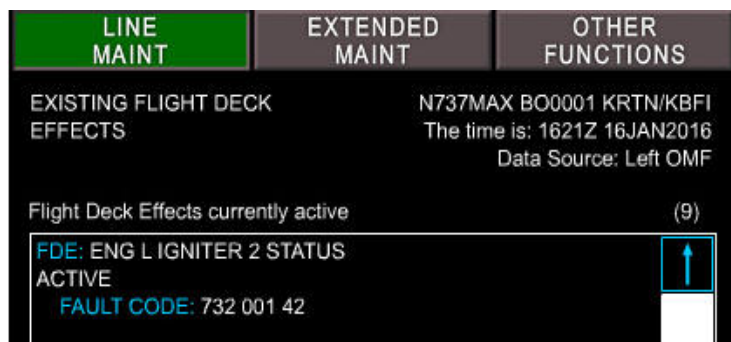
If you remove the inlet cowl when the power plant is on wing, the inlet cowl attach bolt at the 12:00 position is not easy to remove.

If you remove the inlet cowl when the power plant is on the engine stand, the lower inlet cowl bolts are not easy to remove. It is recommended that you remove the inlet cowl lower attach bolts before you lower the power plant from the strut. When the power plant is on the engine stand, remove the remaining bolts to complete the inlet cowl removal.

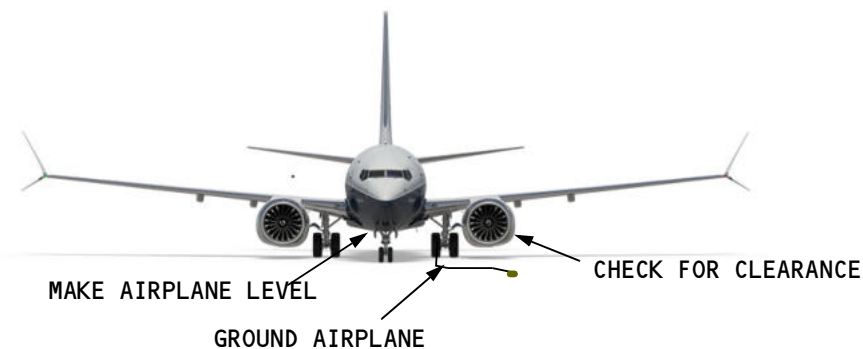
DMC-B737T-A-71-00-00-04A-T_001-00

02-Dec-2020

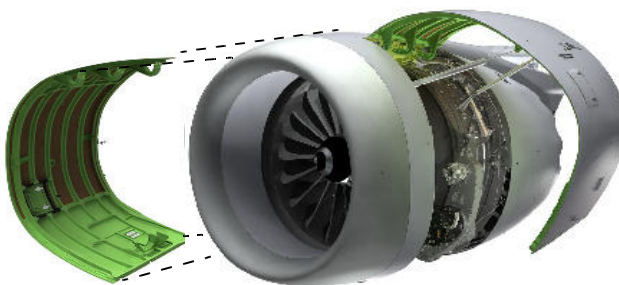
71-00-00



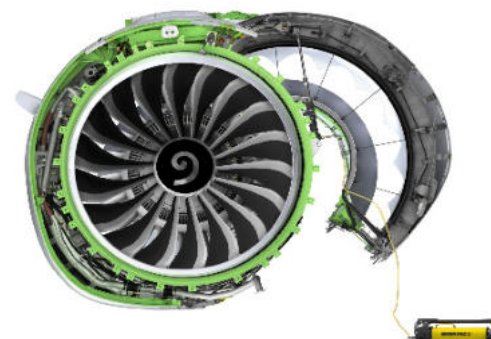
RECORD ENGINE RELATED FAULTS AND VIBRATION HISTORY



OPEN CIRCUIT BREAKERS



REMOVE FAN COWLS



OPEN THRUST REVERSERS

POWER PLANT -- LEAP-1B REMOVAL - GENERAL DESCRIPTION

71-00-00

POWER PLANT -- LEAP-1B REMOVAL - PREPARATION 1

Preparation for Power Plant Removal

The procedures for power plant removal are in the aircraft maintenance manual (AMM) part II practices and procedures. Make sure to follow the cautions/warnings in these procedures to prevent personnel injury and damage to equipment.

The AMM also gives a list of equipment, consumable materials, access and required parts for the power plant removal. Use the AMM and the illustrated parts catalog (IPC) to identify and acquire the items for the task.

The bootstrap kit includes the forward and aft bootstrap, cradles, chains and engine stand. You use lever hoists with the bootstraps to lift and to lower the power plant.

Support equipment includes a hydraulic hand pump and thrust reverser hold open equipment. You use the hand pump to open the thrust reverser and the hold open equipment to keep the reverser open.

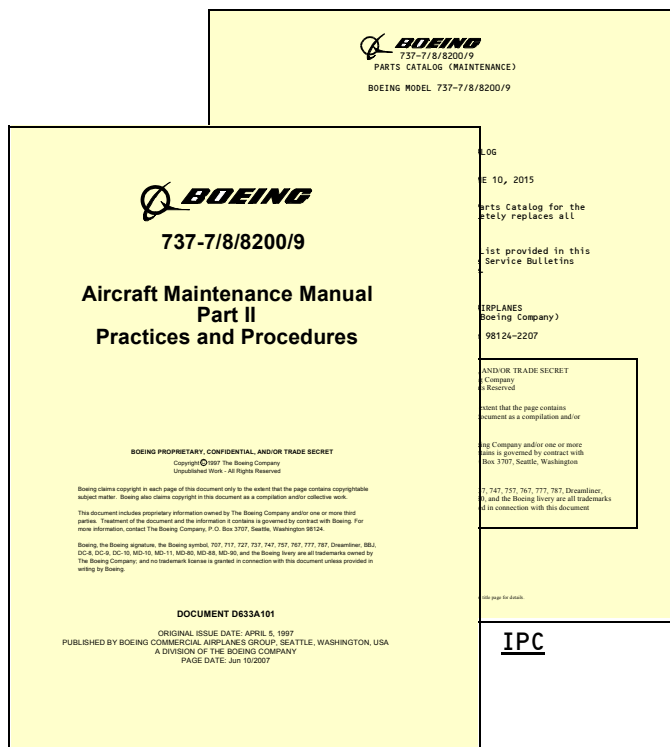
DMC-B737T-A-71-00-00-34A-100A-T_001-00

02-Dec-2020

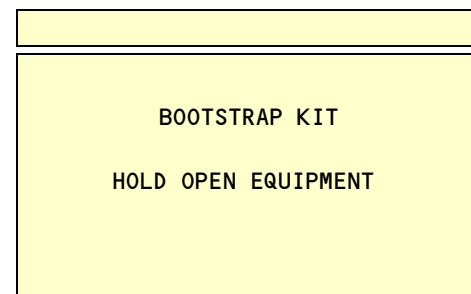
71-00-00



737-7/8/9/10 Maintenance Training Manual



AIRCRAFT MAINTENANCE MANUAL



SUPPORT EQUIPMENT

POWER PLANT -- LEAP-1B REMOVAL - PREPARATION 1

71-00-00

POWER PLANT -- LEAP-1B REMOVAL - PREPARATION 2

Prepare the Airplane for Engine Removal

Make sure the airplane is within 1.5 degrees of a nose-down attitude.

Make sure there is minimum of 5.0 - 8.0 inches (12.7 -20.3 cm) shock strut chrome at the applicable main landing gear oleo. This is one method to provide clearance between the precooler and the aft engine mount after engine removal. If there is not sufficient clearance for the removed engine to move forward, raise the main landing gear strut to increase clearance. An alternate to this procedure is to rotate and tilt the power plant prior to engine movement under the strut.

Make sure that the applicable engine start switch is in auto and the start lever is in the CUTOFF position. Attach DO-NOT-OPERATE tags to the engine start switch and to the start lever.

Pull the applicable engine fire switch handle up, but do not turn it. Attach a DO-NOT-OPERATE tag to the engine fire switch handle.

Make sure that the applicable ENG VALVE CLOSED L (R) and SPAR VALVE CLOSED L (R) lights are ON.

Open the necessary circuit breakers on P6 and P18 circuit breaker panels. This will remove power from the fuel spar valve, engine driven pump supply shutoff, engine ignition, fire protection, start valve, engine fuel high pressure shutoff valve (HPSOV) and the engine electronic controllers. Attach DO-NOT-CLOSE tags on the opened circuit breakers.

Put the HYD PUMPS ENG switch on the P5 overhead panel to the OFF position for the applicable engine to remove hydraulic power.

CAUTION: DO NOT TURN THE FIRE HANDLE AFTER YOU PULLED IT UP. YOU WILL CAUSE THE FIRE BOTTLES

TO RELEASE THEIR CONTENTS IF YOU TURN THE HANDLE.

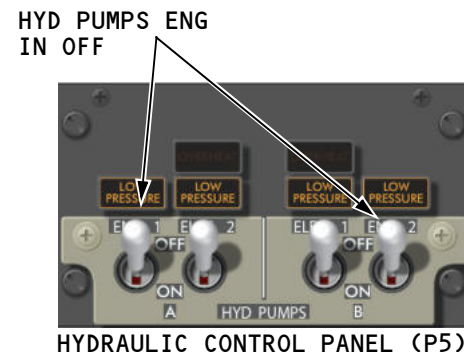
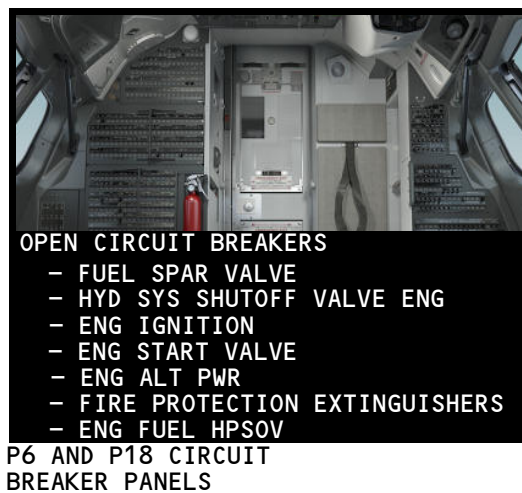
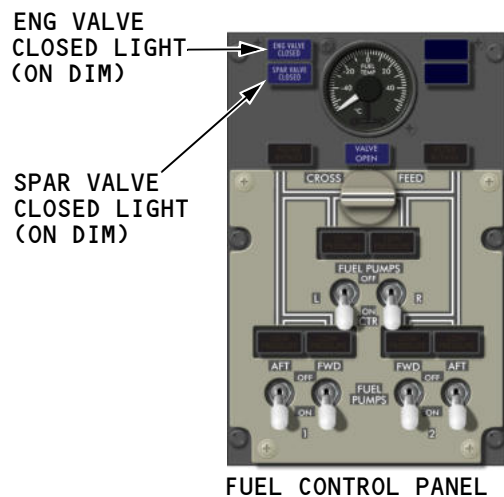
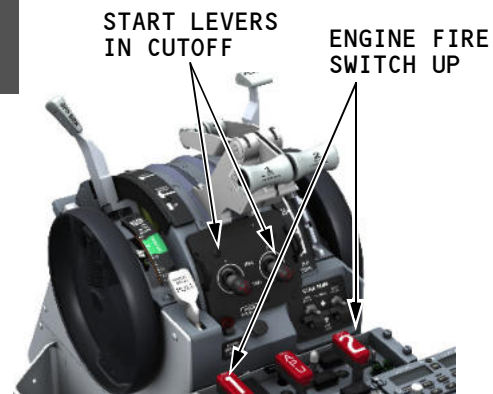
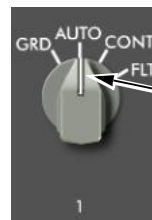
DMC-B737T-A-71-00-00-35A-100A-T_003-00

02-Dec-2020

71-00-00



737-7/8/9/10 Maintenance Training Manual



DMC-B737T-A-71-00-00-35A-100A-T_003-00
ICN-B737T-A-710000-A-81205-00060-A-003-01

POWER PLANT -- LEAP-1B REMOVAL - PREPARATION 2

71-00-00

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - FAN COWL AND THRUST REVERSER

Prepare the Engine for Removal

Remove the left and right fan cowl:

- Unlatch the fan cowl
- Pull up on the bottom of the fan cowl and open
- Disconnect spring door opening system (SDOS) and hold open rod (HOR) from fan cowl and fan case
- Disconnect the fan cowl panel from the hinges and remove it
- Repeat the procedure for the other fan cowl.

Open the thrust reverser C-ducts:

- Unlatch thrust reverser latches from aft to forward
- Use the hydraulic hand pump to open the C-duct. The outboard thrust reverser cowl locks at 55 degrees and the inboard cowl locks at 37 degrees. You hear a click of the lock collar and LOCKED shows on bottom of extended piston
- Install the actuator safety assembly on the extended piston rod to hold the thrust reverser open
- Disconnect the hydraulic hand pump from the thrust reverser opening actuator
- Repeat the procedure for the other thrust reverser C-duct.

Remove electrical power. Install a DO-NOT-OPERATE tag on the BAT switch at the pilots overhead panel, P5.

WARNING: DO THE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSER: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSERS (FOR MAINTENANCE), OPEN THE FAN COWL PANEL. IF YOU DO NOT OBEY THE ABOVE SEQUENCE,

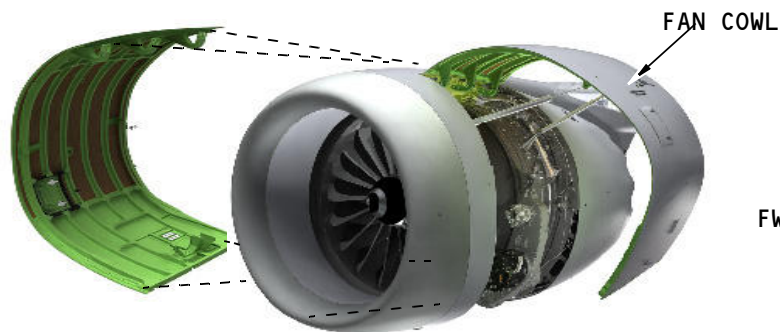
INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING: REMOVE THE ELECTRICAL POWER BEFORE YOU DISCONNECT A FUEL, HYDRAULIC OR ELECTRICAL LINE. AN ACCIDENTAL OPERATION OF THE PRESSURIZED FLUIDS OR ELECTRICAL CIRCUITS CAN CAUSE A FIRE. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

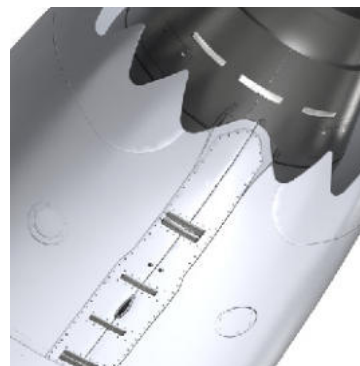
DMC-B737T-A-71-00-00-36A-100A-T_002-00

02-Dec-2020

71-00-00



REMOVE FAN COWLS



OPEN THRUST REVERSER LATCHES

LATCH
HANDLE



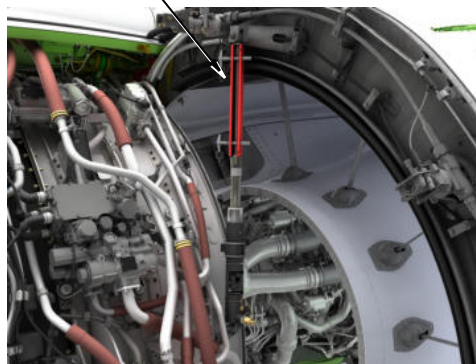
T/R LATCH HANDLE

REMOVE
ELECTRICAL
POWER



ELECTRICAL PANEL (P5)

ACTUATOR
SAFETY ASSEMBLY



OPEN THRUST REVERSER

POWER PLANT -- LEAP-1B REMOVAL - FAN COWL AND THRUST REVERSER

71-00-00

POWER PLANT -- LEAP-1B REMOVAL - INSTALL T/R HOLD OPEN EQUIPMENT

Thrust Reverser Preparation

The hold-open equipment keeps the thrust reverser open when you remove and install the power plant. The hold-open equipment is symmetrical between the inboard and outboard sides.

The hold-open equipment has these components:

- Fitting assemblies (left, right)
- Arm assemblies (inboard, outboard)
- Pins assemblies (lynch, pivot, stop, arm).

Do these steps to install the hold-open equipment:

- Remove the thrust reverser insulation blankets that cover the hold-open equipment attach points
- Back out the drive screws to move the pin holes to lowest position on the left fitting and right fitting assemblies
- Attach the left and right fitting assemblies to the thrust reverser precooler fittings
- Attach the inboard arm assembly from the outboard side to the right fitting assembly
- Attach the outboard arm assembly to the left fitting assembly and to the stop pin and retaining ring of the inboard arm assembly
- Adjust the drive screws until the inboard arm assembly pin hole aligns to the outboard arm assembly pin hole and install the pivot and lynch pins
- Adjust the drive screws until the drive screws take the weight of the thrust reverser.

Lower the thrust reverser on to the hold-open equipment.

Disconnect the manual door opening system (MDOS) actuators and stow on the thrust reverser.

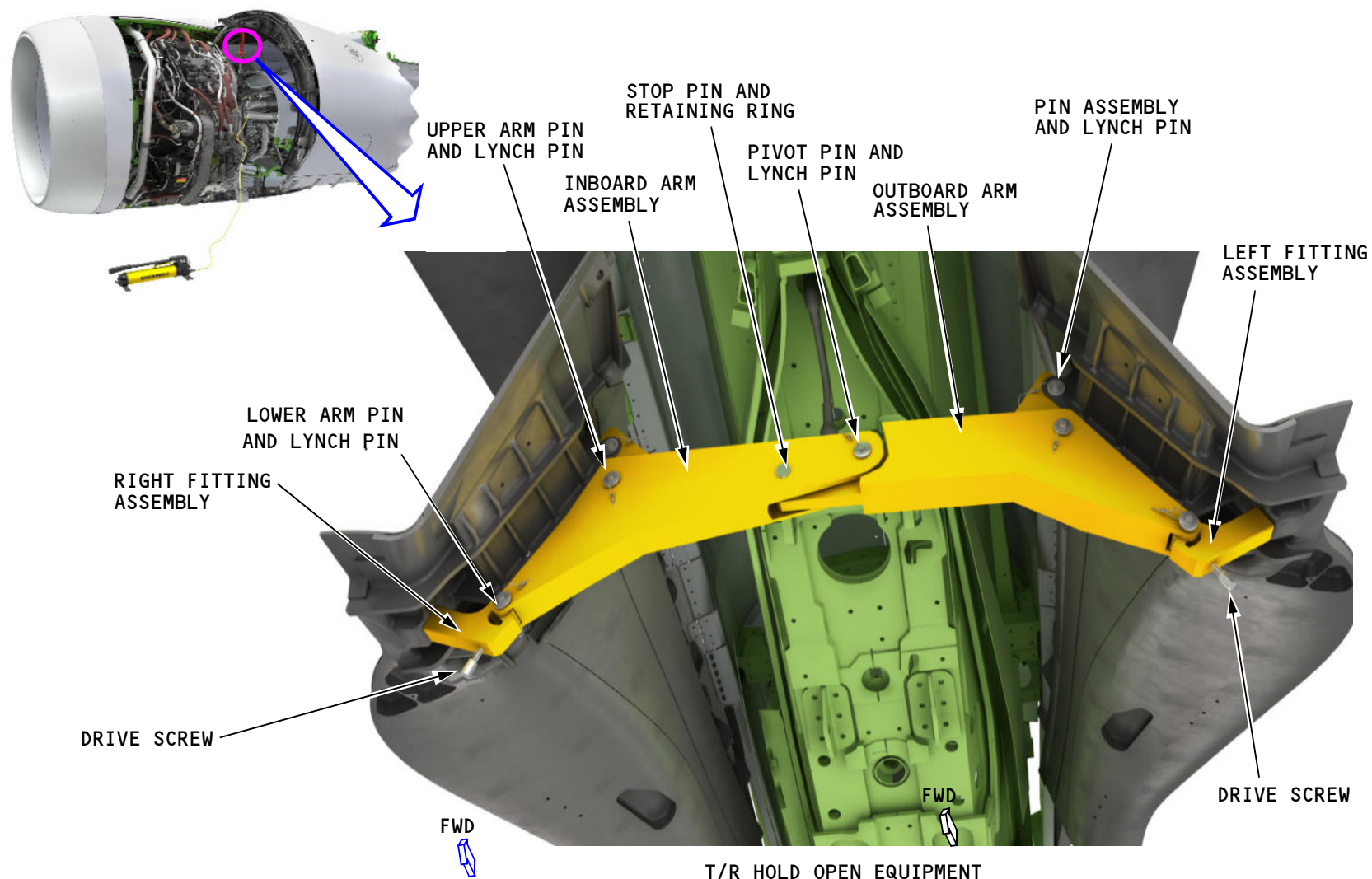
WARNING: DO NOT OPEN THE THRUST REVERSER IF WIND SPEED IS MORE THAN 40 KNOTS. INJURY CAN OCCUR IF THE THRUST REVERSER SUDDENLY MOVES.

WARNING: MAKE SURE ALL THE WEIGHT OF THE THRUST REVERSERS IS HELD BY THE OPENING ACTUATORS. IF THEY ARE NOT, THE THRUST REVERSERS CAN MOVE SUDDENLY WHEN YOU REMOVE THE HOLD-OPEN EQUIPMENT. INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

DMC-B737T-A-71-00-00-37A-100A-T_002-00

02-Dec-2020

71-00-00



DMC-B737T-A-71-00-00-37A-100A-T_002-00
ICN-B737T-A-710000-A-81205-00062-A-003-01

POWER PLANT -- LEAP-1B REMOVAL - INSTALL T/R HOLD OPEN EQUIPMENT

71-00-00

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - DISCONNECT T/R OPENING ACTUATOR

T/R Opening Actuators

Do these steps to disconnect the thrust reverser (T/R) opening actuators:

- Remove the T/R opening actuator safety lock
- Remove the dust cap and connect the pump to the actuator
- Open the return valve on the pump
- Push on the actuator lock collar to disengage the lock
- Let the thrust reverser slowly lower to the thrust reverser hold-open equipment
- Disconnect the pump from the actuator and install the dust cover
- Disconnect the actuator at the lower end
- Temporarily attach the lower end of the actuator to the thrust reverser with a tie
- Repeat the procedure for the other thrust reverser opening actuator.

Put tape around the edge of the thrust reverser latches to keep them in the closed position. This gives clearance for the dolly on the inboard side.

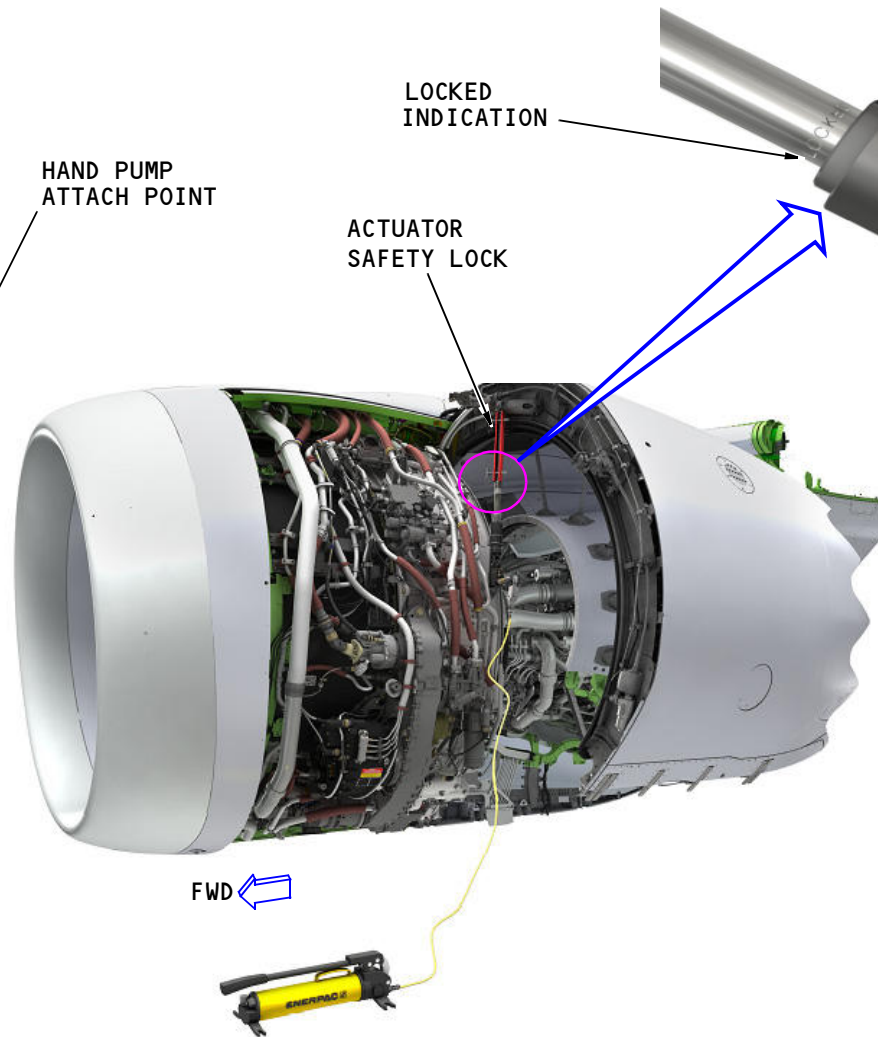
DMC-B737T-A-71-00-00-38A-100A-T_001-00

02-Dec-2020

71-00-00



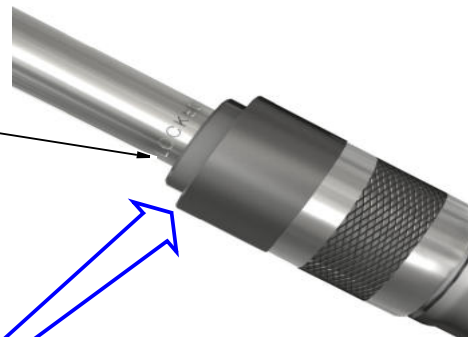
T/R OPENING ACTUATOR



FWD

LOCKED
INDICATION

ACTUATOR
SAFETY LOCK



T/R OPENING
ACTUATOR (STOWED)

DMC-B737T-A-71-00-00-38A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00063-A-001-01

POWER PLANT -- LEAP-1B REMOVAL - DISCONNECT T/R OPENING ACTUATOR

71-00-00

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - FAN CASE DISCONNECT LH

LH Fan Case Disconnects

All fan case disconnects are at the service disconnect panels on the left and right sides of the fan cowl support beam. Disconnect all electrical, pneumatic, hydraulic and fuel lines at the left side of the support beam.

When you disconnect the fuel supply line, it is necessary to drain all remaining fuel in the line. You drain this fuel from the fuel metering unit (FMU) and main fuel pump drain plugs. There is one drain plug on the FMU and three drain plugs on the main fuel pump.

After disconnecting the lines and connectors, install protective covers to protect and to prevent contamination.

EBU Bracket

Remove the engine build up (EBU) bracket from the fan case module. This is the bracket that connects the left fan cowl hold open rod to the fan case.

WARNING: DO NOT GET FUEL IN YOUR MOUTH, OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION: A LARGE VOLUME OF FUEL WILL DRAIN FROM THE FUEL SUPPLY SYSTEM. TO PREVENT A FUEL SPILL, YOU MUST USE A MINIMUM OF TWO CONTAINERS TO DRAIN THE FUEL SYSTEM.

CAUTION: BE CAREFUL WITH THE HYDRAULIC HOSE. DO NOT BEND OR PUT THE HOSE INTO A COIL TOO TIGHTLY.

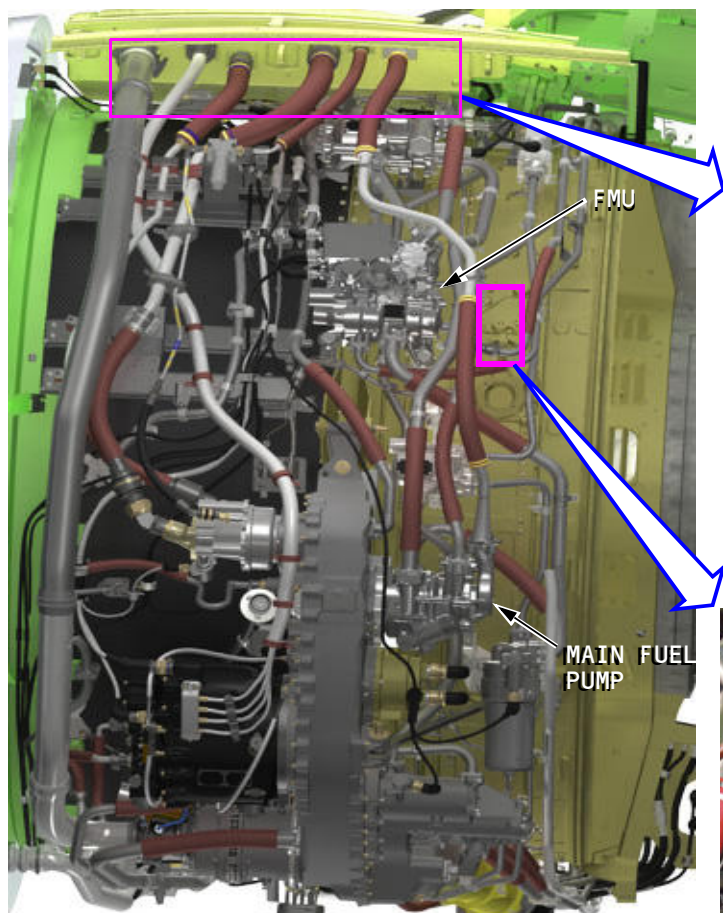
YOU CAN CAUSE A BLOCKAGE OF THE HOSE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE A FAILURE OF THE ENGINE-DRIVEN PUMP (EDP).

CAUTION: DO NOT LET HYDRAULIC FLUID LEAKAGE COLLECT. IF YOU DO NOT MAKE THE AREA CLEAN, DAMAGE CAN OCCUR FROM THE HYDRAULIC FLUID.

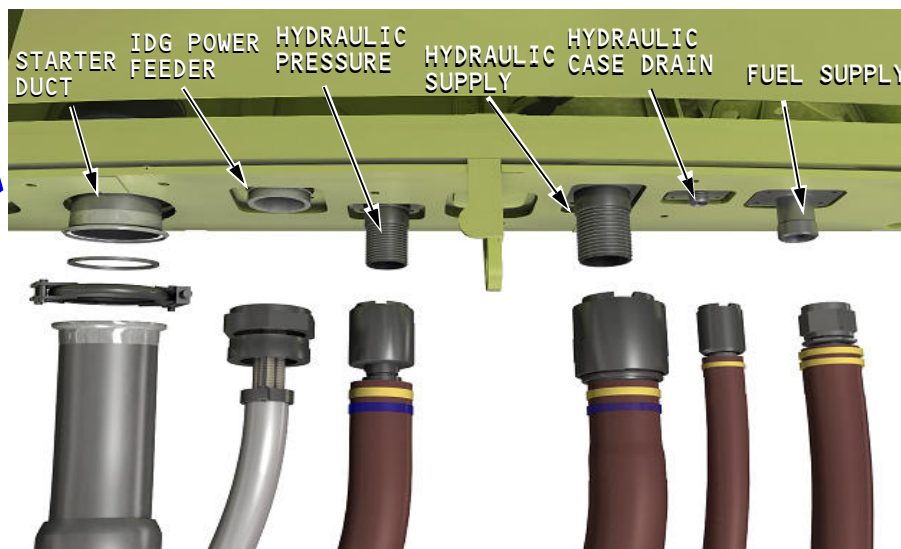
DMC-B737T-A-71-00-00-39A-100A-T_002-00

02-Dec-2020

71-00-00



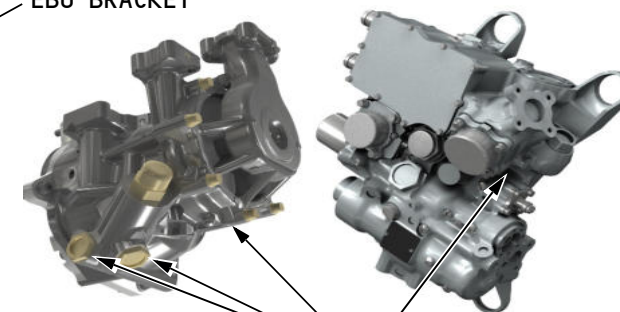
L FAN CASE



L FAN CASE STRUT CONNECTIONS



EBU BRACKET



MAIN FUEL PUMP

FMU

DRAIN PLUGS

POWER PLANT -- LEAP-1B REMOVAL - FAN CASE DISCONNECT LH

71-00-00

DMC-B737T-A-71-00-00-39A-100A-T_002-00
ICN-B737T-A-710000-A-81205-00064-A-002-01

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - FAN CASE DISCONNECT RH

RH Fan Case Disconnects

All electrical disconnects on the right side of the fan case are at the right service disconnect panel on the support beam.

Disconnect the electrical connectors.

After the disconnect, install protective covers on the connectors and receptacles.

Disconnect the strut drain line from the strut and install a protective cover.

Remove the engine build up (EBU) bracket. This is the bracket that connects the right fan cowl hold open rod to the fan case.

CAUTION: MAKE SURE THAT THE ELECTRICAL CONNECTORS ARE CLEAN BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

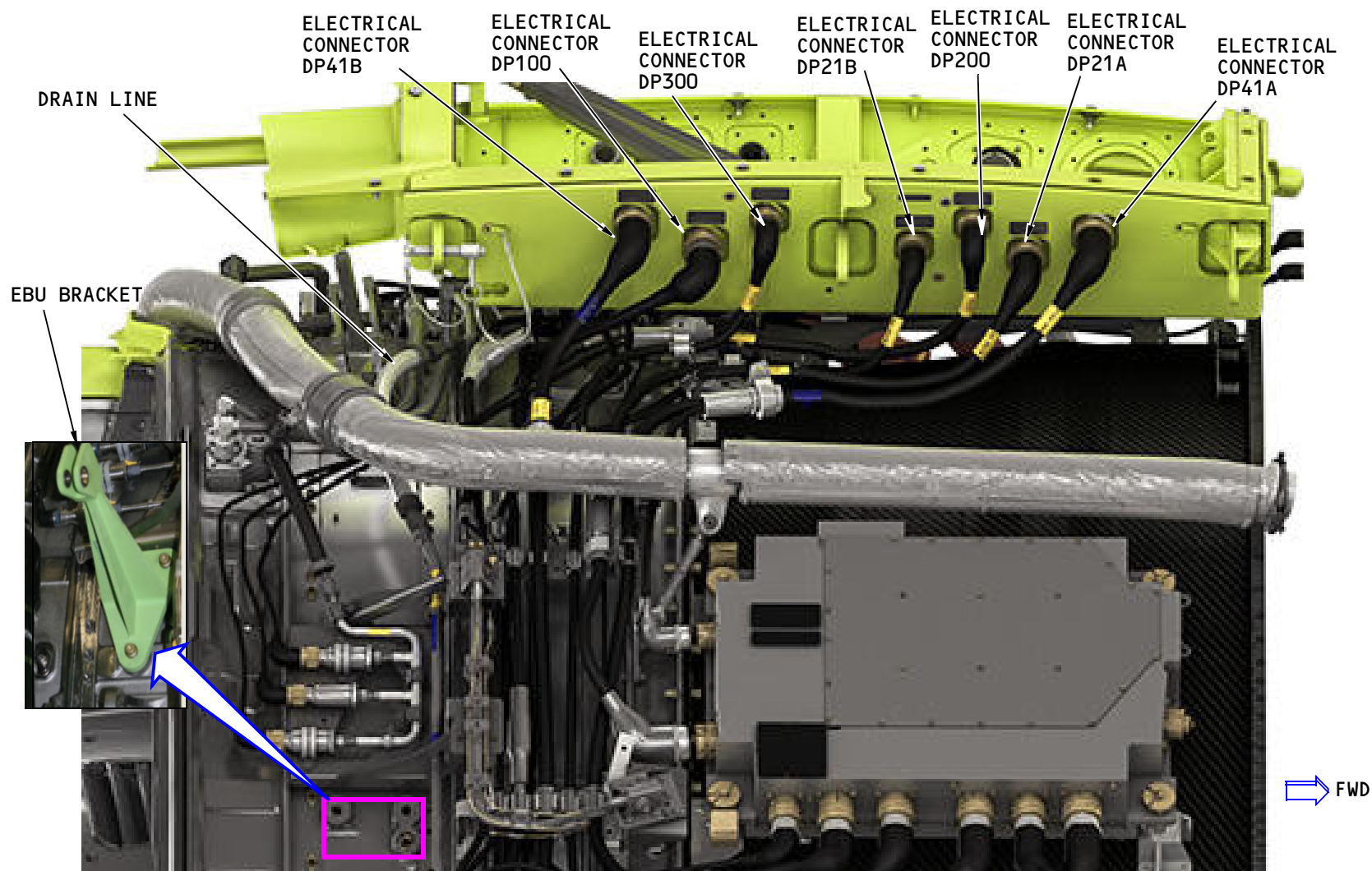
CAUTION: USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS CAN OCCUR.

DMC-B737T-A-71-00-00-40A-100A-T_003-00

02-Dec-2020

71-00-00

DMC-B737T-A-71-00-00-40A-100A-T_003-00
ICN-B737T-A-710000-A-81205-00065-A-004-01



POWER PLANT -- LEAP-1B REMOVAL - FAN CASE DISCONNECT RH

71-00-00

02-Dec-2020

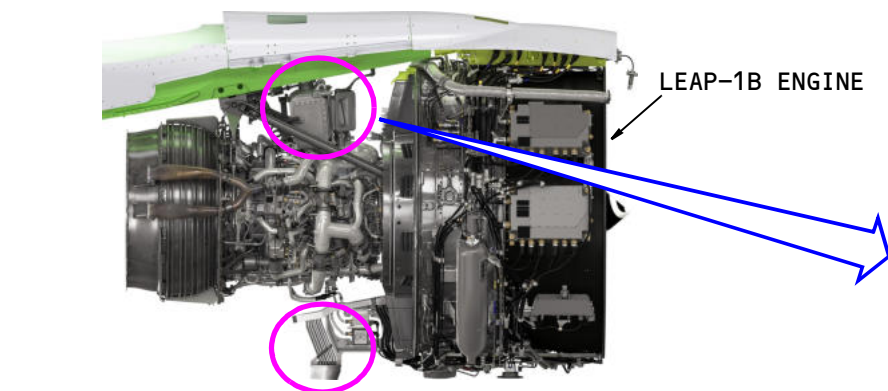
POWER PLANT -- LEAP-1B REMOVAL - ENGINE CORE

Engine Core Disconnect

Remove the crossover duct from the precooler inlet. After removal, install protective caps on the crossover duct and the precooler inlet.

Disconnect the Fan Air Modulating Valve (FAMV) under the precooler on the right side of the core. After the FAMV disconnect, install protective caps on the FAMV line and the hose fitting.

Remove the core drain mast and install protective caps on the drain lines.



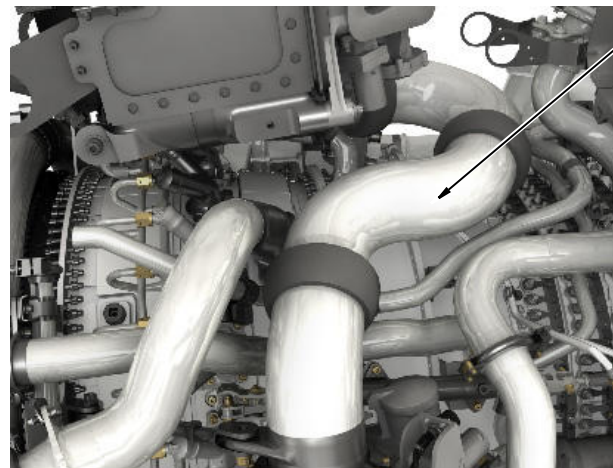
DISCONNECT
- FAMV MUSCLE LINE

REMOVE
- CROSSOVER DUCT
- CORE DRAIN MAST



PRECOOLER AREA (RIGHT SIDE)

⇒ FWD



PRECOOLER AREA (RIGHT SIDE)

⇒ FWD



CORE DRAIN MAST

DMC-B737T-A-71-00-00-42A-100A-T_002-00
ICN-B737T-A-710000-A-81205-00067-A-002-01

POWER PLANT -- LEAP-1B REMOVAL - ENGINE CORE

71-00-00

POWER PLANT -- LEAP-1B REMOVAL - INSTALL FORWARD BOOTSTRAP

Install Forward Bootstrap Equipment

For engine removal, you use the bootstrap system to lower the engine from the strut to the engine dolly. The same bootstrap system is used for engine installation.

The bootstrap system has components that include a forward bootstrap and an aft bootstrap. The bootstrap system components are in the bootstrap kit.

The forward bootstrap has these components:

- Forward center beam assembly
- Forward outboard arm assembly
- Forward inboard arm assembly
- Forward support arm beam assembly
- Dynamometers (2)
- Lever hoists (2)
- Lock pins and retainers.

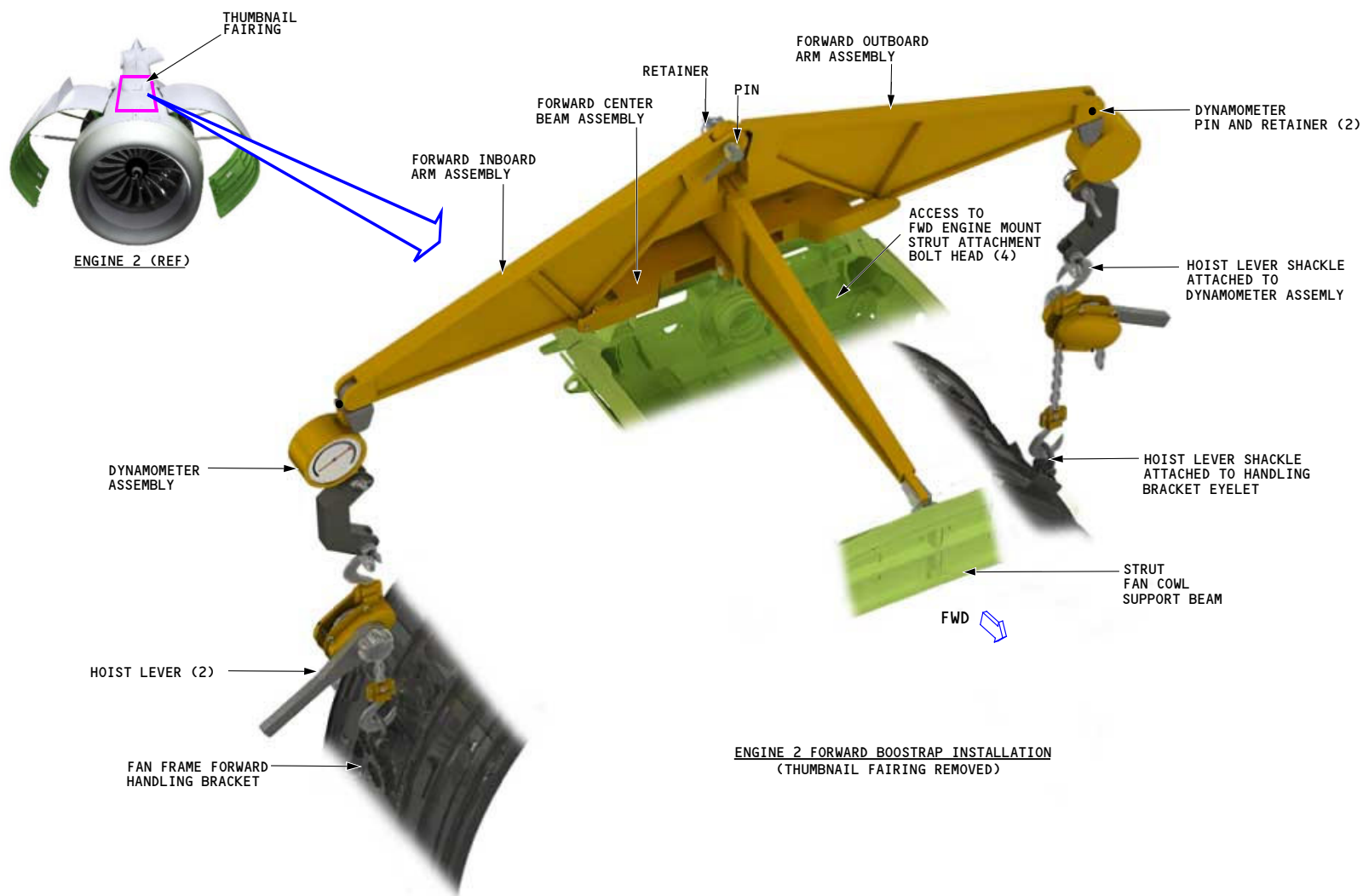
To install the forward bootstrap, remove the forward fairing panel (thumbnail panel) and do the following:

- Use lock pins and retainers to attach the forward center beam assembly to the forward mount brackets on strut
- Use lock pins and retainers to install the forward support arm assembly to the strut fan cowl support beam and to the forward center beam assembly
- Use a lock pin and retainer to attach the inboard and outboard arm assemblies together over the forward support arm assembly.

Install a dynamometer on both ends of the forward outboard arm assembly and the forward inboard arm assembly.

Attach the lever hoists and the two shackles to the dynamometer assemblies.

DMC-B737T-A-71-00-00-43A-100A-T_003-00



POWER PLANT -- LEAP-1B REMOVAL - INSTALL FORWARD BOOTSTRAP

71-00-00

DMC-B737T-A-71-00-00-43A-100A-T_003-00
ICN-B737T-A-710000-A-81205-00068-A-003-01

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - ENGINE HANDLING FWD ADAPTERS AND TOOL

Engine Stand Forward Adapters

You install the two forward engine stand adapters to the mounting point on the engine fan frame.

These adapters are in the storage box on the engine stand cradle. Before using the adapter mounting bolts, always examine, and replace any damaged bolts. These adapters allow the engine fan frame be supported by the engine stand forward support arms when the engine is lowered to the engine stand below.

These adapters also allow the engine to tilt about the pitch axis. The tilt helps get better clearance between the aft engine mount and the precooler when moving the engine on the engine stand forward away from of the engine strut.

Engine Forward Handling Brackets

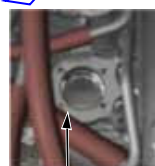
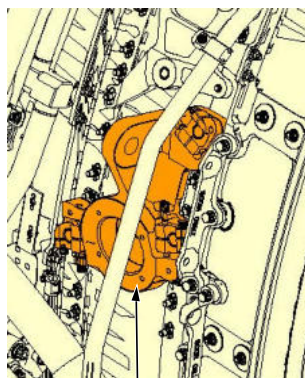
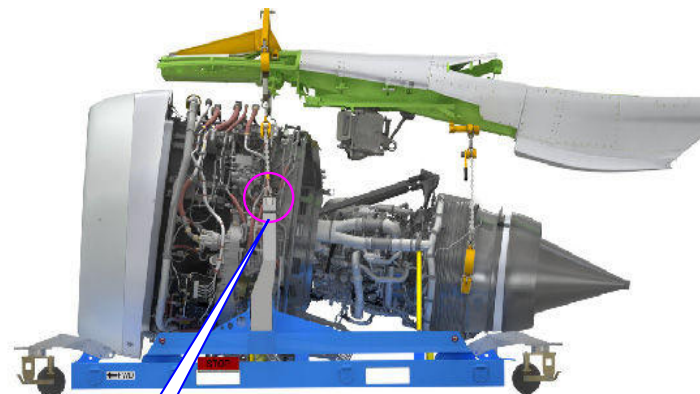
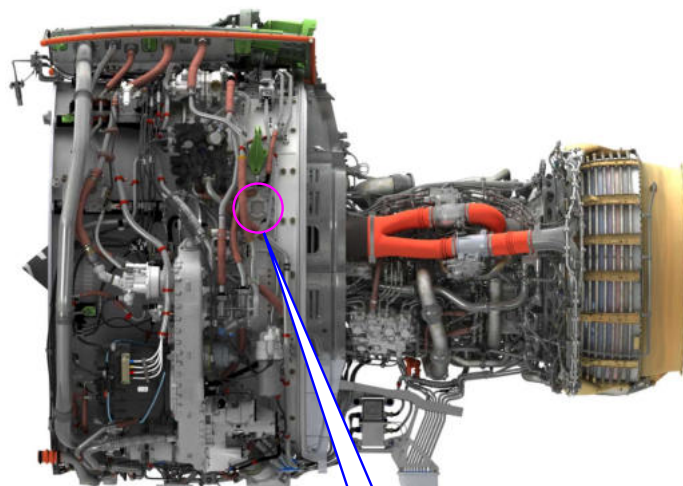
For POST LEAP-1B SERVICE BULLETIN 72-0181 engines, the handling brackets are not a part on the fan frame, but is part of the tool list.

When the handling bracket tool is installed on the fan frame it gives the mounting point for the engine stand forward adapter and the eyelet bracket where the hook of the hoist shackle is attached.

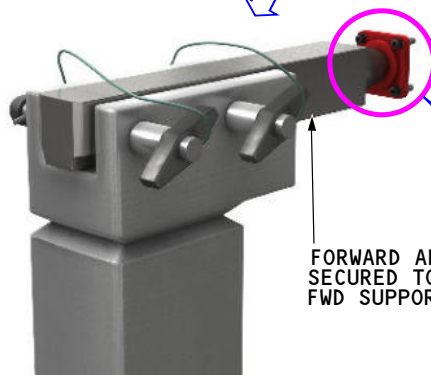
DMC-B737T-A-71-00-00-44A-100A-T_004-00

02-Dec-2020

71-00-00

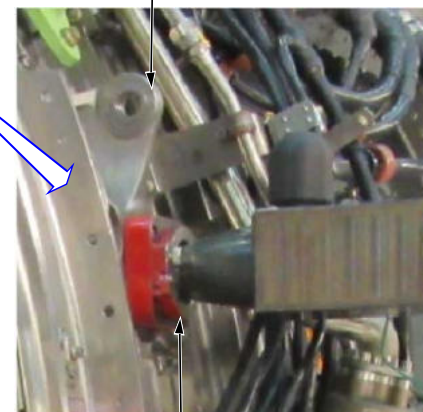


FAN FRAME ENGINE
FORWARD ADAPTER
MOUNT



ENGINE STAND
FWD SUPPORT ARM

FORWARD ADAPTER
SECURED TO
FWD SUPPORT ARM



FWD SHACKLE
LIFTING EYELET

ENGINE STAND FORWARD ADAPTER
ATTACHED TO FAN FRAME MOUNTING

HANDLING BRACKET TOOL INSTALLED
(POST LEAP1B SB 72-0181)

POWER PLANT -- LEAP-1B REMOVAL - ENGINE HANDLING FWD ADAPTERS AND TOOL

71-00-00

DMC-B737T-A-71-00-00-44A-100A-T_004-00
ICN-B737T-A-710000-A-81205-00069-A-004-01

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - INSTALL AFT BOOTSTRAP

Install the Aft Bootstrap

The aft bootstrap has these components:

- Outboard aft crossbeam assembly
- Outboard aft support beam assembly
- Aft torque beam assembly
- Aft outboard arm assembly
- Dynamometer
- Shackles
- Lever hoist.

Install the aft bootstrap on the three pylon attach points as follows:

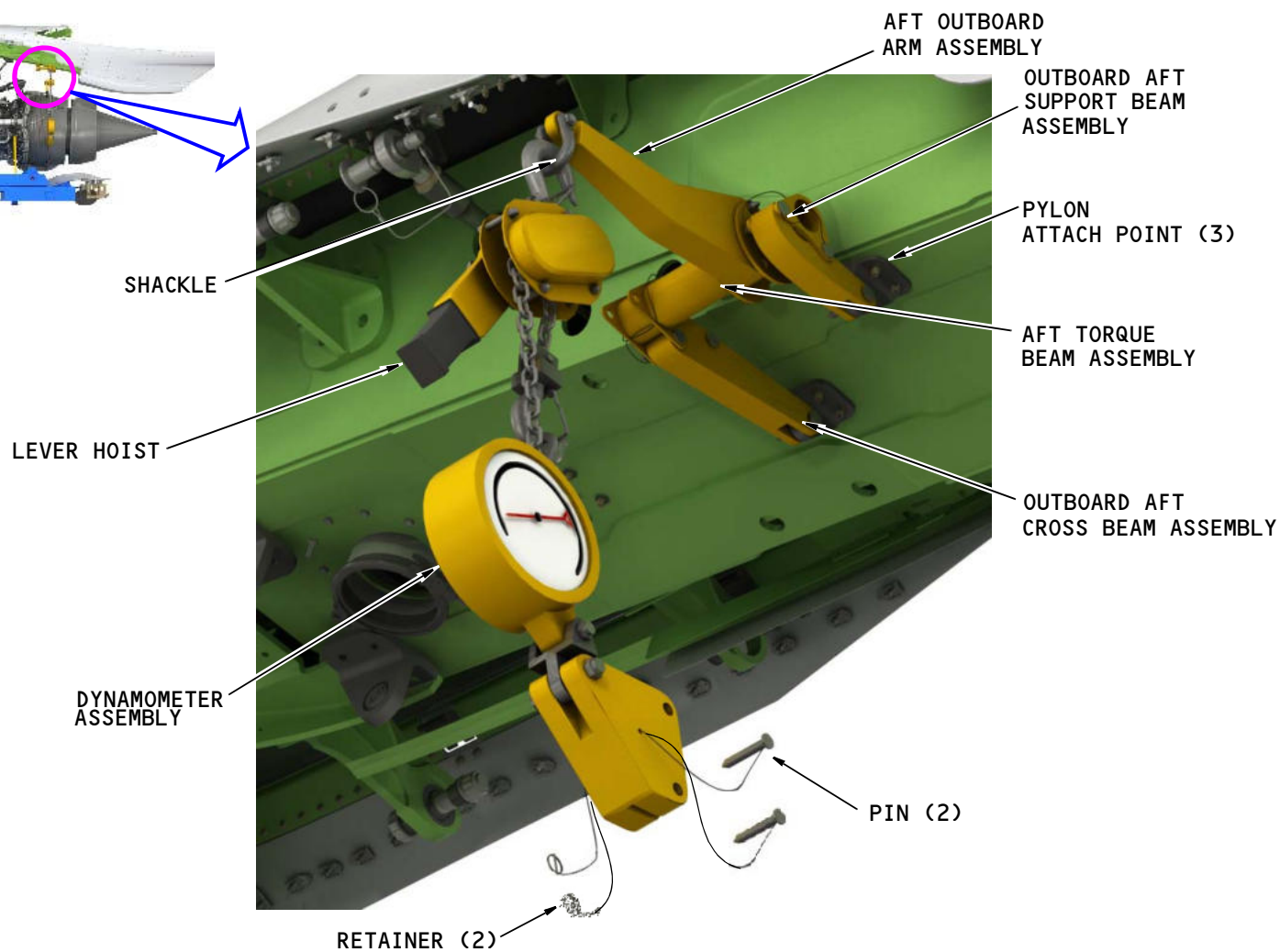
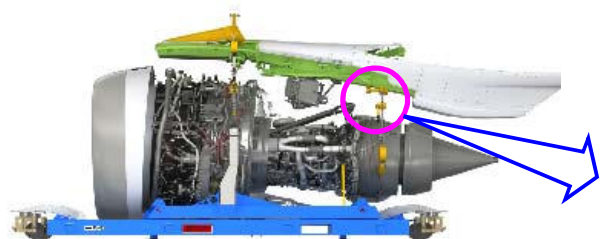
- Install the outboard aft support beam assembly on the outboard forward strut bracket with the pin and retainer
- Install the outboard aft cross beam assembly on the two outboard forward strut brackets with the pins and retainers
- Attach the aft torque beam assembly on the outboard aft cross beam assembly and the outboard aft support beam assembly with the pins and retainers
- Attach the aft outboard arm assembly to the aft torque beam assembly with the pins and retainers
- Install the lever hoist and dynamometer assembly on end of the aft outboard arm assembly with the shackle, bolt, nut and pin.

The aft bootstrap can only be installed with the lever hoist and dynamometer on the outboard location.

DMC-B737T-A-71-00-00-45A-100A-T_001-00

02-Dec-2020

71-00-00



DMC-B737T-A-71-00-00-45A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00070-A-002-01

POWER PLANT -- LEAP-1B REMOVAL - INSTALL AFT BOOTSTRAP

71-00-00

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - AFT BOOTSTRAP TO ENGINE

Install the Aft Bootstrap to Engine

The lever hoist and the dynamometer assembly connects to the aft bootstrap aft outboard arm assembly. The other end of the dynamometer assembly connects to the turbine rear frame (TRF) case bracket.

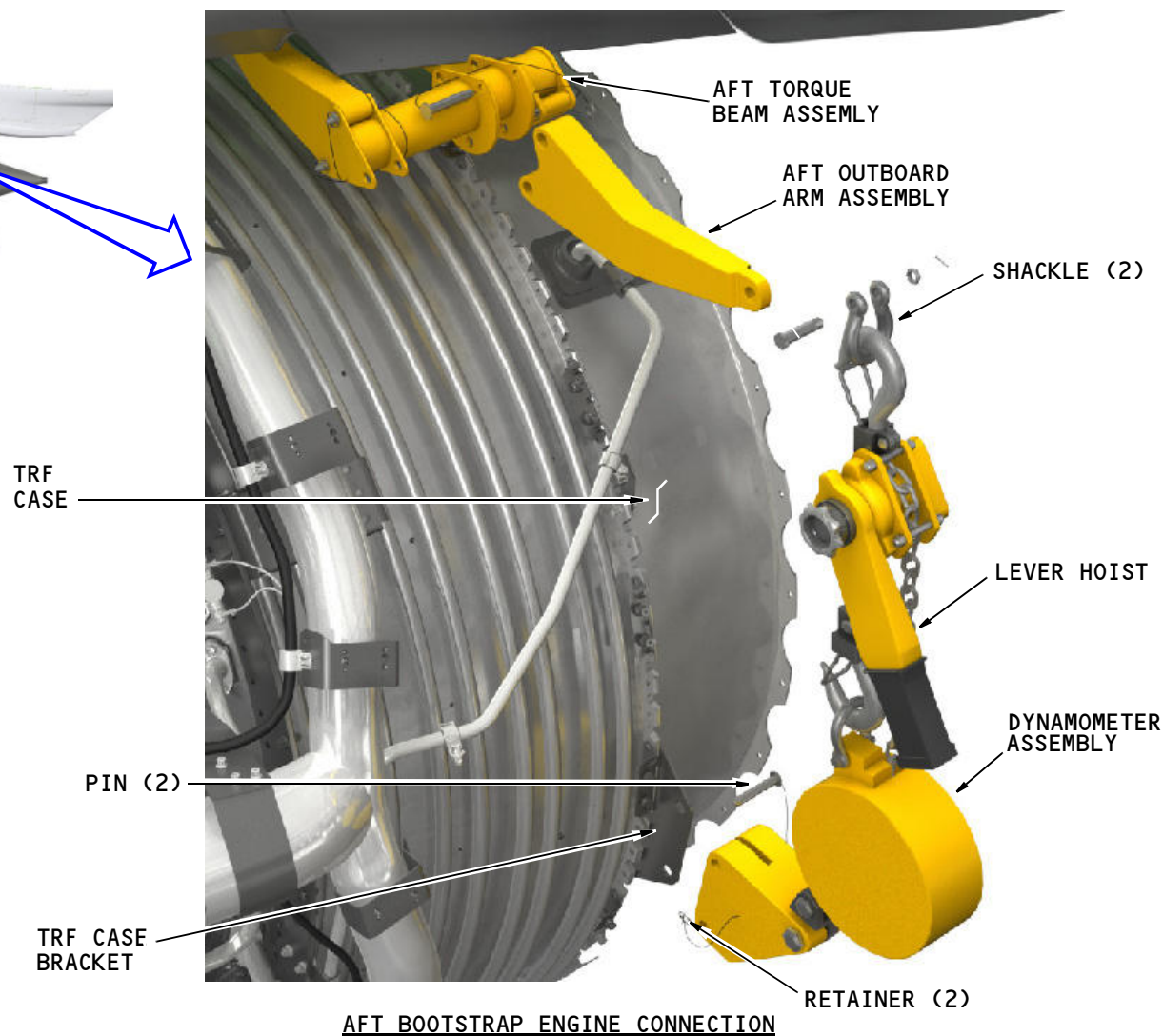
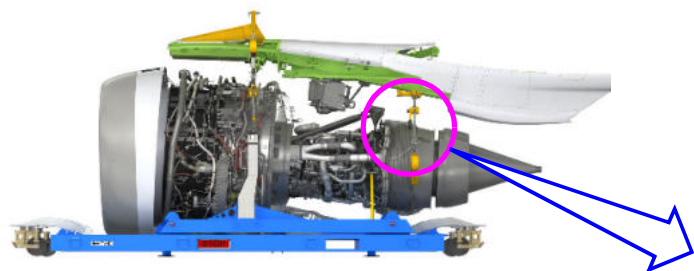
Attach the dynamometer assembly on the TRF case bracket with two pins and two retainers.

Take up the slack in the hoist chain.

DMC-B737T-A-71-00-00-46A-100A-T_001-00

02-Dec-2020

71-00-00



POWER PLANT -- LEAP-1B REMOVAL - AFT BOOTSTRAP TO ENGINE

71-00-00

DMC-B737T-A-71-00-00-46A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00071-A-001-01

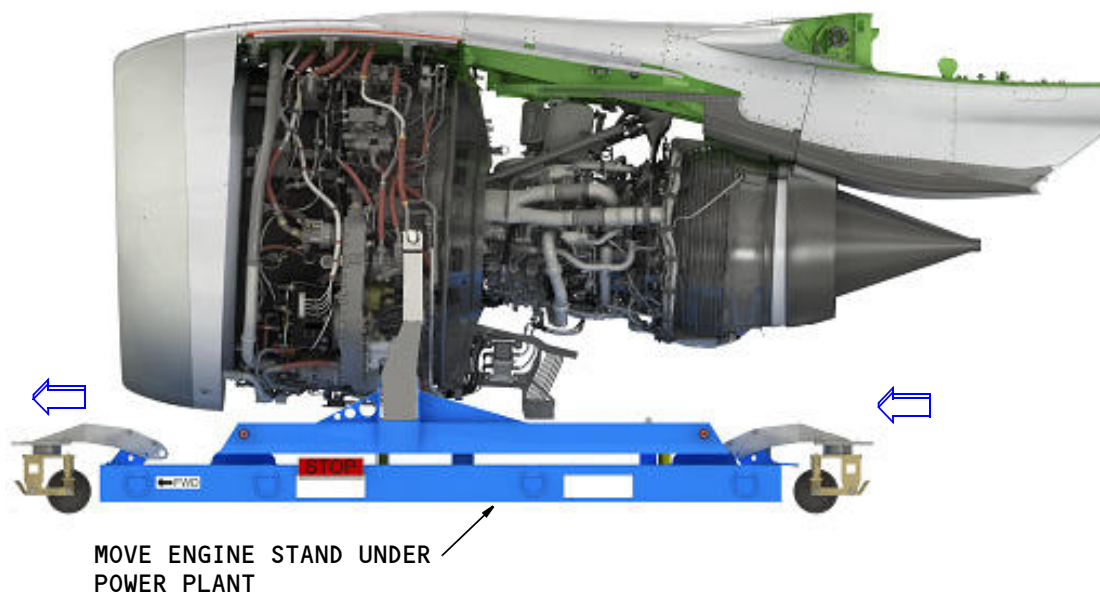
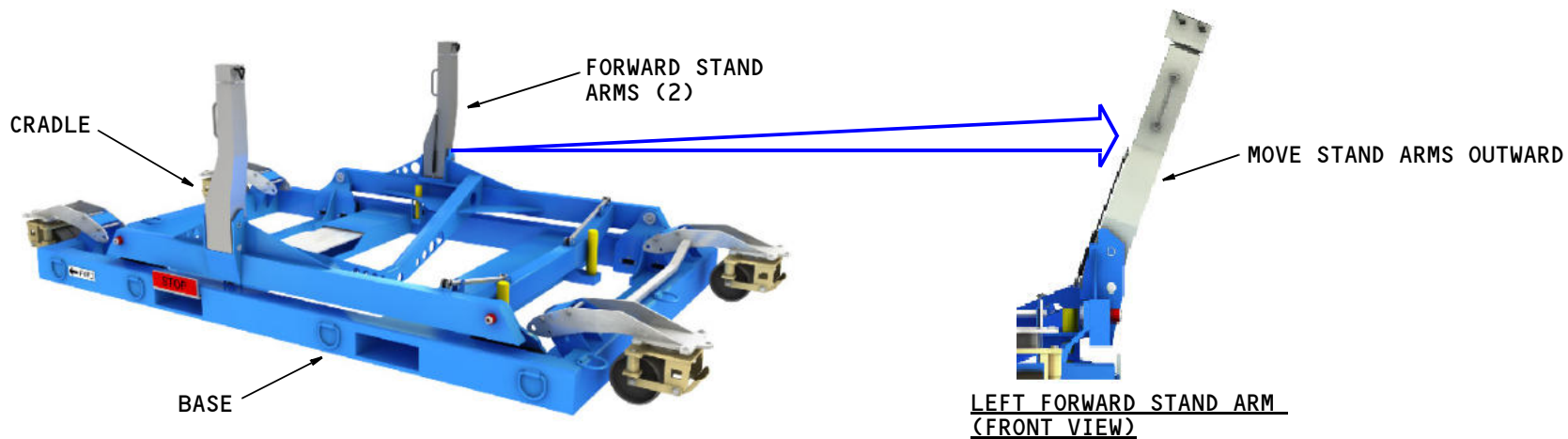
02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - PREPARE TO REMOVE POWER PLANT

Prepare to Remove the Power Plant

Do these steps to prepare the cradle and base for the power plant removal:

- Move the two forward stand arms outward
- Adjust the height of the stand cradle as necessary with the stand hydraulic jack
- Adjust the wheel positions as necessary
- Move the engine stand into its position from rear of the power plant (Use engine stand center red line to center the engine stand under the engine.)
- After the engine stand is in position, move the two forward stand arms inward to close.



POWER PLANT -- LEAP-1B REMOVAL - PREPARE TO REMOVE POWER PLANT

71-00-00

DMC-B737T-A-71-00-00-47A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00072-A-001-01

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - DISCONNECT ENGINE MOUNTS

Apply Preload

When you operate the lever hoists to lower the power plant, continuously monitor all three dynamometers. Operation of one hoist can affect the loads at the other two hoist locations.

Operate the forward and aft lever hoists equally until the dynamometers show the preloads. Be sure that you do not use more force than the preload limits specified in the maintenance manual procedure.

Disconnect the Forward and Aft Engine Mounts

Disconnect the forward and aft engine mounts:

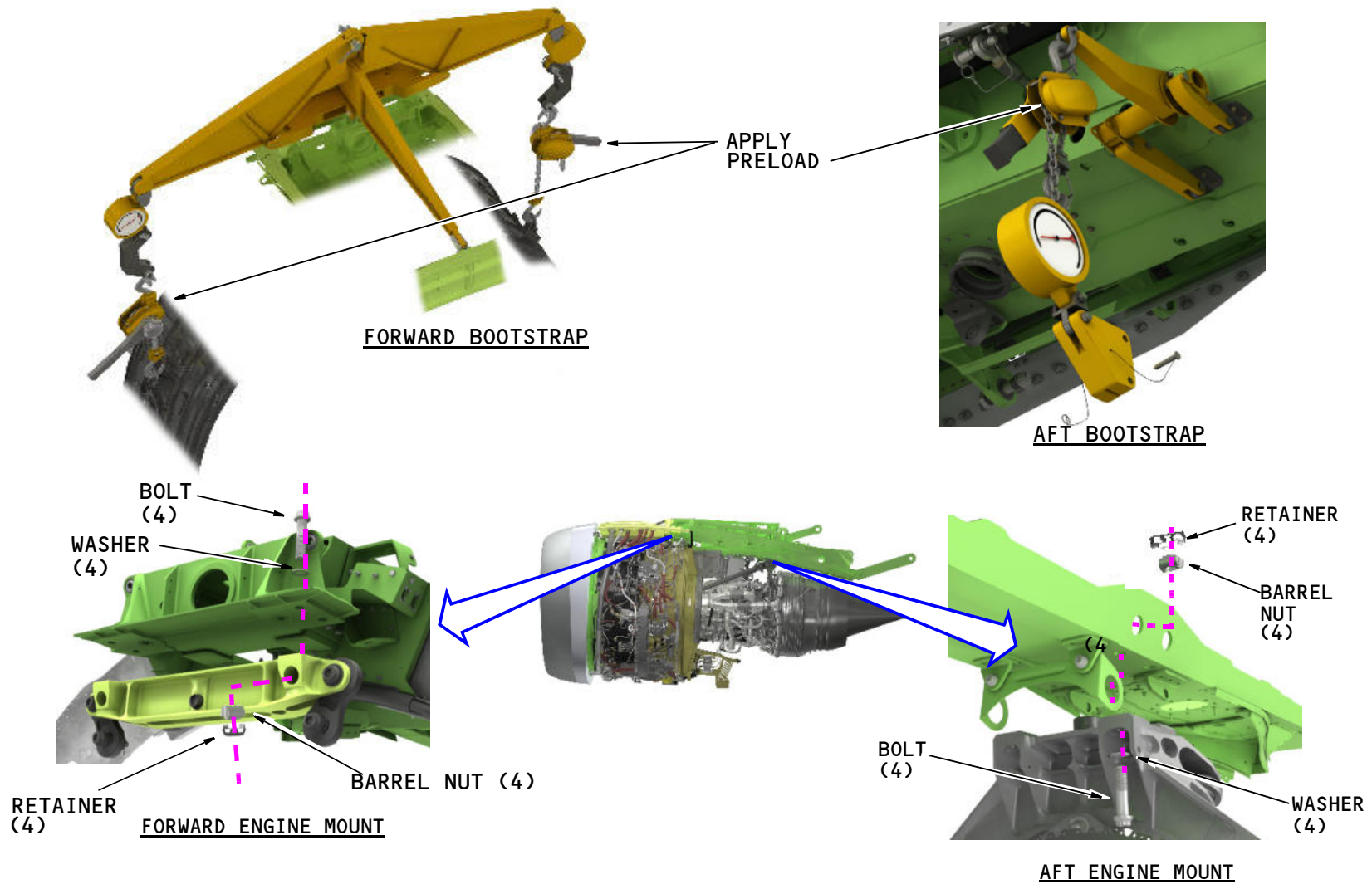
- Loosen all forward and aft engine mount bolts one-half turn
- Check and make sure there is no clearance between the strut surface and the mount surface
- If there is clearance, operate the forward and aft lever hoists to remove the clearance
- With no clearance, the entire weight of the engine is now supported by the bootstrap hoists
- Remove the forward mount bolts and washers from the barrel nuts and barrel nut retainers
- Remove the aft mount bolts and washers from the barrel nuts and barrel nut retainers.

Make sure the engine mount surface continues to touch the related strut surface as you remove the bolts and washers.

CAUTION: DO NOT LOOSEN THE BOLTS MORE THAN ONE-HALF OF A TURN. IF YOU DO NOT OBEY THIS INSTRUCTION, DAMAGE TO THE ENGINE MOUNTS OR THE STRUT CAN OCCUR.

CAUTION: MONITOR THE DYNAMOMETERS ON THE FORWARD AND THE AFT HOISTS. MAKE SURE THAT THE LOADS ARE LESS THAN THE LIMITS SPECIFIED IN THE MAINTENANCE MANUAL PROCEDURE. IF YOU APPLY MORE FORCE THAN THE LIMITS, THE ENGINE CAN FALL. THIS CAN CAUSE INJURIES AND DAMAGE TO EQUIPMENT.

DMC-B737T-A-71-00-00-48A-100A-T_004-00



DMC-B737T-A-71-00-00-48A-100A-T_004-00
ICN-B737T-A-710000-A-81205-00073-A-005-01

POWER PLANT -- LEAP-1B REMOVAL - DISCONNECT ENGINE MOUNTS

71-00-00

02-Dec-2020

POWER PLANT -- LEAP-1B REMOVAL - LOWER THE POWER PLANT

Lower the Power Plant

Do these steps to lower the power plant:

- Using the forward and aft lever hoists together, slowly lower the engine until the engine mount shear pins are clear of the strut mount surface. Note that binding of the aft engine mount can occur if the forward hoists are being lowered too much ahead of the aft hoist.
- Roll the engine by six degrees to align the engine left and right sides to the engine stand arms support. During the roll make sure that the thrust links do not interfere with the precooler, or damage to the precooler can occur.
- Lower the engine equally until the engine is about 4 to 6 in. (10 to 15 cm) above the engine stand. Make the necessary adjustments to align the engine stand forward adapters installed on the engine stand support arms to the fan frame fitting. Attach the engine forward adapters to the fan frame fitting.
- Attach the engine stand aft adapters to the turbine center frame. You raise up the engine stand aft support columns from their stowed position and attach the engine aft support columns to the aft adapters.
- Disconnect the bootstrap attachment from the engine when the engine is safely supported by the engine stand.
- To clear the engine aft mount from interference with the precooler when moving forward from under the strut, you tilt the engine 7 degrees aft down.
- Move the engine stand forward until engine is clear of the strut.

WARNING: LET ONLY APPROVED PERSONS OPERATE THE LEVER HOISTS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

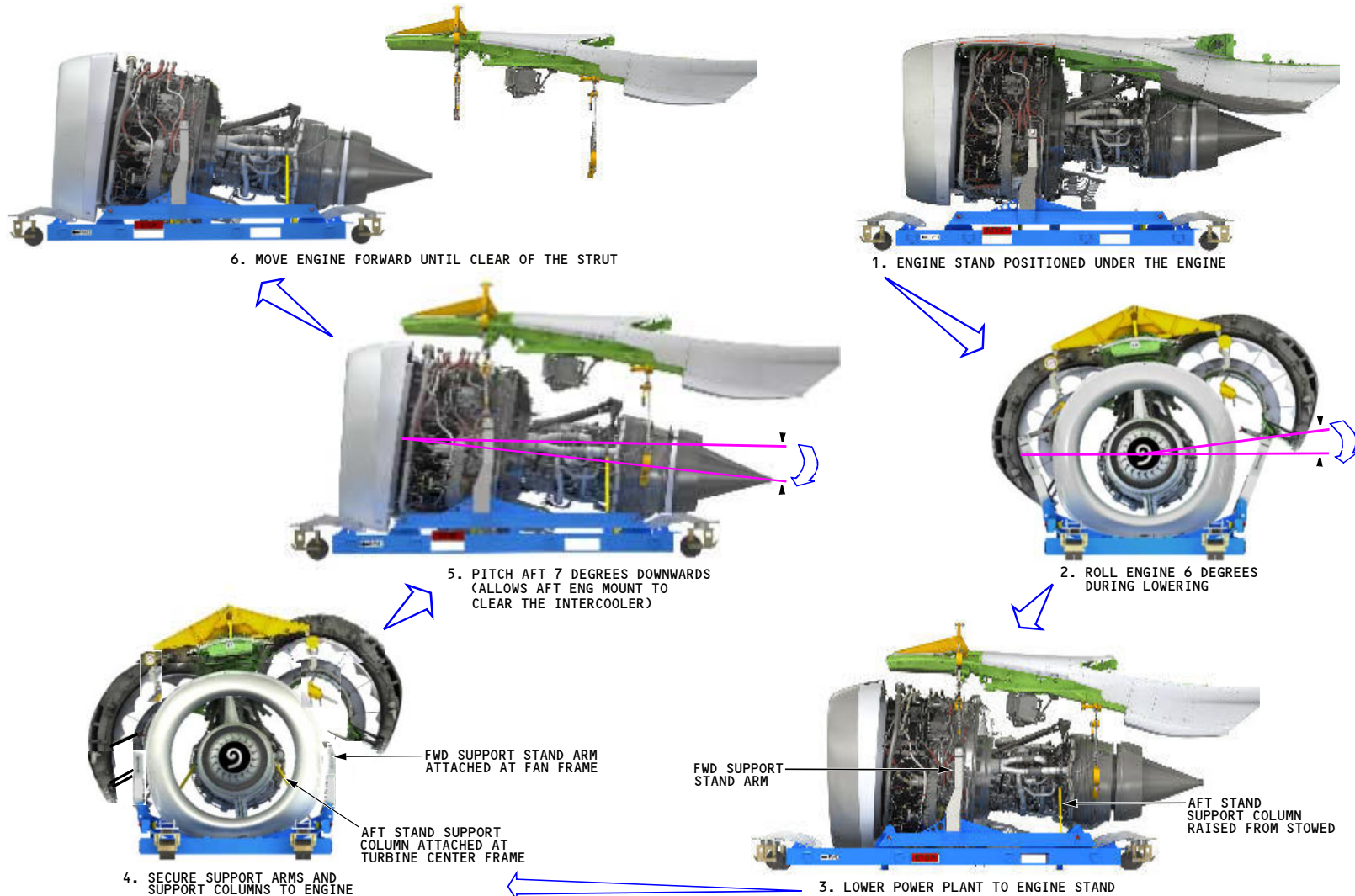
WARNING: MAKE SURE THAT WORKSTANDS AND PERSONNEL ARE AWAY FROM THE ENGINE BEFORE YOU LOWER IT. THE ENGINE CAN MOVE RELATIVE TO THE STRUT WHEN THE SHEAR PINS ARE NOT IN THE UPPER ENGINE MOUNTS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION: MONITOR THE THRUST LINKS WHEN YOU LOWER AND TURN THE ENGINE. MAKE SURE IT DOES NOT INTERFERE WITH THE PRECOOLER OR DAMAGE TO THE PRECOOLER CAN OCCUR.

DMC-B737T-A-71-00-00-49A-100A-T_005-00

02-Dec-2020

71-00-00



POWER PLANT -- LEAP-1B REMOVAL - LOWER THE POWER PLANT

71-00-00

POWER PLANT -- LEAP-1B INSTALLATION - GENERAL

General

You do these tasks to install the power plant:

- Prepare the power plant for installation
- Install the power plant
- Connect fan case hydraulic, fuel, pneumatic and electrical connections to the strut
- Connect and re-install components in the core case
- Install the fan cowls and close the thrust reverser
- Put the airplane back to its usual condition.

The aircraft maintenance manual (AMM) has the procedures, list of materials, and support equipment needed for the installation.

You use the bootstrap kit and hold open equipment that you used to remove the power plant.

LEAP-1B ENGINES



737-7/8/8200/9
AIRCRAFT MAINTENANCE MANUAL

TASK 71-00-02-400-801-G00

3. Power Plant Installation
(Figure 401, Figure 403, Figure 404, Figure 405, Figure 406, Figure 407, Figure 408, Figure 409, Figure 410, and Figure 411)

A. General

- (1) This task provides the instructions on how to install the power plant.
- (2) The power plant weighs approximately 7300 pounds (3318 kilograms). You must include the weight of the ground support equipment when you calculate the load.
- (3) You can install an engine with the inlet cowl or without the inlet cowl installed. Without the inlet cowl installed, the aft loads will be slightly higher than the engine with the inlet cowl installed. This procedure is written for the engine installation with the inlet cowl installed.
- (4) You must do the steps that follow to install the engine: . . .

PREPARE TO INSTALL
INSTALL POWER PLANT
CONNECT TO STRUT
INSTALL COMPONENTS
PUT AIRPLANE BACK TO USUAL CONDITION

AIRCRAFT MAINTENANCE MANUAL

BOOTSTRAP KIT
HOLD OPEN EQUIPMENT



SUPPORT EQUIPMENT

DMC-B737T-A-71-00-00-06A-040A-T_001-00
ICN-B737T-A-710000-A-81205-00076-A-001-01

POWER PLANT -- LEAP-1B INSTALLATION - GENERAL

71-00-00

POWER PLANT -- LEAP-1B INSTALLATION - GENERAL DESCRIPTION

Power Plant Installation

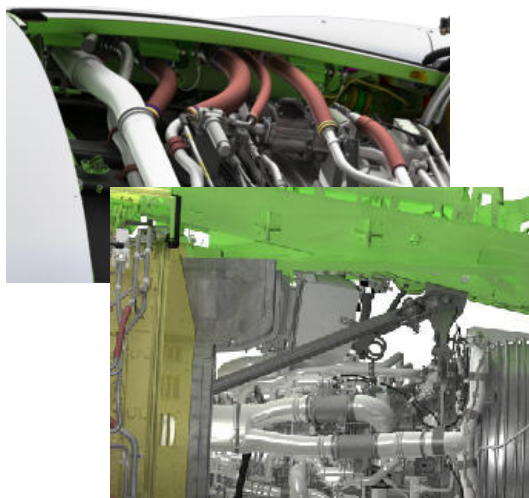
The power plant installation procedure is in the aircraft maintenance manual (AMM).

In general, you will do these steps:

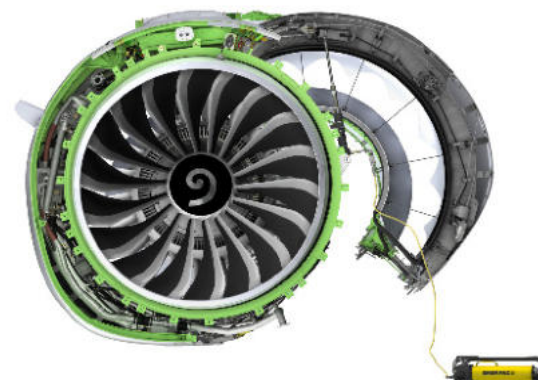
- Prepare to install the power plant
- Install the power plant
- Do the fan case connections
- Do the core area connection
- Remove the support equipment
- Close the thrust reverser
- Install the fan cowls
- Close circuit breakers and make systems active
- Do engine operation.



INSTALL POWER PLANT



FAN CASE/CORE CONNECTIONS



CLOSE THRUST REVERSER
INSTALL FAN COWL



CLOSE CIRCUIT BREAKERS
MAKE SYSTEMS ACTIVE



DO ENGINE OPERATION

POWER PLANT -- LEAP-1B INSTALLATION - GENERAL DESCRIPTION

71-00-00

POWER PLANT -- LEAP-1B INSTALLATION - PREPARATION

Preparation

Prepare the airplane for the power plant installation:

- Make sure the airplane is statically grounded
- Make sure that the airplane is within level of 1.5 degree nose-down attitude
- Make sure that there is minimum of 5.0 to 8.0 inches (12.7 - 20.3 cm) shock strut chrome at the applicable main landing gear oleo
- If you need additional clearance to pull the engine and the transportation stand in and out below the strut, raise the main landing gear strut
- Make sure that the bootstrap and thrust reverser hold-open equipment are properly installed.

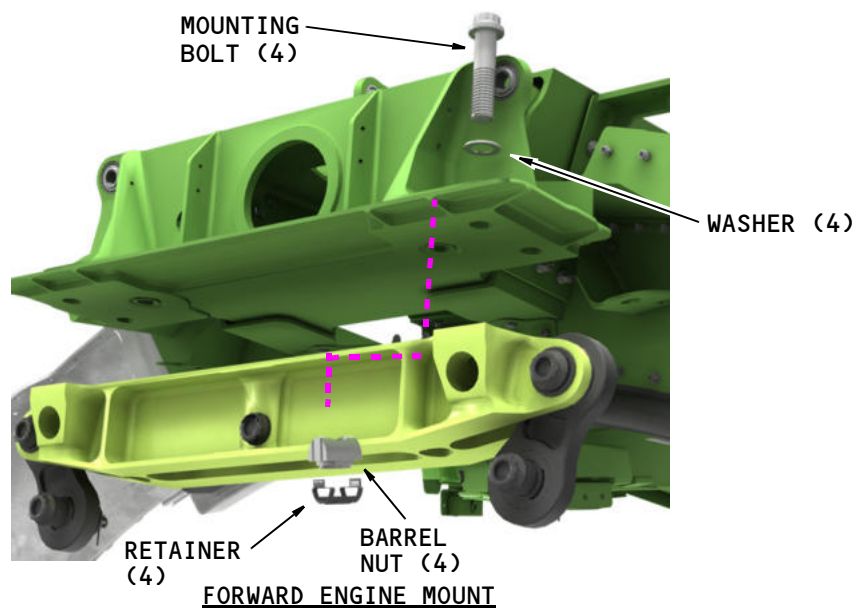
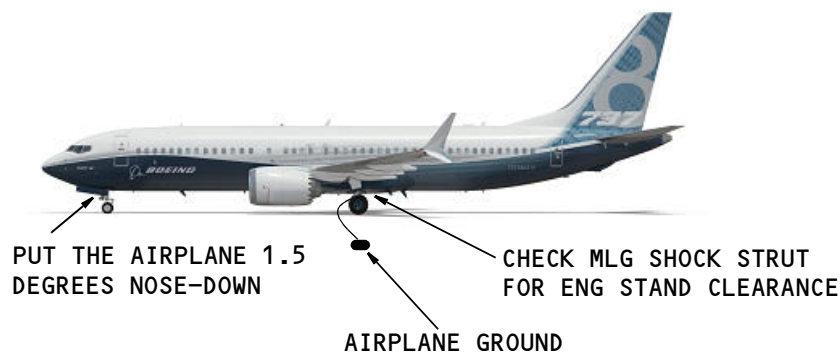
Prepare the forward and aft engine mounts for engine installation:

- If the engine mount bolts and barrel nuts are not new, do A dye penetrant inspection
- Make sure that the bolts have the correct length
- Check the condition of engine and strut mount assemblies and do inspection in accordance with AMM
- Inspect for discoloration, cracks, corrosion, wear, loose fasteners, worn bearings, thrust link clevis bushing damage and signs of adjacent structure failure
- Make sure surface of the engine mounts that will touch strut are free of oil, grease or other unwanted material
- Apply Never-Seez NSBT compound to the shear pins and to mounting bolt shank and threads
- Install the mounting bolts in the captive barrel nuts and do a check of the self-locking torque
- Monitor the torque to turn the bolts and make sure the torque is within limits. If out of limits, replace the barrel nuts.

DMC-B737T-A-71-00-00-50A-100A-T_001-00

02-Dec-2020

71-00-00



POWER PLANT -- LEAP-1B INSTALLATION - PREPARATION

71-00-00

DMC-B737T-A-71-00-00-50A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00077-A-001-01

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - INSTALL PREPARATION 2

Install Preparation

If the landing gear strut was raised, push the engine stand with the engine, from front of the strut towards the rear until the engine stand is aligned under the strut.

Deactivate the cradle damping from the engine stand and make sure that the engine stand wheels are locked.

Install the forward bootstrap lever hoists and dynamometer assemblies to the engine fan case brackets.

Install the aft bootstrap lever hoist and dynamometer assembly to the turbine rear frame (TRF) case bracket.

Operate the lever hoists to take up the slack in the hoist chains.

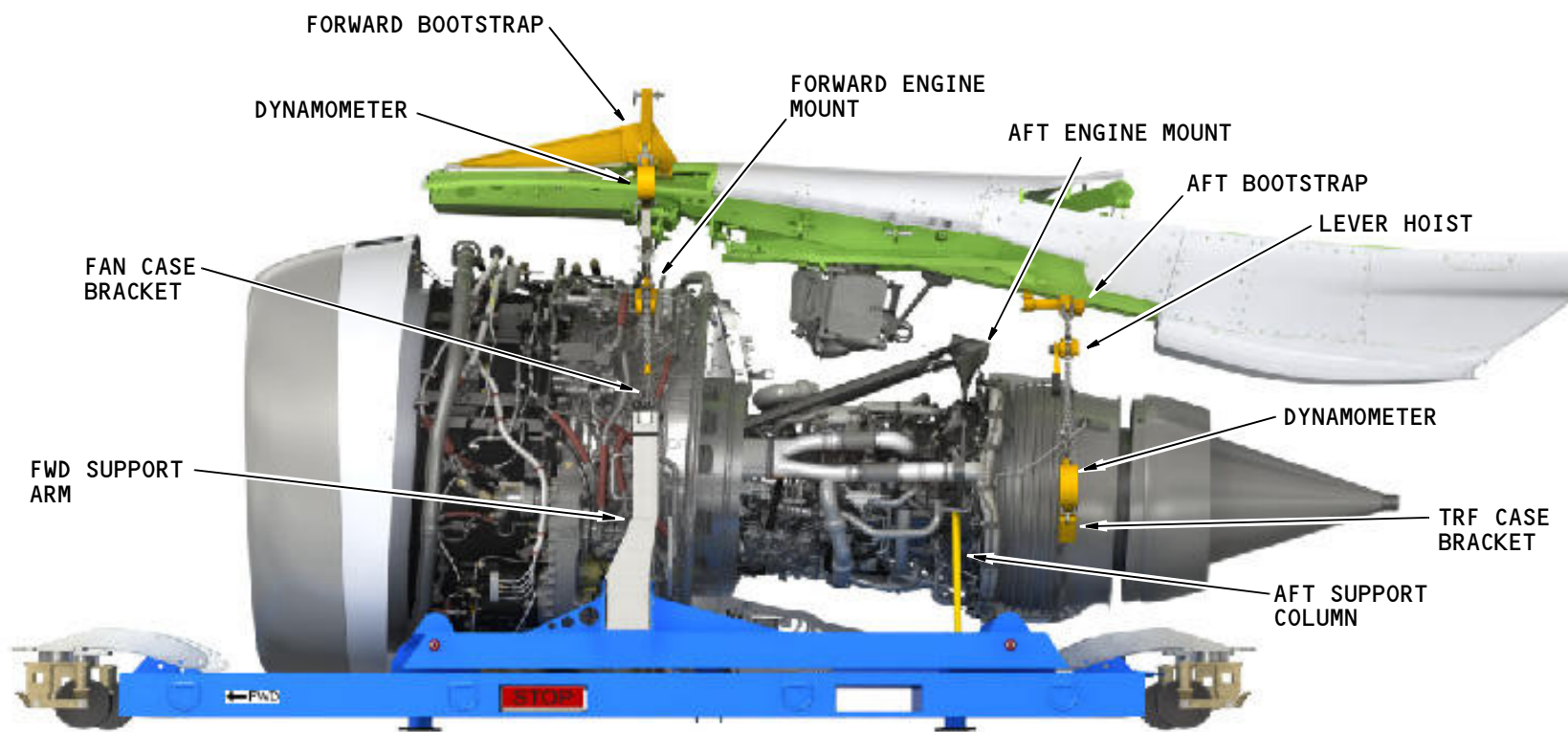
WARNING: LET ONLY APPROVED PERSONS OPERATE THE LEVER HOISTS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING: MONITOR THE DYNAMOMETERS ON THE FORWARD AND THE AFT HOISTS. MAKE SURE THAT THE LOADS ARE LESS THAN THE LIMITS SPECIFIED IN THE MAINTENANCE MANUAL PROCEDURE. IF YOU APPLY MORE FORCE THAN THE LIMITS, THE ENGINE CAN FALL. THIS CAN CAUSE INJURIES AND DAMAGE TO EQUIPMENT.

DMC-B737T-A-71-00-00-51A-100A-T_004-00

02-Dec-2020

71-00-00



DMC-B737T-A-71-00-00-51A-100A-T_004-00
ICN-B737T-A-710000-A-81205-00078-A-003-01

POWER PLANT -- LEAP-1B INSTALLATION - INSTALL PREPARATION 2

71-00-00

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - DISCONNECT ENG ADAPTERS

Install Preparation

If you did not raise the main landing gear strut, do these steps:

- If the engine is not in a 7 degree tilt downward (tail) direction, adjust the cradle of the engine stand and use the lever hoists to tilt the engine to 7 degree downward (tail) direction
- Lower the engine stand base to 2.0 inches (50 mm) from the ground with the stand hydraulic device
- While monitoring the aft engine mount clearance from the precooler, push the engine stand with the engine, from front of the strut towards the rear until the engine stand is aligned under the strut
- Deactivate cradle damping from the engine stand and ensure the engine stand wheels are locked
- Operate the lever hoists to take up the slack in the hoist chains.

Disconnect the engine from the engine stand:

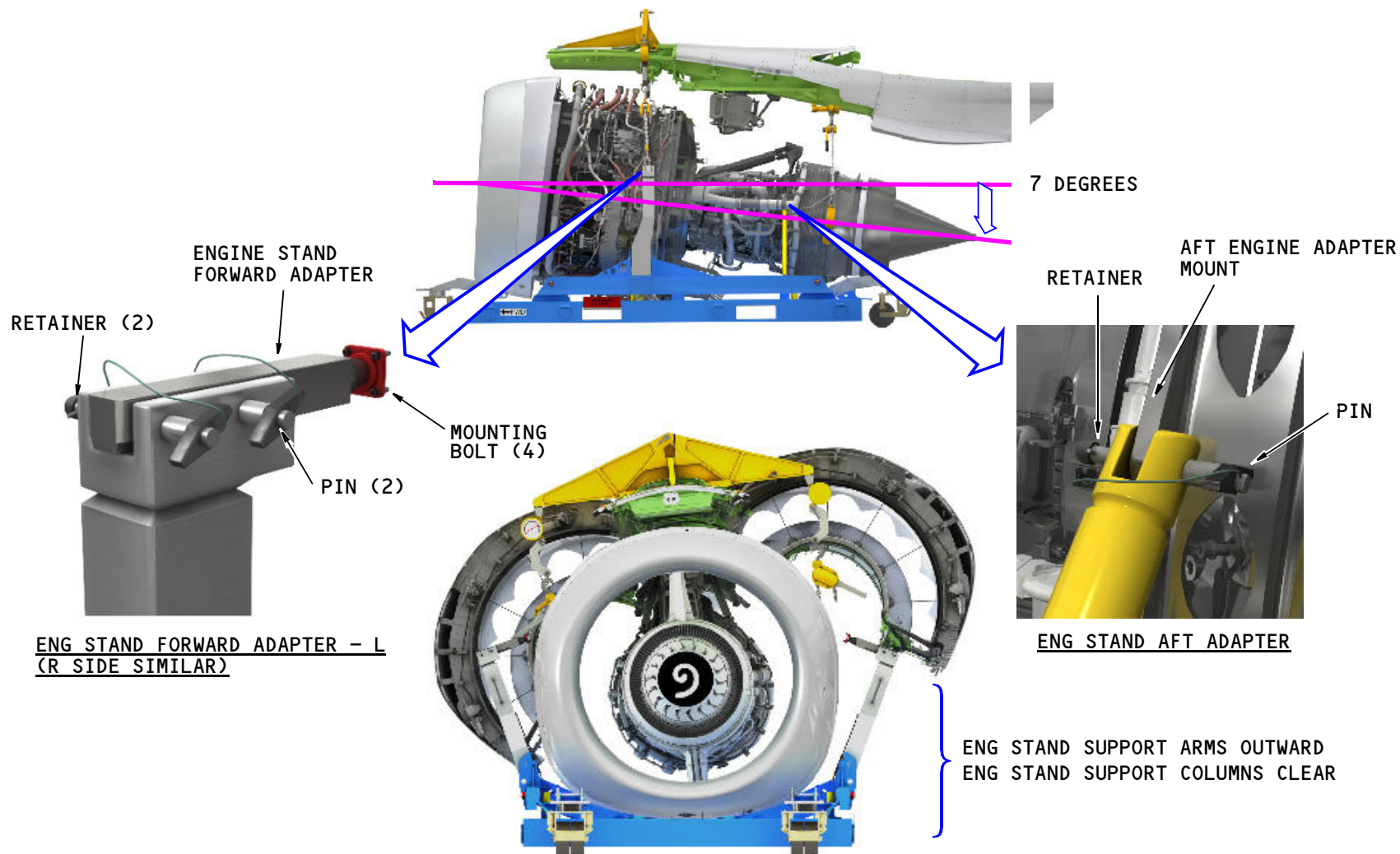
- Disconnect the engine stand forward adapters from the forward engine stand support arms
- Disconnect the engine stand forward adapters from the forward engine adapter mounts
- Disconnect the engine stand aft adapter from the aft engine adapter mount
- Move the two forward stand arms outward and make sure that the aft engine stand support columns are clear from the aft engine adapter mounts.

CAUTION: MONITOR THE THRUST LINKS, THE PRECOOLER, AND THE HINGE BEAM FLANGE WHEN THE ENGINE MOVES OR TURNS. DO NOT MOVE OR TURN THE ENGINE TOO QUICKLY OR THE ENGINE CAN HIT AND CAUSE DAMAGE TO THESE COMPONENTS.

DMC-B737T-A-71-00-00-52A-100A-T_001-00

02-Dec-2020

71-00-00



POWER PLANT -- LEAP-1B INSTALLATION - DISCONNECT ENG ADAPTERS

71-00-00

DMC-B737T-A-71-00-00-52A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00079-A-001-01

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - LIFT THE ENGINE

Lift the Engine

Operate the lever hoists and slowly lift the engine from the engine stand.

Use the aft lever hoist to change the roll of the engine to align the engine mount platform to the strut.

Use the forward lever hoists or the aft lever hoist to change the pitch of the engine.

Monitor the clearance between the engine and the thrust reverser cowls while you lift the engine.

Monitor the thrust links, the precooler and the forward thrust reverser for proper clearance.

Lift the engine until there is approximately 6.0 inches (15.2 cm) between the engine mounts and the strut mounts.

Operate the lever hoists until the clearance is equal and the mounts are aligned.

Lift the engine until you have clearance of 0.50 inch (1.27 cm) between the aft mount and the aft strut mount.

When the shear pin touches the surface of the aft mount strut, level the aft mount platform to the strut platform. If the angular position of the aft mount is not aligned correctly, it will cause binding of the shear pins and lockup.

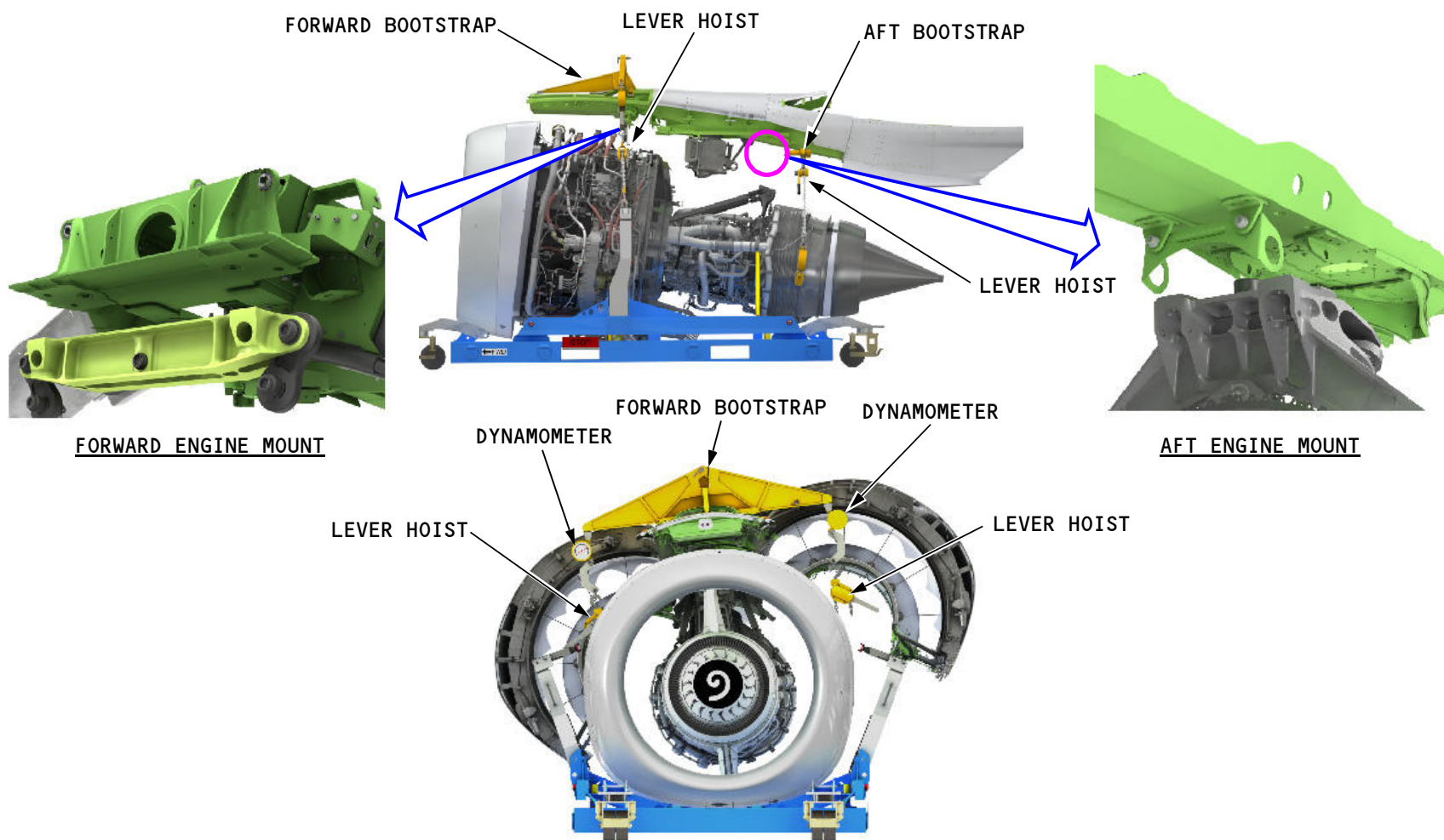
Adjust the forward lever hoists until the forward shear pins engages the mount approximately 0.10 inch (0.25 cm) before the aft shear pin.

Lift the engine until you see no clearance between the upper and lower parts of the engine mounts.

WARNING: LET ONLY APPROVED PERSONS OPERATE THE LEVER HOISTS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING: MONITOR THE DYNAMOMETERS ON THE FORWARD AND THE AFT HOISTS. MAKE SURE THAT THE LOADS ARE LESS THAN THE LIMITS SPECIFIED IN THE MAINTENANCE MANUAL PROCEDURE. IF YOU APPLY MORE FORCE THAN THE LIMITS, THE ENGINE CAN FALL. THIS CAN CAUSE INJURIES AND DAMAGE TO EQUIPMENT.

CAUTION: MONITOR THE THRUST LINKS, THE PRECOOLER, AND THE HINGE BEAM FLANGE WHEN THE ENGINE MOVES OR TURNS. DO NOT MOVE OR TURN THE ENGINE TOO QUICKLY OR THE ENGINE CAN HIT AND CAUSE DAMAGE TO THESE COMPONENTS.



POWER PLANT -- LEAP-1B INSTALLATION - LIFT THE ENGINE

71-00-00

POWER PLANT -- LEAP-1B INSTALLATION - ATTACH THE ENGINE TO STRUT

Attach the Power Plant to the Strut

Remove the load from the lever hoists.

Install the forward engine mount:

- Make sure that the barrel nut retainers, bolts, washers and barrel nuts are not bent, cracked, corroded or show signs of worn areas
- Apply Never-Seez NSBT compound to bolts and exterior cylindrical section of barrel nuts and retainers
- Do not apply the Never-Seez NSBT compound to the barrel nuts thread locking feature
- Install the four bolts, washers, barrel nuts and the barrel nut retainers, with the countersunk side of the washer next to the bolt head
- Tighten each bolt to 50% of final torque
- Tighten each bolt to final torque of 2086 - 2214 inch-pounds (235.7-250.1 Newton-meters)
- Make sure the threads of the barrel nuts are fully engaged and end of mounting bolts chamfer extends out of the nuts.

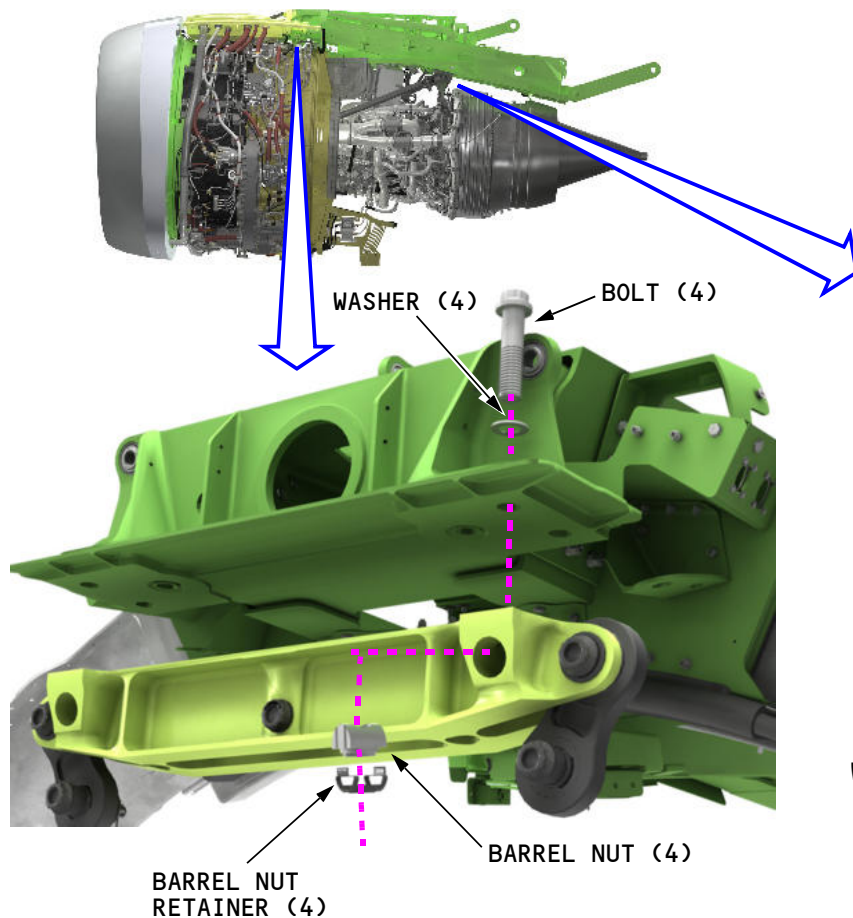
Install the aft engine mount:

- Make sure that the barrel nut retainers, bolts, washers and barrel nuts are not bent, cracked, corroded or show signs of worn areas
- Apply Never-Seez NSBT compound to bolts and exterior cylindrical section of barrel nuts and retainers
- Do not apply the Never-Seez NSBT compound to the barrel nuts thread locking feature
- Install the four bolts, washers, barrel nuts and the barrel nut retainers, with the countersunk side of the washer next to the bolt head
- Tighten each bolt to 50% of final torque
- Tighten each bolt to final torque of 3298 - 3502 inch-pounds (372.6 - 395.7 Newton-meters)
- Make sure the threads of the barrel nuts are fully engaged and end of mounting bolts chamfer extends out of the nuts.

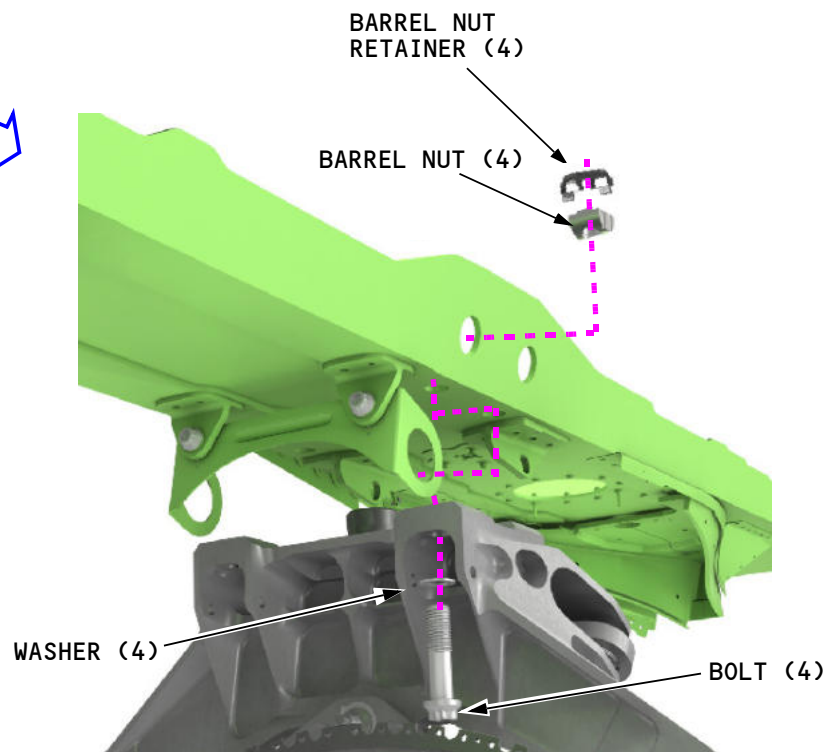
DMC-B737T-A-71-00-00-55A-100A-T_001-00

02-Dec-2020

71-00-00



FORWARD ENGINE MOUNT



AFT ENGINE MOUNT

POWER PLANT -- LEAP-1B INSTALLATION - ATTACH THE ENGINE TO STRUT

71-00-00

DMC-B737T-A-71-00-00-55A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00082-A-001-01

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - REMOVE BOOTSTRAP EQUIPMENT

Disconnect the Forward Bootstrap Equipment

Do the following to remove the forward bootstrap equipment from the power plant:

- Remove the forward lever hoists with the forward dynamometer assemblies and shackles from the engine attach brackets and from the bootstrap equipment
- Remove the forward outboard arm assembly and the forward inboard arm assembly from the bootstrap forward center beam
- Remove the forward support arm assembly from top of the forward center beam
- Remove the forward center beam from top of the forward mount brackets.

Do the following to remove the aft bootstrap equipment from the power plant:

- Remove the aft lever hoists with the aft dynamometer assembly and shackles from the engine attach brackets and from the bootstrap equipment
- Remove the pins and retainers that attach the aft outboard arm assembly to the aft torque beam assembly
- Remove the aft torque beam assembly from the outboard aft cross beam assembly and the outboard aft support beam assembly
- Remove the pins and retainers that attach the outboard aft cross beam assembly to the outboard forward strut brackets
- Remove the pin and retainer that attach the outboard aft support beam assembly to the outboard forward strut bracket.

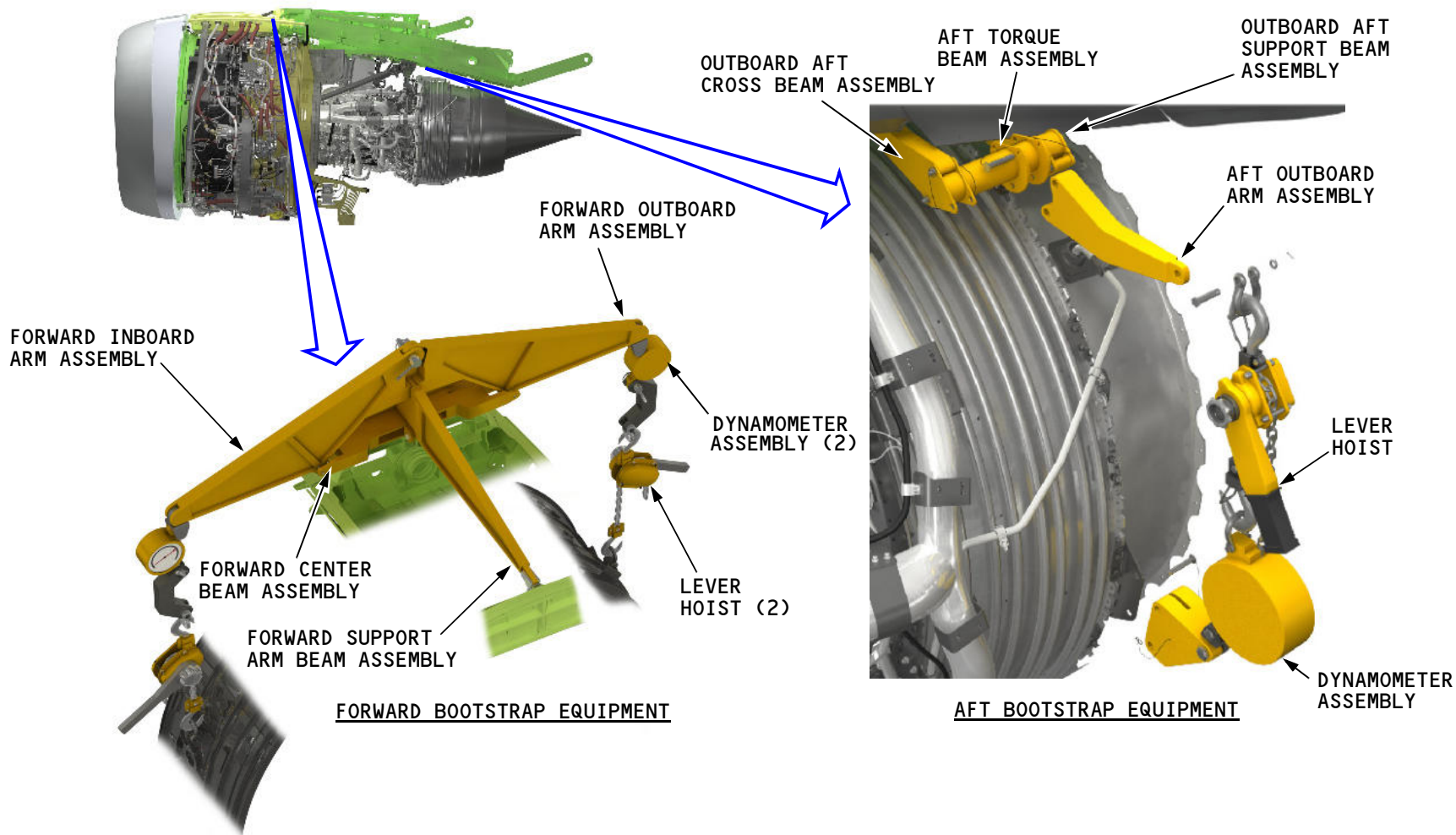
Put all bootstrap equipment into the bootstrap equipment container.

Install the forward fairing panel (thumbnail fairing) on the strut.

DMC-B737T-A-71-00-00-56A-100A-T_001-00

02-Dec-2020

71-00-00



POWER PLANT -- LEAP-1B INSTALLATION - REMOVE BOOTSTRAP EQUIPMENT

71-00-00

POWER PLANT -- LEAP-1B INSTALLATION - LEFT FAN CASE CONNECTIONS

Left Fan Case Connection

All connections on the left side of the fan case are at the service disconnect panel on the left side of the support beam.

Remove the protective caps from the integrated drive generator (IDG) feeder connector and receptacle and connect.

Remove the protective caps from the starter duct and support beam receptacle and install the coupling and the preformed packing.

Remove the protective covers from the hydraulic lines and connect to the fan cowl support beam. The hydraulic hose fittings are self-locking quick disconnect fittings. Hand tighten the hose end until the outer sleeve snaps forward and you hear an audible click sound.

Remove the protective caps from the main fuel supply hose and on the support beam fitting and connect the main fuel supply hose to the fitting.

EBU Bracket

Install the engine build up (EBU) bracket.

CAUTION: BE CAREFUL WITH THE HYDRAULIC HOSE. DO NOT BEND OR PUT THE HOSE INTO A COIL TOO TIGHTLY. YOU CAN CAUSE A BLOCKAGE OF THE HOSE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE A FAILURE OF THE ENGINE-DRIVEN PUMP (EDP).

CAUTION: MAKE SURE THAT YOU TIGHTEN AND FULLY ENGAGE THE COLLAR IN THE RATCHET OF THE SUPPLY DISCONNECT CORRECTLY. THE SUPPLY DISCONNECT CAN LOOSEN AND DECREASE THE

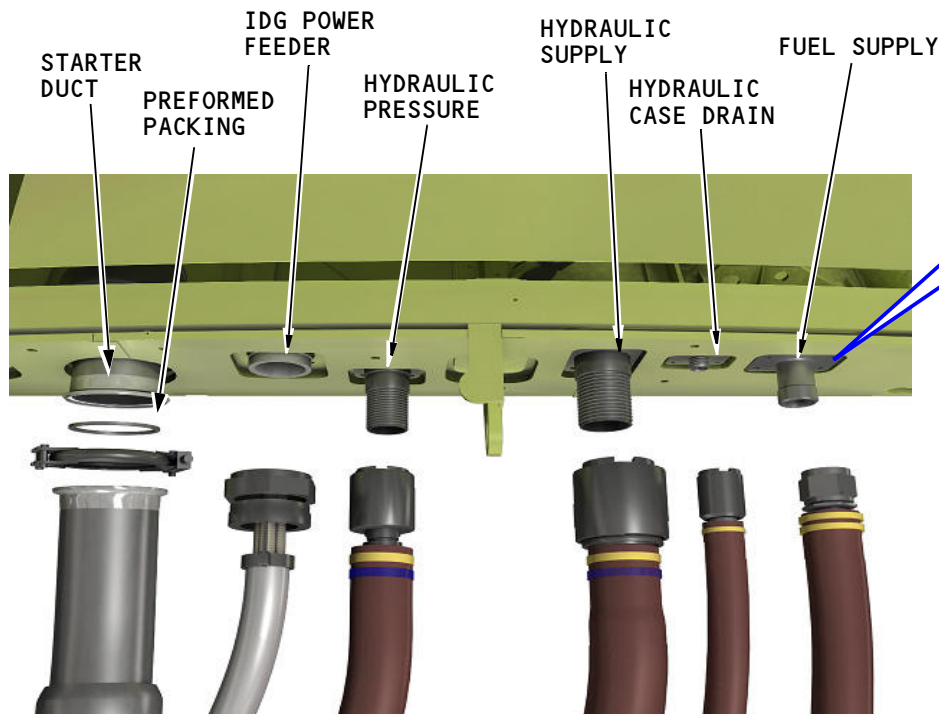
FLOW OF HYDRAULIC FLUID. THIS WILL CAUSE DAMAGE TO THE ENGINE-DRIVEN PUMP.

CAUTION: DO NOT LET HYDRAULIC FLUID LEAKAGE COLLECT. IF YOU DO NOT MAKE THE AREA CLEAN, DAMAGE CAN OCCUR FROM THE HYDRAULIC FLUID.

DMC-B737T-A-71-00-00-57A-100A-T_002-00

02-Dec-2020

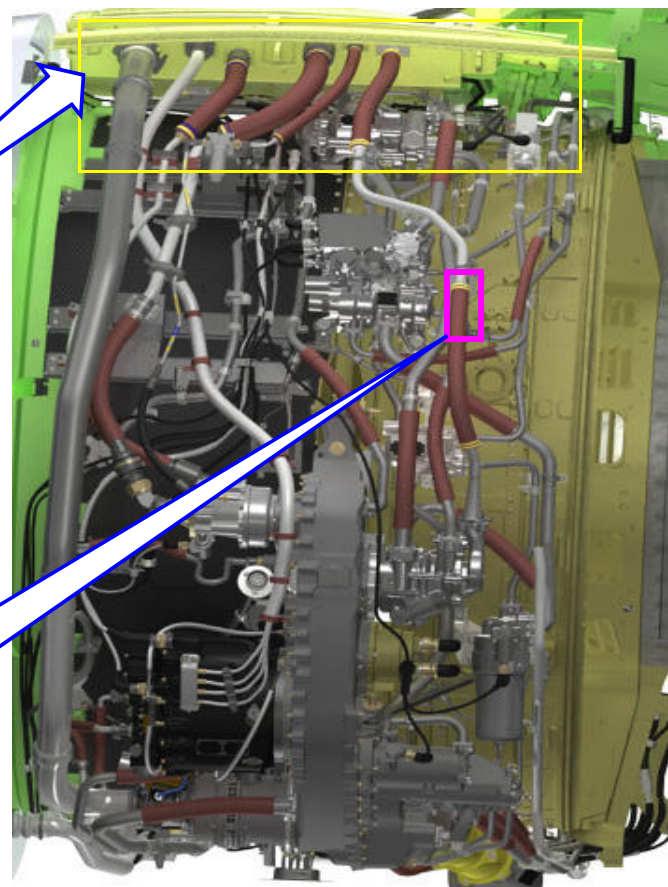
71-00-00



L FAN CASE STRUT CONNECTIONS



EBU BRACKET



FWD

L FAN CASE

DMC-B737T-A-71-00-00-57A-100A-T_002-00
ICN-B737T-A-710000-A-81205-00084-A-003-01

POWER PLANT -- LEAP-1B INSTALLATION - LEFT FAN CASE CONNECTIONS

71-00-00

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - RIGHT FAN CASE CONNECTIONS

Right Fan Case Connections

All right fan case connections are at the service disconnect panel on the right side of the fan cowl support beam. These are the engine plugs:

- DP41B
- DP100
- DP300
- DP21B
- DP200
- DP21A
- DP41A.

These engine plugs connect to receptacles on the strut. The engine plug identifier is the same for the left and right engine. The electrical receptacle identifier is different between the left and right strut. As an example, engine plug DP41B connects to receptacle D30260 on the left strut. This same plug on the right engine connects to receptacle D30460 on the right strut.

Remove the protective covers and connect the electrical harness connectors to their receptacles.

Remove the protective cover from the strut drain line and connect it to the strut fitting.

EBU Bracket

Install the engine build up (EBU) bracket.

CAUTION: MAKE SURE THAT THE ELECTRICAL CONNECTORS ARE CLEAN BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE

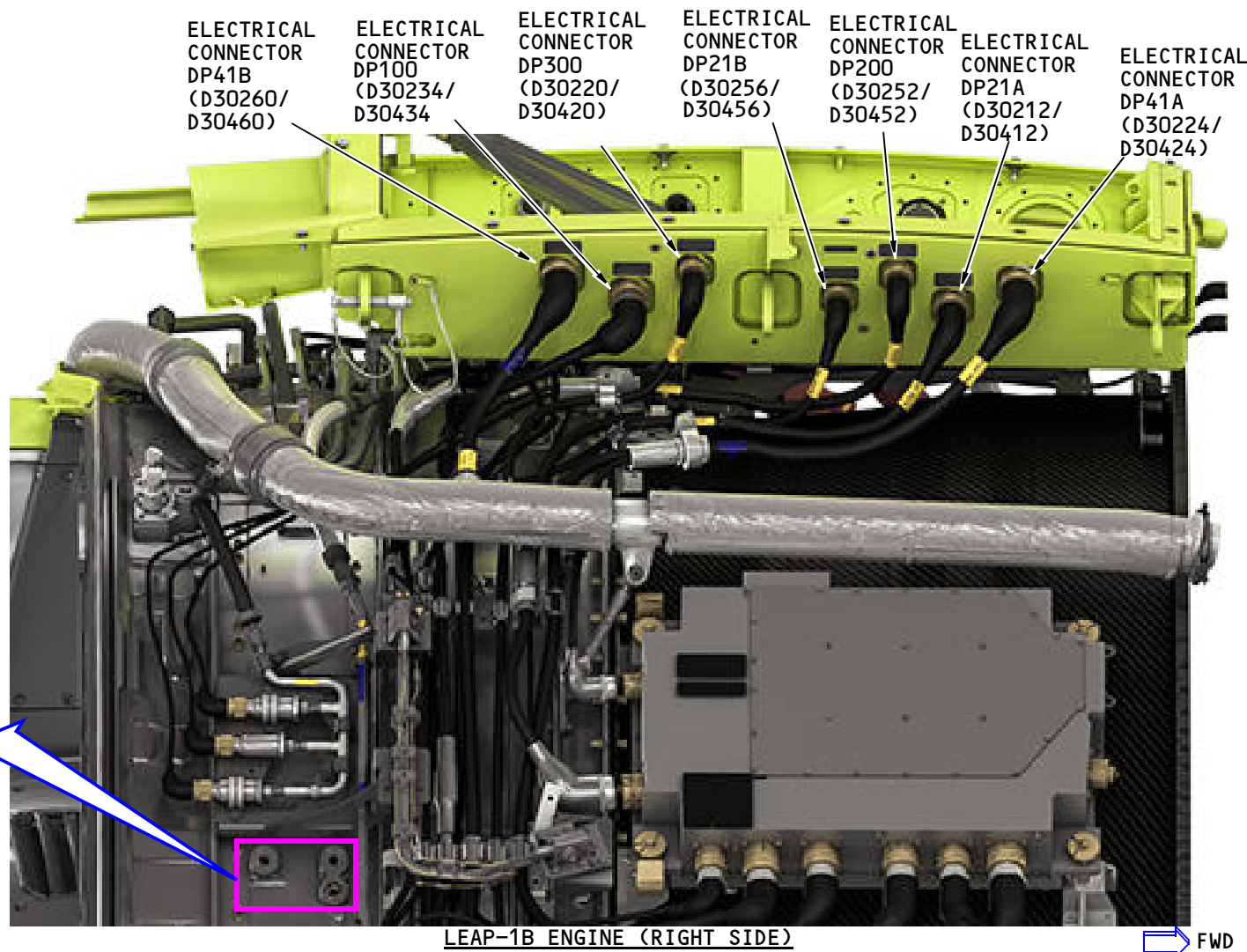
ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO THE EQUIPMENT.

CAUTION: USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

DMC-B737T-A-71-00-00-58A-100A-T_003-00

02-Dec-2020

71-00-00



POWER PLANT -- LEAP-1B INSTALLATION - RIGHT FAN CASE CONNECTIONS

71-00-00

DMC-B737T-A-71-00-00-58A-100A-T_003-00
ICN-B737T-A-710000-A-81205-00085-A-003-01

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - CORE CONNECTIONS

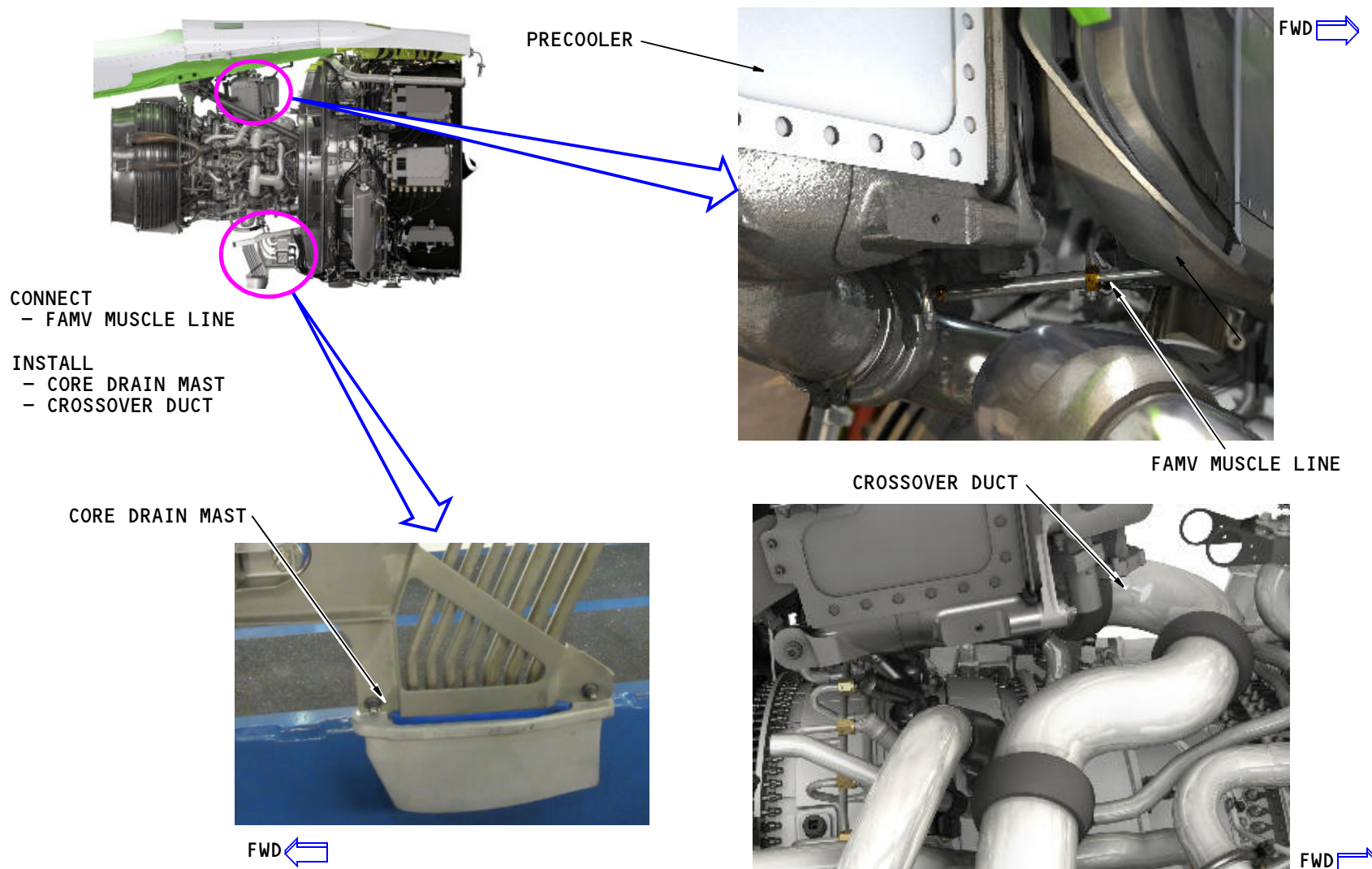
Core Connection

You install the crossover duct and connect the fan air modulating valve (FAMV) line on the right side of the core.

Remove the protective caps from the FAMV muscle line and hose fitting. With caps removed, connect the FAMV line.

Remove the protective caps from the crossover duct and the pre-cooler inlet. Align the crossover duct to the pre-cooler inlet and install.

Install the core drain mast at the bottom of the core.



DMC-B737T-A-71-00-00-60A-100A-T_002-00
ICN-B737T-A-710000-A-81205-00087-A-003-01

POWER PLANT -- LEAP-1B INSTALLATION - CORE CONNECTIONS

71-00-00

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - CONNECT T/R OPENING ACTUATOR

T/R Opening Actuator

Connect the lower fitting of the thrust reverser (T/R) opening actuator to the fan case.

Remove the dust cap from the actuator hand pump connection and connect the hand pump. Close the return to valve on the pump and extend the actuator slightly until the thrust reverser weight is off of the thrust reverser hold-open equipment.

Engage the locking collar on the opening actuator and listen for the audible click sound of the lock collar. Make sure you see the word LOCKED on the bottom of the extended piston.

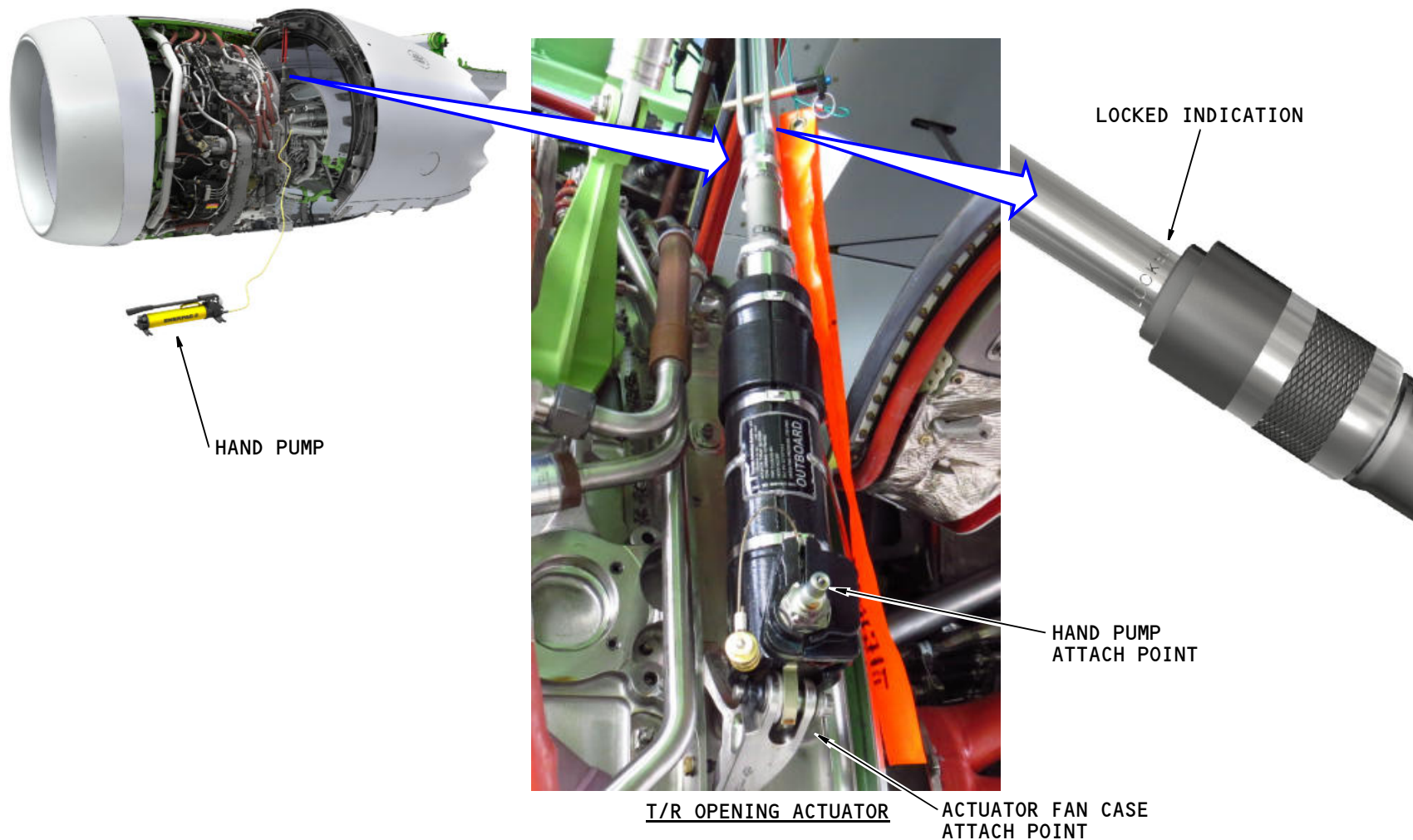
Disconnect the hand pump and install the dust cap.

Repeat the same steps for the opposite side of the engine.

DMC-B737T-A-71-00-00-61A-100A-T_001-00

02-Dec-2020

71-00-00



POWER PLANT -- LEAP-1B INSTALLATION - CONNECT T/R OPENING ACTUATOR

71-00-00

DMC-B737T-A-71-00-00-61A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00088-A-001-01

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - REMOVE HOLD-OPEN EQUIPMENT

Removal of the Hold-Open Equipment

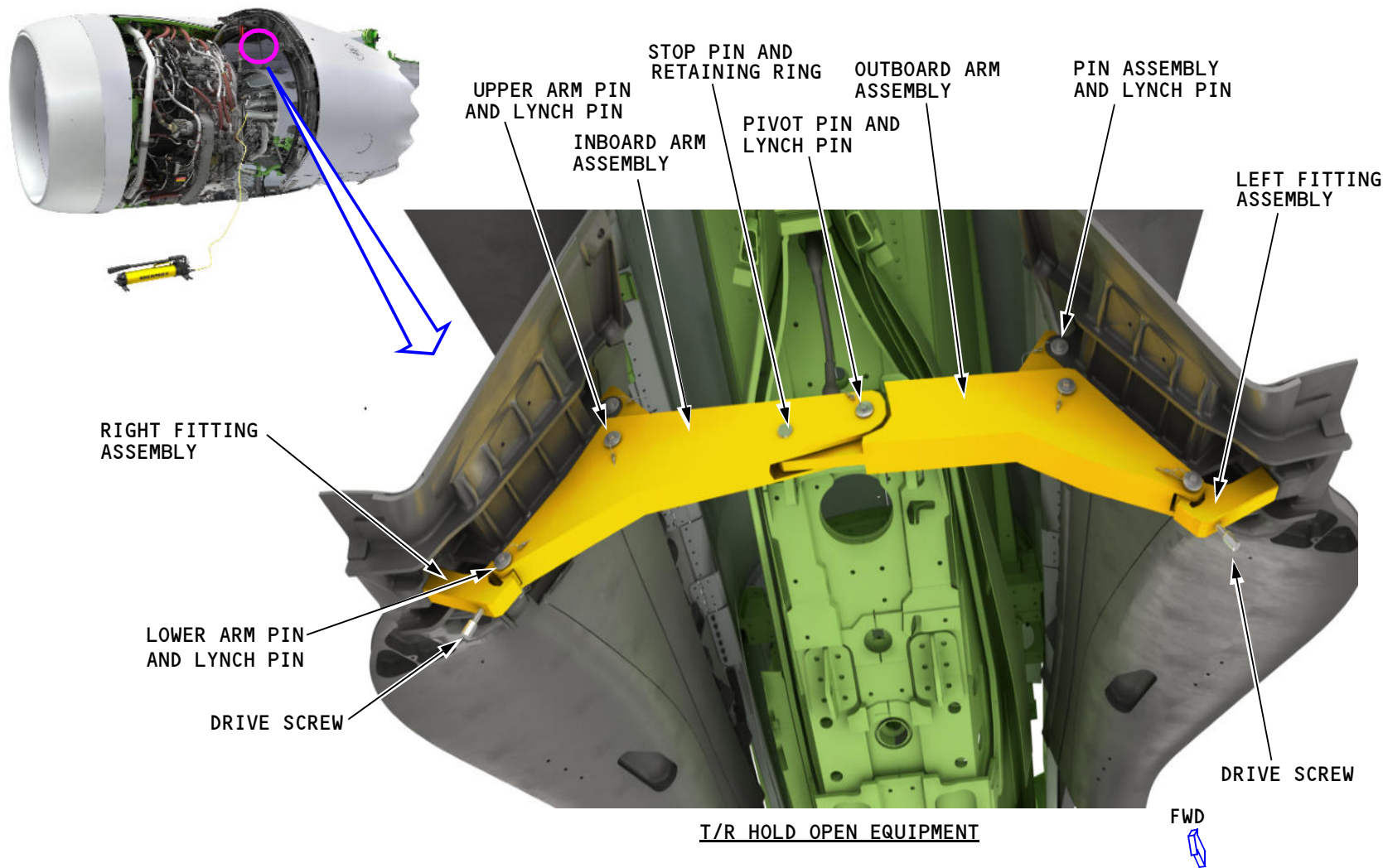
Do these steps to remove the hold-open equipment from the thrust reverser and the strut:

- Make sure the two opening actuators hold the weight of the thrust reverser
- Back out the drive screws to move the pin holes to lowest position on the left and right fitting assemblies
- Disconnect the inboard arm assembly from the outboard arm assembly. You remove the lynch pin from the pivot pin then remove the pivot pin
- Remove the outboard arm assembly from the left fitting assembly and then from the stop pin and retaining ring of the inboard arm assembly
- Remove the inboard arm assembly from the outboard side of the right fitting assembly
- Remove the left fitting assembly and the right fitting assembly from the thrust reverser fittings
- Remove the thrust reverser hold-open equipment.

Install the thrust reverser insulation blankets over the hold-open equipment attach points.

WARNING: MAKE SURE ALL THE WEIGHT OF THE THRUST REVERSER IS HELD BY THE OPENING ACTUATORS. IF THEY ARE NOT, THE THRUST REVERSER CAN MOVE SUDDENLY WHEN YOU REMOVE THE HOLD-OPEN EQUIPMENT. INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

DMC-B737T-A-71-00-00-62A-100A-T_002-00



POWER PLANT -- LEAP-1B INSTALLATION - REMOVE HOLD-OPEN EQUIPMENT

71-00-00

DMC-B737T-A-71-00-00-62A-100A-T_002-00
ICN-B737T-A-710000-A-81205-00089-A-002-01

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - PREPARE FOR SERVICE

Prepare the Power Plant for Service

If the engine preservation procedure was done, do the aircraft maintenance manual (AMM) depreservation of the engine on-wing procedure.

Make sure the servicing of the engine oil system is correct.

Make sure the servicing of the IDG oil is correct.

Close the thrust reverser.

Install and close the fan cowl.

Remove all objects from the engine inlet and work area around the power plant.



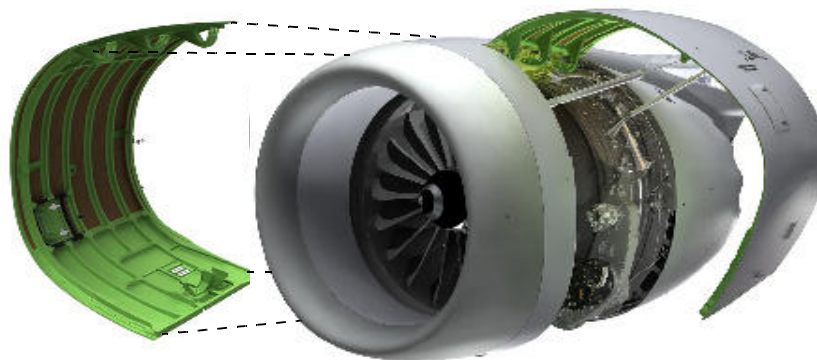
SERVICE ENGINE OIL



SERVICE IDG



CLOSE THRUST REVERSER



INSTALL AND CLOSE FAN COWLS

POWER PLANT -- LEAP-1B INSTALLATION - PREPARE FOR SERVICE

71-00-00

DMC-B737T-A-71-00-00-63A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00090-A-001-01

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - CLOSE CIRCUIT BREAKERS

Prepare the Airplane for Service

Close all the circuit breakers that were opened during engine removal.

These circuit breakers include:

- Fuel Spar Valve Eng
- Fire Protection Extinguishers
- Engine Ignition
- Engine Ignition Power Control
- Engine Start Valve
- Engine Alt Power
- Engine Fuel HPSOV Control
- Engine Fuel Engine HPSOV Indication

Remove the DO-NOT-REMOVE tags from the start levers.

Remove the DO-NOT-REMOVE tags from the HYD PUMPS ENG switch on the pilots overhead panel, P5 forward, for the applicable power plant.

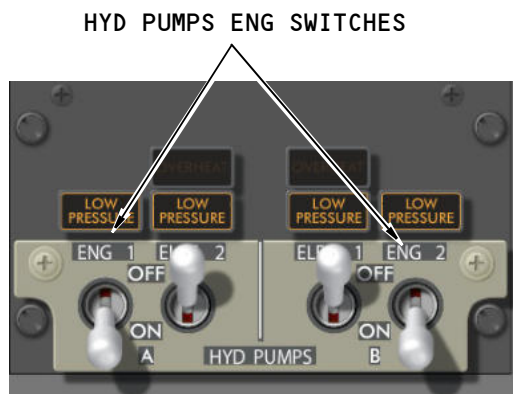
Remove the DO-NOT-OPERATE tag from the BAT switch on the pilots overhead panel, P5 forward. Supply electrical power (AMM 24-22-00).

Apply ground power.

DMC-B737T-A-71-00-00-64A-100A-T_001-00

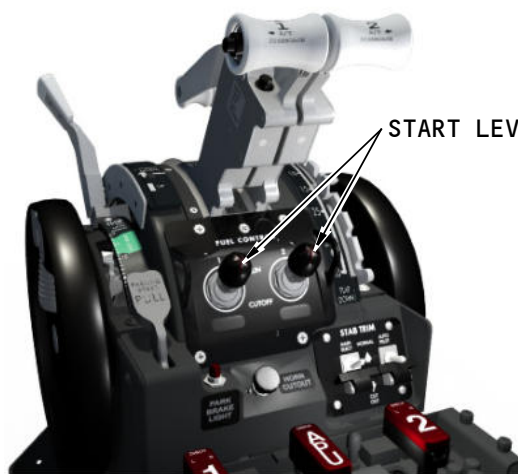


P6 AND P18 CIRCUIT BREAKER PANELS



HYDRAULIC CONTROL PANEL (P5)

BAT SWITCH



CONTROL STAND



EXTERNAL POWER PANEL (P19)



ELECTRICAL POWER PANELS (P5)

POWER PLANT -- LEAP-1B INSTALLATION - CLOSE CIRCUIT BREAKERS

71-00-00

DMC-B737T-A-71-00-00-64A-100A-T_001-00
ICN-B737T-A-710000-A-81205-00091-A-001-01

02-Dec-2020

POWER PLANT -- LEAP-1B INSTALLATION - ENGINE CONFIGURATION

Engine Configuration

Engine configuration data shows on the MAX display system (MDS) or the portable maintenance device (PMD). The configuration data for each propulsion control system includes:

- EEC hardware part number
- EEC serial number
- EEC software location description
- EEC software part numbers
- Engine serial number
- PSS part number
- Engine configuration
- Engine rating
- Bump
- N1 trim
- EEC software versions.

Make sure the new engine serial number and the engine thrust rating are correct.

If the replacement engine is different from the removed engine, install the new software part number.

If necessary, load the proper trim balance data for the replacement engine.

Check the EECs software configuration. If not correct, install new software.

On the MAINT DATA PGS, ENGINE EXCEEDANCE page, check for recorded exceedance. If you find exceedances for the replaced engine position, erase the exceedance for the applicable engine..

DMC-B737T-A-71-00-00-65A-100A-T_001-00

02-Dec-2020

71-00-00

SYSTEM CONFIGURATION

ATA SYSTEM CONFIGURATION

71 - 80 Left Engine

ELECTRONIC ENGINE CONTROL (EEC)-1A

Hardware Part Number:

Serial Number:

Software Location Description:

EEC-1A

73 EEC OPS Part Number:

GEC4A-7500-18ZZ

77 EEC AVM Part Number:

GEC50-7440-0406

77 EEC EHM Part Number:

GEC46-7582-96ZZ

77 EEC AML Part Number:

GEC40-7567-3001

Options:

Engine Serial Number:

000-000

PSS Part Number:

2474M65P06

71 - 80 Left Engine

Engine Serial Number:

000-000

PSS Part Number:

2474M65P06

Engine Configuration:

0001

Engine Rating:

1B28B2

Bump:

Y

N1 Trim:

0

73 EEC OPS Version:

0060

77 EEC AVM Version:

0304

77 EEC EHM Version:

0060

77 EEC AML Version:

0060

SYSTEM CONFIGURATION

SYS MENU STATUS MAINT DATA PGS MAINT CTRL PGS ONBD MAINT

ENGINE 1 EXCEEDANCE AUTO

DATE: XX XXX XX TIME: XX:XX:XX N1 AMBER

DATE: XX XXX XX TIME: XX:XX:XX N2 REDLINE

DATE: XX XXX XX TIME: XX:XX:XX N1 REDLINE

DATE: XX XXX XX TIME: XX:XX:XX N2 AMBER

DATE: XX XXX XX TIME: XX:XX:XX EGT AMBER

DATE: XX XXX XX TIME: XX:XX:XX EGT REDLINE

DATE: XX XXX XX TIME: XX:XX:XX EGT START

DATE: XX XXX XX TIME: XX:XX:XX EGT AMBER

PREV MENU PRINT SEND ERASE PREV PAGE NEXT PAGE

ENGINE EXCEEDANCE PAGE

POWER PLANT -- LEAP-1B INSTALLATION - ENGINE CONFIGURATION

71-00-00

POWER PLANT -- LEAP-1B - PRESERVATION/DEPRESERVATION

General

Preservation procedures for the power plant give protection from environmental damage when the airplane is not in use for an extended period of time. The procedure protects the engine against corrosion, fluids, and debris entering the engine. It may also consist of preparing the engine oil system for storage.

The procedure for preservation will vary depending upon the duration of inactivity and if the engine is operable or non-operable. Other conditions such as high humidity, large temperature changes, and locations near salt water can affect the procedures used.

Depreservation of the power plant returns power plant systems to normal operation after the airplane has not been used for an extended period of time.

The Aircraft Maintenance Manual (AMM) has the following preservation schedule based with the engine on wing:

- Up to 10 days
- Up to 30 days
- Up to 90 days
- 30- 365 days
- 365 days and beyond.

DMC-B737T-A-71-00-00-66A-100A-T_002-00

02-Dec-2020

71-00-00



737-7/8/9/10 Maintenance Training Manual



737-7/8/8200/9

Aircraft Maintenance Manual Part II Practices and Procedures

BOEING PROPRIETARY, CONFIDENTIAL, AND/OR TRADE SECRET
Copyright © 2016 The Boeing Company
Unpublished Work - All Rights Reserved

Boeing claims copyright in each page of this document only to the extent that the page contains copyrightable subject matter. Boeing also claims copyright in this document as a compilation and/or collective work.
This document includes proprietary information owned by The Boeing Company and/or one or more third parties. Treatment of the document and the information it contains is governed by contract with Boeing. For more information, contact The Boeing Company, P.O. Box 3707, Seattle, Washington 98124.
Boeing, the Boeing logo, the Boeing symbol, 737, 717, 727, 737, 747, 757, 767, 777, 787, Dreamliner, BBJ, DC-8, DCR, DC-10, MD-10, MD-11, MD-80, MD-88, MD-90, and the Boeing family are all trademarks owned by The Boeing Company, and no trademark license is granted in connection with this document unless provided in writing by Boeing.

ORIGINAL ISSUE DATE: APRIL 5, 2016
PUBLISHED BY BOEING COMMERCIAL AIRPLANES GROUP, SEATTLE, WASHINGTON, USA
A DIVISION OF THE BOEING COMPANY
PAGE DATE: Jun 10/2018

PRESERVATION SCHEDULE 737-7/8/8200/9

UP TO 10 DAYS

UP TO 30 DAYS

UP TO 90 DAYS

30 - 365 DAYS

365 DAYS AND BEYOND

AIRCRAFT MAINTENANCE MANUAL

POWER PLANT -- LEAP-1B - PRESERVATION/DEPRESERVATION

71-00-00

ENGINE -- INTRODUCTION

Purpose

This chapter gives general data about the modules and internal components of the LEAP-1B engine. The LEAP-1B is a high bypass dual rotor axial flow turbofan engine. The LEAP-1B is a CFM engine. CFM is a 50/50 cooperative venture between General Electric (GE) aviation and SAFRAN Aircraft Engines (formerly SNECMA).

SAFRAN Aircraft Engines is responsible for these parts of the engine:

- Air inlet section
- Low pressure turbine (LPT) section
- Oil system
- Accessory drives
- Engine indicating.

GE Aviation is responsible for these parts of the engine:

- High pressure compressor (HPC) section
- High pressure turbine section (HPT) section
- Combustion diffuser section
- Fuel and control system
- Air system
- Ignition system
- Starting system.

The engine supplies power to these airplane systems:

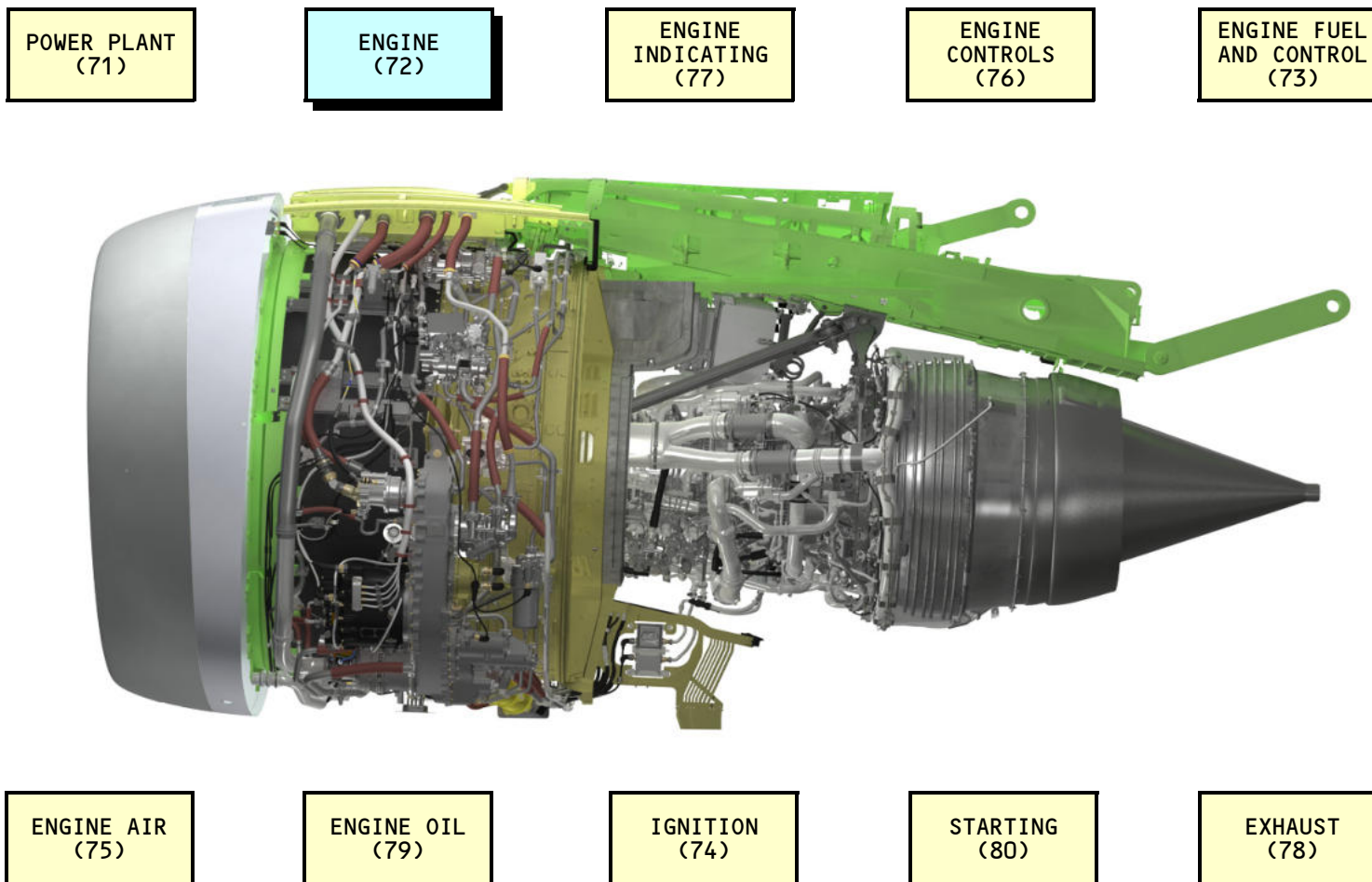
- Electrical
- Hydraulic
- Pneumatic.

Abbreviations and Acronyms

AGB	- accessory gearbox
ALF	- aft looking forward
BLISK	- bladed disk (single component)

CDP	- compressor discharge pressure
EDP	- engine driven pump
EEC	- electronic engine control
EGT	- exhaust gas temperature
HP	- high pressure
HPC	- high pressure compressor
HPT	- high pressure turbine
IDG	- integrated drive generator
IGB	- inlet gear box
LEAP	- leading edge aviation propulsion
LPC	- low pressure compressor
LPT	- low pressure turbine
MFP	- main fuel pump
OPT	- oil pressure and temperature
P	- pressure
PAS	- pneumatic air starter
PMA	- permanent magnet alternator
RDS	- radial drive shaft
SFD	- squeeze film damper
T	- temperature
TGB	- transfer gear box
TS	- transfer shaft

DMC-B737T-A-72-00-00-02A-018A-T_002-00



ENGINE -- INTRODUCTION

72-00-00



ENGINE -- NAMEPLATE DATA

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE -- NAMEPLATE DATA

General

General data for the LEAP-1B engine shows on the nameplate. The engine nameplate is on the right fan case aft of the oil tank.

This data is on the engine nameplate:

- Regulatory agency
- Engine manufacturer and serial number
- Engine model configuration
- Type certificate number
- Maximum continuous thrust
- Takeoff thrust
- Date of manufacture.

Engine assembly data shows in the regulatory agency data blocks. The engines assembled by the General Electric company use the three top right blocks. The engines assembled by SAFRAN Aircraft Engines use the three top left blocks. The serial number shows for all types.

The first line of the seven blocks is recorded at the assembly.

From left to right the blocks give RATED TO MODEL CONFIGURATION information that includes:

- CONFIG - (model configuration of the engine)
- RATED TO in daN (rated takeoff thrust in metric decanewtons)
- MAX CONT in daN (maximum continuous thrust in metric decanewtons)
- TAKEOFF THRUST in LB (takeoff thrust in pounds)
- MAX CONT THRUST in LB (maximum continuous thrust in pounds)
- N1 TRIM (low pressure compressor/turbine trim rating)
- SERV BUL (service bulletins applied to this engine)

Six additional rows are available to document this (rated to model) configuration data if it needs to be changed for the engine.

The lower two blocks show the manufacturer data. The first block shows the manufacturer of the engine. For engines assembled by General Electric, the block shows G.E. CO. For engines assembled by SAFRAN, the block shows SAFRAN. The second block shows the date of manufacture.

Engine Thrust Ratings and Airplane Model Application

There are five basic engine thrust rating configurations for the 737 MAX. The maximum thrust and takeoff thrust limits are based on three causes. They are airplane weight, elevator/rudder control limits and customer option. The longer-fuselage 737 MAX-8 and 737 MAX-9 models are able to operate at the maximum thrust capability of the LEAP-1B engine. The lower thrust ratings may not be sufficient for the 737 MAX-8, 737 MAX-9. Bump ratings are also available as customer options. The Boeing Equivalent Thrust and Usage Chart shows the basic engine thrust ratings for the 737 MAX models. Boeing Equivalent Thrust is the thrust each engine configuration will make when it is attached onto the airplane's strut. This is usually less than the OEM thrust ratings.

The engine data entry plug gives the selected thrust rating data to the electronic engine controls (EECs). The data entry plug is on the pressure subsystem.

Airplane Models

The normal models are 737 MAX-7, -8, and -9. Some examples of other variations are -8 200, and Boeing Business Jet (BBJ).



ENGINE -- NAMEPLATE DATA

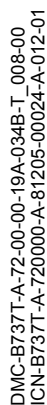
General Engine Data

The general engine data applies to all engine configurations 1B21 - 1B28.

N1 Trim

The EEC uses an N1 trim level that adjusts the engine fan speed for engine thrust variations. After engine assembly, a test cell measures the actual engine thrust. Each engine must produce 27,900 lbs of thrust. Some engines produce more than 27,900 lbs of thrust. These engines are derated using an N1 trim level. The data entry plug gives the EECs the N1 trim level for this engine. The data entry plug has a lanyard that stays with the engine if you change the pressure sub-system.

The N1 trim levels are 0 through 7. Level 0 is no trim, and level 7 is the maximum trim of 1.51%.



BOEING EQUIVALENT THRUST AND USAGE CHART
NOTE: NOT ALL MODELS/RATINGS SHOWN

ENGINE -- NAMEPLATE DATA



ENGINE -- GENERAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE -- GENERAL DESCRIPTION

General

The LEAP-1B is a modular concept design engine. The engine has three major modules; 20 minor modules and an accessory drive module. The engine fan diameter is 69.4 inches (1.76 meters). The bare engine weight is 5830 pounds (2644 kilograms).

The 3 major modules are:

- The fan major module
- The core major module
- The low pressure turbine major module.

These are some of the minor engine modules:

- Low pressure compressor (LPC)
- High pressure compressor (HPC)
- Combustion section
- High pressure turbine (HPT)
- Low pressure turbine (LPT)
- Accessory drive (The accessory drive section is three minor modules, an inlet gearbox, a transfer gearbox and an accessory gearbox).

The LPC and the LPT are on a common shaft which is called N1.

The HPC and the HPT are on a common shaft which is called N2.

Low Pressure Compressor (Fan and Booster)

The LPC is a four-stage compressor. The fan is the first stage of compression. The fan increases the speed of the air. The booster is a three-stage compressor. The booster increases the pressure of this air and sends it to the HPC.

The splitter fairing divides the air into these two air flows:

- Primary
- Secondary.

The primary air flow goes into the core of the engine to be compressed and used for combustion. This airflow produces about 20 percent of the thrust.

The secondary air flow goes into the fan duct. The shape of the fan duct makes the pressure in the duct change. This pressure change makes thrust. The secondary airflow makes 80 percent of the engine thrust.

High Pressure Compressor (HPC)

The HPC is a ten-stage compressor. It increases the pressure of the air from the LPC and sends it to the combustor. The HPC also supplies bleed air for the aircraft pneumatic system and cooling for the engine.

Combustion Section

The combustion section mixes air from the compressors and fuel from the fuel nozzles. This mixture of air and fuel burns in the combustion chamber to make hot gases. The hot gases go to the HPT.

High Pressure Turbine (HPT)

The HPT is a two-stage turbine. It changes the energy of the hot gases into mechanical energy. The HPT uses this mechanical energy to turn the HPC rotor and the accessory drive.

Low Pressure Turbine (LPT)

The LPT is a five-stage turbine. It also changes the energy of the hot gases into mechanical energy. The LPT uses this mechanical energy to turn the fan and booster (LPC).

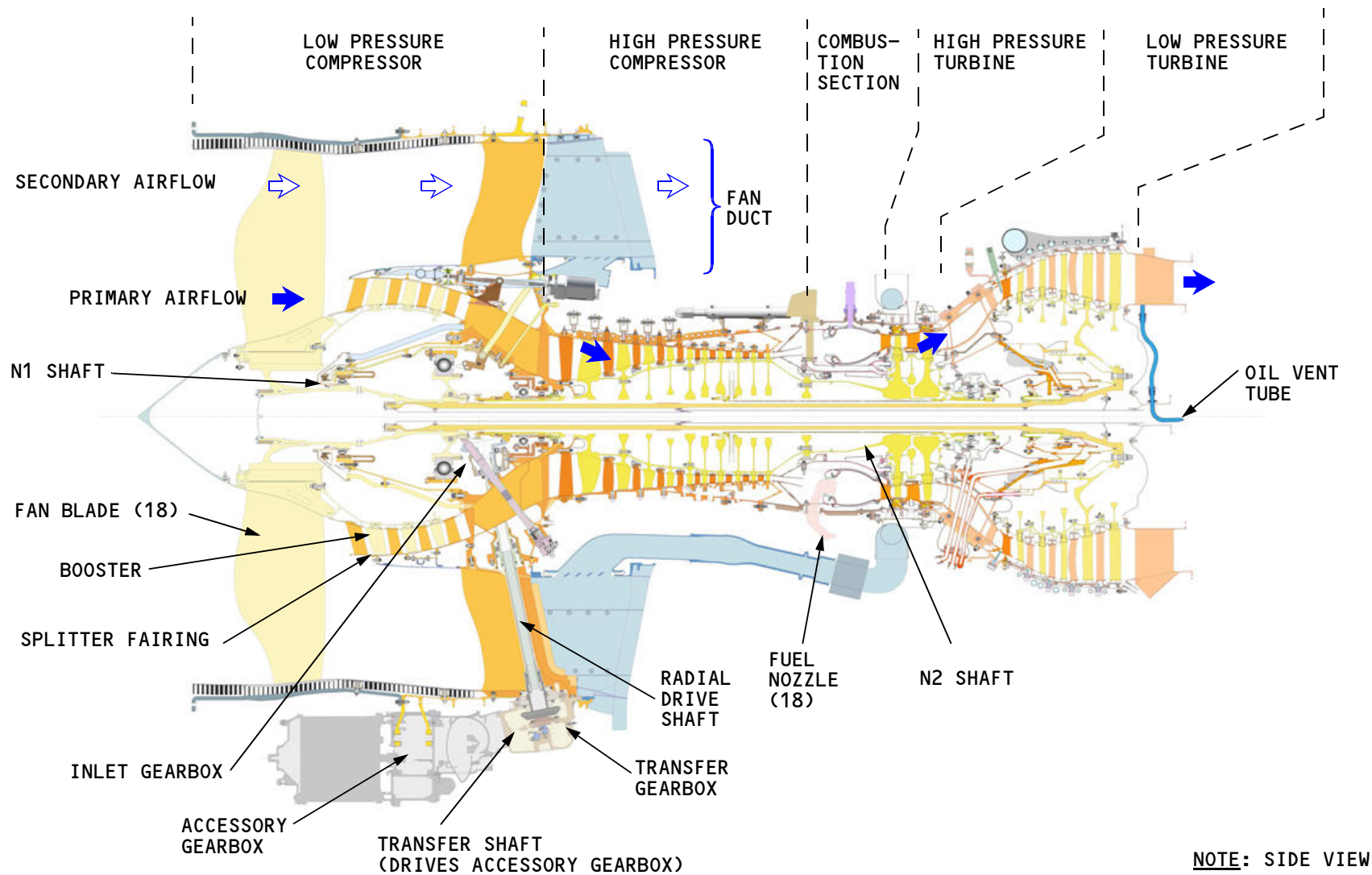
DMC-B737T-A-72-00-00-02A-040A-T_002-00

ENGINE -- GENERAL DESCRIPTION

Accessory Drive

The N2 shaft turns the AGB through the accessory drive. The accessory drive has these minor modules and assemblies:

- Inlet gear box (IGB)
- Transfer gearbox (TGB)
- Accessory gearbox (AGB)
- Number 3 bearing assembly
- Radial driveshaft (RDS)
- Transfer shaft (TS).



ENGINE -- GENERAL DESCRIPTION

72-00-00

DMC-B737T-A-72-00-00-02A-040A-T_002-00
ICN-B737T-A-720000-A-81205-00025-A-001-01

02-Dec-2020

ENGINE -- MAIN ENGINE BEARINGS

General

Five main engine bearings hold the N1 shaft and the N2 shaft. The bearings are numbered from 1 to 5. The ball bearings absorb the axial (thrust) and radial loads from the shafts. Roller bearings absorb radial loads. The main engine bearings are in three oil sumps. The oil sumps are A, B and C from front to rear.

Main Engine Bearings

The A sump has two ball bearings (number 2B and 3B) and two roller bearings (number 1R and 3R).

The B sump has 1 roller bearing (number 4).

The C sump has 1 roller bearing (number 5).

Bearings number 1, 2 and 5 hold the low pressure rotor (N1) .

Bearings number 3 and 4 hold the high pressure rotor (N2).

The number 3 bearing assembly has two bearings, the forward bearing is a ball bearing and the aft bearing is a roller bearing.

Sump Seals

Sump seals prevent oil leaks from the engine sumps.

The front of the A sump is sealed with a segmented radial (carbon) seal (SRS) and a labyrinth seal. An abradable seal upstream of the number 3 bearing seals the middle of the A sump. An SRS seal downstream of the number 3 bearing seals the aft end of the A sump. The A sump is vented to the oil vent tube.

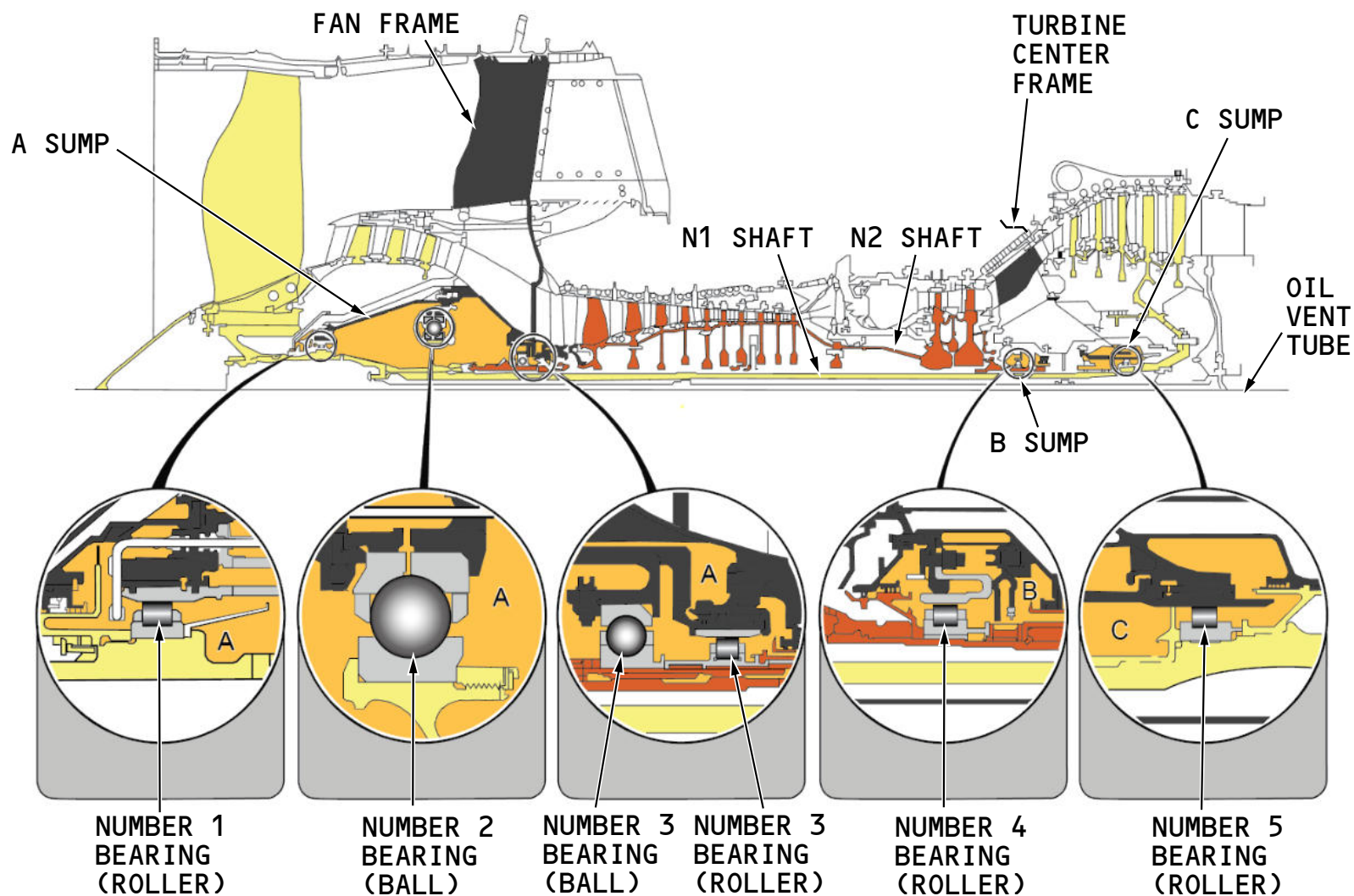
An labyrinth seal upstream and an SRS seal downstream of the number 4 bearing seals the B sump.

An SRS seal upstream and an labyrinth seal downstream of the number 5 bearing seals the C sump.

The B and C sumps are not vented.

Frames

The fan frame gives support for the number 1, 2 and 3 bearings. The turbine center frame gives support for the number 4 and 5 bearings.



ENGINE -- MAIN ENGINE BEARINGS

72-00-00

DMC-B737T-A-72-00-00-20A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00026-A-002-01

ENGINE -- AERODYNAMIC STATIONS

General

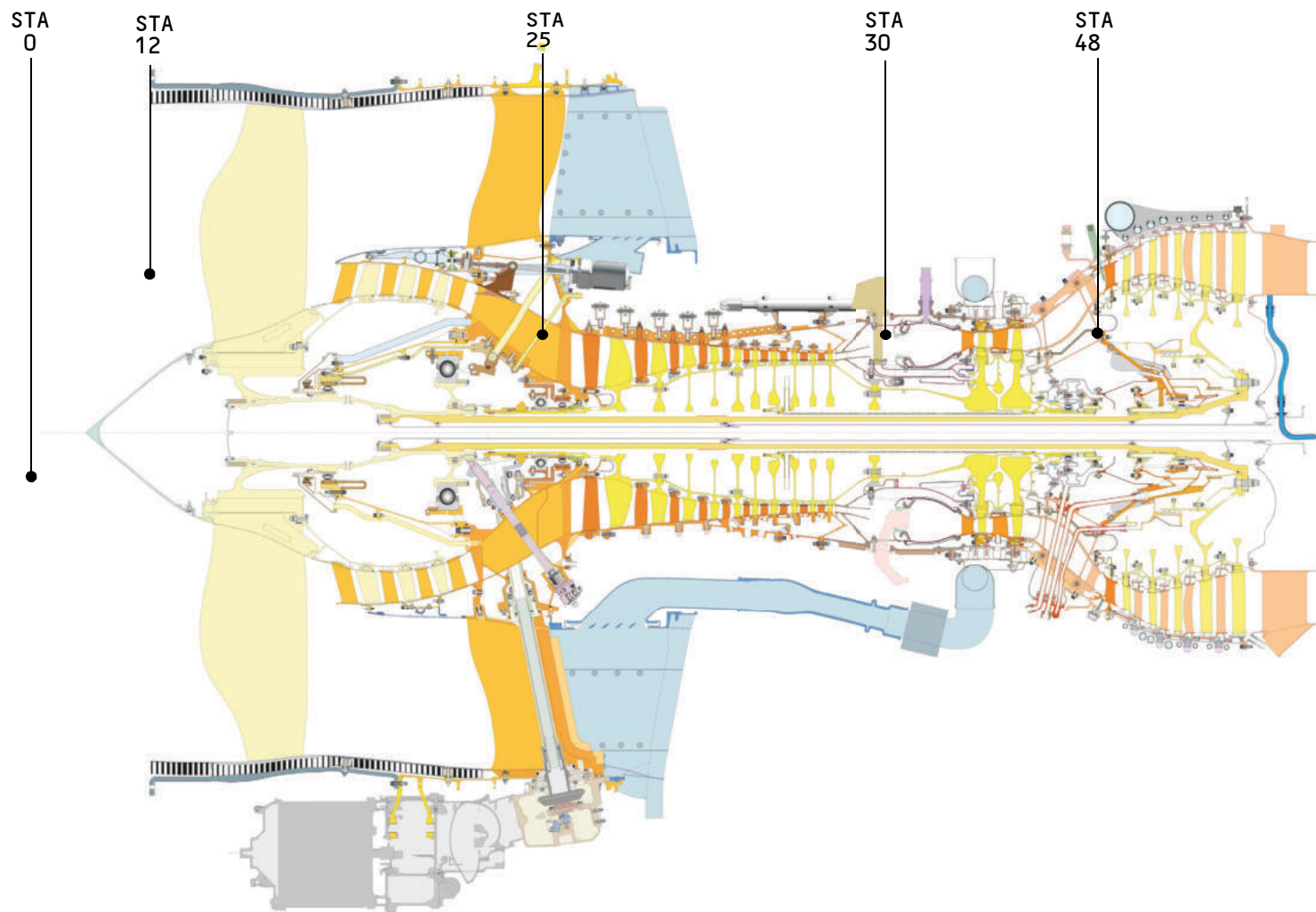
There are probes or sensors at these 5 engine aerodynamic stations:

- Station 0. The P0 sensor on the pressure subsystem (PSS) box measures ambient air pressure.
- Station 12. The T12 sensor in the engine inlet measures fan inlet temperature. The T12 sensor is installed on the inlet cowl, at the 11:00 position, from the aft looking forward.
- Station 25. The T25 sensor measures high pressure compressor inlet temperature. The T25 sensor is installed on the fan hub frame, at the 3:00 position, from the aft looking forward.
- Station 30. The PS3 and T3 sensors measure high pressure compressor discharge pressure and temperature. The PS3 outlet port is found on the combustion case, at the 10:00 position, from the aft looking forward. Tubes connect the PS3 outlet port to the PSS. The T3 sensor is installed on the high pressure compressor case, at the 10:00 position, from the aft looking forward.
- Station 48. The eight T48 (EGT) probes measure temperature at the inlet of the low pressure turbine. The EGT probes are attached to the turbine center frame (TCF).

DMC-B737T-A-72-00-00-21A-034B-T_002-00

02-Dec-2020

72-00-00



ENGINE -- AERODYNAMIC STATIONS

72-00-00

DMC-B737T-A-72-00-00-21A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00027-A-002-01

02-Dec-2020

ENGINE -- MAJOR MODULES

Engine Modules

The LEAP-1B is a modular engine. There are 3 major modules and an accessory drive.

The 3 major modules are the:

- Fan major module
- Core major module
- Low pressure turbine major module.

The fan major module is the front of the engine. The inlet cowl attaches to the front of the fan major module. The aft end of the fan major module attaches to the core major module.

The core major module is the center of the engine. The front of the core major module attaches to the aft of the fan major module. The aft end of the core major module attaches to the low pressure turbine major module.

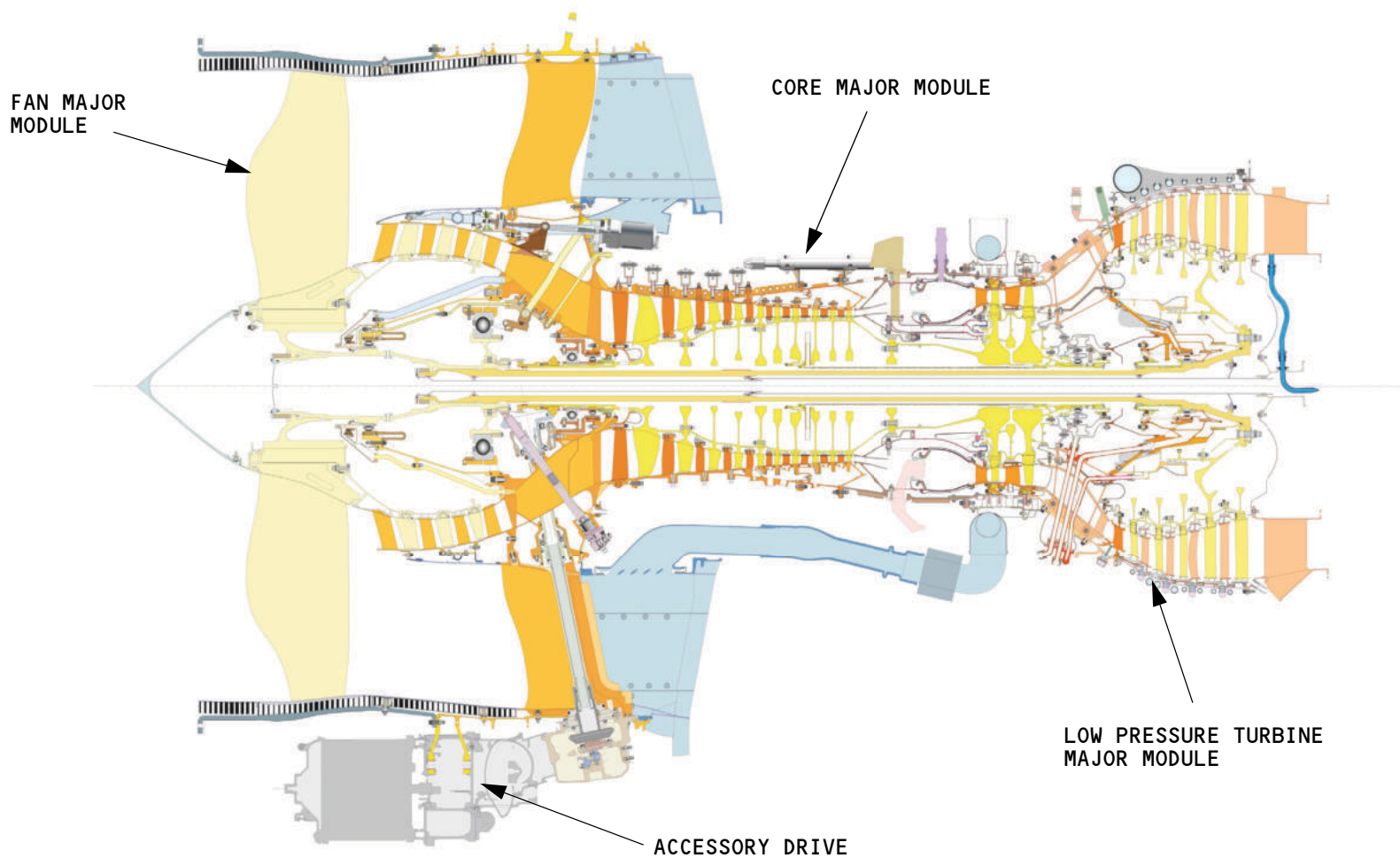
The low pressure turbine major module is the aft end of the engine. The front of the low pressure turbine major module attaches to the aft end of the core major module.

The 3 major modules and the accessory drive section are divided into 20 minor modules. With this modular concept, it is possible to replace damaged minor or major modules. This type of a program gives the operator more satisfactory maintenance of the engine and better turn around time.

DMC-B737T-A-72-00-00-22A-034B-T_002-00

02-Dec-2020

72-00-00



ENGINE -- MAJOR MODULES

DMC-B737T-A-72-00-00-22A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00028-A-002-01

02-Dec-2020

72-00-00

ENGINE -- FAN MAJOR MODULE

General

The fan major module is at the front of the engine, downstream from the inlet cowl.

The fan major module has these 5 modules:

- Fan and booster
- Fan frame
- Fan case
- Number 1 and 2 bearing support
- Inlet gearbox (IGB) and Number 3 bearing support.

The main functions of the fan modules are to give:

- Approximately 90% of the engine thrust
- Attachment to the engine strut/pylon
- Structural rigidity in the front section
- Containment for front section major deterioration and/or damage
- Noise reduction for the fan section
- Attach points for gearbox and nacelle equipment
- Attachment to the core engine.

The splitter fairing divides the primary and secondary airflow of the engine.

Fan and Booster

The functions of the fan and booster are to:

- Increase air velocity, which makes thrust
- Increase the pressure of the air that goes to the high pressure compressor (HPC).

Fan Frame Module

The fan frame module has these major assemblies:

- Fan hub frame
- Outlet guide vanes
- Upper and lower bifurcation struts
- Variable bleed valve (VBV) door system.

Fan Case

The fan case is made of composite material. Six aft acoustic panels are found here. These panels decrease noise and give a smooth aerodynamic surface for secondary airflow.

Number 1 and 2 Bearing Support Module

The number 1 and 2 bearing support module is part of the fan major module. Its functions are to:

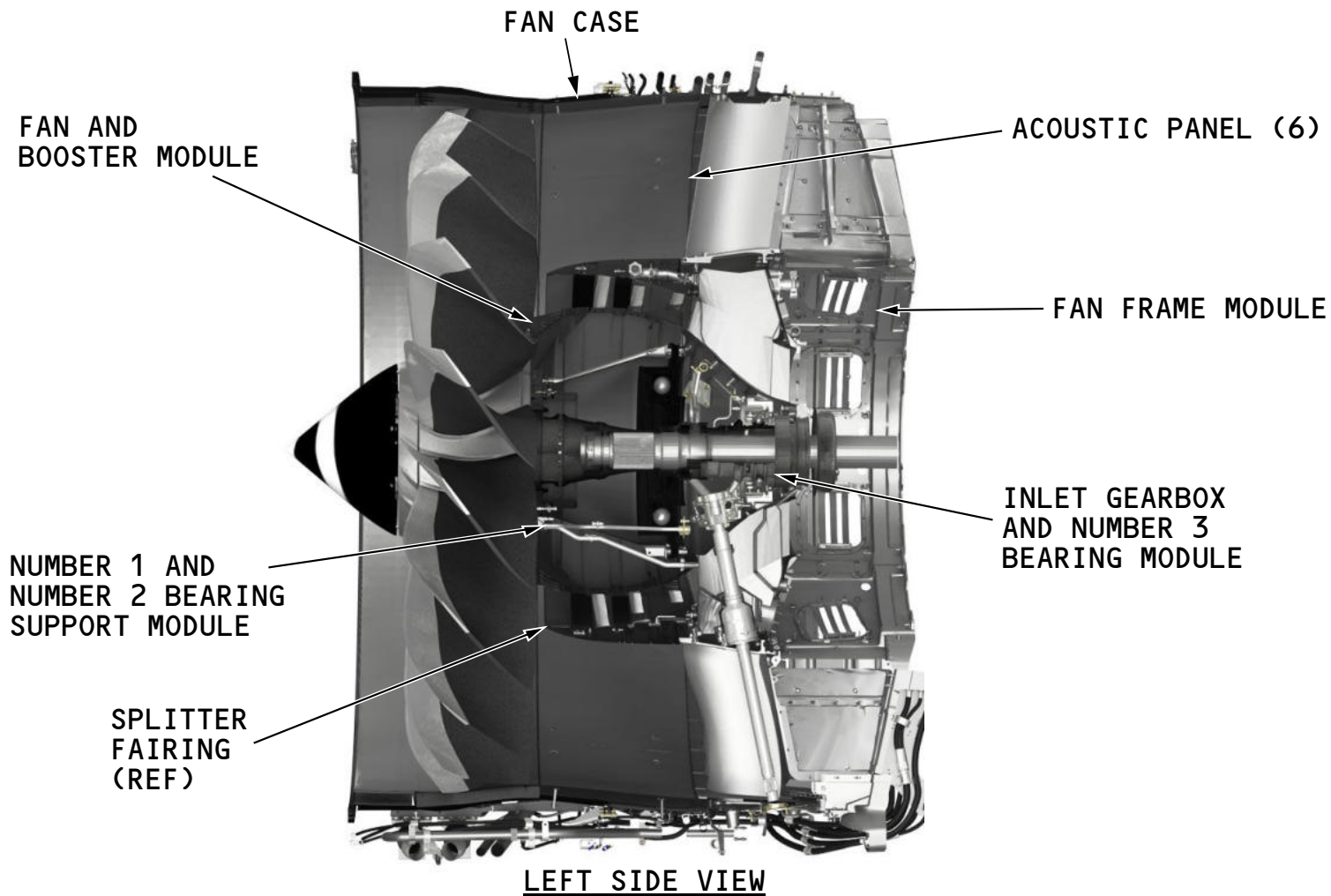
- Give support to the low pressure compressor shaft
- Go around the front section of the forward oil sump
- Give support to the N1 speed sensor
- Give a vent for the forward sump
- Give support to the forward vibration sensor
- Give bearing lubrication.

The bearing 1 and 2 support module is in the forward sump between the fan and booster module and the fan frame module.

Inlet Gearbox (IGB) and Number 3 Bearing

The IGB transmits mechanical energy between the HPC front shaft and the radial drive shaft.

The IGB and number 3 bearing assembly is located in the forward sump of the engine on the fan frame module.



ENGINE -- FAN MAJOR MODULE

72-00-00

DMC-B737T-A-72-00-00-23A-034B-T_005-00
ICN-B737T-A-720000-A-81205-00029-A-003-01

02-Dec-2020

ENGINE -- FAN AND BOOSTER

Fan and Booster

The fan and booster are at the front of the engine, downstream from the air inlet cowl.

This module has the:

- Front spinner cone
- Platform front shroud
- Fan disk
- Fan blades
- Fan blade platform
- Booster.

The fan and booster assembly connects to the N1 shaft, and the fixed assembly attaches to the fan frame.

Front Spinner Cone

The aerodynamic front spinner cone gives a smooth surface for airflow to the fan stage. It keeps ice build-up to a minimum and decreases noise level.

Platform Front Shroud

The platform front shroud is an aerodynamic fairing which directs engine inlet airflow. The platform front shroud is made up of titanium and attaches to the fan disk. Balance weights attach to the platform front shroud between the front spinner cone and the fan blades.

Fan Disk

The fan disk function is to give the mechanical coupling for the fan blades. It is a titanium alloy and has 18 dovetail slots for the installation of the fan blades. Each dovetail slot has a fan disk shim installed to prevent wear when the fan blade is installed.

Fan Blades

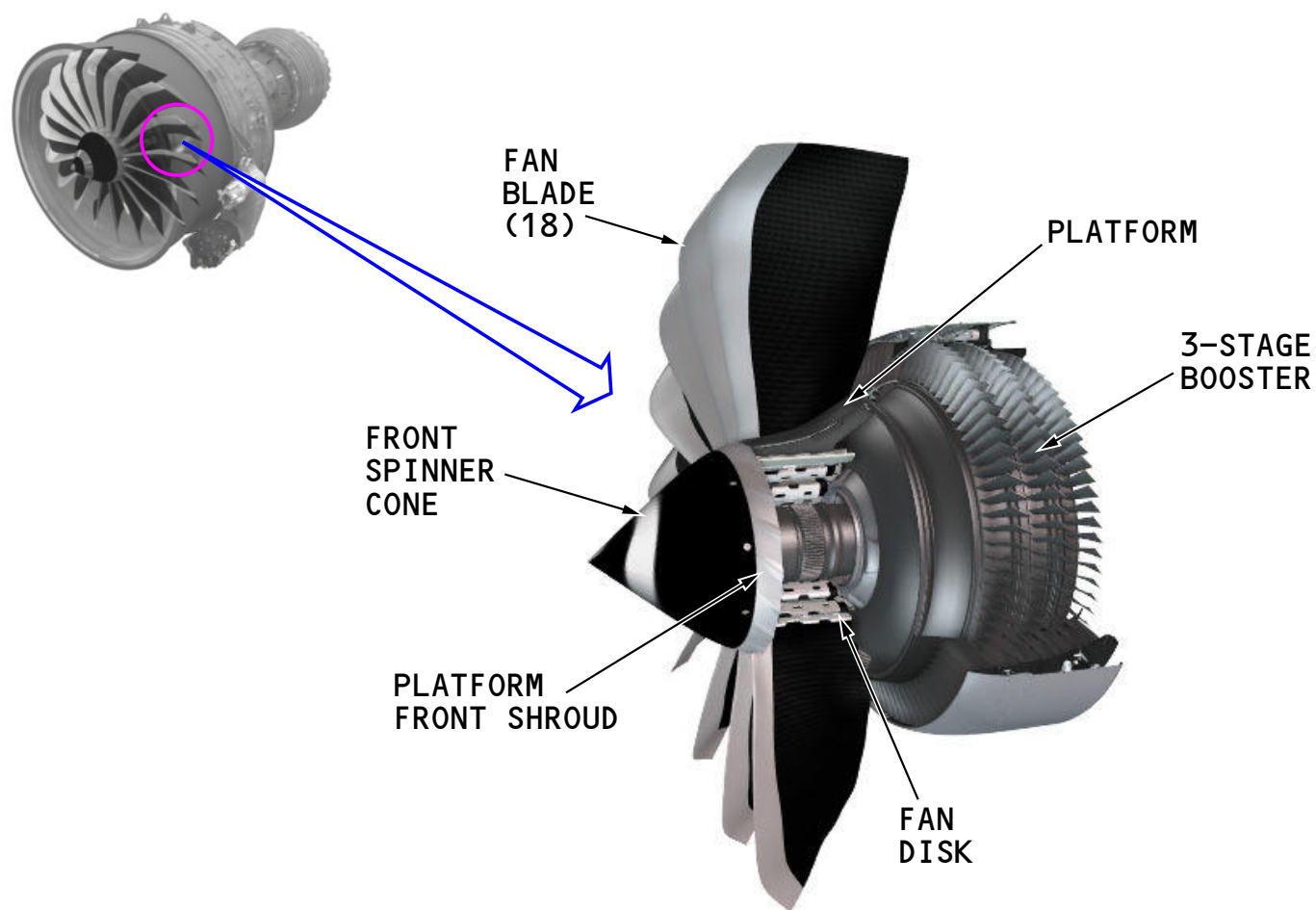
There are 18 wide chord composite blades with titanium leading edges. They are the first stage of the LPC. The LPC increases the velocity of the air that goes into the engine from the inlet cowl.

Fan Blade Platform

There are 18 fan blade platforms. They seat in between each fan blade.

Booster

The 3 stage booster increases the velocity and compresses the air as it goes through each stage. The booster is located aft of the fan disk. The booster is a three stage rotor with blades and four stages of stator vanes. The four stages of vanes change the velocity of the air and increase the pressure. The booster attaches to the rear flange of the fan disk. The booster interfaces with the fan disk and the fan frame.



ENGINE -- FAN AND BOOSTER

72-00-00

PLATFORM FRONT SHROUD AND FAN BLADE INSTALLATION -- COMPONENT DESCRIPTION

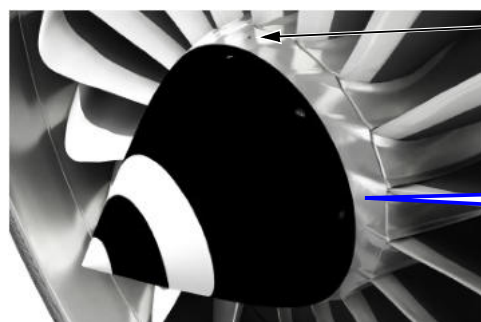
Platform Front Shroud Installation

The platform front shroud helps you install the spinner cone and the platform front shroud without errors. The platform front shroud has a 12 o'clock locator hole for the spinner. This hole is not in the center of the flange as the other holes are. Because this hole is not in the center of the flange, you must correctly align the spinner for installation.

The platform front shroud also has a mark on its outer surface. This mark is a small circular dimple. When you correctly install the platform front shroud, this dimple helps you find fan blade number one.

Fan Disk Installation And Description

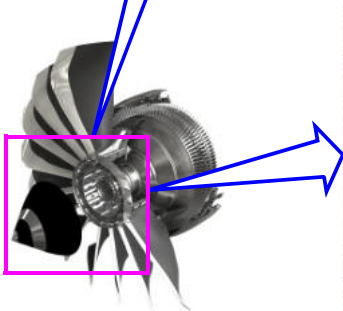
The fan disk is not a line replaceable unit and is only installed or replaced at an approved overhaul shop. The fan disk has dovetail slots for fan blade installation. The front of the fan disk has numbers to help you find the slots for blades one and five. There are no other fan blade numbers on the front of the fan disk.



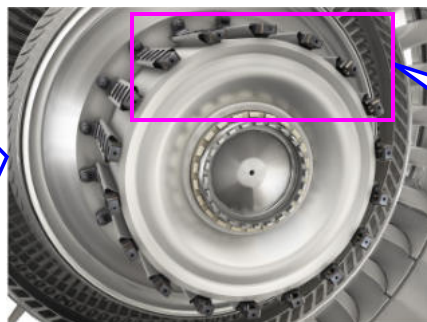
MARK

12 O'CLOCK
OFFSET HOLE

SPINNER AND PLATFORM FRONT SHROUD



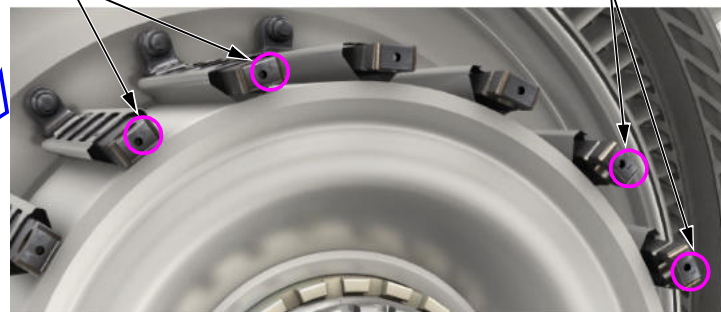
ENGINE INLET AND LOW
PRESSURE COMPRESSOR



FAN DISK (FAN BLADES REMOVED)

FAN DISK
NUMBERS
(BLADE 5)

FAN DISK
NUMBERS
(BLADE 1)



DMC-B737T-A-72-00-00-45A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00077-A-002-01

PLATFORM FRONT SHROUD AND FAN BLADE INSTALLATION -- COMPONENT DESCRIPTION

72-00-00

02-Dec-2020

ENGINE -- FAN BLADES AND SPINNER

Fan Blades

The 18 fan blades are wide chord carbon fiber composite blades with titanium leading edges. The base of each fan blade has a dovetail, that goes into a recess on the fan disk. Composite wear strips interface between the fan blade and the fan disk shields. It is not necessary to do regular re-lubrication of the fan blade root. Spacers hold each fan blade in the correct radial position. The spacers are made of composite material. They hold the fan blades in the static position. Titanium and honeycomb fan blade locks hold the blades axially. The fan blade locks are at the front of the fan blade root. Platforms are between each fan blade. The platforms have a pin that goes into the fan disk at the rear. The platform front shroud retains the front of the platforms.

You must remove the front spinner and the platform front shroud before you can remove one or more of the fan blades. Position the appropriate fan blade at the 6:00 position. You then remove the fan blade spacer. A special tool may be necessary to remove the fan blade spacer from the fan disk. The fan blade spacer keeps the fan blade lock in place, so the lock can then be removed. Remove the fan blade platforms on each side of the fan blade. The fan blade can then be removed from the fan disk. Always refer to the appropriate AMM procedures for fan blade removal and installation procedures.

You install balance screws and weights in the platform front shroud. There are 60 possible positions for the balance screws and weights. The positions are numbered on the platform front shroud. You cannot see the balance screws and nuts if the front spinner is installed. Balance procedures for the fan blades use these configurations:

- Balance screw
- Balance screw and nut
- No balance screw or nut.

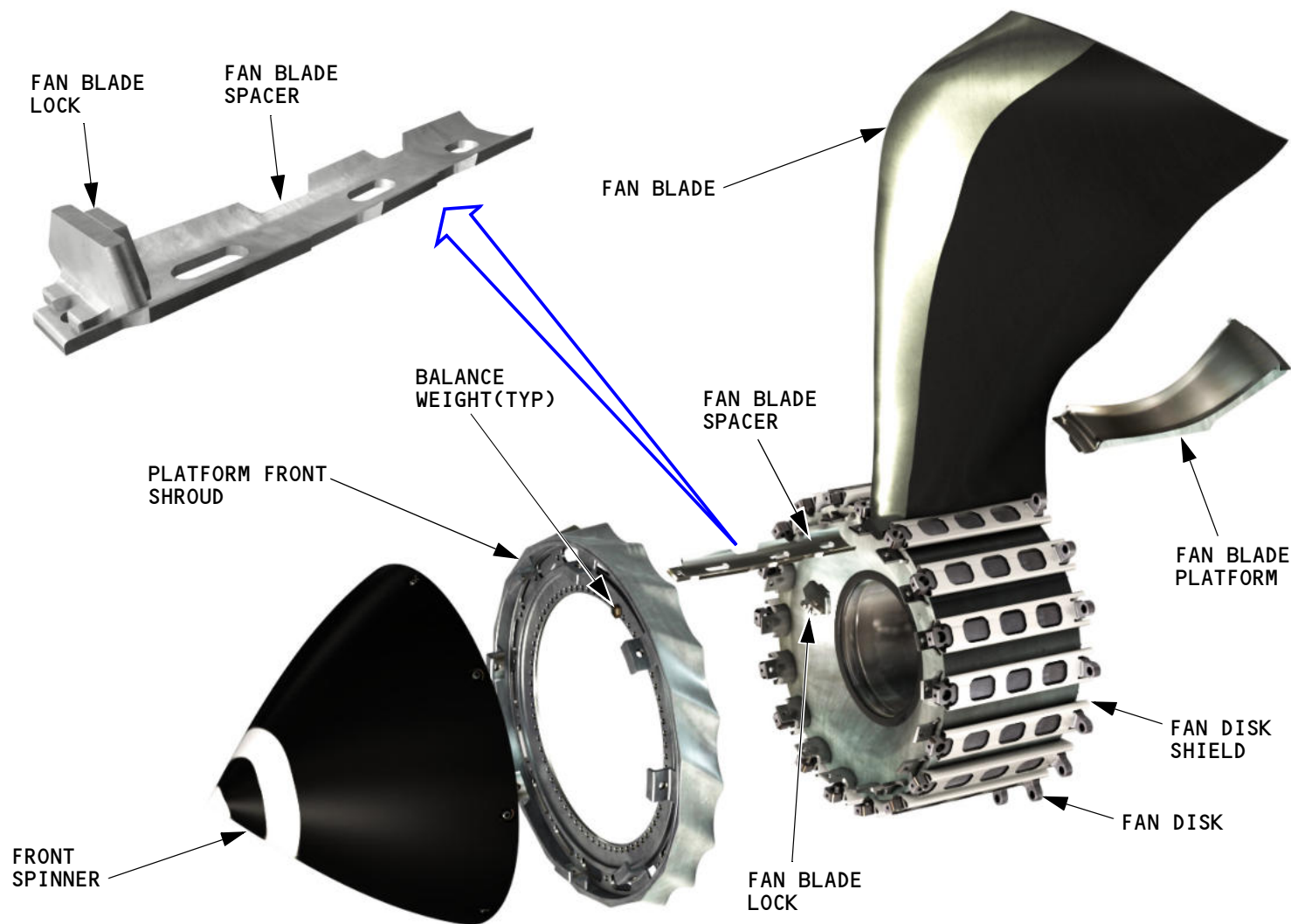
Looking from the engine inlet, the engine fan blades, platforms and weights are numbered in a counterclockwise direction. All components including spacers and locks, are typically numbered and returned to their original locations.

This information is engraved on each blade :

- Manufacturing S.N. (Bottom of metal leading edge close to fan blade platform - convex suction side)
- Blade Serial number (Top of metal leading edge and on the side of the blade shank - concave pressure side)
- Moment weight (Blade shank - not visible when fan blade platform is installed - concave pressure side).

Spinner

The front spinner cone is an aerodynamic fairing that gives a smooth surface for airflow to the fan stage. The spinner cone keeps ice build-up to a minimum and decreases noise level. It is an anodized aluminum alloy. The front spinner cone attaches to the platform front shroud with 9 screws. The front spinner uses torx plus fasteners. During installation it is necessary that the front spinner cone, the platform front shroud and the fan disk align.



ENGINE -- FAN BLADES AND SPINNER

72-00-00

DMC-B737T-A-72-00-00-34A-034B-T_005-00
ICN-B737T-A-720000-A-81205-00042-A-002-01

02-Dec-2020

ENGINE -- CORE MAJOR MODULE

General

The core major module is a high-pressure, high-speed, gas generator that supplies the power to operate the engine. This section is sometimes called the core engine.

The core major module has these modules:

- High Pressure Compressor (HPC)
- Combustion section
- High Pressure Turbine (HPT)

The HPC receives air from the fan and booster module. The HPC compresses the air and sends it to the combustion chamber. The HPT nozzles point the combustion air onto the HPT rotor to turn the HPT. The energy not used by the HPT rotor moves the low pressure turbine (LPT), fan rotor, and booster.

The number 3 bearing is in the fan frame module, and gives support for the forward end of the HPC.

The number 4 bearing is on the HPT rear shaft, and gives support for the aft end of the HPC.

The HPC has these assemblies:

- HPC rotor
- HPC front stator
- HPC rear stator.

The combustion section has these assemblies:

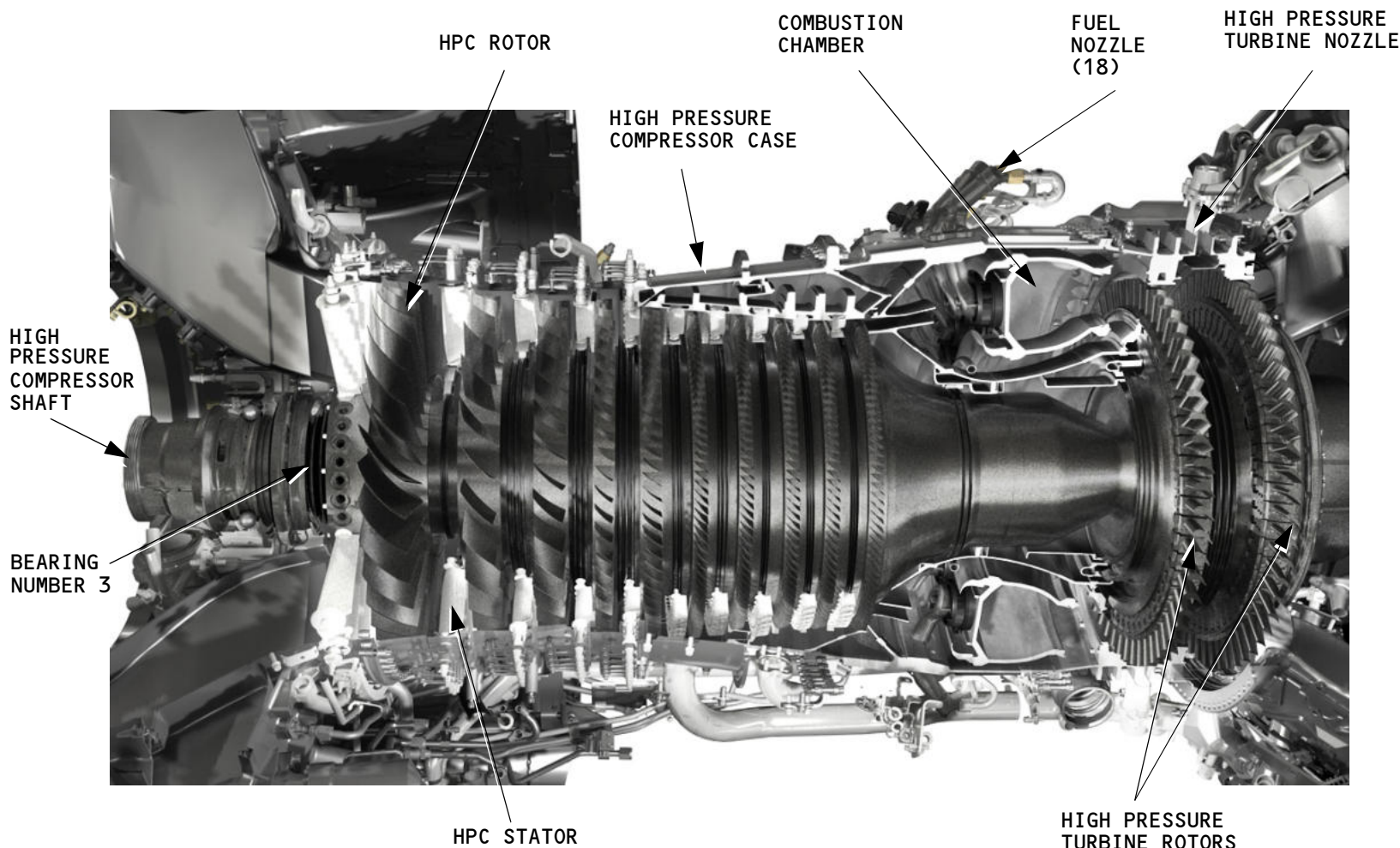
- Combustion chamber and case
- Combustion diffuser nozzle.

The HPT has these assemblies:

- Stage 1 HPT nozzle

- Two stage HPT rotor
- HPT case and stage 2 nozzle.

DMC-B737T-A-72-00-00-25A-034B-T_002-00



ENGINE -- CORE MAJOR MODULE

72-00-00

DMC-B737T-A-72-00-00-25A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00031-A-001-01

02-Dec-2020

ENGINE -- HIGH PRESSURE COMPRESSOR (HPC)

The High Pressure Compressor (HPC)

The HPC is a 10 stage axial flow compressor. The HPC is between the low pressure compressor (LPC) and the combustor. Its primary function is to increase the pressure of the air for combustion. The HPC air also supplies cooling air for internal engine components, air for the environmental control system (ECS) and other customer bleed systems. The first five stages of the HPC rotor use blisk technology (bladed disk). Stages 6-10 are one compressor rotor spool. The spool has dovetail grooves that hold dovetail blades for each stage. The stator system has one stage of inlet guide vanes (IGVs), four stages of variable stator vanes (VSVs) in the front stator case and six stages of fixed stators in the rear stator case of the HPC.

These stages give air for cooling and customer bleed systems:

- HPC stage 4 - ECS, internal cooling of the turbine center frame and the low pressure turbine (LPT)
- HPC stage 7 - nacelle/booster anti-ice, eductor valve, cooling of the high pressure turbine (HPT) stage 2 nozzles and HPT internal cooling
- HPC stage 10 - transient bleed valve and ECS.

The HPC has these modules:

- Rotor
- Front stator case
- Rear stator case.

Rotor

The HPC rotor increases the velocity of the air from the booster module. Stators pressurize the air before it goes into the combustion section.

The HPC is in the compressor case. It gets support from the number 3 bearing.

The rear shaft of the HPC attaches with bolts to the HPT front shaft through the compressor discharge pressure (CDP) air seal.

The rear arm of the stage 1 compressor rotor blisk is bolted to the front side of the stage 2 compressor rotor blisk. The rear side of the stage 2 compressor rotor blisk is bolted to the front arm of the stage 3-4 compressor rotor blisk. The rear arm of the stage 3-4 compressor rotor blisk is bolted to the front side of the stage 5 compressor rotor blisk.

The stage 6-10 compressor rotor spool is one component. The rotor spool has dovetail grooves that hold the dovetail blades.

Front Stator Case

The HPC front stator case has an upper and lower case. The cases attach together at the 3 o'clock and 9 o'clock positions.

The front stator assembly has these:

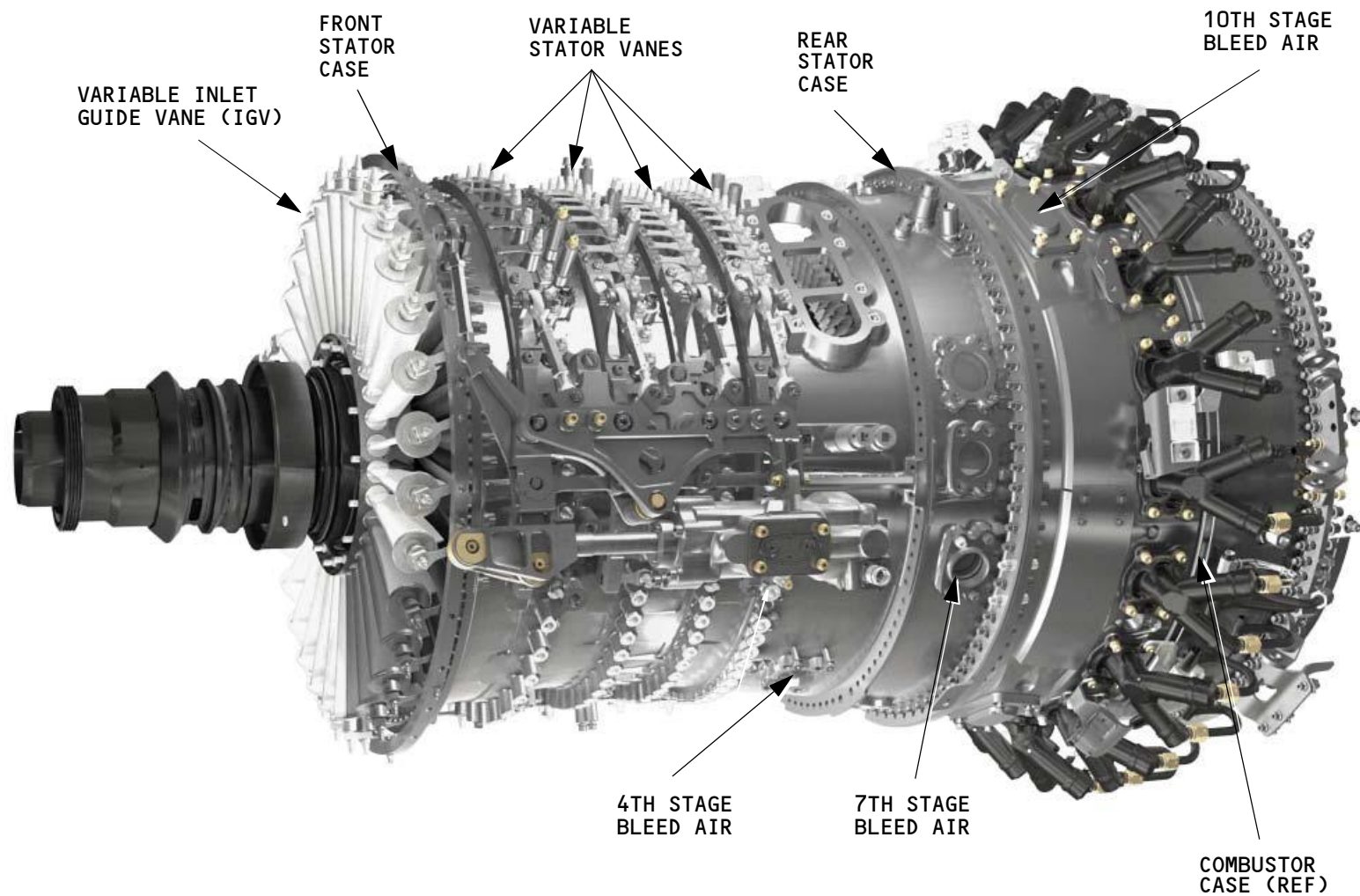
- Stator case halves (2)
- IGVs
- VSV stages 1, 2, 3 and 4
- VSV actuation system.

Rear Stator Case

The HPC rear stator case changes airspeed to pressure and points the air to the next higher stage.

The rear stator has six individual stages.

The HPC case at stage 7 contains the 7th stage bleed port.



ENGINE -- HIGH PRESSURE COMPRESSOR (HPC)

72-00-00

DMC-B737T-A-72-00-00-26A-034B-T_003-00
ICN-B737T-A-720000-A-81205-00032-A-004-01

02-Dec-2020

ENGINE -- COMBUSTION SECTION

Combustion Section

The Combustion section has these two assemblies:

- Combustor diffuser nozzle (CDN)
- Combustion chamber.

The CDN case is between the high pressure compressor (HPC) and the high pressure turbine (HPT). It provides the structural load path between the rear HPC stator case and HPT stator case.

The CDN case supports the combustion chamber assembly inner and outer liners and provides a mounting location for the eighteen fuel nozzles and the two igniters.

The abradable compressor discharge pressure (CDP) stator seal and the HPT stage 1 nozzle guide vane (NGV) support are also attached to the diffuser with nut-and-bolt attachment points.

Combustion Case Assembly

The combustion case assembly is a weldment structure that gives the structural interface between the HPC, the combustor, and the LPT. It transmits the engine axial loads and supplies 10th stage bleed air for engine and airplane use.

The combustion case assembly includes the compressor outlet guide vanes (OGVs) and diffuser, which decreases velocity of the HPC air flow before it goes into the combustion chamber. This makes combustion efficiency better.

Twin annular premixing swirler (TAPS) fuel nozzles attach to pads around the outer case of the CDN. Two (2) igniters attach at the 5:30 and 6:00 position on the CDN.

The fuel nozzles receive fuel from this equipment, which attaches to the fan case and engine core:

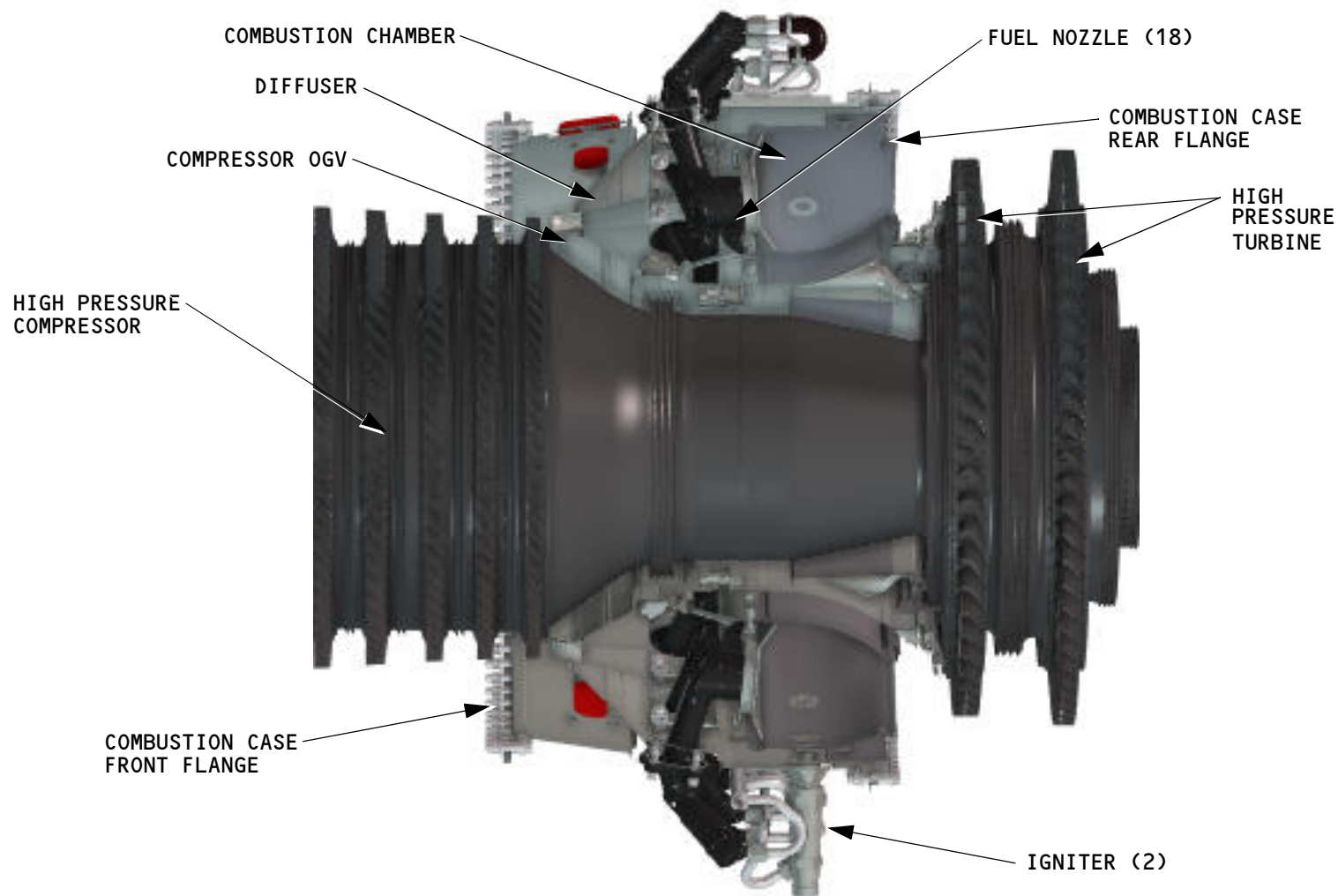
- Fuel metering unit - fan case
- Split control unit (SCU) - engine core
- 3 fuel manifolds - engine core.

Combustion Chamber

The combustion chamber is a short annular structure in the CDN. It is between the HPC stator stage 10 and the HPT nozzle.

The combustion chamber has these:

- 18 TAPS Fuel nozzles
- Combustion chamber dome
- Inner and outer liners
- Radial mixer
- Inner and outer baffle.



ENGINE -- COMBUSTION SECTION

72-00-00

DMC-B737T-A-72-00-00-27A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00033-A-001-01

02-Dec-2020

ENGINE -- HIGH PRESSURE TURBINE (HPT)

High Pressure Turbine (HPT)

The HPT changes the energy of gasses from the combustion chamber into mechanical energy to turn the high pressure compressor (HPC) and accessory drive. The forward flange of the HPT attaches to the outer surface of the combustion case with bolts. The aft flange attaches to the turbine center frame (TCF) forward flange with bolts. The HPT is a two-stage air-cooled assembly that has these:

- Stage 1 HPT nozzle assembly
- 2 stage HPT rotor
- HPT case and stage 2 nozzle assembly.

Stage 1 HPT Nozzle Assembly

The stage 1 HPT nozzle assembly supplies the gas flow from the combustion chamber to the blades of the stage 1 HPT rotor at an angle that gives the best performance during all conditions of operation. The HPT stage 1 nozzle is attached to the aft end of the combustion case.

The stage 1 HPT nozzle assembly has these:

- 18 nozzle segments of 2 vanes each
- Forward inner nozzle support
- Aft outer stationary seal.

2 stage HPT Rotor

The two-stage HPT rotor receives gas flow from the combustion chamber through the HPT stage 1 and stage 2 nozzles. The nozzles and rotor change the energy of the gas flow into mechanical energy for the HPT rotor to turn the HPC and accessory drive.

The two-stage HPT rotor assembly receives cooling air from the HPC stages 7 and 10. Each HPT rotor blade in stage 1 and stage 2 contains holes for cooling each HPT rotor. The rotor assembly is supported by

the number 4 roller bearing. The stage 1 rotor receives compressor discharge air for cooling from the modulated turbine cooling (MTC) system. The stage 2 rotor is cooled by HPC 7th stage impeller tubes.

The 2 stage HPT rotor has these:

- Front shaft
- 2 disks
- Stage 1 and stage 2 rotor blades
- Rear shaft.

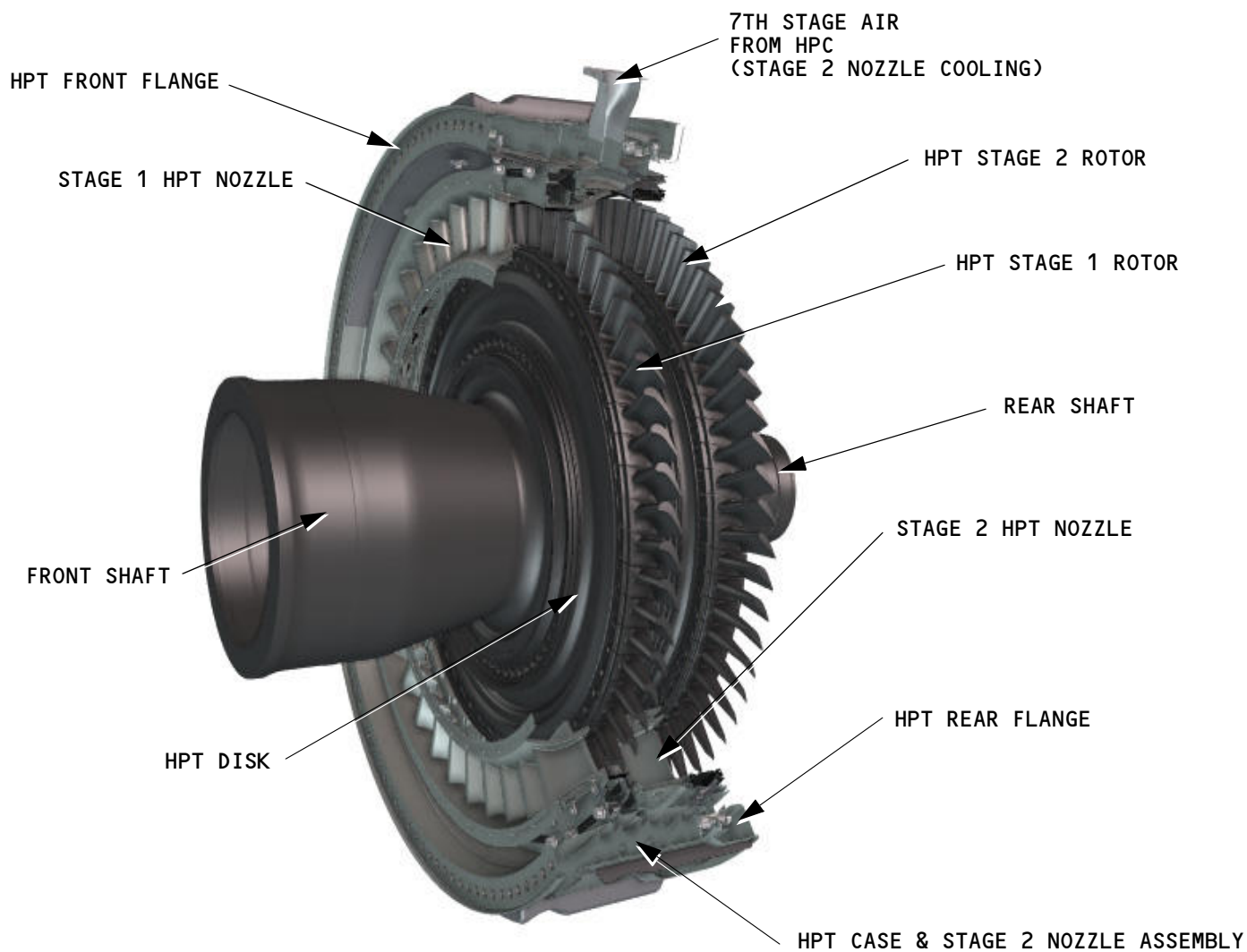
HPT Case and Stage 2 Nozzle Assembly

The HPT case and stage 2 nozzle assembly are the connection between the core section and the TCF module of the engine. The HPT case contains the high pressure and temperature gas coming from the combustion chamber. The HPT case and stage 2 HPT nozzle assembly are the fixed part of the HPT. The stage 2 HPT nozzle assembly directs the stage 1 air to the stage 2 HPT rotor blades. The HPT stage 2 nozzle assembly uses 7th stage HPC bleed air to cool the HPT stage 2 nozzle assembly.

The stage 2 HPT nozzle has these:

- 36 single segment nozzle vanes
- Stage 2 stator hangers and stator shrouds.

The high pressure turbine active clearance control (HPTACC) system cools the outer case. See ATA 75-20-00 for more information about the HPTACC system.



ENGINE -- HIGH PRESSURE TURBINE (HPT)

72-00-00

DMC-B737T-A-72-00-00-03A-040A-T_003-00
ICN-B737T-A-720000-A-81205-00034-A-001-01

02-Dec-2020

ENGINE -- LOW PRESSURE TURBINE (LPT) MAJOR MODULE

General

The LPT is a 5 stage turbine. It is located between the turbine center frame (TCF) assembly and the turbine rear vane (TRV) module.

The functions of the LPT major module are to:

- Change the energy of gasses leaving the high pressure turbine (HPT) into mechanical energy to turn the fan and booster
- Give a rear support for the LP rotors.

The LPT major module is at the rear of the engine and has these modules:

- LPT stage 1 nozzle module
- LPT shaft
- LPT rotor and stator module.

LPT Stage 1 Nozzle Module

The LPT stage 1 nozzle module function moves the airflow from the LPT transition duct to the first stage of the LPT rotor at the angle that gives the best performance during all conditions of operation.

The LPT stage 1 nozzle module has these:

- 40 stage 1 nozzle segments
- Nozzle shroud
- 8 EGT temperature sensors.

LPT Shaft Module

The LPT shaft module transmits mechanical energy from the LP turbine to the fan and booster module. It also provides radial support to the rotating assembly.

The shaft gives a vent for the engine forward and rear sump through a center vent tube.

The LPT shaft module is attached to the number 1 and number 2 bearing support module at the front, and the number 5 bearing support module at the rear.

LPT Rotor and Stator Module

The LPT rotor and stator module changes the energy of gasses from the HPT into mechanical energy that turns the fan and booster.

The LPT rotor and stator module is a five-stage axial flow turbine that has these:

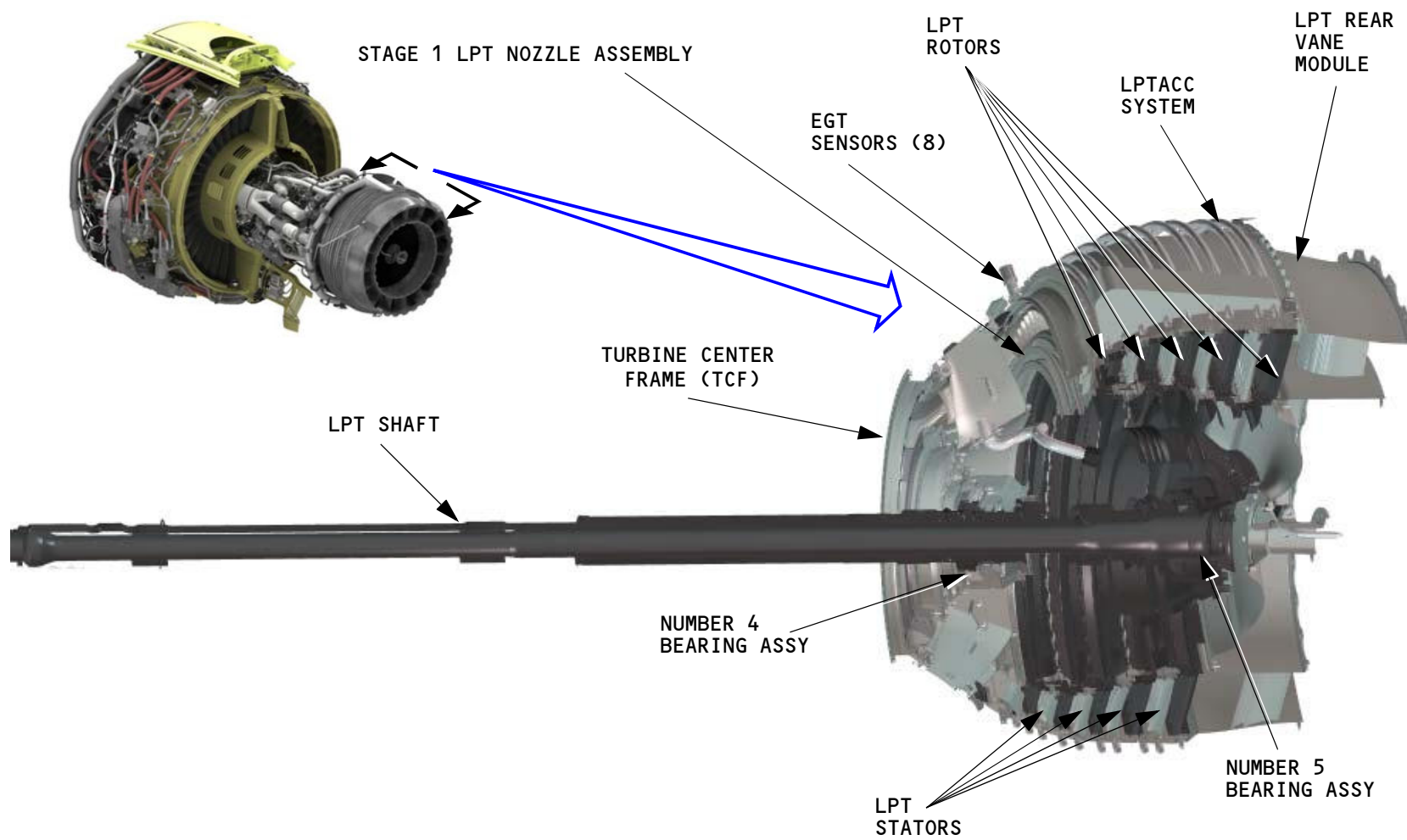
- Rotor assembly
- Stator assembly
- Low pressure turbine active clearance control (LPTACC) system.

LPT Cooling

HPC 4th stage air cools the LPT case front flange with air from the TCF. Holes in the TCF and the front flange of the LPT case allow cooling air to flow.

HPC 4th stage air cools the LPT stage 1-4 disks. This air also comes from the TCF. The LPT disks and rotating rings allow the air to flow.

The low pressure turbine active clearance control system (LPTACC) cools the outer case. See ATA 75-20-00 for more information about the LPTACC system.



ENGINE -- LOW PRESSURE TURBINE (LPT) MAJOR MODULE

72-00-00

DMC-B737T-A-72-00-00-04A-040A-T-004-00
ICN-B737T-A-720000-A-81205-00035-A-004-01

02-Dec-2020

ENGINE -- LPT SHAFT MODULE

Low Pressure Turbine (LPT) Shaft Module

The functions of the LPT shaft module are:

- Transmit the torque from the LPT to the fan and booster assembly
- Provide radial support to the rotating assembly
- Ventilate the forward (A) sump
- Ensure oil containment.

The LPT shaft module is attached to:

- The number 1 and no. 2 bearing support module at the front
- The number 5 bearing support module and the turbine rear frame at the rear.

The LPT shaft module has these components:

- The LPT shaft
- A center vent tube
- Number 5 bearing (roller).

LPT Shaft

The LPT shaft is a high strength steel. The forward end of the LPT shaft has outer splines that engage into inner splines on the fan shaft. It holds the forward and rear ventilation tubes (center vent tube). The rear support is the number 5 roller bearing.

Center Vent Tube

The center vent tube gives an overboard vent for the engine forward (A) sump.

The center vent tube is a titanium alloy. It installs in the center of the LPT shaft. It is in two parts, the front vent tube and a rear vent tube. An inter-tube o-ring is the seal between the front and rear vent tubes.

Shaft dampers decrease the vibration between the LPT shaft and the center vent tube.

Number 5 Bearing

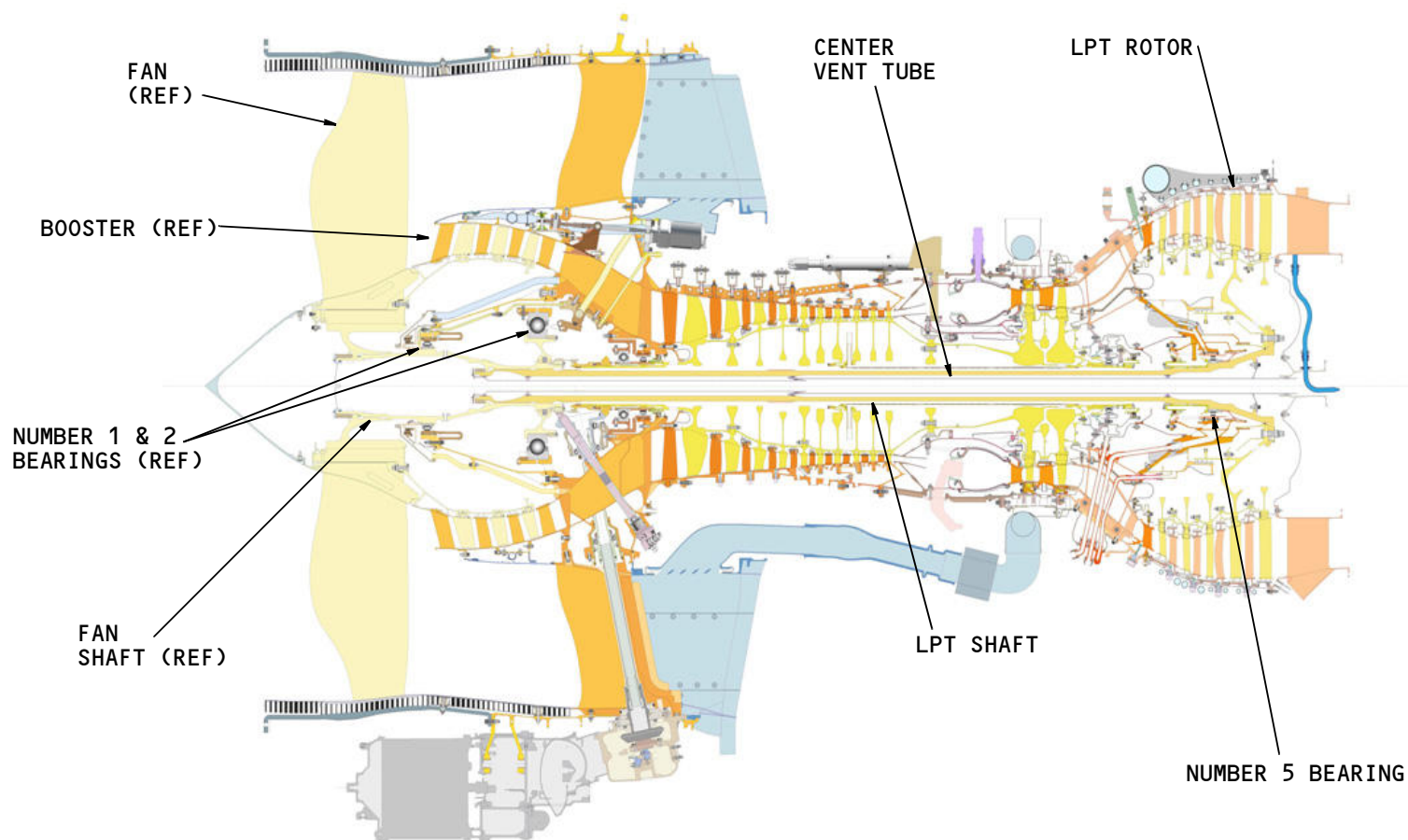
The number 5 bearing gives radial support to the LPT rotor to decrease vibrations.

The number 5 bearing is an oil-damped roller bearing at the rear of the LPT shaft. Oil-damping decreases the vibration level of the assembly as it turns.

The number 5 bearing is in the rear sump, between a scoop support and the rear rotating seal. The outer race is on the number 5 bearing support and attaches with bolts and nuts.

Bearing Lubrication

The number 5 oil nozzle lubricates the number 5 roller bearing. The forward and aft rotating seals are in contact with a segmented radial seal and a labyrinth seal. The segmented radial seal is at the forward end of the bearing chamber and the labyrinth seal is at the aft end of the chamber.



ENGINE -- LPT SHAFT MODULE

72-00-00

DMC-B737T-A-72-00-00-28A-034B-T_001-00
ICN-B737T-A-720000-A-81205-00036-A-001-01

02-Dec-2020

ENGINE -- LPT ROTOR AND STATOR MODULE

Low Pressure Turbine (LPT) Rotor Assembly

The LPT rotor assembly is in the LPT case and has five stages. The function of the LPT is to change the energy of the airflow into mechanical energy. The LPT rotor assembly has these components:

- Stages 1 to 5 rotor blades
- Four (4) disks supported by the LPT stage 4 stubshaft disk
- LPT shaft.

The blades are solid wide chord. Dovetail slots hold the blades radially. The stage 1 to 4 blades are nickel alloy. The stage 5 blades are titanium alloy. The disks and stubshaft disk are nickel alloy.

LPT Stator Assembly

The stator assembly increases the speed of airflow that comes from the high pressure turbine (HPT). It has these components:

- LPT case
- Stage 2 to 5 nozzle assemblies
- Low pressure turbine active clearance control (LPTACC) system.

LPT Case

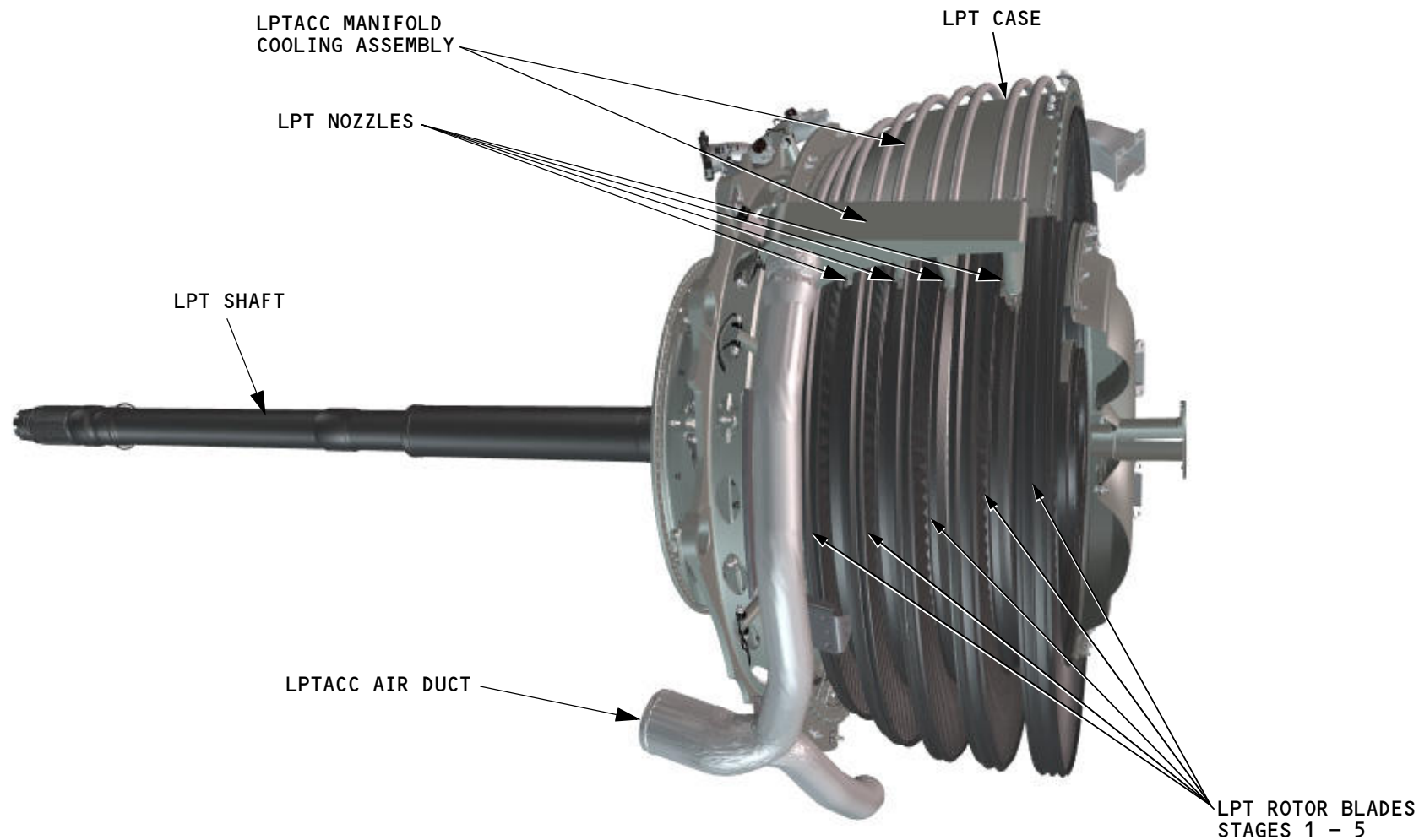
The LPT case gives radial support and makes a smooth aerodynamic surface for the air flow between the LPT stage 1 nozzle and the turbine rear vane. The front flange of the LPT case attaches to the LPT stage 1 nozzle with bolts. The rear flange of the LPT case attaches to the turbine rear vane module with bolts. The LPT case holds outer stationary air seals and the nozzles in grooves. The LPT case gives support for the stator assembly. The case is a nickel-chrome alloy. The front flange of the LPT case attaches to the rear flange of the combustor case. The rear flange attaches to the outer front flange of the turbine center frame (TCF).

Stage 2 to 5 Nozzles

The LPT stage 2 to 5 nozzles direct the airflow onto the LPT blades.

LPTACC Air Duct and Manifold Cooling Assemblies

The low pressure turbine active clearance control (LPTACC) system puts cooling air on the outer surface of the LPT case. This system controls blade tip clearance and increases fuel economy. An LPTACC valve controls the quantity of fan air supplied to the LPTACC air duct. The LPTACC manifold cooling assemblies are on supports, held by brackets around the LPT case. There are two LPTACC manifold cooling assemblies, one upper and one lower. The duct and manifolds are a steel alloy.



ENGINE -- LPT ROTOR AND STATOR MODULE

72-00-00

DMC-B737T-A-72-00-00-29A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00037-A-001-01

02-Dec-2020

ENGINE -- TURBINE CENTER FRAME (TCF)

General

The turbine center frame (TCF) is located between the high pressure turbine (HPT) and the low pressure turbine (LPT).

Purpose

The TCF is the second structural part of the engine. The structure gives support for minimum blade tip clearance. The TCF struts also give radial support at the aft end of the HPT rotor for the number 4 bearing. The TCF struts are in the LPT transition duct fairing to give a cool area for the oil tubes. The structure holds the LPT transition duct, which gives a smooth aerodynamic air flow between the HPT and LPT. In the LPT transition duct there are 12 fairings which direct air flow to the LPT.

Interface

The TCF contains the LPT stage 1 nozzles.

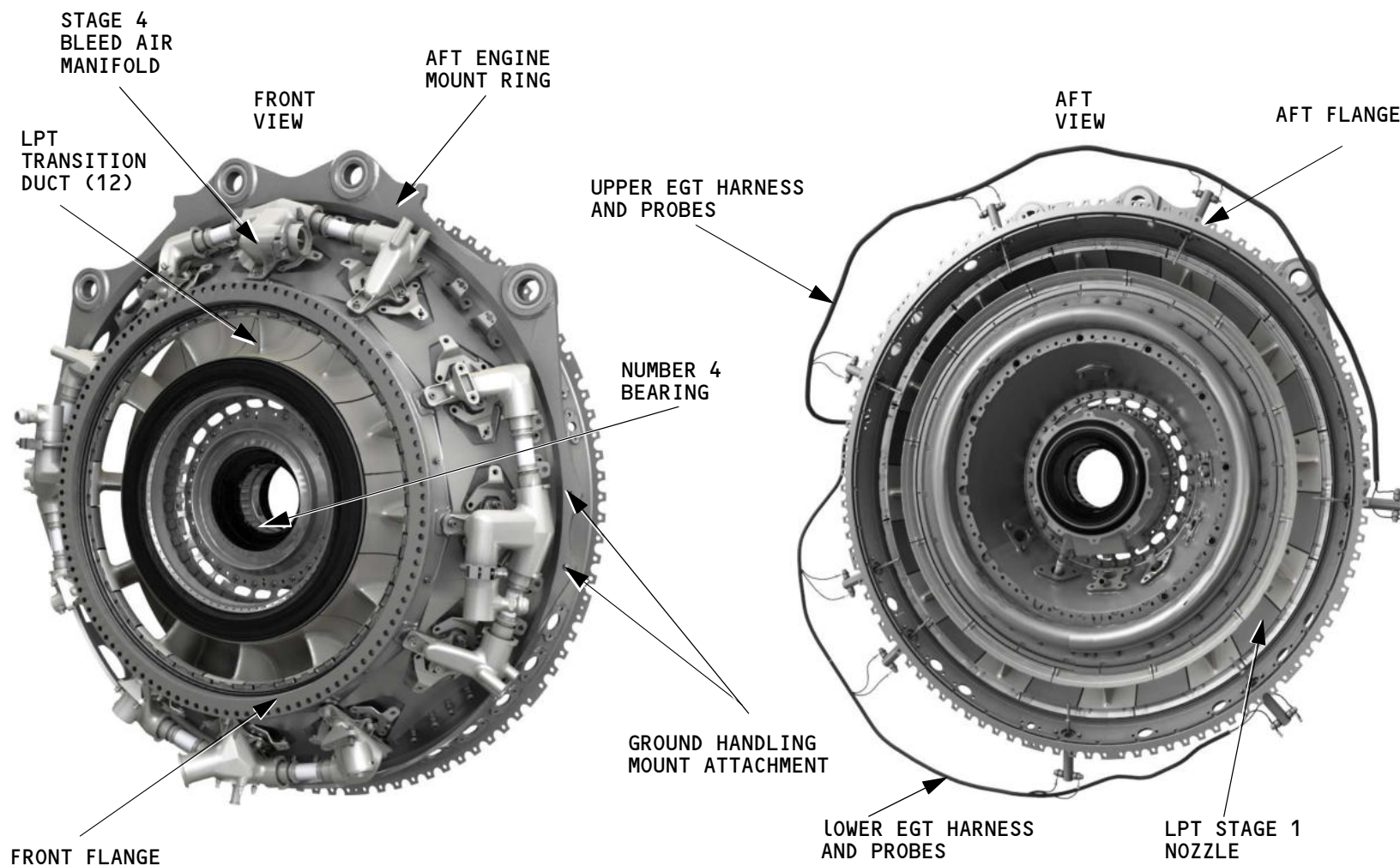
The aft engine mount attaches to the aft section of the TCF. Four spherical bearings interface with the rear engine mount. The rear engine mount interfaces with the aft pylon. Ground handling points are at the 3 and 9 o'clock position.

The exhaust gas temperature (EGT) sensors (8) are connected to the TCF external case. The function of the EGT sensors is to measure the temperature of HPT output.

Bleed air from stage 4 of the HPC is used for internal cooling the TCF and the LPT.

Number 4 Bearing

The TCF inner hub structure (B-Sump) contains the number 4 roller bearing. The hub structure also is the support for the oil tubes. The TCF bearing B-Sump is a sealed compartment. It does not have a vent. The rotating and stationary oil seals for the number 4 bearing compartment contain the oil in the compartment. These oil seals also prevent air leakage into the compartment.



DMC-B737T-A-72-00-00-05A-040A-T_002-00
ICN-B737T-A-720000-A-81205-00046-A-003-01

ENGINE -- TURBINE CENTER FRAME (TCF)

02-Dec-2020

72-00-00

ENGINE -- TURBINE REAR VANE MODULE

Turbine Rear Vane Module

The turbine rear vane module attaches between the low pressure turbine (LPT) module and the exhaust nozzle module. It also has an interface with the center vent tube.

The front section of the turbine rear vane module attaches to the rear flange of the LPT module with bolts. The rear section attaches to the exhaust nozzle and exhaust plug.

The main functions of the turbine rear vane module are to:

- Direct and ensure a smooth aerodynamic airflow
- Support the exhaust system module and center vent tube
- Ventilate the front sump
- Seal the rear of the engine.

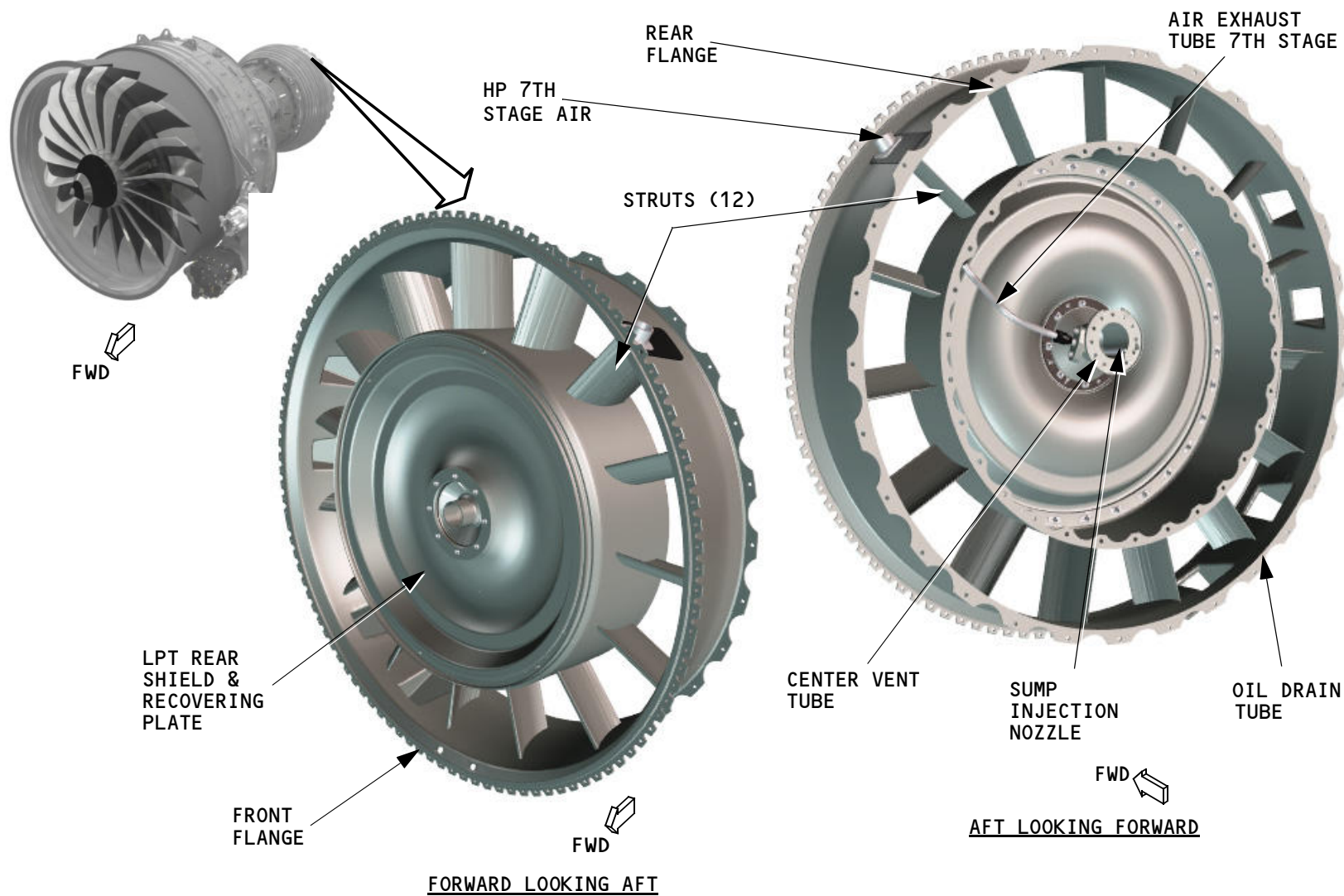
LPT Rear Vane

The LPT rear vane is a nickel alloy compound and has twelve (12) struts. All of the struts give structural support to the module. At the front flange of the LPT rear vane is the LPT rear shield and recovering plate attached with bolts. The recovering plate lets a small flow of purge air release into the primary airflow while the rear shield continues to seal. The rear shield also supports the center vent tube. Inside the center vent tube is the sump injection nozzle for the sump eductor system.

The sump eductor system increases the airflow through the A sump at low engine speeds. Air from the HPC 7th stage goes into the air exhaust tube at the 12 o'clock position through an injection nozzle when the sump eductor valve is open. Airflow through the injection nozzle creates a lower pressure inside the center vent tube, which then pulls more air through the A sump. The additional air pulled through

the A sump improves the function of the seals and helps to reduce the oil consumption.

An oil drain tube is connected to the lowest point of the LPT case to catch oil leakage from the rear sump. The drain tube attaches to the LPT rear vane by a support tube.



ENGINE -- TURBINE REAR VANE MODULE

72-00-00

DMC-B737T-A-72-00-00-33A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00041-A-002-01

02-Dec-2020

ENGINE -- ACCESSORY DRIVE

General

The accessory drive transmits mechanical energy from the N2 shaft to the accessory gearbox. The inlet gearbox (IGB) turns the radial drive shaft (RDS). The RDS turns the transfer gearbox (TGB). The TGB turns the transfer shaft (TS). The TS turns the accessory gearbox (AGB). The transfer of mechanical energy reverses when the starter operates.

Inlet Gearbox (IGB)

The IGB transmits mechanical energy from the high pressure compressor front shaft to the radial drive shaft. It also holds the front of the HPC rotor.

The IGB contains these parts:

- Horizontal bevel gear with coupling/locking nut
- Pinion bevel gear
- Number 3 bearing (roller and ball)
- Number 3 bearing rotating seal.

Radial Drive Shaft (RDS)

The RDS transmits mechanical energy from the IGB to the transfer gearbox. The assembly is located at the 6:00 position (aft looking forward).

The RDS has an inner radial drive shaft and housing, a shaft mid-length ball bearing, and an outer radial drive shaft and housing. The shaft mid-length bearing gives correct centering of the RDS in its housing.

Transfer Gearbox (TGB)

The TGB transmits mechanical energy from the RDS to the transfer drive shaft.

The TGB is on the fan frame module at the 6:00 position and has:

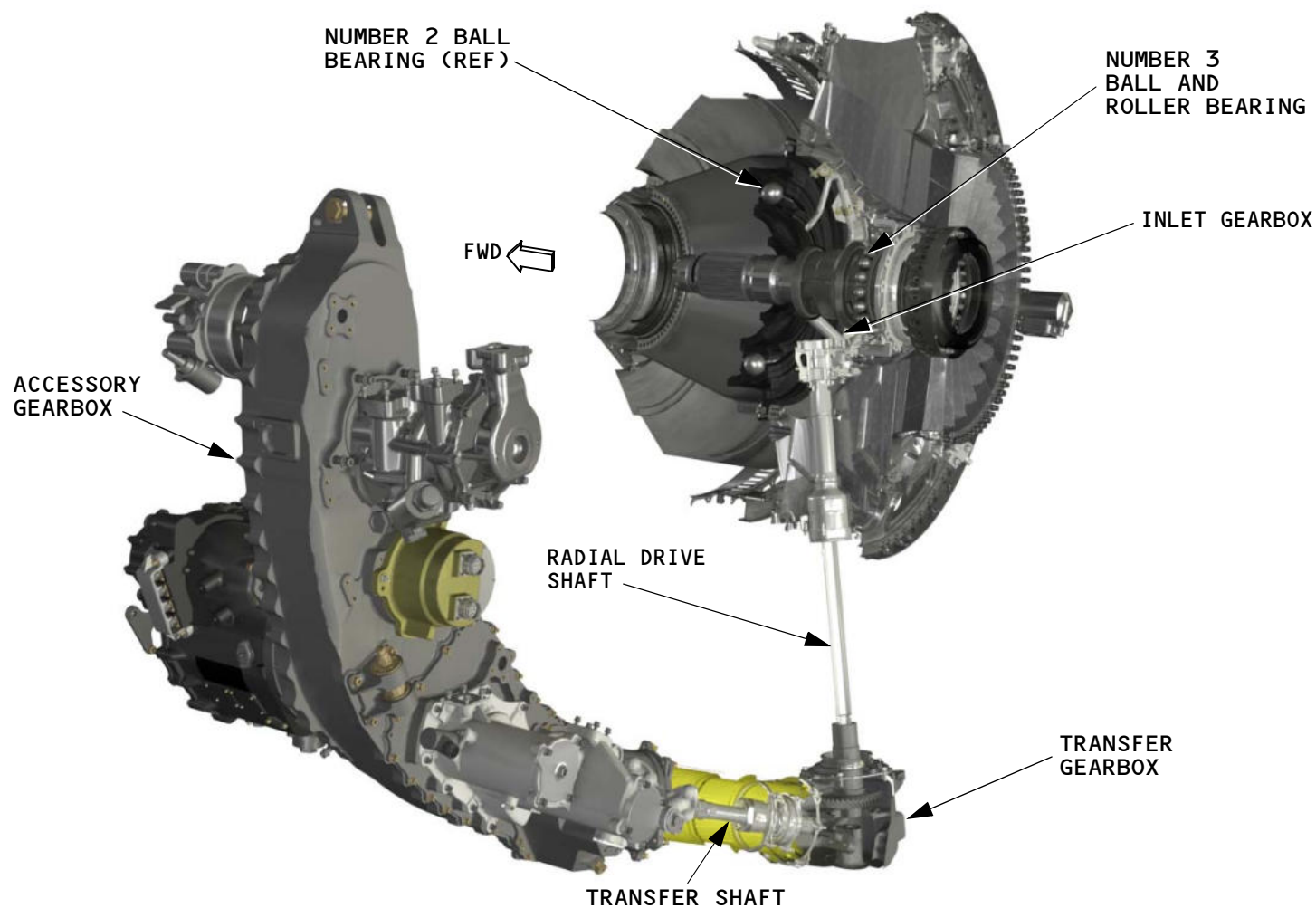
- Transfer gearbox housing
- Input bevel gear
- Horizontal bevel gear and transfer shaft assembly.

Transfer Shaft (TS) and Transfer Tube

The TS has a spline at each end and transmits mechanical energy from the TGB to the accessory gearbox. The transfer shaft is inside the transfer tube.

The TGB transfer tube is made of an aluminium alloy. The TGB transfer tube ensures oil containment between the TGB housing and the AGB housing. It also houses the TGB horizontal bevel gear assembly. It provides a bearing support for two TGB roller bearings and one TGB ball bearing for the TGB horizontal bevel gear assembly.

The transfer tube attaches to the TGB housing with bolts, and makes the connection with the AGB.



ENGINE -- ACCESSORY DRIVE

72-00-00

DMC-B737T-A-72-00-00-30A-034B-T_004-00
ICN-B737T-A-720000-A-81205-00038-A-003-01

02-Dec-2020

ENGINE -- ACCESSORY GEARBOX (AGB)

Accessory Gearbox (AGB)

The AGB holds and turns airplane and engine accessories. The internal gear train reduces and increases the rotational speed to meet the specific drive requirements of each accessory. The housing is made of an aluminium alloy.

When the engine is running, the AGB transmits torque to engine and aircraft accessories. The torque comes from the engine HPC front shaft to the AGB through the RDS and the TGB assembly.

When you start the engine, the AGB transmits the torque from the pneumatic air starter (on the AGB assembly) to the engine HPC front shaft through the TGB assembly and the RDS. You use the handcranking drive pad on the AGB to rotate the HPC during borescope inspection.

The AGB assembly is on the fan frame module of the engine at the 8:00 position. It is attached with 4 link assemblies: one lower (link 1), one lateral (link 2), one upper (link 3) and one in the middle of the AGB housing (link 4).

The AGB front face has mounting pads for these:

- N2 speed sensor (bottom end of housing next to starter)
- Handcranking drive pad (the handcranking drive pad has a cover installed) - rotating carbon seal
- Starter - V-band clamp
- Integrated drive generator (IDG) - quick attach detach (QAD) - rotating carbon seal
- Hydraulic pump - rotating carbon seal
- Oil pressure/temperature (OPT) sensor.

The AGB rear face connects with the transfer shaft (TS) sleeve. It has the mounting pads for these:

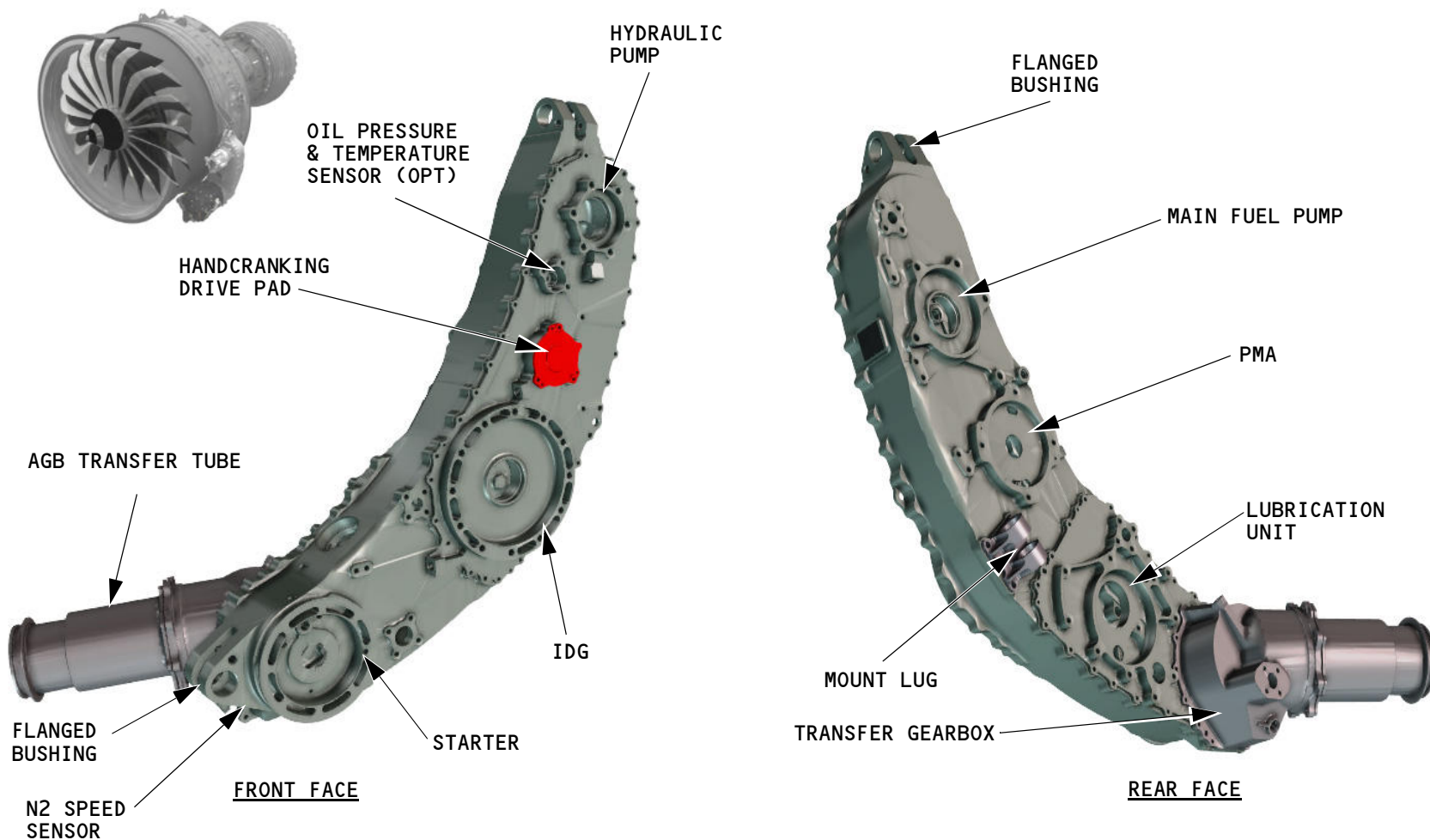
- Main fuel pump - V-band clamp - rotating carbon seal
- Permanent magnet alternator (PMA) - EEC power supply
- Lubrication unit with oil filter cartridge and magnetic scavenge screen plugs.

Some of the mounting pads have an overboard drain.

DMC-B737T-A-72-00-00-31A-034B-T_002-00

02-Dec-2020

72-00-00



DMC-B737T-A-72-00-00-31A-034B-T_002-00
ICN-B737T-A-720000-A-81205-00039-A-001-01

ENGINE -- ACCESSORY GEARBOX (AGB)

02-Dec-2020

72-00-00

ENGINE -- ACCESSORIES

General

Engine and airplane accessories are on the accessory gearbox (AGB). The AGB is on the left side of the engine fan case. The AGB is powered by the N2 rotor through gear boxes and drive shafts. Open the left fan cowl to access the AGB and the accessories.

Accessories

These engine and airplane accessories are on the front face of the AGB:

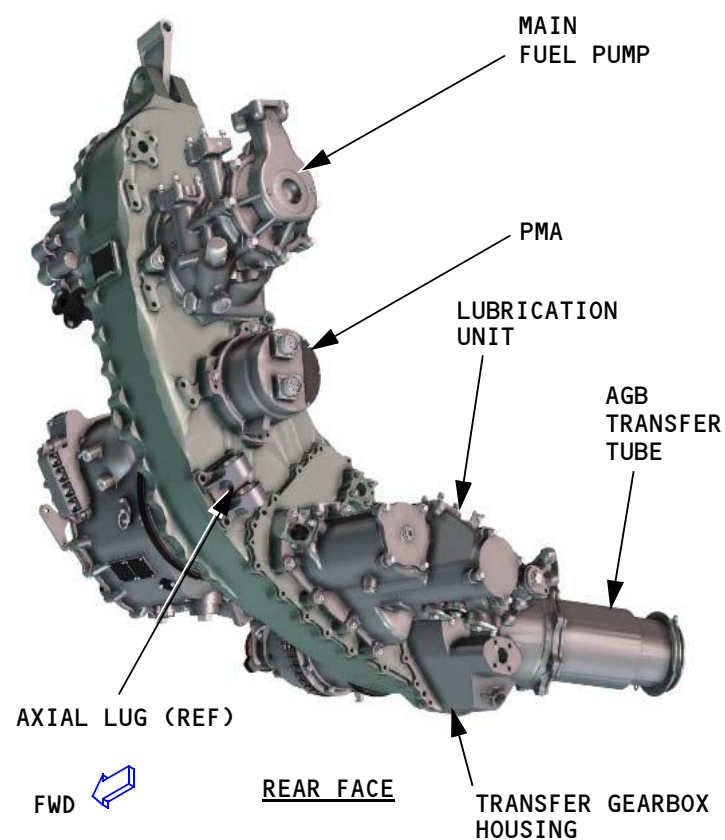
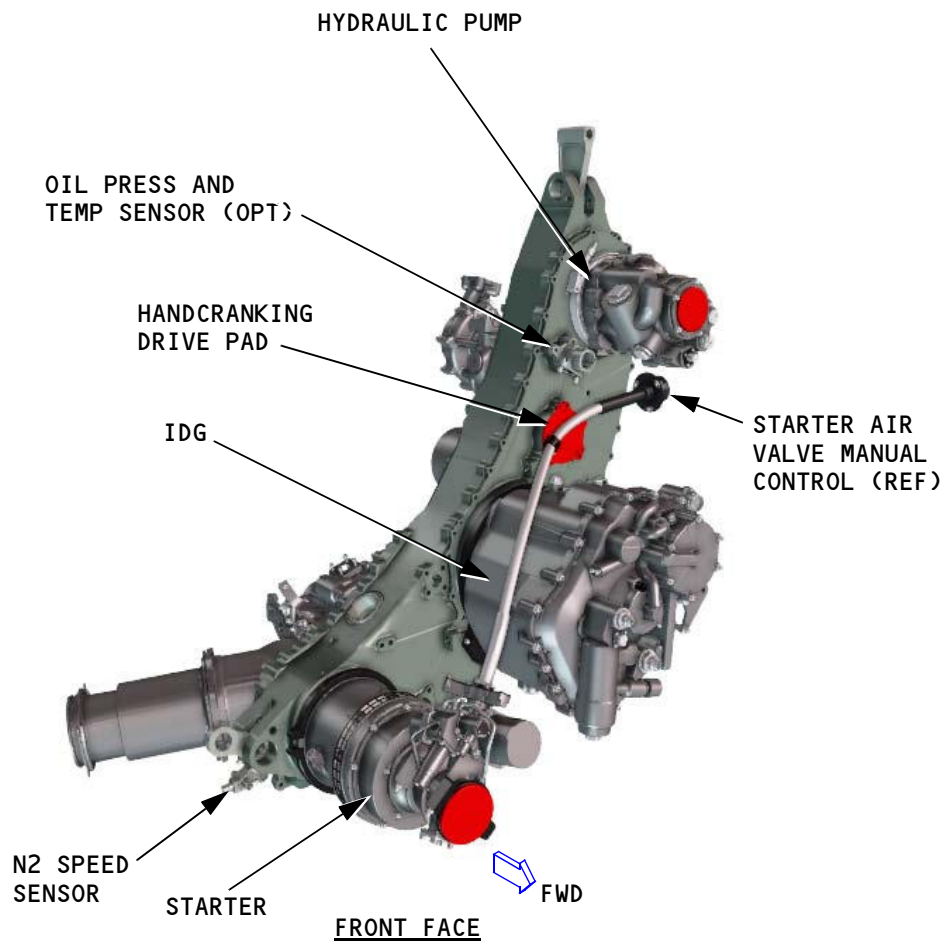
- Hydraulic pump
- Oil pressure/temperature sensor (OPT)
- Handcranking drive pad
- Integrated drive generator (IDG)
- Starter
- N2 speed sensor (bottom end of AGB housing next to starter).

These engine accessories are on the rear face of the AGB:

- Main fuel pump
- Permanent magnet alternator (PMA)
- Lubrication unit.

Training Information Point

You remove the cover, and use the handcranking drive pad to turn the N2 rotor during borescope inspection.



ENGINE -- ACCESSORIES

72-00-00

DMC-B737T-A-72-00-00-32A-034B-T_001-00
ICN-B737T-A-720000-A-81205-00040-A-002-01

02-Dec-2020

ENGINE -- BORESCOPE INSPECTION PORTS - LOCATION

Purpose

There are 22 borescope ports, identified as A to V. These ports give access for the visual inspection of the engine internal parts.

Borescope Inspection Ports

The borescope ports are on the outer case of the engine. The inspection ports for rotating parts (A - N and P - V) are on the right side of the engine. One port (O) is on the left side of the combustion chamber. All ports have self-locking plugs except the low pressure compressor (LPC). The LPC (A-B) has no plugs.

Low Pressure Compressor (LPC) (Booster)

The 2 LPC ports (A and B) are approximately at the 3:00 o'clock position through the fan flow path fairing.

High Pressure Compressor (HPC)

The 11 HPC ports (C to M) are between 3:00 and 5:00 o'clock position at each HPC stage on the right side of the engine.

Combustion Chamber

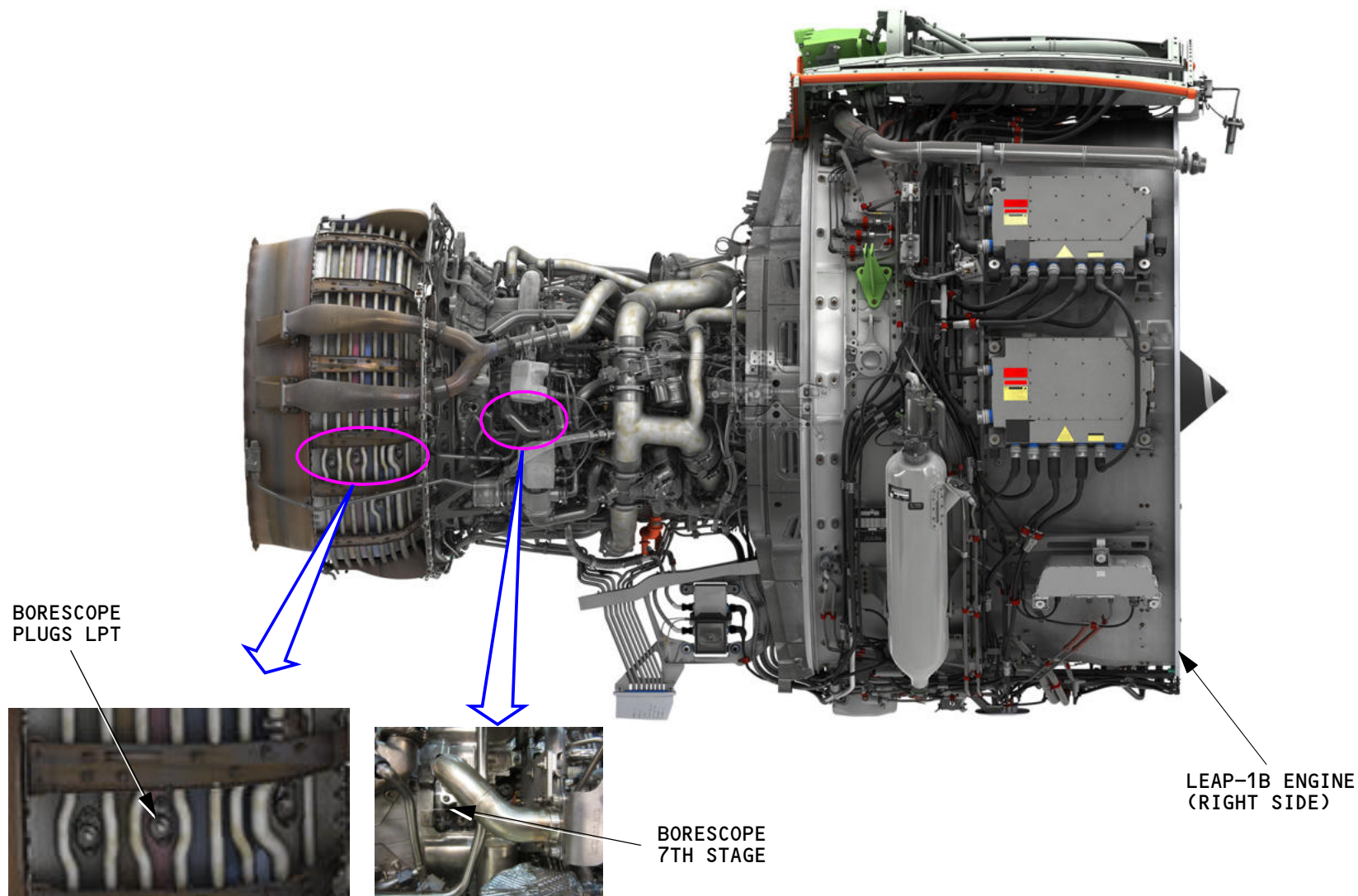
The 2 ports for the combustion chamber are N at 10 o'clock, and O at 2 o'clock. The inspection of the combustion chamber is also possible through the igniter ports.

High Pressure Turbine (HPT)

The 2 HPT ports (P and Q) are at 3:00 o'clock.

Low Pressure Turbine (LPT)

The 5 LPT ports (R to V) are at 4:00 o'clock.



DMC-B737T-A-72-00-00-03A-300A-T_005-00
ICN-B737T-A-720000-A-81205-00043-A-006-01

ENGINE -- BORESCOPE INSPECTION PORTS - LOCATION

72-00-00

02-Dec-2020

ENGINE -- BORESCOPE INSPECTION PORTS - GENERAL DESCRIPTION

General

The LEAP-1B uses on-condition maintenance. The engine has no periodic overhaul schedules. The engine can stay on wing until a non-normal condition occurs, if the engine has had all scheduled inspections completed. Some rotating parts have lifetime cycle limits. Refer to the engine manufacturer for life limited parts information.

Borescope inspection lets you check the condition of engine internal parts. You can not get access to engine internal parts, but you can do a visual inspection with a borescope probe put in ports on the engine outer casing.

On-Condition Maintenance

On-condition maintenance will use these maintenance practices:

- Physical parts/engine system inspections
- Borescope inspections
- Oil circuit monitoring, metallic particles indication
- Engine performance monitoring
- Built-in test (BITE) message troubleshooting.

These maintenance practices prevent the requirement for the scheduled overhaul of accessories or engine modules.

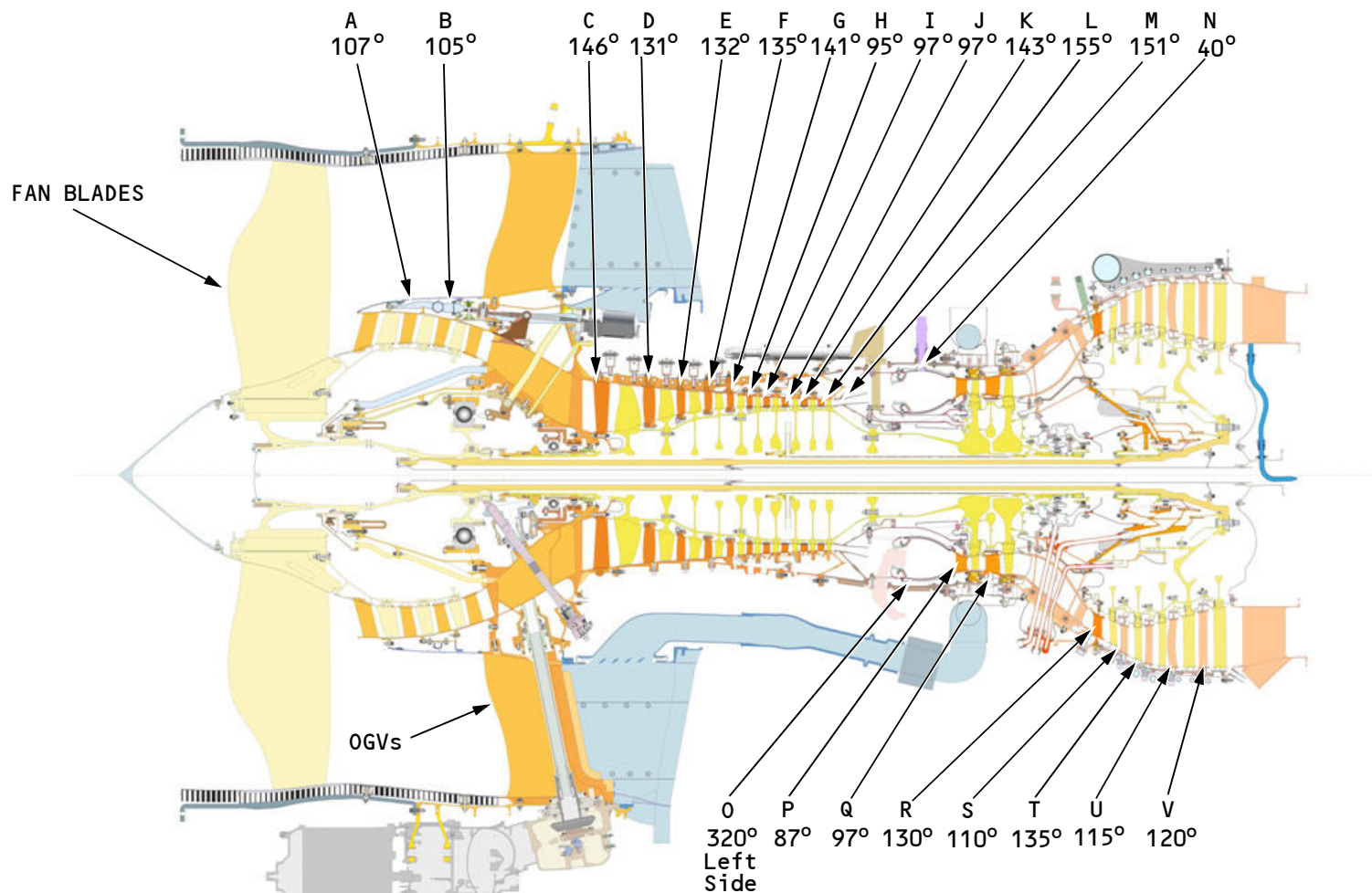
Other Engine Health Monitoring Tools

Engine performance trend monitoring evaluates engine deterioration over a period of use. The airplane condition monitoring system (ACMS) records many engine parameters each flight such as exhaust gas temperature.

The engine vibration monitoring system (AVM) uses sensors in different positions in the engine to measure vibration levels. You can also use the system to do onboard engine balancing.

A spectrometric oil analysis program (SOAP) does an analysis of all microscopic (smaller than 10 microns) metal particles in an oil sample from the oil tank. The type and concentration of metal found shows the start of parts damage.

Lubrication particles analysis does a check of the lubrication oil in the oil system. Filters and magnetic chip detectors collect visible-to-the-eye particles (larger than 10 microns) from worn engine parts for visual inspection and analysis. There are six magnetic chip detectors on the lubrication unit.



PLUG THREADS C AND D ARE LUBRICATED WITH ENGINE OIL,
ALL OTHER PLUGS ARE COATED WITH A PURE NICKEL SPECIAL COMPOUND.

RADIAL DEGREES ARE VIEWED IN A CLOCKWISE
DIRECTION, AFT LOOKING FORWARD.
ALL PORTS ARE ON RIGHT SIDE EXCEPT PORT O

ENGINE -- BORESCOPE INSPECTION PORTS - GENERAL DESCRIPTION

72-00-00

ENGINE INDICATING -- INTRODUCTION

Purpose

The engine indicating system supplies engine data to the MAX display system (MDS). The engine indicating system has these subsystems:

- Low pressure rotor tachometer (N1)
- High pressure rotor tachometer (N2)
- Exhaust gas temperature (EGT)
- Airborne vibration monitoring (AVM).

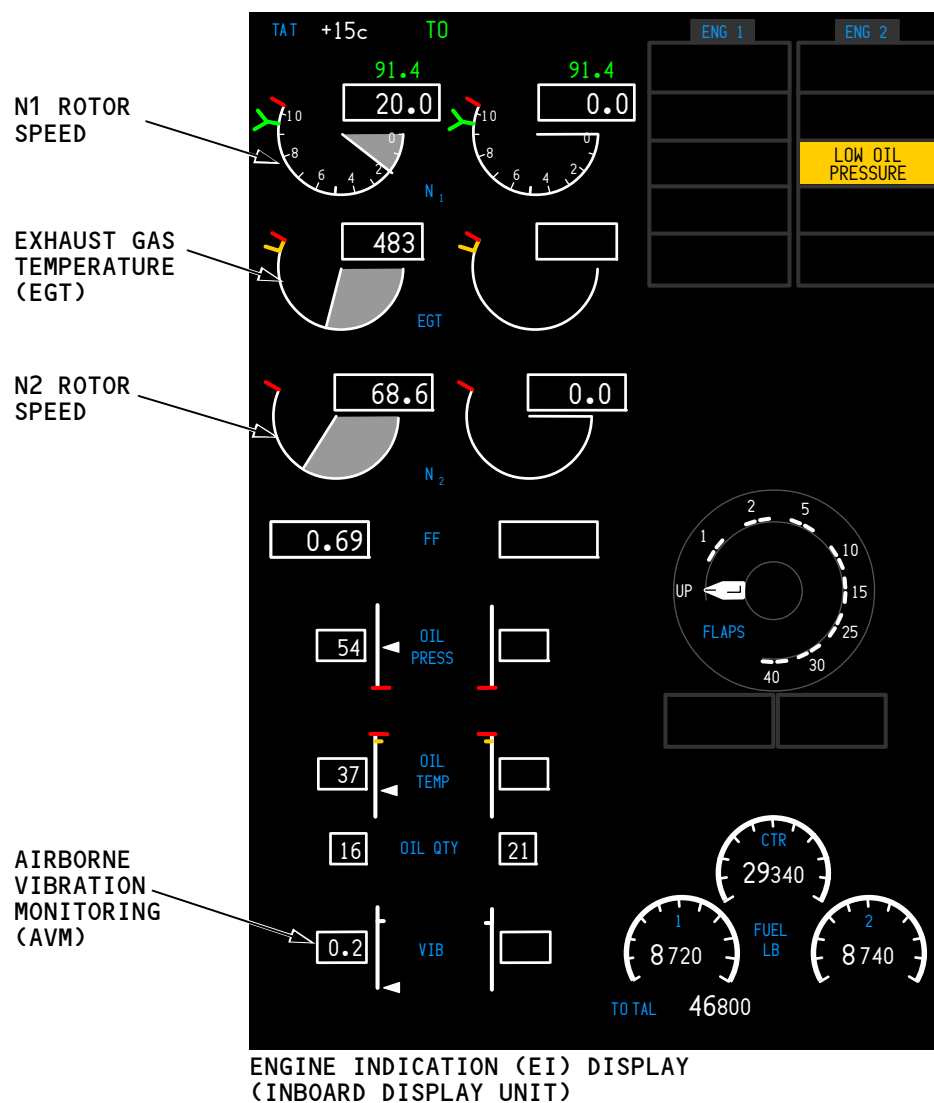
The MDS usually shows engine indication (EI) data on the left or right inboard display unit (DU). EI data can also show on one of the outboard DUs if the inboard display unit on that side has a failure.

Abbreviations and Acronyms

altn	- alternate
ACMS	- airplane condition monitoring system
AGB	- accessory gearbox
AMP	- amplitude
AVM	- airborne vibration monitoring
BB	- broadband
CONT	- continuous
CU	- cockpit unit
DFDAU	- digital flight data acquisition unit
DPC	- display processing computer
DSP	- digital signal processor
DU	- display unit
EEC	- electronic engine control
EGT	- exhaust gas temperature
EHM	- engine health management
EI	- engine indication
EPCS	- electronic propulsion control system
FDR	- flight data recorder

FMCS	- flight management computer system
GRD	- ground
HPC	- high pressure compressor
HPT	- high pressure turbine
LPC	- low pressure compressor
LPT	- low pressure turbine
MCDU	- multi-purpose control display unit
MFP	- multi-function panel
MDS	- MAX display system
N1	- low pressure rotor
N2	- high pressure rotor
NFS	- network file server
NOB	- number one bearing
NVM	- non-volatile memory
ONS	- onboard network system
PMA	- permanent magnet alternator
PMD	- portable maintenance device
SPD REF	- speed reference
TCF	- turbine center frame
VIB	- vibration

DMC-B737T-A-77-00-00-02A-018A-T_001-00



ENGINE INDICATING -- INTRODUCTION

77-00-00



ENGINE INDICATING -- GENERAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE INDICATING -- GENERAL DESCRIPTION

General

The engine indicating system shows these parameters for each engine:

- Low pressure rotor speed (N1)
- High pressure rotor speed (N2)
- Exhaust gas temperature (EGT)
- Engine vibration (VIB).

Electronic Engine Control

The electronic engine controls (EECs) receive an analog input from these engine sensors:

- N1 speed sensor
- N2 speed sensor
- EGT probes (T48)
- Number 1 bearing accelerometer
- Turbine center frame (TCF) accelerometer.

The EECs change the analog signals to digital signals. The EECs send the digital signals on an ARINC 429 data bus to two display processing computers (DPC)s for the engine indication. Auxiliary analog N1 and N2 parameters go directly to the DPCs. They do not go through the EECs.

The EECs also have an Ethernet (ARINC 615A) interface with the network file server (NFS) for the engine health management (EHM) functions. The EHM uses engine configuration data and flight data to calculate the trim balance solution for the engine. The trim balance procedure uses the vibration data from the No. 1 bearing and the turbine center frame accelerometers. The EHM is only available on the portable maintenance device (PMD). Access to the EHM is through the OTHER FUNCTIONS menu in the ONBD MAINT function. More

information about the EHM is in ATA 46-13 onboard network system (ONS).

The EECs attach to the right side of the engine fan case. The EECs are part of the engine fuel and control system. More information about the EEC is in ATA 73.

Display Processing Computer

The two display processing computers (DPCs) use digital input from the EECs to show these engine parameters on the MAX display system (MDS):

- N1
- N2
- EGT
- VIB.

The DPCs send the data to the MDS through a fiber-optic cable.

The DPCs use their analog N1 and N2 signals as auxiliary inputs when the EECs do not have electrical power. The EGT and the engine vibration indication shows only when the EECs have electrical power.

The DPCs have an Ethernet (ARINC 615A) interface with the ONS in the NFS, for the software data loading function. The DPCs have an ARINC 429 interface with the digital flight data acquisition unit (DFDAU). The engine indication (EI) application in the DPCs transmits many engine parameters to the DFDAU. The DFDAU records these parameters. The DFDAU also supplies data back to the DPCs through the ARINC 429 bus.

DMC-B737T-A-77-00-00-02A-040A-T_003-00

ENGINE INDICATING -- GENERAL DESCRIPTION

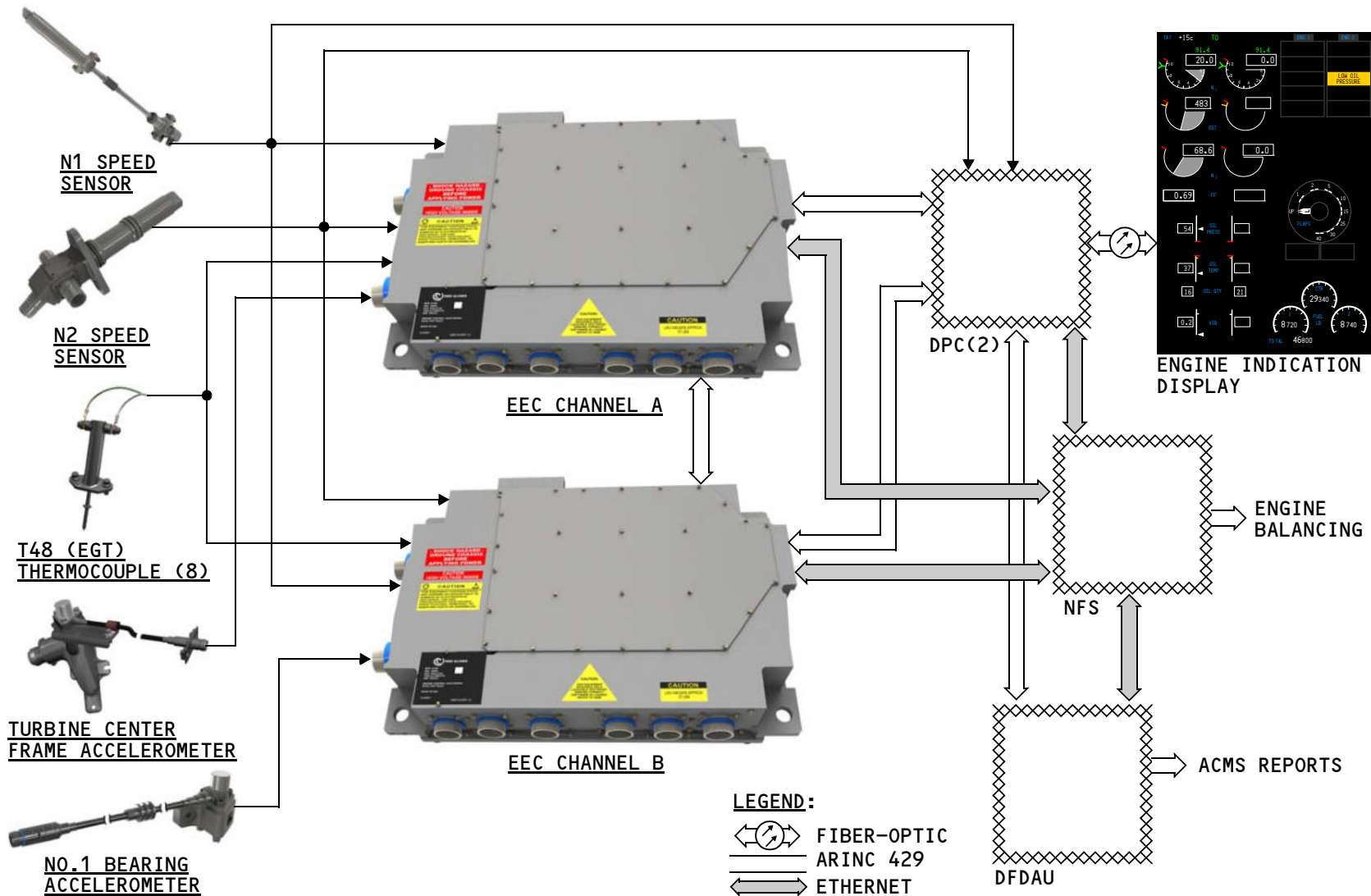
Network File Server

The network file server (NFS) has an Ethernet (ARINC 615A) interface with the DFDAU for the airplane condition monitoring system (ACMS). The ACMS collects maintenance data and makes reports that let the airline do trend analysis of the data. These are the baseline ACMS reports:

- Engine exceedance
- Engine fault snapshot
- Engine in-flight failure
- Engine performance
- Engine start.

This data also lets the airline do troubleshooting of airplane problems with real-time flight data.

You use the multi-purpose control display unit (MCDU) to get access to the ACMS functions.



ENGINE INDICATING -- GENERAL DESCRIPTION

77-00-00

ENGINE TACHOMETER SYSTEM -- GENERAL DESCRIPTION

Purpose

The engine tachometer system supplies the engine low pressure rotor (N1) and the engine high pressure rotor (N2) speed signals to these components:

- Electronic engine controls (EECs)
- Display processing computers (DPCs).

General Description

Each EEC channel gets an analog signal from the N1 and the N2 speed sensors. The EECs change these analog signals to digital signals.

The EECs use the two signals for channel A and channel B operation. Each channel sends data to each DPC on an ARINC 429 data bus.

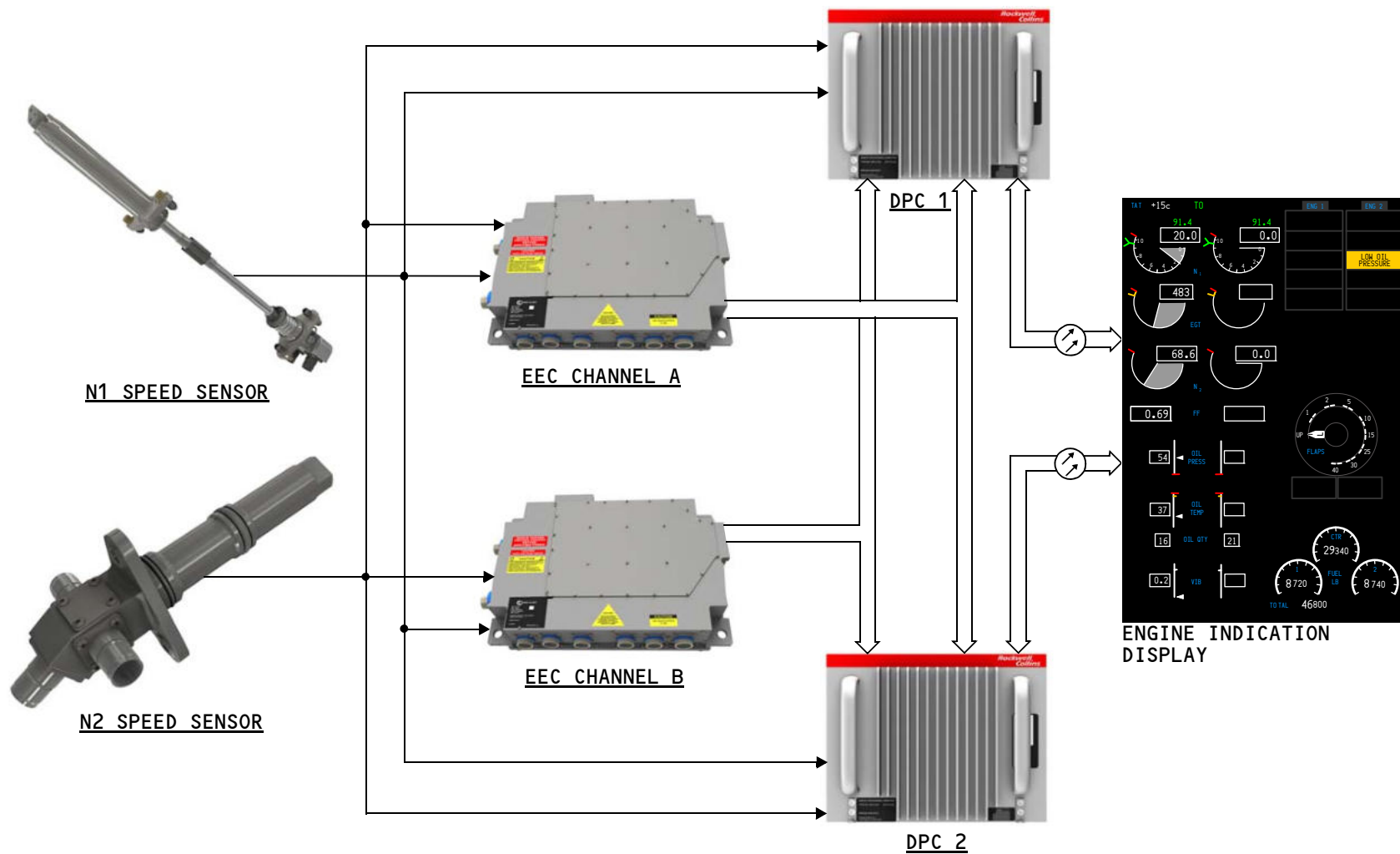
The AVM function in the EECs receives an analog input from the speed sensors to help calculate vibration levels.

Usually, the DPCs use inputs from the EECs to show N1 and N2 on the MAX display system (MDS). The DPCs can also use the analog input directly from the speed sensors to show N1 and N2 indications. The DPCs send the data to the display unit through a fiber-optic cable. N1 and N2 speeds show on the engine indication (EI) display of the MDS. N1 and N2 speeds also show on the electronic propulsion control system (EPCS) maintenance data pages and the PERFORMANCE maintenance data page.

DMC-B737T-A-77-11-00-02A-040A-T_002-00

02-Dec-2020

77-11-00



DMC-B737T-A-77-11-00-02A-040A-T_002-00
ICN-B737T-A-771100-A-81205-00005-A-002-01

ENGINE TACHOMETER SYSTEM -- GENERAL DESCRIPTION

77-11-00

02-Dec-2020

ENGINE TACHOMETER SYSTEM -- N1 SPEED SENSOR

Purpose

The N1 speed sensor measures the rotational speed of the low pressure (LP) rotor for:

- Engine control
- Vibration monitoring
- Flight deck indication.

The procedure for the fan trim balance also uses the signal from the N1 speed sensor.

The sensor supplies the speed signal to the electronic engine controls (EECs) and the display processing computers (DPCs).

Component Location

The N1 speed sensor is installed in a guide tube, in a strut at the 5 o'clock position of the fan hub. You can see the sensor housing with the electrical connectors, with the sensor installed in the engine.

Physical Description

The sensor end of the N1 speed sensor has three isolated sensing elements. Each sensing element has a pole piece and an electrical winding around a magnet.

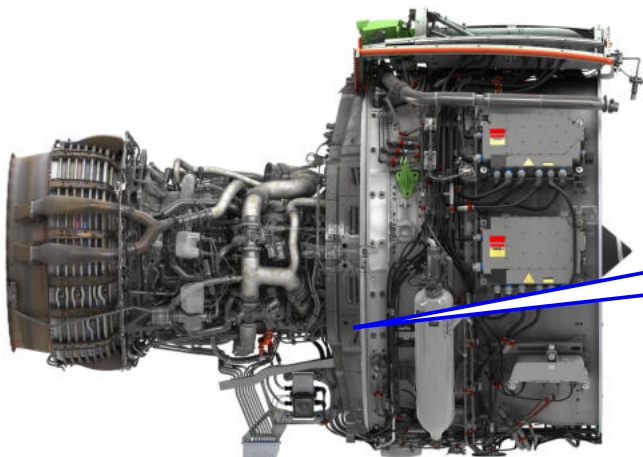
The N1 speed sensor has three electrical connectors. One connector is for the DPCs, one is for EEC channel A, and one is for EEC channel B.

The housing of the speed sensor touches the end of the guide tube when it is installed. When the installation bolts are tightened which attach the speed sensor into its position, a spring in the speed sensor

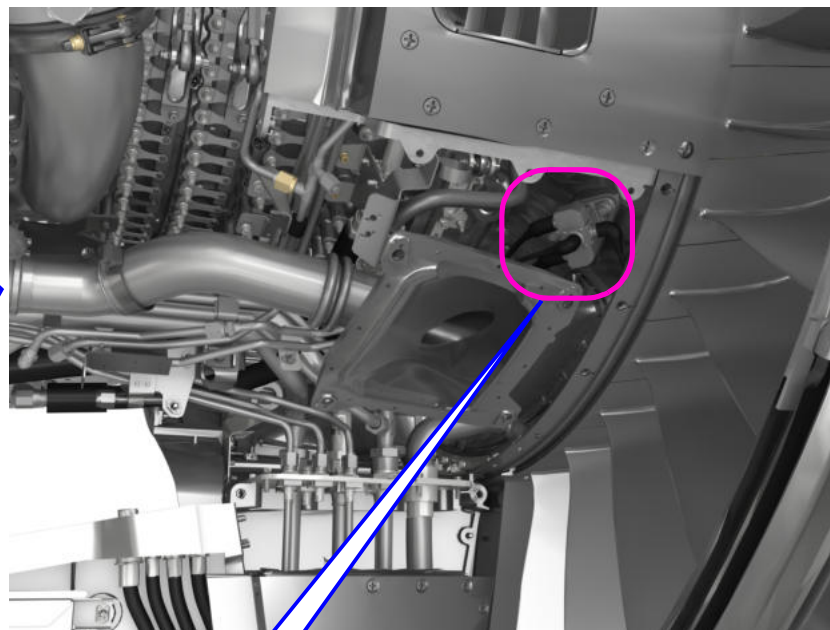
housing is compressed. The spring pressure on the speed sensor decreases speed sensor vibration.

Also, there is a damper on the speed sensor shaft to prevent sensor vibration.

DMC-B737T-A-77-11-00-03A-034B-T_002-00

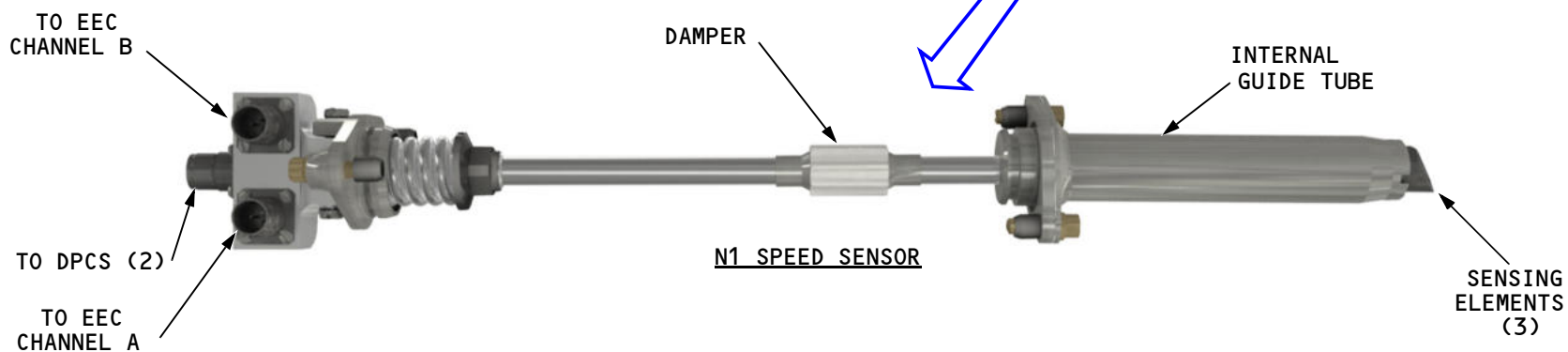


LEAP-1B ENGINE
(RIGHT SIDE)



FAN HUB FRAME
5:00 POSITION

NOTE:
INNER CORE FAIRING
SHOWN REMOVED



ENGINE TACHOMETER SYSTEM -- N1 SPEED SENSOR

77-11-00

DMC-B737T-A-77-11-00-03A-034B-T_002-00
ICN-B737T-A-77-1100-A-81205-00006-A-003-01

02-Dec-2020

ENGINE TACHOMETER SYSTEM -- N2 SPEED SENSOR

Purpose

The N2 speed sensor measures the rotational speed of the high pressure (HP) rotor for:

- Engine control
- Vibration monitoring
- Over-speed detection
- Flight deck indication.

The sensor supplies the high pressure rotor speed signal to the electronic engine controls (EECs) and the display processing computers (DPCs).

Component Location

Open the left fan cowl to get access to the N2 speed sensor. The N2 speed sensor is on the engine accessory gearbox (AGB) at the 7 o'clock position.

Physical Description

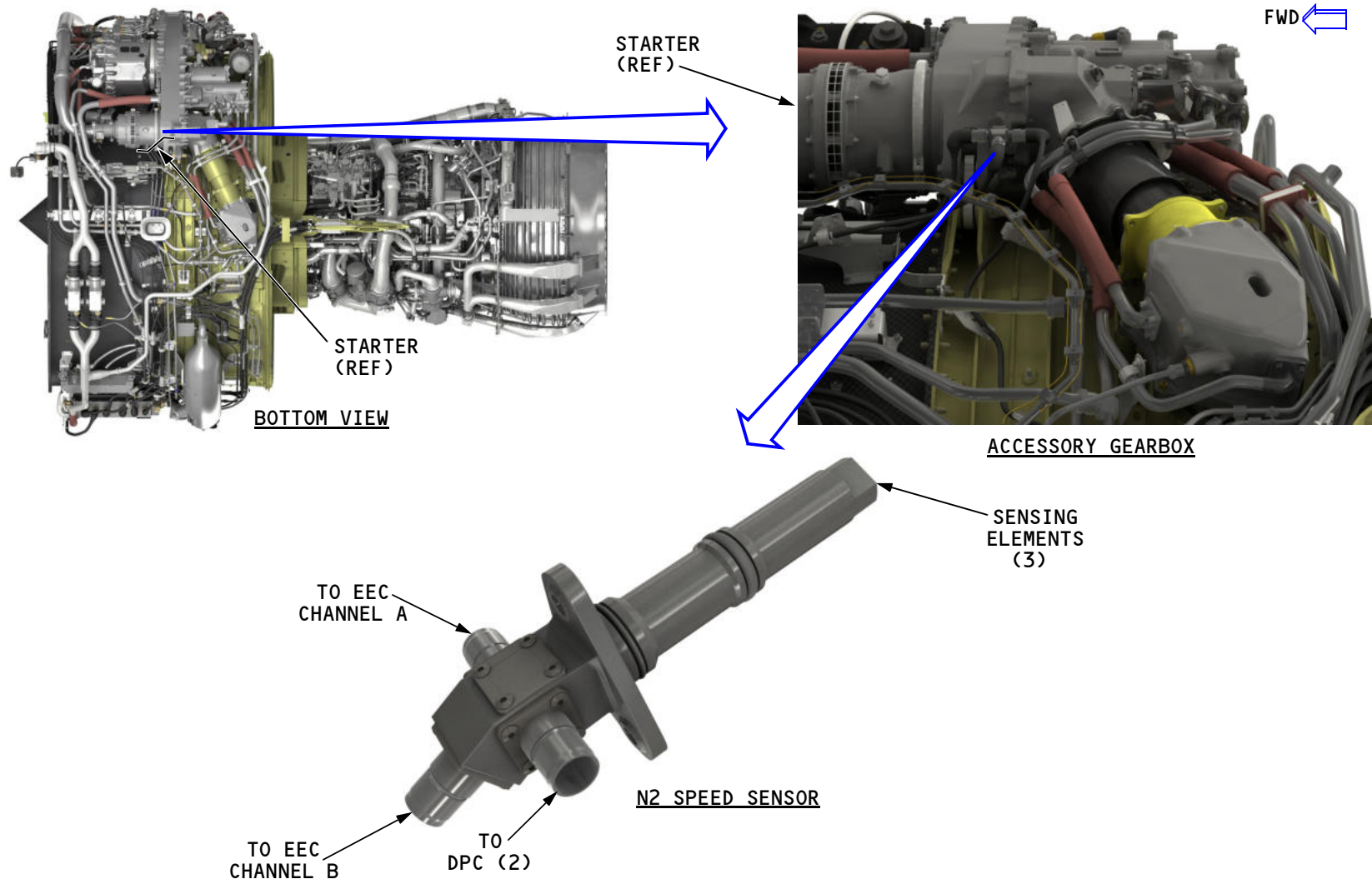
The sensor end of the N2 sensor has three isolated sensing elements. Each sensing element has a pole piece and an electrical winding around a magnet.

The N2 sensor has three electrical connectors. One connector is for the DPCs, one is for EEC channel A, and one is for EEC channel B.

DMC-B737T-A-77-11-00-04A-034B-T_001-00

02-Dec-2020

77-11-00



ENGINE TACHOMETER SYSTEM -- N2 SPEED SENSOR

77-11-00

DMC-B737T-A-77-11-00-04A-034B-T_001-00
ICN-B737T-A-77-1100-A-81205-00007-A-001-01

02-Dec-2020

ENGINE TACHOMETER SYSTEM -- FUNCTIONAL DESCRIPTION

General

The N1 indication is the speed of the engine low pressure rotor, and shows in percent N1.

The N2 indication is the speed of the engine high pressure rotor, and shows in percent N2.

Usually, the display processing computers (DPCs) use inputs from the electronic engine controls (EECs) to show N1 and N2. The DPCs use the analog signals directly from the speed sensors if the EECs do not supply a digital input.

N1 Actual (Digital Display and Pointer)

The N1 digital display shows the current engine fan speed. It shows in units of percent. The N1 digital display and the box around the digital display are white when N1 is less than the N1 redline.

A pointer on a circular dial also shows N1 speed. A shaded area follows this pointer. The pointer is usually white. The shaded area is usually gray.

These indications change to red when N1 is more than the N1 redline:

- N1 digital display
- Box around the N1 digital display
- N1 pointer
- Shaded area.

When N1 speed goes below the redline limit, the indication goes back to the usual color.

You will know that there has been an N1 exceedance when all of these are true:

- All engines are shut down; and

- The box around the N1 digital display turns to red.

N1 Redline

The N1 redline shows the maximum approved engine low pressure rotor speed for the LEAP-1B engine. The EECs supply the redline values to the DPCs. The redline shows in red.

N1 Command Sector

The command sector shows when there is a difference between the current N1 speed and the commanded N1 speed. The EECs use the thrust lever angle position to set the commanded N1.

The commanded N1 shows at the top edge of the command sector when the thrust lever position is more than the current engine speed. It shows at the lower edge of the command sector when the thrust lever position is less than the current engine speed.

The command sector and N1 command are white.

N1 Reference Bug

The N1 reference bug shows the N1 thrust target position. The N1 reference bug is green. The flight management computers (FMCs) usually calculate the bug position and send it to the DPCs.

You can also set the position of the N1 reference bug manually. You use the N1/SPD REF (speed reference) SET page to enter the desired N1. The N1/SPD REF SET display format shows on the Max display system (MDS) when you push the INFO button.

The N1 reference digital display also shows the N1 thrust target position. The display is green and shows above the N1 digital display.



ENGINE TACHOMETER SYSTEM -- FUNCTIONAL DESCRIPTION

See the FMCS section for more information about the N1 target.

N2 Digital Display and Pointer

The N2 digital display and the box around the digital display are white when N2 is less than the N2 redline.

A pointer on a circular dial also shows N2 speed. A shaded area follows this pointer. The pointer is usually white. The shaded area is usually gray.

These indications change to red when N2 is above the N2 redline:

- N2 digital readout
- Box around the N2 digital readout
- N2 pointer
- Shaded area.

When N2 speed goes below the redline limit, the indication goes back to the usual color.

You will know that there has been an N2 exceedance when all of these are true:

- All engines are shut down; and
- The box around the N2 digital display turns to red.

N2 Redline

The N2 redline shows the maximum approved engine high pressure rotor speed for the LEAP-1B engine. The EECs supply the redline value to the DPCs. The redline shows in red.

N1 and N2 Exceedances

The DPCs calculate and keep N1 and N2 exceedance data. Red line exceedances and the time of the exceedances show on the engine exceedance maintenance data page.

Maintenance Data Pages

N1 data and N2 data for each engine shows on the performance maintenance data page. Only the data from the EEC channel in control shows on the performance maintenance page.

N1 data and N2 data for each engine also shows on the electronic propulsion control system (EPCS) maintenance data pages. Data from the 2 EEC channels shows on the EPCS pages.

THRUST Alert Message

In flight, if the engine does not give the commanded thrust, the EECs send a signal to the DPCs. This causes the amber THRUST alert message to show. The EEC sends a high engine thrust or a low engine thrust discrete on the ARINC 429 data bus. The THRUST alert flashes for 10 seconds, then shows continuously. Also, the adjacent message blocks for the same engine flash but do not show text. The message stops when the over-thrust or the under-thrust condition stops.

The DPCs inhibit the flash mode for the takeoff and the landing. The THRUST message shows but the indication does not flash.

There is one THRUST alert message for each engine. Alert messages show in the top right corner of the engine indication (EI) display.

ENGINE TACHOMETER SYSTEM -- FUNCTIONAL DESCRIPTION

FUEL FLOW Alert Message

The EECs send the fuel flow data from the fuel flow meter to the DPCs for the flight deck indication. The fuel flow data also goes to the FMCS. The FMCS continuously calculate the fuel used by each engine and the APU. It also calculates the total fuel used, the total fuel that remains and an estimate of the fuel that remains at the end of the flight.

In cruise flight, the FMC compares the indicated fuel flow and the expected fuel flow. If the difference is more than the threshold limit for 5 minutes, the FMC supplies the logic for the FUEL FLOW alert message to show. The FUEL FLOW alert message flashes for 10 seconds then shows continuously. Also, the adjacent message blocks for the same engine flash but do not show text. The message stops when the fuel flow difference stops.

The DPCs inhibit the flash mode for the takeoff and the landing. When the inhibit is active, the FUEL FLOW message shows but the indication does not flash.

There is one FUEL FLOW alert message for each engine. Alert messages show in the top right corner of the EI display.

Bowed Rotor Start (BRS) Motoring Annunciation

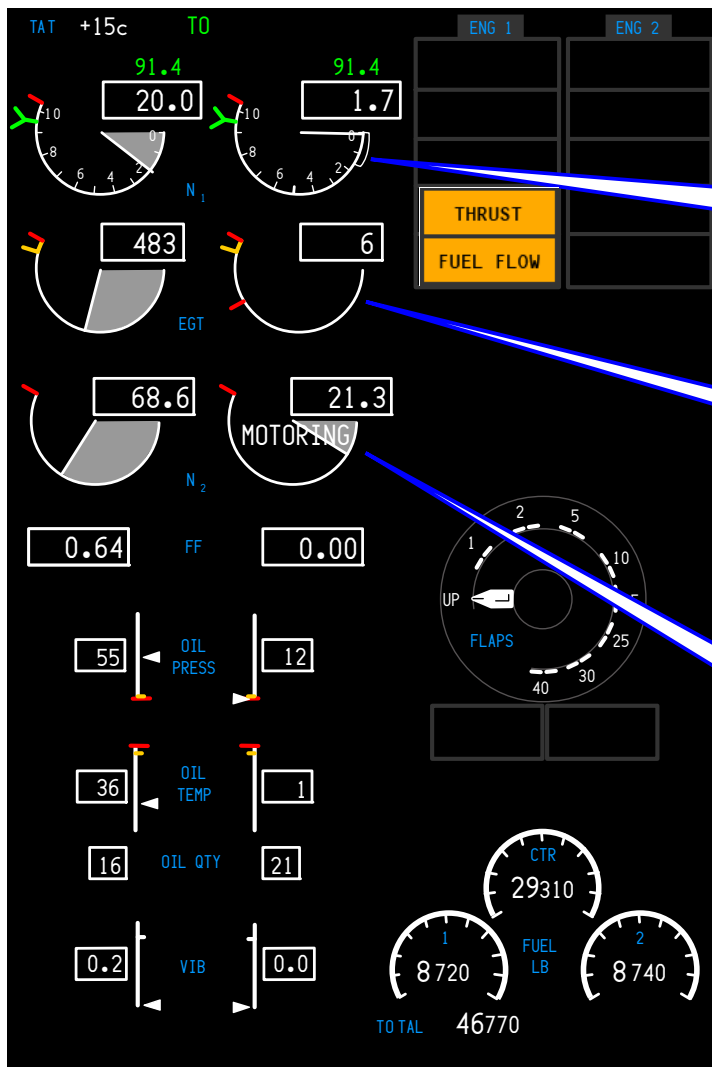
The white MOTORING message shows on the related N2 display when the engine is in the start sequence. The engine does a dry-motor operation without fuel or ignition in the initial part of the start sequence, to correct for a bowed rotor condition. More information on the bowed rotor condition and the MOTORING message is in the ATA 80 Engine Starting section.

Engine Failure Annunciation

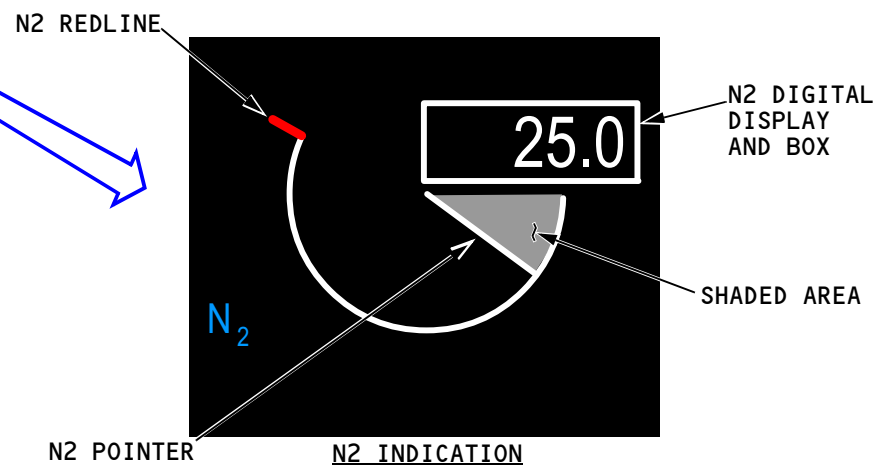
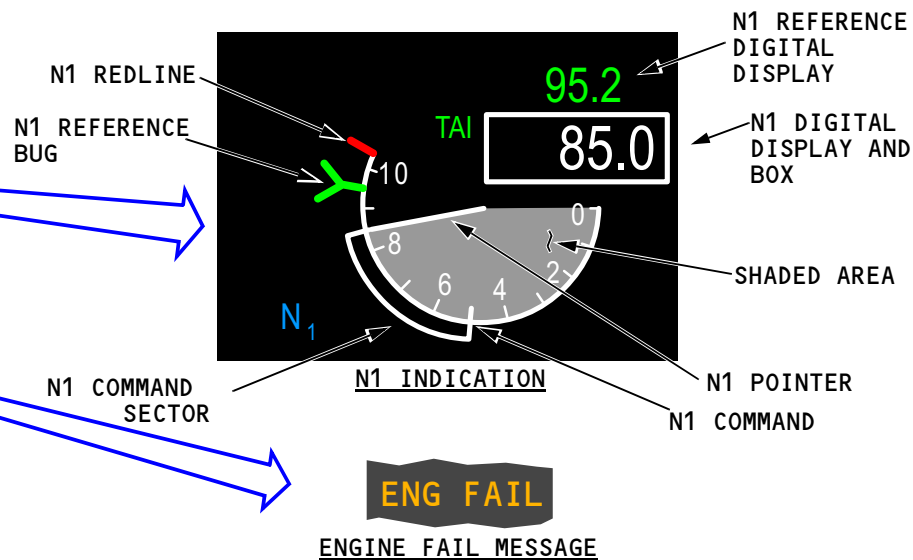
The amber ENG FAIL message shows on the related EGT display if all of these conditions occur:

- The 2 engine speeds are at or above idle
- The 2 engine fire handles are stowed
- The 2 start levers are in the idle position
- One of the 2 engine N2 speed then goes below idle.

The alert stops when the engine speed goes to idle or more, or the crew moves the engine start lever to the CUTOFF position.



ENGINE INDICATION (EI) DISPLAY



ENGINE TACHOMETER SYSTEM -- FUNCTIONAL DESCRIPTION

77-11-00

EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- GENERAL DESCRIPTION

Purpose

The exhaust gas temperature (EGT) indication system monitors the exhaust gas temperature at the stage one nozzle of the low pressure turbine.

General Description

The EGT system has eight sensors and two EGT harness assemblies. Each assembly has four of the sensors and supplies input to the electronic engine controls (EECs).

The EECs use the EGT signals to measure the T48 temperature at:

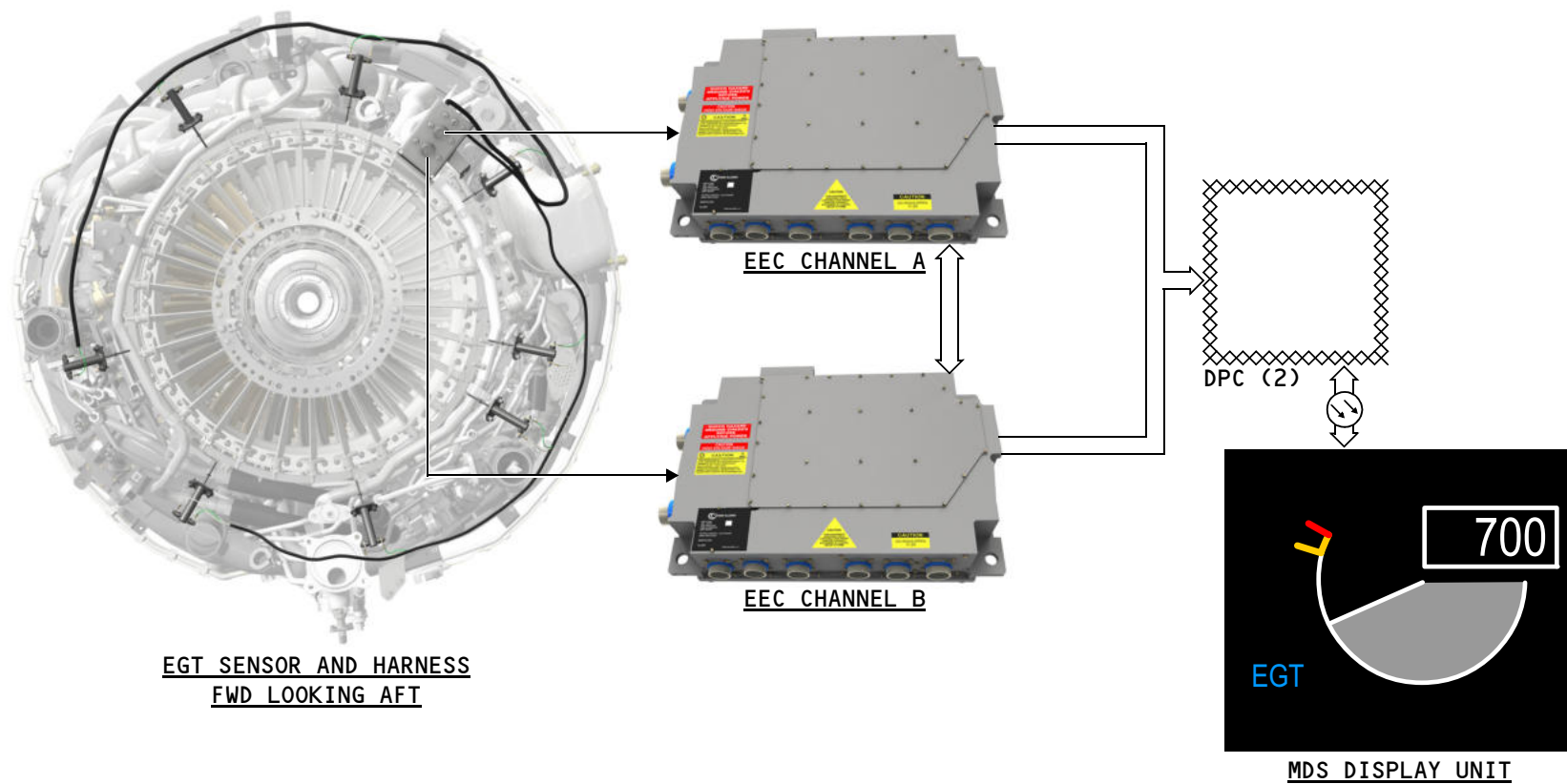
- Engine start
- Engine re-light
- Engine normal operation.

The EECs also use the signals to calculate the EGT margin. The EGT margin is the difference between the maximum permitted EGT (the EGT red-line) and the highest EGT that occurs on the engine during the takeoff. The EGT margin is an indicator of engine performance deterioration.

The EECs send the EGT data to the display processing computers (DPCs) on an ARINC 429 bus.

The DPCs are part of the MAX display system (MDS). The DPCs send the EGT data to show on the engine indication (EI) page. The EI page usually shows on the inboard display unit (DU). EGT data also shows on the electronic propulsion control system (EPCS) maintenance pages and on the PERFORMANCE maintenance page.

DMC-B737T-A-77-21-00-02A-040A-T_001-00



DMC-B737T-A-77-21-00-02A-040A-T_001-00
ICN-B737T-A-772100-A-81205-00004-A-001-01

EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- GENERAL DESCRIPTION

77-21-00

02-Dec-2020

EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- EGT SENSORS AND HARNESSSES

General

There are eight EGT (T48) sensors and two EGT harnesses on each engine. The 2 EGT harnesses connect the sensors to 2 engine harnesses. The engine harnesses connect to the 2 EECs.

Purpose

The EGT sensors supply analog signals in proportion to the exhaust gas temperature.

The EGT harnesses send the sensor signals to the EECs. The EECs use these signals for the engine control and indication.

Component Locations

The sensors measure the temperature at the stage one nozzle of the low pressure turbine (LPT). The turbine center frame (TCF) holds the sensors in their position. Each EGT sensor has one sensing element that has 2 conductive wires. A housing contains the conductive wires in a protective coil. The protective coil has the shape of a spring to push and seal the interface between the EGT sensor and the stage one nozzle. This makes sure that there is the correct depth of the sensor in the engine gaspath.

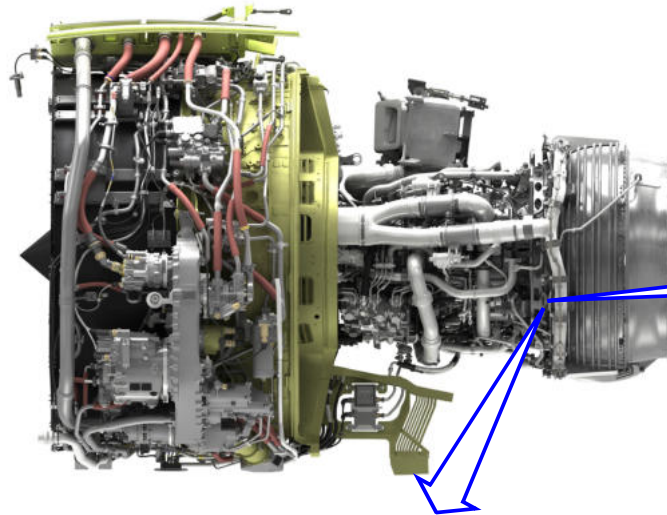
The sensors are numbered from one to eight, clockwise, aft looking forward. Sensor number 1 is installed at the 1 o'clock position. Sensor number 8 is installed at the 12 o'clock position. All of the EGT sensors are the same and are interchangeable.

There are two EGT harnesses, a left harness and a right harness. The left harness connects EGT sensor number 1, 8, 7 and 6 to EEC channel A. The right harness connects EGT sensor number 2, 3, 4 and 5 to EEC channel B. Wires at the top of the sensor housing connect the sensor to the EGT harness.

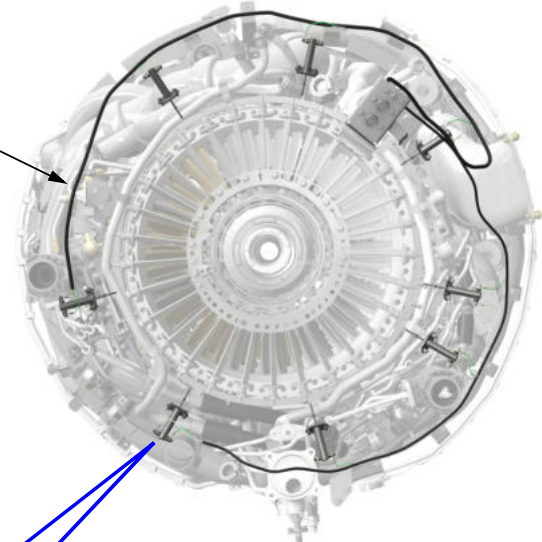
Training Information Point

The wires at the top of the sensor housing connect to the harness with captive nuts. Do not remove the captive nuts from the wires. Damage can occur to the wires or the captive nuts.

DMC-B737T-A-77-21-00-02A-034B-T_001-00



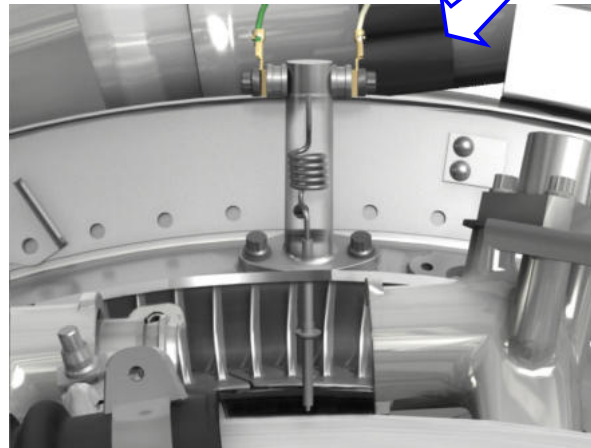
EGT HARNESS (2)



TURBINE CENTER FRAME
(FWD LOOKING AFT)



EGT (T48) SENSOR (8)



EGT SENSOR – CUTAWAY VIEW
TURBINE CENTER FRAME

DMC-B737T-A-77-21-00-02A-034B-T_001-00
ICN-B737T-A-772100-A-81205-00005-A-001-01

EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- EGT SENSORS AND HARNESSES

77-21-00

02-Dec-2020



EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK

EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- FUNCTIONAL DESCRIPTION

General

The electronic engine controls (EECs) send the exhaust gas temperature (EGT) data to the MAX display system (MDS) display processing computers (DPCs). The temperature that shows is an average from the eight EGT sensors on the engine.

When the EECs are de-energized, they do not send the EGT data. The digital display, pointer, and shaded area then go blank.

The EECs energize when any one of these conditions occurs:

- The engine start lever is in the IDLE position
- The engine start switch is in the ground (GRD) position
- The engine start switch is in the continuous (CONT) position
- The EEC MAINT POWER switch function on the MISC SYSTEM CONTROLS (MSC) MDS page is in the TEST position
- N2 speed is >8.5%; when the permanent magnet alternator (PMA) supplies power to the EECs. .

EGT Digital Display and Pointer

The digital display shows EGT in degrees Celsius (C). The digital display and the box around the digital display are usually white.

A pointer on a circular dial also shows EGT. The pointer is usually white and the area behind the pointer is usually shaded gray.

The EGT indication changes color when the temperature is above one of these limits:

- The EGT start redline limit of 753 degrees C
- The EGT maximum continuous amber limit of 1013 degrees C
- The EGT redline limit of 1038 degrees C.

These EGT indications change to a red color or an amber color when the temperature is above one of the limits:

- The EGT digital display
- The box around the digital display
- The pointer
- The shaded area.

When the EGT decreases to the correct range, the digital display and the box around the digital display are again white.

EGT Start Redline

The EGT start redline (753 degrees C) is the maximum limit for the EGT during an engine start on the ground. The EGT start redline shows as a red radial line on the EGT indication during the engine pre-start. The start redline goes blank when the engine is at the idle speed.

During an engine start when the airplane is on the ground, the EEC will do a check for a possible hot start. If the EEC senses a hot start, the white box around the digital display flashes on and off 2 times for each second. It continues to flash until the start lever is moved to the cutoff position.

If there was an EGT start redline exceedance (>753 degrees C), the box around the digital display will be red. It will be red only after the airplane is on the ground and the EECs on the two engines are de-energized. The box will stay red until reset with a selection on the engine exceedance maintenance page.

EGT Maximum Continuous Limit and Amber Band

The EGT maximum continuous limit (1013 degrees C) is the start of the EGT caution range. Continuous operation of an engine with EGT more than this value could cause damage to the engine. The EEC sends the EGT maximum continuous limit value to the DPCs. The limit shows as an amber color.

EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- FUNCTIONAL DESCRIPTION

The EGT caution range shows as an arc with an amber color. The range is between the maximum continuous limit and the EGT redline. This is an advisory range and not an exceedance. When the EGT goes below 1013 degrees C, the EGT indications go back to the usual indications.

During the takeoff, the amber color change is inhibited for five minutes or until the takeoff is completed, whichever comes first. If one engine has a failure during the inhibit period, then the amber color change inhibit extends to ten minutes for the engine that operates.

EGT Redline

The EGT redline (1038 degrees C) is the approved engine operation limit for EGT. The EGT redline shows as a red radial line on the EGT indication.

If there was an EGT redline exceedance (>1038 degrees C), the box around the digital display will be red. It will be red only after the airplane is on the ground and the EECs on the two engines are de-energized. The box will stay red until reset with a selection on the engine exceedance maintenance page.

Engine Failure Message

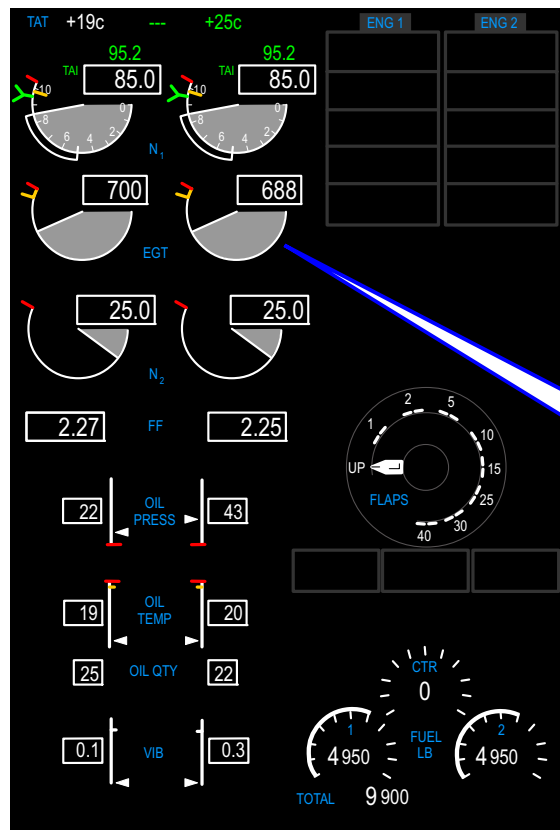
The amber ENG FAIL message shows on the related EGT display if these conditions occur:

- The speed of the two engines are at or above idle.
- The two engine fire handles are stowed
- The two engine start levers are in the idle position
- The N2 speed of one of the two engines then goes below idle.

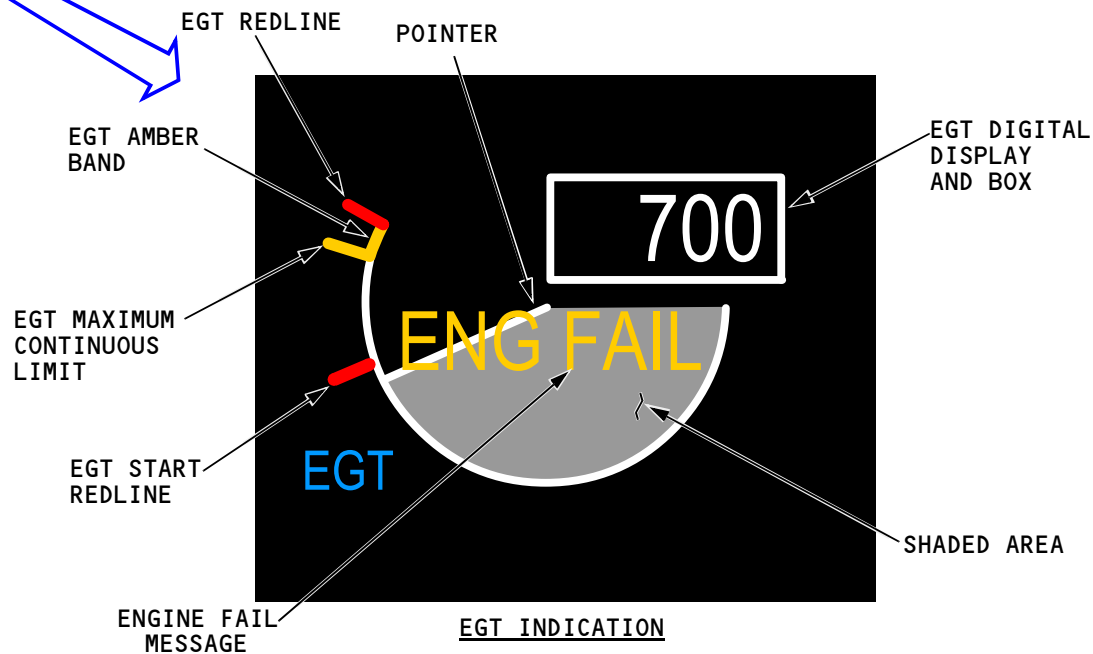
DMC-B737T-A-77-21-00-02A-042A-T_003-00

02-Dec-2020

77-21-00



ENGINE INDICATION DISPLAY



EXHAUST GAS TEMPERATURE (EGT) INDICATING SYSTEM -- FUNCTIONAL DESCRIPTION

77-21-00

ENGINE INDICATING -- TRAINING INFORMATION POINTS - ENGINE EXCEEDANCES

General

The engine indicating system can show exceedance data for these engine parameters:

- N1
- N2
- EGT operating limit
- EGT start limit.

Exceedance Indication

If there is an exceedance for the current flight leg, the box around the N1, N2, or EGT digital indication can show a red color. The red color shows when these conditions occur:

- The airplane is on the ground
- All EEC's are off.

Engine Exceedance Data

If an exceedance occurs, the recorded exceedance data helps you select the necessary maintenance procedure. You use this information to select the correct maintenance procedure:

- Highest value (maximum) of the parameter during the exceedance
- Time above the redline limit.

Maintenance Data Pages menu

You use the MAINT DATA PGS menu to show ENGINE 1 EXCEEDANCE or ENGINE 2 EXCEEDANCE data. The DISPLAY prompt does not show if there are no exceedances in memory.

Engine Exceedance Maintenance Page

You use the engine exceedance maintenance page to show the engine exceedance(s). The page can show redline exceedances of N1, N2, and EGT. The EGT exceedance indication can show for an exceedance during the engine operation or for an exceedance during the engine start.

Exceedances show for the last six flight legs (the current flight leg shows as flight leg zero). Each flight leg can record a maximum of 10 exceedances for each engine.

Footer Menu

These functions are available at the bottom of the page:

- PREV MENU
- PRINT
- SEND
- ERASE
- PREV PAGE (cyan if only one page)
- NEXT PAGE (cyan if only one page).

Select PREV MENU to return to the previous menu.

Select PRINT to send the page data to an installed printer.

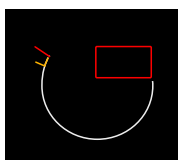
Select SEND to use ACARS to transmit the page data to a ground station.

Select ERASE to erase the exceedance and reset the digital red box on the engine indicating system.

Select PREV or NEXT PAGE (if available) to navigate a maintenance page with more than 1 page.



ENGINE INDICATION



EXCEEDANCE INDICATION

SYS MENU	STATUS	MAINT DATA PGS	MAINT CTRL PGS	ONBD MAINT
DISPL SELECT N	PRINT SELECT N	DATALIN X SELECT N	ERASE SELECT N	
ATA SYSTEM	REAL	MANUAL	AUTO	
23 AIR CONDITIONING	D ISPL A Y	SHOW LIST	SHOW LIST	
27 FLIGHT CONTROL	D ISPL A Y	SHOW LIST	SHOW LIST	
27 SHED	D ISPL A Y	SHOW LIST	D ISPL A Y	
31 MOD	D ISPL A Y			
32 PSEU	D ISPL A Y	SHOW LIST	SHOW LIST	
36 AIR SUPPLY	D ISPL A Y	D ISPL A Y	SHOW LIST	
49 APU	D ISPL A Y	SHOW LIST	SHOW LIST	
71 PERFORMANCE	D ISPL A Y	SHOW LIST	SHOW LIST	
71 EPCS	D ISPL A Y	SHOW LIST	SHOW LIST	
71 ENGINE 1 EXCEEDANCE	D ISPL A Y			
71 ENGINE 2 EXCEEDANCE	D ISPL A Y			

MAINTENANCE DATA PAGES

SYS MENU	STATUS	MAINT DATA PGS	MAINT CTRL PGS	ONBD MAINT
ENGINE 1 EXCEEDANCE		AUTO		
DATE: 17 FEB 16 TIME: 16:31:06 N2 REDLINE	DATE: 17 FEB 16 TIME: 16:31:06 N1 REDLINE	DATE: TIME:	DATE: TIME:	
123.6 117.5 00:08.0	129.0 104.3 00:00.5			
DATE: 17 FEB 16 TIME: 16:31:07 EGT REDLINE	DATE: 17 FEB 16 TIME: 16:31:07 EGT START	DATE: TIME:	DATE: TIME:	
1112.3 1070.0 1063.0 00:01.2 1058.0 00:03.5 1053.0 00:07.8 1048.0 00:25.6 1043.0 00:35.5 1038.0 00:40.0 1033.0 00:46.7 1028.0 00:57.1 1023.0 01:12.6 1013.0 02:37.9	843 840.0 00:00.9 830.0 00:01.2 820.0 00:02.7 810.0 00:03.5 800.0 00:03.8 790.0 00:05.6 780.0 00:07.5 770.0 00:10.0 760.0 00:11.7 750.0 00:12.1			
DATE 17 FEB 16 UTC 16:31:55				
PREV MENU	PRINT	SEND	ERASE	PREV PAGE NEXT PAGE

ENGINE EXCEEDANCE MAINTENANCE PAGE

ENGINE INDICATING -- TRAINING INFORMATION POINTS - ENGINE EXCEEDANCES

77-00-00

ENGINE INDICATING -- TRAINING INFORMATION POINTS - PERFORMANCE MAINTENANCE PAGE

General

The performance maintenance page shows real-time engine performance data for the two engines. The electronic engine controls (EECs) send the engine data to the display processing computers (DPCs) through an ARINC 429 digital databus.

These parameters can show in units selected by a customer option:

- GROSS WEIGHT
- FF (fuel flow)
- OIL QTY.

The units of these parameters will agree with the units that show on the EI (engine indication) display.

Maintenance Data Page Snapshot

You can record the data that shows on a maintenance data page as a snapshot. This records the current display data. The display processing computers (DPCs) record and keep the snapshots in non-volatile memory (NVM).

The flight or maintenance crew can do a manual snapshot or the DPCs can do an automatic event snapshot. You select the RECORD digital switch at the bottom of the maintenance data page to do a manual snapshot. The DPCs do an automatic event snapshot when specified conditions occur. Engine exceedances or system status messages can cause an automatic event snapshot. The system can keep a maximum of 5 manual and 5 automatic event snapshots.

To see a manual or automatic event snapshot, you select the DISPLAY or SHOW LIST digital switch that shows on the DISPLAY SELECTION menu page. If there is only 1 snapshot recorded, the menu shows a DISPLAY digital switch. If there are 2 or more

snapshots recorded (manual or automatic event snapshots), the menu shows a SHOW LIST digital switch. After you select the SHOW LIST digital switch, you can select which of the snapshots to see.

After you see the snapshot, you can erase it if it is not necessary to keep it. To erase the snapshot, select the ERASE digital switch at the bottom of the snapshot. The ERASE digital switch shows on the snapshot in the same area that the RECORD digital switch shows at the bottom of a maintenance data page.

You can also print a maintenance data page or send it by datalink to the ground. You select the PRINT digital switch to print the page on the flight compartment printer, if a printer is installed. You select the SEND digital switch to send the data to the ground by datalink.



SYS MEN U		STATUS		MAIN T DATA PGS		MAIN T CTR L PGS		ONBD MAINT	
PERFORMANCE									
SAT (C)		+3.3		CAS		45		GROSS WT (LBS)	
GS		0.0		MACH		0.150		TAT	
								ALT	
								282	
ENGINE 1					ENGINE 2				
90.9		N1 MAX		90.9					
19.2		N1 CMD IND		20.0					
14.6		N1 CMD		14.6					
19.2		N1 SEL IND		20.0					
19.2		N1 SEL		20.0					
472		EGT IND		478					
472		EGT		478					
68.8		N2		68.8					
611		FF (LBS/HR)		612					
58.5		PS3		58.8					
36.3		TRA		36.1					
-0.9		T/R SLEEVE L		-1.0					
-0.5		T/R SLEEVE R		-1.1					
38.2		OIL PRESS IND		39.5					
38.2		OIL PRESS		39.5					
+72		OIL TEMP IND		+72					
+72		OIL TEMP		+72					
89		OIL QTY (%)		89					
DATE 06 JAN 16 UTC 16:42:57									
PRE V MEN U		PRINT		SEN D		REC ORD		PRE V PAGE	
								NEX T PAGE	

PERFORMANCE MAINTENANCE DATA PAGE

ENGINE INDICATING -- TRAINING INFORMATION POINTS - PERFORMANCE MAINTENANCE PAGE

77-00-00

DMC-B737T-A-77-00-00-05A-300A-T_003-00
ICN-B737T-A-770000-A-81205-00009-A-005-01

02-Dec-2020

ENGINE INDICATING -- TRAINING INFORMATION POINTS - EPCS MAINTENANCE PAGES

GENERAL

The electronic propulsion control system (EPCS) maintenance pages supply maintenance data to the technicians and engineers. You also use the EPCS pages when you do some AMM maintenance procedures. There are 4 pages of EPCS maintenance pages.

Pages 1 and 2 show engine sensor data from the electronic engine controls (EECs). The EEC channel in control for engine 1 and engine 2 shows in a white box, near the top of the page.

Page 3 shows vibration-related data in the top section of the page. The highest of four vibration parameters (N1 FAN VIBE, N1 LPT VIBE, N2 HPC VIBE, or N2 HPT VIBE) shows in a white box. Engine speed bin indications and N1 speed bin targets show in the lower section of the page. The speed bins will show a FULL indication in a white box. The FULL indication shows when there is sufficient data recorded at the N1 speed targets to let the EECs calculate a balance solution. You use this page when you do the engine trim balance procedure.

Page 4 shows digital discrete data from the EECs, isolated into 3 different status words. Also, the lower section of the page shows the analog N1, N2 and oil quantity parameters that are connected directly to the digital processing computers (DPCs).

Maintenance Data Page Snapshot

You can record the data that shows on a maintenance data page as a snapshot. This records the current display data. The display processing computers (DPCs) record and keep the snapshots in non-volatile memory (NVM).

The flight or maintenance crew can do a manual snapshot or the DPCs can do an automatic event snapshot. You select the RECORD digital

switch at the bottom of the maintenance data page to do a manual snapshot. The DPCs do an automatic event snapshot when specified conditions occur. Engine exceedances or system status messages can cause an automatic event snapshot. The system can keep a maximum of 5 manual and 5 automatic event snapshots.

To see a manual or automatic event snapshot, you select the DISPLAY or SHOW LIST digital switch that shows on the DISPLAY SELECTION menu page. If there is only 1 snapshot recorded, the menu shows a DISPLAY digital switch. If there are 2 or more snapshots recorded (manual or automatic event snapshots), the menu shows a SHOW LIST digital switch. After you select the SHOW LIST digital switch, you can select which of the snapshots to see.

After you see the snapshot, you can erase it if it is not necessary to keep it. To erase the snapshot, select the ERASE digital switch at the bottom of the snapshot. The ERASE digital switch shows on the snapshot in the same area that the RECORD digital switch shows at the bottom of a maintenance data page.

You can also print a maintenance data page or send it by datalink to the ground. You select the PRINT digital switch to print the page on the flight compartment printer, if a printer is installed. You select the SEND digital switch to send the data to the ground by datalink.



ENGINE 1

19.1	19.1	N1
68.8	68.8	N2
36.3	36.3	TRA
-0.9	-0.9	T/R SLEEVE L
-0.4	-0.7	T/R SLEEVE R
14.5	14.5	P0
58.5	58.5	PS3
55.6	55.6	P38
0.0	0.0	BLEED FLOW
+4.6	+4.7	T12
+7	+7	T25
+203	+202	T3
95.7	93.5	VBV
94.5	94.5	VBV CMD
19.5	19.5	NIK12
87.5	87.2	VSV
87.3	87.3	VSV CMD
69.8	69.8	N2K25
2.0	1.6	FUWL STR DP
7.7	6.3	FCDC DP
0.7	0.6	DUEL FLT DP
32.0	31.9	FMV
31.9	31.9	FMV CMD
617	614	FF (LBS/HR)
+25	+25	PMV
+25	+25	PMV CMD
160	159	PFMIN

DATE 06 J

ENGINE 1

19.1	19.1	N1
68.8	68.8	N2
-0.3	-0.2	TBV
-13.0	-13.0	TBV CMD
9.1	9.3	HPTACC
9.2	9.2	HPTACC CMD
19.8	18.1	LPTACC
17.9	17.9	LPTACC CMD
-0.1	-0.1	MTC
0.0	0.0	MTC CMD
-0.2	-0.2	BAI/SBV
-13.0	-13.0	BAI/SBV CMD
14.5	14.5	SAV DP
0.0	0.0	SAV CMD
+22	+21	EEC INT TEMP
+71	+71	OIL T
38.3	38.0	OIL P
4.0	4.2	OIL FLT DP
0	0	DMS
472	473	EGT (un)
485	458	EGT 1
457	466	EGT 2
485	468	EGT 3
461	498	EGT 4
-5	-5	REGULATOR
15	15	IDLE IN CTRL

DATE 06

ENGINE 1

19.1	N1
68.8	N2
839	N1 TACH
11804	N2 TACH
0.0	NOB N1 AMP
0.00	NOB N2 AMP
0.06	NOB BB AMP
0.2	TCF N1 AMP
0.13	TCF N2 AMP
0.15	TCF BB AMP
0.0	N1 FAN VIBE
0.0	N1 LPT VIBE
0.0	N2 HPC VIBE
0.1	N2 HPT VIBE
147	FAN PH
47	LPT PH

GROUND RUN SPEED BIN STATUS

1	2	3	4	5	6	7
ENG 1 N1	38	45	50	54	59	63
ENG 2 N1	38	45	50	54	59	63

DATE 06 JAN 16

ENGINE 1

DPC-L	DPC-R	N1 ANALOG	DPC-L	DPC-R
19.2	19.2	20.0	68.8	68.8
89	89	OIL QTY (%)	89	89

DATE 06 JAN 16 UTC 16:43:42

ENGINE INDICATING -- TRAINING INFORMATION POINTS - EPCS MAINTENANCE PAGES

77-00-00

AIRBORNE VIBRATION MONITORING (AVM) SYSTEM -- GENERAL DESCRIPTION

Purpose

The airborne vibration monitoring (AVM) system supplies N1 and N2 rotor vibration data for the flight crew display and engine trim balancing. The engine vibration indication shows on the engine indication (EI) display as a secondary engine parameter.

The DPCs also send the vibration data to the digital flight data acquisition unit (DFDAU). The DFDAU collects the vibration data for the airplane condition monitoring system (ACMS) report.

General Description

The AVM system does these operations:

- Calculates the vibration for each engine rotor (N1 and N2)
- Selects the highest value for each engine
- Shows the highest values on the MAX display system (MDS).

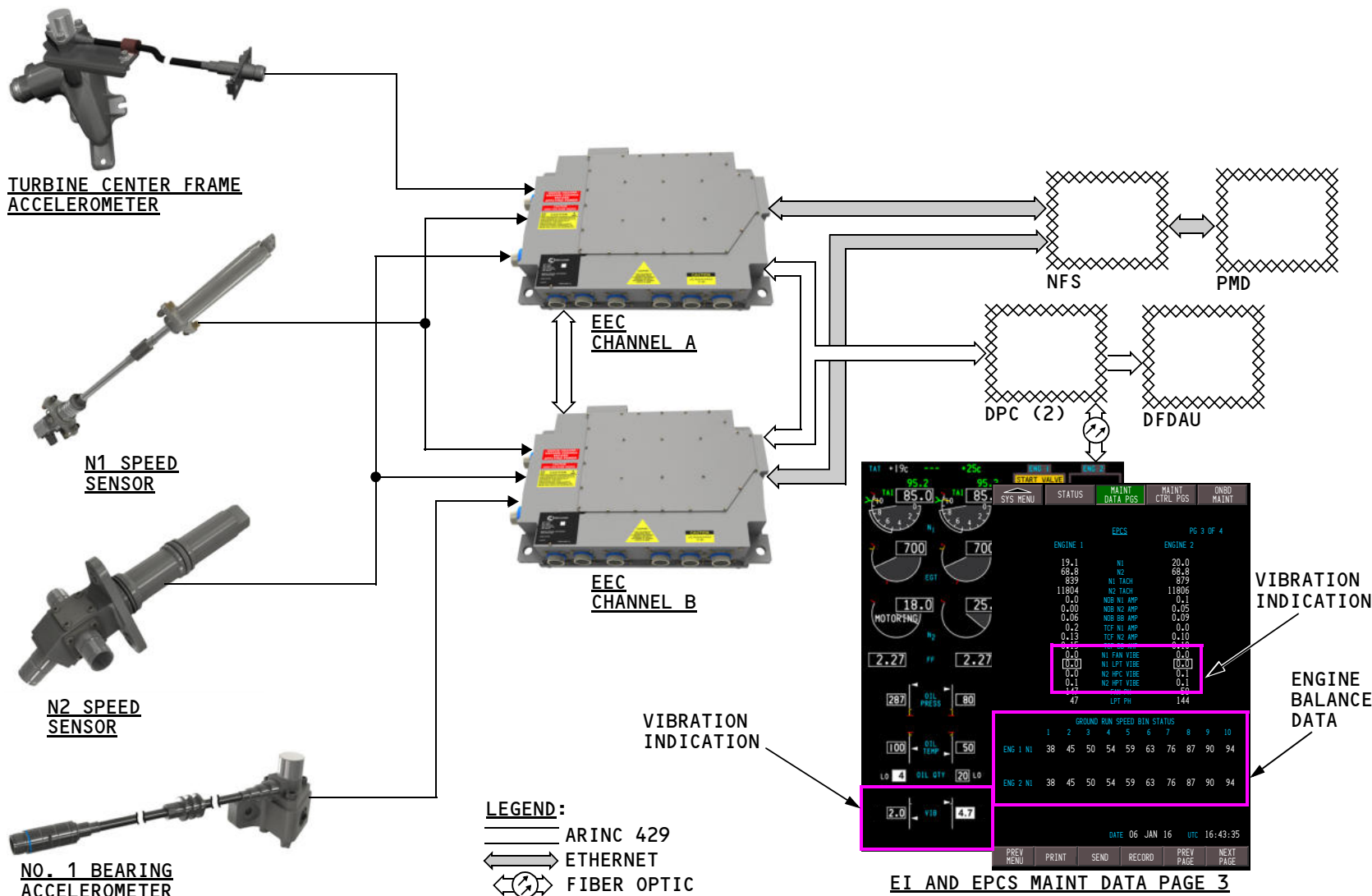
Vibration signals go from two accelerometers on the engine to the electronic engine controls (EECs). Each of the accelerometers give inputs to different channels of the EEC. The turbine center frame (TCF) accelerometer data goes to EEC channel A. The No. 1 bearing accelerometer data goes to EEC channel B. The AVM system also uses the data from the N1 and N2 speed sensors to calibrate the accelerometer data.

The AVM logic is in the EECs. The EECs calculate the data and change the vibration analog data into digital data. The EECs then send the calculated vibration data to the display processing computers (DPCs) in the MDS. The DPCs send the vibration data to show on the engine indication (EI) display. The vibration data also shows on page 3 of the electronic propulsion control system (EPCS) maintenance page.

The EECs also send vibration data to the onboard network system (ONS) for the engine trim balance operations. You use a portable maintenance device (PMD) to access the menu selections on the engine health management (EHM) page, for the engine trim balance operations.

DMC-B737T-A-77-31-00-02A-040A-T_002-01

DMC-B737T-A-77-31-00-02A-040A-T_002-01
ICN-B737T-A-773100-A-81205-00012-A-002-01



AIRBORNE VIBRATION MONITORING (AVM) SYSTEM -- GENERAL DESCRIPTION

02-Dec-2020

77-31-00

AVM SYSTEM -- COMPONENT LOCATION

General

The airborne vibration monitoring (AVM) system has two permanent vibration sensors (accelerometers) and one alternate accelerometer on each engine. These are the names of the accelerometers:

- The number 1 bearing accelerometer
- The turbine center frame (TCF) accelerometer.

The accelerometers are piezoelectric devices with attached leads.

The AVM system uses the signals from the TCF accelerometer and the number 1 bearing accelerometer to calculate the vibration level. The vibration indication shows on the engine indication (EI) display in the flight compartment. Vibration data also shows on page 3 of the electronic propulsion control system (EPCS) maintenance page.

The engine health management (EHM) application also uses the signals from the TCF accelerometer and the number 1 bearing accelerometer. It uses the two inputs to calculate the fan trim balance solution.

Number 1 Bearing Accelerometer

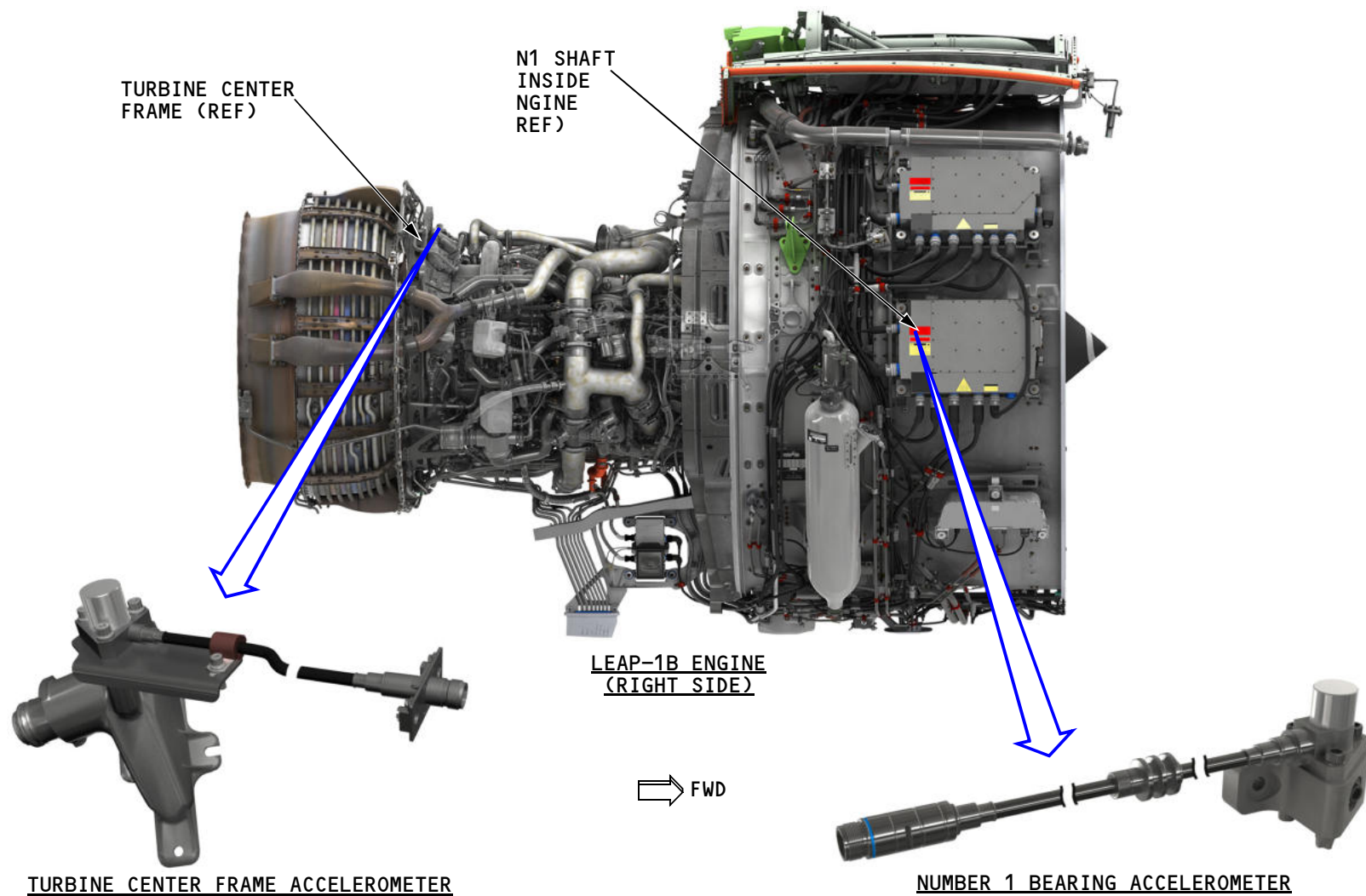
The number 1 bearing accelerometer is internal to the engine, at a forward location of the N1 shaft, near the number 1 bearing. The number 1 bearing accelerometer is sensitive to fan unbalance conditions. Since it is internal to the engine, it is not an LRU. You do maintenance on the number 1 bearing accelerometer only during the engine overhaul. An electrical connector attaches to the sensor wire bundle at the fan hub frame at the 2 o'clock position. Open the right thrust reverser cowl to get access to this area.

Turbine Center Frame (TCF) Accelerometer

The turbine center frame accelerometer installs at the aft end of the engine on the turbine center frame (TCF). The TCF accelerometer is sensitive to unbalance conditions of the:

- High-pressure compressor (HPC)
- High-pressure turbine (HPT)
- Low-pressure turbine (LPT)
- Fan.

You must open the left thrust reverser cowl to get access to the TCF accelerometer.



DMC-B737T-A-77-31-00-02A-055A-T_004-00
ICN-B737T-A-773100-A-81205-00013-A-002-01

AVM SYSTEM -- COMPONENT LOCATION

77-31-00

02-Dec-2020



AIRBORNE VIBRATION MONITORING SYSTEM -- VIBRATION INDICATIONS

THIS PAGE INTENTIONALLY LEFT BLANK

AIRBORNE VIBRATION MONITORING SYSTEM -- VIBRATION INDICATIONS

General

Real-time engine vibration shows on the engine indication (EI) display. The display shows vibration data in terms of Cockpit Unit (CU) values graduating from zero (0) to five (5). Each EEC channel sends vibration data to the MAX display system (MDS). Only the highest vibration value from one of the two accelerometers shows for each engine.

Normal Indication

The airborne vibration monitoring (AVM) logic in the EECs collects these 4 vibration parameters from the 2 accelerometers:

- N1 narrow band from the number 1 bearing (NOB) accelerometer
- N1 narrow band from the turbine center frame (TCF) accelerometer
- N2 narrow band from the turbine center frame (TCF) accelerometer
- N2 narrow band from the number 1 bearing (NOB) accelerometer.

EEC logic then changes the above data into CUs, and then selects the highest of the four vibration values for the flight compartment display. If the logic cannot find if the N1 or the N2 vibration is higher, it sends broadband (BB) vibration data to the display. BB is an indication of the unfiltered engine vibration.

The engine vibration level shows as an arrow pointer on a vertical column. There is also a digital number display in a box. The column starts at 0 CU and extends to 5 CU. The maximum value that shows in the number display is 5.0 CU. The threshold limit of the engine vibration shows as a white line on the column scale.

Non-Normal Indication

A high vibration alert shows when the value of the highest vibration (N1, N2 or BB) is above the limit of 4.0 CUs. The alert changes the colors of the digital number and the box around the number. The color of the number in the box changes to black and the color of the background of the box changes to white. The color change occurs when either one of the two EEC channels senses that the engine vibration is above the limit. When a high vibration alert occurs, the EI shows all the secondary engine parameters if they are blank.

During the engine start, the engine vibration level as found by an EEC can be more than the limit, because of a bowed rotor condition. The EECs inhibit the alert during the engine start when the airplane is on the ground. This prevents any action by the flight crew that is not necessary. After the engine is at idle, the EECs remove the inhibit on the alert. If high vibration continues after the inhibit period, the non-normal indication shows.

Maintenance Page Indication

Vibration data also shows on page 3 of the electronic propulsion control system (EPCS) maintenance data page. These are the amplitude (AMP) data inputs to the EECs from each accelerometer that show:

- NOB N1 AMP
- NOB N2 AMP
- NOB BB AMP
- TCF N1 AMP
- TCF N2 AMP
- TCF BB AMP.

Below those indications, these are the vibration parameters that show (display shows in CUs):

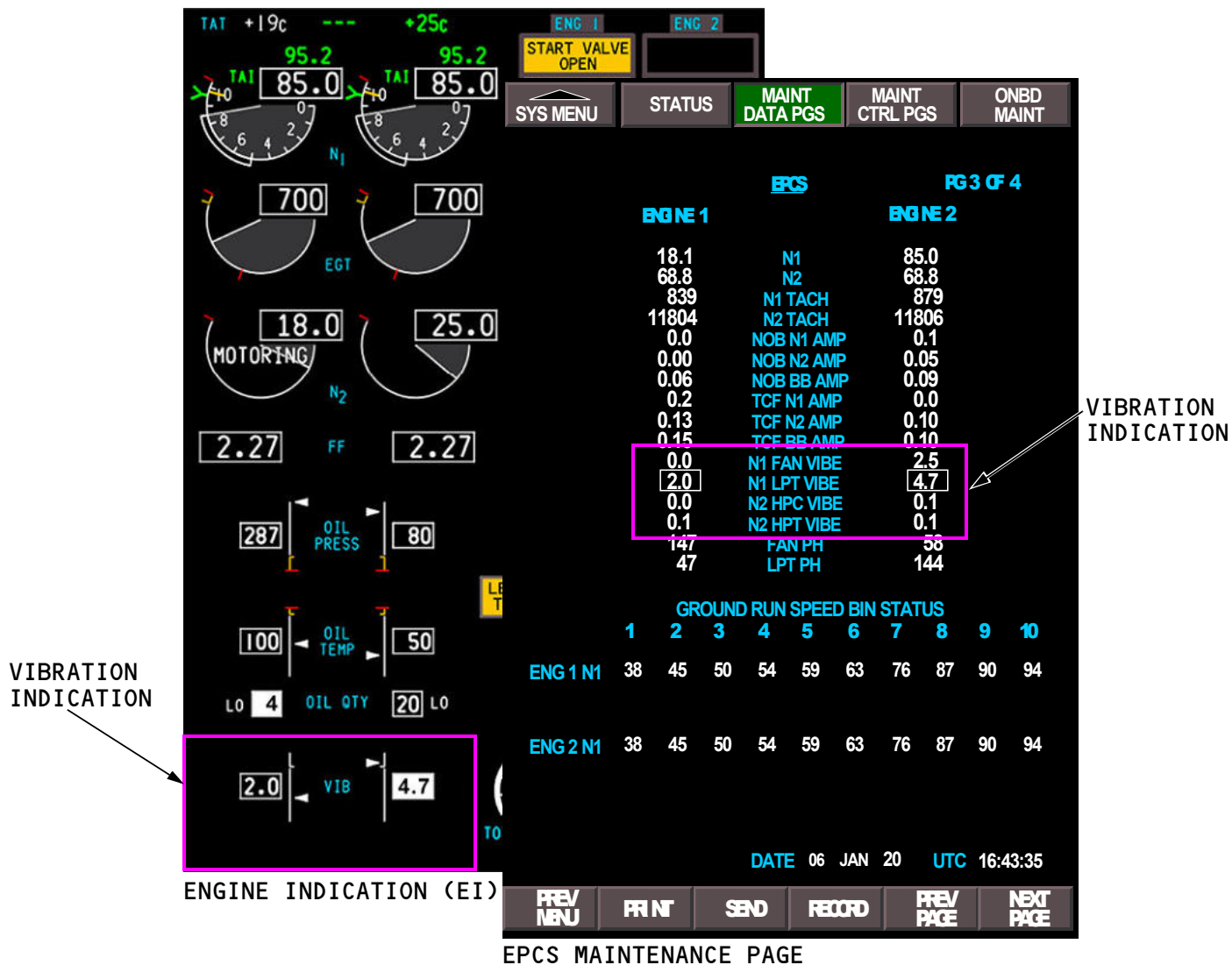
AIRBORNE VIBRATION MONITORING SYSTEM -- VIBRATION INDICATIONS

- N1 FAN VIBE
- N1 LPT VIBE
- N2 HPC VIBE
- N2 HPT VIBE.

The highest parameter of the four shows with a white box around the digital number. This number is the same number as the number that shows on the EI display.

Below those indications, the phase angle of the unbalance condition for the fan and the LPT shows.

At the bottom of the page, the ground run speed bin status shows. These indications show when there is sufficient data collected at each of the N1 speed targets to calculate a fan trim balance solution.



AIRBORNE VIBRATION MONITORING SYSTEM -- VIBRATION INDICATIONS

77-31-00



AIRBORNE VIBRATION MONITORING (AVM) SYSTEM -- FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK

AIRBORNE VIBRATION MONITORING (AVM) SYSTEM -- FUNCTIONAL DESCRIPTION

General

The airborne vibration monitoring (AVM) system uses these inputs to calculate the engine vibration levels:

- N1 speed sensor
- N2 speed sensor
- No. 1 bearing accelerometer (NOB) or alternate No. 1 bearing accelerometer
- Turbine center frame (TCF) accelerometer.

The N1 and N2 speed sensors are part of the engine tachometer system. More information about the engine tachometer system is in ATA 77-11.

The AVM system does these functions:

- Calculates the vibration for each engine rotor (N1 and N2)
- Selects the highest value (N1, N2, or BB) for each engine
- Shows the highest value on the engine indication (EI) display
- Sends specified vibration data to show on page 3 of the electronic propulsion control system (EPCS) maintenance page
- Interfaces with the engine health management (EHM) function in the onboard network system (ONS) for the engine trim balance function.

If one of the engine rotor speed inputs (N1 or N2) has a failure, the system calculates a broadband (BB) vibration indication. Broadband vibration is an indication of the unfiltered engine vibration.

Accelerometers

The accelerometers use piezoelectric crystals. The accelerometers supply an output that is a small electrical signal. The output level changes when the engine structure moves in a radial direction. The output is in proportion to the vibration level of the engine.

The location of the No. 1 bearing accelerometer is near the N1 shaft, at a forward location near the fan. It is sensitive to an unbalance of the fan. This accelerometer supplies a low pressure (LP) rotor vibration signal to electronic engine control (EEC) channel B.

You can only replace a defective No. 1 bearing accelerometer during an engine overhaul. If the No. 1 bearing accelerometer has a failure, install the alternate No. 1 bearing accelerometer. This accelerometer uses the same electrical connector as the No. 1 bearing accelerometer. It is installed in a location with sufficient sensitivity to be used for fan trim balance.

The location of the TCF accelerometer is at the 12 o'clock position on the turbine center frame. This accelerometer is sensitive to vibration at the high-pressure compressor (HPC), the high-pressure turbine (HPT), the low-pressure turbine (LPT), and to an unbalance of the fan. It supplies the vibration signal to EEC channel A.

Electronic Engine Controls (EECs)

The number 1 bearing accelerometer sends the vibration data to EEC channel B only. The TCF accelerometer sends the vibration data to EEC channel A only. EEC channel B sends the number 1 bearing vibration data to EEC channel A through the cross-channel data link. The AVM logic is in the EECs. The AVM function processes the accelerometer signals from both EEC channels. It selects the highest value to show on the EI display.

The EECs record the amount of time and the N1 speed at which the maximum vibration level occurred for both the Number 1 bearing accelerometer and the TCF accelerometer. The EECs also record the amount of time and the N2 speed at which the maximum vibration level occurred for both the Number 1 bearing accelerometer and the TCF

AIRBORNE VIBRATION MONITORING (AVM) SYSTEM -- FUNCTIONAL DESCRIPTION

accelerometer. These 4 values and their related data are kept for every flight history leg or ground run, to a maximum of 10 flight legs.

EEC channel A (only) also uses the vibration data for the functions related to the engine trim balance system. EEC channel A interfaces with the engine health management (EHM) application in the ONS through an Ethernet connection. You connect a portable maintenance device (PMD) to the ONS to do the engine trim balance functions.

Vibration Indication

The highest vibration signal continuously goes to the display processing computers (DPCs) on an ARINC 429 databus. This data shows on the EI display of the MAX display system (MDS). The engine vibration indication is a real-time display of the sensed engine vibration. A non-normal indication shows on the EI display when the value of the highest vibration is above the vibration limit.

Vibration data also shows on page 3 of the electronic propulsion control system (EPCS) maintenance page. The speed bins for the engine trim balance function also show on page 3 of the EPCS page.

Engine Health Management (EHM)

The AVM system in EEC channel A supplies the vibration data for the EHM functions. The EHM function is in the onboard network system (ONS), on the network file server (NFS). You use the EHM function to do the engine trim balance function. The EHM function is the interface between EEC channel A and the portable maintenance device (PMD).

These are the EHM sub-functions:

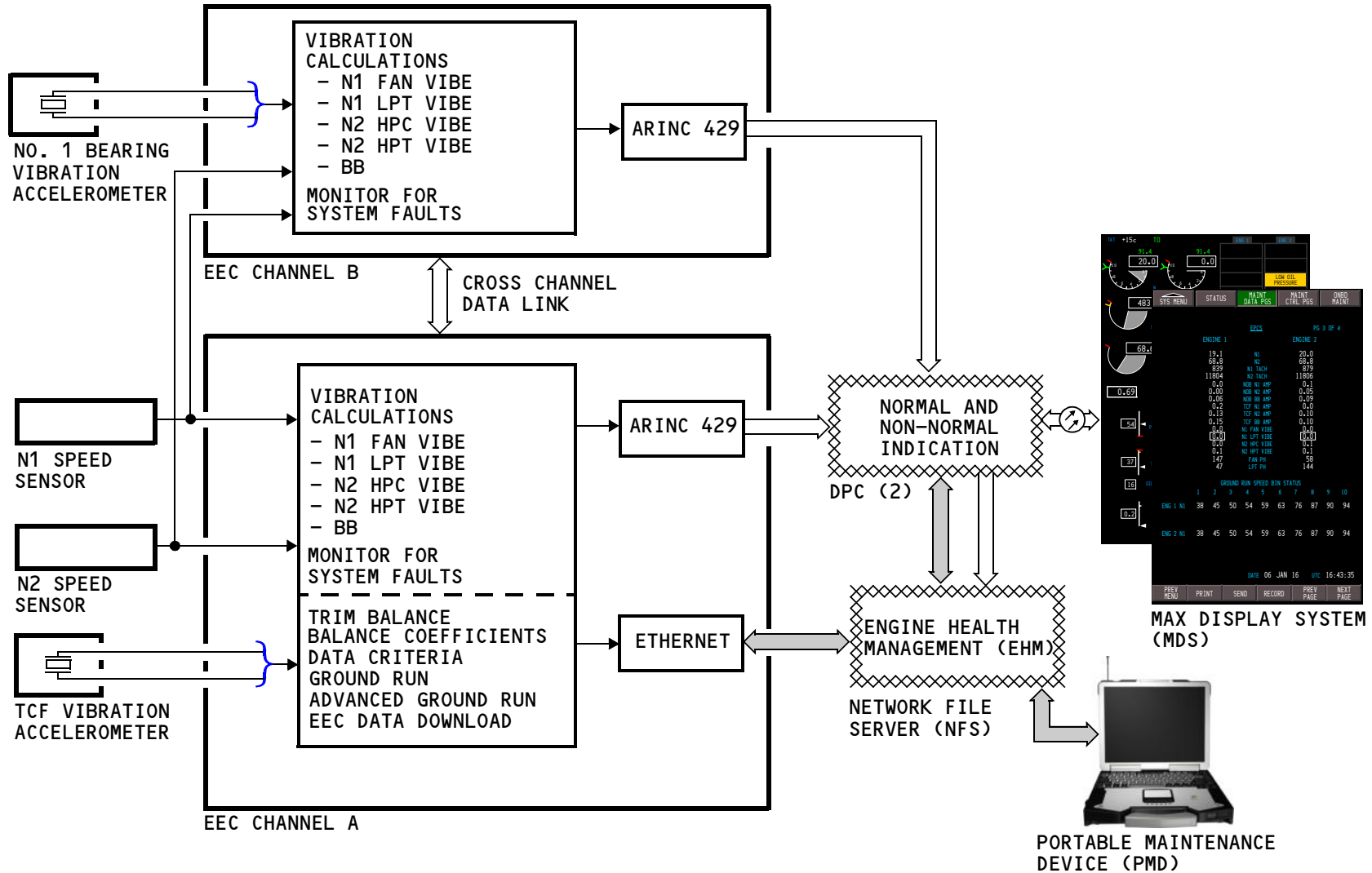
- Trim balance
- Balance coefficients
- Data criteria

- Ground run
- Advanced ground run
- EEC data download

More information about the EHM sub-functions is on the ATA 77-31 Operation page.

Portable Maintenance Device (PMD)

To do the engine trim balance procedure, you use the EHM screens that show in the onboard maintenance function (OMF) of the ONS. You must use a portable maintenance device (PMD) to get access to the EHM function.



AIRBORNE VIBRATION MONITORING (AVM) SYSTEM -- FUNCTIONAL DESCRIPTION

77-31-00

AIRBORNE VIBRATION MONITORING SYSTEM -- OPERATION

General

An unbalanced low pressure (N1) rotor can cause engine vibration and cabin noise. An N1 rotor unbalance condition can occur if you install an untested engine or if you replace some components on an engine. You use the engine trim balance function to decrease the engine vibration and airplane cabin noise.

There are procedures in the airplane maintenance manual (AMM) to look for high vibration from the high pressure rotor (HPC/HPT), or the low pressure rotor (fan and low pressure turbine). There is also an AMM procedure to calculate a new fan trim balance solution.

High Pressure System (HPC/HPT) Vibration Survey

The high pressure system vibration survey supplies vibration data from the high pressure compressor (HPC) and the high pressure turbine (HPT) to the electronic engine controls (EECs). An inspection of the high pressure vibration data can help isolate the cause of high engine vibration to the low pressure or high pressure rotor system. This data is for information only. You cannot balance the high pressure system with the engine installed. It is necessary to remove and disassemble the engine to balance the high pressure system.

Low Pressure System (Fan/LPT) Vibration Survey

You do the low pressure system (fan/LPT) vibration survey if you do this engine maintenance:

- You install an engine that was not pre-tested and pre-trimmed at an approved test facility
- You replace a full set of fan blades on the engine
- You replace 3 or more pairs of fan blades on the engine.

The vibration survey makes sure that the engine balance is satisfactory. An inspection of the low pressure vibration data can help isolate the cause of high engine vibration.

You do the vibration survey to supply the low pressure system vibration data to the EECs. The EECs use the vibration data to calculate a new engine balance solution.

Low Pressure System (Fan/LPT) Trim Balance (With ONS)

You can do the engine trim balance procedure with the engine health management (EHM) function in the onboard network system (ONS).

Engine Health Management (EHM) Function

You do the vibration surveys and the engine trim balance procedure through the EHM function. The EHM function is in the ONS, on the network file server (NFS).

The EHM function has an interface with the airborne vibration monitoring (AVM) system in channel A of the EEC. The EHM function tells EEC channel A to calculate vibration data and send the data to the ONS.

You use a portable maintenance device (PMD) to get access to the EHM functions. From the onboard network system (ONS), select the Other Functions menu. From the Other Functions menu, select the EHM function. Then you select an engine, and then you can get access to the other sub-functions from the Functions menu.

The EHM function has these sub-functions:

- Trim balance
- Balance coefficients
- Data criteria
- Ground run



AIRBORNE VIBRATION MONITORING SYSTEM -- OPERATION

- Advanced ground run
- EEC data download.

Trim Balance

The engine trim balance function helps to decrease the N1 vibration levels on the engine. The engine trim balance function uses engine operation data and the engine weight configuration to calculate a new engine balance solution. The new balance solution is calculated with data from ground engine operation, or from a selection of flight legs or an average of recent flight legs.

After you select the trim balance function, the flight history shows. The flight history function records fan (N1) and core (N2) vibration data at specified engine speeds. The function can record the data during flight or ground operations. It uses this data for the engine trim balance.

The name for the specified engine speeds is "speed bins". The applicable speed bins give smaller increments in the fan mode. The fan mode is the speed at which the highest amplitude of vibration occurs. Thus, it is the point which is most sensitive for the engine balance solution. It is necessary to use accurate data for the fan mode, to get a good balance solution. You can set the precision level of the data with the data criteria selections. You use the specified speed bins to collect the vibration data.

You make adjustments to the engine trim balance when you change the configuration of the balance weight screws. There is a weight table which is a list of the weights to remove or to install to balance the engine. You remove and install fan weights in the platform front shroud, for the one-plane solution.

The engine balance function can also do a two-plane solution. For the two-plane solution, you also attach clips onto the blades of the last

stage of the low-pressure turbine (LPT). It is more usual to do a one-plane balance procedure. It is less usual to do a two-plane balance procedure.

The EEC keeps the engine weight configuration in memory. This is the list of the fan weights and LPT clips and their locations as installed on the engine.

Balance Coefficients

The balance solution uses a set of balance coefficients to calculate an initial solution that will decrease the vibration levels. Initially, the EEC has a set of generic coefficients that it keeps in storage. But, in some conditions, specific coefficients are necessary to get a good balance solution.

To make the specific coefficients, you make known out-of-balance conditions on the engine. Then you do a minimum of one engine ground operation with each of the different balance weight configurations. This lets the EEC calculate a set of specific coefficients. To calculate these coefficients, it is necessary to have flight or ground operation data with two different configurations of engine trim balance weights.

Data Criteria

You use the data criteria function to select the level of data recorded for the flight history data and ground operation data. The EEC uses this data to calculate the engine trim balance solutions.

The criteria selections are: HIGH, MEDIUM, LOW, and NONE. The HIGH selection uses the highest criteria level to collect the data. The MEDIUM selection uses a medium criteria level to collect the data. The LOW selection uses the lowest criteria level to collect the data. The

AIRBORNE VIBRATION MONITORING SYSTEM -- OPERATION

NONE selection uses no stability criteria to collect the data. When you select a new data criteria to record the data, it will replace the flight history and ground operation data criteria with the new criteria.

Ground Run

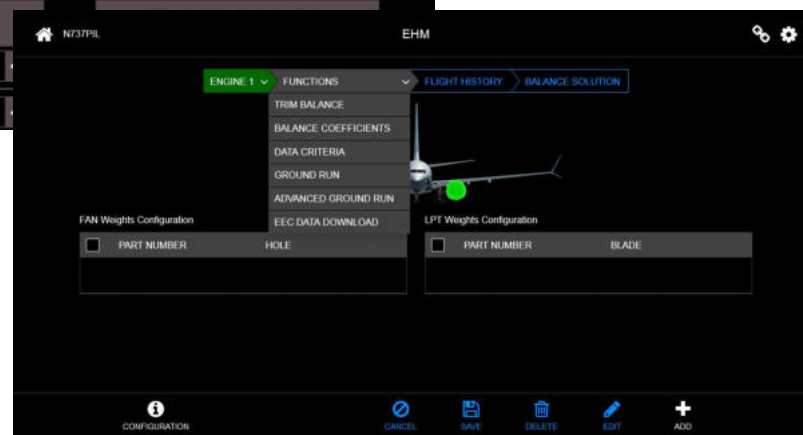
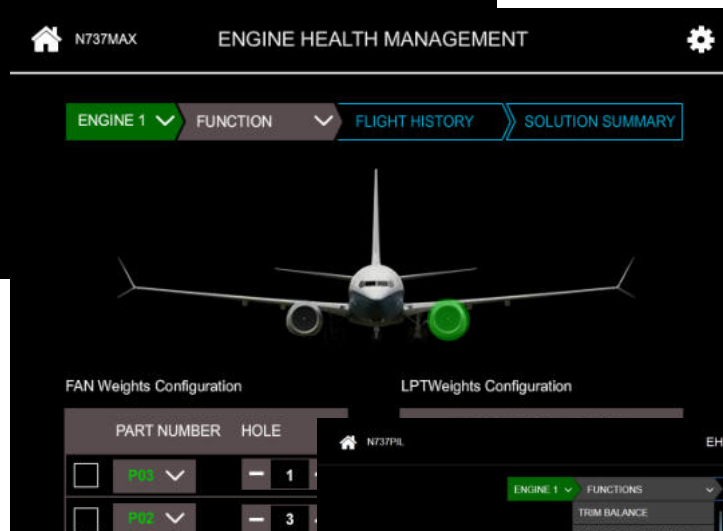
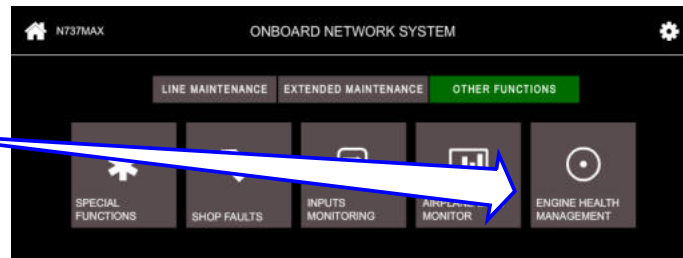
You use the ground run function to cause the EEC to record vibration data from an engine ground operation. You can do ground runs by sweeping the throttle up and down or by moving the throttle in steps.

Advanced Ground Run

You use the advanced ground run function to do an engine ground operation to collect advanced vibration data. Advanced ground run will collect high bandwidth data for spectral analysis.

EEC Data Download

You use the EEC data download function to tell the EEC to move the flight history, the balance weight configuration, or the balance solution to a PMD. You can also use the EEC data download function to move shop faults, syslogs, full flight data, or other EEC data to the file download manager. You get access to the download manager through the Line Maintenance menu in the ONS.



DMC-B737T-A-77-31-00-01A-100A-T_004-00
ICN-B737T-A-773100-A-81205-00098-A-001-01

AIRBORNE VIBRATION MONITORING SYSTEM -- OPERATION

02-Dec-2020

77-31-00



ENGINE CONTROLS -- INTRODUCTION

Purpose

The engine control system supplies most of the signals to control the engine thrust. It also supplies signals to other airplane systems that use engine control status.

The engine control system has these components:

- Thrust lever assemblies
- Thrust lever resolvers
- Engine start levers
- Thrust lever interlock solenoids.

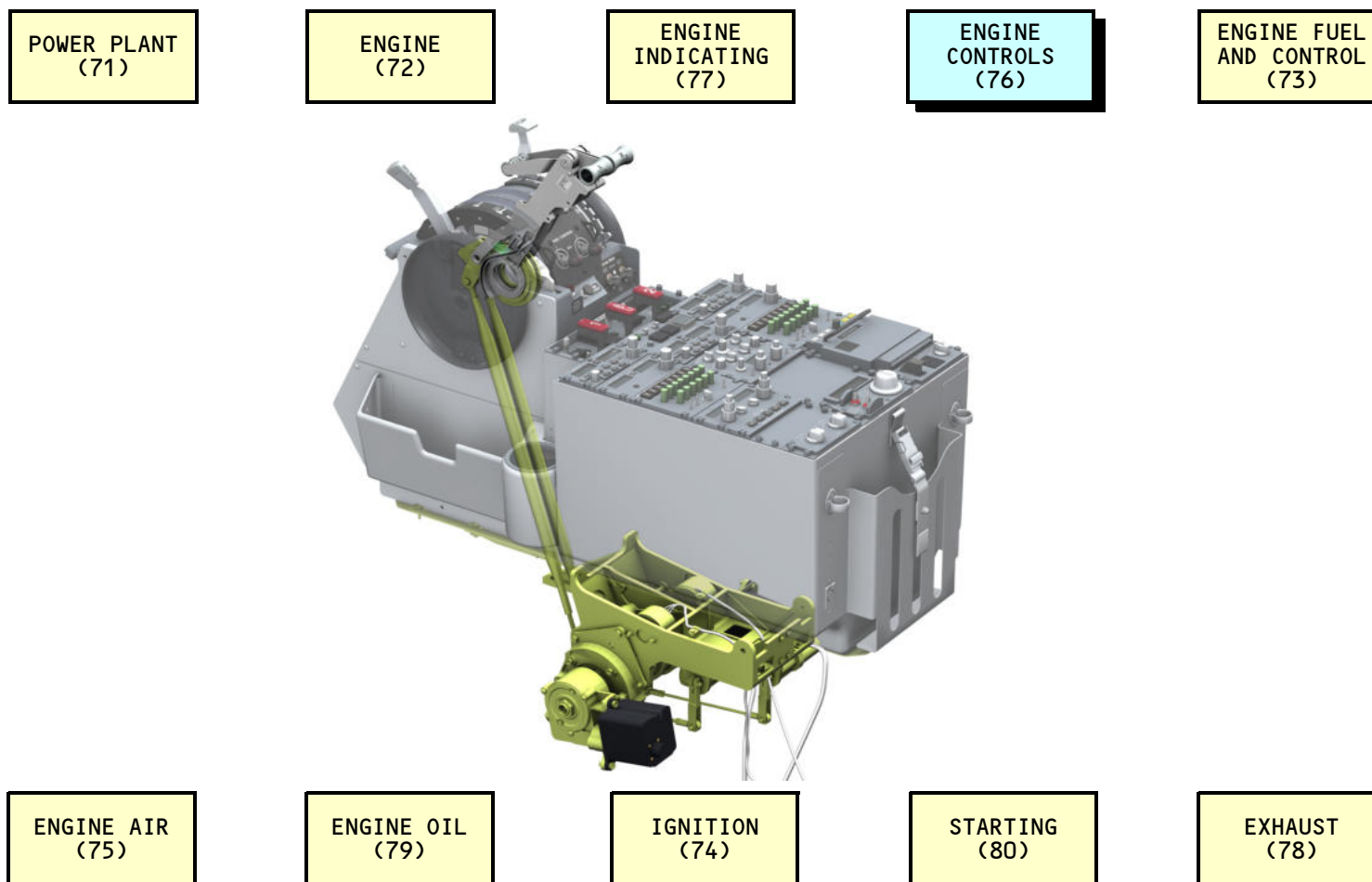
Abbreviations and Acronyms

AGB	- accessory gearbox
AMM	- aircraft maintenance manual
ASM	- autothrottle servomotor
DPC	- display processor computer
EEC	- electronic engine control
FDAU	- flight data acquisition unit
IDG	- integrated drive generator
LVDT	- linear variable differential transformer
RLA	- reverse thrust lever angle
TLA	- thrust lever angle
TO/GA	- takeoff/go-around
TRA	- thrust lever resolver angle
T/R	- thrust reverser

DMC-B737T-A-76-00-00-02A-018A-T_001-00

02-Dec-2020

76-00-00



ENGINE CONTROLS -- INTRODUCTION

76-00-00

DMC-B737T-A-76-00-00-02A-018A-T_001-00
ICN-B737T-A-760000-A-81205-00010-A-001-01

02-Dec-2020

ENGINE CONTROLS -- GENERAL DESCRIPTION

General

The engine control system supplies manual and automatic control inputs to operate the engine. The engine control system has these components:

- Thrust levers (forward and reverse)
- Thrust lever resolvers
- Engine start levers
- Thrust lever interlock solenoids.

Thrust Levers

Use the thrust levers to supply the manual inputs to the engine control system. There are two thrust lever assemblies, one for each engine. For each engine, there is a forward thrust lever and a reverse thrust lever. The reverse thrust lever is on the forward thrust lever.

For each engine, the thrust levers supply a thrust command signal to the electronic engine controls (EECs) through the thrust lever resolver. Each thrust lever assembly connects mechanically to the resolver through an adjustable rod.

A lock pawl prevents the operation of the forward thrust lever and the reverse thrust lever at the same time.

Thrust Lever Resolver

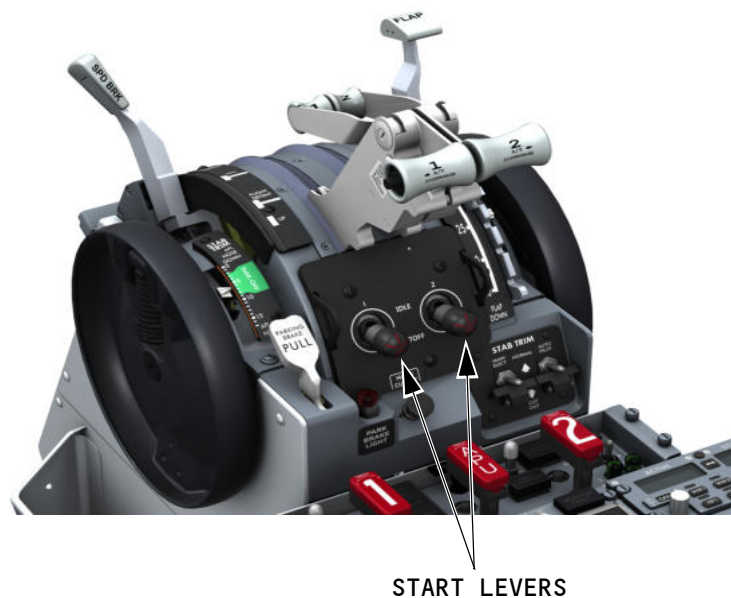
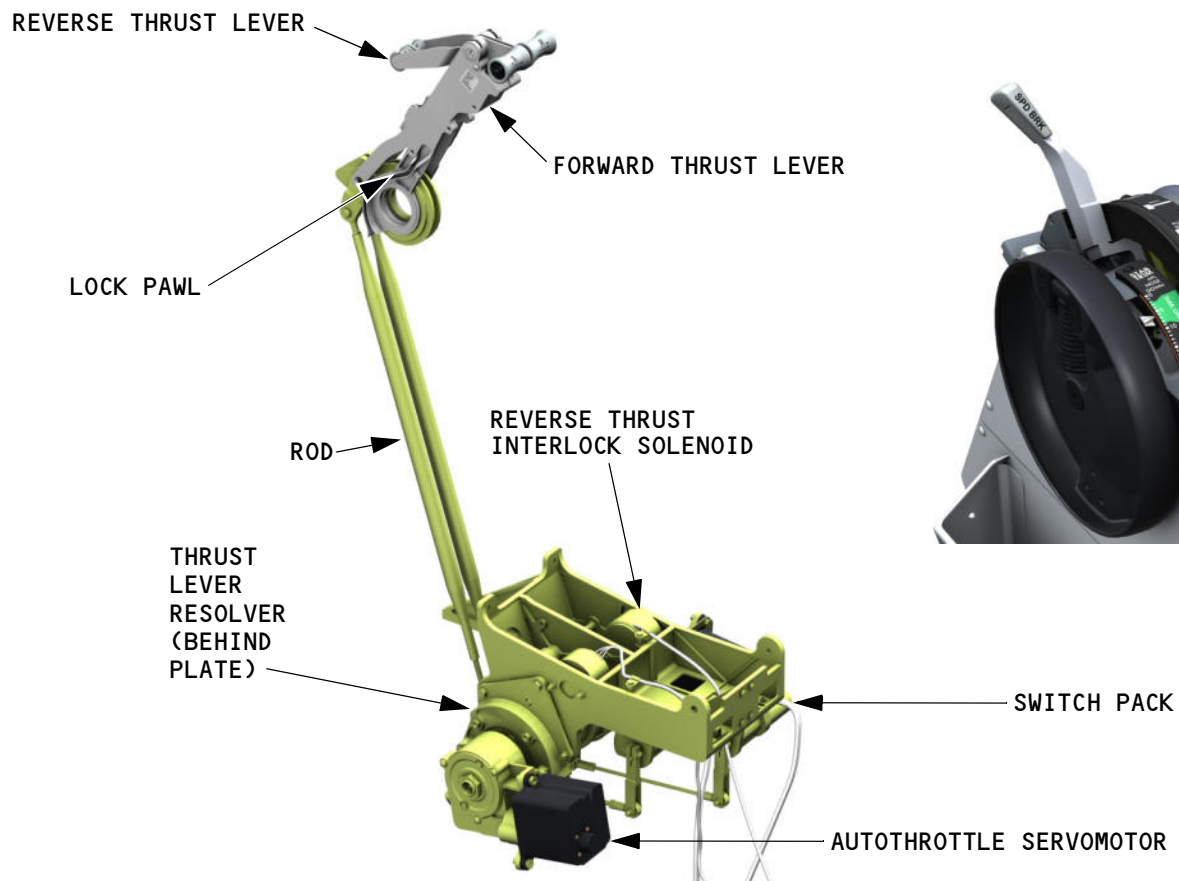
There are two thrust lever resolver assemblies, one for each engine. Each thrust lever resolver assembly has two separate resolver outputs, one for EEC channel A and one for EEC channel B. The thrust lever resolvers change the mechanical forward and reverse thrust lever positions to an analog thrust lever resolver angle (TRA) signal. The EECs use the TRA signal to control the engine thrust setting.

Start Levers

There are two start levers, one for each engine. The engine start lever controls the start and shutdown sequence. The start lever also supplies signals to different aircraft and engine systems and components.

Reverse Thrust Interlock Solenoids

There are two reverse thrust interlock solenoids, one for each engine. Each reverse thrust interlock solenoid limits the range of motion of a reverse thrust lever. The thrust reverser can be deployed, but engine reverse thrust can not go above idle until the thrust reverser sleeves are near the full deployed position. EEC channel A operates the solenoids. The thrust lever interlock solenoids are above the autothrottle assembly.



ENGINE CONTROLS -- GENERAL DESCRIPTION

76-00-00

DMC-B737T-A-76-00-00-02A-040A-T_003-00
ICN-B737T-A-760000-A-81205-00011-A-004-01

02-Dec-2020

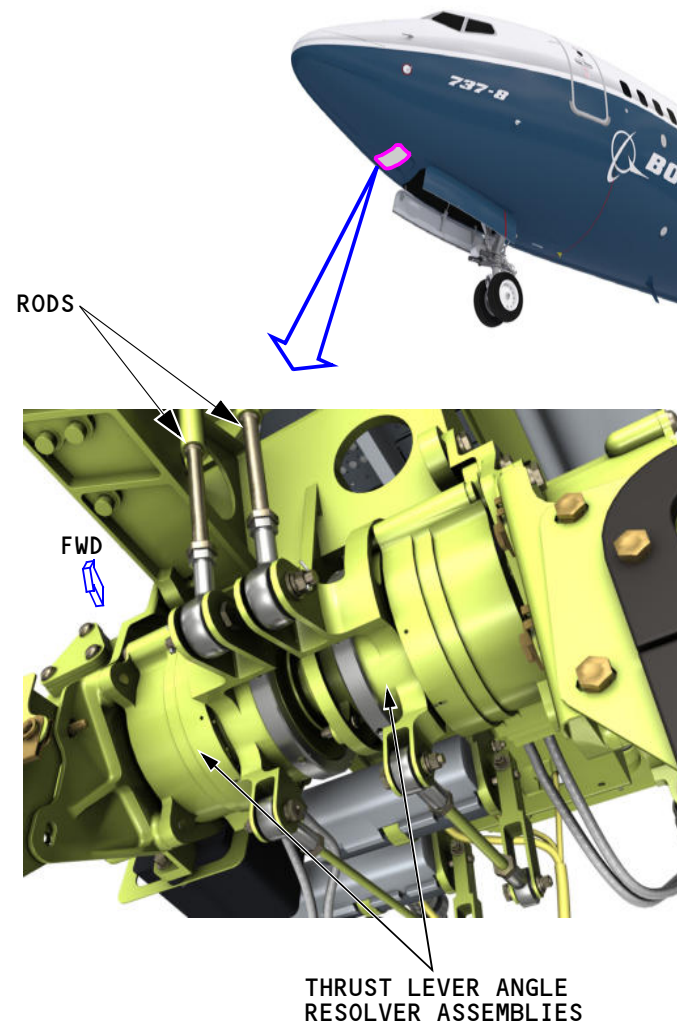
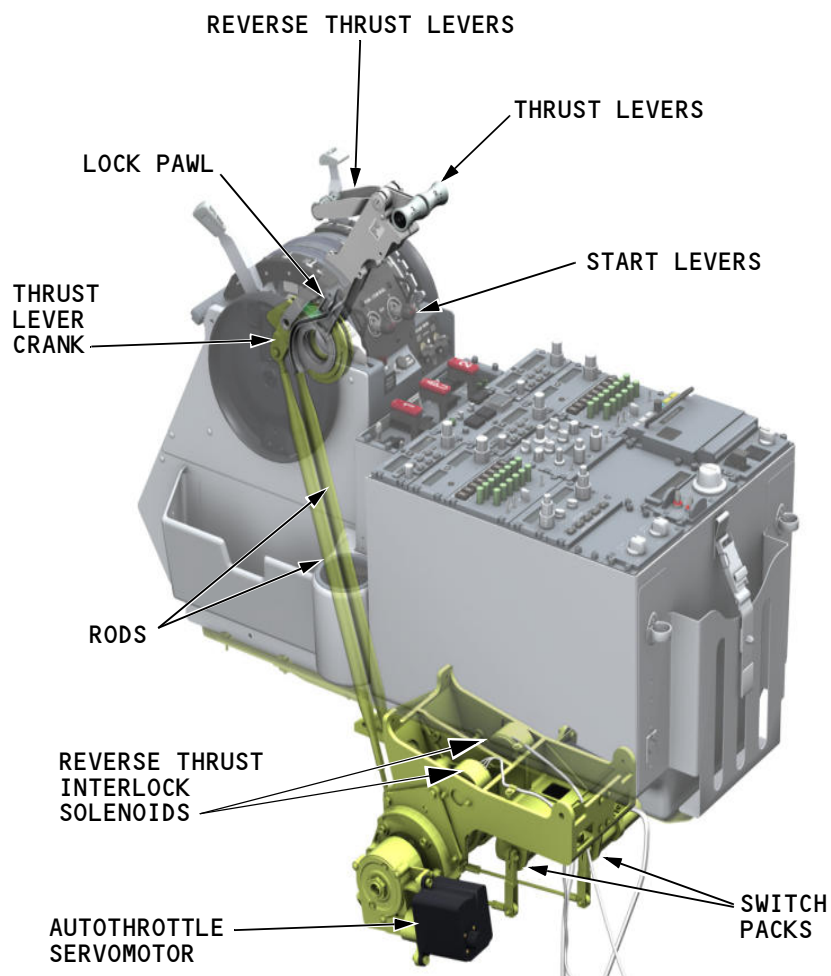
ENGINE CONTROLS -- COMPONENT LOCATION

Component Locations

The thrust levers and the start levers are on the control stand in the flight compartment. Remove panels from the side of the aisle stand to get access to these components:

- Start levers
- Lock pawls
- Thrust lever cranks and rods.

The reverse thrust interlock solenoids and thrust lever resolvers are in the autothrottle assembly. The autothrottle assembly is under the flight compartment floor in the forward equipment compartment. Enter the forward equipment compartment through the access door, forward of the nose landing gear.



DMC-B737T-A-76-00-00-02A-055A-T_002-00
ICN-B737T-A-760000-A-81205-00012-A-003-01

ENGINE CONTROLS -- COMPONENT LOCATION

76-00-00

02-Dec-2020

ENGINE CONTROLS -- INTERFACES

General

These equipment and mechanisms related to the engine control system are on the control stand and below the flight compartment floor in the autothrottle assembly:

- Resolvers
- Clutch pack
- Autothrottle switch packs
- Autothrottle servomotor (ASM) and gearbox assembly
- Take off/go-around (TO/GA) switches and autothrottle disengage switches.

Resolvers

There are two resolver assemblies, one for each engine. Each thrust lever assembly contains two resolvers. The resolver changes the forward and reverse thrust lever positions to an analog electrical signal. This signal goes to the EECs.

Clutch Pack

The clutch pack gives a friction connection between these mechanisms:

- Thrust levers
- ASM and gearbox assembly.

The clutch lets the autothrottle system move the resolver and the thrust levers. The clutch also gives a friction force to prevent the free movement of the thrust levers. The clutch lets the pilot move the thrust levers independently of the autothrottle system.

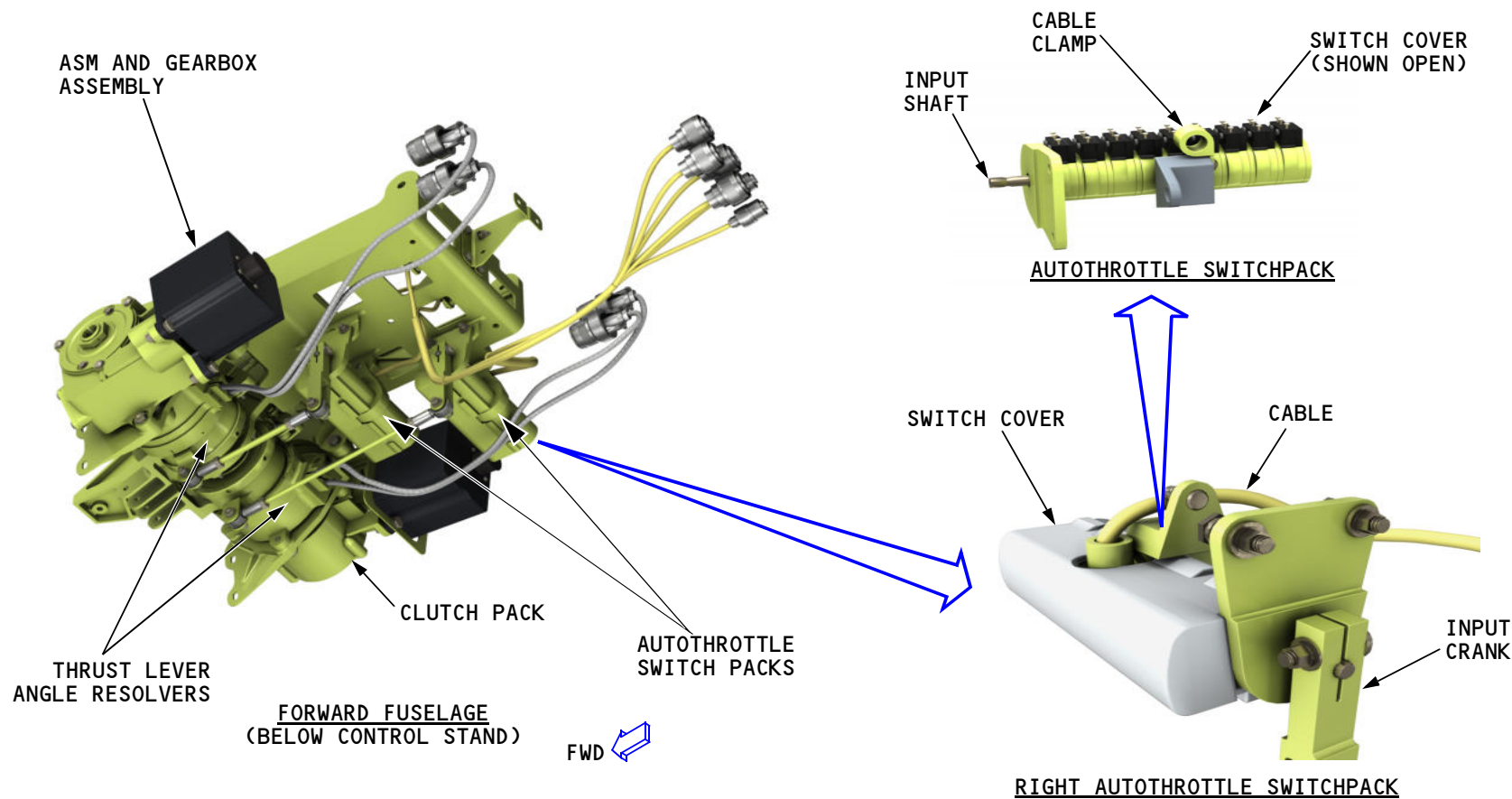
Autothrottle Switch Packs

There are two autothrottle switch packs, one for each thrust lever assembly. Each thrust lever assembly moves a switch pack through a mechanical linkage. Each switch pack has nine switches that supply discrete thrust lever position signals to various systems. These are the switches and the function of each switch:

- S1 - auto ground speedbrake control and landing gear warning
- S2 - autobrake system
- S3 - autobrake system
- S4 - thrust reverser sync lock control
- S5 - No Connection
- S6 - No Connection
- S7 - wing thermal anti-ice system
- S8 - aural warning - takeoff warning and weather radar and tail skid (option)
- S9 - landing gear warning.

Autothrottle Servomotor (ASM) and Gearbox Assembly

The thrust management function of the autothrottle system supplies the thrust command signal to the EECs. To do this, the ASM and gearbox assembly moves the TLA resolvers and the thrust levers through the clutch pack.



DMC-B737T-A-76-00-00-06A-110A-T_003-00
ICN-B737T-A-760000-A-81205-00013-A-002-01

ENGINE CONTROLS -- INTERFACES

02-Dec-2020

76-00-00

ENGINE CONTROLS -- THRUST LEVER

General

The thrust levers and the resolvers in the autothrottle assembly work together to supply a thrust command to the EEC. The autothrottle system makes the automatic inputs. The thrust levers are used to make manual inputs.

Thrust Lever

There are two thrust lever assemblies, one for each engine. A thrust lever assembly has many parts. These parts mechanically transmit the thrust command to the resolver assembly:

- Forward thrust lever
- Reverse thrust lever
- Control link
- Crank
- Rod.

These parts operate the crank:

- Forward thrust lever
- Reverse thrust lever
- Control link.

The crank connects with the clutch pack and the resolver in the autothrottle assembly through the rod. The forward thrust lever and the crank are on the same shaft, but they move independently. The forward thrust lever holds the reverse thrust lever. The control link directly connects the reverse thrust lever and the crank. The control link moves up when the reverse thrust lever is raised. The control link moves down when the reverse thrust lever is lowered.

When the forward thrust lever is moved, the position of the reverse thrust lever locks the control link onto the forward thrust lever. The force goes to the crank through the control link.

When the reverse thrust lever is moved, the force goes to the crank through the control link.

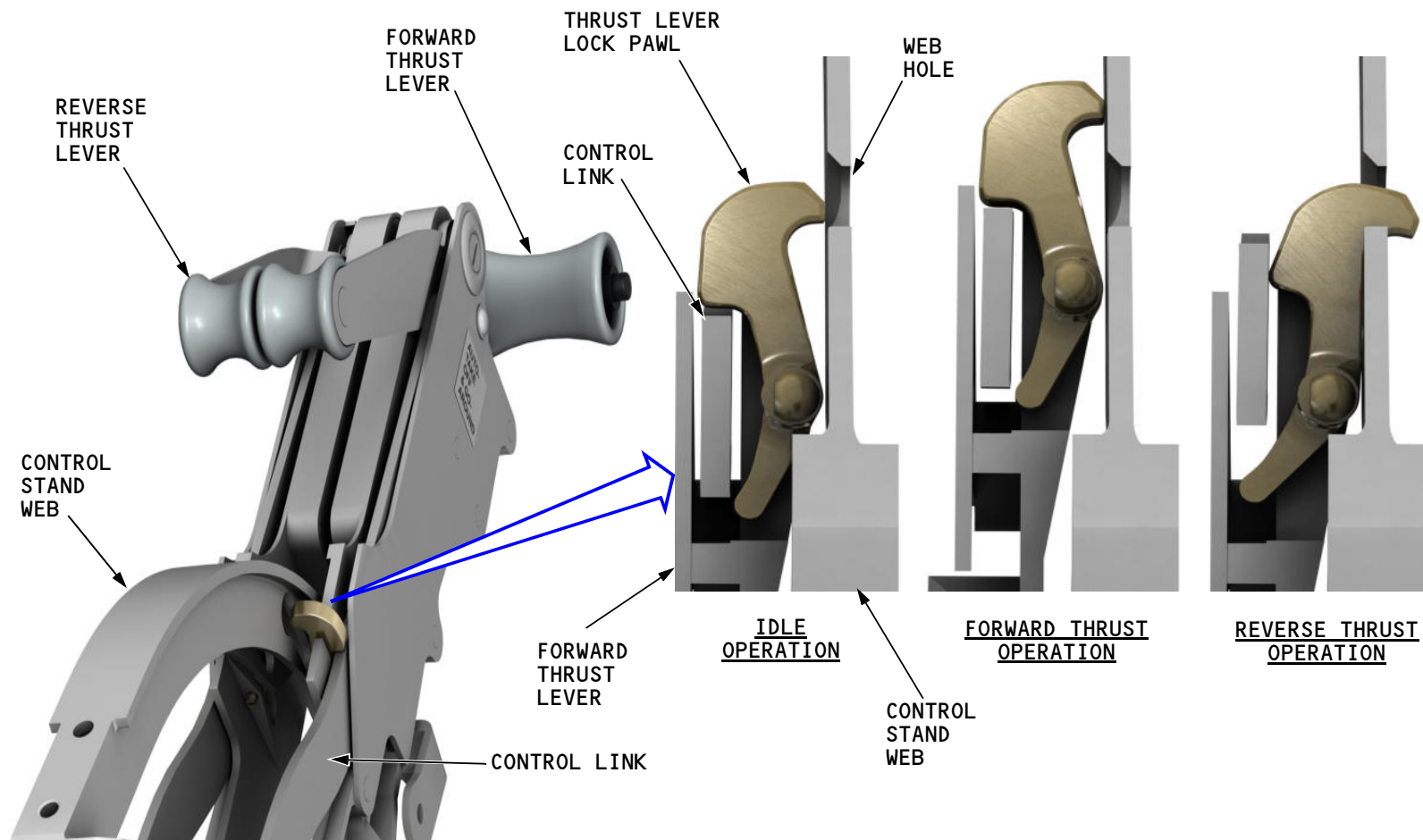
The thrust lever lock pawl prevents the operation of the forward thrust lever and the reverse thrust lever at the same time. The pawl must move into a hole on the control stand web for the reverse thrust lever to move. The pawl can move into the hole only when the forward thrust lever is in the idle position.

When the forward thrust lever is not at idle, the pawl locks the reverse thrust lever at the STOW position. This prevents the movement of the reverse thrust lever when the forward thrust lever is not in the idle position.

When the forward thrust lever is at idle, the lock pawl aligns with the web hole. Movement of the reverse thrust lever will engage lock pawl into the web hole. When the reverse thrust lever moves, the pawl locks the forward thrust lever.

Training Information Point

Adjust the length of the rod to change the relationship between the thrust lever position and the resolver position.



DMC-B737T-A-76-00-00-07A-110A-T_001-00
ICN-B737T-A-760000-A-81205-00014-A-002-01

ENGINE CONTROLS -- THRUST LEVER

02-Dec-2020

76-00-00

ENGINE CONTROLS -- THRUST LEVER ANGLE RESOLVER

General

The thrust lever angle resolvers change the mechanical thrust lever position to an analog position signal. Excitation for the resolvers comes from the electronic engine controls (EECs). The analog position signal goes to the EECs. The EECs use these signals to control the engine thrust.

Thrust Lever Angle Resolvers

There are two resolver assemblies in the auto throttle assembly, one for each engine. Each resolver assembly has a clutch and a resolver. Each resolver sends two thrust lever resolver angle (TRA) signals to the EECs, one for EEC channel A and one for EEC channel B. These components are on the same shaft but turn independently:

- ASM and gearbox assemblies (2)
- Resolvers (2)
- Clutch assemblies (2).

Functional Description

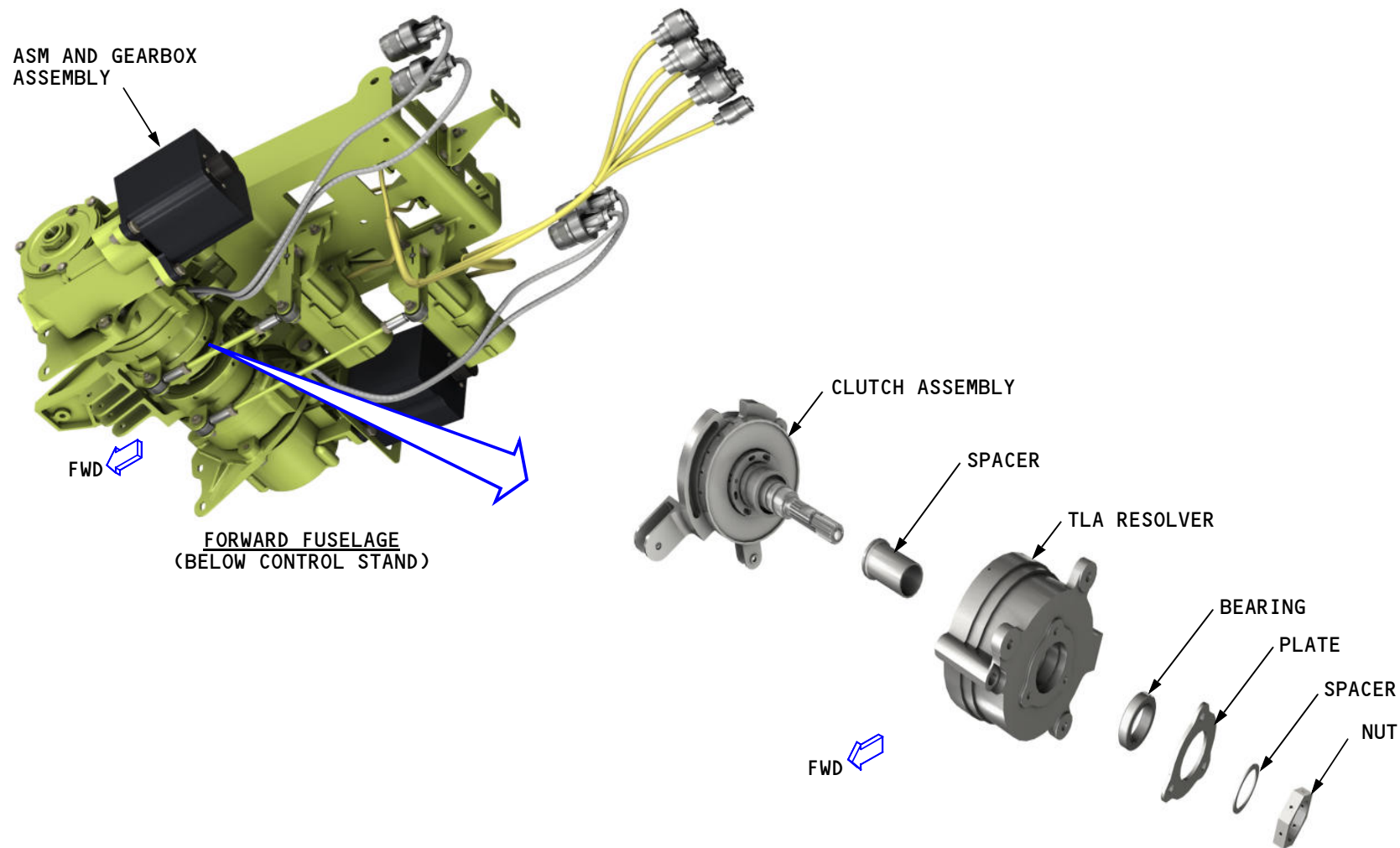
The thrust lever angle resolvers get ac excitation signals from the EECs. Each resolver sends two analog TRA signals back to the EECs, one for EEC channel A and one for EEC channel B. The EECs monitor the TRA signals. If the TRA signals for one of the engines do not agree or if one of the signals go out of range, the EEC stores a BITE fault in EEC memory.

Thrust lever angle (TLA) is the thrust lever position on top of the control stand. Reverse thrust lever angle (RLA) is the position of the reverse lever when it is pulled up. Thrust lever resolver angle (TRA) is the analog signal that comes from the resolver. TRA and the TLA are not the same. The table shows different thrust settings and the related TLA, RLA, and TRA.

TRA to TLA and RLA Relationship			
Description	TLA	RLA	TRA
Out of range low	0.0	-	< 3.0
Aft Mechanical stop	0.0	109.0	6.0
Maximum Reverse Thrust	0.0	103.8	8.0
Reverse part-power (number 2 detent)	0.0	97.8	10.3
Reverse Idle	0.0	62.2	24.0
Thrust reverser interlock block	0.0	56.5	26.0
Idle stop rig point	0.0	0.0	36.0
Idle flat upper end	2.4	0.0	38.0
Maximum climb thrust	44.0	0.0	72.0
Maximum takeoff thrust	52.0	0.0	78.0
Maximum forward thrust	58.3	0.0	82.5
Forward mechanical stop	60.6	0.0	84.0
Out of range high	> 65.4	0.0	> 87.0

Training Information Point

A special spanner wrench is necessary to replace the thrust lever angle resolver.



DMC-B737T-A-76-00-00-08A-110A-T_002-00
ICN-B737T-A-760000-A-81205-00015-A-001-01

ENGINE CONTROLS -- THRUST LEVER ANGLE RESOLVER

76-00-00

02-Dec-2020

ENGINE CONTROLS -- START LEVER

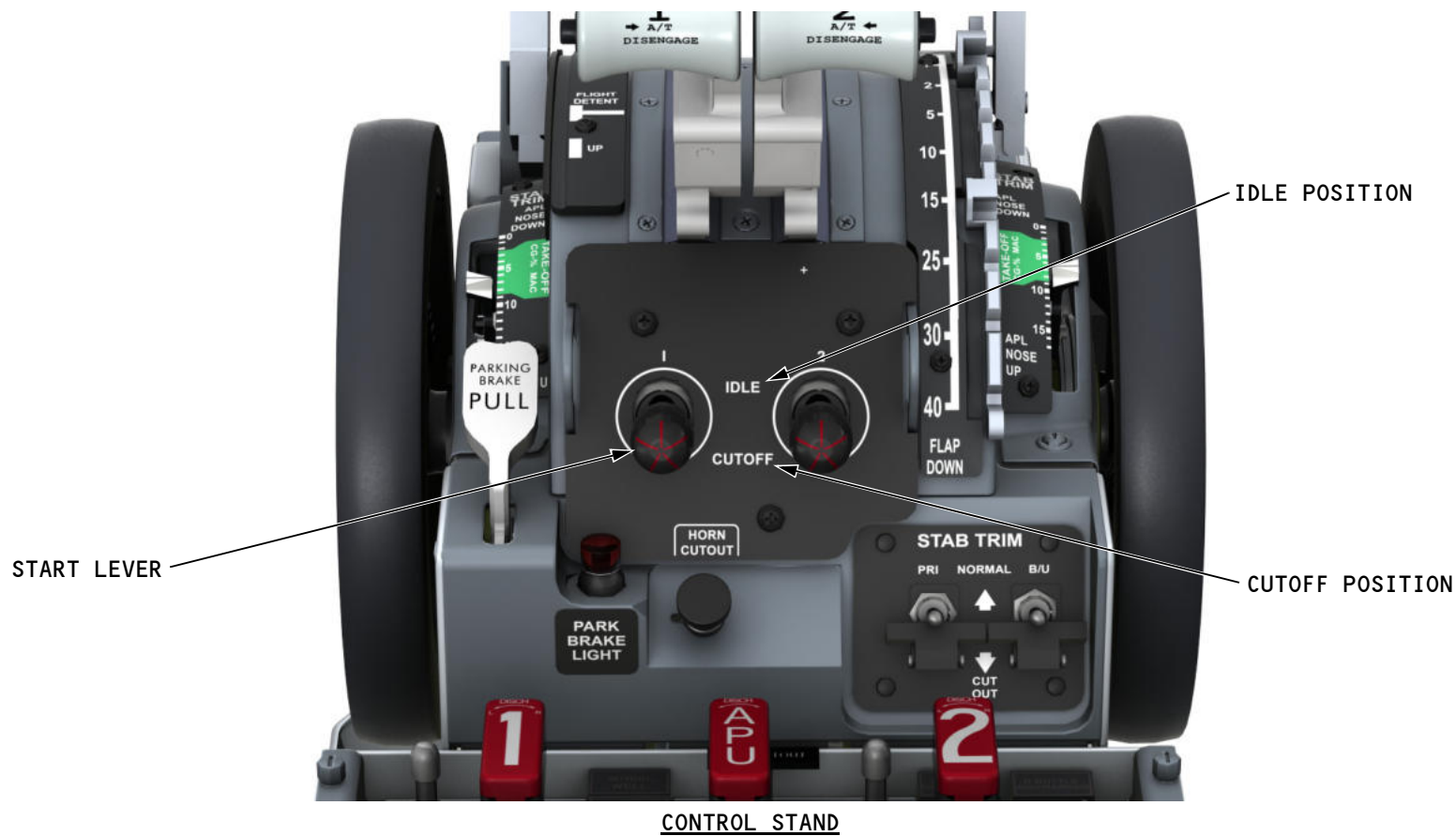
General

The start lever must be raised during an engine start sequence. The start lever is lowered to shut down the engine. The start lever sends lever position signals to interfacing systems.

Start Lever

There are two start levers, one for each engine. The start lever has two positions, IDLE and CUTOFF. Pull the lever out of the detent to move it from one position to the other.

Each start lever operates 6 internal switches. Two of the switches send signals to the EEC channels A and B. Two of the switches interface with the engine ignition system. The other two switches send signals to valves in the engine fuel feed system.



ENGINE CONTROLS -- START LEVER

76-00-00

DMC-B737T-A-76-00-00-09A-110A-T_001-00
ICN-B737T-A-760000-A-81205-00016-A-002-01

02-Dec-2020

ENGINE CONTROLS -- START LEVER - FUNCTIONAL DESCRIPTION

Functional Description

Each engine start lever contains six internal switches. The operation of these switches send signals to other systems.

These actions happen when the start lever is moved to the IDLE position and the internal switches move to the IDLE position:

- Fuel control panel receives start lever position for indication logic
- The engine fuel spar valve opens
- Ignition power (115v ac) goes to the EECs
- Two engine start lever relays move to the IDLE position
- Integrated drive generator (IDG) manual disconnect circuit arms
- Idle (engine run) position is sent to the digital flight data acquisition unit (DFDAU)
- Idle (engine run) position is sent to the DPCs.

These actions happen when the start lever is moved to the CUTOFF position and the internal switches move to the CUTOFF position:

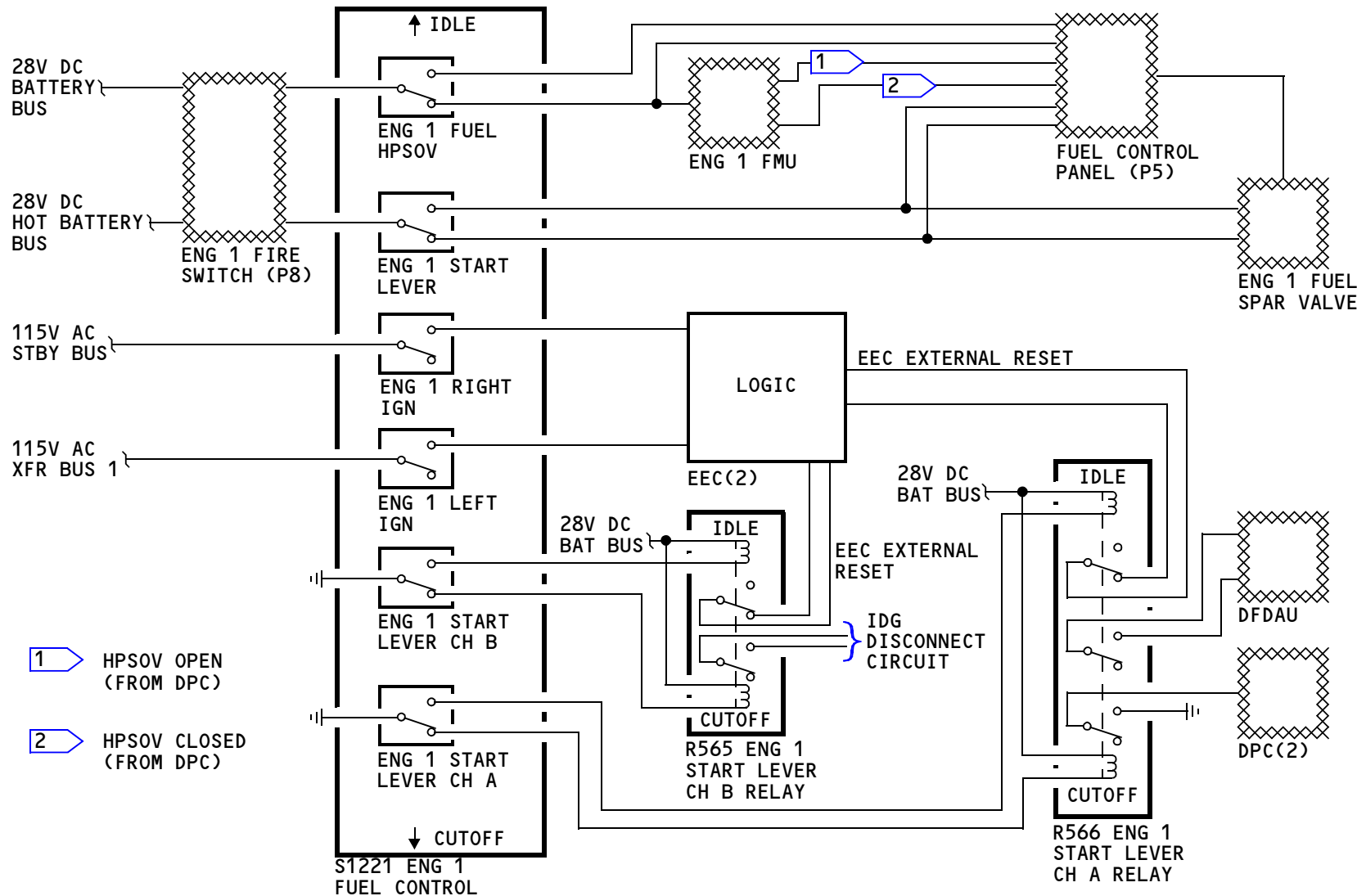
- Fuel control panel receives start lever position for indication logic
- The engine fuel spar valve closes
- Ignition power is removed from the EECs
- Two engine start lever relays move to the cutoff position
- The high pressure shutoff valve (HPSOV) in the fuel metering unit (FMU) closes
- EEC channels A and B reset.

NOTE: The EEC reset feature allows EEC channels A & B to alternate between active and standby modes of operation.

NOTE: The fuel control panel logic monitors high pressure shutoff valve (HPSOV) position in the fuel metering unit (FMU), and the ENG 1 FUEL HPSOV switch position for operation of the Engine Valve Closed light.

NOTE: The fuel control panel logic monitors the fuel spar valve position and the ENG 1 START LEVER switch position for operation of the Spar Valve Closed light.

DMC-B737T-A-76-00-00-02A-042A-T_004-00



ENGINE CONTROLS -- START LEVER - FUNCTIONAL DESCRIPTION

76-00-00

ENGINE CONTROLS -- REVERSE THRUST INTERLOCK SOLENOIDS

General

The reverse thrust interlock solenoid and interlock latch prevent reverse thrust lever movement beyond the reverse idle position, until the T/R sleeves have deployed. When the sleeves have deployed, the solenoid and latch operate to permit movement of the reverse thrust lever beyond the reverse idle position.

Reverse Thrust Interlock Solenoids

There are two reverse thrust interlock solenoids, one for each thrust lever assembly. They are a rotary type solenoid. Each reverse thrust interlock solenoid uses a rod to operate a latch. When the reverse thrust lever is moved to the deploy position, the latch contacts a stop on the brake mechanism. This keeps the engine at idle speed and allows the reverser sleeves to move to the deployed position. Electronic engine control (EEC) channel A energizes the reverse thrust interlock solenoid, which disengages the interlock latch. When the latch disengages, the reverse thrust lever can then be operated in the direction of full reverse thrust.

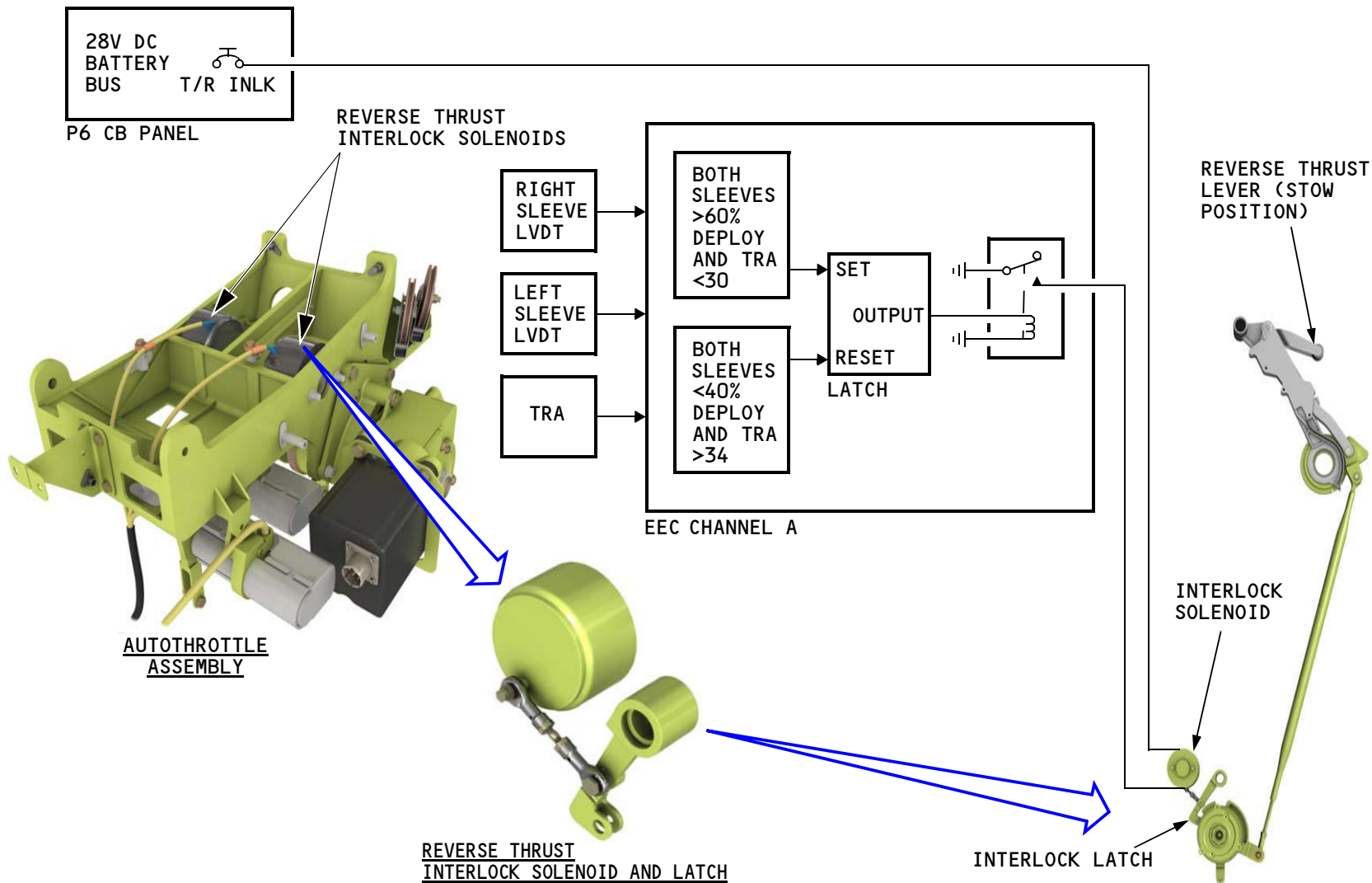
Functional Description

Each interlock solenoid is connected to EEC channel A for the respective engine. EEC channel A receives the T/R translating sleeve position from the linear variable differential transformer (LVDT) of each sleeve. When both sleeves are more than 60% deployed, EEC channel A energizes the interlock solenoid, and the interlock latch retracts. The reverse thrust lever can now move past the idle deploy position to increase reverse thrust. When the thrust lever is stowed and the sleeves have retracted past 40% deployed, EEC channel A deenergizes the interlock solenoid and the interlock latch moves back to the normal position.

DMC-B737T-A-76-00-00-10A-110A-T_003-00

02-Dec-2020

76-00-00



ENGINE CONTROLS -- REVERSE THRUST INTERLOCK SOLENOIDS

76-00-00



ENGINE FUEL AND CONTROL -- INTRODUCTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE FUEL AND CONTROL -- INTRODUCTION

General

The engine fuel and control system gives the quantity of fuel necessary to make the commanded thrust. It meters the fuel and injects it into the combustor. The system also sends the necessary fuel pressure to operate the engine air system.

The engine fuel and control system has three subsystems:

- Fuel distribution
- Fuel control
- Fuel indication.

Abbreviations And Acronyms

ACS	-	air conditioning system
ADIRU	-	air data inertial reference unit
AGB	-	accessory gearbox
ASM	-	autothrottle servo-motor
A/T	-	autothrottle
BAI	-	booster anti-ice
BET	-	Boeing equivalent thrust
BITE	-	built-in test equipment
BV	-	bypass valve
CCDL	-	cross channel data link
CDS	-	common display system
CDU	-	control display unit
CONT	-	continuous
DEP	-	data entry plug
ΔP	-	differential pressure
DFDAU	-	digital flight data acquisition unit
DPC	-	display processing computer
DU	-	display unit
EAI	-	engine anti-ice
EEC	-	electronic engine control

EECS	-	electronic engine control system
EGT	-	exhaust gas temperature
EHSV	-	electro-hydraulic servo valve
EOS	-	electronic overspeed
EPCS	-	electronics propulsion control system
FADEC	-	full authority digital electronic control
FCC	-	flight control computer
FF	-	fuel flow
FIM	-	fault isolation manual
FMC	-	flight management computer
FMU	-	fuel metering unit
FMV	-	fuel metering valve
FOD	-	foreign object debris
FWD	-	forward
GRD	-	ground
HP	-	high pressure
HPC	-	high pressure compressor
HPSOV	-	high pressure shutoff valve
HPT	-	high pressure turbine
HPTACC	-	high pressure turbine active clearance control
IDG	-	integrated drive generator
I/O	-	input/output
J	-	junction
LG	-	landing gear
LP	-	low pressure
LPT	-	low pressure turbine
LPTACC	-	low pressure turbine active clearance control
LRU	-	line replaceable unit
LVDT	-	linear variable differential transformer
mA	-	milliamp
MDS	-	max display system
MEL	-	minimum equipment list
MHX	-	main fuel/oil heat exchanger

DMC-B737T-A-73-00-00-02A-018A-T_006-00

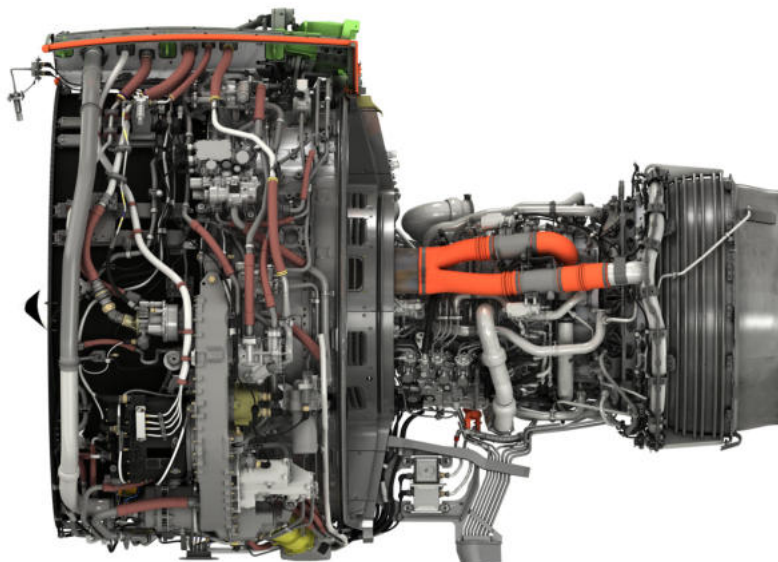
ENGINE FUEL AND CONTROL -- INTRODUCTION

MLG	- main landing gear	TV	- throttle valve
MTC	- modulated turbine cooling	VBV	- variable bleed valve
N1	- low pressure compressor rotor speed	VSV	- variable stator vane
N2	- high pressure compressor rotor speed		
NFS	- network file server		
OMF	- onboard maintenance function		
ONS	- onboard networking system		
PDL	- portable data loader		
PMA	- permanent magnet alternator		
PMV	- pilot metering valve		
P0	- ambient static pressure		
PPME	- pilot primary main enriched		
PPMNE	- pilot primary main non-enriched		
PR	- pressure regulator		
PSEC	- pilot secondary		
PSS	- pressure sub-system		
PT	- total pressure		
RVDT	- rotary variable differential transducer		
SAC	- single annular combustor		
SAV	- starter air valve		
SB/BAI	- start bleed/booster anti-ice		
SCU	- split control unit		
SFH	- servo fuel heater		
TAPS	- twin annular premix swirler		
TAT	- total air temperature		
TBV	- transient bleed valve		
TCMA	- thrust control malfunction accommodation		
TGB	- transfer gearbox		
T/L	- thrust lever		
TLA	- thrust lever angle		
TLR	- thrust lever angle resolver		
T/R	- thrust reverser		
TRA	- thrust resolver angle		

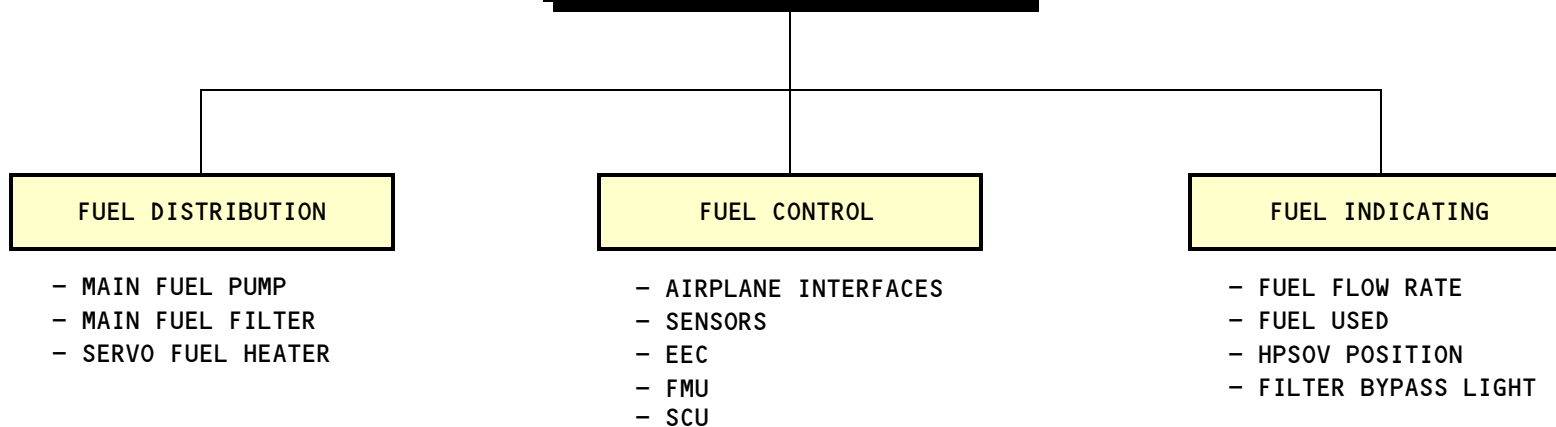
DMC-B737T-A-73-00-00-02A-018A-T_006-00

02-Dec-2020

73-00-00



ENGINE FUEL AND CONTROL SYSTEM



ENGINE FUEL AND CONTROL -- INTRODUCTION

73-00-00



ENGINE FUEL AND CONTROL -- GENERAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE FUEL AND CONTROL -- GENERAL DESCRIPTION

General

All engine fuel and control components are on the engine. The airplane fuel system supplies fuel to the engine fuel and control system. The airplane also gives and receives digital and analog control data to and from the engine fuel and control system. The engine fuel and control system uses this data to control the engine and give engine status to other airplane systems.

Electronic Engine Controls (EECs)

The EECs control the engine fuel and control system. There are two EECs, channel A and channel B. A different channel is in control for each flight. If the channel in control cannot control engine operation, the EECs switch to the other channel. The EECs use input data to calculate the engine fuel and control outputs to operate the engine.

Airplane Fuel System

The airplane fuel system supplies pressurized fuel from the center or main tank. The fuel goes from the tank through a boost pump and a spar valve before the fuel goes to the engine.

Engine Strut

The engine strut has signature pins that supply the airplane model data to the EECs. The EECs use this to find the maximum certified thrust for the airplane. This is different from the rated thrust. The maximum certified thrust is the thrust the engines can supply if the thrust levers are put against the forward stops. The rated thrust is the maximum autothrottle thrust. The strut signature pins also give the EECs the engine position.

Thrust Reverser System

The EECs get thrust reverser translating sleeve position from a linear variable differential transformer (LVDT). The EECs use this data to put a limit on engine thrust while the thrust reverser translating sleeves move. The EECs also set the engine to idle if it senses a thrust reverser sleeve is not fully closed in flight. The EECs send thrust reverser position to the display processor computers (DPCs) for thrust reverser indication on the display units (DUs).

Aisle Control Stand

The EECs use the thrust lever resolvers (TLRs) in the aisle stand to get thrust resolver angle (TRA). The EECs use this data to get the commanded engine thrust. The EEC also sends TRA to the flight control computers (FCCs).

The start levers and fire handle send signals directly to the HPSOV in the fuel metering unit (FMU). This lets the flight crew shut down the engine in normal or emergency situations.

Each start lever sends its position to the DPCs. The start levers also send ignition power to the EECs. The EECs operate the engine ignition system.

ENG VALVE CLOSED Light

The ENG VALVE CLOSED light shows the position of the HPSOV. The ENG VALVE CLOSED light comes on bright when the HPSOV is in transit or is not in the commanded position. This light is on dim when the HPSOV is closed and has a close command. The ENG VALVE CLOSED light is off when the HPSOV is open and has an open command.



ENGINE FUEL AND CONTROL -- GENERAL DESCRIPTION

Display Processing Computers (DPCs)

The EECs send and receive data from airplane systems and components through the two DPCs:

The EECs send input data from many of the engine sensors to the DPCs. The DPCs send some of this data to the display units (DUs). The engine indication display normally shows on one of the inboard DUs.

ENGINE CONTROL Light and EEC Switches

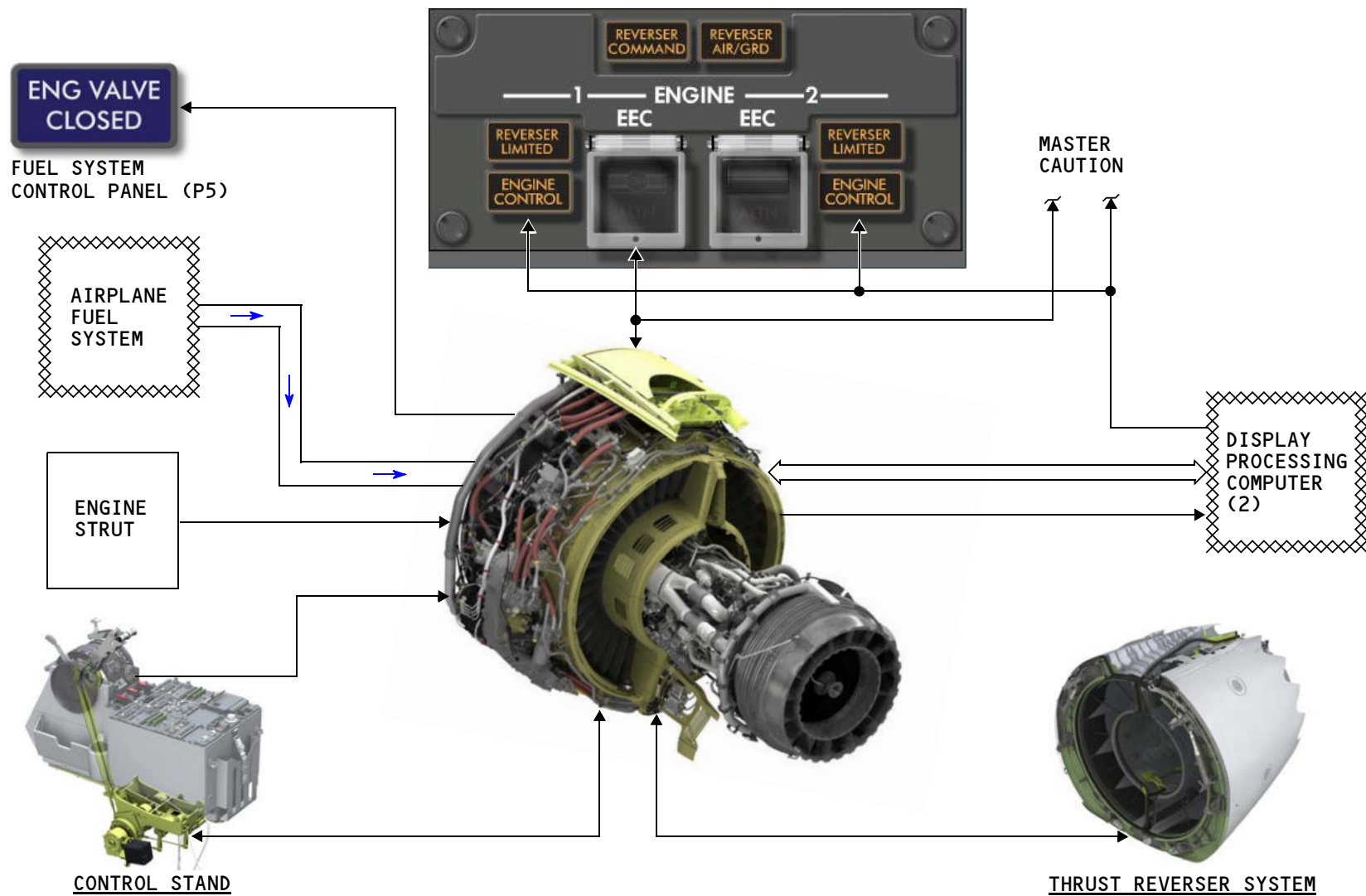
The EECs send a signal to the ENGINE CONTROL light on the engine control panel through the DPCs for some faults found by the EECs. The airplane can not be dispatched when the ENGINE CONTROL light comes on.

The EECs send a signal to the engine control panel when an EEC is in a soft or hard alternate mode of operation. The ALTN light is on when the EECs operate in an alternate mode. When the ENGINE CONTROL or ALTN light comes on, the MASTER CAUTION and ENG annunciator lights also come on.

DMC-B737T-A-73-00-00-02A-040A-T_004-00

02-Dec-2020

73-00-00



DMC-B737T-A-73-00-00-02A-040A-T_004-00
ICN-B737T-A-730000-A-81205-00004-A-003-01

ENGINE FUEL AND CONTROL -- GENERAL DESCRIPTION

73-00-00

02-Dec-2020

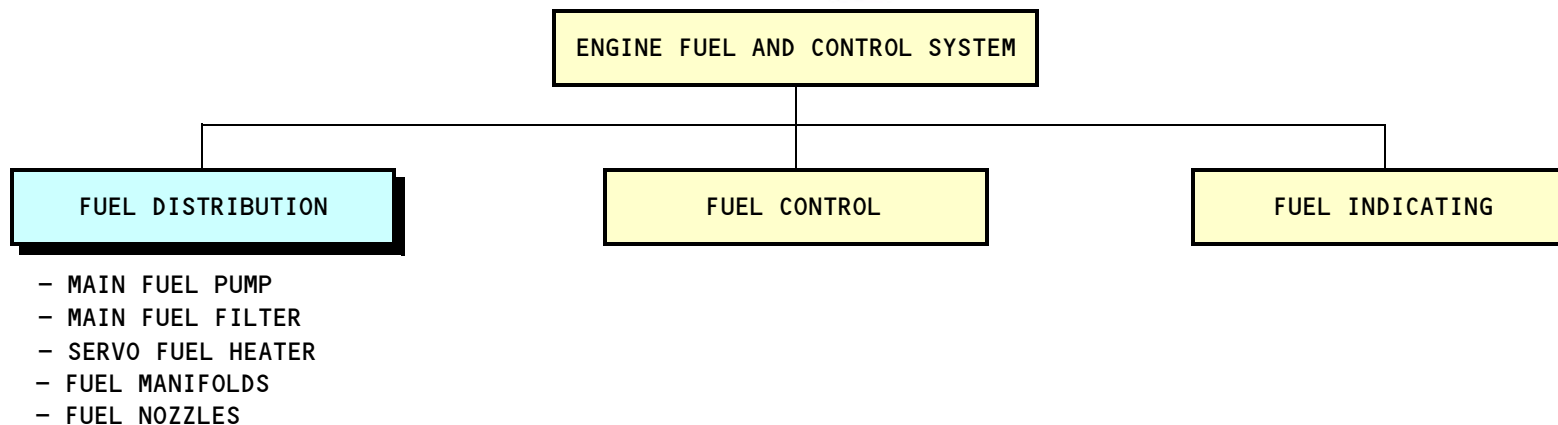
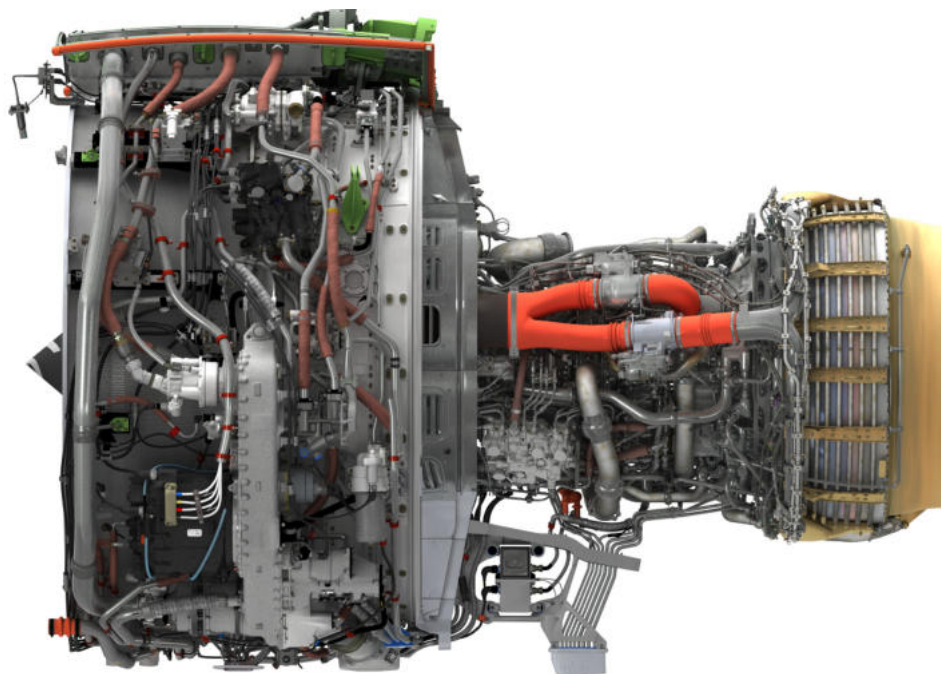
ENGINE FUEL AND CONTROL -- DISTRIBUTION - INTRODUCTION

Purpose

The airplane fuel system supplies fuel to the engine fuel distribution system. The engine fuel distribution system supplies fuel to the engine for combustion and servo system operation.

These are the major components of the engine fuel distribution system:

- Main fuel pump
- Main fuel filter
- Servo fuel heater
- Main fuel/oil heat exchanger
- Fuel manifolds
- Fuel nozzles.



ENGINE FUEL AND CONTROL -- DISTRIBUTION - INTRODUCTION

73-11-00

DMC-B737T-A-73-11-00-02A-018A-T_001-00
ICN-B737T-A-731100-A-81205-00007-A-002-01

02-Dec-2020

ENGINE FUEL AND CONTROL -- DISTRIBUTION - GENERAL DESCRIPTION

General Description of Fuel Flow

1. The main fuel pump receives fuel from the airplane fuel system.
2. The main fuel pump low pressure centrifugal pump section sends fuel to the fuel metering unit (FMU).
3. The main fuel pump always supplies more combustion fuel to the FMU than the FMU can use. The FMU adds this bypass fuel to the low pressure fuel and sends it to the main fuel/oil heat exchanger.
4. The intermediate pressure fuel (bypass and low pressure fuel) from the main fuel/oil heat exchanger returns to the fuel pump.
5. The main fuel pump high pressure gear stage section sends high pressure fuel to the fuel filter.
6. From the fuel filter, high pressure fuel goes to the FMU and the servo fuel heater.
7. High pressure fuel gets additional heat from engine oil in the servo fuel heater, then goes to the FMU and split control unit (SCU). Servo fuel goes from the SCU to operate the EEC servo controlled air system functions.
8. Metered (heated and filtered) fuel goes from the FMU to the fuel flow transmitter. The EECs control the metering valve in the FMU.
9. Metered fuel goes through the fuel flow transmitter to the SCU. The EECs control the splitting of the metered fuel into three manifolds in the SCU.
10. Each of the fuel nozzles gets two fuel manifold inputs from the SCU. The fuel nozzles supply atomized fuel to the engine combustion chamber.

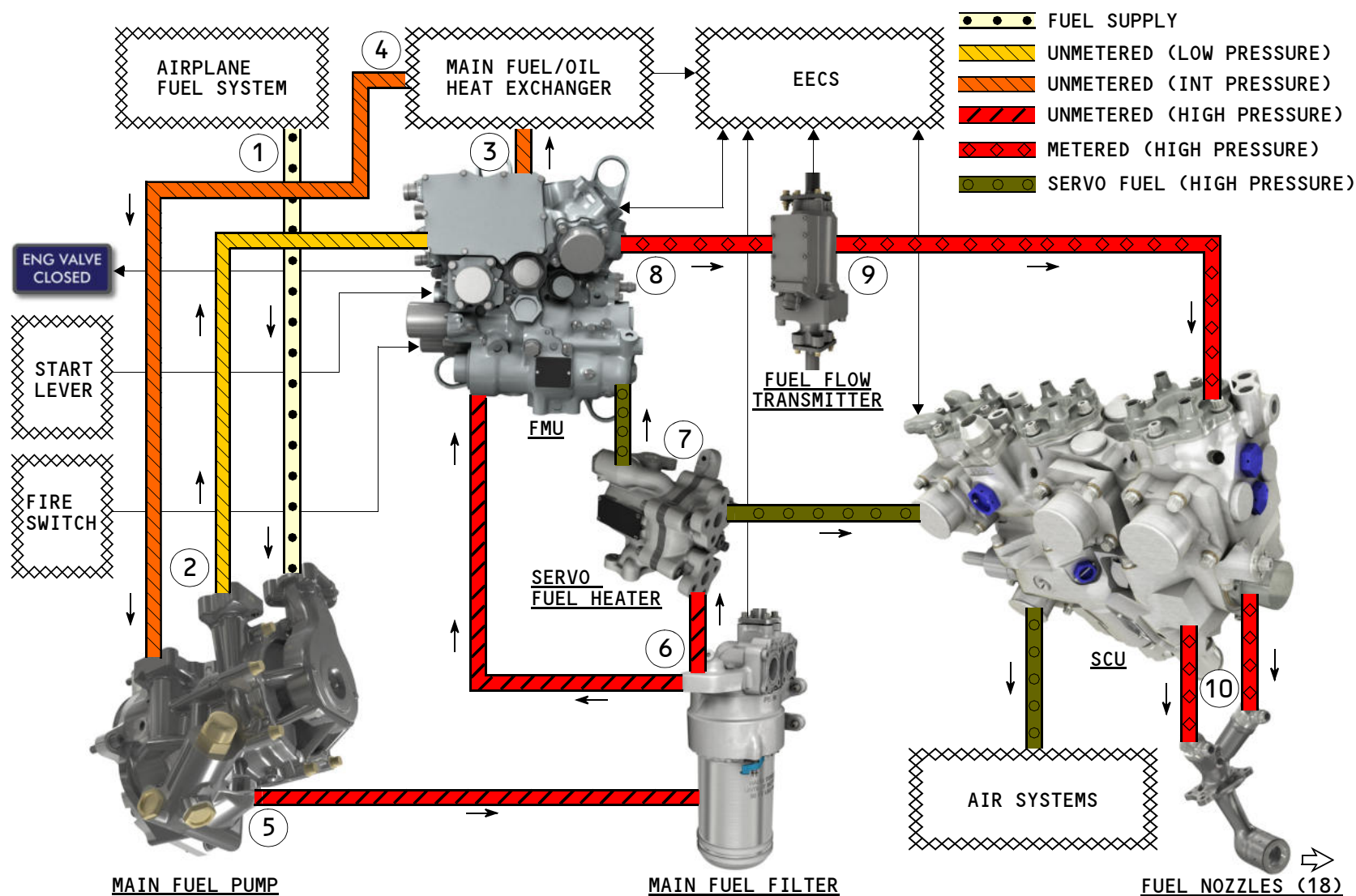
The start lever and fire switch in the flight compartment control the high pressure shutoff valve (HPSOV) in the FMU. The engine valve closed light shows the HPSOV position.

Differential pressure sensors on the FMU, main fuel filter, and main fuel/oil heat exchanger give bypass indications to the EECs.

DMC-B737T-A-73-11-00-02A-040A-T_001-00

02-Dec-2020

73-11-00



ENGINE FUEL AND CONTROL -- DISTRIBUTION - GENERAL DESCRIPTION

73-11-00

ENGINE FUEL AND CONTROL -- DISTRIBUTION - COMPONENT LOCATION

General

These are the engine fuel distribution system components:

- Main fuel pump
- Main fuel filter
- Servo fuel heater
- Main fuel/oil heat exchanger
- Fuel manifolds
- Fuel nozzles.

Component Locations

The main fuel pump is on the aft side of the Accessory Gearbox (AGB) at the 8 o'clock position. It is secured with bolts.

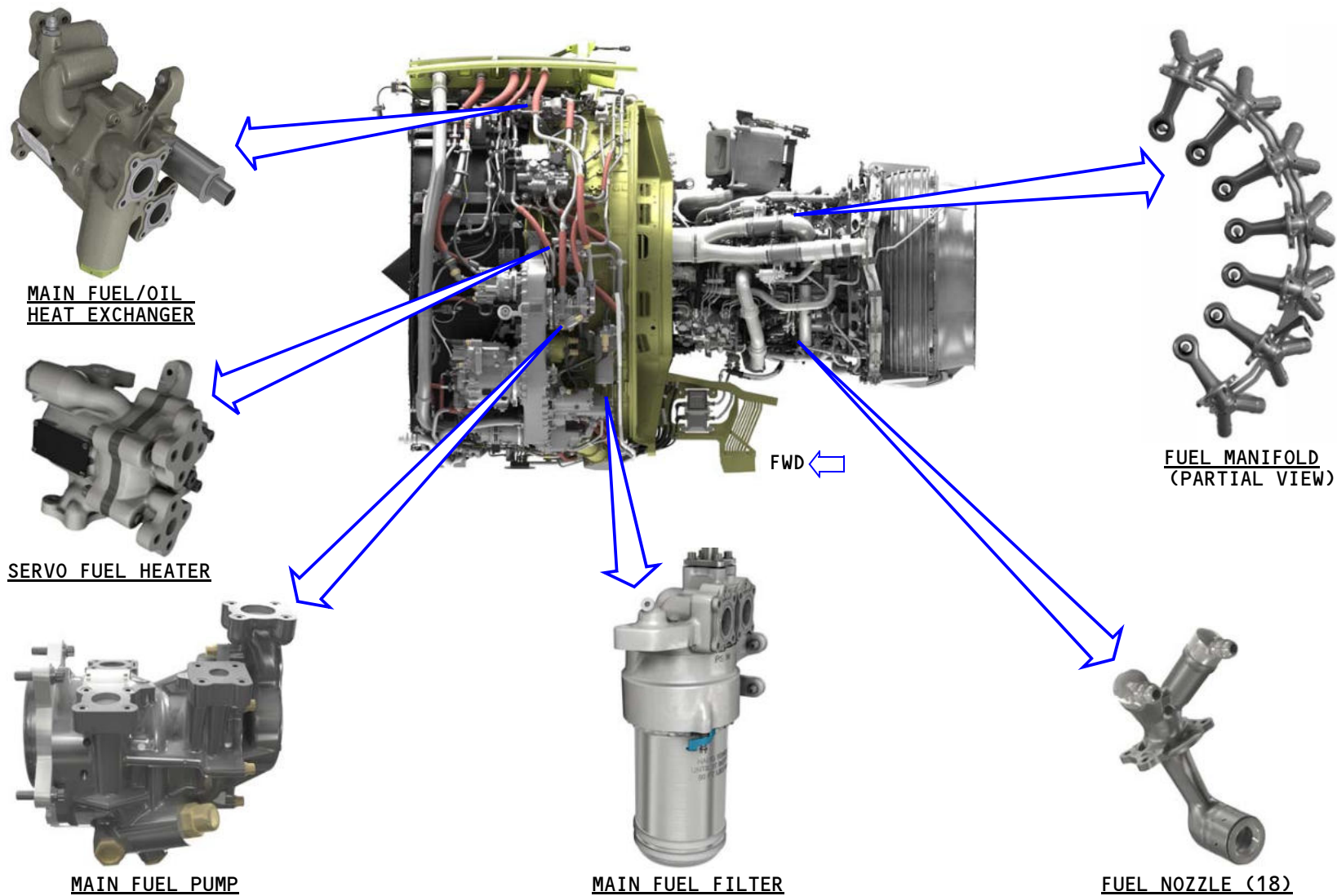
The main fuel filter assembly is on the air inlet section at 8 o'clock, just behind the permanent magnet alternator (PMA). The main fuel filter contains the filter housing and the fuel filter cartridge.

The servo fuel heater is on the air inlet section at the 9 o'clock position, just aft of the AGB.

The main fuel/oil heat exchanger is on the air inlet section at the 11 o'clock position, just below the engine strut.

The three piece fuel manifold goes around the outside of the forward section of the combustor case.

The 18 fuel nozzles are equally distributed around the combustion section.



DMC-B737T-A-73-11-00-02A-055A-T_003-00
ICN-B737T-A-731100-A-81205-00009-A-003-01

ENGINE FUEL AND CONTROL -- DISTRIBUTION - COMPONENT LOCATION

73-11-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- DISTRIBUTION - MAIN FUEL PUMP

General

The main fuel pump supplies pressurized fuel for engine operation. There is a low pressure fuel pump and a high pressure fuel pump inside the main fuel pump. The main fuel pump supplies high pressure fuel for the engine combustion system and the servo system.

Low Pressure Fuel Pump

The airplane fuel feed manifold usually supplies pressurized fuel to the low pressure pump. If the airplane cannot supply pressurized fuel, the fuel pump can do a suction feed from the related main fuel tank.

The low pressure fuel pump is a centrifugal impeller pump. This type of pump can operate at low fuel inlet pressure and with fuel that is part liquid and part vapor. Low pressure fuel goes from the low pressure pump outlet to the fuel metering unit (FMU) jet pump. The jet pump mixes the low pressure fuel with high pressure fuel not used by the engine. The name of this mixture is intermediate pressure fuel. Intermediate pressure fuel goes to the MHX. Intermediate pressure fuel has less pressure than high pressure fuel. The lower fuel pressure lets the MHX be lighter and give good operation.

High Pressure Fuel Pump

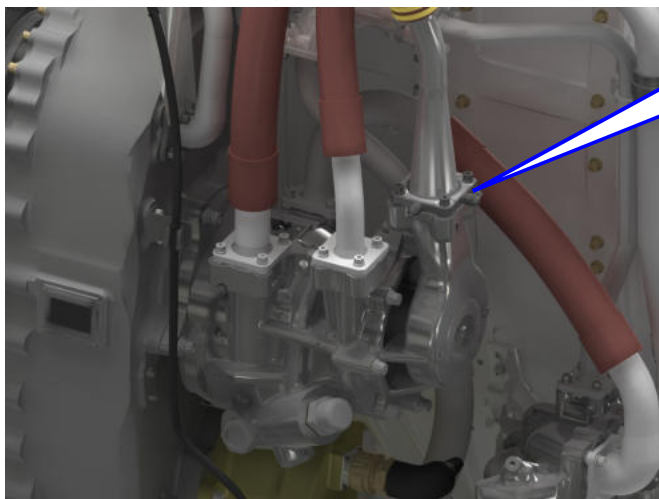
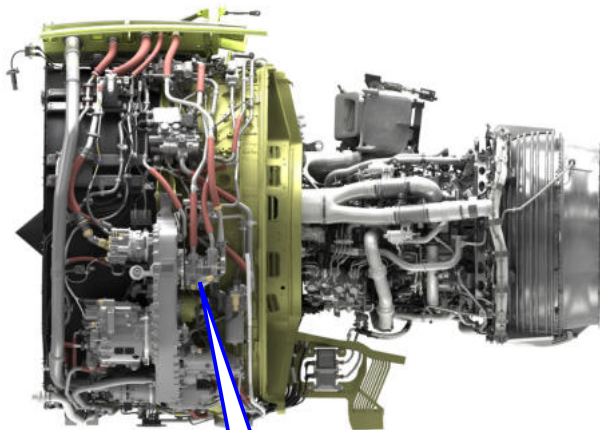
Intermediate pressure fuel goes from the main fuel/oil heat exchanger (MHX) to the high pressure pump inlet. The high pressure pump is a single element, positive displacement gear pump. This type of pump produces high fuel pressure. High pressure is necessary to give a strong combustor fuel spray pattern and to operate the engine actuators that are part of the servo system. High pressure fuel goes from the high pressure pump outlet to the main fuel filter.

The engine accessory gearbox (AGB) turns a drive shaft to operate the main fuel pump. The main fuel pump attaches to the aft side of the AGB with 5 bolts. The main fuel pump weighs 40.38 lbs / 18.3 kg.

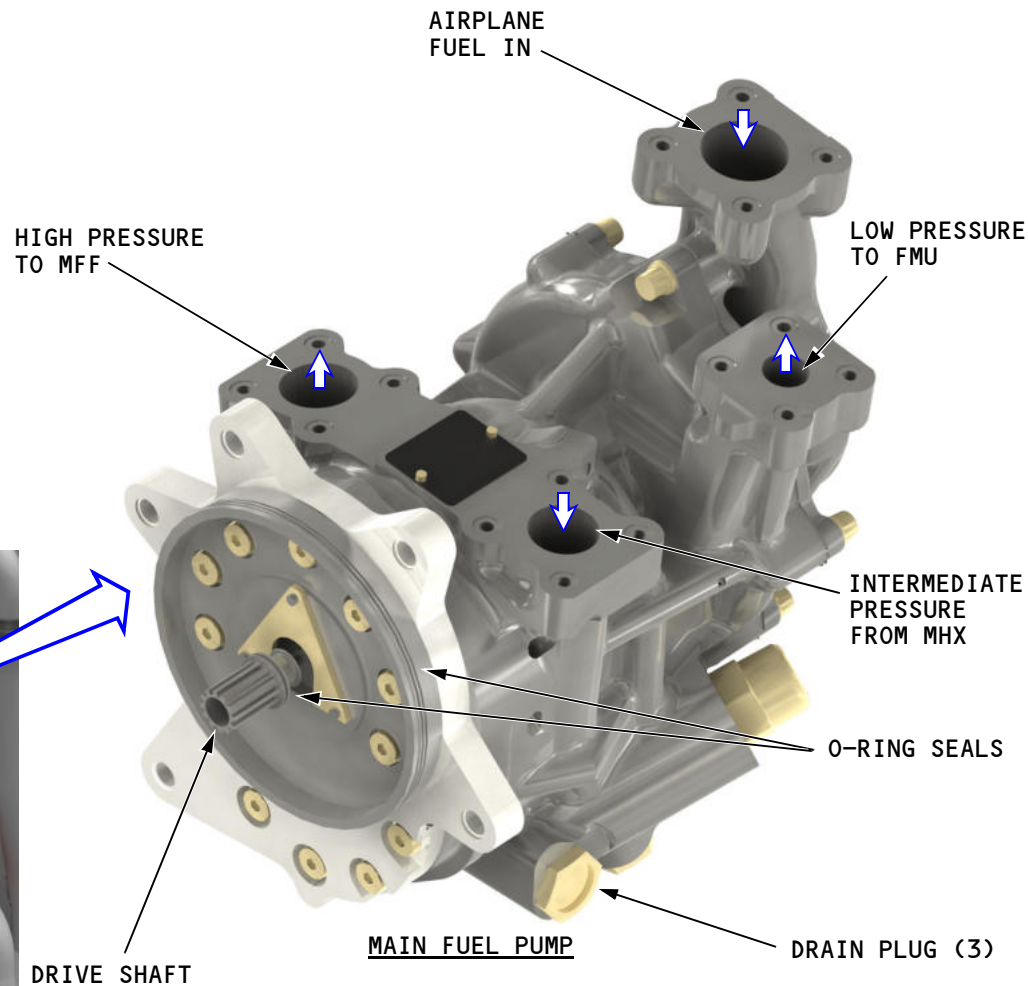
Training Information Point

During the removal of the main fuel pump, you must remove 3 drain plugs to drain remaining fuel.

DMC-B737T-A-73-11-00-03A-110A-T_003-00



MAIN FUEL PUMP (INSTALLED)



ENGINE FUEL AND CONTROL -- DISTRIBUTION - MAIN FUEL PUMP

73-11-00

DMC-B737T-A-73-11-00-03A-110A-T_003-00
ICN-B737T-A-731100-A-81205-00010-A-005-01

02-Dec-2020

ENGINE FUEL AND CONTROL -- DISTRIBUTION - MAIN FUEL/OIL HEAT EXCHANGER

General

The main fuel/oil heat exchanger (MHX) heats combustion fuel to prevent ice and cools the engine oil.

to the OPV outlet. OPV outlet oil goes to the oil dampened bearings. The oil bypass valve bypasses the heat exchanger core if it is blocked or plugged.

Location

The MHX is on the fan case at the 10 o'clock position. You open the left fan cowl to get access to the MHX.

Description

The MHX has these parts:

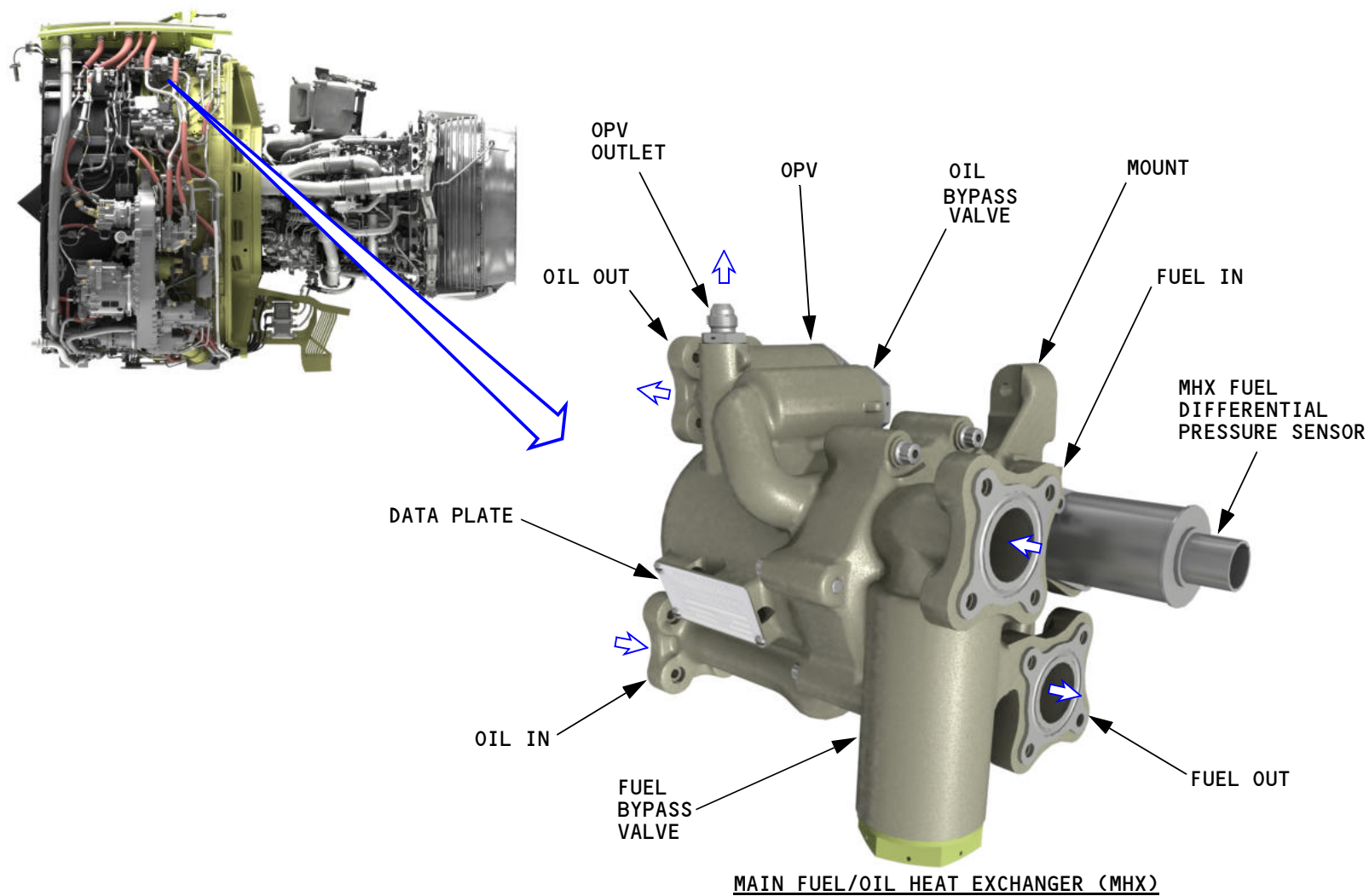
- Housing
- Data plate
- Mount
- Fuel and oil connections
- Exchanger core (inside the housing)
- Oil pressurizing valve (OPV)
- Oil bypass valve
- MHX differential pressure sensor
- Fuel bypass valve.

The MHX receives fuel from the fuel metering unit (FMU). The MHX heats the fuel, and then sends it to the high pressure pump in the main fuel pump. The fuel bypass valve bypasses the heat exchanger core if it is blocked or plugged.

The MHX differential pressure sensor measures the MHX fuel circuit for a bypass condition.

The surface air cooled oil coolers (SACOC) send engine oil to the MHX. The MHX cools the oil, and then sends most of this oil to the engine for lubrication. Some of the oil goes through the OPV and then

DMC-B737T-A-73-11-00-02A-034B-T_003-00



ENGINE FUEL AND CONTROL -- DISTRIBUTION - MAIN FUEL/OIL HEAT EXCHANGER

73-11-00

DMC-B737T-A-73-11-00-02A-034B-T_003-00
ICN-B737T-A-731100-A-81205-00050-A-001-01

02-Dec-2020

ENGINE FUEL AND CONTROL -- DISTRIBUTION - MAIN FUEL FILTER

General

The main fuel filter receives high pressure fuel from the main fuel pump. The main fuel filter divides the fuel into two circuits. The two circuits are combustion fuel and servo fuel. Combustion fuel goes to the fuel metering unit (FMU). Servo fuel goes to the servo fuel heater.

Main Fuel Filter

The main fuel filter has a line replaceable filter cartridge. To remove and replace the filter cartridge, first remove the drain plug to drain the fuel from the filter bowl. Then, push and hold the latch while you turn the fuel filter bowl. The latch has a spring to hold the latch against the fuel filter bowl teeth. This locks the fuel filter bowl to the main fuel filter housing. If necessary, put a wrench or socket on the hex on the bottom of the filter bowl. Remove and replace the preformed packing from the drain plug. A groove in the main fuel filter housing holds the 2 back-up rings and the packing. Remove and replace the back-up rings and packing. Remove and replace the fuel filter cartridge.

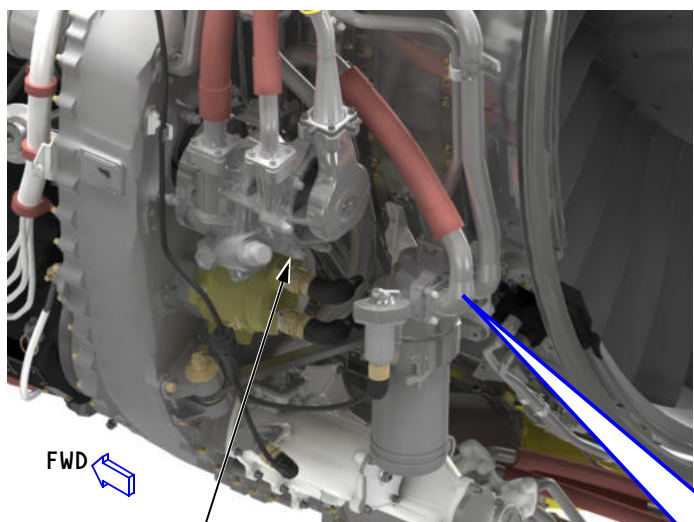
Installation is the opposite of the removal.

Always follow the procedures contained in approved maintenance documents.

DMC-B737T-A-73-11-00-02A-200A-T_003-00

02-Dec-2020

73-11-00



FWD

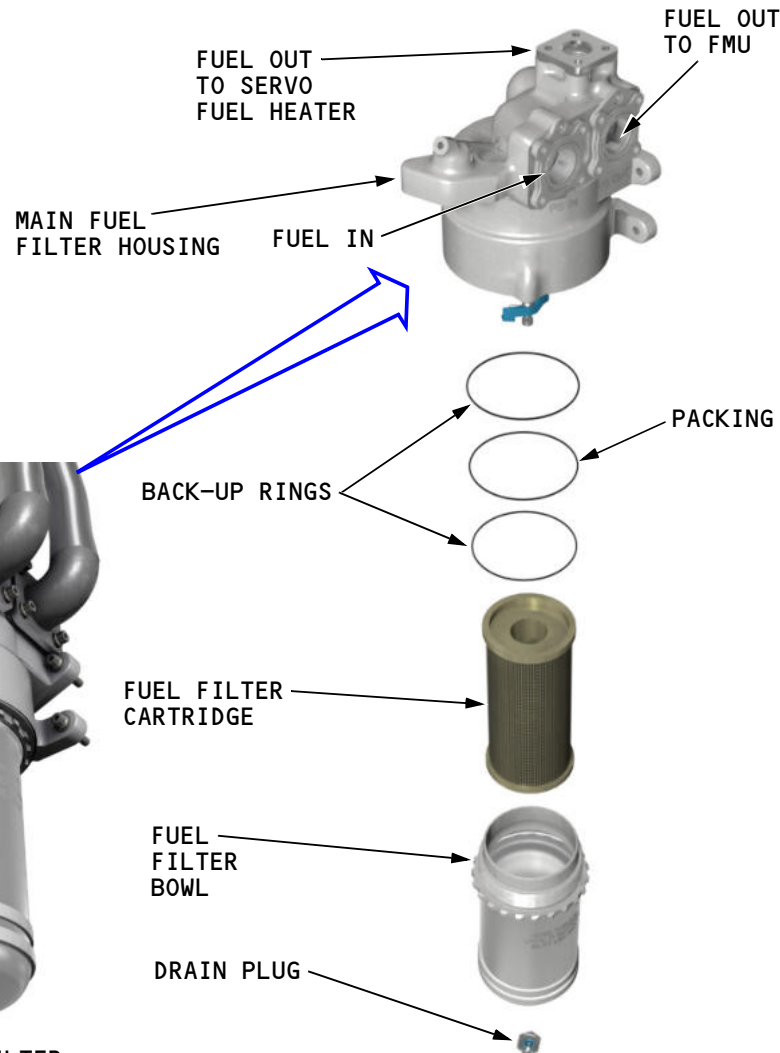
AGB AFT SIDE

MAIN FUEL PUMP
(REF)

FUEL FILTER
DIFFERENTIAL
PRESSURE SENSOR
(REF)

LATCH

MAIN FUEL FILTER



DMC-B737T-A-73-11-00-02A-200A-T_003-00
ICN-B737T-A-731100-A-81205-00043-A-003-01

ENGINE FUEL AND CONTROL -- DISTRIBUTION - MAIN FUEL FILTER

73-11-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- DISTRIBUTION - FUEL MANIFOLDS

General

The fuel manifolds supply fuel to the fuel nozzles. The EECS schedule the fuel flow between the two fuel manifolds. The EECS monitor engine parameters and calculate the fuel flow schedule.

Fuel Manifolds

The engine has 2 manifolds and each manifold has 3 separate sections.

The split control unit (SCU) supplies fuel to the 2 manifolds. The 2 fuel manifolds are pilot primary main and pilot secondary. The pilot primary main manifold is larger in diameter than the pilot secondary manifold.

The pilot primary main manifold has 2 different circuits, pilot primary main non-enriched (PPMNE) and pilot primary main enriched (PPME). The PPMNE circuit is only in the left and right manifold sections. The PPMNE circuit supplies fuel to 14 fuel nozzles. The PPME circuit is only in the bottom manifold section. The PPME circuit supplies fuel to 4 fuel nozzles, which are nearest to the igniter plugs. The PPME circuit gives more fuel to its fuel nozzles during the engine start to aid with the light-off.

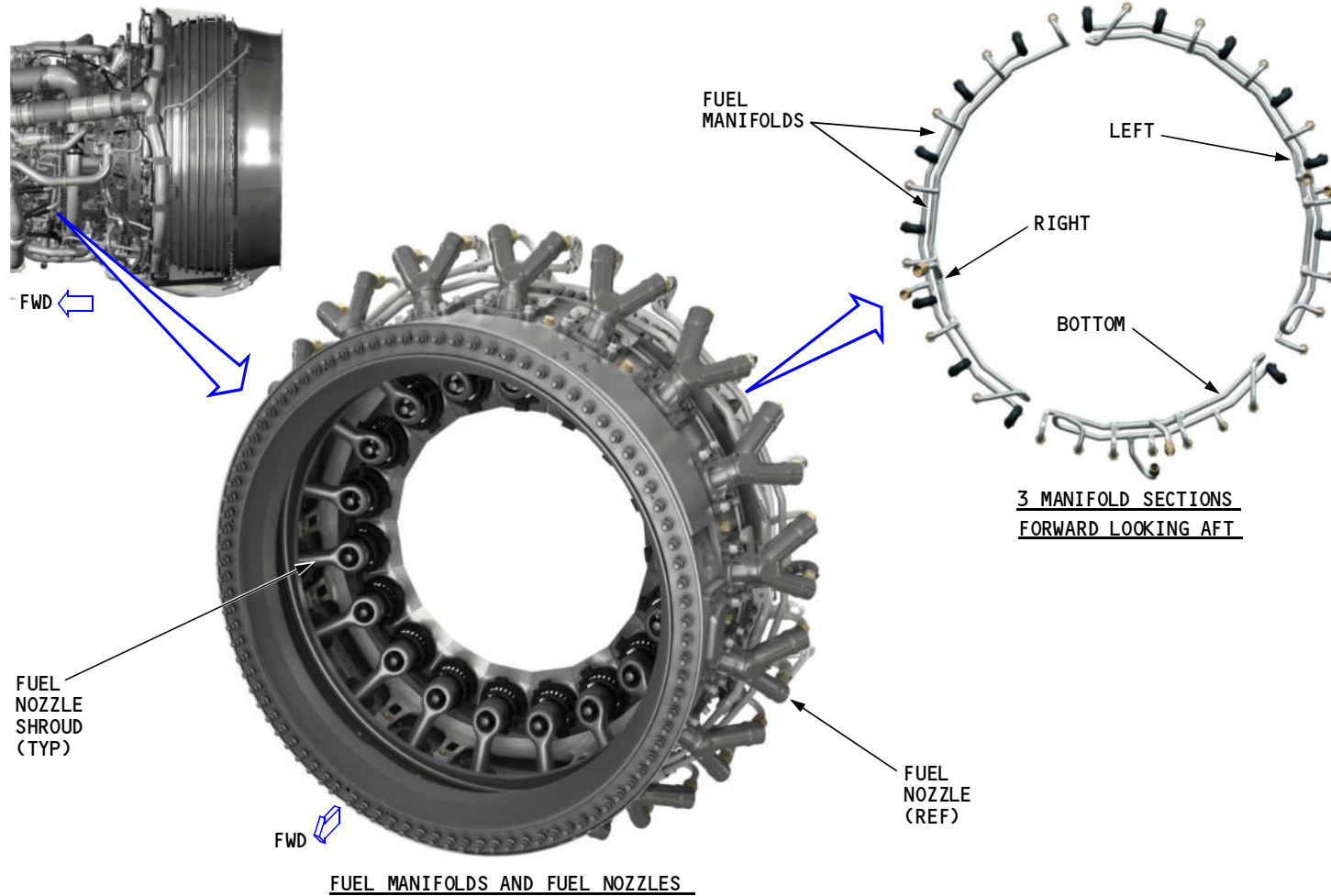
The SCU supplies the pilot secondary (PSEC) manifold in all 3 sections. The PSEC manifold supplies fuel to all 18 fuel nozzles.

Some areas of the fuel manifold sections are exposed to high temperatures. The PSEC fuel flow is low in some thrust configurations. Low fuel flow in a manifold exposed to high temperatures can cause coking. Coking is the deposit that forms when kerosene is exposed to high temperatures (pyrolysis).

DMC-B737T-A-73-11-00-04A-110A-T_002-00

02-Dec-2020

73-11-00



ENGINE FUEL AND CONTROL -- DISTRIBUTION - FUEL MANIFOLDS

73-11-00

DMC-B737T-A-73-11-00-04A-110A-T_002-00
ICN-B737T-A-731100-A-81205-00011-A-005-01

02-Dec-2020

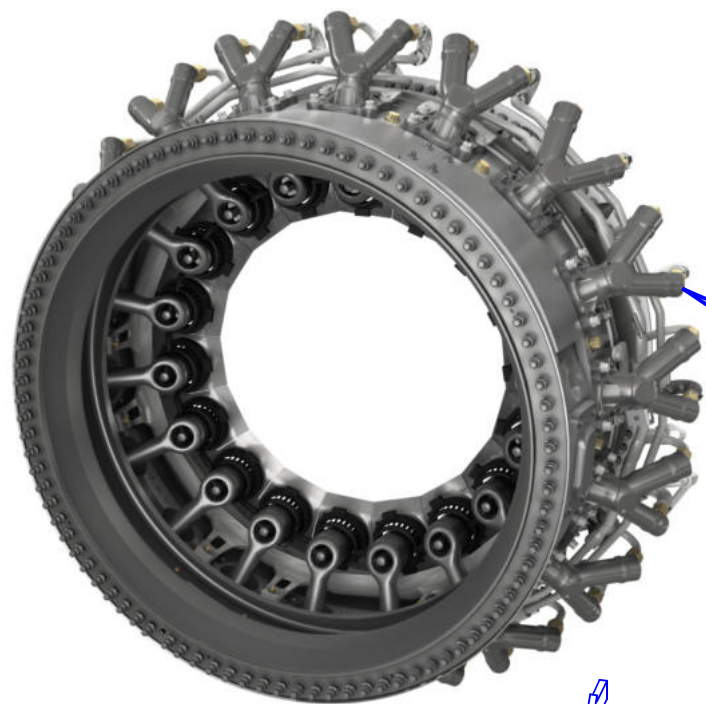
ENGINE FUEL AND CONTROL -- DISTRIBUTION - FUEL NOZZLES

General

The fuel manifolds supply fuel to the fuel nozzles. There are 18 fuel nozzles. The fuel nozzles send a spray of fuel into the combustion chamber.

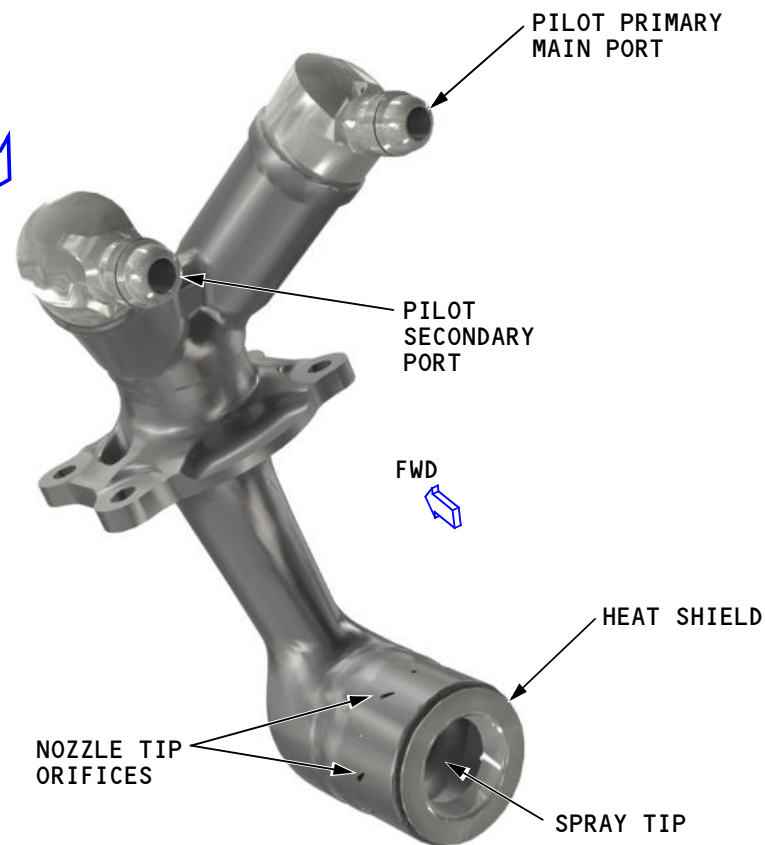
Fuel Nozzles

The fuel nozzles are a twin annular premixing swirler (TAPS) design. All of the fuel nozzles are the same part number. Each fuel nozzle receives pilot primary main and pilot secondary fuel flow. The heat shield gives protection to the fuel nozzle from heat in the combustion chamber. In the fuel nozzle, pilot primary main fuel separates into pilot primary and main fuel. The pilot primary fuel supplies the spray tip. Main fuel supplies the nozzle tip orifices. The spray tip fuel mixes with pilot swirler air flow and the nozzle tip orifices mix fuel with main premixer air flow.



ENGINE DIFFUSER SECTION

FWD



FUEL NOZZLE

ENGINE FUEL AND CONTROL -- DISTRIBUTION - FUEL NOZZLES

73-11-00

DMC-B737T-A-73-11-00-04A-040A-T_001-00
ICN-B737T-A-731100-A-81205-00045-A-001-01

02-Dec-2020



ENGINE FUEL AND CONTROL -- DISTRIBUTION - FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE FUEL AND CONTROL -- DISTRIBUTION - FUNCTIONAL DESCRIPTION

General

The main fuel pump gets fuel from the airplane fuel system(1). The fuel is pressurized (2), and heated (3 and 4). The fuel is then pressurized to high pressure and filtered (5). The fuel filter assembly divides the fuel into combustion fuel and servo fuel (6).

Combustion fuel goes to the fuel metering unit (FMU). The FMU sends metered fuel to the split control unit (SCU) through the fuel flow transmitter (8 and 9). The SCU supplies fuel to 18 fuel nozzles (10).

Servo fuel goes to the FMU and the SCU (7). The FMU servo fuel goes to the fuel metering valve (FMV) and the high pressure shutoff valve (HPSOV). The servo valves in the SCU supply pressurized fuel to the servo controlled air systems (chapter 75).

Fuel System

Fuel from the airplane goes to the main fuel pump. The main fuel pump has two pumps. There is a low pressure (LP) pump that has a centrifugal impeller. There is also a high pressure (HP) pump that uses two constant displacement gears.

Airplane fuel goes to the LP pump first. LP fuel then goes to the FMU jet pump. There is a bypass valve that sends combustion bypass fuel to the jet pump. The jet pump uses combustion bypass fuel to make an intermediate fuel pressure. Fuel from the jet pump goes through the FMU strainer. The strainer collects large debris to protect the high pressure gear pump.

The FMU supplies fuel to the main fuel/oil heat exchanger (MHX). If the MHX is blocked or plugged, the bypass valve lets the fuel bypass the MHX. The MHX has a differential pressure sensor.

The MHX supplies fuel to the HP pump in the main fuel pump. The main fuel pump has a bypass valve to protect the system from too much pressure.

The HP pump increases fuel pressure for servo system operation and for combustion. The HP pump supplies fuel to the main fuel filter, which cleans the fuel. The main fuel filter bypass valve opens when the fuel filter becomes blocked or plugged. The fuel filter differential pressure sensor monitors the condition of the main fuel filter.

After the main fuel filter, fuel goes thru the wash filter. The wash filter uses a tube shaped self-cleaning filter. The wash filter cleans the servo fuel when it goes through the side of the wash filter. Unfiltered combustion fuel flows through the center of the wash filter. Combustion fuel cleans the servo fuel contamination from the wash filter and sends it to the fuel nozzles. The wash filter has a bypass valve. The bypass valve opens when the wash filter becomes blocked.

Combustion Fuel

Combustion fuel goes to the FMU and the internal bypass valve (BYP VLV). The bypass valve supplies combustion fuel to the fuel metering valve (FMV). The FMV supplies fuel to the high pressure shutoff valve (HPSOV). The fuel system is designed to supply more fuel to the FMV than the engine will use. The bypass valve returns the excess fuel to HP pump through the jet pump. The FMV gets command signals from the EEC channel in control.

The FMU combustion fuel goes to the fuel flow meter, then to the split control unit (SCU).

The combustion components in the SCU are the pilot meter valve (PMV) and the throttle valve (TV).

ENGINE FUEL AND CONTROL -- DISTRIBUTION - FUNCTIONAL DESCRIPTION

The SCU supplies combustion fuel to the PMV. The PMV divides the combustion fuel into primary and secondary fuel. The PMV divides primary fuel into pilot primary main enriched (PPME) fuel and pilot primary main non-enriched (PPMNE) fuel. The PMV supplies secondary fuel to the TV. The TV regulates the pressure drop through the PSEC and ensures the correct fuel split. A servo in the SCU controls the PMV and the TV. A PPME pressure sensor sends fuel manifold pressure data to the EECs.

Servo Fuel

Servo fuel supplies the control pressure to move the valves in the engine air system (chapter 75).

The fuel filter assembly supplies servo fuel to the servo fuel heater (SFH). The servo fuel heater uses engine oil to heat the servo fuel. This makes sure that water in the fuel will not freeze in the servo system. The SFH supplies fuel to the SCU and the FMU.

The SCU supplies servo fuel to the electro-hydraulic servo valves (EHSV), then to the pressure regulator (PR), PMV, and TV.

The FMU divides the servo fuel into FMV and HPSOV servo fuel.

The FMU pressure regulator supplies pressure regulated servo fuel to move the FMV.

The HPSOV stops metered fuel flow when it closes. The control signal to close the HPSOV usually comes from the start lever. The fire handle switch can override the start lever to close the HPSOV. When the start lever and the fire switch give an open signal, the HPSOV opens with fuel pressure from the FMV.

Training Information Point

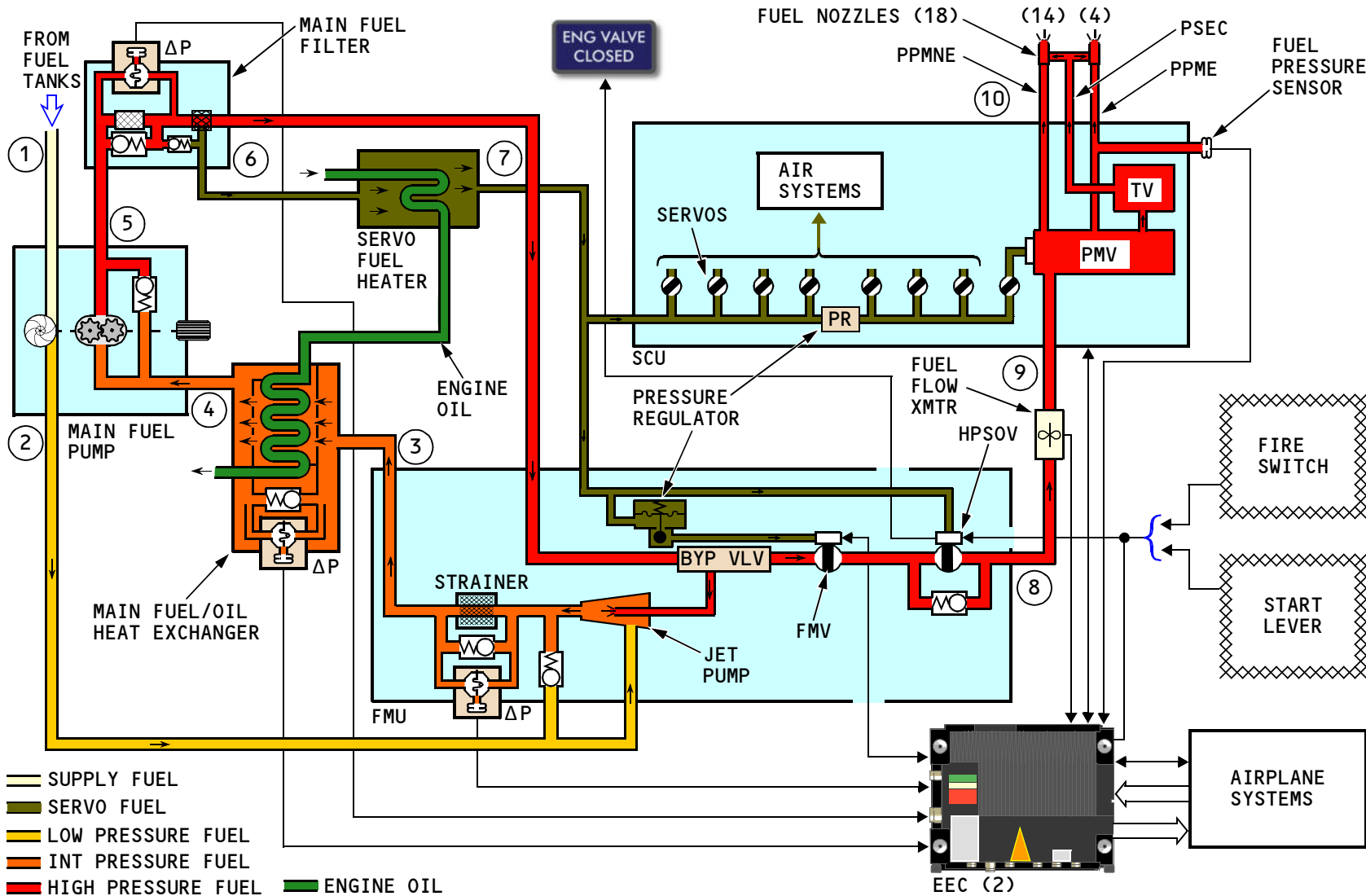
The EECs use the fuel flow transmitter signal for engine control when the FMV position signal is not valid.

The EECs control these engine protections:

- Electronic overspeed (EOS); and
- Thrust control malfunction accommodation (TCMA)

If an overspeed or TCMA condition is sensed, an EEC channel sends a signal to the HPSOV to close. One or the other channel of the EEC can send the signal to cause the shutdown. It can be the control channel or it can be the standby channel. When the HPSOV closes, fuel flow stops, and the engine stops.

DMC-B737T-A-73-11-00-02A-042A-T_005-00



ENGINE FUEL AND CONTROL -- DISTRIBUTION - FUNCTIONAL DESCRIPTION

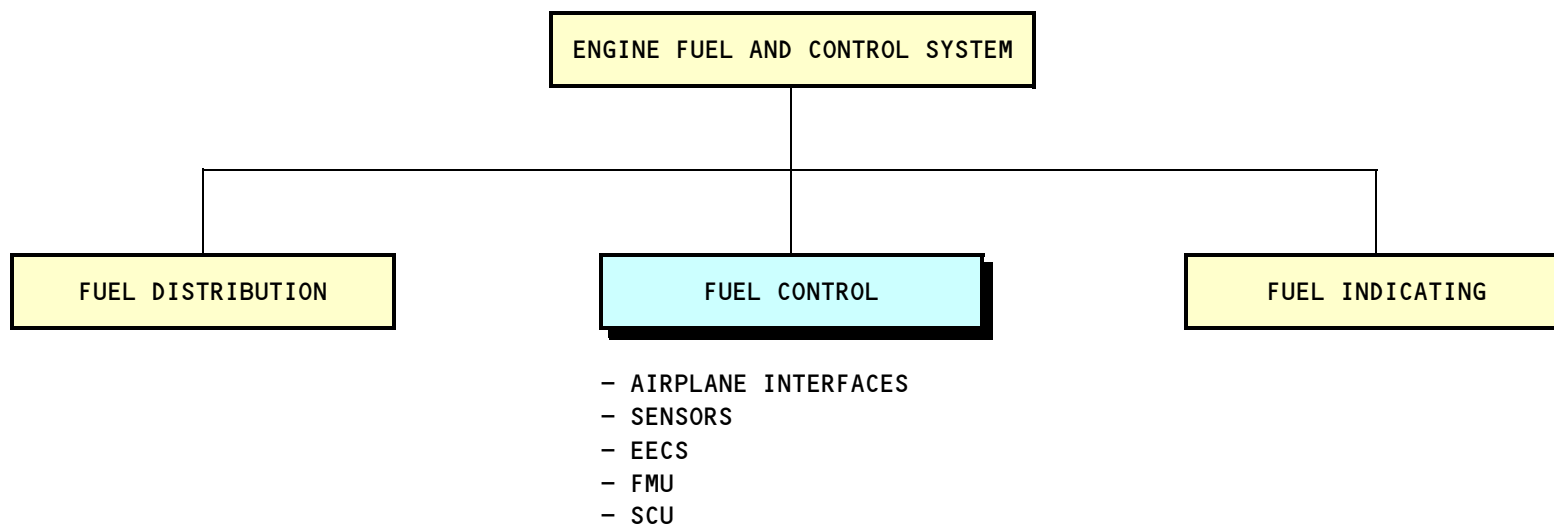
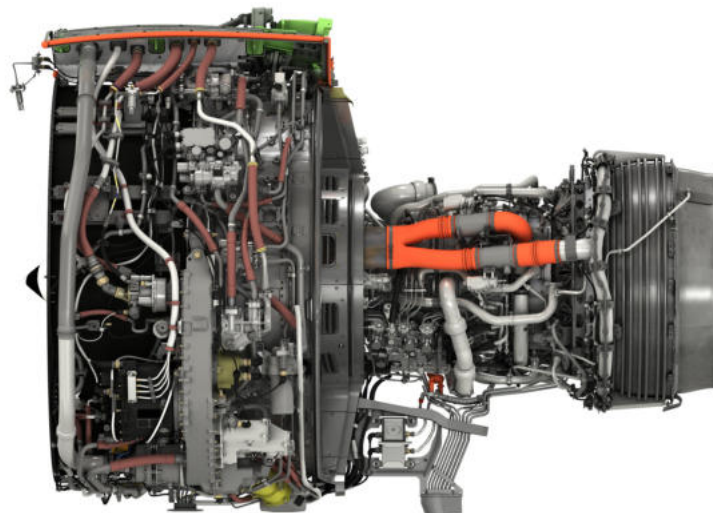
73-11-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - INTRODUCTION

General

The engine control system controls fuel flow for engine operation.
These are the major engine control system components:

- Airplane interfaces
- Sensors
- Electronic engine controls (EECs)
- Fuel metering unit (FMU)
- Split control unit (SCU).



DMC-B737T-A-73-21-00-02A-018A-T_002-00
ICN-B737T-A-732100-A-81205-00029-A-002-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - INTRODUCTION

73-21-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - GENERAL DESCRIPTION

Electronic Engine Controls (EECs)

The engine control system has two electronic engine controls (EECs); channel A and channel B. Each channel is the same part number. Only one of the channels controls the engine at a time. The channel in control normally switches to the other EEC during the next startup after each flight. The channels communicate with each other through a cross channel data link (CCDL). If the channel in control is unable to maintain a critical control system function, control switches to the other channel for the rest of the flight. Some control system functions, such as thrust reverser control, are only done by one of the EEC channels.

Power for the EECs usually comes from the permanent magnet alternator (PMA) on the accessory gearbox. The PMA has two isolated windings, one for each EEC channel. Power for each EEC can also come from the airplane. Airplane power also goes to the EECs for start, ignition, and thrust reverser control.

The EECs control these functions:

- Fuel flow to meet thrust lever angle resolver position (TRA)
- Command signals to the electro hydraulic servo valves (EHSV) in the split control unit (SCU) and fuel metering unit (FMU)
- Interface with engine sensors, engine control panel, and display processing computers (DPCs)
- Maintenance monitoring of all engine systems/sensors, and interface with the network file server (NFS).

TRA command

The EECs send excitation to, and receive angle position data from the thrust lever angle resolvers for engine thrust command. The resolvers have two isolated windings, one for each EEC channel.

The EECs use these and other sensors to calculate the proper fuel flow to meet the commanded thrust (TRA) :

- Inlet temperature (T12)
- Low pressure compressor discharge temperature (T25)
- High pressure compressor discharge temperature (T3).

Some engine pressures and the engine data entry plug data comes from the pressure sub-system (PSS).

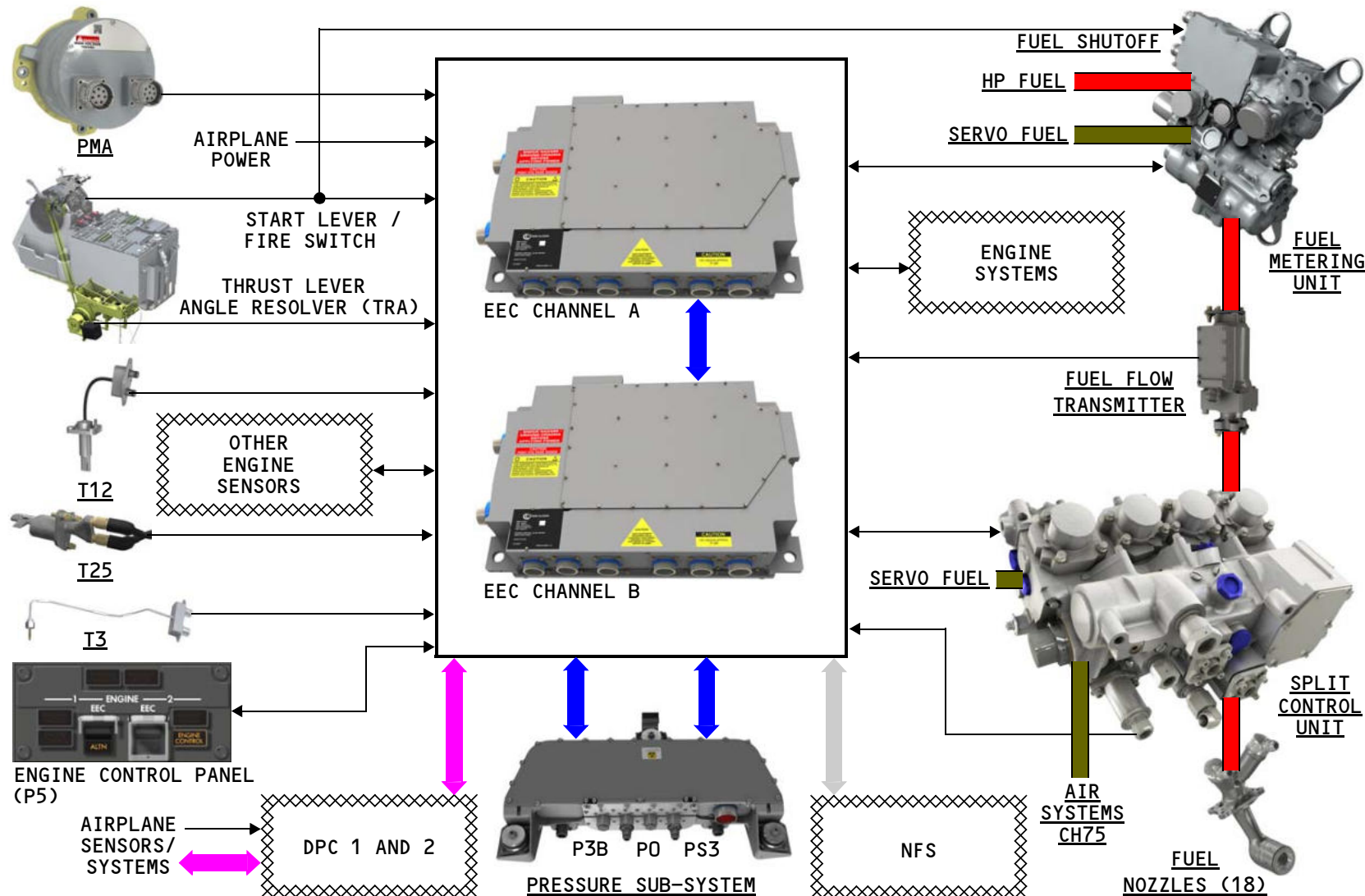
The EECs have normal and alternate modes of operation. If air data parameters from the DPCs become unavailable, the EECs go to a soft alternate mode. You push the guarded switch on the engine control panel to put the engines in the hard alternate mode for the rest of the flight, or before dispatch. In hard alternate mode, air data parameters come from stored data in the EECs.

FMU and SCU

The FMU has the high pressure shutoff valve (HPSOV) and the fuel metering valve. Flight compartment controls operate the HPSOV, the EEC channel in control operates the fuel metering valve. The channel in control sends signals to the servos in the SCU for air system and fuel control. Metered fuel comes from the fuel flow transmitter, and then gets split into two fuel manifolds feeding 18 fuel nozzles.

Display Processing Computer 1 and 2

The EECs receive data from many airplane systems through the display processing computers (DPCs). The DPCs get display and fault data from the EECs.



DMC-B737T-A-73-21-00-02A-040A-T_006-00
ICN-B737T-A-732100-A-81205-00030-A-005-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - GENERAL DESCRIPTION

73-21-00

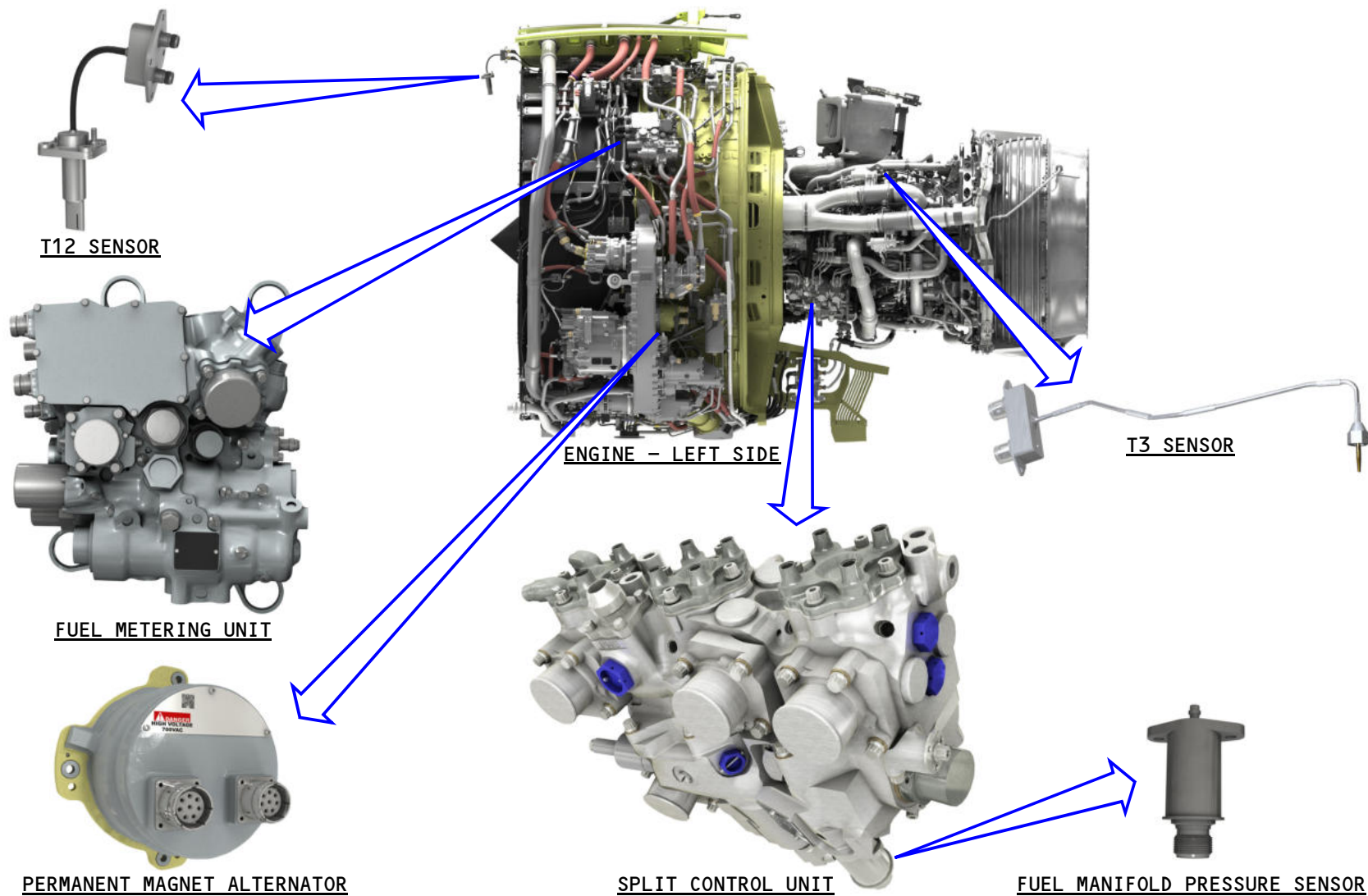
02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - COMPONENT LOCATION - LEFT SIDE

Engine Left Side

These components of the engine fuel and control system are on the left side of the engine:

- T12 sensor (inlet cowl)
- Fuel metering unit (FMU)
- Permanent magnet alternator (PMA)
- Split control unit (SCU)
- Fuel manifold pressure sensor
- T3 sensor.



DMC-B737T-A-73-21-00-03A-055A-T_002-00
ICN-B737T-A-732100-A-81205-00032-A-002-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - COMPONENT LOCATION - LEFT SIDE

73-21-00

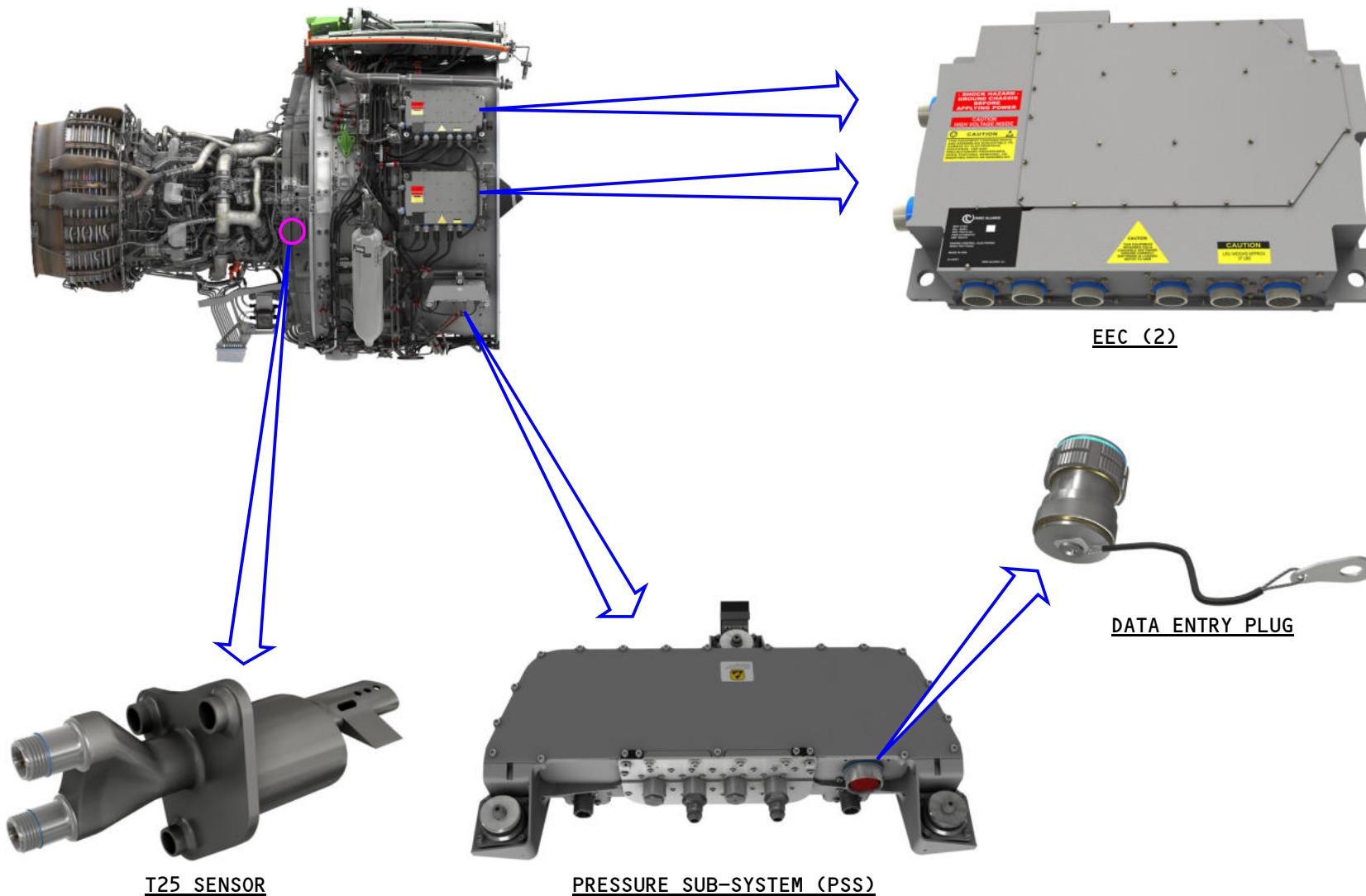
02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - COMPONENT LOCATION - RIGHT SIDE

Engine Right Side

These components of the engine fuel and control system are on the right side of the engine:

- T25 Sensor
- Pressure sub-system (PSS)
- Data entry plug
- Electronic engine controls (EECs).



DMC-B737T-A-73-21-00-04A-055A-T_002-00
ICN-B737T-A-732100-A-81205-00033-A-002-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - COMPONENT LOCATION - RIGHT SIDE

73-21-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ANALOG INTERFACES

General

The engine fuel and control system has interfaces with many systems and components. There are analog interfaces between the engine fuel and control system, the engine systems, and the airplane systems.

The engine control system has dual redundant, line replaceable, electronic engine control (EEC) channels (A and B). Channels A and B are connected through a cross-channel datalink (CCDL). Each EEC uses analog and digital input data to calculate the engine fuel and control outputs to operate the engine. This graphic shows the analog interfaces, the next graphic shows the digital interfaces.

Engine Fuel And Control System To Engine Connection

The EECs have interfaces with these engine systems and components:

- Engine speed sensors
- Engine temperature sensors
- Fuel flow transmitter
- Engine strut
- Permanent magnet alternator (PMA)
- TLA resolvers
- Airplane power
- Engine pressure sensors and pressure sub-system (PSS)
- Fire switch and start lever
- Thrust reverser
- Oil system sensors
- Ignition system
- Start valve
- Fuel metering unit
- Split control unit (SCU)
- Engine air control system.

Engine Sensors

The EECs use input data from engine sensors to calculate the engine fuel and control outputs for engine operation. These are the analog engine sensor inputs:

- N1, low pressure rotor speed
- N2, high pressure rotor speed
- T12, inlet total temperature
- T25, high pressure compressor inlet temperature
- T3, high pressure compressor discharge temperature
- T48, first stage low pressure turbine nozzle temperature.

Fuel Flow Transmitter

The fuel flow transmitter sends fuel flow information to the EECs. The EECs send this information to the display processor computers (DPCs). The display unit (DU) then shows fuel flow along with other engine parameters.

Engine Strut

Jumper wires in the engine strut provide airplane model and engine position data to the EECs. The EECs use airplane model information to set allowable engine thrust ratings. The EECs use engine position to format the engine maintenance message numbers that show on the max display system (MDS).

Permanent Magnet Alternator (PMA)

The PMA is the normal electrical power supply for the EECs when the N2 rotor speed is above 8.5%.



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ANALOG INTERFACES

Thrust Command

Moving the thrust lever sends an engine thrust command to the EECs. The EECs get this command from the thrust lever angle (TLA) resolver. Power for the resolver comes from each of the EECs.

AC Transfer Bus 1 or 2

An EEC can use the AC transfer bus power when it does not receive power from the PMA. The EECs for engine 1 get power from AC transfer bus 1. Engine 2 EECs get power from AC transfer bus 2.

AC transfer bus power is the normal source of electrical power for the left ignition system.

AC Standby Bus

AC standby bus power is the normal source of electrical power for the right ignition system. AC standby bus power is an alternate source of electrical power to engine #1 EEC channel A, when transfer bus 1 is not energized.

Engine pressures and Pressure Sub-System (PSS) box

The engine has differential pressure sensors on the oil and fuel filters, and heat exchangers, and also supplies PS3 pneumatic pressures to the PSS box. The PSS box also has an ambient pressure sensor. The data entry plug connects to the PSS. The PSS connects to the EECs with a dedicated data bus.

Start Lever

The EECs monitor the start lever position. The start lever has 2 positions; cutoff and idle. In the cutoff position the start lever supplies

power to close the HPSOV in the FMU. The cutoff position also provides a reset signal to the EECs.

Engine Fire Switches

The engine fire switch has 2 positions; normal and fire. In the fire position the engine fire switch supplies a close command to the HPSOV. The engine fire switch has priority over the start lever and will close the HPSOV when the start lever is in the idle position. Combustion fuel flow stops when the HPSOV is closed.

Thrust Reverser

The EEC uses the thrust reverser (T/R) linear variable differential transformers (LVDTs) to monitor the T/R position. Power for the LVDTs comes from the EECs. The EEC also controls the thrust reverser control valve and interlock solenoid during reverse thrust operation.

Oil indication system

This is the analog data the EECs receive from the oil system sensors:

- Oil pressure
- Oil temperature
- Debris monitoring system (DMS).

Ignition System

The EECs control AC power from the airplane electrical power system to operate the left and/or right ignition systems on the engine.

Start valve

The EECs control electrical power to open the start valve. The start valve has pressure sensors to indicate valve position.

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ANALOG INTERFACES

Fuel Metering Unit (FMU)

The FMU supplies metered fuel for combustion. The FMU gets electrical commands from the EECs for engine fuel control. The FMU also gets commands from the airplane start lever and fire handle switch to control the high pressure shutoff valve (HPSOV).

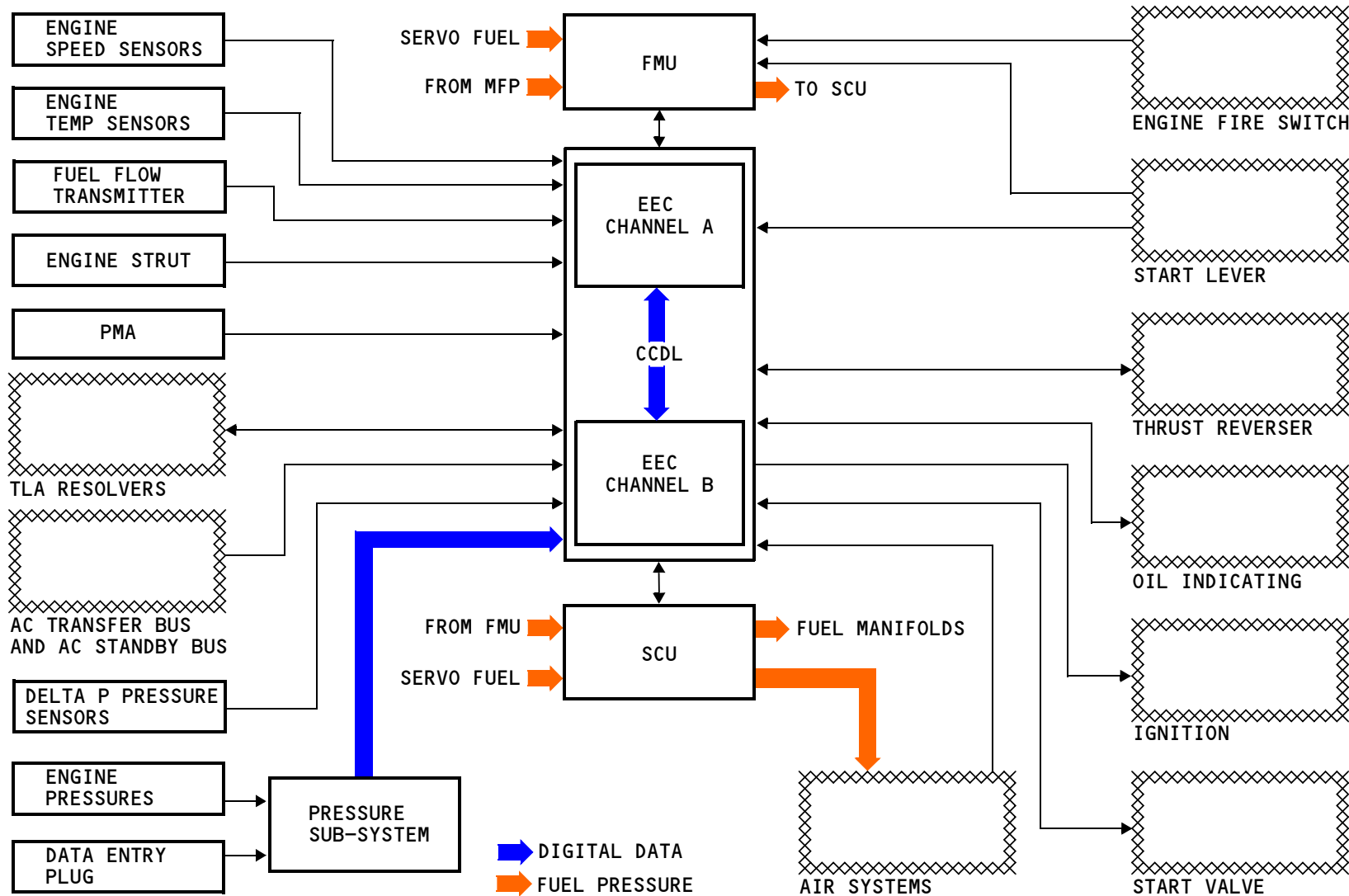
Split Control Unit (SCU)

The SCU splits the metered fuel into two manifolds for combustion. A fuel manifold pressure sensor on the SCU sends fuel pressure data to the EECs. The SCU also has the servos that control the servo fuel pressure for engine air systems operation. The SCU gets electrical commands from the EECs for the servo control.

Engine Air Control

The EECs get LVDT/RVDT position feedback from the air system actuators and valves. This is done for control loop monitoring and fault detection. These are the servo fuel systems the EECs control through the SCU:

- Variable stator vanes (VSVs)
- Variable bleed valves (VBVs)
- Transient bleed valve (TBV)
- Modulated turbine cooling (MTC)
- Start bleed/booster anti-ice (SB/BAI)
- Low pressure turbine active clearance control (LPTACC)
- High pressure turbine active clearance control (HPTACC).



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ANALOG INTERFACES

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - DIGITAL INTERFACES

General

The engine fuel and control system has interfaces with many systems and components. There are digital and analog interfaces between the engine fuel and control system, the engine systems, and the airplane systems. This graphic only shows the digital interfaces.

The electronic engine controls (EECs) are dual redundant line replaceable units (LRUs), identified as channel A and channel B. The two channels communicate with each other and the pressure subsystem (PSS) with a cross channel data link (CCDL). The EECs use the analog and digital input data to calculate the engine fuel and control outputs to operate the engine. Engine data is also used by many other systems.

Display Processing Computer (DPC)

The EECs have a digital connection to the two display processing computers (DPCs). The DPCs have digital and analog connections with these systems and components:

- Air data inertial reference units (ADIRUs)
- Flight management computers (FMCs)
- Digital flight data acquisition unit (DFDAU)
- Other airplane systems
- Proximity switch electronics unit (PSEU)
- Radio altimeters (RA)
- Flight control computer (FCC) A
- Network file server (NFS)

ADIRU (2)

The ADIRUs give temperature and pressure data to the EECs through the DPCs.

FMC (2)

The FMCs contain the data to control the engines during a flight. The FMCs are connected to the DPCs and also the FCC A.

DFDAU

The DPCs collect engine data. They send this data to the DFDAU. The DFDAU creates airplane condition monitoring system (ACMS) reports.

Other Airplane Systems

Many other airplane systems send and receive data to/from the DPCs. Some of this data comes from, and goes to the EECs.

PSEU

The PSEU sends analog and digital position data to the DPCs. Some of this data goes to the EECs.

RA

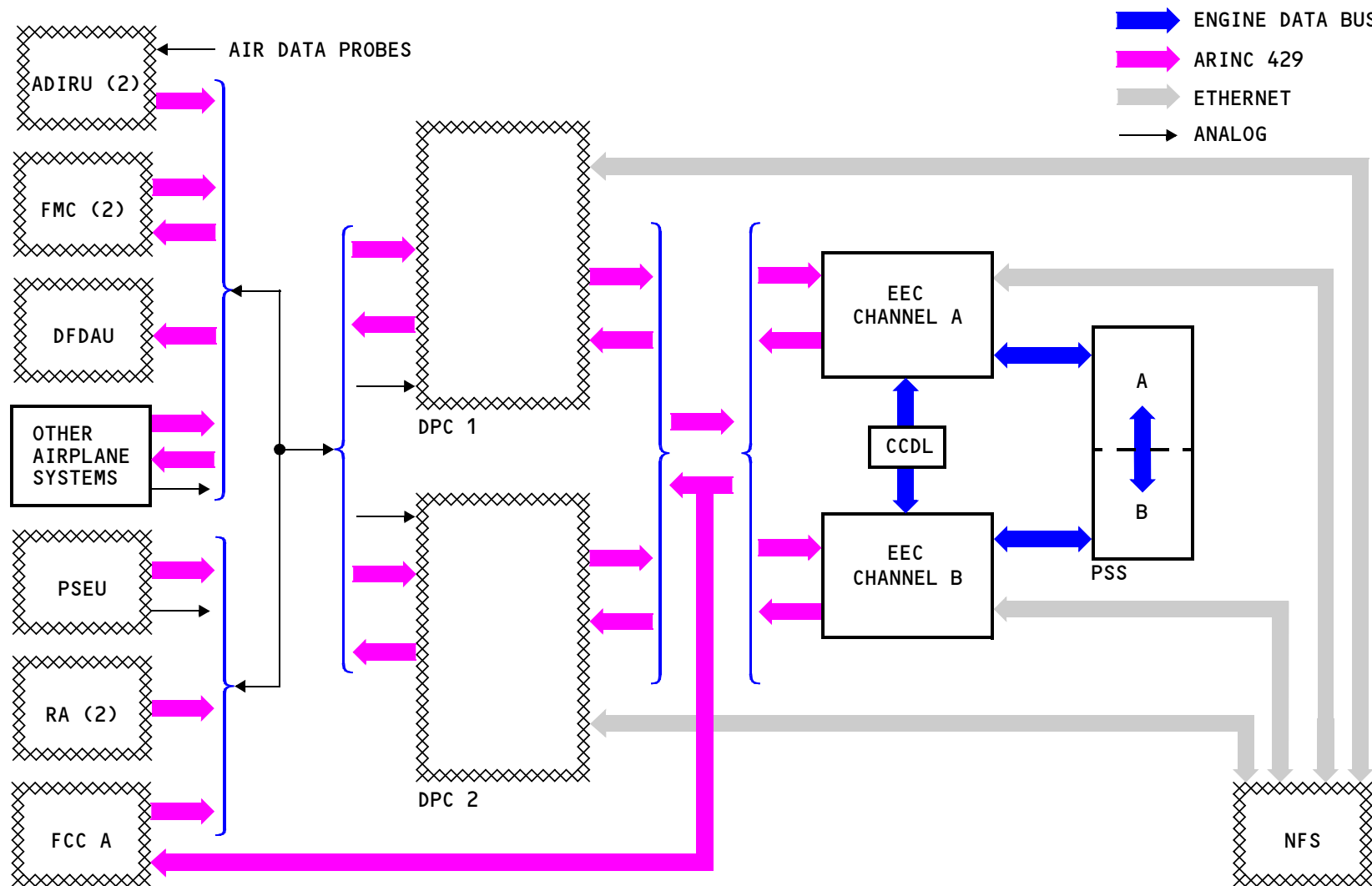
The radio altimeters send data to the DPCs. The EECs use this data to calculate the air/ground condition.

FCC A

The FCC A has the auto throttle (A/T) function. The A/T controls the engines with commands through the DPC to the EECs.

NFS

The NFS has a ethernet type digital interface with the EECs. The NFS provides the onboard maintenance function. The NFS is also the interface for the engine health monitoring function and data loading.



DMC-B737T-A-73-21-00-30A-110A-T_001-00
ICN-B737T-A-732100-A-81205-00153-A-001-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - DIGITAL INTERFACES

02-Dec-2020

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ELECTRONIC ENGINE CONTROL

General

The LEAP-1B engine control system is a full authority digital electronic control (FADEC) system. The primary control unit for the engine is the electronic engine control (EEC) .

The engine control system has two EEC computers that attach to the right side of the engine fan case. Each channel can control the engine (active mode). Only one channel is in the active mode at a time. The other channel is in the standby mode.

The EECs use digital and analog signals from the engine and other airplane systems to control and monitor the engine. The EECs send engine data to other airplane systems.

EEC Electrical Connections

There are electrical connectors on the EECs. The EECs use these connectors to receive and send data to the airplane and engine. The connectors are J1 to J9.

EEC Installation

EEC channel A attaches to the engine fan case at the 2:00 position, aft looking forward. EEC channel B attaches to the engine fan case at the 4:00 position. The EECs attach to the fan case with four shock mounts. The EECs are grounded to the engine with bonding straps. Each EEC channel weighs 34.8 lb / 15.8 kg. You can use ground support equipment for the installation and the removal of the EEC channels.

Functional Description

The goal of the FADEC system is to keep total engine function at all times. Usually one input fault will not change the engine's operation. With two or more input faults the FADEC system will assume a default

or calculated condition. The engine continues to give the highest level of controllable thrust within certificated limits.

The two EEC channels communicate through a cross channel data link (CCDL).

The engine control system has two modes of operation, dual channel operation or single channel operation. The EECs usually operate in the dual channel mode.

Dual channel operation is when the active channel reads input data from the opposite channel with the cross channel data link. In dual channel operation the active channel compares its input data with the opposite channel data. The active channel can select the best signal. The active channel can also use an average of the signals to calculate the value that it uses to control the engine. Channels A and B alternate being the active channel and the standby channel. With each engine start the EECs compare the health of each channel. The channel with the best health becomes the active channel.

Single channel operation is when the active channel of the EECS uses only its sense circuits to control the engine. Single channel operation occurs when the EEC channels can not communicate with each other (the CCDL does not operate).

These are the major functions of the EECs:

- Input signal validation and processing
- Fault Accommodation
- Starting, ignition, and shutdown control
- Thrust and power management
- Reverse thrust control
- Engine operational limits protection
- Flight deck display and indication
- Diagnostics and maintenance operation.



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ELECTRONIC ENGINE CONTROL

Input Signal Validation and Processing

The EECs get digital and analog signals from the engine and other airplane systems. The EECs process, and select these input signals. Some of these signals have more than one source for the same data. This improves engine reliability because if one data source is inoperative the EECs can use the other data sources. If the active channel finds that all data sources are valid, it uses the best data to control the engine. An example of this is T48. T48 is the same as exhaust gas temperature (EGT). Each EEC channel gets four EGT signals. If all four signals are valid, the EEC uses the average temperature as the selected EGT. If one of the signals is out of range, the average of the other three EGTs is used to control the engine.

If all sources of a given parameter are not valid, the EECs can use a default value to operate the engine.

If the EECs find a signal is not valid, it stores a message in BITE.

Fault Accommodation

Accommodation of input signal faults is by EEC input signal processing logic. Most input signals are dual redundant. When an input signal has a fault or is out of range, the EECs will use the other signal. When no valid signal exists, the EECs will use a fail-safe value, simulate the missing value, or hold the last valid value. When the active channel finds that it cannot control the engine, the EECs will automatically change the active channel to the opposite channel. The EECs constantly monitor channel health to know which channel has the best health for engine operation.

Starting, Shutdown, and Ignition Control

The EECS has enhanced manual start capability. Enhanced manual start capability means that the active EEC controls the engine start after the start initiation. Use the start switch and the start lever to initiate an engine start. The EEC then controls all other aspects of the start sequence. This includes control of the Starter Air Valve (SAV) and the starter cutout, ignition system control, and the fuel schedule and control.

Thrust and Power Management

Engine thrust is selected by moving the thrust lever. The EECs get thrust lever angle (TLA) from the thrust lever resolvers. The resolvers send thrust lever resolver angle (TRA) to the EECs. The active EEC controls the engine thrust by increasing or decreasing N1 speed.

TRA angles show on max display system (MDS) electronic propulsion control system (EPCS) maintenance page.

Reverser Thrust Control

On the ground, the active EEC uses thrust reverser translating sleeve position to limit reverse thrust to idle speed until the both sleeves are more than 60% deployed.

Engine Operational Limits Protection

The EECs include protection against the exceedance of engine operational limits. The active EEC keeps these engine parameters in limits:

- N1 speed
- N2 speed
- PS3 (HPC static pressure)

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ELECTRONIC ENGINE CONTROL

- Fuel flow
- Maximum EGT.

The EECs have an electronic overspeed (EOS) function. The EOS gives a backup signal to prevent an N2 overspeed condition.

Flight Deck Display and Indication

The EECs send analog and digital data to the airplane MAX display system (MDS). The inboard display units (DUs) can show the engine systems information.

For each engine indication, the MDS will use the active channel digital data parameter if it is valid. If it is not valid, the MDS will use the digital data parameter from the opposite EEC channel. If the digital-data from both channels is not valid, the MDS will select the analog input. When there is no analog-data, or if the analog-data is out of range, then the MDS display is blank.

When the active channel or the standby channel senses an engine alert or an exceedance, it shows on the MDS.

Diagnostics and maintenance operation

The EECs include diagnostics and maintenance logic. This logic gives fault detection, isolation, and annunciation.

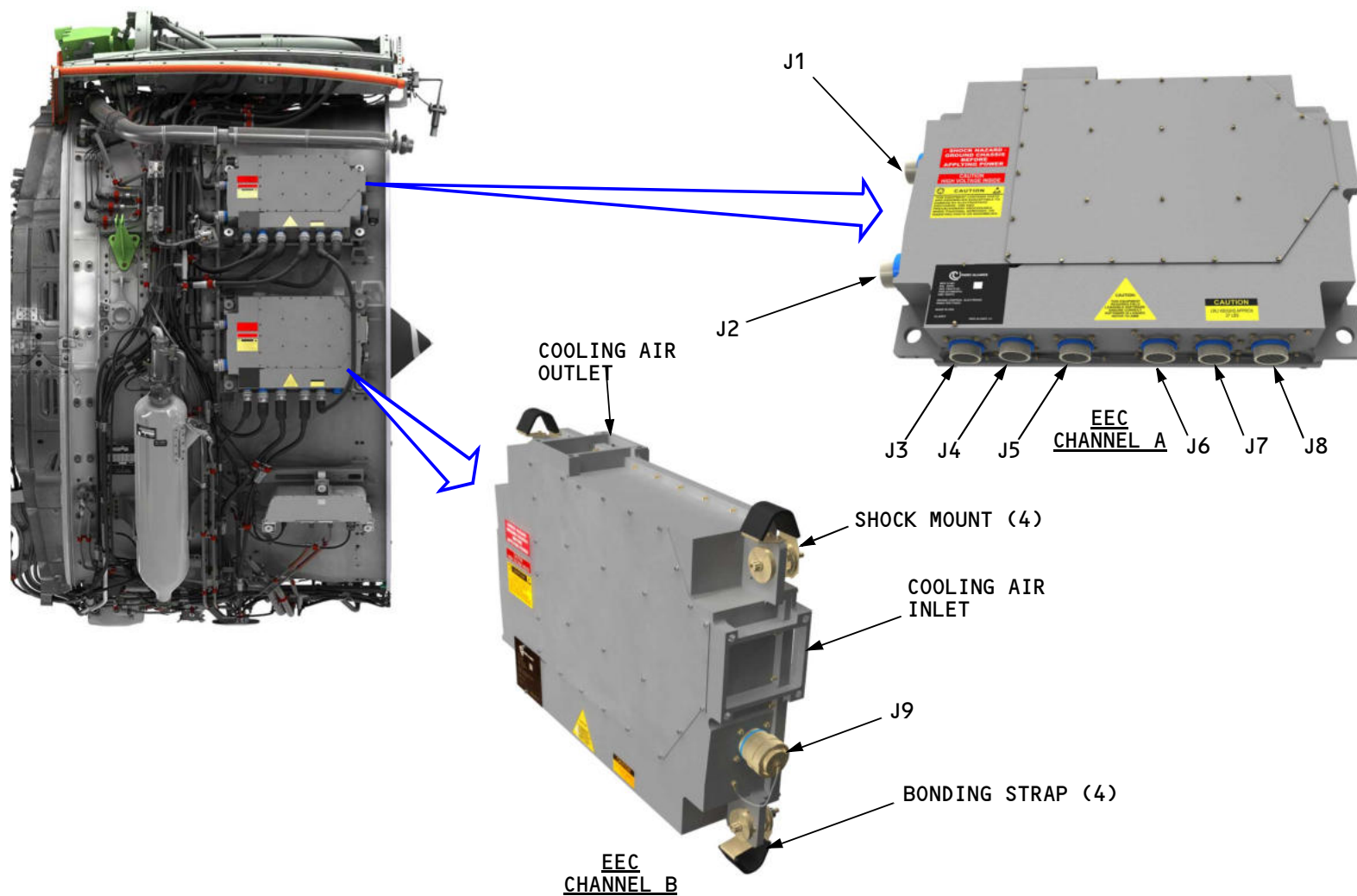
The EECs have a ground maintenance mode. The ground maintenance mode has these functions:

- Recall stored data
- Enter new data
- Special functions
- System tests.

DMC-B737T-A-73-21-00-14A-110A-T_004-00

02-Dec-2020

73-21-00



DMC-B737T-A-73-21-00-14A-110A-T_004-00
ICN-B737T-A-732100-A-81205-00037-A-004-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ELECTRONIC ENGINE CONTROL

73-21-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - PRESSURE SUBSYSTEM

General

The pressure subsystem (PSS) supplies engine pressures and data entry plug data to the electronic engine controls (EECs). The PSS is a line replaceable unit (LRU).

Location

The right fan cowl must be open to access the PSS. The PSS attaches to the engine fan case at the 4:00 position, aft looking forward. The PSS weighs 10 lb / 4.5 kg.

The PSS attaches to the fan case with 3 shock mounts. The PSS has 1 bonding strap.

Functional Description

The PSS has 2 isolated internal channels that independently connect to EEC channel A and EEC Channel B. The 2 PSS channels are connected with an internal cross channel data link (CCDL). Each channel measures 3 pneumatic pressure inputs. The LEAP-1B does not use the 4th pressure (PS12) input, it is plugged.

The PSS has 3 functions. The first function is to change the pressure inputs to a digital electronic signal. The PSS sends the digital signals to the EECs. The second function of the PSS is to read the data entry plug data and send the data to the EEC channels. The third function is to be a backup CCDL for the EECs. If the CCDL between the EEC channels does not operate, the EECs will use the PSS as a CCDL.

The PSS has these connections:

- Channel B
- P3B
- P0
- PS3

- Channel A
- Data entry plug.

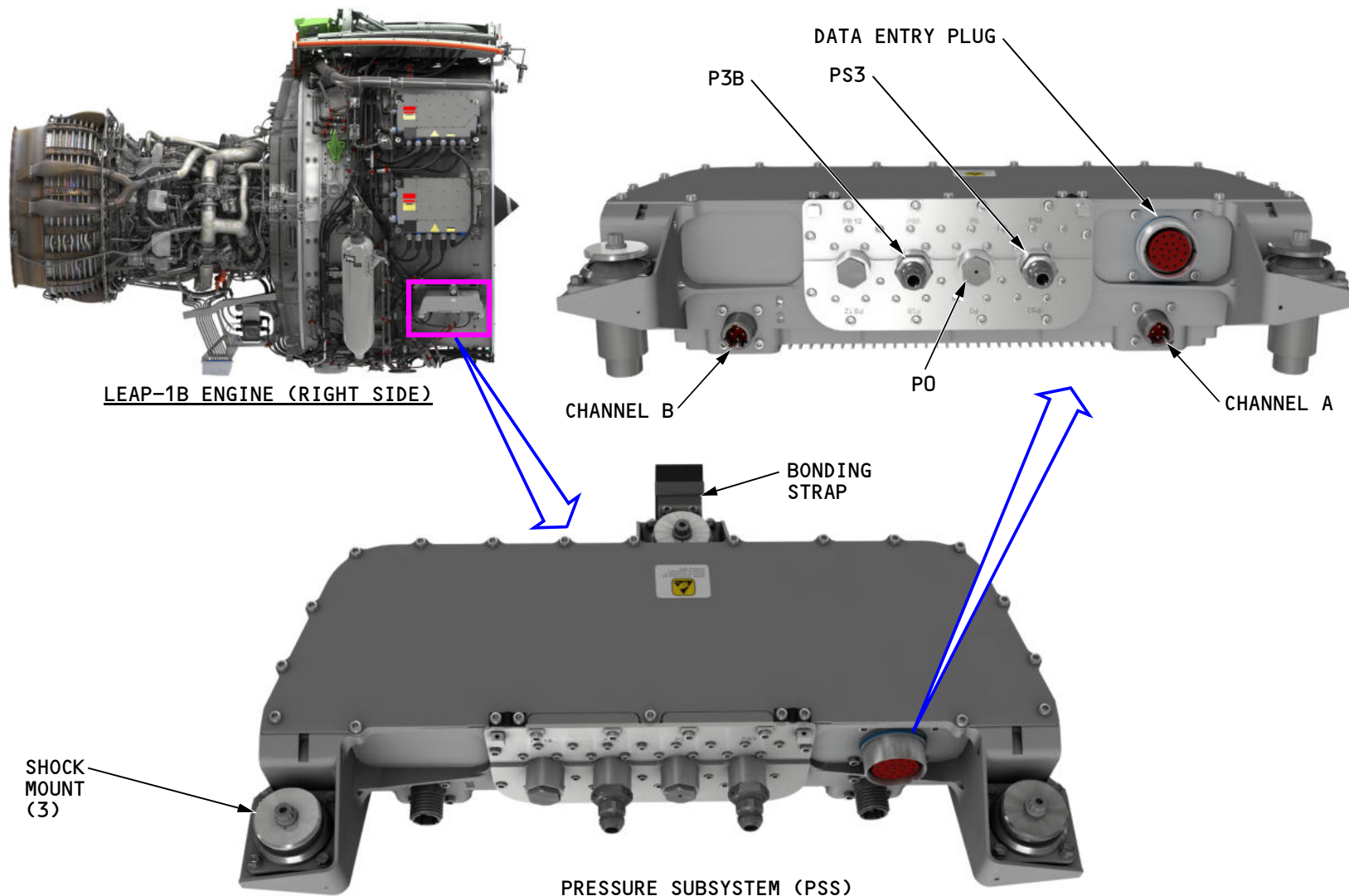
Each PSS channel has an input/output (I/O) connection with the related EEC channel. Each EEC channel sends electrical power to the related PSS channel. Each PSS channel converts and sends the pressure signals for P3B, P0, and PS3 to the related EEC channel.

P3B is the environmental control system (ECS) high pressure compressor (HPC) bleed pressure.

P0 is the ambient pressure in the undercowl compartment of the fan case. The EECs correct the P0 to the ambient pressure external to the nacelle.

PS3 is the HPC discharge pressure.

The data entry plug (DEP) gives the engine thrust rating, engine thrust bump, N1 modifier, and engine configuration to the PSS.



DMC-B737T-A-73-21-00-04A-040A-T_003-00
ICN-B737T-A-732100-A-81205-00155-A-002-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - PRESSURE SUBSYSTEM

73-21-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - DATA ENTRY PLUG

Purpose

The data entry plug (DEP) gives the following data to the pressure subsystem (PSS):

- Engine thrust rating
- Engine thrust bump
- N1 trim
- Engine configuration.

Location

The DEP is on the right side of the engine fan case. The DEP connects to PSS connector J1, also known as the DEP connector. To get access to the PSS and the DEP, open the right fan cowl.

Function

The electronic engine controls (EECs) read the DEP data during the EECs power-up initialization.

The DEP contains 9 fusible pins that are permanently set by the manufacturer. The fusible pins set the engine thrust rating and the engine thrust bump.

The DEP contains 11 push-pull pins. The push-pull pins are used to set the N1 trim and the engine configuration. Maintenance can change the position of the push-pull pins.

Engine Thrust Rating

The DEP sets the engine thrust between 25,000 lb - 29,300 lb BET and must be compatible with the airplane model.

Engine Thrust Bump

The engine thrust bump is a customer option that gives more thrust than the normal thrust rated maximum-takeoff reference-speed. The flight crew selects the thrust bump function from the N1 limit page in the flight management computer (FMC). The airplane uses the engine thrust bump for that takeoff only. The thrust bump is only available on some airplane model and engine thrust combinations.

N1 Trim

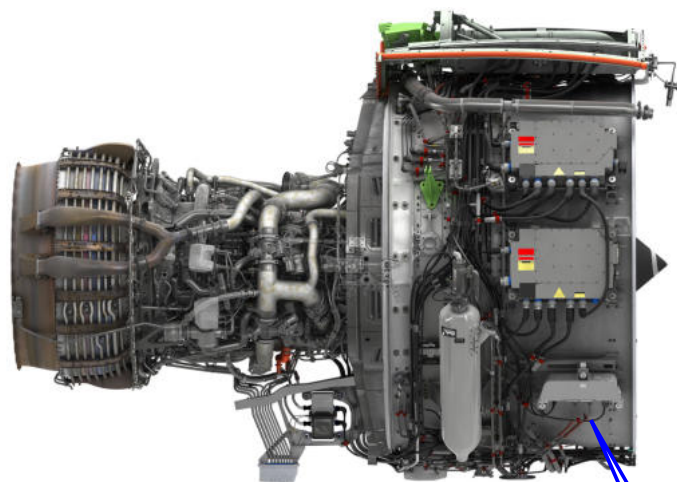
The EECs use N1 trim to derate engine thrust to a nominal 29,300 lb BET. This makes sure all engine produce the same reference thrust.

Engine Configuration

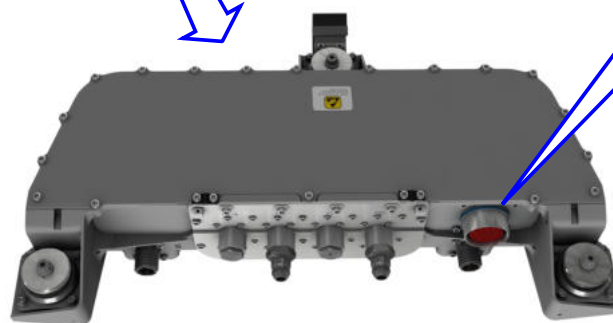
The engine configuration discretes send engine hardware data to the EECs.

Training Information Point

The DEP plug permanently attaches to the engine fan case with a lanyard. The lanyard keeps the DEP with the engine when you remove the PSS.



LEAP-1B ENGINE (RIGHT SIDE)



PRESSURE SUBSYSTEM (PSS)



DATA ENTRY PLUG

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - DATA ENTRY PLUG

73-21-00

DMC-B737T-A-73-21-00-15A-110A-T_004-00
ICN-B737T-A-732100-A-81205-00038-A-002-01

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ENGINE CONTROL LIGHT AND EEC SWITCHES

Purpose

These engine control status indications are on the engine control panel :

- Electronic engine control (EEC) switch ON indication (white light)
- EEC ALTN light (amber)
- ENGINE CONTROL light (amber).

The white ON light and the amber ALTN light are in the EEC switches. The EEC switches have a clear plastic switch guard. When the EECs are in the normal mode (switch in), the white ON light shows, and the ALTN light is off. When the EECs are in an alternate mode, the ALTN light is on.

There are two alternate modes for EEC operation. They are soft alternate and hard alternate.

The EECs usually receive total air pressure (PT) from the air data inertial reference units (ADIRUs). When PT is not valid, the EECs go to the soft alternate mode, and the ALTN light comes on. The white ON indication also shows during soft alternate mode operation.

When you move the EEC switch to the alternate position (switch out), the EECs on that engine go to the hard alternate mode and the ALTN light comes on. The white ON indication goes off when the hard alternate mode is selected.

When an EEC finds a no-dispatch engine failure condition, it sends a signal to the display processing computers (DPCs). The DPCs control the amber ENGINE CONTROL light. The ENGINE CONTROL light comes on when all of these conditions occur:

- Airspeed is less than 80 knots or the airplane is on the ground for more than 30 seconds

- EEC is energized (engine starting, running, or EEC energized for maintenance)
- A no-dispatch engine fault occurs.

The ENGINE CONTROL light does not come on in flight.

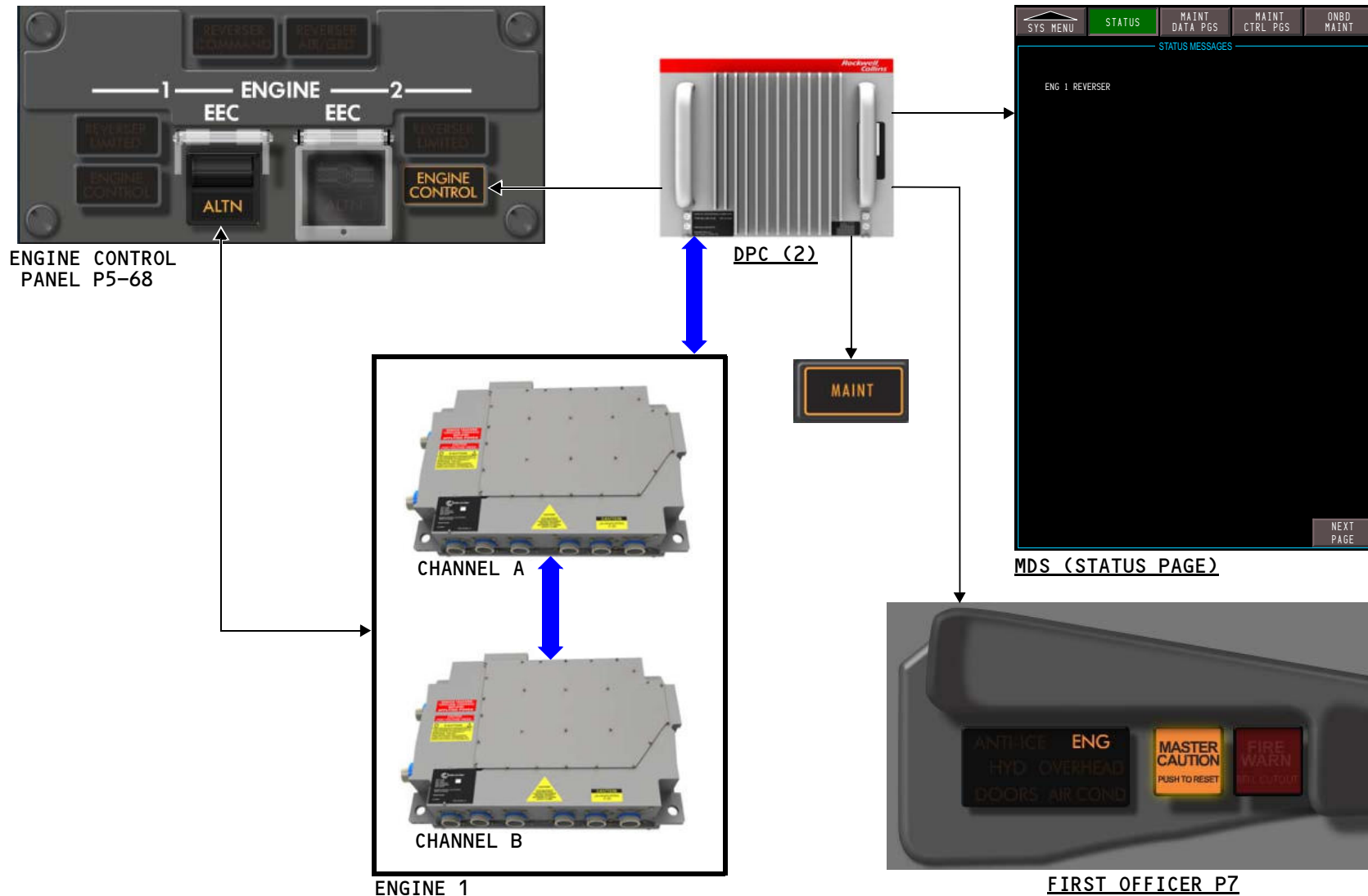
When an EEC finds an engine failure condition that affects aircraft dispatch, it sends a status message signal to the display processing computers (DPCs). The DPCs control the amber MAINT light. The MAINT light comes on when these conditions occur:

- Airspeed is less than 80 knots or the airplane is on the ground for more than 30 seconds
- A new status message occurs.

The MAINT light does not come on in flight.

Training Information Point

The ENGINE CONTROL light comes on to indicate a no dispatch condition. You use the max display system (MDS), onboard maintenance function (OMF), fault isolation manual (FIM), and the airplane maintenance manual (AMM) to find and repair the fault. Other engine faults can cause a status message. The MAINT (amber) light comes after landing to show a new status message is active. You use the dispatch deviation guide (DDG) to determine the impact on dispatch from engine status messages.



DMC-B737T-A-73-21-00-18A-110A-T_004-00
ICN-B737T-A-732100-A-81205-00043-A-003-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - ENGINE CONTROL LIGHT AND EEC SWITCHES

02-Dec-2020

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - FUEL METERING UNIT

General

The fuel metering unit (FMU) supplies fuel for engine combustion. The FMU receives fuel metering signals from the electronic engine controls (EECs) and fuel shutoff signals from the airplane.

Location

The FMU attaches to the engine fan case at the 10 o'clock position. Open the left fan cowl to get access to the FMU.

Physical Description

The FMU has electrical and fuel connections.

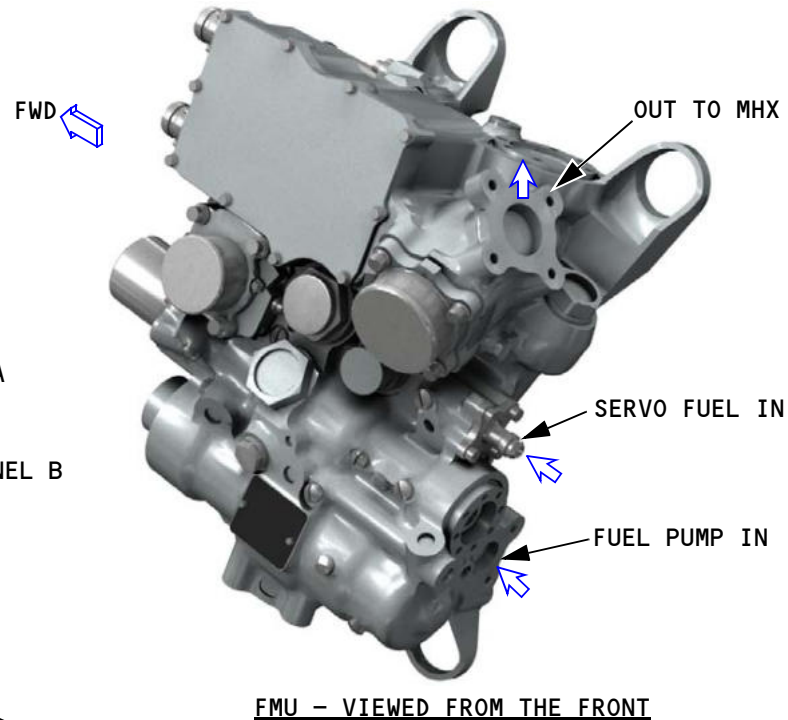
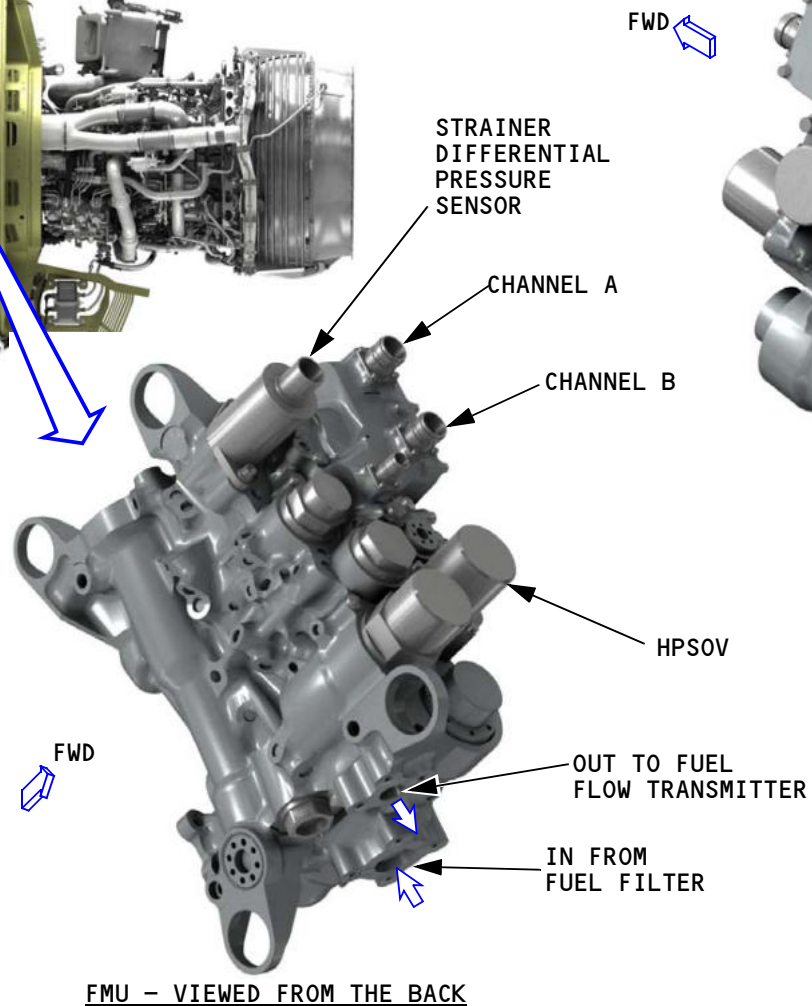
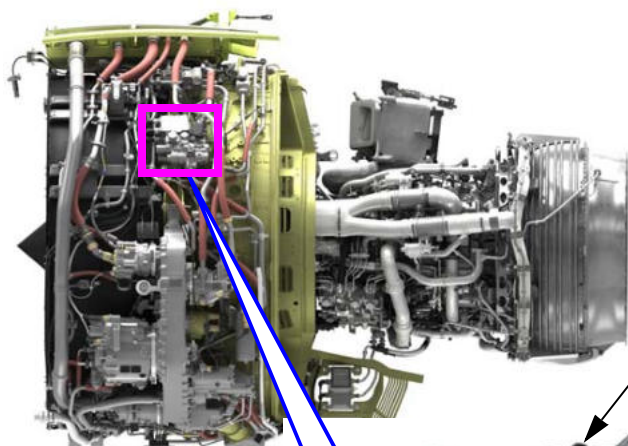
These are the FMU electrical connections:

- Strainer differential pressure sensor
- EEC channel A
- EEC channel B
- High pressure shutoff valve (HPSOV) solenoid.

These are the FMU hydraulic (fuel) connections:

- Low pressure (LP) fuel in from the fuel pump assembly
- Intermediate pressure (IP) fuel out to the main fuel/oil heat exchanger (MHX)
- High pressure (HP) in from the fuel filter
- HP servo fuel in from the servo fuel heater
- HP combustion fuel out to the fuel flow transmitter.

DMC-B737T-A-73-21-00-17A-110A-T_002-00



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - FUEL METERING UNIT

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - SPLIT CONTROL UNIT

General

The split control unit (SCU) supplies fuel for engine servo system operation and combustion. The SCU receives fuel metering signals and engine servo command signals from the electronic engine controls (EECs).

Location

The SCU attaches to the engine high pressure compressor case at the 7 o'clock position aft looking forward. Open the left fan cowl and left thrust reverser cowl to get access to the SCU.

Physical Description

The SCU has electrical and fuel connections.

These are the SCU electrical connections:

- Fuel pressure sensor
- EEC channel A
- EEC channel B.

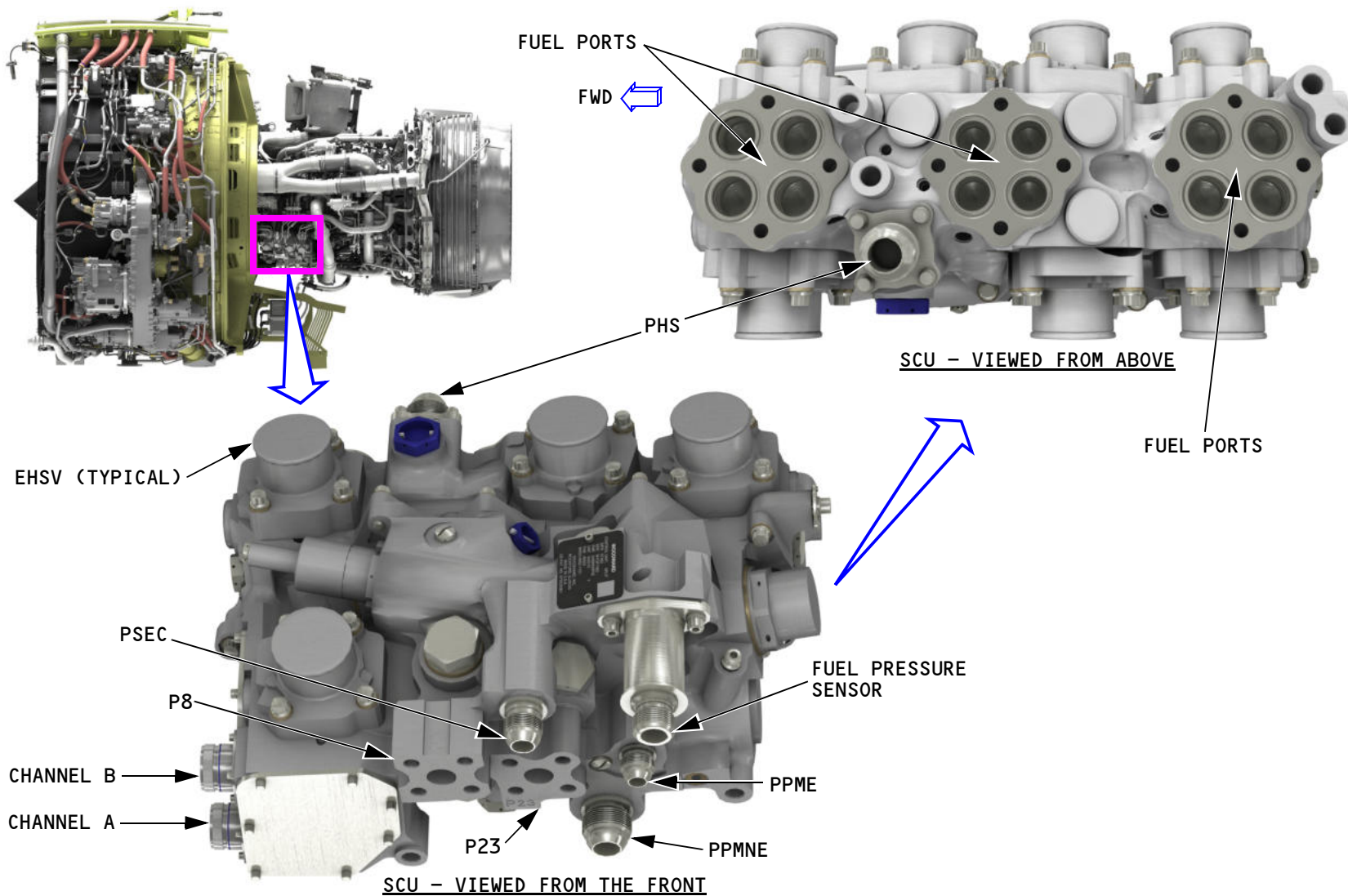
These are the SCU hydraulic (fuel) connections:

- P23 Combustion fuel supply (IN)
- PHS Servo fuel supply (IN)
- P8 SCU fuel return (OUT) to the fuel metering unit (FMU) jet pump
- Pilot primary main enriched (PPME) to the fuel nozzles
- Pilot primary main non-enriched (PPMNE) to the fuel nozzles
- Pilot secondary (PSEC) to the fuel nozzles
- Fuel ports.

The electro hydraulic servo valves (EHSV) control the fuel ports.

These are the fuel ports:

- Variable bleed valve (VBV) head
- Variable bleed valve (VBV) rod
- Variable stator vane (VSV) head
- Variable stator vane (VSV) rod
- Modulated turbine cooling (MTC)
- High pressure turbine active clearance control (HPTACC)
- Low pressure turbine active clearance control (LPTACC)
- Case pressure regulated (PC/REG)
- Booster anti-ice/start bleed valve (BAI/SBV) C2
- Booster anti-ice/start bleed valve (BAI/SBV) C1
- Transient bleed valve (TBV) C1
- Transient bleed valve (TBV) C2.



DMC-B737T-A-73-21-00-01A-034B-T_002-00
ICN-B737T-A-732100-A-81205-00160-A-001-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - SPLIT CONTROL UNIT

73-21-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - T12 SENSOR

General

The T12 sensor supplies fan inlet total temperature data to the electronic engine controls (EECs) for engine control when the airplane is on the ground and for five minutes after takeoff. Five minutes after takeoff, the EECs use these to calculate the selected total temperature:

- T12 channel A
- T12 channel B
- Total air temperature from ADIRU 1
- Total air temperature from ADIRU 2.

The EECs use total temperature to control these:

- Engine thrust management functions
- Fuel control functions
- EGT functions
- Variable bleed valve (VBV)
- High pressure turbine clearance control (HPTACC)
- EEC blowers
- Booster anti-ice (BAI)
- Starter air valve (SAV) pressure.

Location

The T12 sensor is at the 11:00 position in the engine inlet cowl (aft looking forward). The T12 sensor probe extends into the air that flows into the fan.

An access panel on the top of the inlet cowl supplies access to the T12 sensor.

Two electrical connectors connect the T12 sensor to the EECs. The connectors are on the aft side of the inlet cowl under the left fan cowl.

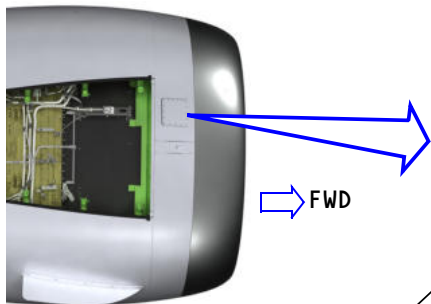
Functional Description

There are two platinum elements in the sensor assembly. One element is for channel A, the other is for channel B.

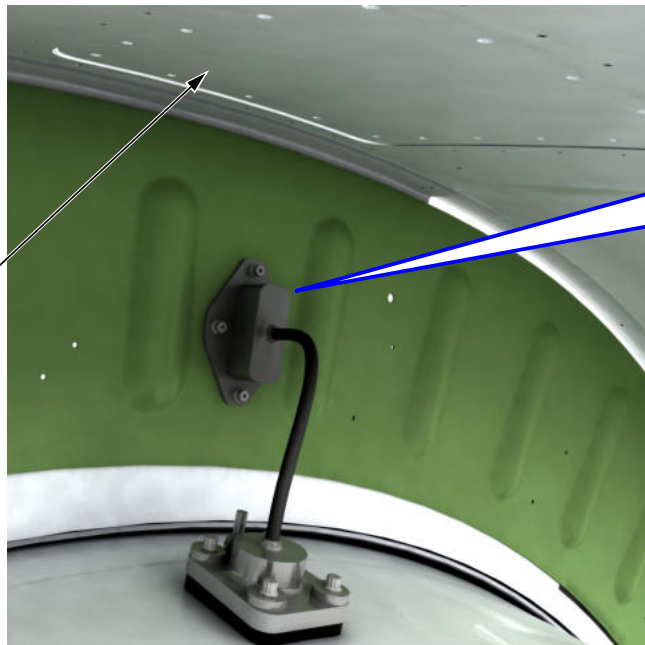
The EECs supply a constant voltage power supply to the T12 platinum sensor elements. The elements change resistance in proportion to air temperature. The output from the platinum elements goes back to the EECs. The EECs measure the difference between the constant output voltage and input from the sensor. The EECs convert the difference into a fan inlet temperature.

The housing is designed to prevent ice accretion, noise, and turbulence.

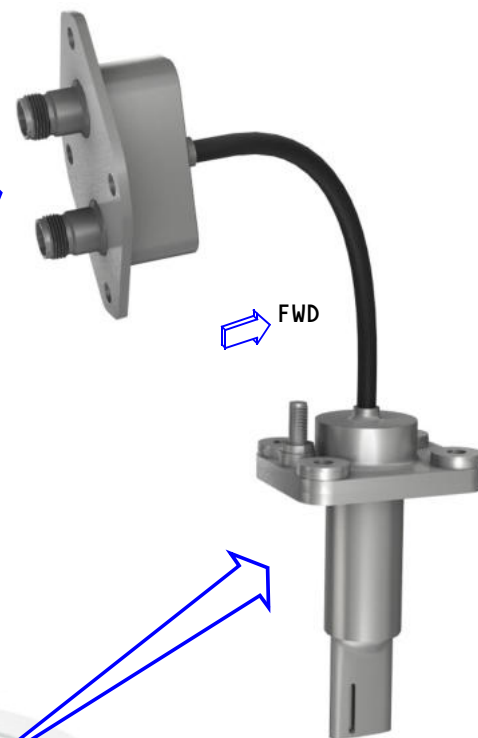
DMC-B737T-A-73-21-00-12A-110A-T_001-00



T12 ACCESS PANEL

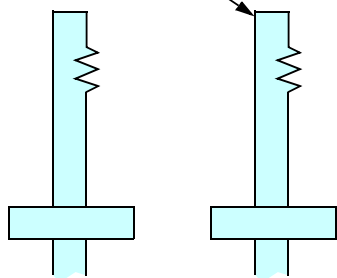


INSIDE INLET COWL



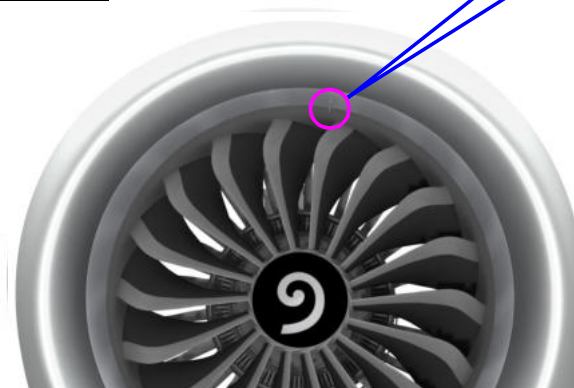
T12 SENSOR

T12 PLATINUM
ELEMENT
(TYP)



CHANNEL A

CHANNEL B



INLET COWL

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - T12 SENSOR

73-21-00

DMC-B737T-A-73-21-00-12A-110A-T_001-00
ICN-B737T-A-732100-A-81205-00035-A-001-01

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - T25 SENSOR

Purpose

The T25 temperature sensor supplies high pressure compressor (HPC) inlet temperature data to the electronic engine controls (EECs). The EECs use the T25 temperature information to control these components:

- Transient bleed valve (TBV)
- Variable bleed valve (VBV)
- Variable stator vanes (VSV)
- Booster anti-ice (BAI)
- High pressure turbine clearance control (HPTACC)
- Low pressure turbine clearance control (LPTACC).

The T25 sensor has two separate electrical outputs, one for each EEC channel.

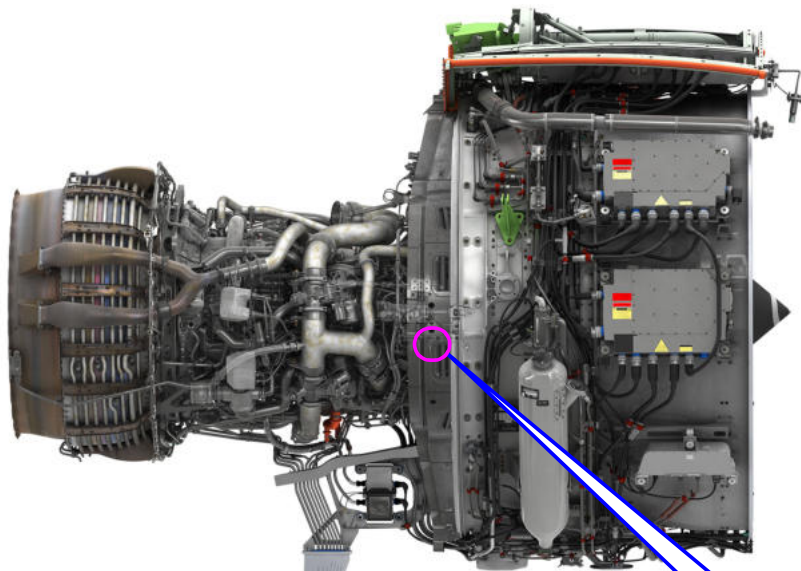
Location

The T25 sensor is on the fan hub frame between 3 and 4 o'clock aft looking forward. You remove the inner core fairing to get access to the sensor.

Functional Description

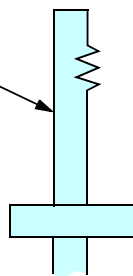
There are two platinum elements in the T25 sensor assembly. One element is for channel A, the other is for channel B. The EECs supply a constant voltage power supply to the T25 sensor elements. The elements change resistance to the electrical current in proportion to air temperature. The different output voltages from the platinum elements goes back to the EECs. The EECs measure the difference between the constant output voltage and input voltage from the sensor. The EECs convert the difference into a HPC inlet temperature.

The housing is designed to prevent ice accretion, noise, and turbulence. The sensor is not heated.

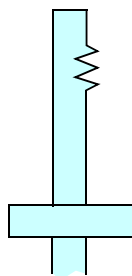


LEAP-1B ENGINE (RIGHT SIDE)

T25 PLATINUM
ELEMENTS
(TYPICAL)



CHANNEL A



CHANNEL B



T25 SENSOR

FWD

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - T25 SENSOR

73-21-00

DMC-B737T-A-73-21-00-11A-110A-T_003-00
ICN-B737T-A-732100-A-81205-00034-A-002-01

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - T3 SENSOR

General

The T3 temperature sensor is a thermocouple that measures 10th stage compressor discharge air temperature. The T3 sensor sends this temperature data to EEC channel A and EEC channel B.

The EECs use T3 temperature data to control:

- Fuel control functions
- Engine starting functions
- Ice crystal detection
- High pressure turbine active clearance control (HPTACC)
- Low pressure turbine active clearance control (LPTACC).

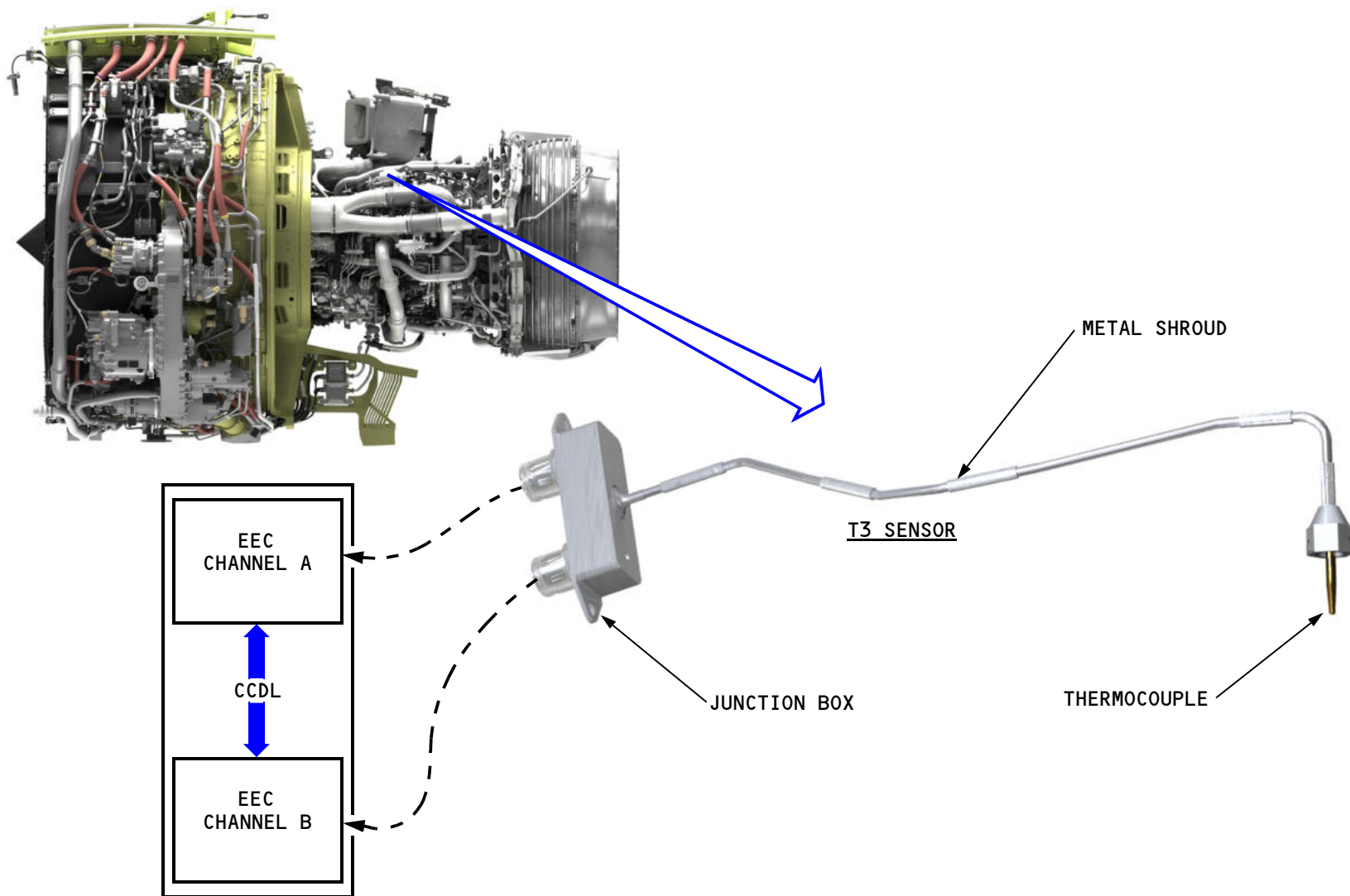
Functional Description

The dual bi-metallic (K type thermocouple) sensor makes a milliamp (mA) current proportional to temperature. When temperature increases, the current increases. When temperature decreases, the current decreases. The EECs convert the current to a digital temperature reading.

DMC-B737T-A-73-21-00-13A-110A-T_001-00

02-Dec-2020

73-21-00



DMC-B737T-A-73-21-00-13A-110A-T_001-00
ICN-B737T-A-732100-A-81205-00036-A-001-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - T3 SENSOR

73-21-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - PERMANENT MAGNET ALTERNATOR

General

The permanent magnet alternator (PMA) usually supplies electrical power to the electronic engine control (EEC) channels. The PMA is the primary electrical power source for the EEC channels.

NOTE: AC transfer bus 1 is another source of power for the engine 1 EECS channels. AC transfer bus 2 is another source of power for the engine 2 EECS channels. The ac standby bus is an additional source of power for the engine 1 channel A.

Location

The PMA is on the rear side of the accessory gearbox (AGB) at the 8:00 position. To access the PMA, open the left fan cowl.

Physical Description

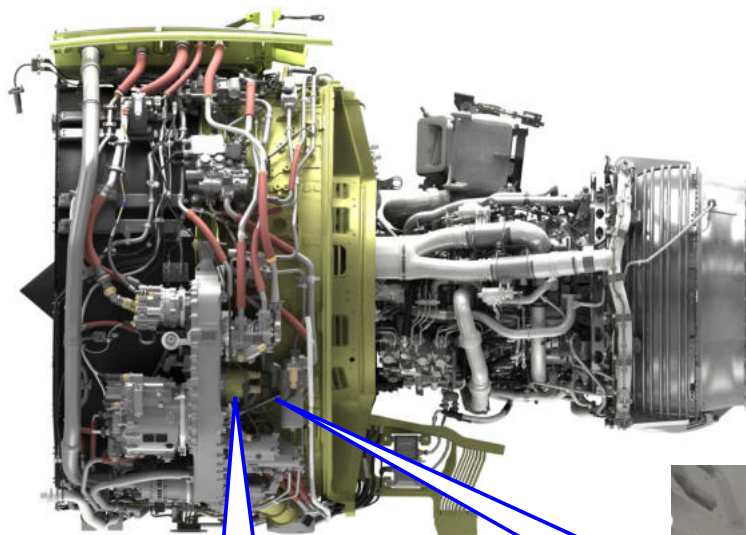
The PMA contains a housing and stator assembly, and a rotor. The stator has two separate sets of windings. One set of windings gives power to channel A and the other set gives power to channel B.

Functional Description

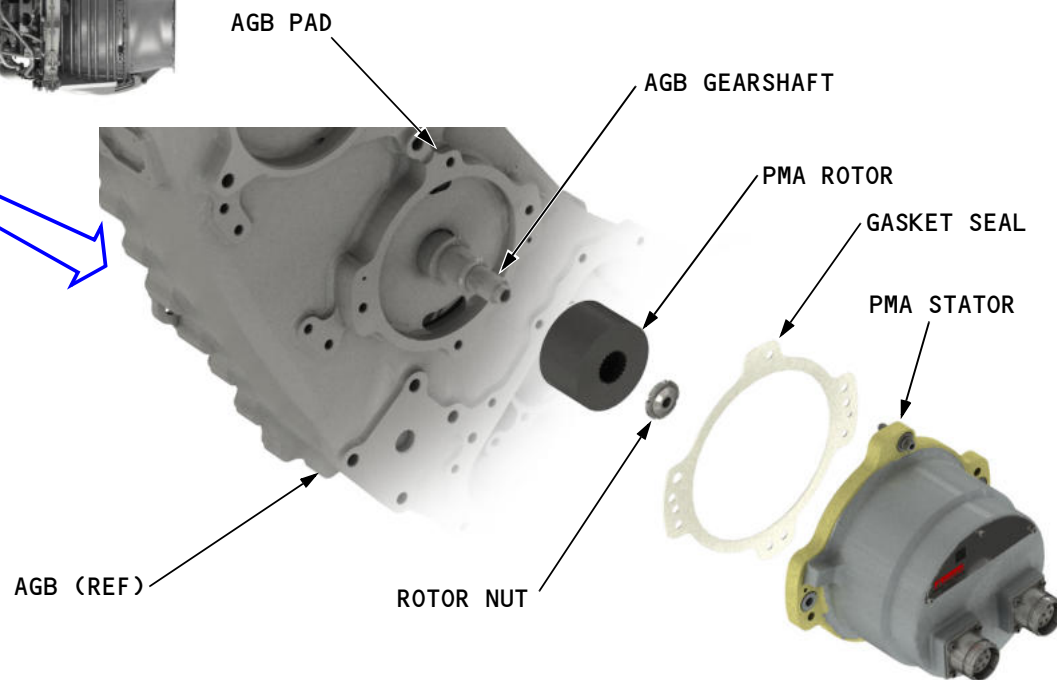
The PMA supplies electrical power to channel A and channel B of the EECS. When N2 speed is approximately 8.5% or more, the PMA can supply the EEC channels with power.

If the PMA cannot supply power to one of the channels, the related ac transfer bus will supply power to that channel. When the PMA and ac transfer bus are not available, the ac standby bus can supply engine 1 channel A with power. The ac standby bus is not available to power channel B.

CAUTION: DO NOT LET THE ROTOR AND STATOR TOUCH WHEN YOU REMOVE THE STATOR. DAMAGE TO THE ROTOR AND THE STATOR CAN OCCUR IF THEY TOUCH. THE MAGNETS ON THE ROTOR APPLY A STRONG MAGNETIC FORCE ON THE STATOR. INSTALL THE GUIDE PINS BEFORE REMOVING OR INSTALLING THE PMA HOUSING.



PERMANENT MAGNET ALTERNATOR



EXPLODED VIEW

DMC-B737T-A-73-21-00-16A-110A-T_002-00
ICN-B737T-A-732100-A-81205-00039-A-003-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - PERMANENT MAGNET ALTERNATOR

73-21-00

02-Dec-2020

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - EECS ENGINE 1 POWER SUPPLY- FUNCTIONAL DESCRIPTION

Purpose

The permanent magnet alternator (PMA) is the primary ac power supply for the electronic engine controls (EECs) operation. The 115v ac transfer bus 1 can also supply power to the EECs for engine 1. The 115v ac standby bus can supply power to EEC channel A when no other power is available.

Functional Description

The PMA has two, three-phase 115v ac power outputs. One output supplies channel A and the other supplies channel B.

The airplane supplies 115v single phase ac power to each engine EEC channel. 115v ac transfer bus 1 supplies phase A power to EEC channel A. 115V ac transfer bus 1 supplies phase B power to EEC channel B.

The engine 1 alternate power relay R576 is in junction box J22. This relay is controlled by the display processing computers (DPCs). There are two contacts in this relay, one for channel A and the other for channel B. The alternate power relay must energize to supply the EEC channels with transfer bus 1 power.

Any of these conditions cause the DPCs to energize the engine 1 alternate power relay R576:

- Engine start lever set to idle
- Engine start switch set to ground (GRD)
- Engine start switch set to continuous (CONT)
- Max display system (MDS) EEC maintenance switch is set to on.

The alternator speed logic sensor in the EEC is an additional source for N2 speed.

Operation

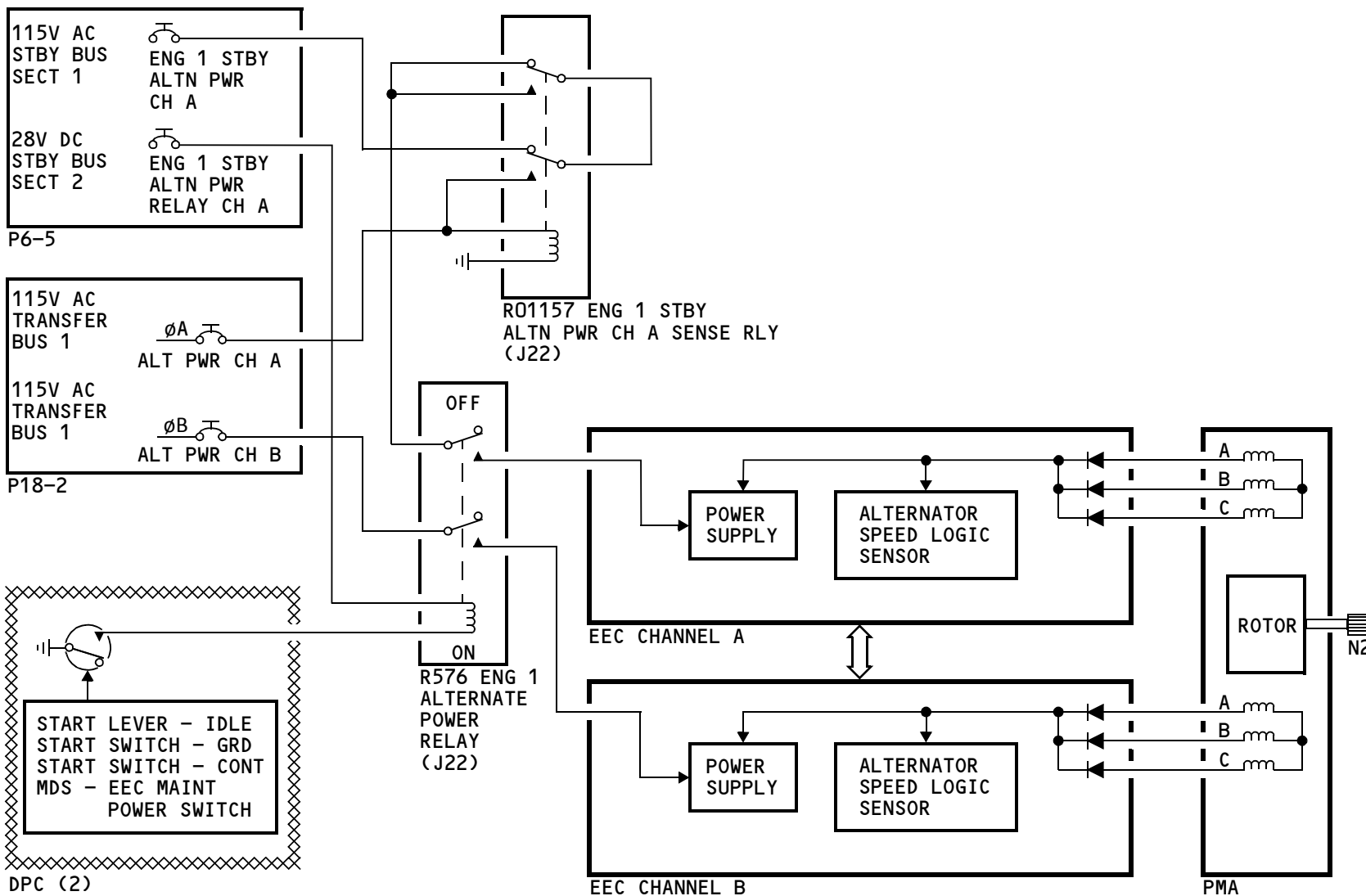
The 115v ac transfer bus 1 supplies power to the EEC when the engine 1 alternate power relay R576 is energized. The EEC uses 115v ac transfer bus power when N2 speed is less than 8.5%. The EEC uses PMA power when N2 speed is more than 8.5% and PMA power quality is good. The EEC does not parallel the AC power inputs.

If the PMA power supply has a failure, that EEC channel receives power from the 115v ac transfer bus. The other EEC channel continues to receive power from the PMA.

If the two PMA power outputs fail, the EEC channels A and B receive power from the 115v ac transfer bus.

The 115v ac standby bus supplies power to EEC channel A when no other power source is available. When 115v ac transfer bus 1 has no power, the engine 1 standby alternate power channel A sense relay R01157 is not energized. When R01157 is not energized, 115v ac standby bus power goes to EEC channel A.

DMC-B737T-A-73-21-00-17A-042A-T_003-00



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - EECs ENGINE 1 POWER SUPPLY- FUNCTIONAL DESCRIPTION

02-Dec-2020

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - EECS ENGINE 2 POWER SUPPLY - FUNCTIONAL DESCRIPTION

Purpose

The permanent magnet alternator (PMA) is the primary ac power supply for the electronic engine controls (EECs) operation. The 115v ac transfer bus 2 can also supply power to the EECs for engine 2.

Functional Description

The PMA has two, three-phase 115v ac power outputs. One output supplies channel A and the other supplies channel B.

The airplane supplies 115v single phase ac power to each EEC channel. 115v ac transfer bus 2 supplies phase A power to EEC channel A. 115V ac transfer bus 2 supplies phase B power to EEC channel B.

The engine 2 alternate power relay R575 is in junction box J24. This relay is controlled by the display processing computers (DPCs). There are two contacts in this relay, one for channel A and the other for channel B. The alternate power relay must energize to supply the EEC channels with transfer bus 2 power.

The PMA power has a higher priority than transfer bus power. The power supply in the EEC channel selects PMA power if it is available. PMA power is available when N2 speed is more than 8.5%.

Any of these conditions cause the DPCs to energize the engine 2 alternate power relay R575:

- Engine start lever set to idle
- Engine start switch set to ground (GRD)
- Engine start switch set to continuous (CONT)
- Max display system (MDS) EEC maintenance switch is set to on.

The alternator speed logic sensor in the EECs is an additional source for N2 speed.

Operation

The ac transfer bus 2 supplies power to the EECs when the engine 2 alternate power relay R575 is energized. The EECs use transfer bus power when N2 speed is less than 8.5%. The EECs use PMA power when N2 speed is more than 8.5% and PMA power quality is good. The EEC does not parallel the ac power inputs.

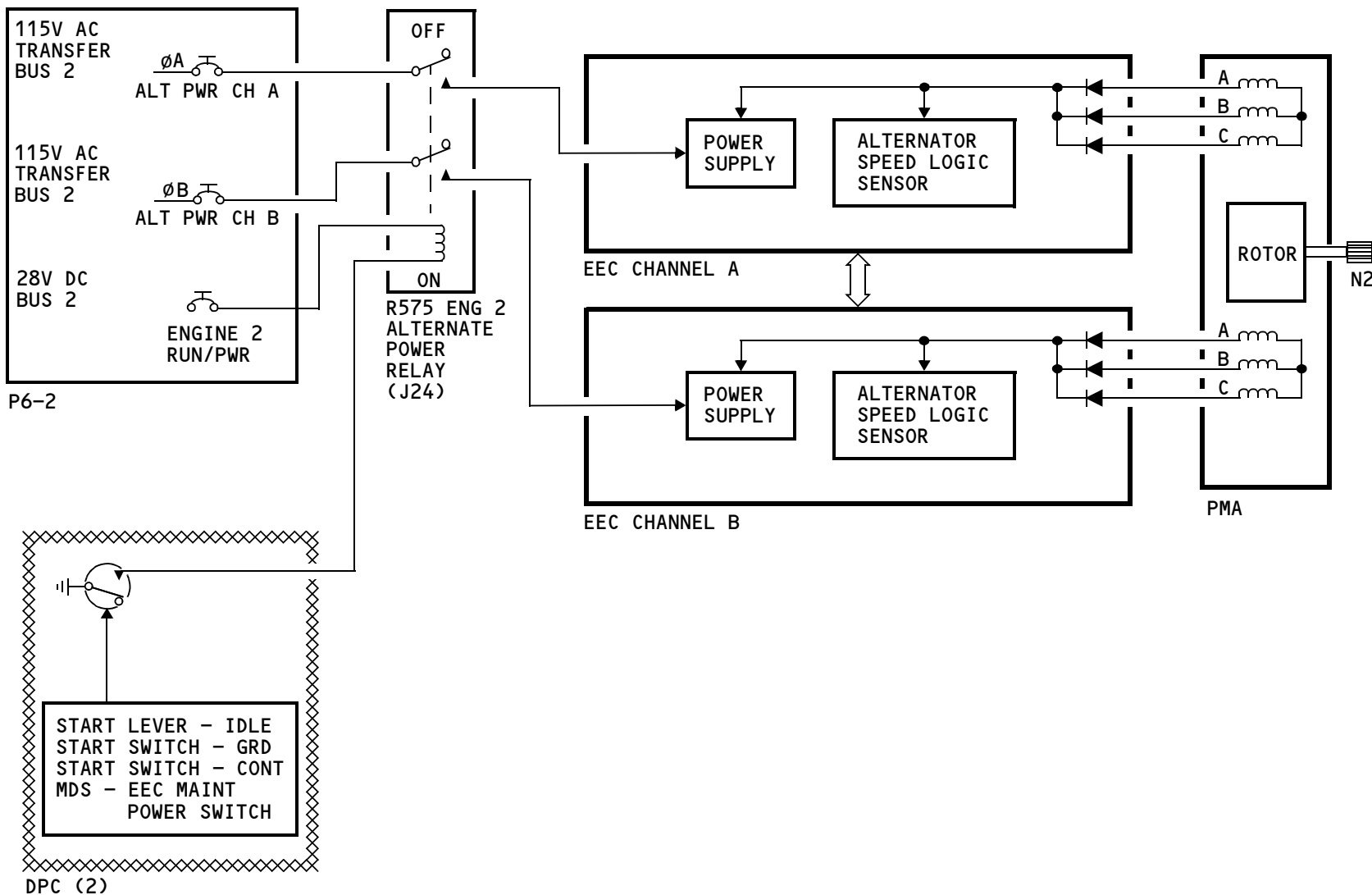
If the PMA power supply has a failure, that EEC channel receives power from the ac transfer bus. The other EEC channel continues to receive power from the PMA.

If the two PMA power outputs fail, the EEC channels A and B receive power from the ac transfer bus.

DMC-B737T-A-73-21-00-06A-042A-T_001-00

02-Dec-2020

73-21-00



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - EECs ENGINE 2 POWER SUPPLY - FUNCTIONAL DESCRIPTION

02-Dec-2020

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - IDLE CONTROL - FUNCTIONAL DESCRIPTION

Engine Idle Control

The electronic engine controls (EECs) control the engine idle speeds (N2 speed). The engine has these idle modes:

- Ground
- Flight
- Approach.

Ground Idle

The ground idle speed that the EEC selects is the one that requires the highest N2 speed. The EEC sets ground idle speed based on the following requirements:

- Minimum N2 speed for IDG operation is 64.5%
- Minimum threshold for PS3
- Minimum threshold for N2.

The EEC commands ground idle 1 second after touchdown, unless the reverse thrust lever has been moved from the stow to the deploy position. With the reverse thrust lever in the deployed position, approach idle is maintained for 8 seconds after touchdown. After 8 seconds the engine goes to ground idle.

Flight Idle

Flight idle is when the airplane is in the air and all of these conditions are true:

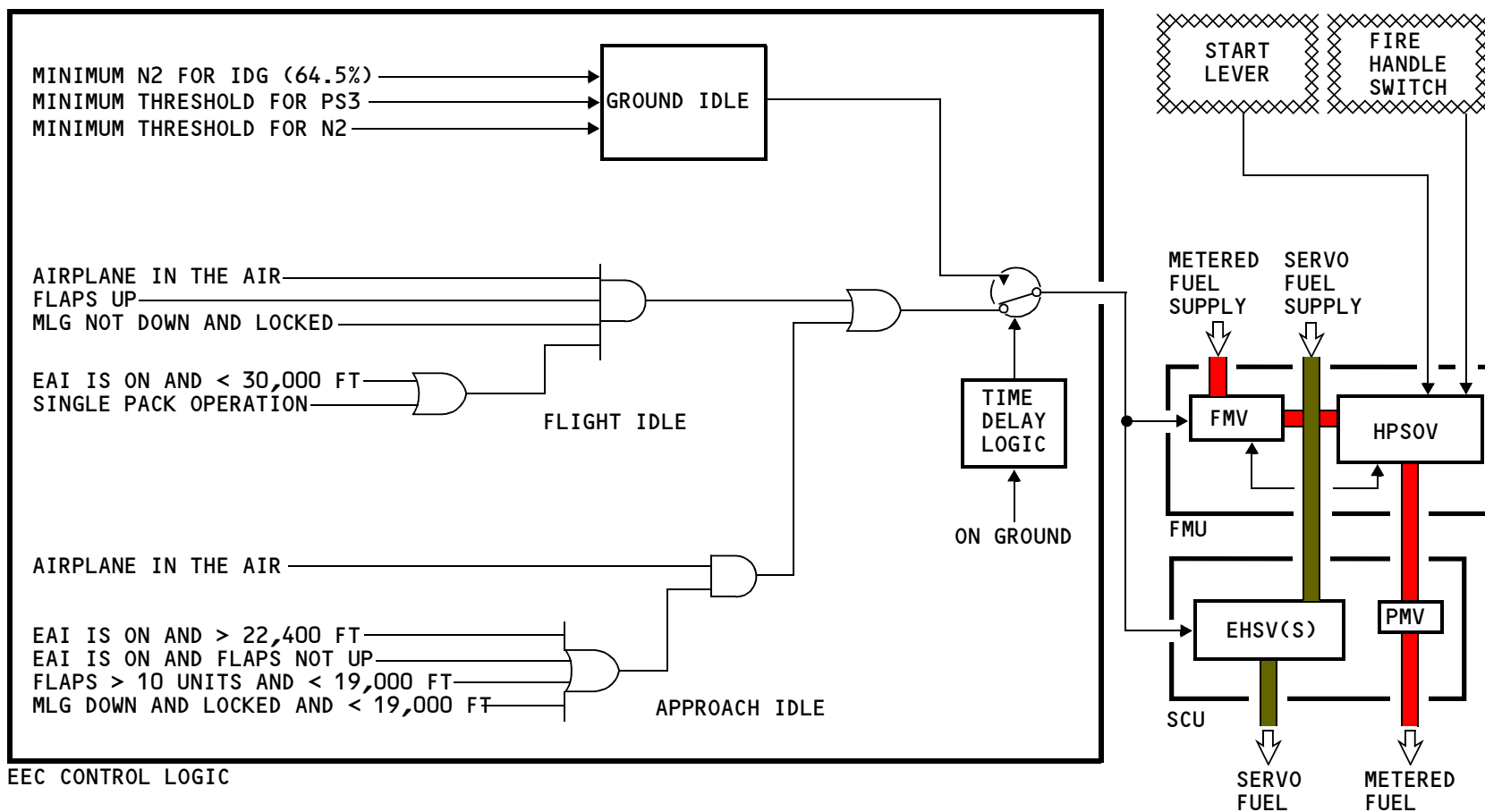
- Flaps are up
- Main landing gear (MLG) is not down and locked
- One of the airplanes engine anti-ice (EIA) systems is ON and the altitude is less than 30,000 feet, OR
- A single air conditioning pack is on (at all altitudes).

Approach Idle

Approach idle keeps the N2 speed high enough to allow quick acceleration of the engine. Approach idle is when the airplane is in the air and one or more of these conditions are true:

- Engine anti-ice (EAI) is ON and the altitude is more than 22,400 feet
- EAI is ON and the flaps are not up
- Flaps are more than 10 units and the altitude is less than 19,000 feet
- MLG is down and locked and the altitude is less than 19,000 feet.

NOTE: There are no mechanical adjustments for idle speeds.



DMC-B737T-A-73-21-00-09A-042A-T_002-00
ICN-B737T-A-732100-A-81205-00045-A-003-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - IDLE CONTROL - FUNCTIONAL DESCRIPTION

02-Dec-2020

73-21-00



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - THRUST CONTROL - FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - THRUST CONTROL - FUNCTIONAL DESCRIPTION

Engine Thrust Control

The electronic engine controls (EECs) use N1 speed to control engine thrust. The EECs have six N1 reference speeds. The N1 reference speeds are at the same thrust lever resolver and thrust lever angles for all engine thrust ratings and airplane models. These are the N1 reference speed names, the thrust lever resolver angles (TRA), thrust lever angles (TLAs), and reverse lever angles (RLA) (note: the angles are in degrees):

- Maximum reverse thrust (8 TRA, 104 RLA)
- Idle reverse (24 TRA, 56 RLA)
- Forward idle stop (36 TRA, 0 TLA)
- Maximum climb (72 TRA, 45 TLA)
- Maximum take-off/go-around rated thrust (78 TRA, 53 TLA)
- Maximum certified thrust (82.5 TRA, 58 TLA).

The EPCS maintenance data page shows the TRA angles.

The EEC channel in control sends command signals to the fuel metering unit (FMU) and the split control unit (SCU). Then the FMU and the SCU operate the engine at commanded N1. The EECs calculate the commanded N1 speed based on this data:

- TRA
- N1 trim
- Engine bleed configuration
- Ambient static pressure (P0)
- Engine thrust rating
- Airplane model
- Mach.

TRA comes from the thrust lever resolvers (TLRs).

N1 trim comes from the data entry plug.

Engine bleed configuration, or the amount of bleed air the airplane takes from the engine comes from the display processing computers (DPCs). If bleed air demand increases, N1 speed decreases to compensate for the additional load. This keeps engine hot section in limits for the current engine thrust rating.

P0 comes from the air data inertial reference units (ADIRUs) and the P0 pressure sensors in the pressure subsystem (PSS).

Engine thrust rating comes from the data entry plug.

The airplane model comes from program pins in the engine strut connector.

Mach is the speed of the air going through the engine divided by the speed of sound at the current ambient conditions.

EEC Mode Description

These are the three EEC operation modes:

- Normal mode
- Soft alternate mode
- Hard alternate mode.

The EECs have a two position mode selection switch which is on the aft P5 panel. The switch positions are ON and alternate.

Normal Mode

The EECs are in normal mode when these conditions are true:

- Total pressure (PT) is valid
- EEC SWITCH on the P5 aft overhead panel is in the ON position.

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - THRUST CONTROL - FUNCTIONAL DESCRIPTION

PT is valid when these conditions are all true:

- PT signal from both ADIRUs are in limits
- PT signals agree
- Pitot probe heat for at least one PT probe is on, or
- Pitot probe heat off, airplane is on the ground and TRA is less than 53 degrees.

In the normal mode the EECs calculate Mach with:

- The two PT values from the ADIRUs
- P0 from the ADIRUs and from the P0 port on the pressure sub-system box.
- The difference between the current ambient temperature and the standard day temperature.

When the airplane is on the ground and for the first 5 minutes of flight the EECs use T12 for current ambient temperature. When the airplane has been in flight for 5 minutes the EECs use T12 and TAT (from the ADIRUs) to calculate current ambient temperature.

The EECs have 2 alternate modes of operation, soft and hard.

Soft Alternate Mode

The EECs go to soft alternate mode when PT is not valid. In the soft alternate mode, the EECs use this data to synthesize mach:

- Total air temperature (TAT)
- P0
- Last valid difference between the current ambient temperature and the standard day temperature.

Hard Alternate Mode

The EECs go to the hard alternate mode when one of these occur:

- EEC is in the soft alternate mode (ALTN light on) and the thrust resolver angle is less than 51.6 degrees
- EEC switch is in the alternate position.

In hard alternate mode, the EECs use this data to synthesize Mach:

- P0
- Assumed corner point temperature
- Engine bleed configuration.

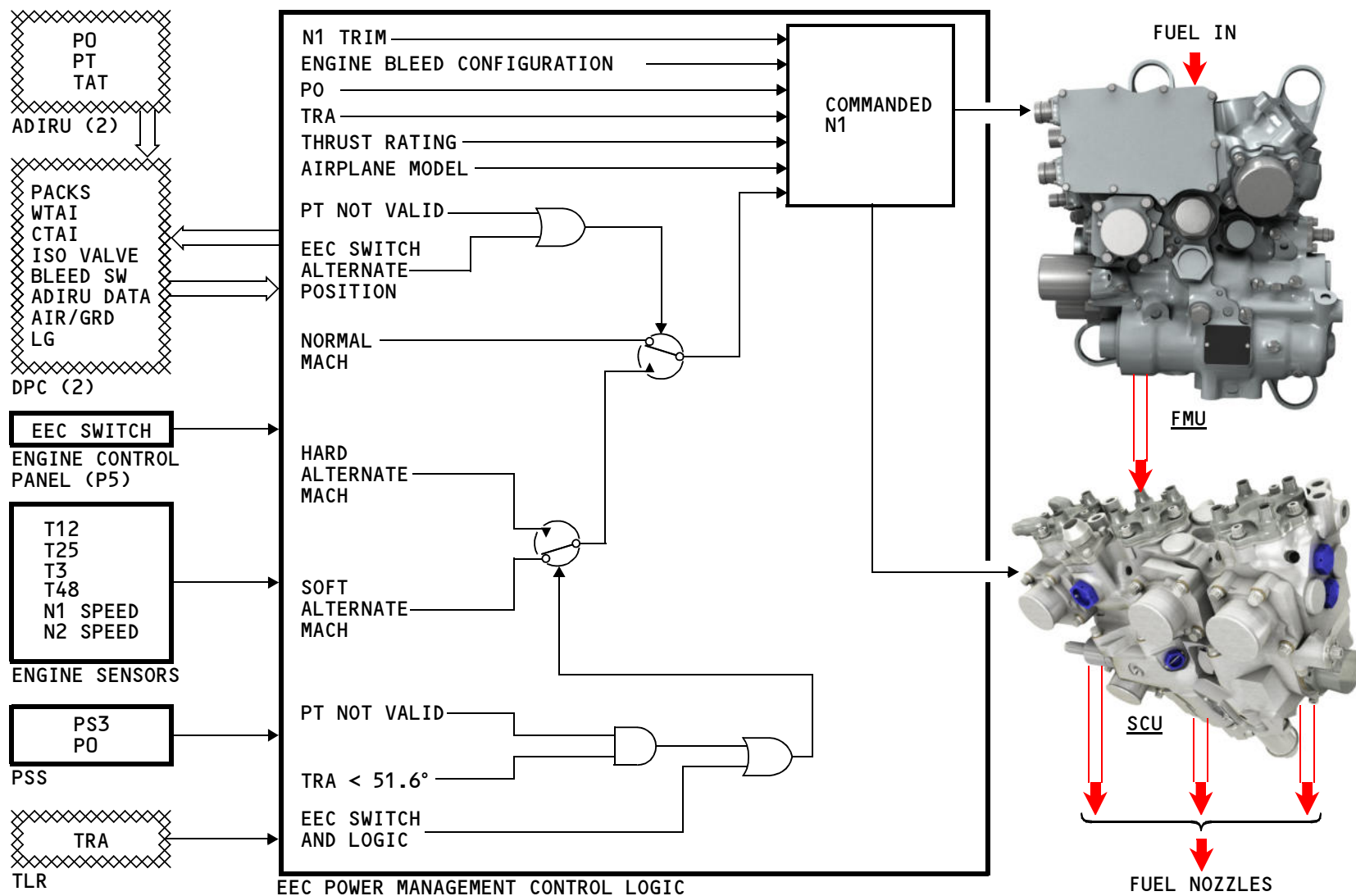
The corner point temperature is standard day temperature plus 15°C or 10°C. The EECs use TRA to select 15°C or 10°C. The corner point temperature is the highest ambient temperature at which the engine can make rated thrust. When the ambient temperature is higher than the corner point temperature, engine thrust will automatically be reduced to protect the engine from exceeding the EGT limit.

In the hard alternate mode, the EECs use the corner point temperature and P0 to synthesize mach. In this mode, EGT exceedances during hot day conditions are possible.

The EECs return to the normal mode when the EEC switch is in the ON position and one of these is true:

- PT is valid and the engine is < idle, OR
- PT is valid and the engine is at or above idle for < 2 seconds and TRA is < 51.6°.

NOTE: If one engine EEC is in the normal mode and the other engine EEC is in an alternate mode, it can cause thrust lever stagger. When this condition occurs, the pilots put both engine EEC switches in the alternate position.



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - THRUST CONTROL - FUNCTIONAL DESCRIPTION

02-Dec-2020

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - EOS AND TCMA

General

The electronic engine controls (EECs) contain the engine overspeed (EOS) and thrust control malfunction accommodation (TCMA) logic. An EOS occurs when the N2 is more than the maximum limit. A TCMA is an uncommanded or uncontrolled high thrust in which the thrust lever command cannot decrease thrust. These protections are active in forward and reverse thrust.

Engine Overspeed (EOS)

An EOS is when the N2 is more than 120% or RPM is more than 20,574. The permanent magnet alternator (PMA) gives an N2 signal through an or gate to the high side switch. The N2 speed sensor gives an N2 signal to an or gate through the low side switch. When the high side and low side switches close, the EEC sends power to the EOS/TCMA electro hydraulic servo valve (EHSV). The EOS/TCMA EHSV uses servo fuel to close the high pressure shutoff valve (HPSOV). When the HPSOV closes the engine stops.

Thrust Control Malfunction Accommodation (TCMA)

The TCMA prevents high asymmetric thrust conditions when the airplane is on the ground. High asymmetric thrust can cause lateral runway excursions.

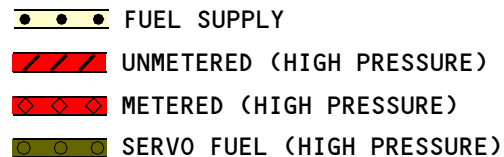
The EEC detects a TCM when:

- Airplane is on ground
- Altitude is less than 17,500 feet
- Calibrated airspeed (CAS) is less than 200 knots
- Selected N1 is more than the TCMA threshold.

When a TCMA is detected the EEC TCMA logic gives a TCMA output to close the high and low side switches. When the high side and low

side switches close, the EEC sends power to the EOS/TCMA EHSV. Then the EOS/TCMA EHSV uses servo fuel to close the HPSOV. When the HPSOV closes the engine stops.

The A/C shutoff solenoid gives servo fuel to the HPSOV to close the valve. The A/C shutoff solenoid receives open/close commands from the start lever and the fire switch.



NOTE: EEC CHANNEL B IS SIMILAR
TO EEC CHANNEL A

DMC-B737T-A-73-21-00-18A-042A-T_003-00
ICN-B737T-A-732100-A-81205-00161-A-003-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - EOS AND TCMA

73-21-00



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - SCU - FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - SCU - FUNCTIONAL DESCRIPTION

General

The split control unit (SCU) is on the engine core at the 7:00 position aft looking forward. The SCU contains the components to control the fuel flow to the fuel manifolds and fuel nozzles. The SCU also contains the electro-hydraulic servo valves (EHSV) that operate the engine air systems.

The electronic engine controls (EECs) receives thrust commands from the thrust lever angle resolver. The EEC channel in control sends control signals to the EHSV (servos) in the SCU. The EHSV in the SCU change the electrical control signals to servo fuel pressure which moves these components:

- Pilot metering valve (PMV)
- High pressure turbine active clearance control (HPTACC) valve
- Low pressure turbine active clearance control (LPTACC) valve
- Modulated turbine cooling (MTC) valve
- Transient bleed valve (TBV)
- Start bleed/booster anti-ice (SB/BAI) valve
- Variable stator vanes (VSVs)
- Variable bleed valves (VBVs).

Each of the above have a position feedback (LVDT/RVDT) to EEC channels A and B.

Some of the actuators and the SCU have a return fuel connection which goes to the jet pump in the fuel metering unit (FMU).

Pilot Metering Valve (PMV)

The PMV gets servo fuel pressure from the PMV EHSV. The EECs are the power supply for the linear variable differential transformer (LVDT) in the PMV. The LVDT sends the PMV position to EEC channels A and B. The PMV divides the combustion fuel into fuel circuits. The EECs

monitor engine data and calculate the correct PMV position. The PMV adjusts the pilot primary and the pilot secondary fuel flows to make the engine combustion more efficient.

The PMV makes:

- Pilot primary main enriched (PPME) fuel
- Pilot primary main non-enriched (PPMNE) fuel
- Pilot secondary (PSEC) fuel.

The PMV supplies PPME and PSEC fuels to 4 enriched fuel nozzles. The 4 enriched nozzles are near the 2 engine igniter plugs. The fuel pressure sensor monitors PPME fuel pressure and supplies this data to the EECs.

The PMV supplies PPMNE and PSEC fuels to 14 non-enriched fuel nozzles. The throttle valve (TV) on the PSEC controls the PSEC pressure to make sure that the fuel splits accurately.

HPTACC and LPTACC

The HPTACC and LPTACC EHSV control the servo fuel pressure for the HPTACC and LPTACC valve operation. The HPTACC and LPTACC systems supply cooling air to the related turbine cases. This optimizes the turbine blade tip to turbine case clearances and increases engine efficiency.

Modulated Turbine Cooling (MTC)

The MTC EHSV controls the servo fuel pressure for the MTC actuators to operate. The MTC supplies cooling air from the high pressure compressor (HPC) stage ten to the high pressure turbine (HPT) blades to optimize cooling air use.

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - SCU - FUNCTIONAL DESCRIPTION

Transient Bleed Valve (TBV)

The TBV EHSV controls the flow of servo fuel pressure to the TBV. The TBV opens HPC stage 10 bleed air into the exhaust to improve the compressor stall margin.

Start Bleed/Booster Anti-ice (SB/BAI)

The SB/BAI EHSV controls the servo fuel pressure for the SB/BAI valve to operate. The SB/BAI controls HPC 7th stage bleed air. For the engine start the SB/BAI uses 7th stage air to improve the stall margin. The SB/BAI also controls the BAI airflow to the engine booster splitter to prevent ice.

Variable Bleed Valves (VBVs)

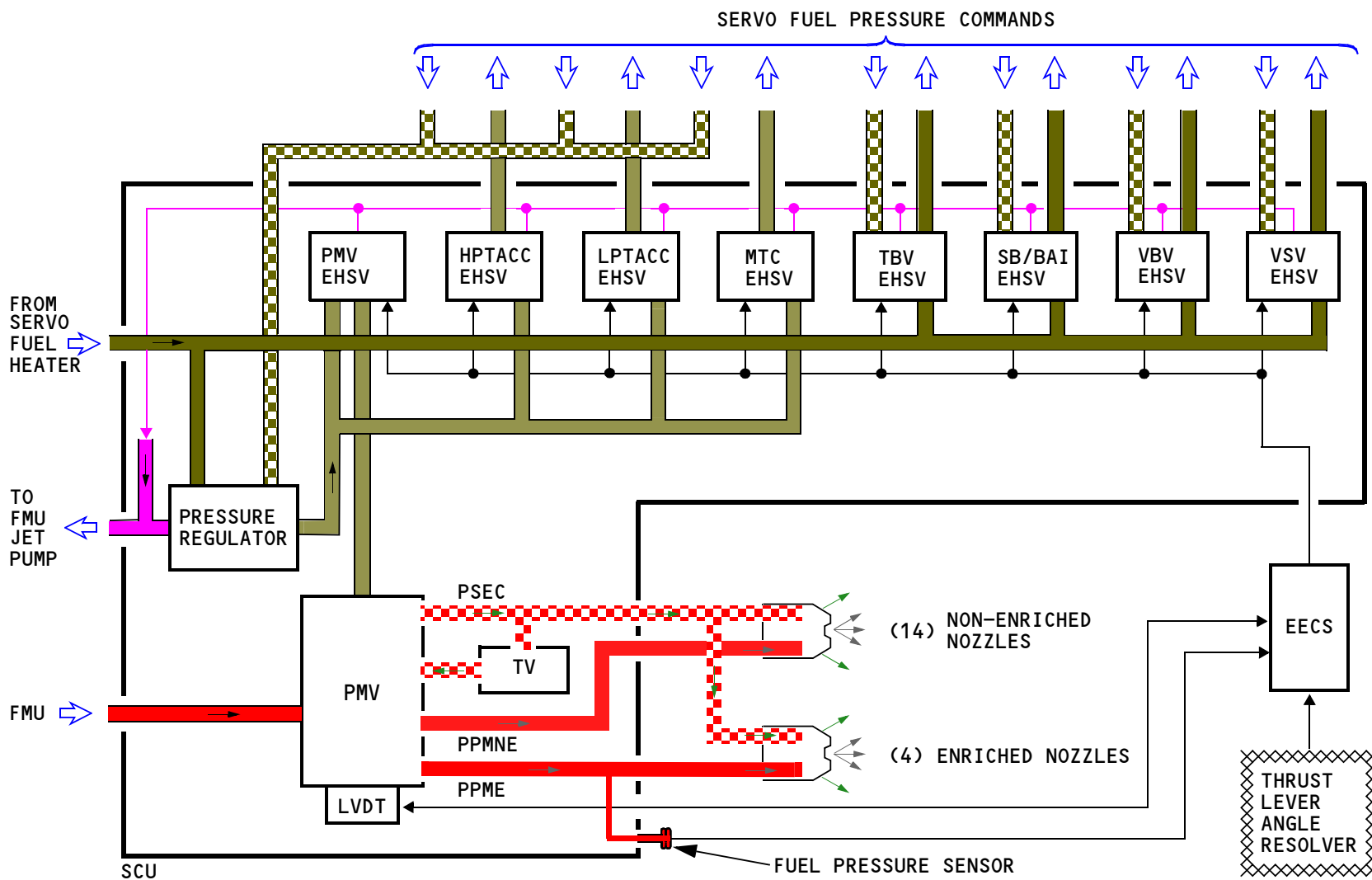
The VBV EHSV controls the servo fuel pressure for the VBV actuators operation. The VBV system:

- Prevents low pressure turbine (LPT) stalls
- Prevents ice accretion and ingestion when the airplane goes through ice crystal icing conditions
- Sends foreign object debris (FOD) overboard to prevent HPC damage.

Variable Stator Vanes (VSVs)

The VSV EHSV controls the servo fuel pressure for the VSV actuators operation. The VSV system controls the quantity of airflow through the HPC. This gives smooth turbulent free airflow through the HPC for all engine speeds, temperatures, and altitudes.

DMC-B737T-A-73-21-00-07A-042A-T_001-00



DMC-B737T-A-73-21-00-07A-042A-T_001-00
ICN-B737T-A-732100-A-81205-00042-A-001-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - SCU - FUNCTIONAL DESCRIPTION

02-Dec-2020

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - FUNCTIONAL DESCRIPTION

General

The electronic engine controls (EECs) control engine operation. This graphic shows the primary interfaces and functions of EEC channel A.

EEC

The EECs are dual redundant line replaceable units (LRUs), channel A and channel B. The two channels (A and B) operate independently and sequentially, and are connected by a cross channel data link (CCDL). Only one channel is in control of the engine. The channel in control alternates between the EECs every flight. The other channel (not in control) is in a standby mode. Signals go between the EEC channels to determine control system health. The standby channel becomes the active channel if a critical fault occurs in the channel in control.

EEC channel A receives these inputs:

- Permanent magnet alternator (PMA) power
- Transfer bus 1 power
- AC standby bus power (for engine 1 EEC channel A power only, also available for engine 1/2 R ignition power)
- Thrust resolver angle (TRA)
- Engine temperatures
- Engine rotor speeds
- Pressure sub-system (PSS) (engine pressures and data entry plug data)
- EEC cross channel data
- Servo system positions
- Fuel flow
- Engine control panel switch positions
- Fuel metering valve (FMV) position
- Thrust reverser (T/R) position
- Start valve position.

EEC channel A send these outputs:

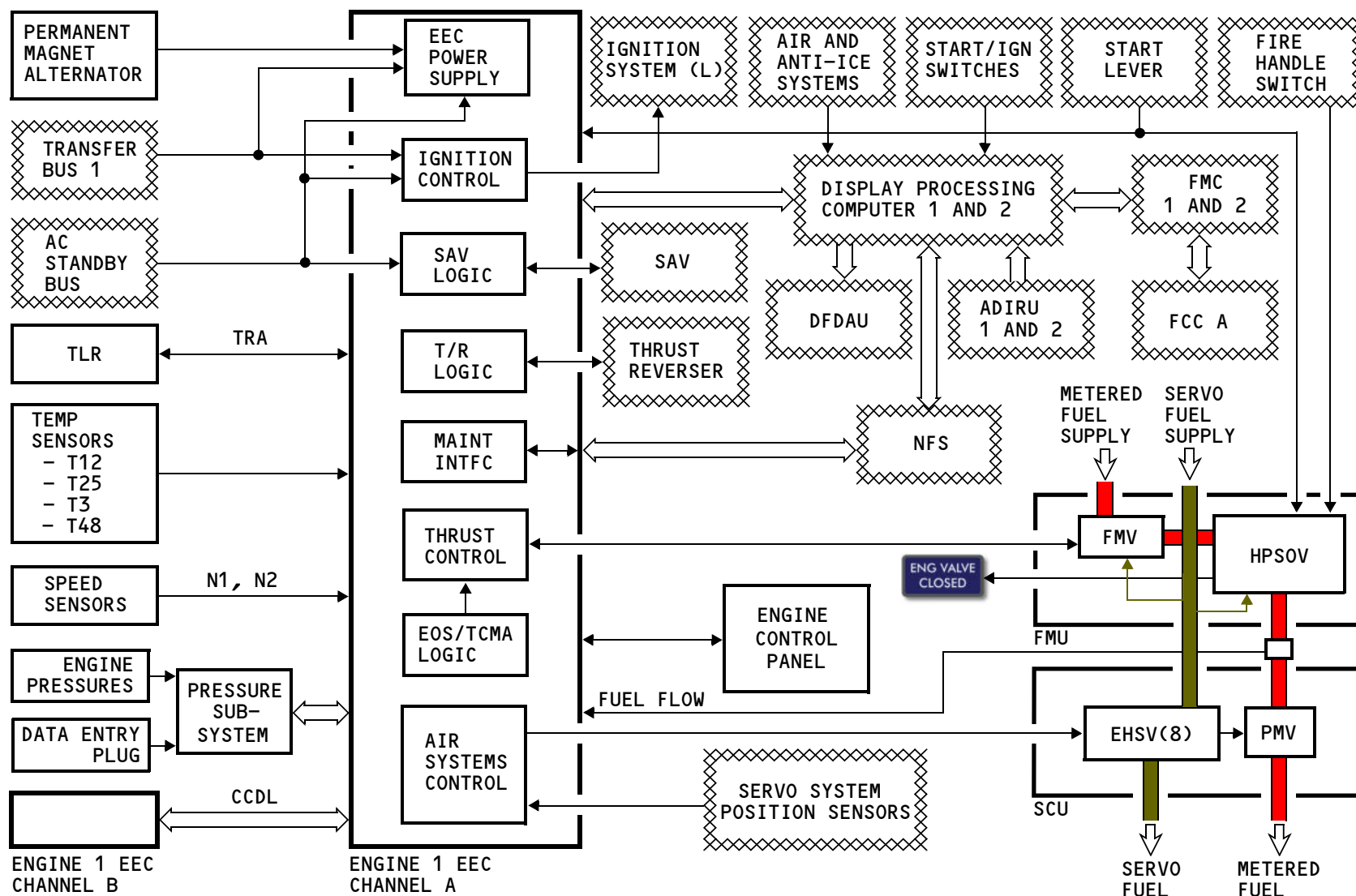
- Ignition system power
- Start valve power
- T/R 1 control valve module command signals and T/R position sensor power.
- Command signals to the electro hydraulic servo valves (EHSVs).

EEC channel A digital interfaces:

- Network file server NFS
- Display processing computer (DPC) 1 and 2.

The EEC channels have control logic for these engine operational functions:

- EEC power control
- Ignition system control
- Start valve control
- T/R control logic (is only in channel A for engine 1 and 2)
- Maintenance interface
- Thrust control
- Engine overspeed and thrust control malfunction accommodation (EOS/TCMA)
- Engine air systems control.



ENGINE FUEL AND CONTROL -- ENGINE CONTROL - FUNCTIONAL DESCRIPTION

73-21-00

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - TRAINING INFORMATION POINT - EEC BITE - INDICATIONS

Fault Information

When an engine electronic control (EEC) finds a fault, the EECs assign it a dispatch level. There are five dispatch levels:

- ENGINE CONTROL light
- ALTERNATE MODE light
- STATUS Message
- Scheduled maintenance task (SMT) Message
- Un-correlated maintenance message.

The EECs set a maintenance message for all detected faults. The onboard maintenance function (OMF) correlates maintenance messages with the appropriate flight deck effect (FDE). An FDE can be a MAX display system (MDS) message or a light in the flight compartment.

ENGINE CONTROL Light Faults

The ENGINE CONTROL light, MASTER CAUTION and ENG annunciator lights come on when an EEC finds a no-dispatch level fault.

Alternate (ALTN) Light Faults

The ALTN light, MASTER CAUTION and ENG annunciator lights come on when an EEC is in an alternate mode. The EECs go to an alternate mode when:

- The pressure total (PT) is not valid, or
- The EEC switch is in the alternate position.

Check the minimum equipment list (MEL)/dispatch deviation guide (DDG) for requirements to dispatch with the ALTN light on.

Status Messages

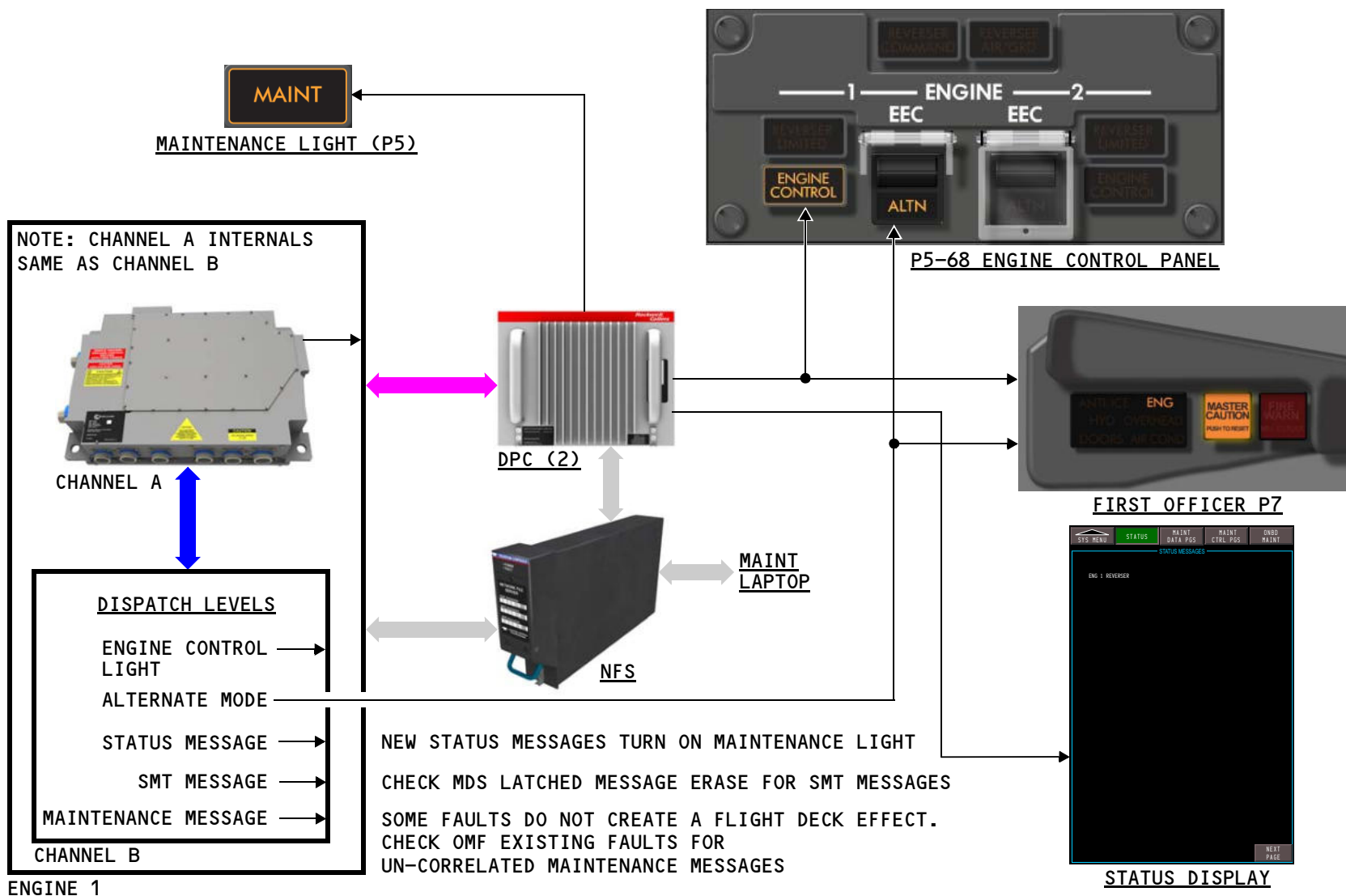
The EECs set a status message when there is a fault in engine operation that may allow continued operation under the restrictions listed in the MEL/DDG. The amount of time allowed for continued operation and any impact on maintenance or flight operations is listed for each status message in the related MEL item. Some status messages may also indicate a no dispatch condition. Status messages show on the MDS status display.

SMT Messages

The EECs set an SMT message when there are faults in engine operation that may allow continued operation under the restrictions listed in the Maintenance Planning Document (MPD). The amount of time allowed for continued operation before a check for SMT messages is necessary is specified in the MPD. SMT messages show on the MDS LATCHED MESSAGE ERASE display, which is one of the MAINT CTRL PGS pages.

Maintenance Messages

The EECs set only a maintenance message when there are faults that allow unrestricted continued operation of the engine. You can only see these un-correlated maintenance messages in the OMF EXISTING FAULTS function (under the EXTENDED MAINT menu). There are no other flight deck effects for these kinds of faults.


DMC-B737T-A-73-21-00-13A-300A-T_004-00
ICN-B737T-A-732100-A-81205-00049-A-004-01

ENGINE FUEL AND CONTROL -- ENGINE CONTROL - TRAINING INFORMATION POINT - EEC BITE - INDICATIONS

02-Dec-2020

73-21-00

ENGINE FUEL AND CONTROL -- OMF - OPERATIONAL TEST

Operational Tests

Operational tests are a type of ground test. You access engine operational tests through the onboard maintenance function (OMF).

Prepare for Operation Tests

To prepare for an engine operational test, select the EEC MAINT POWER soft switch to TEST. You must also set the ground test switch to ENABLE.

Operational Test Access

To access the operational tests:

- Select SYS on the multi-function display (MFD) panel
- The systems display then shows on the inboard display unit (DU)
- Select the MAINT soft key to show the main menu
- Then select ONBD MAINT
- Then select LINE MAINT
- Then select GROUND TESTS
- Then select an ATA System
- Then select OPERATIONAL TEST
- Then select a Test Type.

In this example you select 71-80 Left Engine. The display shows 2 operational tests.

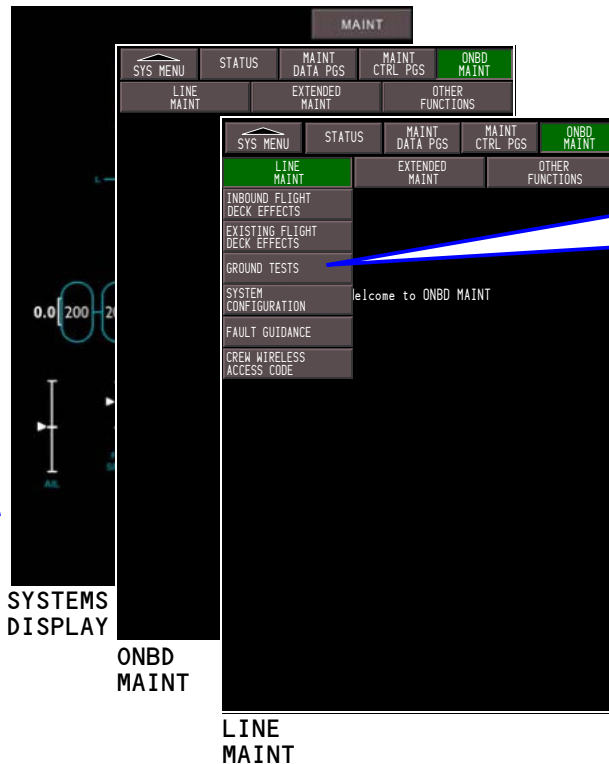
NOTE: The aircraft maintenance manual (AMM) gives instructions to follow for operational tests.

SELECT THE
EEC MAINT POWER
SOFT SWITCH TO
TEST

MOVE THE GND TEST
SWITCH TO
ENABLE



MULTI-FUNCTION PANEL
MFD SWITCHES

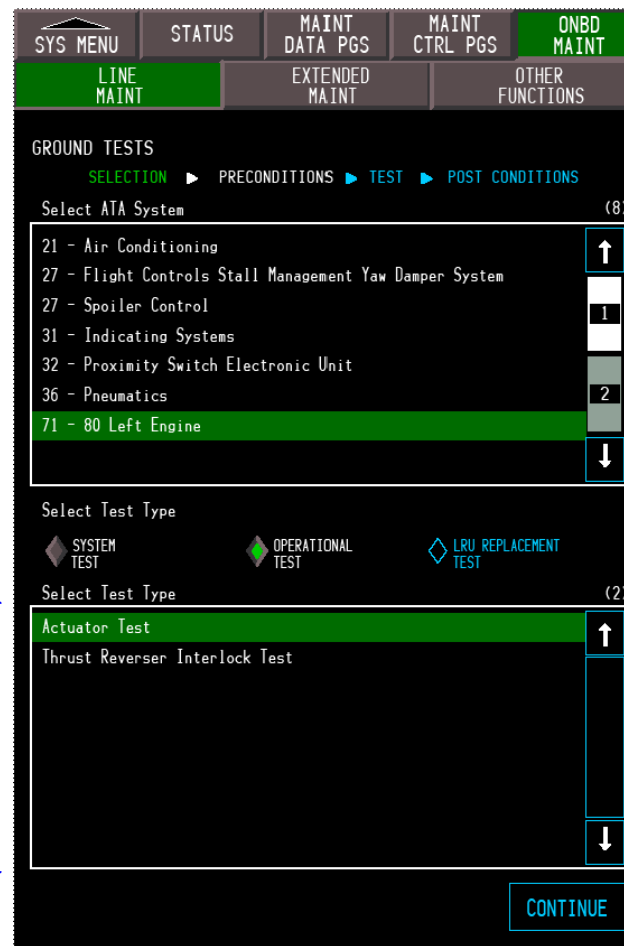


SYSTEMS
DISPLAY

ONBD
MAINT

LINE
MAINT

SELECT
TEST
TYPE



GROUND
TESTS

ENGINE FUEL AND CONTROL -- OMF - OPERATIONAL TEST

73-21-00

ENGINE FUEL AND CONTROL -- OMF - SYSTEM TESTS

System Tests

System tests are a type of ground test. You access engine system tests through the onboard maintenance function (OMF).

Prepare for System Tests

To prepare for an engine system test, select the EEC MAINT POWER soft switch to TEST. You must also set the ground test switch to ENABLE.

System Test Access

To access the system tests:

- Select SYS on the multi-function display (MFD) panel
- The systems display then shows on the inboard display unit (DU)
- Select the MAINT soft key to show the main menu
- Then select ONBD MAINT
- Then select LINE MAINT
- Then select GROUND TESTS
- Then select an ATA System
- Then select SYSTEM TEST
- Then select a Test Type.

In this example you select 71-80 Left Engine. The display shows 7 system tests.

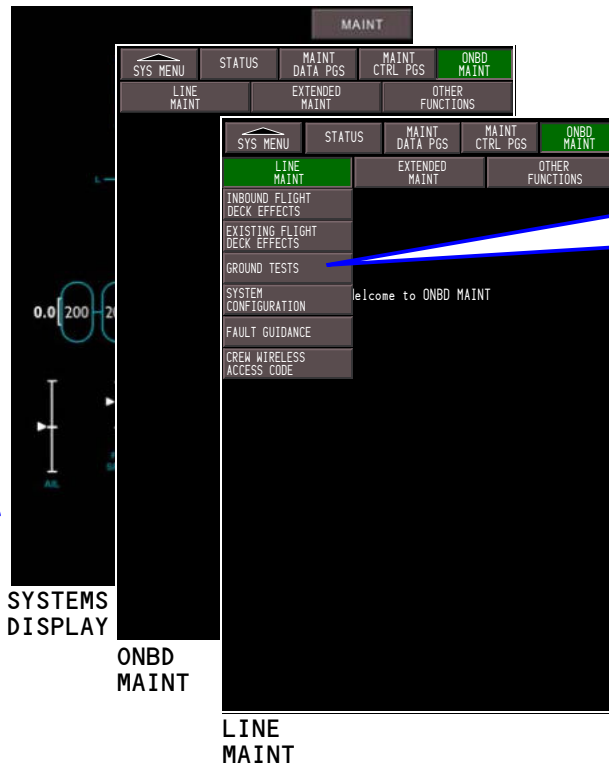
NOTE: The aircraft maintenance manual (AMM) gives instructions to follow for system tests.

SELECT THE
EEC MAINT POWER
SOFT SWITCH TO
TEST

MOVE THE GND TEST
SWITCH TO
ENABLE



MULTI-FUNCTION PANEL
MFD SWITCHES

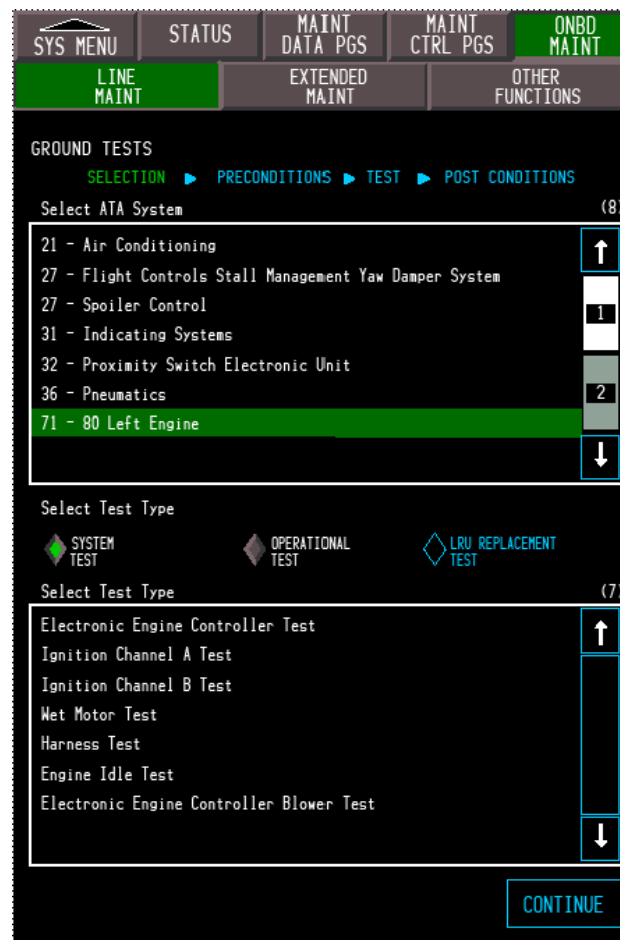


SYSTEMS
DISPLAY

ONBD
MAINT

LINE
MAINT

SELECT
TEST
TYPE



GROUND
TESTS

ENGINE FUEL AND CONTROL -- OMF - SYSTEM TESTS

73-21-00

ENGINE FUEL AND CONTROL -- OMF - OTHER FUNCTIONS

Engine Special Functions

Engine special functions give access to different maintenance tasks. You access engine special functions through the onboard maintenance function (OMF).

- Debris monitoring system chip count entry channel B.

NOTE: Chapter 71 of the AMM gives instructions to follow for each of these special functions.

Prepare for Special Functions

To prepare for an engine special function task, select the EEC MAINT POWER soft switch to TEST. You must also set the ground test switch to ENABLE.

Other Functions

In this example you select 71-80 Left Engine.

These are the available engine special functions on Page 1:

- Fuel system preservation
- Water wash
- Engine serial number entry
- Erase all channel A faults
- Erase all channel B faults
- Thrust reverser cycle
- Engine running simulation
- Lamp check out.

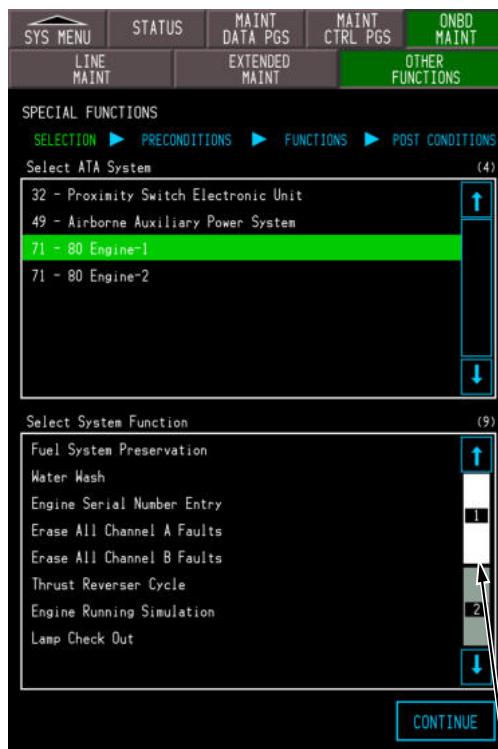
These are the available engine special functions on Page 2, with the EEC OFF:

- Starter air valve manual start.

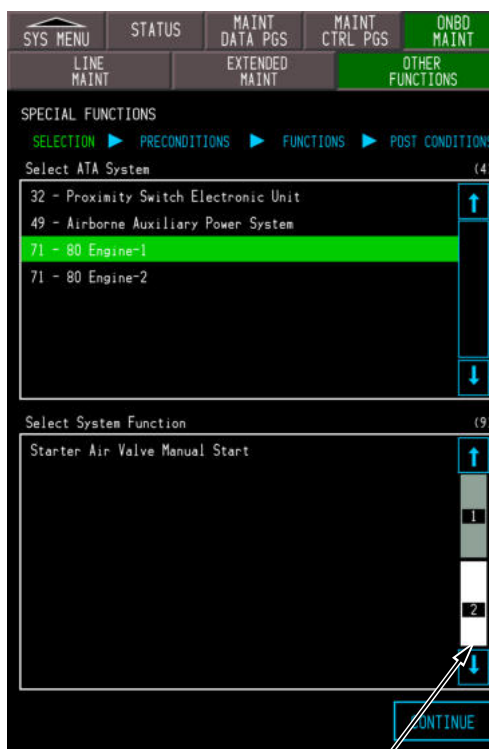
These are the available engine special functions on Page 2, with the EEC ON:

- Starter air valve manual start
- Menu mode hardware code entry

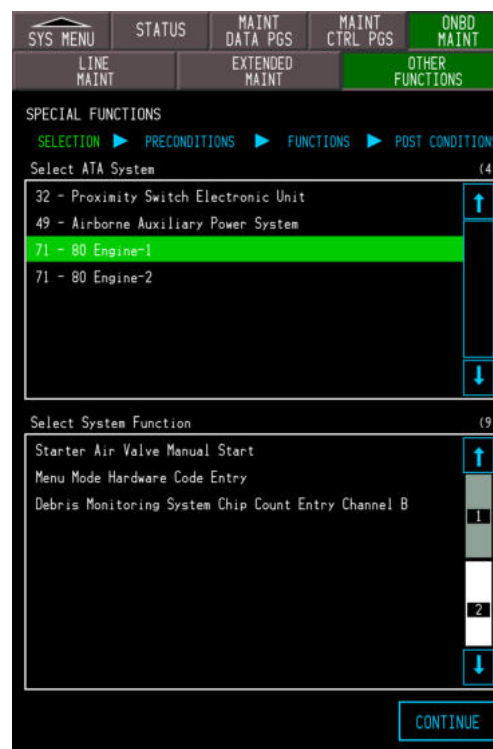
DMC-B737T-A-73-21-00-11A-300A-T_004-00



OTHER
FUNCTIONS
(EEC-ON AND EEC-OFF)



OTHER
FUNCTIONS
(EEC-OFF)



OTHER
FUNCTIONS
(EEC-ON)

PAGE 1 OF 2 PAGE 2 OF 2

ENGINE FUEL AND CONTROL -- OMF - OTHER FUNCTIONS

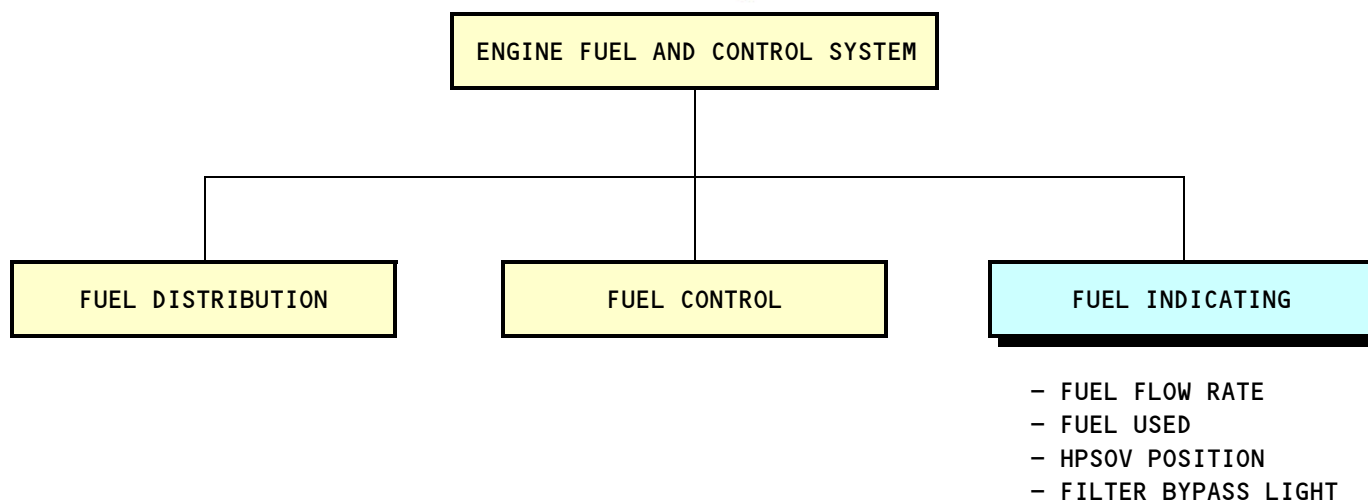
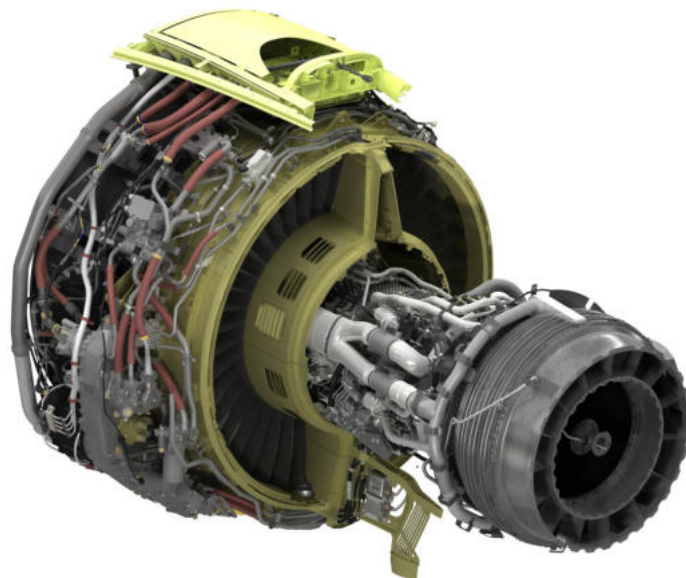
73-21-00

ENGINE FUEL AND CONTROL -- FUEL INDICATING - INTRODUCTION

General

The fuel indication system has these indications:

- Fuel flow rate
- Fuel used
- High pressure fuel shutoff valve (HPSOV) position
- Fuel filter bypass light.



ENGINE FUEL AND CONTROL -- FUEL INDICATING - INTRODUCTION

73-30-00

DMC-B737T-A-73-30-00-02A-018A-T_001-00
ICN-B737T-A-733000-A-81205-00005-A-001-01

02-Dec-2020

ENGINE FUEL AND CONTROL -- FUEL INDICATING - GENERAL DESCRIPTION

Fuel Indicating System Components

These are some of the fuel indicating system components:

- Fuel flow transmitter
- FUEL FLOW display switch
- Fuel filter differential sensor
- High pressure shutoff valve (HPSOV).

The flow transmitter supplies fuel flow data to the electronic engine controls (EECs), which then send the data to the display processing computers (DPCs). The EECs calculate the fuel flow rate, and the DPCs calculate fuel used. The fuel flow data shows on the engine indication display. The engine indication display usually shows on the inboard side of the left inboard display unit (DU).

The FUEL FLOW display switch is on the multi-function panel. The multi-function panel is on the P9 panel in the flight compartment. These are the 3 positions of the FUEL FLOW display switch:

- RESET (momentary)
- RATE (neutral)
- USED (momentary).

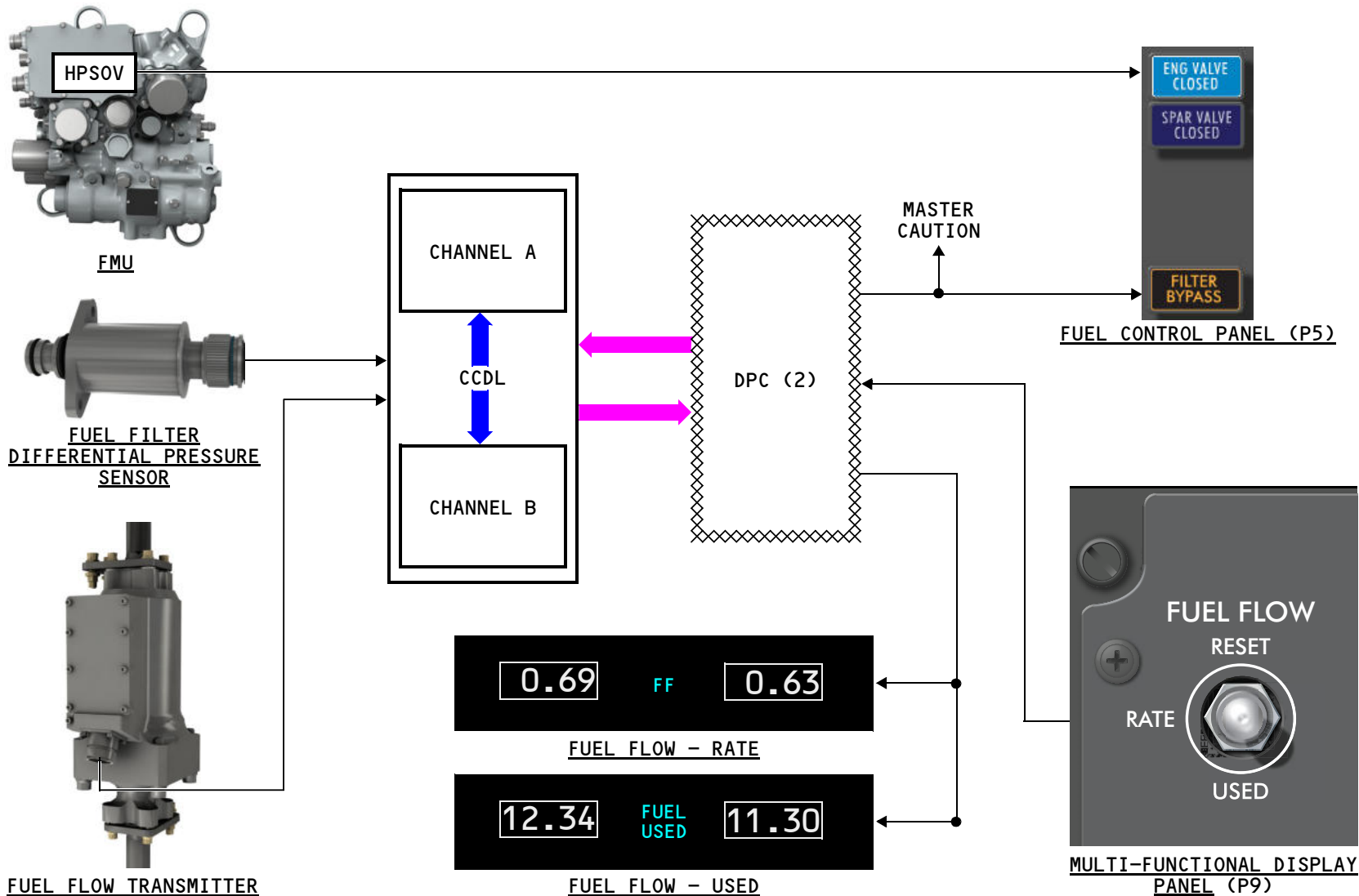
The fuel flow indication switch is usually in the center or RATE position. A spring sets the switch back to RATE from the other 2 positions.

The RATE position shows real-time fuel flow data. The USED position shows the quantity of fuel used since the last reset. The used indication goes back to the rate indication 10 seconds after the indication control switch goes back to RATE. If the switch is held for more than 30 seconds, the display changes back to fuel flow.

The RESET position sets the fuel used display digital counter to zero. The counter also sets to zero when electrical power to the DPCs is removed and reconnected.

The EECs monitor fuel filter differential pressure with the fuel filter differential sensor. Filter differential pressure is the difference between the filter inlet pressure and outlet pressure. If the differential pressure is too high, a bypass signal goes to the EECs, and then to the DPCs. The DPCs cause the amber FILTER BYPASS, MASTER CAUTION and FUEL annunciator lights to come on.

The blue ENGINE VALVE CLOSED light shows the position of the HPSOV. With the valve closed, the light is on dim. When the valve is open, the light is off. The light comes on bright when valve position and switch position disagree.



DMC-B737T-A-73-30-00-02A-040A-T_003-00
ICN-B737T-A-733000-A-81205-00006-A-003-01

ENGINE FUEL AND CONTROL -- FUEL INDICATING - GENERAL DESCRIPTION

02-Dec-2020

73-30-00

ENGINE FUEL AND CONTROL -- FUEL INDICATING - FUEL FILTER DIFFERENTIAL PRESSURE SENSOR

General

The fuel filter differential pressure sensor measures the main fuel filter (MFF) differential pressure. The main fuel oil/heat exchanger (MHX) delta pressure sensor measures the fuel differential pressure across the heat exchanger. The electronic engine controls (EECs) monitor the sensors for a bypass signal. The EECs send a bypass signal to the display processing computers (DPCs). The DPCs cause the amber FILTER BYPASS, MASTER CAUTION, and FUEL lights to come on.

Location

The fuel filter differential pressure sensor is on the MFF assembly. The MHX delta pressure sensor is on the MHX. You open the left fan cowl to access the MFF and the MHX assemblies.

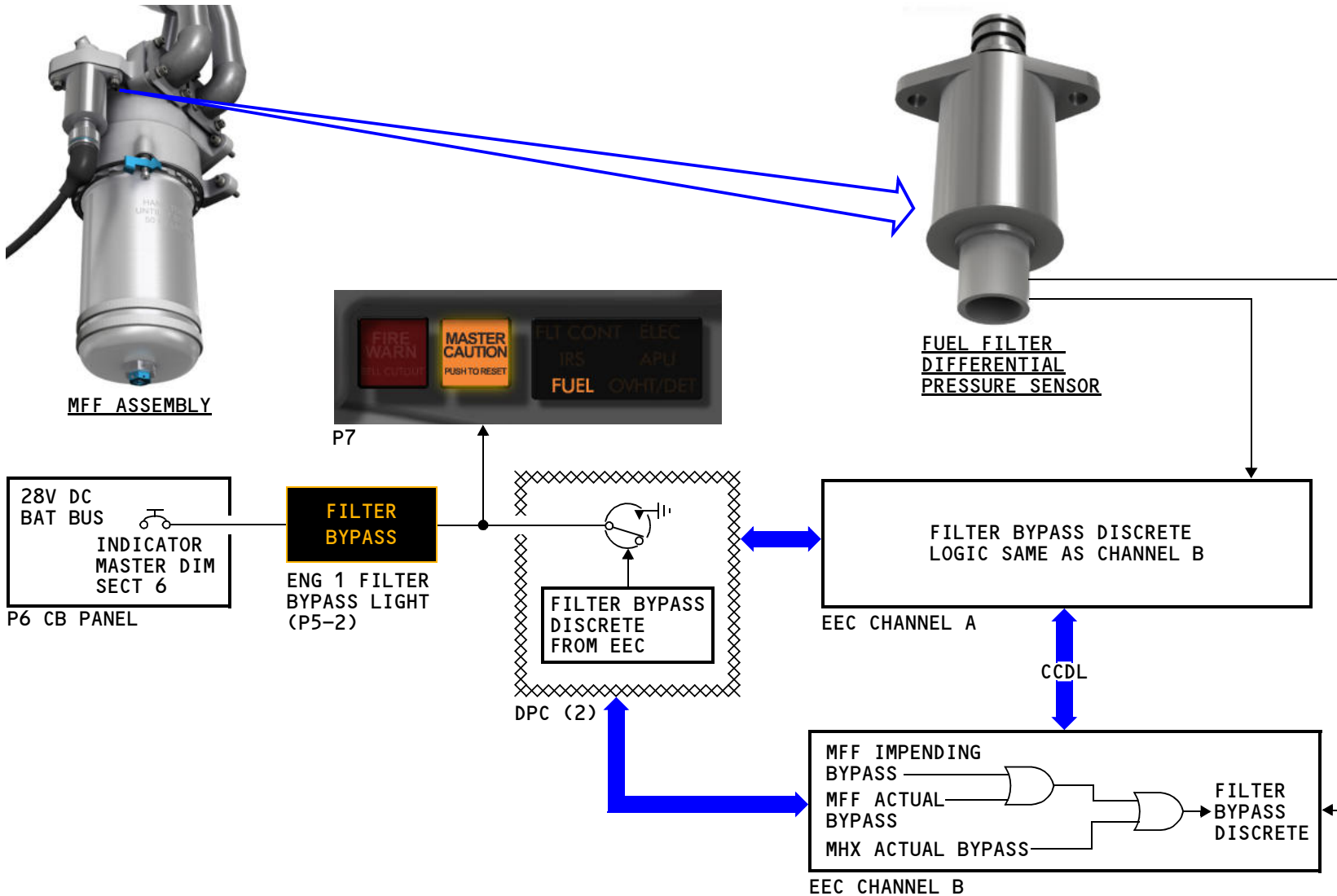
Functional Description

The EECs sense an impending fuel filter bypass when the differential pressure is between 16.5 psid and 47.4 psid. The EECs sense an actual fuel filter bypass when the differential pressure is >47.4 psid. The EECs sense a MHX bypass when the delta pressure is >22.6 psid. For any of these conditions, the EECs send a signal to the DPCs. The DPCs supply a ground which causes the amber FILTER BYPASS, MASTER CAUTION, and FUEL lights to come on.

DMC-B737T-A-73-30-00-04A-110A-T_002-00

02-Dec-2020

73-30-00



ENGINE FUEL AND CONTROL -- FUEL INDICATING - FUEL FILTER DIFFERENTIAL PRESSURE SENSOR

73-30-00

ENGINE FUEL AND CONTROL -- FUEL INDICATING - FUEL FLOW TRANSMITTER

Purpose

The fuel flow transmitter measures the fuel mass-flow that goes to the split control unit (SCU). The SCU gives fuel to the fuel manifolds and fuel nozzles. This data goes to the electronic engine controls (EECs). The EECs send the data to the display processing computers (DPCs).

Location

The fuel flow transmitter is on the fan case at the 11:00 position. To get access to the fuel flow transmitter, open the left fan cowl.

Functional Description

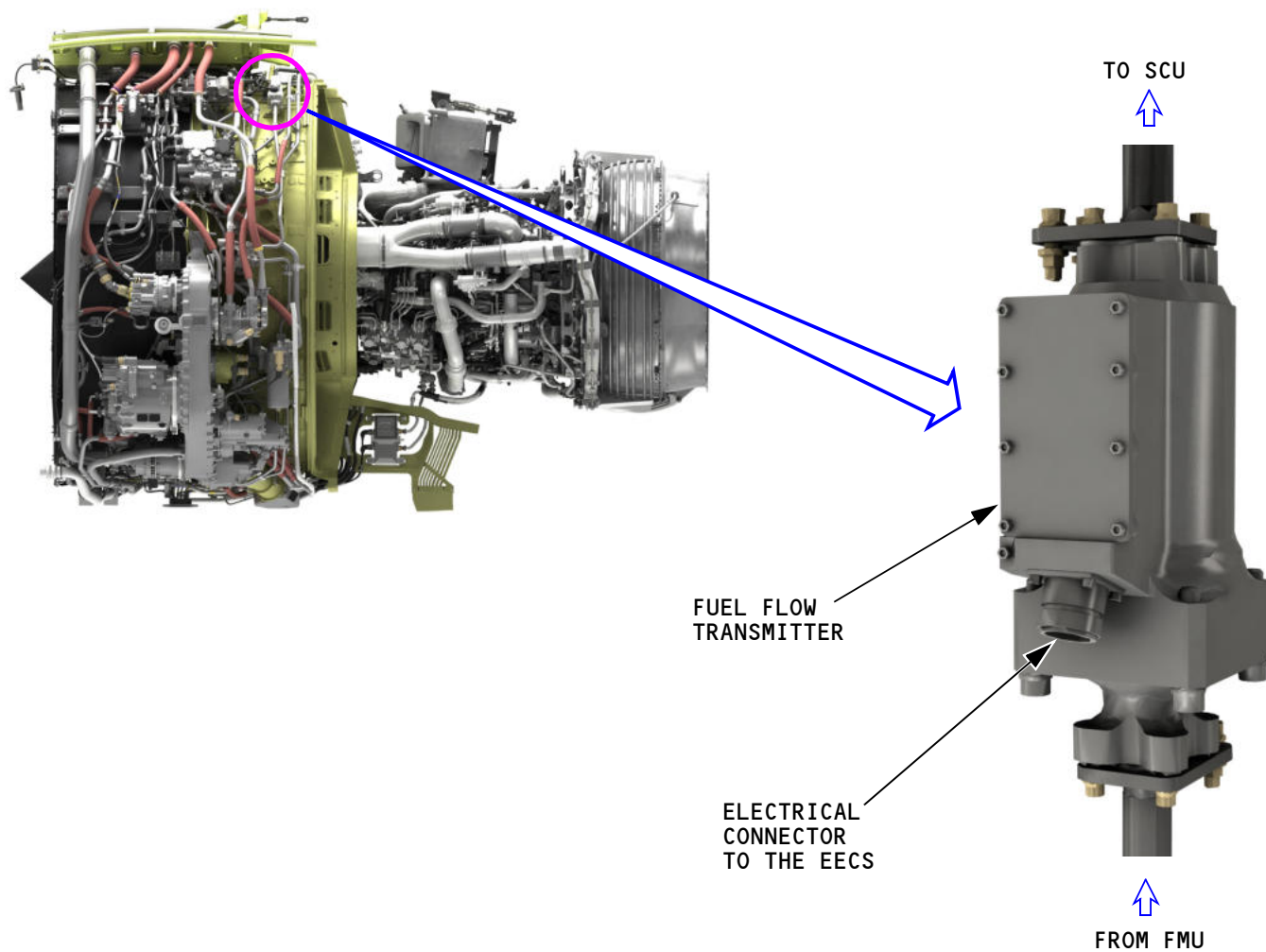
Fuel flows from the fuel metering unit (FMU) to the fuel flow transmitter. The fuel flow rotates the aperture in the fuel flow transmitter. There are 2 magnets that attach to the aperture. The fuel flow transmitter has 2 separate windings. The rotating magnets make an electric pulse in each of the windings. The fuel flow transmitter sends the pulses to the EECs. The EECs measure the time interval between the pulses to calculate fuel flow.

NOTE: When the DPCs cannot find fuel flow data, or the fuel flow data is not correct, the fuel used stays at the last calculated value and fuel flow blanks.

DMC-B737T-A-73-30-00-03A-110A-T_002-00

02-Dec-2020

73-30-00



DMC-B737T-A-73-30-00-03A-110A-T_002-00
ICN-B737T-A-733000-A-81205-00007-A-001-01

ENGINE FUEL AND CONTROL -- FUEL INDICATING - FUEL FLOW TRANSMITTER

73-30-00

02-Dec-2020

ENGINE AIR -- INTRODUCTION

Purpose

The engine air system has these control functions:

- Engine cooling
- Compressor airflow.

Engine Cooling

Engine cooling has these functions:

- High pressure turbine active clearance control (HPTACC)
- Low pressure turbine active clearance control (LPTACC)
- Modulated turbine cooling (MTC).

Compressor Airflow Control

The compressor airflow control system has these functions:

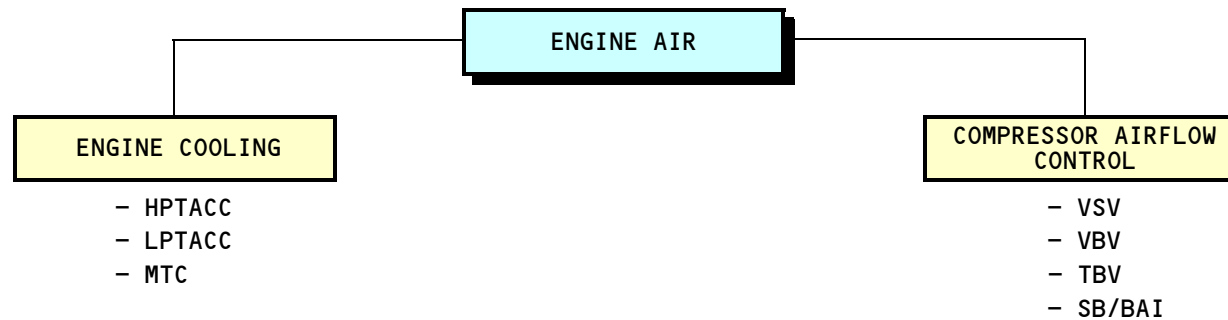
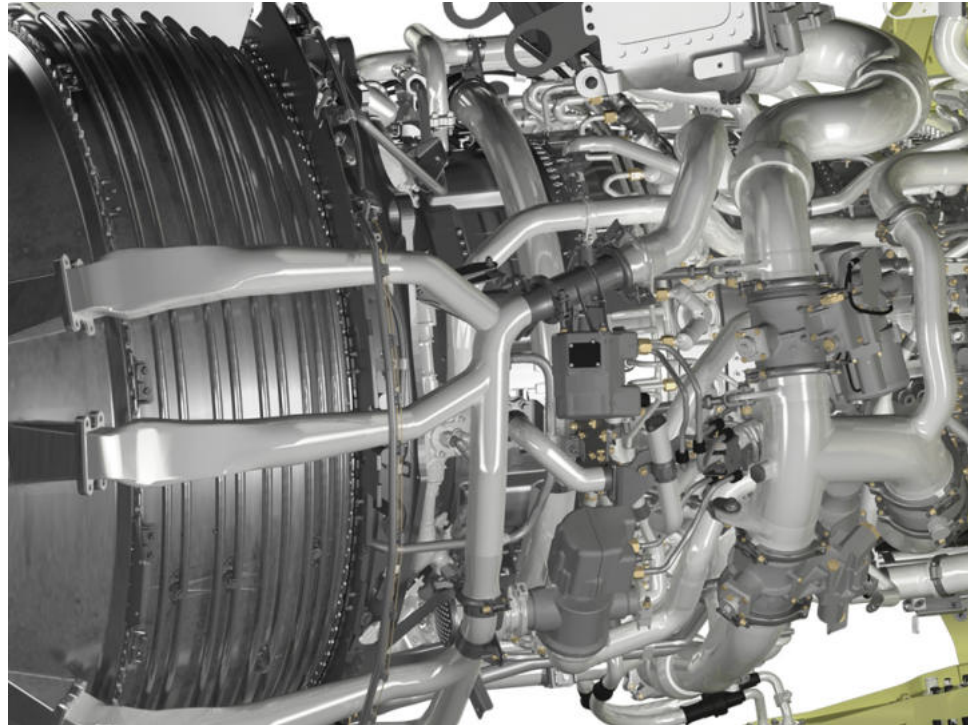
- Variable stator vanes (VSVs)
- Variable bypass valves (VBVs)
- Transient bleed (TBV)
- Start bleed/booster anti-ice (SB/BAI).

Abbreviations and Acronyms

ADIRU	- air data inertial reference unit
CCC	- core compartment cooling
DPC	- display processing computer
EEC	- electronic engine control
EGT	- exhaust gas temperature
HPC	- high pressure compressor
HPT	- high pressure turbine
HPTACC	- high pressure turbine active clearance control
IGV	- inlet guide vane
LPC	- low pressure compressor

LPT	- low pressure turbine
LPTACC	- low pressure turbine active clearance control
LVDT	- linear variable differential transformer
MTC	- modulated turbine cooling
P0	- aircraft static air pressure
PT	- aircraft total air pressure
RVDT	- rotary variable differential transformer
SB/BAI	- start bleed/booster anti-ice
SCU/SVA	- split control unit/servo valve assembly
TAT	- aircraft total air temperature
TBV	- transient bleed valve
TRA	- thrust lever resolver angle
T3	- compressor discharge air temperature
T25	- HPC inlet air temperature
VBVs	- variable bleed valves
VSVs	- variable stator vanes

DMC-B737T-A-75-00-00-02A-018A-T_003-00



ENGINE AIR -- INTRODUCTION

75-00-00

DMC-B737T-A-75-00-00-02A-018A-T_003-00
ICN-B737T-A-750000-A-81205-00004-A-002-01

02-Dec-2020

ENGINE AIR -- GENERAL DESCRIPTION

General

The EECs send signals to the servos in the split control unit (SCU) to control the engine air system. There are different servos for each of the engine air functions. The SCU servos use fuel pressure to move the engine air system valves and actuators. LVDTs or RVDTs give actuator position information to the EECs for system control feedback. Position data shows on the MDS EPCS maintenance data pages.

The engine air system has two primary subsystems:

- Engine cooling
- Compressor airflow control.

The engine cooling subsystem has these functions:

- High pressure turbine active clearance control (HPTACC)
- Low pressure turbine active clearance control (LPTACC)
- Modulated turbine cooling (MTC).

The compressor airflow control subsystem has these functions:

- Variable bleed valves (VBV)
- Variable stator vanes (VSV)
- Transient bleed (TBV)
- Start bleed/booster anti-ice (SB/BAI).

Engine Cooling Subsystem

The HPTACC system controls cooling air flow to the external parts of the high pressure turbine (HPT) case. This cooling air adjusts the clearance between the HPT shrouds and rotor blades. The smaller clearance helps the engine use less fuel and controls EGT increase during fast acceleration.

The LPTACC system controls cooling air flow to the external parts of the low pressure turbine (LPT) case. This cooling air keeps the

clearance between the LPT shrouds and rotor blades small. The smaller clearance helps the engine use less fuel.

The MTC system controls cooling air flow to the HPT first stage to help the engine use less fuel. The MTC system sends maximum cooling air to the HPT first stage during high-power operations. During lower-power operations, the MTC system modulates cooling air flow.

Compressor Airflow Control Subsystem

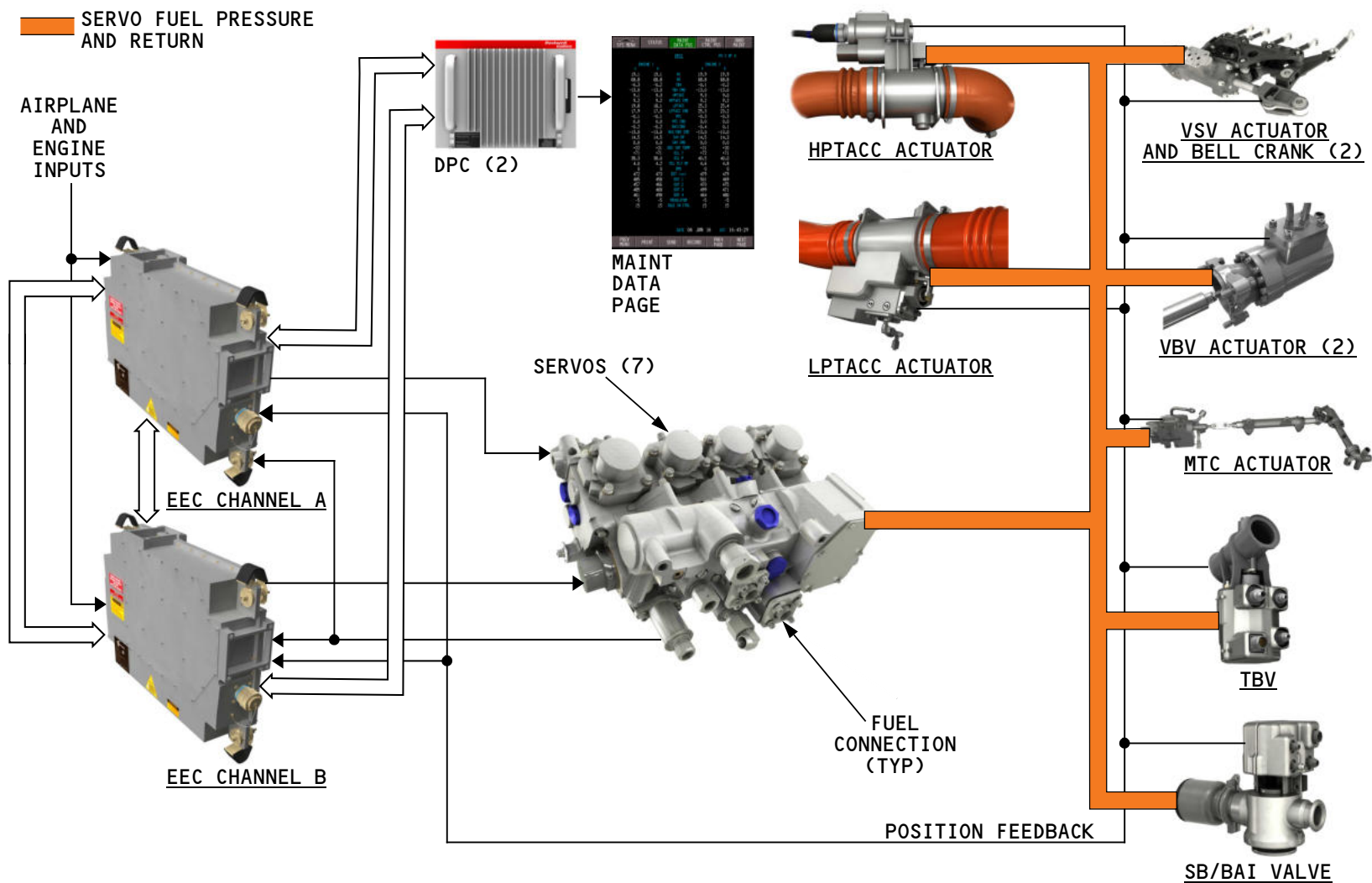
The VBV system controls the amount of air in the primary gaspath. This provides the correct amount of air for combustion. To do this the system uses 2 actuators and 8 doors.

The VSV system makes sure that airflow through the HPC is smooth at all engine speeds. The system controls the first 5 stages of HPC stator vanes (this includes the IGVs). The system also uses 2 actuators and 2 bell crank assemblies.

The TBV system lets air from the 10th stage of compression go to the engine exhaust. This ensures proper operability at all engine speeds.

The SB/BAI valve releases 7th stage compressor air during engine air starts (SB). This decreases air loads for the N2 rotor. The valve also provides anti-ice air to the engine booster (LPC) inlet (BAI).

DMC-B737T-A-75-00-00-02A-040A-T_002-00



DMC-B737T-A-75-00-00-02A-040A-T_002-00
ICN-B737T-A-750000-A-81205-00005-A-005-01

ENGINE AIR -- GENERAL DESCRIPTION

02-Dec-2020

75-00-00

ENGINE AIR -- LEAP-1B ENGINE - COMPONENT LOCATION - LEFT SIDE

Active Clearance Control (ACC) Valves

These active clearance control (ACC) valves are on the left side of the engine core section:

- High pressure turbine active clearance control (HPTACC) valve
- Low pressure turbine active clearance control (LPTACC) valve

Compressor Airflow Control Components

These compressor airflow control components are on the left side of the engine:

- Variable bleed valve (VBV) actuator (7:00 position of the fan hub rear face)
- Variable stator vane (VSV) actuator (9:00 position of the engine compressor case).

The VBVs and the VSVs are on the engine's circumference, perpendicular to its longitudinal axis. The VBVs are in the engine fan hub area. The VSVs are on the front of the high pressure compressor case.

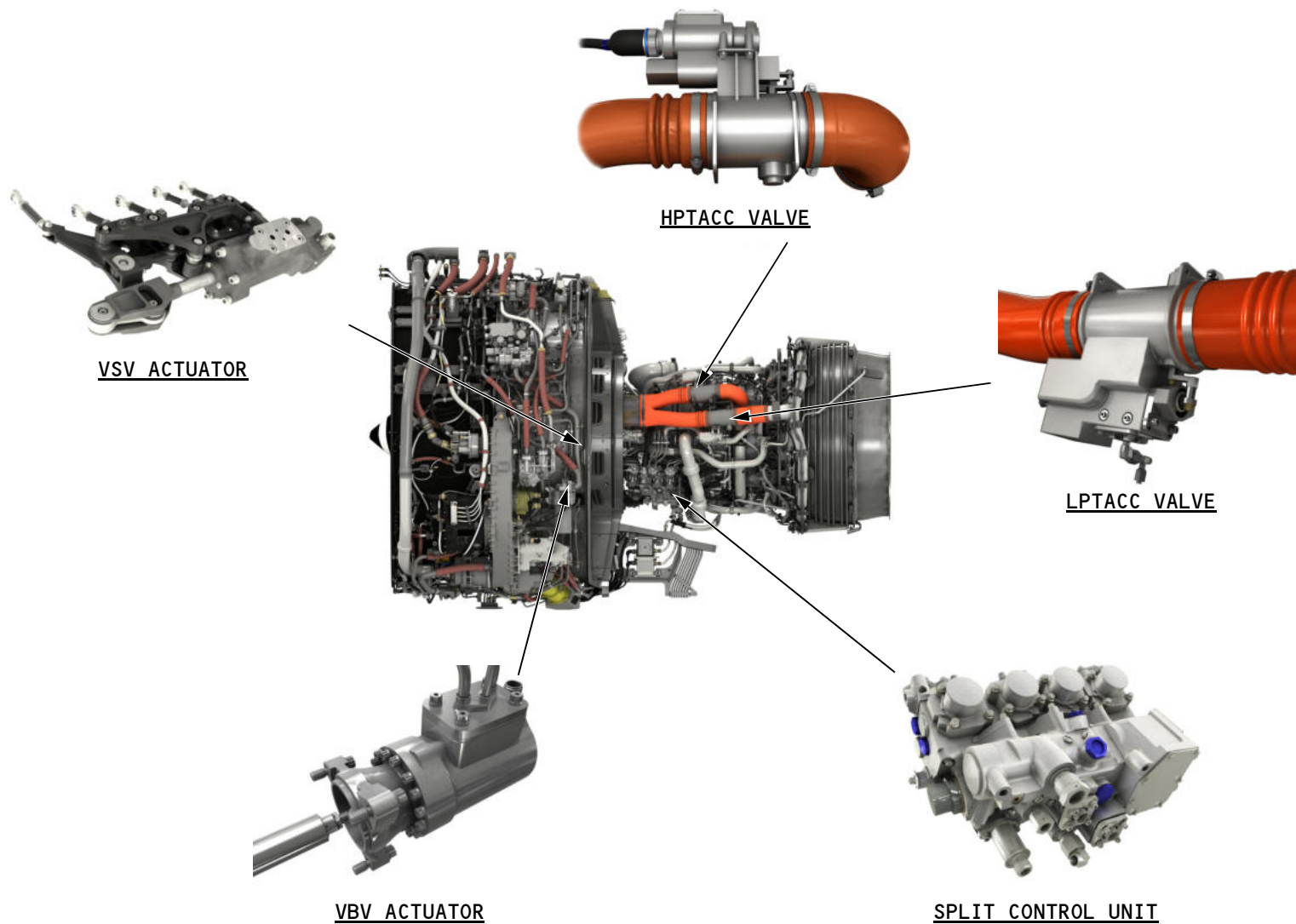
Modulated Turbine Cooling (MTC) Actuator and Valve Assembly

One of the two MTC actuator and valve assemblies is on the left side of the engine core at 8:00. You cannot easily see the valve because it is in the engine.

Split Control Unit (SCU)

The SCU is on the left side of the engine core at the 8:00 position.

DMC-B737T-A-75-10-00-01A-055A-T_002-00



ENGINE AIR -- LEAP-1B ENGINE - COMPONENT LOCATION - LEFT SIDE

75-10-00

DMC-B737T-A-75-10-00-01A-055A-T_002-00
ICN-B737T-A-751000-A-81205-00002-A-002-01

02-Dec-2020

ENGINE AIR -- LEAP-1B ENGINE - COMPONENT LOCATION - RIGHT SIDE

Compressor Airflow Control Components

These compressor airflow control components are on the right side of the engine:

- Variable bleed valve (VBV) actuator (3:00 position of the fan hub rear face)
- Variable stator vane (VSV) actuator (3:00 position of the engine compressor case).

The VBVs and the VSVs are on the engine's circumference, perpendicular to its longitudinal axis. The VBVs are in the engine fan hub area. The VSVs are on the front of the high pressure compressor case.

Modulated Turbine Cooling (MTC) Actuator and Valve Assembly

One of the two MTC actuator and valve assemblies is on the right side of the engine core at 2:00. You cannot easily see the valve because it is in the engine.

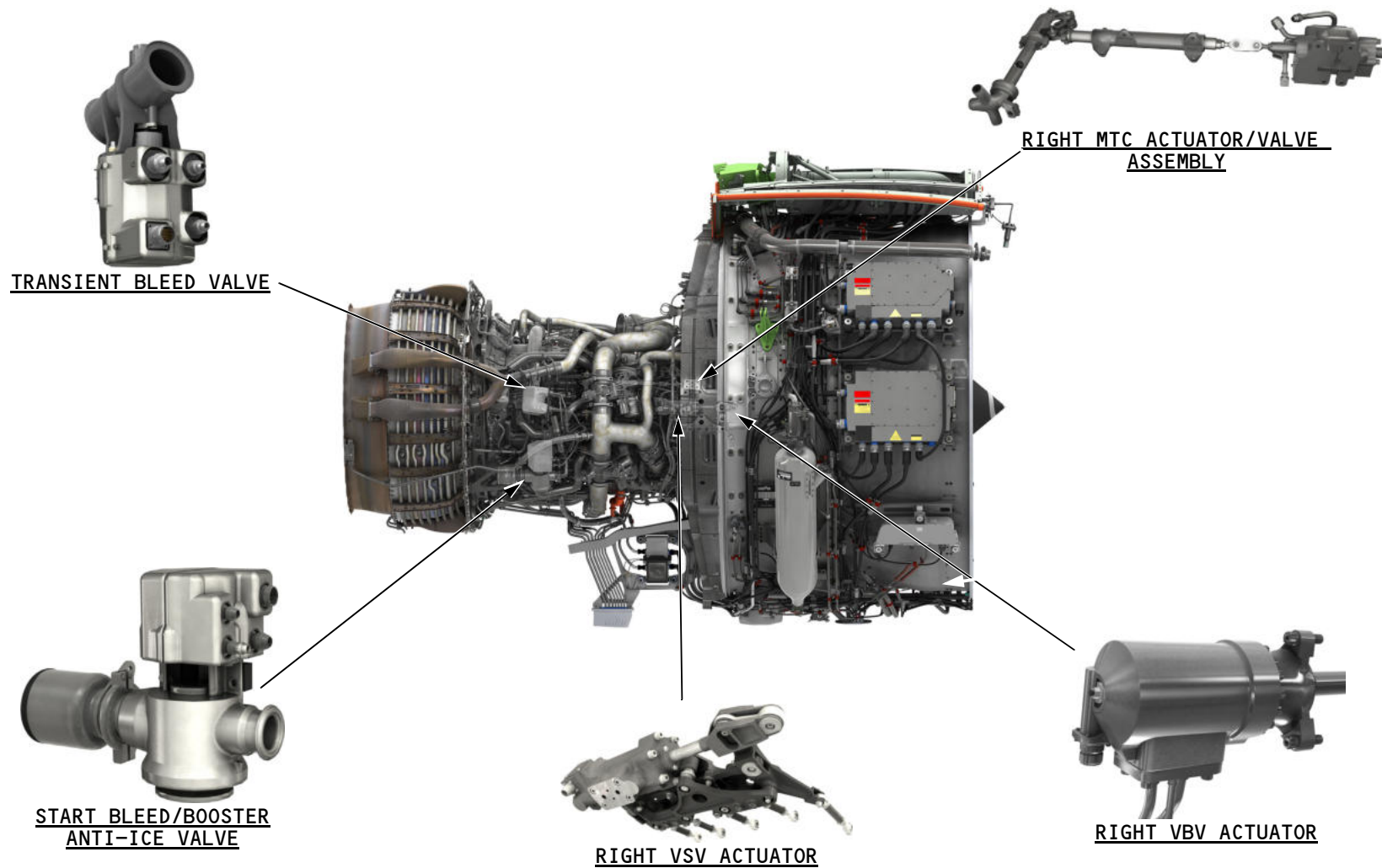
Transient Bleed Valve (TBV)

The TBV is on the right side of the engine core. It is forward of the high pressure turbine case at 3:00.

Start Bleed/Booster Anti-Ice (SB/BAI) Valve

The SB/BAI valve is on the right side of the engine core. It is forward of the high pressure turbine case at 5:00.

DMC-B737T-A-75-10-00-02A-055A-T_002-00



DMC-B737T-A-75-10-00-02A-055A-T_002-00
ICN-B737T-A-751000-A-81205-00001-A-003-01

ENGINE AIR -- LEAP-1B ENGINE - COMPONENT LOCATION - RIGHT SIDE

75-10-00

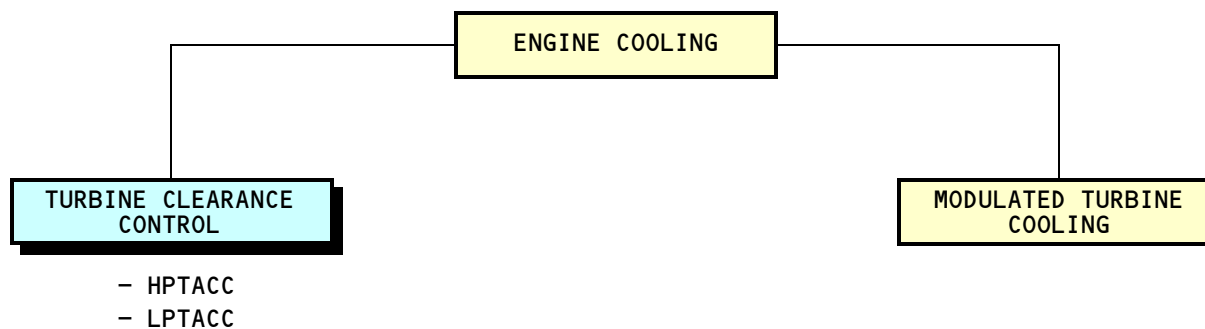
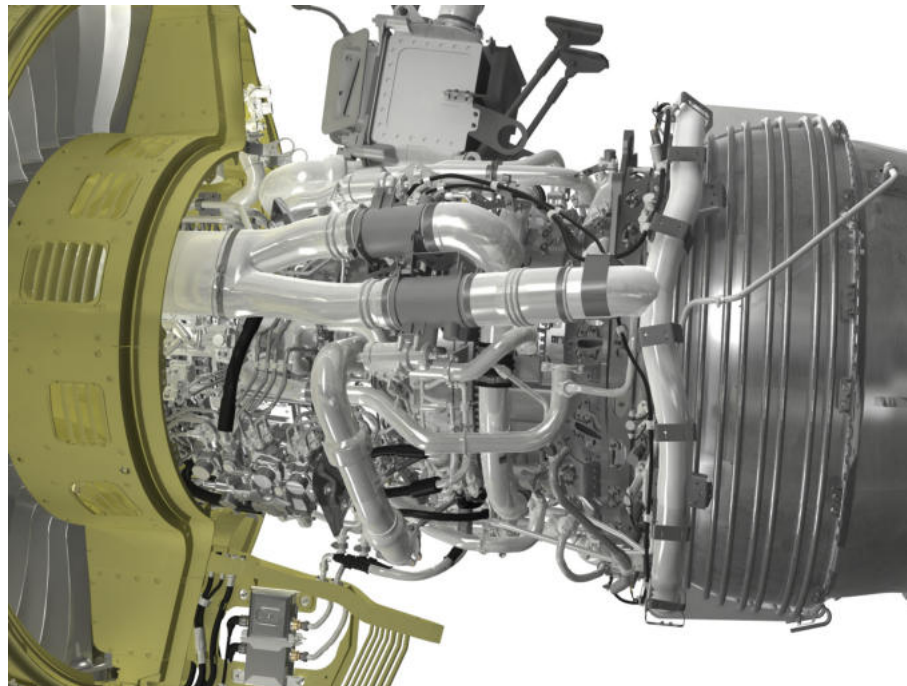
02-Dec-2020

ENGINE AIR -- TURBINE CLEARANCE CONTROL - INTRODUCTION

General

The engine cooling system has these two turbine clearance control subsystems:

- High pressure turbine active clearance control (HPTACC)
- Low pressure turbine active clearance control (LPTACC).



ENGINE AIR -- TURBINE CLEARANCE CONTROL - INTRODUCTION

75-20-00

DMC-B737T-A-75-20-00-02A-018A-T_003-00
ICN-B737T-A-752000-A-81205-00002-A-002-01

02-Dec-2020

ENGINE AIR -- HPTACC - COMPONENT LOCATION

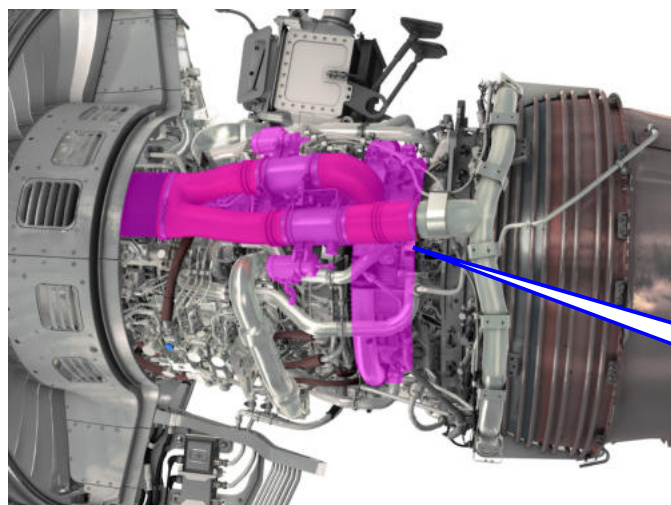
Component Locations

The ACC system components are on the left side of the engine. The HPTACC system and the LPTACC system use these components of the ACC system:

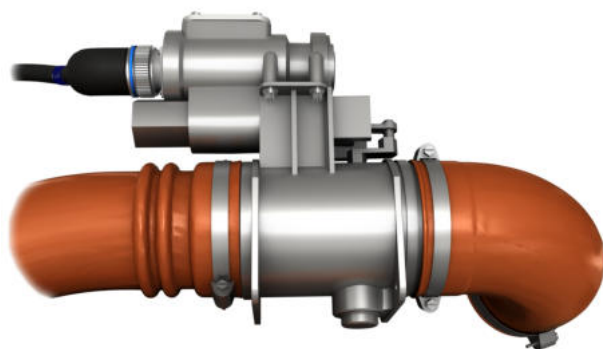
- ACC inlet louver
- ACC inlet duct
- ACC forward wye duct

The HPTACC valve is on the left side of the engine. HPT air manifold 1 and 2 are on the left side of the engine aft of the HPTACC valve. HPT air manifold 3 and 4 are on the right side of the engine aft of the HPTACC valve.

Open the left fan cowl and thrust reverser to get access to the ACC system components and the HPTACC valve.

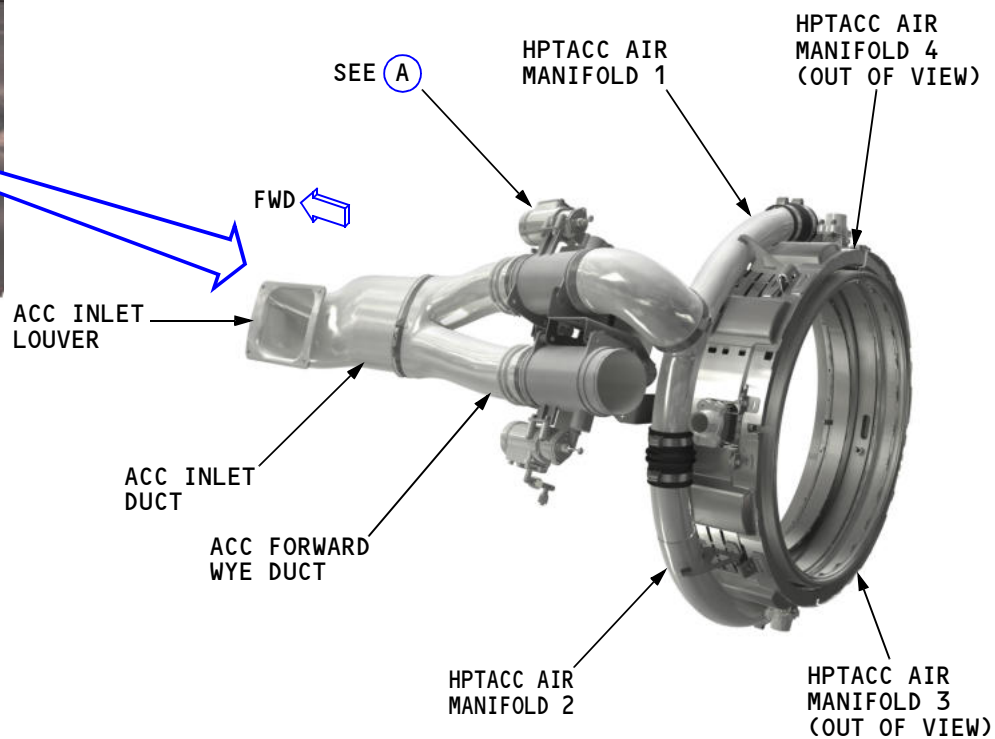


ENGINE (LEFT SIDE VIEW)



HPTACC VALVE ASSEMBLY

(A)



ENGINE AIR -- HPTACC - COMPONENT LOCATION

75-21-00

ENGINE AIR -- HPTACC - VALVE ASSEMBLY

Purpose

The high pressure turbine active clearance control (HPTACC) valve controls the amount of fan discharge air delivered to the high pressure turbine cooling manifold. This reduces the HPT blade tip clearance which improves fuel economy.

Operation

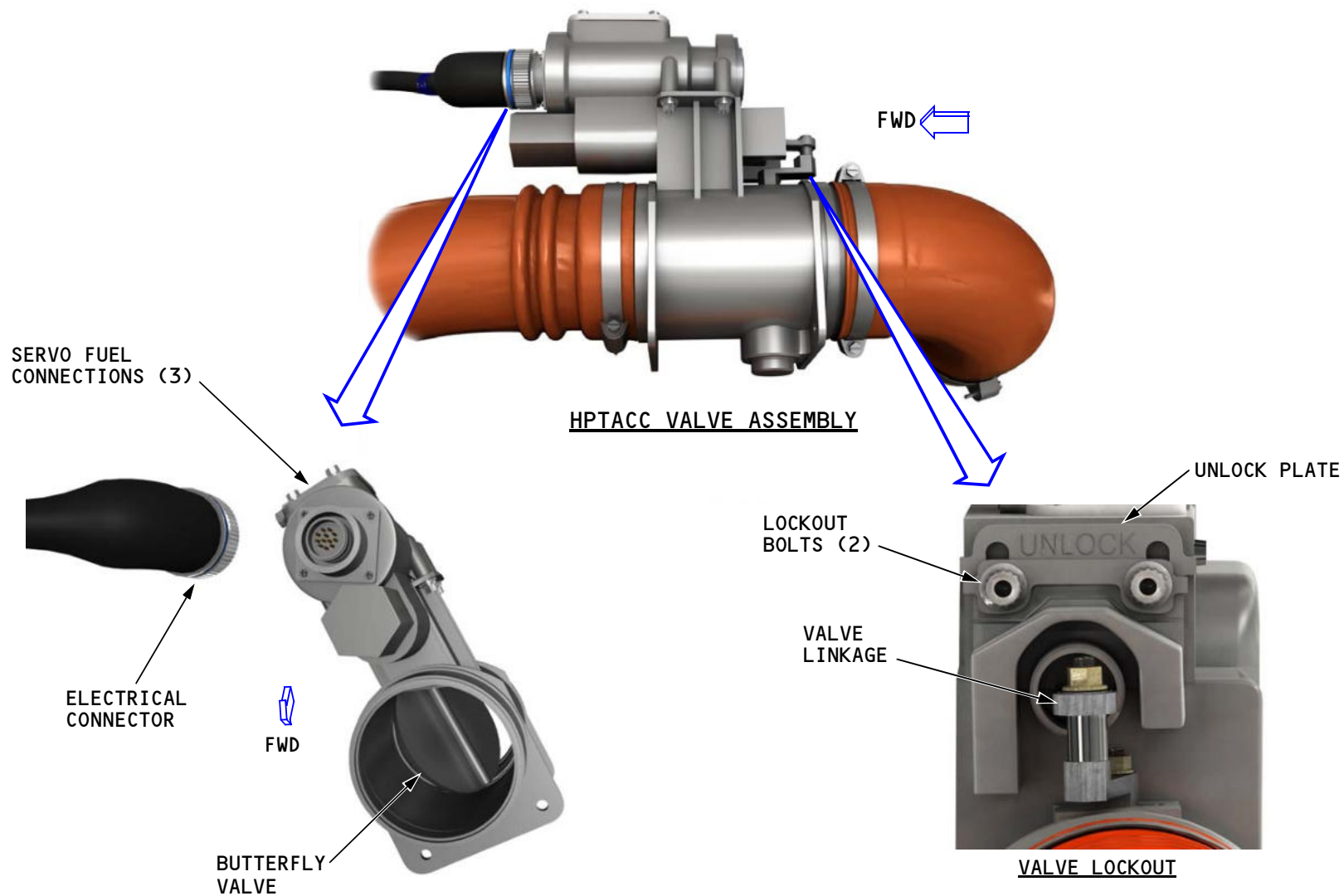
The EEC channel in control operates the HPTACC valve. The EEC sends commands to the split control unit (SCU), which then sends servo fuel pressure to control the HPTACC valve. The servo fuel operates an actuator which turns the butterfly valve shaft. Two LVDTs attach to the actuator. Each LVDT gives valve position feedback through an isolated cable to a separate EEC channel.

You can lock the HPTACC valve closed for dispatch when necessary by doing these steps:

- Turn the valve linkage to the CLOSED position
- Loosen the two lock bolts
- Move the unlock plate until you see the word "LOCK"
- Re-tighten the lock bolts.

Always be sure to follow the appropriate airplane maintenance manual procedure when you do this task.

DMC-B737T-A-75-21-00-02A-034B-T_001-00



ENGINE AIR -- HPTACC - VALVE ASSEMBLY

75-21-00

DMC-B737T-A-75-21-00-02A-034B-T_001-00
ICN-B737T-A-752100-A-81205-00010-A-002-01

02-Dec-2020

ENGINE AIR -- HPTACC - FUNCTIONAL DESCRIPTION

Control

The EECs use airplane and engine data to control the high pressure turbine active clearance control (HPTACC) valve. Fan air decreases the temperature of the HPT shroud support and increases engine efficiency.

The EEC uses this data to control the HPTACC system:

- High pressure compressor inlet temperature
- High pressure compressor discharge temperature
- N1 RPM
- N2 RPM
- Ambient pressure
- HPTACC valve position.

The EEC channel in control sends a command signal to a servo in the split control unit (SCU). The SCU sends servo fuel pressure to move the HPTACC valve. This controls the amount of air that goes to the HPT air manifold.

The HPTACC valve assembly has two LVDTs. One LVDT sends an electrical signal to EEC channel A. The other LVDT sends an electrical signal to EEC channel B. The EEC uses signals from the LVDTs to monitor the position of the HPTACC valve.

Air Flow

Fan air enters the HPTACC system through these components:

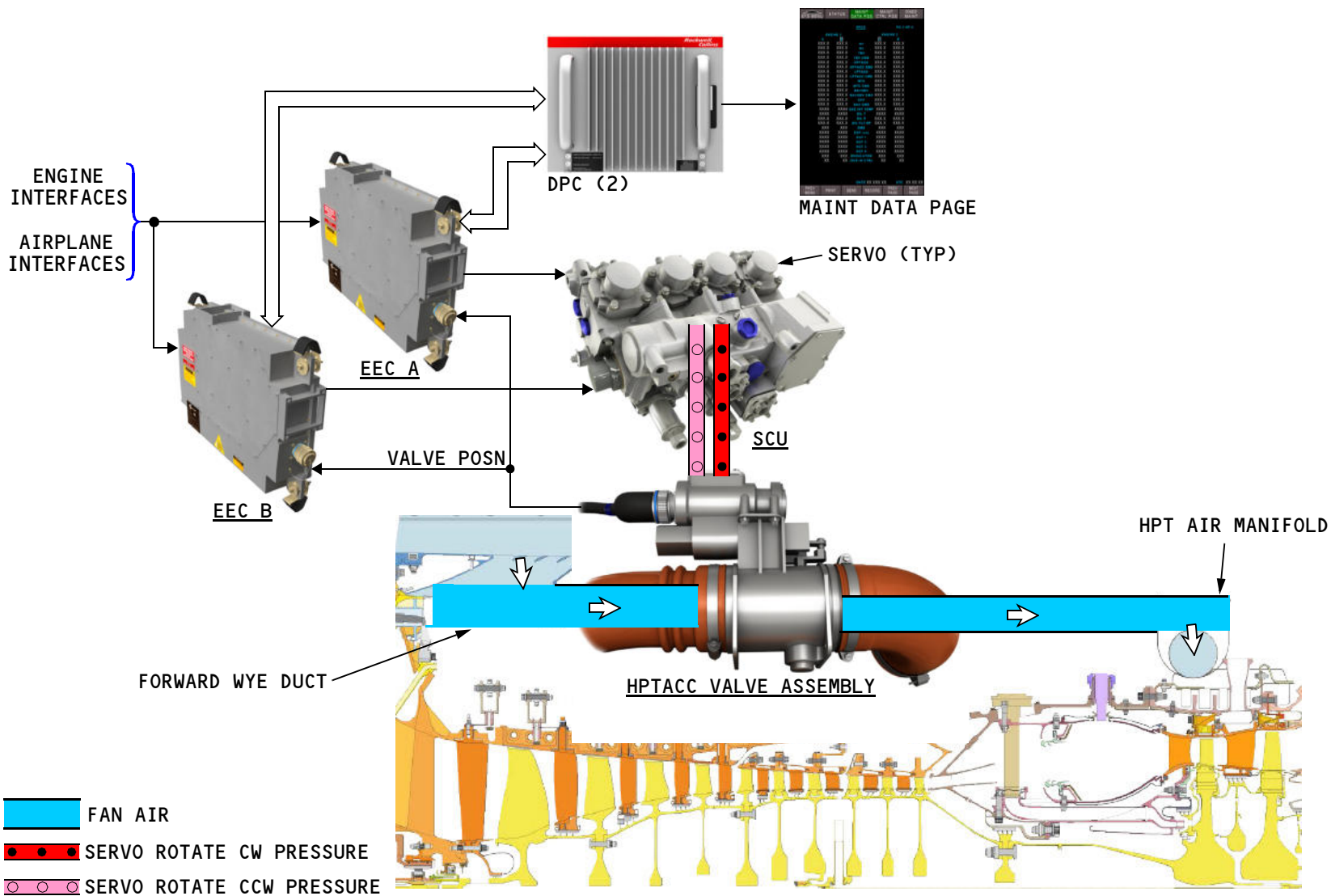
- ACC inlet louver
- ACC inlet duct
- ACC forward wye duct.

Hose clamps connect the individual HPTACC components together. The four segments of the HPTACC manifold connect with HPTACC connect boots.

Training Information Point

The HPTACC position shows on the EPCS maintenance data page. You access the page using the MAINT DATA PGS menu of the MDS.

DMC-B737T-A-75-21-00-02A-042A-T_001-00



DMC-B737T-A-75-21-00-02A-042A-T_001-00
ICN-B737T-A-752100-A-81205-00004-A-001-01

ENGINE AIR -- HPTACC - FUNCTIONAL DESCRIPTION

75-21-00

02-Dec-2020

ENGINE AIR -- LPTACC - COMPONENT LOCATION

Component Locations

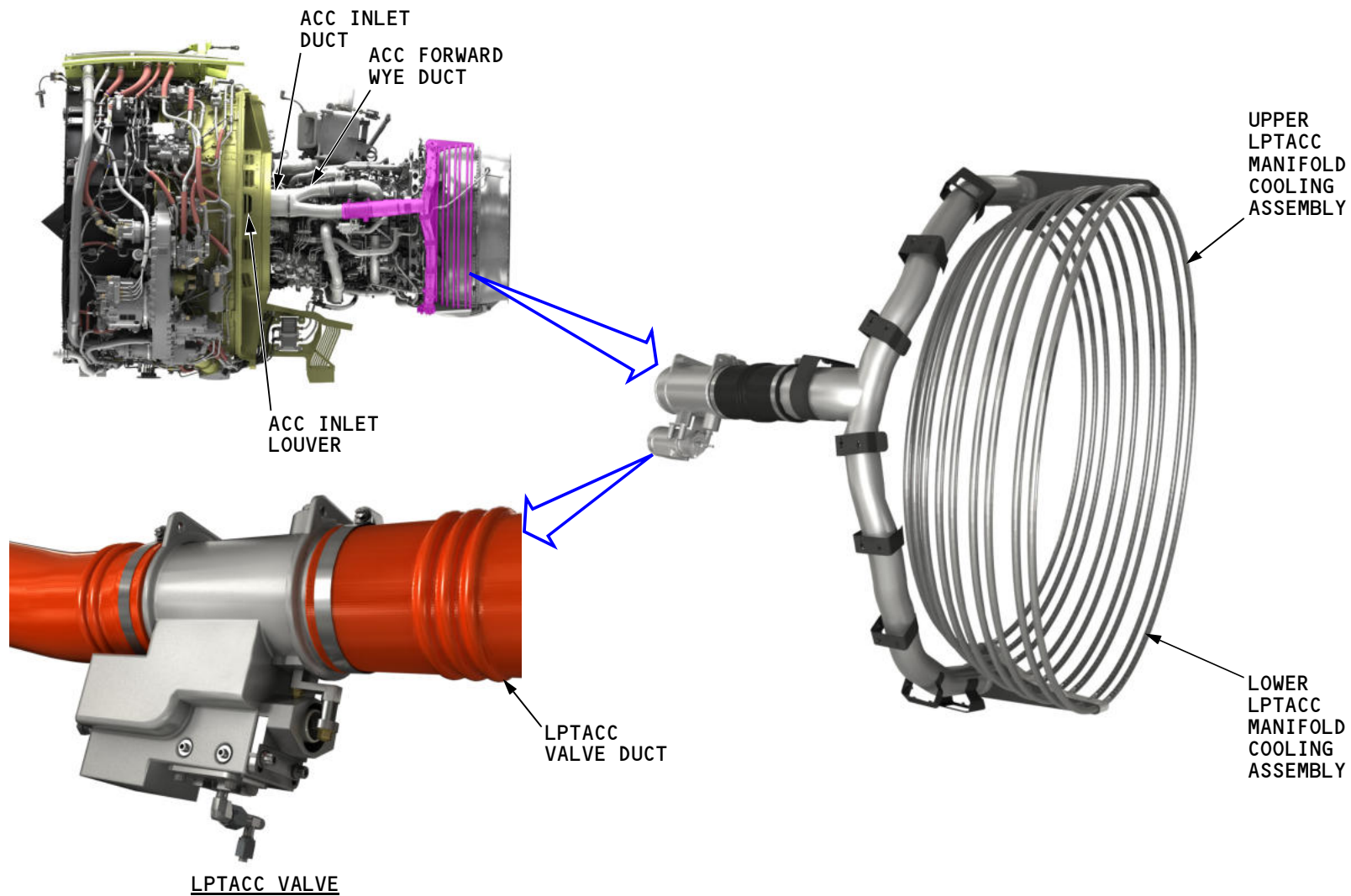
Most of the components of the low pressure turbine active clearance control (LPTACC) system are on the left side of the engine. They are on the high pressure compressor (HPC) case. Aft looking forward, these components are at the 9:00 position:

- Inlet louver
- Inlet duct
- Forward wye duct
- LPTACC valve
- LPTACC valve duct.

There are two manifold cooling assemblies around the low pressure turbine case:

- Upper manifold assembly
- Lower manifold assembly

Open the left fan cowl and thrust reverser to get access to the LPTACC system components.



DMC-B737T-A-75-22-00-02A-055A-T_001-00
ICN-B737T-A-752200-A-81205-00003-A-001-01

ENGINE AIR -- LPTACC - COMPONENT LOCATION

75-22-00

02-Dec-2020

ENGINE AIR -- LPTACC - VALVE ASSEMBLY

Purpose

The low pressure turbine active clearance control (LPTACC) valve controls the amount of fan discharge air delivered to the low pressure turbine air manifold. This reduces LPT blade tip clearance which improves fuel economy.

Operation

The EEC channel in control operates the LPTACC valve. The EEC sends commands to the split control unit (SCU), which then sends servo fuel pressure to control the LPTACC valve. The servo fuel operates an actuator which turns the butterfly valve shaft. Dual LVDTs attach to the actuator. Each LVDT gives valve position feedback through an isolated cable to a separate EEC channel.

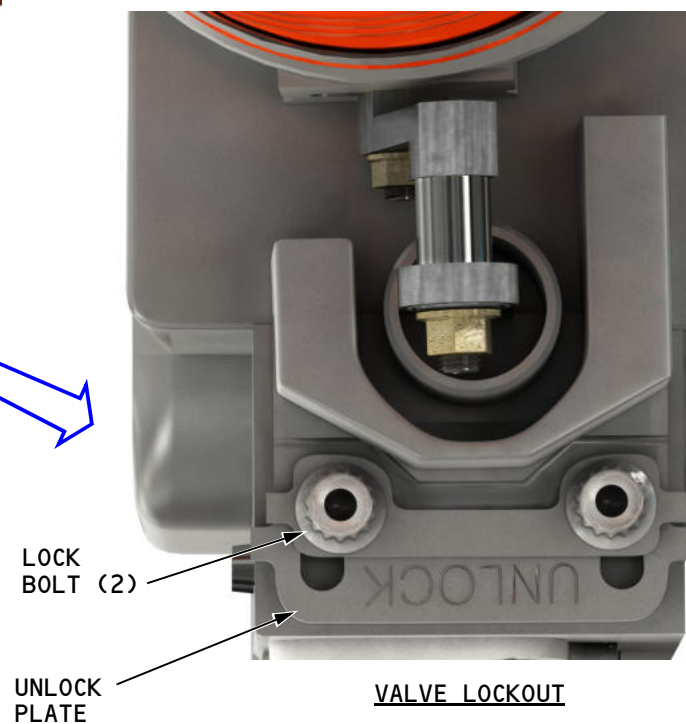
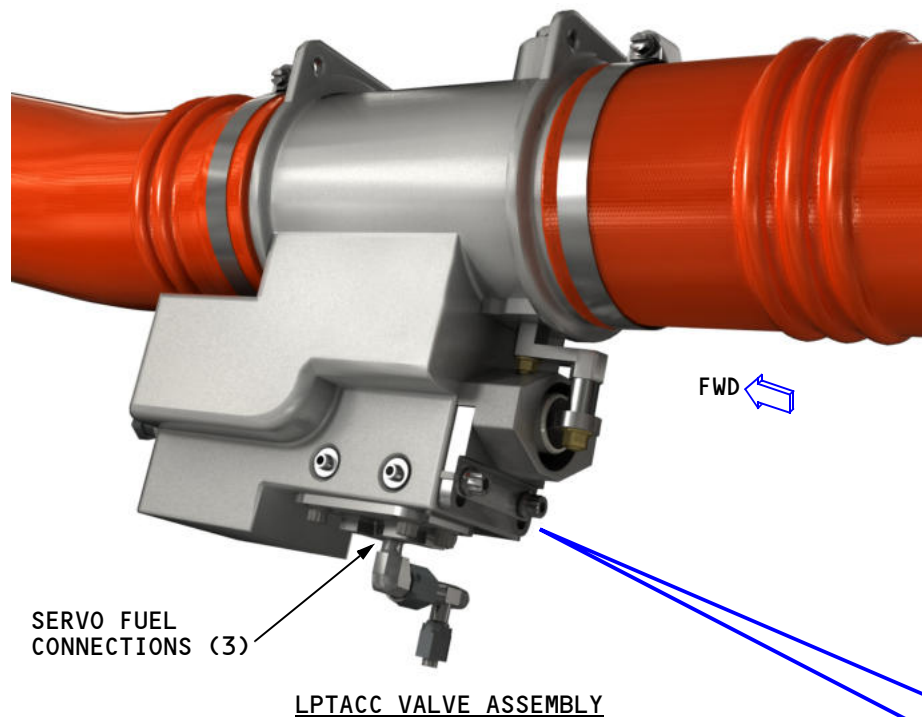
You can lock the LPTACC valve closed for dispatch when necessary by doing these steps:

- Rotate the valve linkage to the CLOSED position
- Loosen the two lock bolts
- Move the unlock plate until you see the word "LOCK"
- Re-tighten the lock bolts.

DMC-B737T-A-75-22-00-02A-034B-T_001-00

02-Dec-2020

75-22-00



ENGINE AIR -- LPTACC - VALVE ASSEMBLY

75-22-00

DMC-B737T-A-75-22-00-02A-034B-T_001-00
ICN-B737T-A-752200-A-81205-00010-A-001-01

02-Dec-2020

ENGINE AIR -- LPTACC - FUNCTIONAL DESCRIPTION

General

The low pressure turbine active clearance control (LPTACC) system controls the low pressure turbine (LPT) blade tip clearance. The LPTACC valve increases or decreases the amount of fan air that goes to the LPT case. This air cools the LPT case. Cooling the LPT case keeps the clearance with the LPT blade tips to a minimum. This increases fuel efficiency.

The electronic engine controls (EECs) use this data to control the LPTACC valve:

- Total air pressure (PT)
- Ambient pressure (P0)
- Total air temperature (TAT)
- N1
- Exhaust gas temperature (EGT).

Control

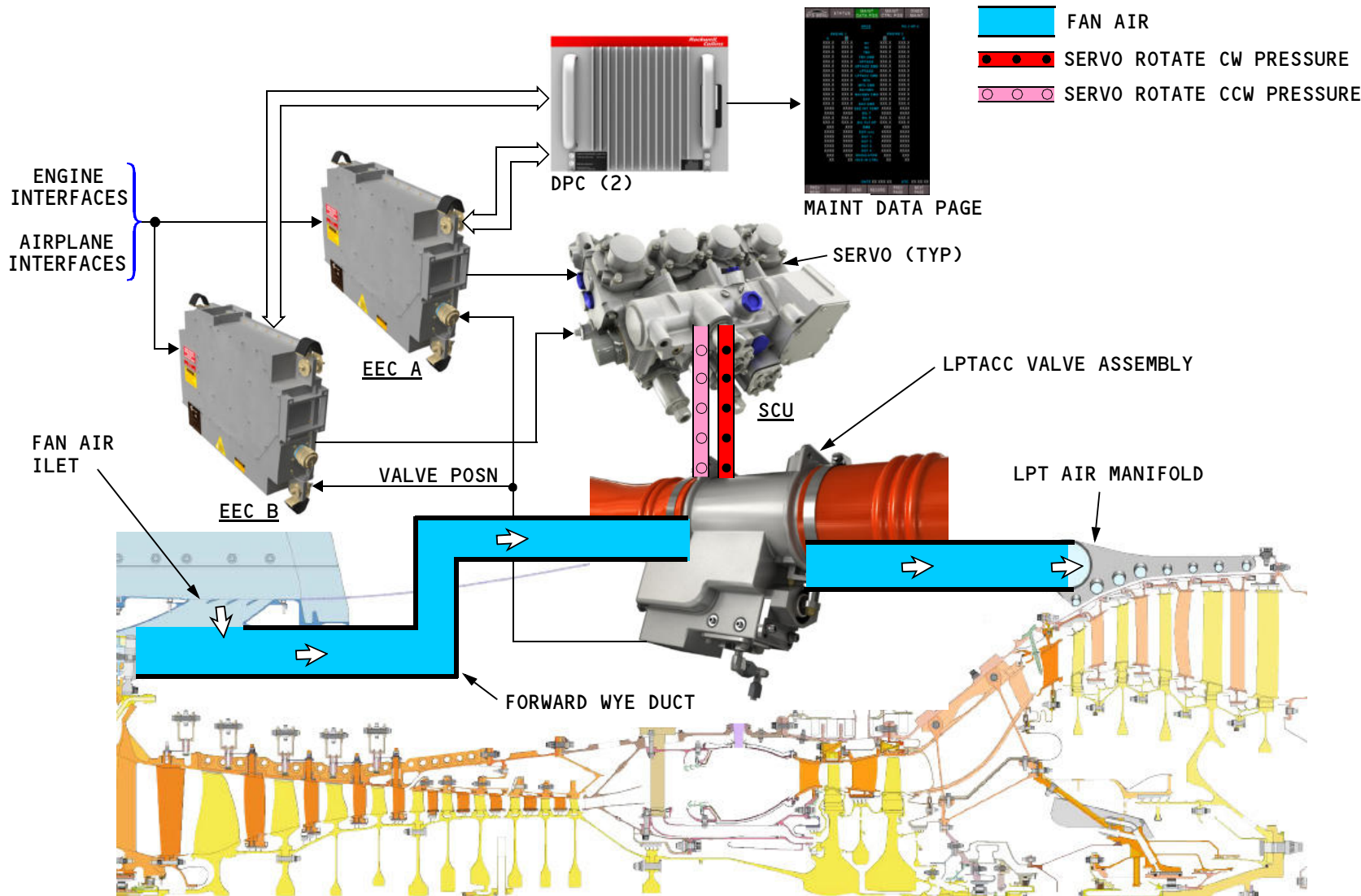
The EEC channel in control sends a signal to a servo in the split control unit (SCU). The SCU send servo fuel pressure to the LPTACC valve to open or close the valve.

The LPTACC valve has two LVDTs. The EECs use the LVDTs to monitor the position of the LPTACC actuator. One LVDT sends the signal to EEC channel A. The other LVDT sends the signal to EEC channel B.

Training Information Point

The LPTACC valve position shows on the EPCS maintenance data pages of the MAX display system (MDS).

DMC-B737T-A-75-22-00-02A-042A-T_002-00



DMC-B737T-A-75-22-00-02A-042A-T_002-00
ICN-B737T-A-752200-A-81205-00004-A-002-01

ENGINE AIR -- LPTACC - FUNCTIONAL DESCRIPTION

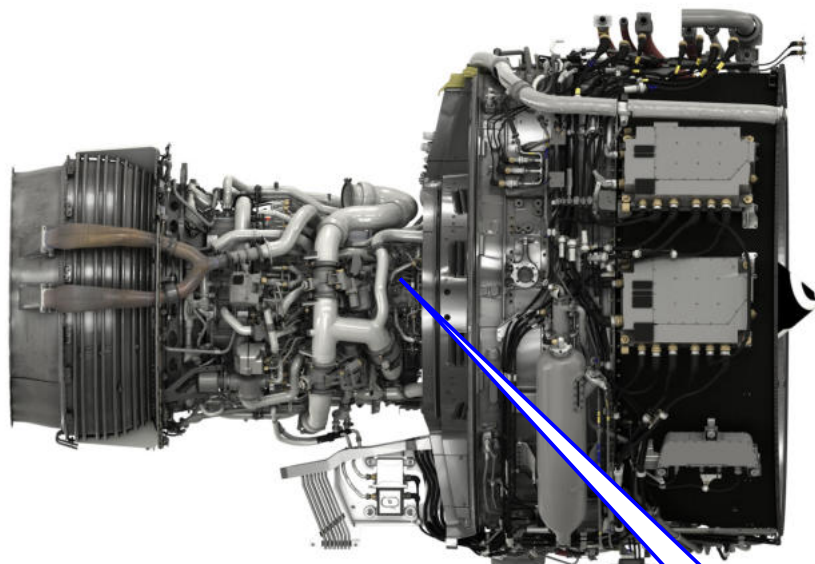
75-22-00

02-Dec-2020

ENGINE AIR -- MTC - COMPONENT LOCATION

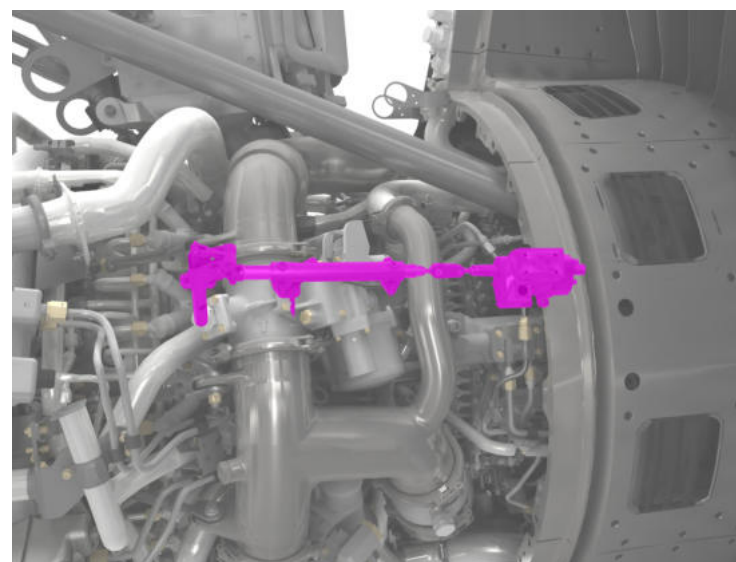
Component Locations

There is one modulated turbine cooling (MTC) actuator and one MTC valve. The actuator is under the thrust reverser cowl over the front high pressure compressor case. It is at the 2:00 position when you see the engine from the rear. The valve is on the combustor case, also at the 2:00 position. Components downstream of the valve are inside the engine.



LEAP-1B ENGINE (RIGHT SIDE)

⇒ FWD



MTC ACTUATOR ASSEMBLY

ENGINE AIR -- MTC - COMPONENT LOCATION

75-23-00

DMC-B737T-A-75-23-00-04A-055A-T_002-00
ICN-B737T-A-752300-A-81205-00009-A-003-01

02-Dec-2020

ENGINE AIR -- MTC - VALVE/ACTUATOR ASSEMBLY

Purpose

The modulated turbine cooling (MTC) valve/actuator assembly controls the amount of cooling air flow to the stage 1 HPT stator blades. Reduced cooling airflow improves fuel economy. There is one MTC valve/actuator. It is on the right side of the engine.

General Description

The EEC channel in control operates the MTC actuator. The EEC sends commands to the split control unit (SCU), which then sends servo fuel pressure to the actuator. The servo fuel pressure moves the actuator which then operates the valve inside the engine.

Each actuator has an LVDT that gives position feedback to the EEC.

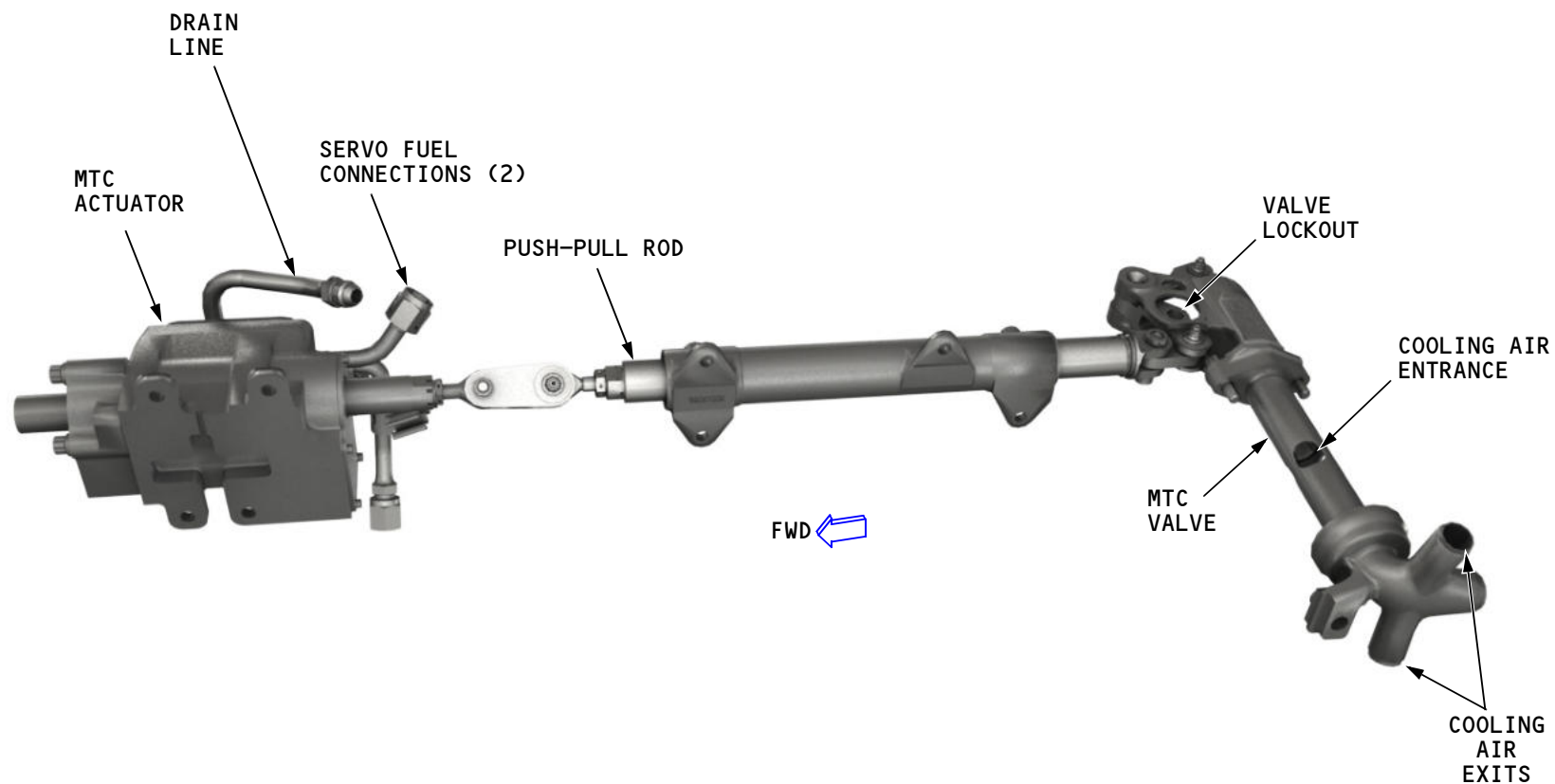
The actuator connects to a push-pull rod mechanism, which then moves the valve inside the engine. The valve inside the engine is a cylinder shaped, gate-type valve that has openings. When the valve moves towards the open position, more air from the 10th stage of the HPC goes through the openings. When the valve moves towards the closed position, the openings are blocked, which reduces the cooling air flow. From the openings, the cooling air goes to the stage 1 HPT rotor and stators.

The valve linkage has a lockout hole that allows dispatch with the system inoperative. The rod end bolt moves from the rod to the linkage, which locks the valve in the open position. Always refer to the appropriate airplane maintenance manual for instructions.

DMC-B737T-A-75-23-00-03A-034B-T_002-00

02-Dec-2020

75-23-00



DMC-B737T-A-75-23-00-03A-034B-T_002-00
ICN-B737T-A-752300-A-81205-00012-A-001-01

ENGINE AIR -- MTC - VALVE/ACTUATOR ASSEMBLY

75-23-00

02-Dec-2020

ENGINE AIR -- MTC - FUNCTIONAL DESCRIPTION

Cooling Air Flow

The modulated turbine cooling (MTC) system uses compressor discharge air to cool the high pressure turbine (HPT). The MTC valve controls the amount of cooling air flow. The MTC valve assembly has one actuator that operates it. At high power settings, the engine makes the most internal heat. At low engine power settings, the engine makes less internal heat. The MTC system gives less cooling air flow at low engine power settings.

Operation

The MTC system gives the most cooling air to the HPT stage 1 blades during these phases of flight:

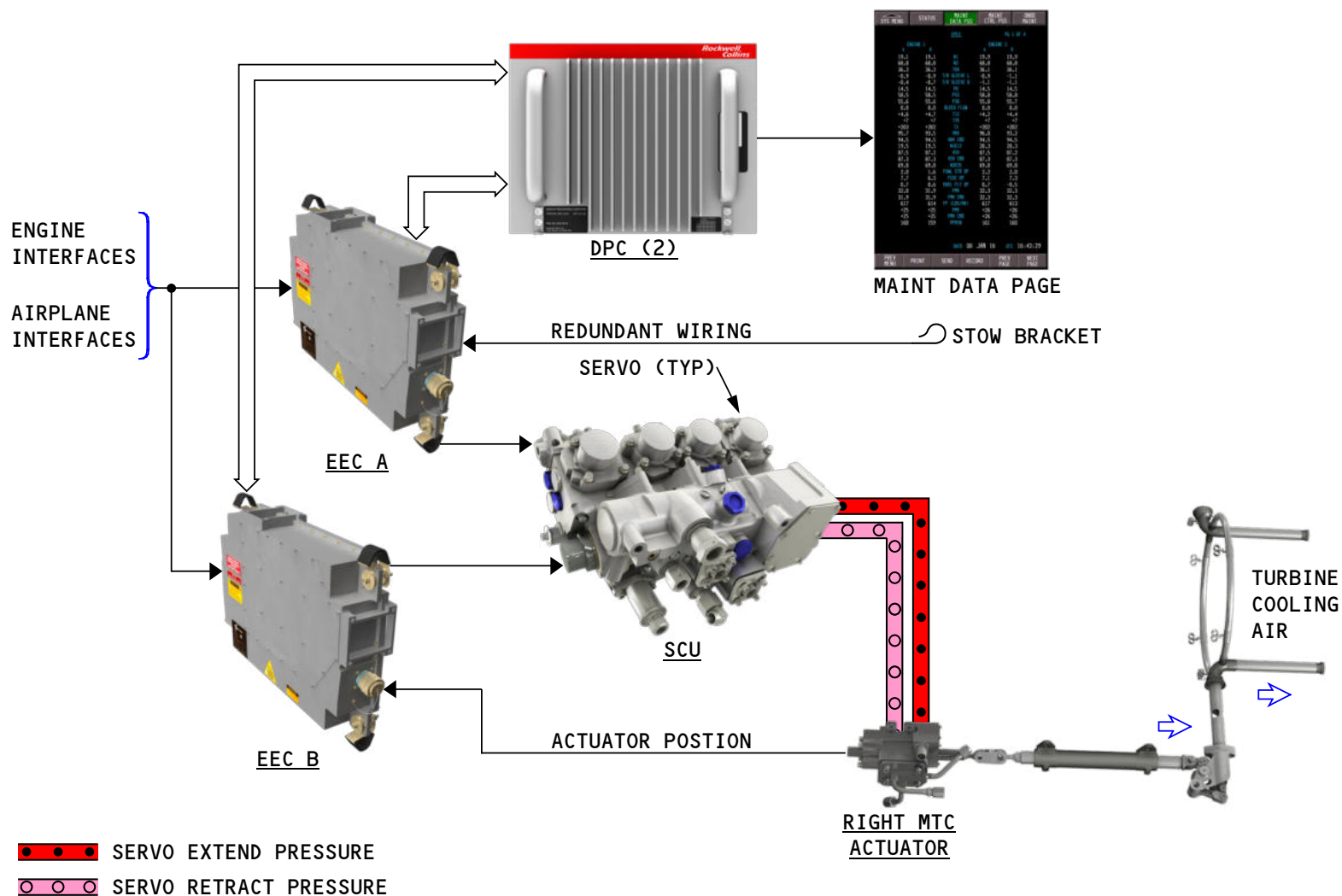
- Takeoff
- Climb.

The EEC sends a signal to the servo in the split control unit (SCU). The SCU then gives servo fuel pressure to the MTC actuator. The actuator extends and retracts to control how much cooling air goes to the first stage of the HPT.

When the MTC valve is open, valve openings let compressor discharge air in. This gives cool air to the HPT stage 1 blades. When the MTC valve is closed, it stops MTC air. The actuator has an LVDT that gives actuator position feedback to an EEC.

You can see the MTC actuator position on the EPCS maintenance data page.

DMC-B737T-A-75-23-00-04A-042A-T_003-00



DMC-B737T-A-75-23-00-04A-042A-T_003-00
ICN-B737T-A-752300-A-81205-00010-A-003-01

ENGINE AIR -- MTC - FUNCTIONAL DESCRIPTION

75-23-00

02-Dec-2020



ENGINE AIR -- COMPRESSOR CONTROL - INTRODUCTION

Purpose

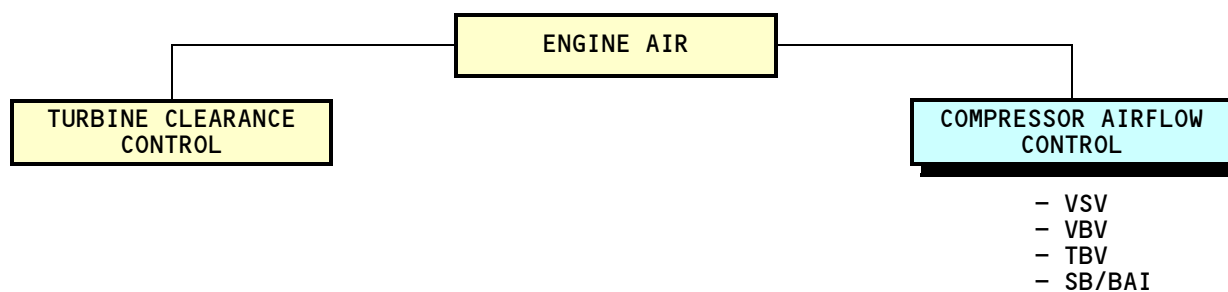
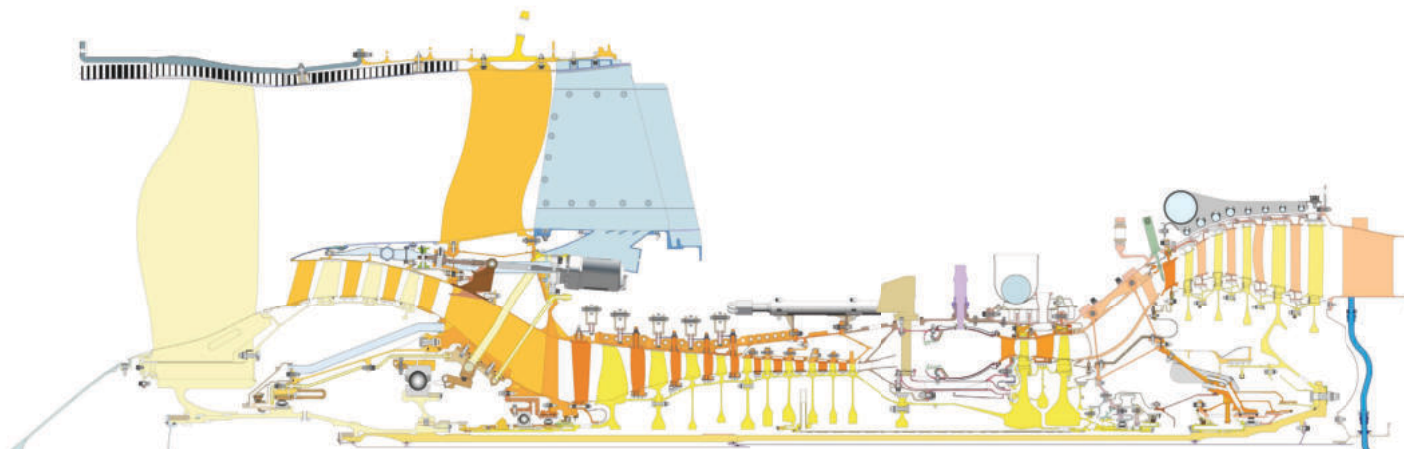
The compressor airflow control system has these functions:

- Variable stator vanes (VSVs)
- Variable bleed valves (VBVs)
- Transient bleed (TBV)
- Start bleed/booster anti-ice (SB/BAI).

DMC-B737T-A-75-30-00-02A-018A-T_001-00

02-Dec-2020

75-30-00



ENGINE AIR -- COMPRESSOR CONTROL - INTRODUCTION

75-30-00

DMC-B737T-A-75-30-00-02A-018A-T_001-00
ICN-B737T-A-753000-A-81205-00002-A-001-01

02-Dec-2020

ENGINE AIR -- VSV SYSTEM - COMPONENT LOCATION

Component Locations

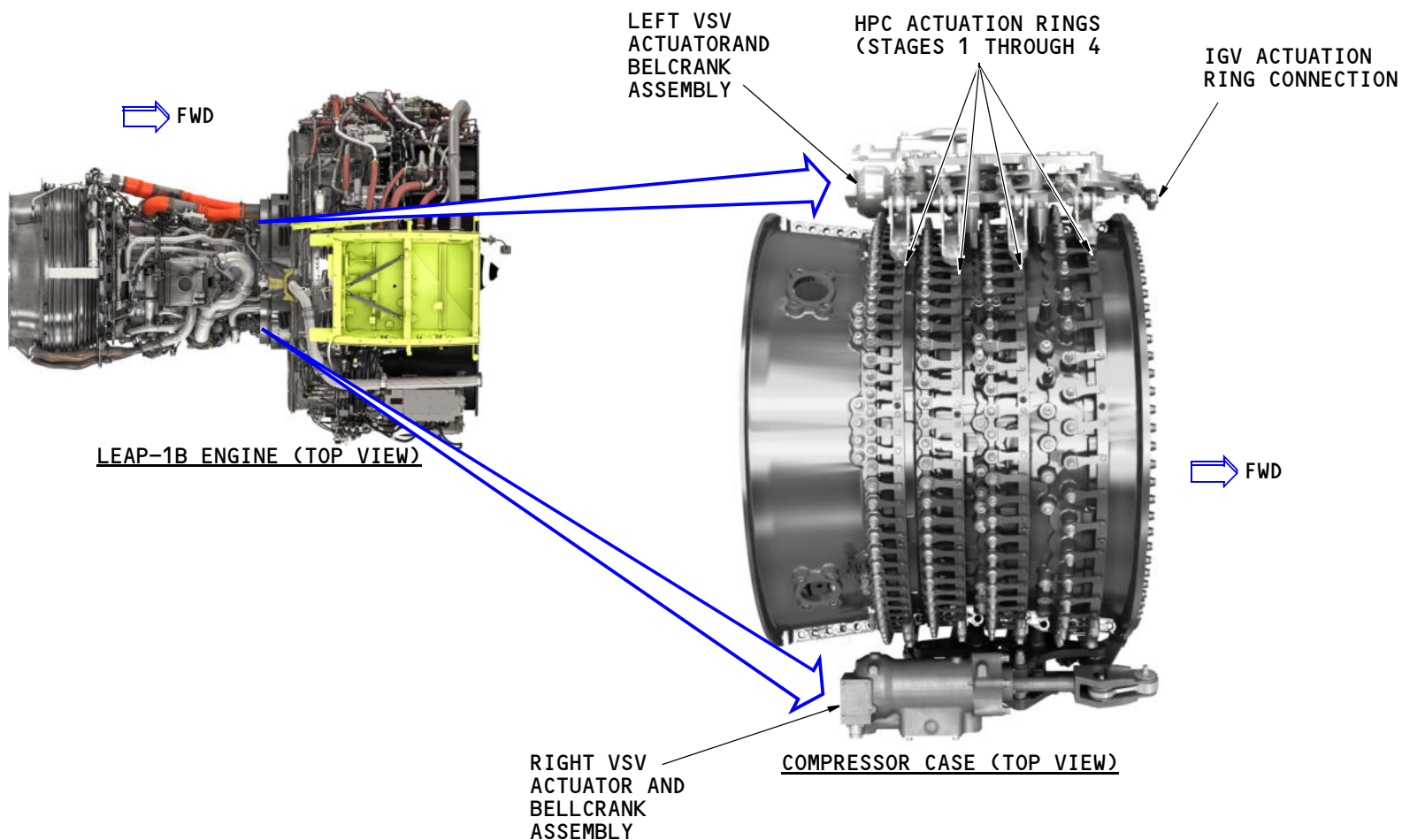
The VSV system has 2 VSV actuators and 2 bellcrank assemblies. One actuator and one bellcrank are on the HPC case at these positions:

- 3:00 position
- 9:00 position.

The actuators and bellcranks are connected to actuation rings. The actuation rings go around the HPC case. The actuation rings connect to and adjust the angles of these parts of the HPC:

- Inlet guide vanes (IGVs)
- Stage 1 stator vanes
- Stage 2 stator vanes
- Stage 3 stator vanes
- Stage 4 stator vanes.

Open the fan cowls and thrust reversers to get access to the VSV system components.



ENGINE AIR -- VSV SYSTEM - COMPONENT LOCATION

75-31-00

DMC-B737T-A-75-31-00-02A-055A-T_002-00
ICN-B737T-A-753100-A-81205-00003-A-002-01

02-Dec-2020

ENGINE AIR -- VSV SYSTEM - VSV ACTUATORS

Purpose

The VSV actuators give the force necessary to change the position of these HPC components:

- Inlet guide vanes
- Stage 1 through stage 4 stator vanes.

General Description

The EEC channel in control sends commands to the split control unit (SCU), which then sends servo fuel pressure to the VSV actuators. The servo fuel pressure moves the VSV actuator rod.

The VSV actuator connects to a mechanical linkage system. The VSV mechanical linkage system has these parts:

- Bellcrank assembly
- Turnbuckles (5).

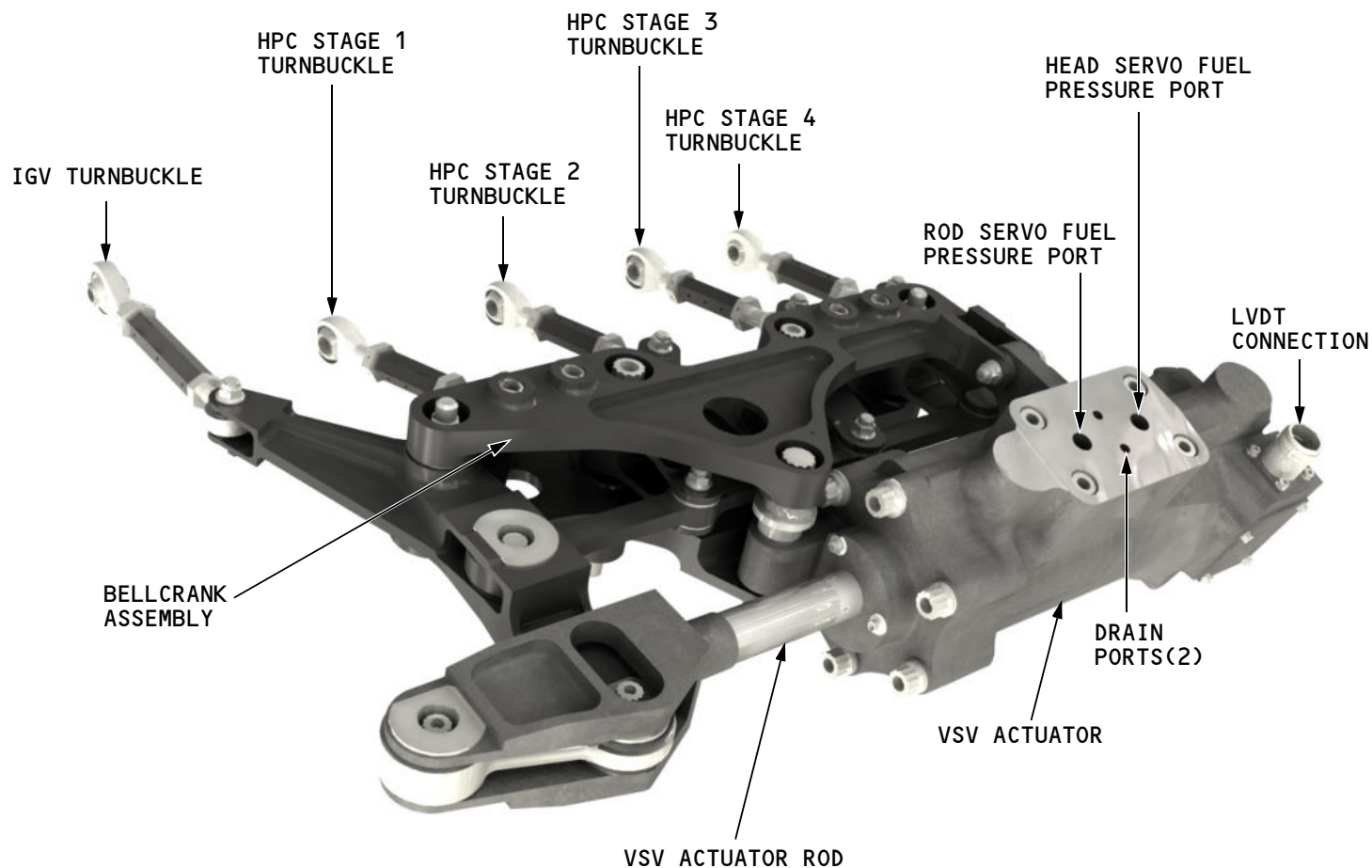
The VSV actuator rod gives a linear movement. The VSV mechanical linkage system changes the linear movement into a circular movement in the actuation rings. This circular movement controls the angle of the inlet guide vanes and the stage 1 through stage 4 stator vanes.

Each VSV actuator has an LVDT that gives position feedback to the related EEC channel.

DMC-B737T-A-75-31-00-02A-034B-T_001-00

02-Dec-2020

75-31-00



ENGINE AIR -- VSV SYSTEM - VSV ACTUATORS

75-31-00

ENGINE AIR -- VSV SYSTEM - FUNCTIONAL DESCRIPTION

General

The variable stator vanes (VSVs) system controls the angular position of these parts of the engine:

- Variable stator vanes
- Inlet guide vanes.

The VSV system adjusts the angle of air flow between the vanes and the rotating blades in the HPC. This gives better compressor operation and helps to prevent an HPC stall.

The electronic engine controllers (EECs) use this data to control the VSVs position:

- Ambient pressure (P0)
- N2 speed
- HPC inlet air temperature (T25)
- VSV actuator position feedback.

Control

The EEC usually gets P0 from the pressure subsystem box, and engine data from engine sensors. These inputs are used to calculate the commanded VSVs position. The EEC channel in control gives a signal to the split control unit (SCU). The SCU gives servo fuel pressure to the two VSV actuators. Each actuator connects to a bellcrank assembly. The two actuator and bellcrank assemblies operate together to move the HPC variable stator vanes through 4 actuation rings. They also move the IGVs through one actuation ring. Each actuator has an LVDT. The EECs use the LVDTs to monitor the position of the VSV actuators. One LVDT gives an electrical signal to channel A of the EEC. The other LVDT gives an electrical signal to channel B.

Operation

The VSVs and IGVs are in their closed position when N2 is at idle. They move to a more open position when N2 increases. If N2 speed decreases, the EEC makes the VSVs and IGVs go to a more closed position.

Training Information Point

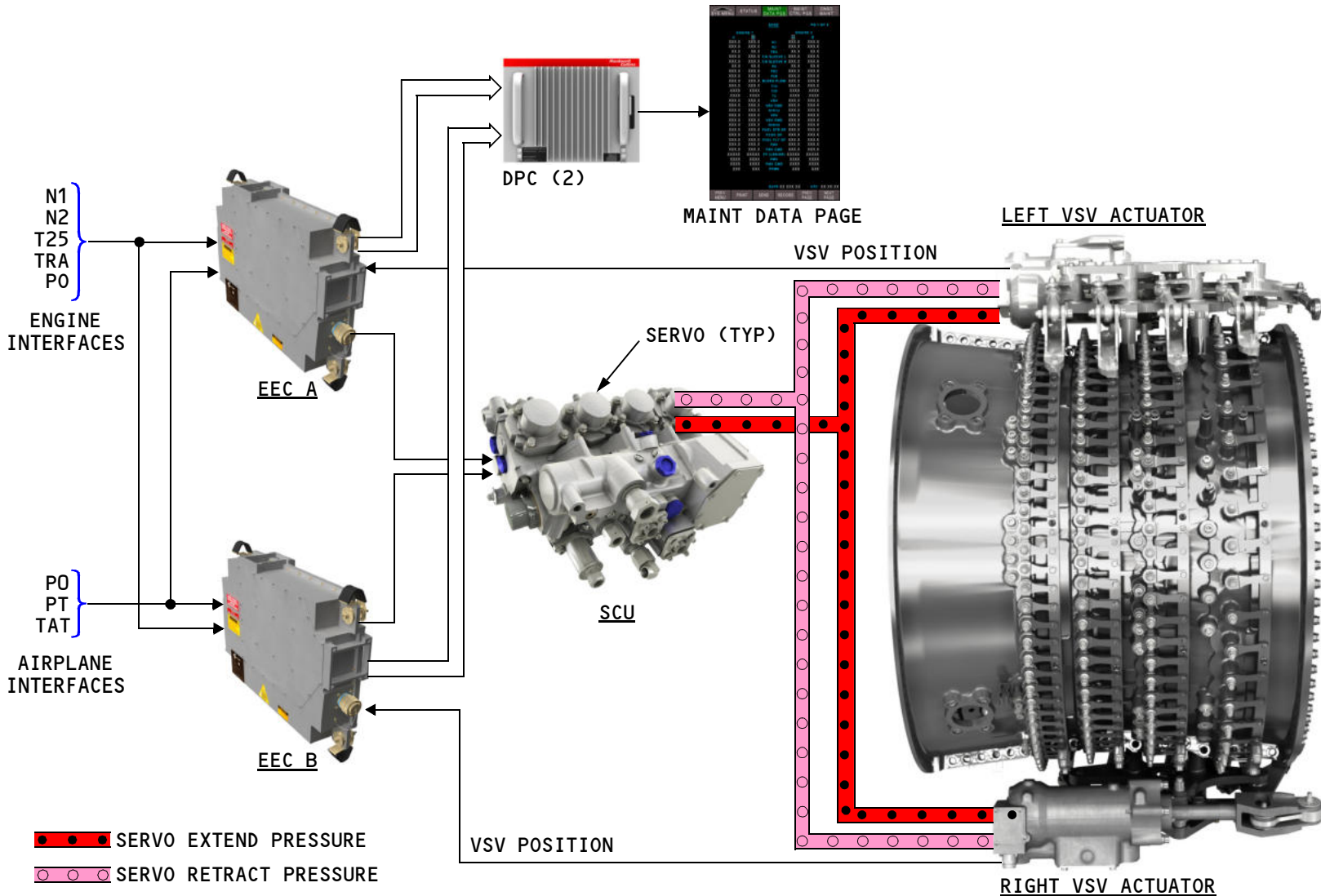
You can see the VSV position on the electronic propulsion control system (EPCS) maintenance data page. The VSV position data includes the IGV position. The EPCS pages are in the MAINT DATA PGS menu of the Max display system.

Training Information Point

The VSV actuators are interchangeable.

Training Information Point

After you replace a VSV actuator, it is not necessary to adjust it.



DMC-B737T-A-75-31-00-02A-042A-T_001-00
ICN-B737T-A-753100-A-81205-00004-A-001-01

ENGINE AIR -- VSV SYSTEM - FUNCTIONAL DESCRIPTION

02-Dec-2020

75-31-00

ENGINE AIR -- VBV SYSTEM - COMPONENT LOCATION

Component Locations

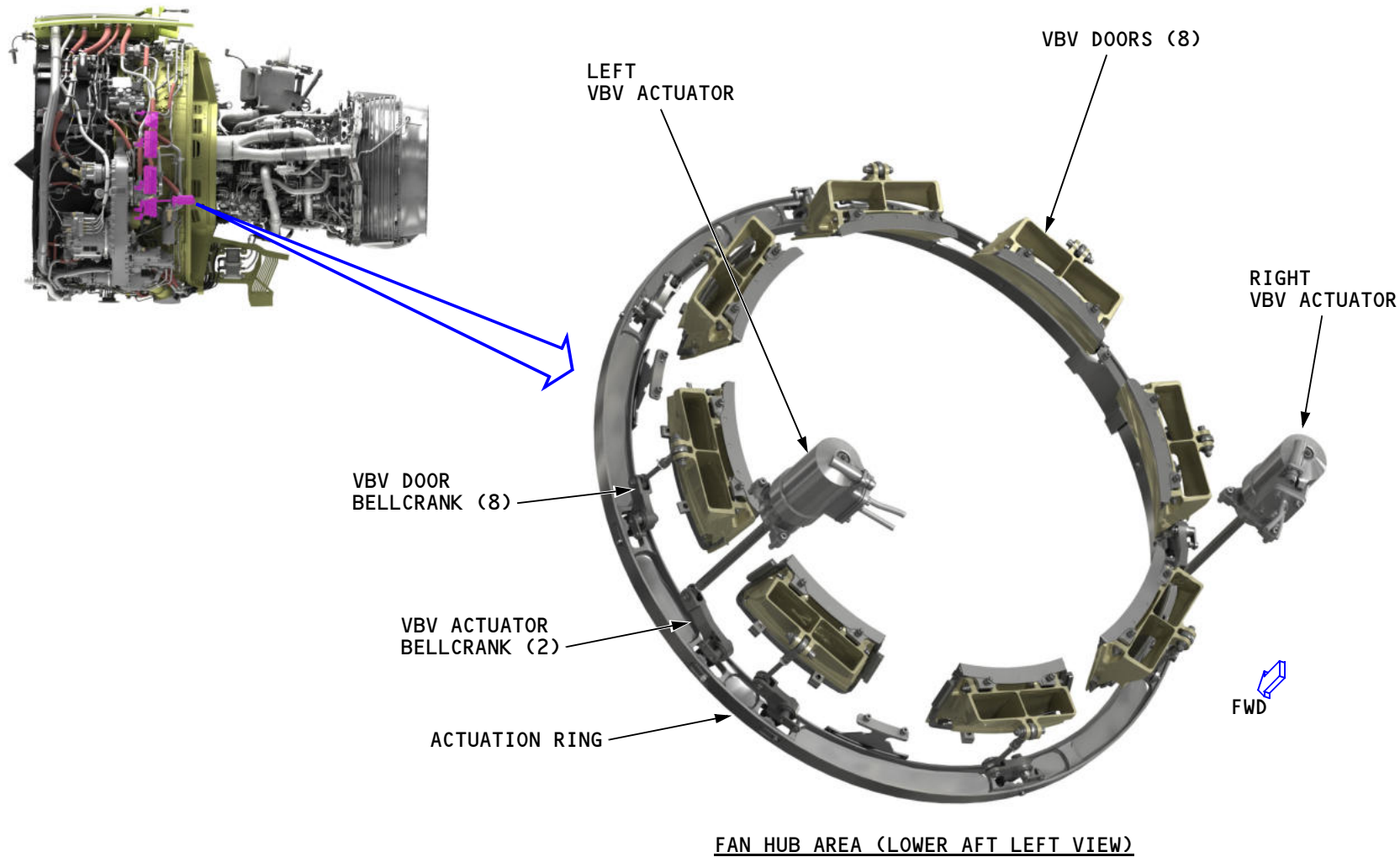
The right variable bleed valve (VBV) actuator is on the rear face of the fan hub at the 3:00 position.

The left VBV actuator is on the rear face of the fan hub at the 7:00 position.

These VBV components are also in the fan hub area:

- Actuation ring
- VBV door (8)
- VBV door bellcrank (8)
- VBV actuator bellcrank (2).

Open the two fan cowl and thrust reverser cowl to get access to the VBV system components.



ENGINE AIR -- VBV SYSTEM - COMPONENT LOCATION

75-32-00

DMC-B737T-A-75-32-00-02A-055A-T_001-00
ICN-B737T-A-753200-A-81205-00004-A-001-01

02-Dec-2020

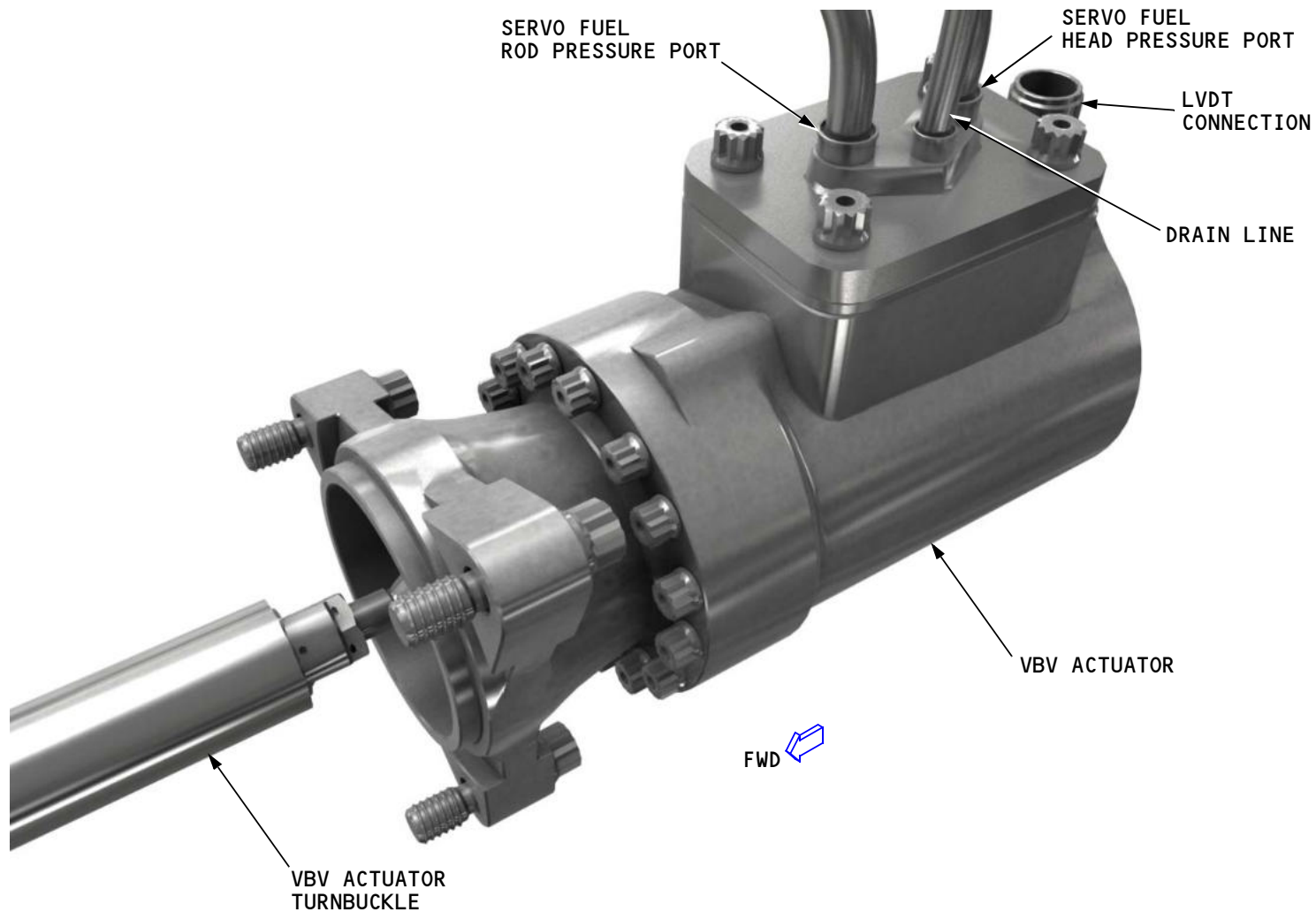
ENGINE AIR -- VBV SYSTEM - VBV ACTUATORS

Purpose

Two VBV actuators control the movement of the VBV doors. One VBV actuator is on each side of the engine.

General Description

The EEC channel in control sends commands to the split control unit (SCU), which then sends servo fuel pressure to the VBV actuators. The SCU gives servo fuel pressure to both the head and rod ports of each VBV actuator. The port that receives the highest pressure controls the direction of movement. Each actuator has an LVDT that gives actuator position feedback to the related EEC. The VBV actuators mount to the fan frame with three bolts. The actuators connect to the VBV doors through the VBV actuator turnbuckle.



DMC-B737T-A-75-32-00-05A-034B-T_002-00
ICN-B737T-A-753200-A-81205-00012-A-002-01

ENGINE AIR -- VBV SYSTEM - VBV ACTUATORS

75-32-00

02-Dec-2020



ENGINE AIR -- VBV SYSTEM - DOORS

General Description

The VBV doors control the quantity of low pressure compressor (LPC) air that mixes with the fan discharge air flow.

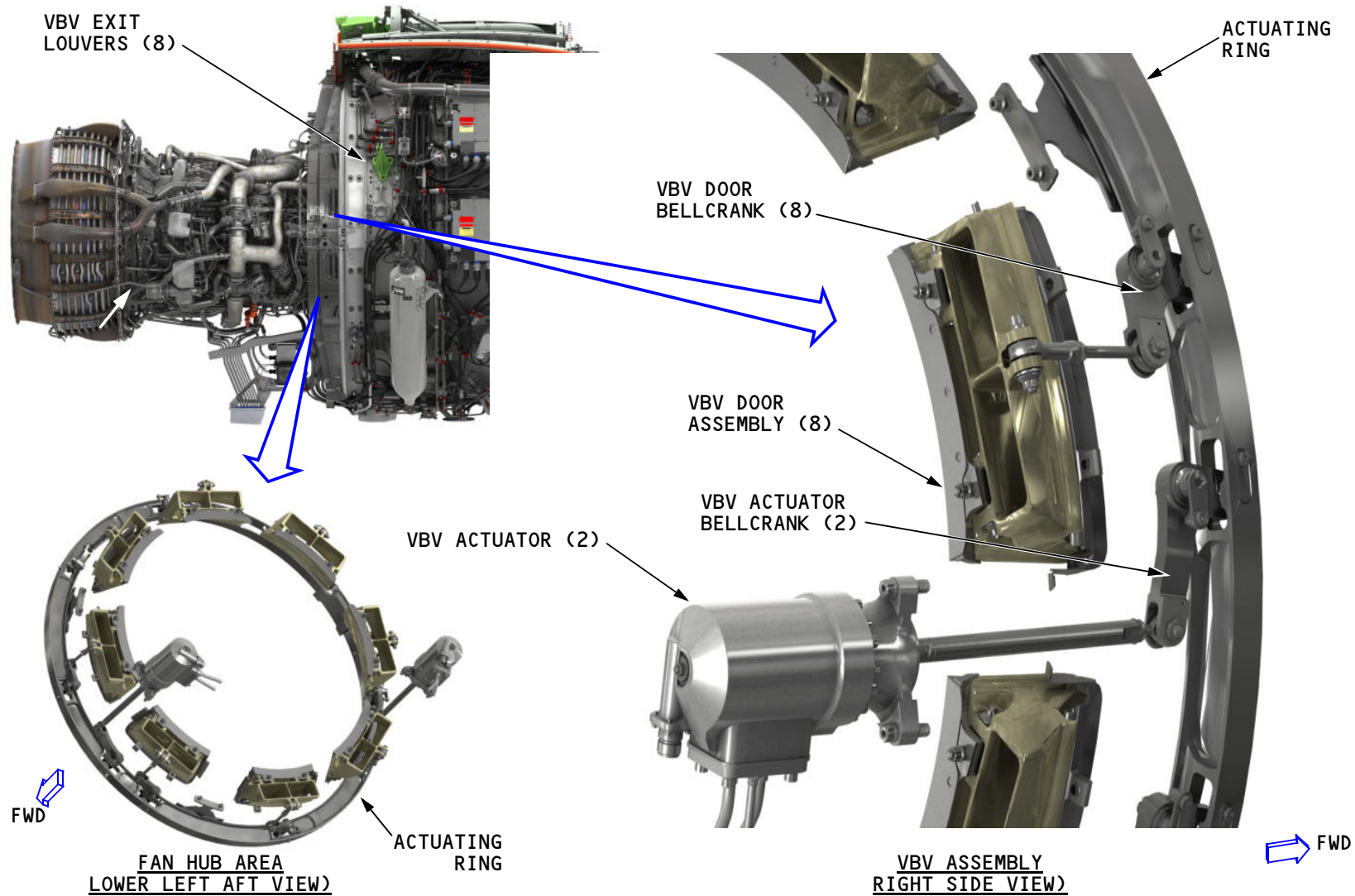
Physical Description

There are 2 VBV actuators. There are also 2 VBV actuator bellcranks. Each VBV actuator connects to an VBV actuator bellcrank. The VBV actuator bellcranks connect to the VBV actuating ring. 8 VBV door bellcranks connect to the actuating ring. 1 VBV door assembly connects to each VBV door bellcrank.

DMC-B737T-A-75-32-00-02A-034B-T_002-00

02-Dec-2020

75-32-00



DMC-B737T-A-75-32-00-02A-034B-T_002-00
ICN-B737T-A-753200-A-81205-00005-A-002-01

ENGINE AIR -- VBV SYSTEM - DOORS

02-Dec-2020

75-32-00

ENGINE AIR -- VBV SYSTEM - FUNCTIONAL DESCRIPTION

General

The variable bleed valve (VBV) system lets part of the low pressure compressor (LPC) discharge air bypass into the secondary airflow. This controls the amount of air that goes into the high pressure compressor (HPC). During a fast deceleration, the VBV system increases the LPC stall margin. At low engine speed and during the thrust reverser operation, the VBV system keeps foreign object debris (FOD) out of the high pressure compressor (HPC). This prevents damage to the engine and improves engine stability.

The electronic engine controls (EECs) use these inputs to control the VBVs :

- P0
- PT
- TAT
- T25
- VSV position
- N1 speed
- N2 speed
- Thrust lever resolver angle (TRA).

Control

The EECs normally get P0, PT and TAT from the ADIRUs through the DPCs. The EECs get N1, N2, T25, and VSV position from engine sensors. TRA comes from the thrust lever resolver. P0 data also comes from the pressure subsystem box. The EECs use this data to schedule an angular position for the VBV doors. The EEC channel in control sends a command signal to the split control unit (SCU). The SCU sends servo fuel pressure to move the two VBV actuators. The actuators connect with the eight VBV doors through an actuation ring.

Each actuator has an LVDT. The EEC uses the LVDTs to monitor the position of the actuators. One LVDT sends an electrical signal to channel A of the EEC. The other LVDT sends an electrical signal to channel B.

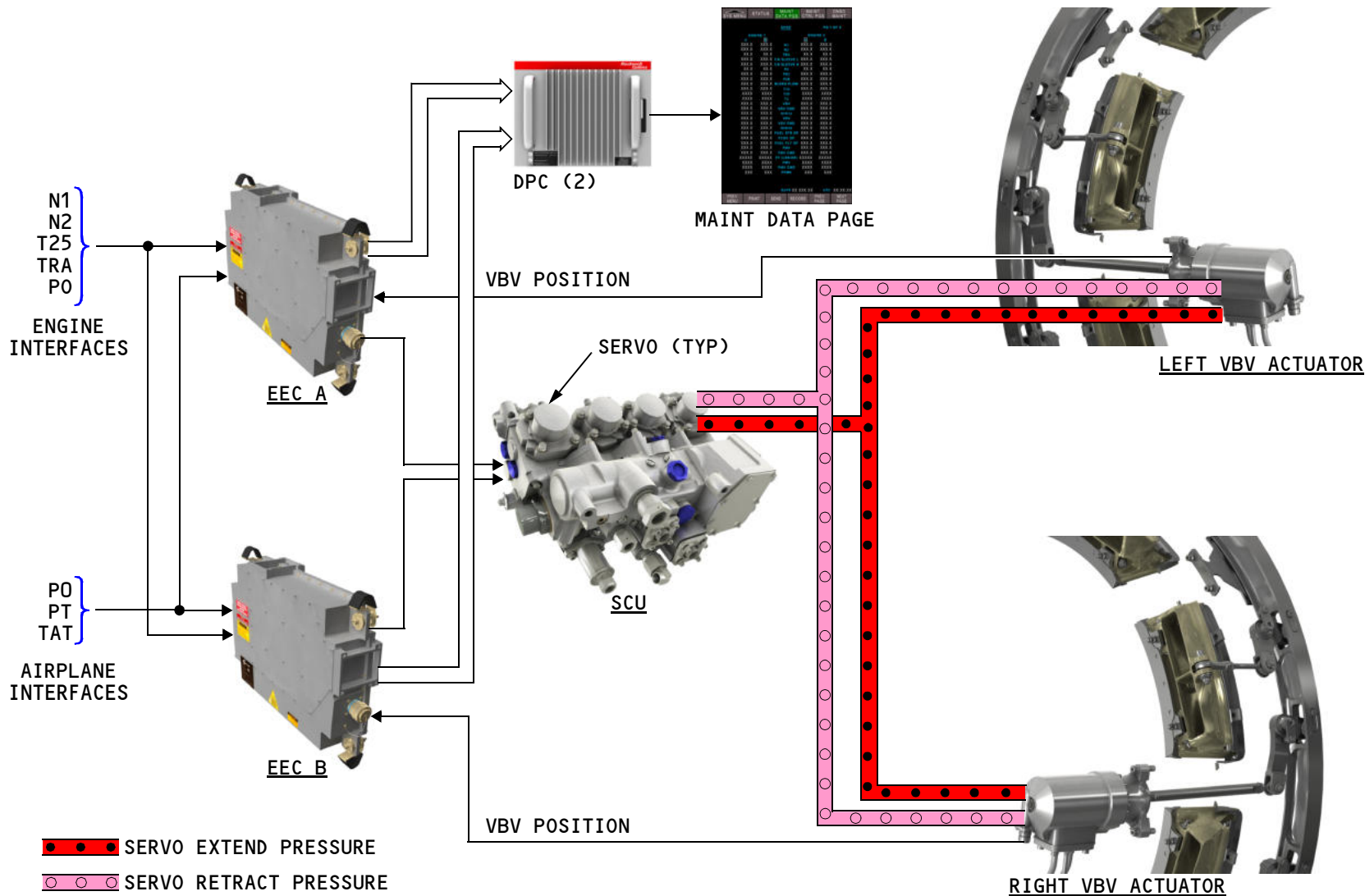
Operation

During steady state operation, the VBV doors go more closed as N1 increases. The VBV doors are closed when N1 is above 80%.

The EEC commands the VBV doors to be more open during engine deceleration and thrust reverser operation.

Training Information Point

VBV door position shows on the EPCS maintenance data page of the max display system (MDS).



ENGINE AIR -- VBV SYSTEM - FUNCTIONAL DESCRIPTION

75-32-00

02-Dec-2020

ENGINE AIR -- TBV - COMPONENT LOCATION

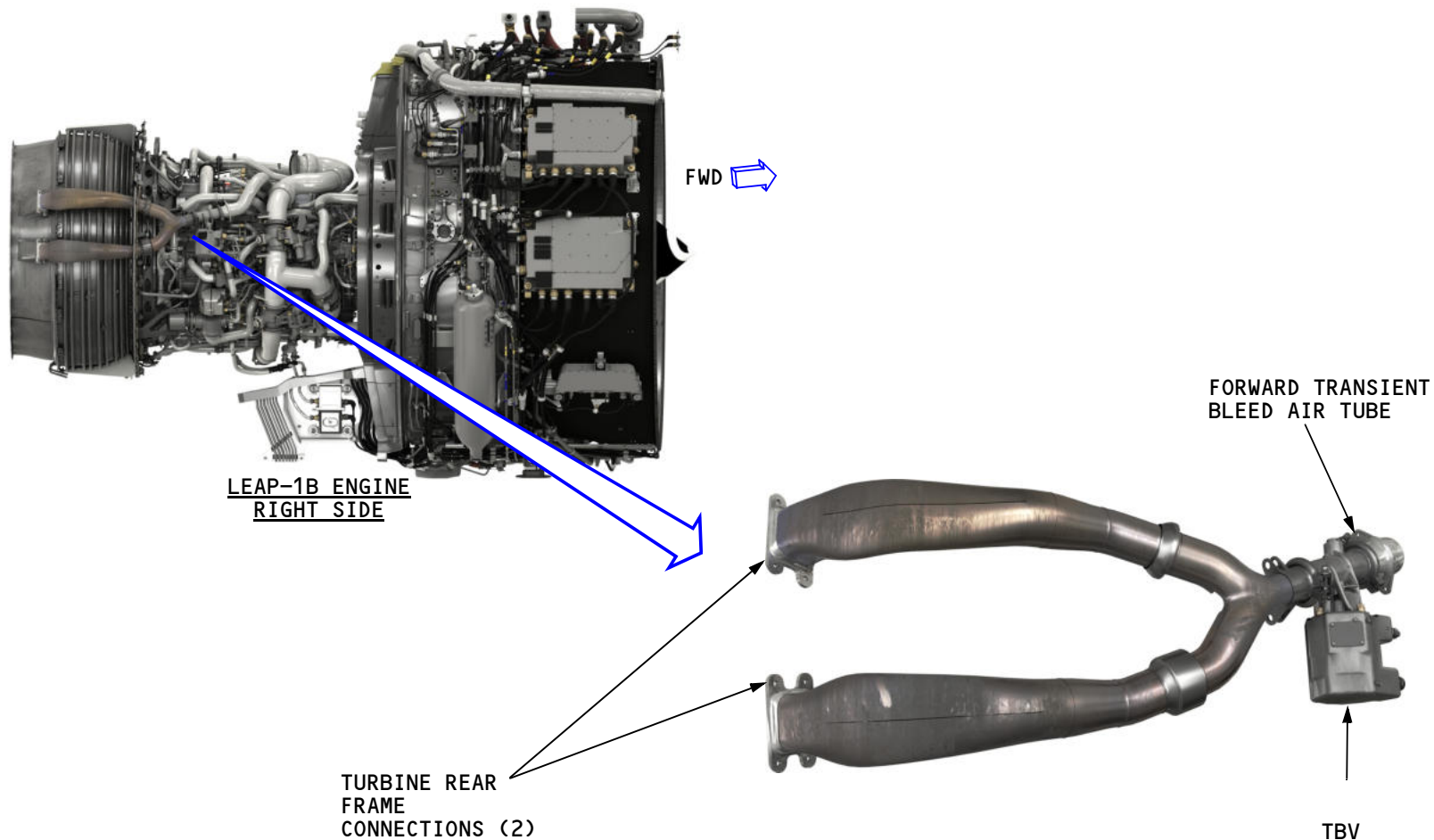
Component Locations

Most of the transient bleed valve (TBV) system components are on the right side of the engine. You open the right fan duct cowl and right thrust reverser cowl to get access to the TBV system components.

These are the components of the TBV system:

- Forward transient bleed air tube
- TBV
- Turbine rear frame connections.

The forward end of the forward transient bleed air tube connects near the top of the combustor case. The aft end of the forward transient bleed air tube connects to the TBV. The TBV is on the right side of the combustor case. It gives the air to the two TBV ducts. The two turbine rear frame connections are on the aft end of the TBV ducts. They are on the right side of the exhaust section. They connect to the TBV outlet.



ENGINE AIR -- TBV - COMPONENT LOCATION

75-23-00

DMC-B737T-A-75-23-00-02A-055A-T_003-00
ICN-B737T-A-752300-A-81205-00003-A-002-01

02-Dec-2020

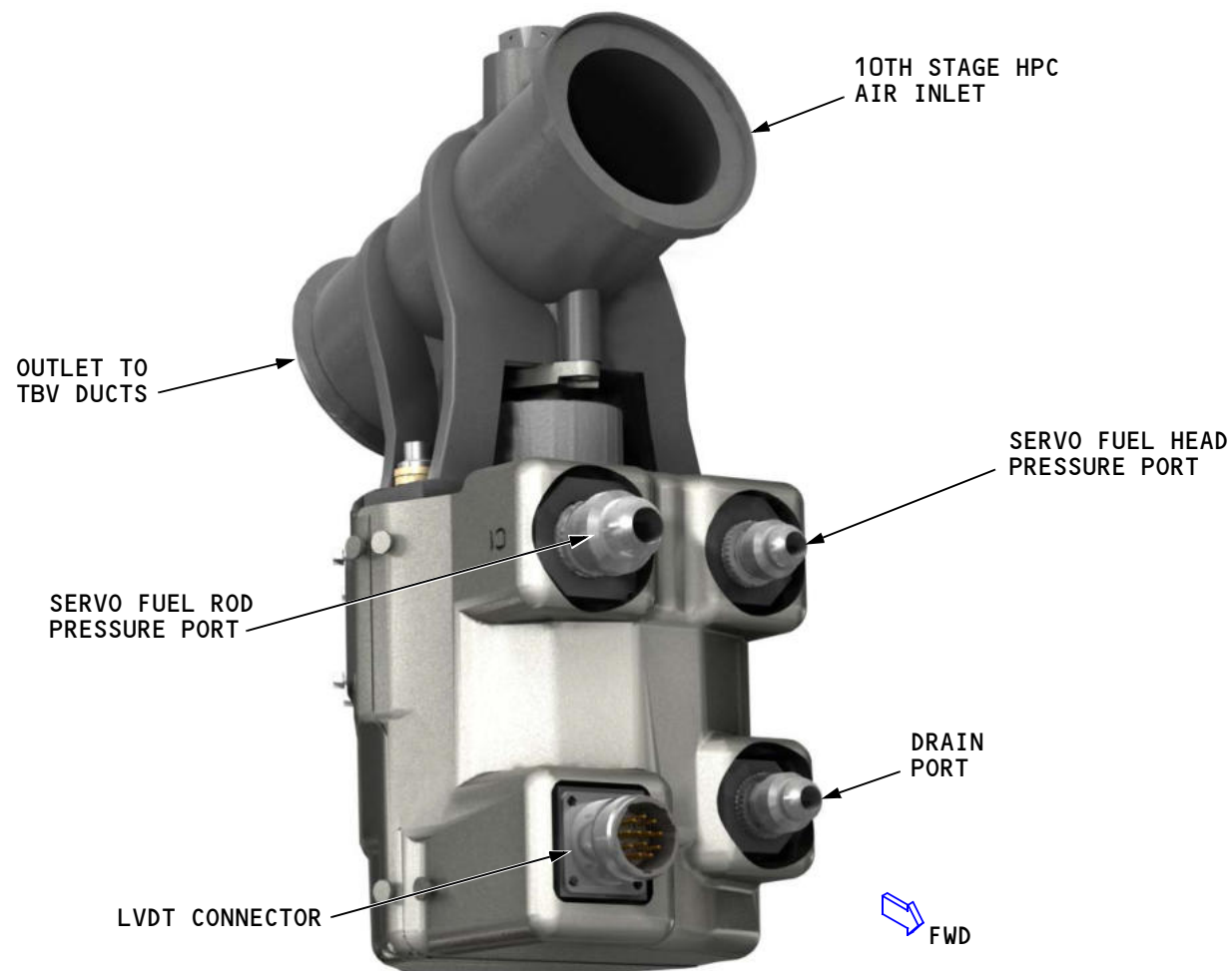
ENGINE AIR -- TBV - VALVE ASSEMBLY

Purpose

The transient bleed valve (TBV) opens during engine deceleration and other conditions to help prevent compressor stalls.

General Description

The TBV assembly has an actuator that moves a butterfly valve. When the valve opens, it sends 10th stage HPC air to the engine exhaust gaspath. The EEC channel in control sends commands to the split control unit (SCU), which then sends servo fuel pressure to the TBV. Servo fuel pressure operates the actuator. The actuator moves the butterfly valve and an LVDT position feedback sensor. The LVDT gives actuator position feedback to both of the EEC channels.



ENGINE AIR -- TBV - VALVE ASSEMBLY

75-23-00

DMC-B737T-A-75-23-00-02A-034B-T_002-00
ICN-B737T-A-752300-A-81205-00011-A-003-01

02-Dec-2020

ENGINE AIR -- TBV - FUNCTIONAL DESCRIPTION

General

The transient bleed valve (TBV) lets air from the high pressure compressor (HPC) 10th stage go to the turbine rear frame, into the exhaust gas path. The TBV system helps to prevent an HPC stall during engine starts and during engine acceleration/deceleration.

The electronic engine controls (EECs) use these parameters to control the transient bleed valve (TBV) position:

- N2 (corrected)
- Pressure at engine station 3 (PS3)
- Total air temperature (TAT)
- Fuel flow.

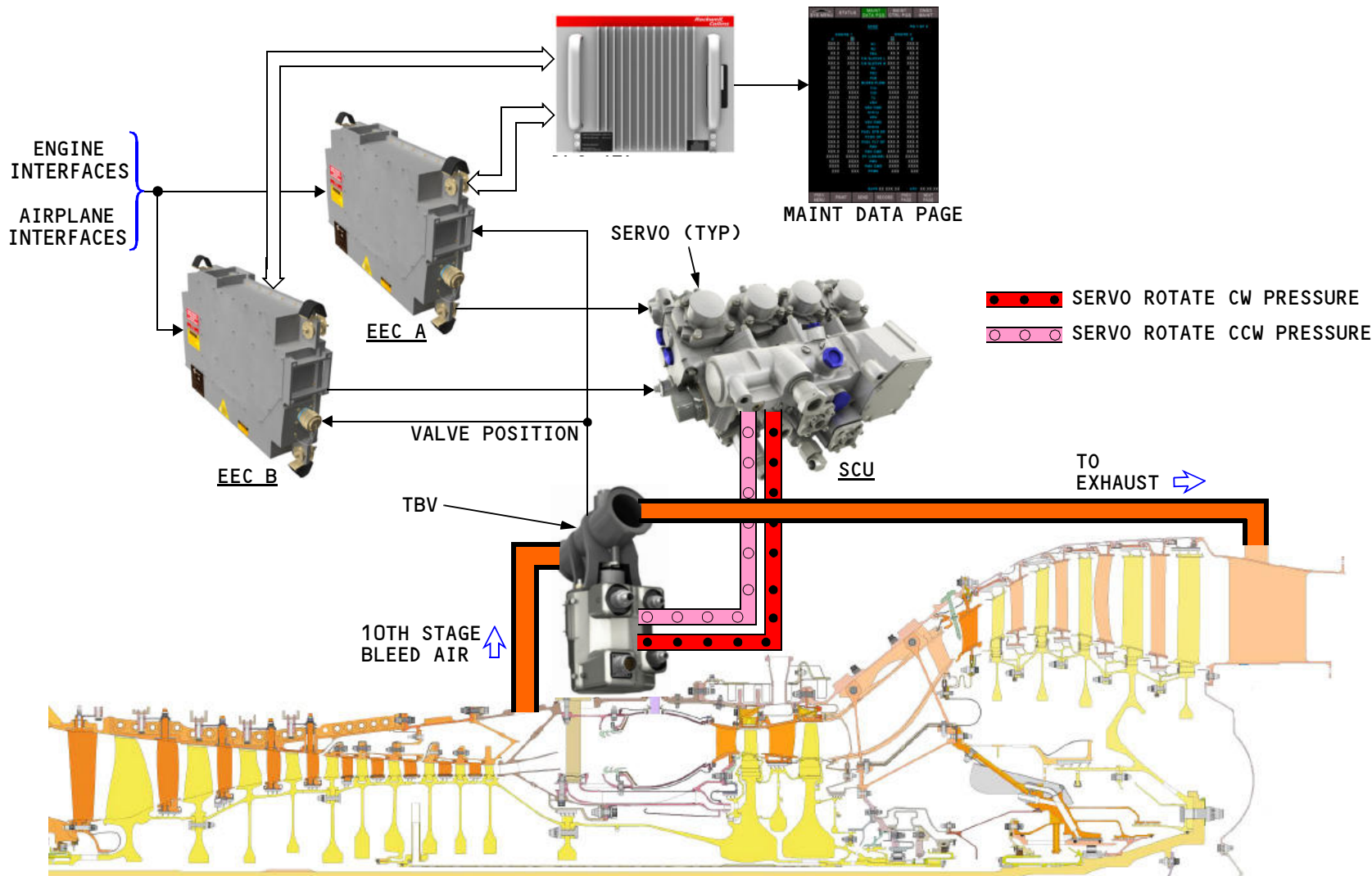
Operation

The EEC channel in control sends a signal to the servo in the split control unit (SCU). The servo sends fuel pressure to rotate the valve to the open or closed position.

Position sensors in the TBV send valve position back to the EECs.

Training Information Point

The TBV position shows on the EPCS maintenance data page.



ENGINE AIR -- TBV - FUNCTIONAL DESCRIPTION

75-23-00

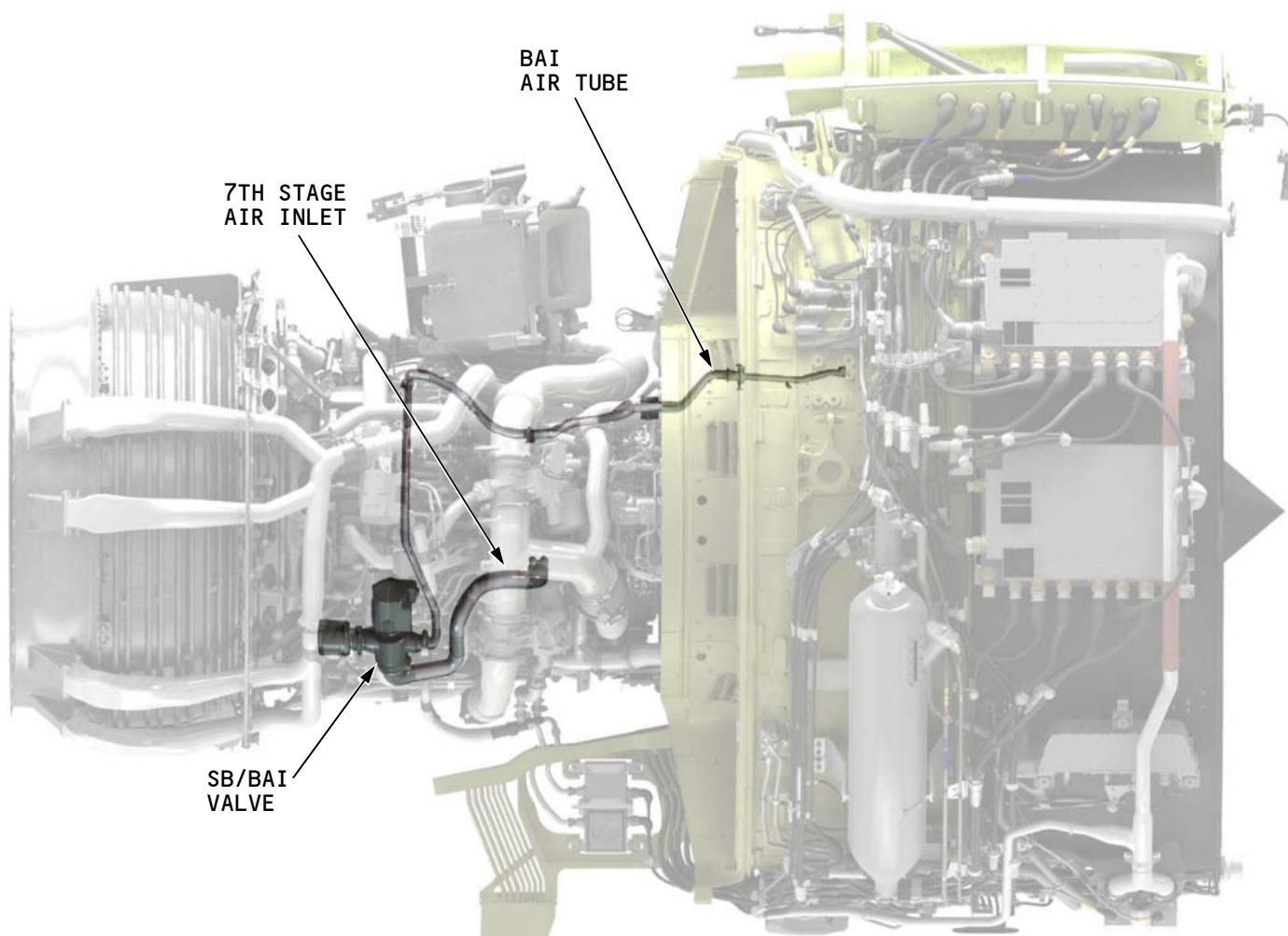
DMC-B737T-A-75-23-00-02A-042A-T_004-00
ICN-B737T-A-752300-A-81205-00004-A-001-01

02-Dec-2020

ENGINE AIR -- START BLEED/BOOSTER ANTI-ICE VALVE - COMPONENT LOCATION

General

The start bleed/booster anti-ice valve is on the right side of the combustor case at the 4:00 position. The forward end of the start bleed tube duct connects to the right side of the compressor case at the 3:00 position. The aft end of the start bleed tube duct connects to the bottom of the SB/BAI valve. The aft end of the booster anti-ice air tube connects to the SB/BAI valve forward side. The tube then goes to the top of the engine, where it turns forward. The forward end of the booster anti-ice air tube connects to a manifold on the low-pressure compressor booster.



DMC-B737T-A-75-33-00-02A-034B-T_002-00
ICN-B737T-A-753300-A-81205-00004-A-002-01

ENGINE AIR -- START BLEED/BOOSTER ANTI-ICE VALVE - COMPONENT LOCATION

75-33-00

02-Dec-2020

ENGINE AIR -- START BLEED/BOOSTER ANTI-ICE VALVE - VALVE ASSEMBLY

Purpose

The start bleed/booster anti-ice (SB/BAI) valve has these two functions:

- Decreases HPC air loads during engine start
- Prevents ice formation on the booster (LPC) inlet.

General Description

The SB/BAI valve assembly has an actuator and a valve. The valve has one air inlet from 7th stage HPC air. There are two air outlets:

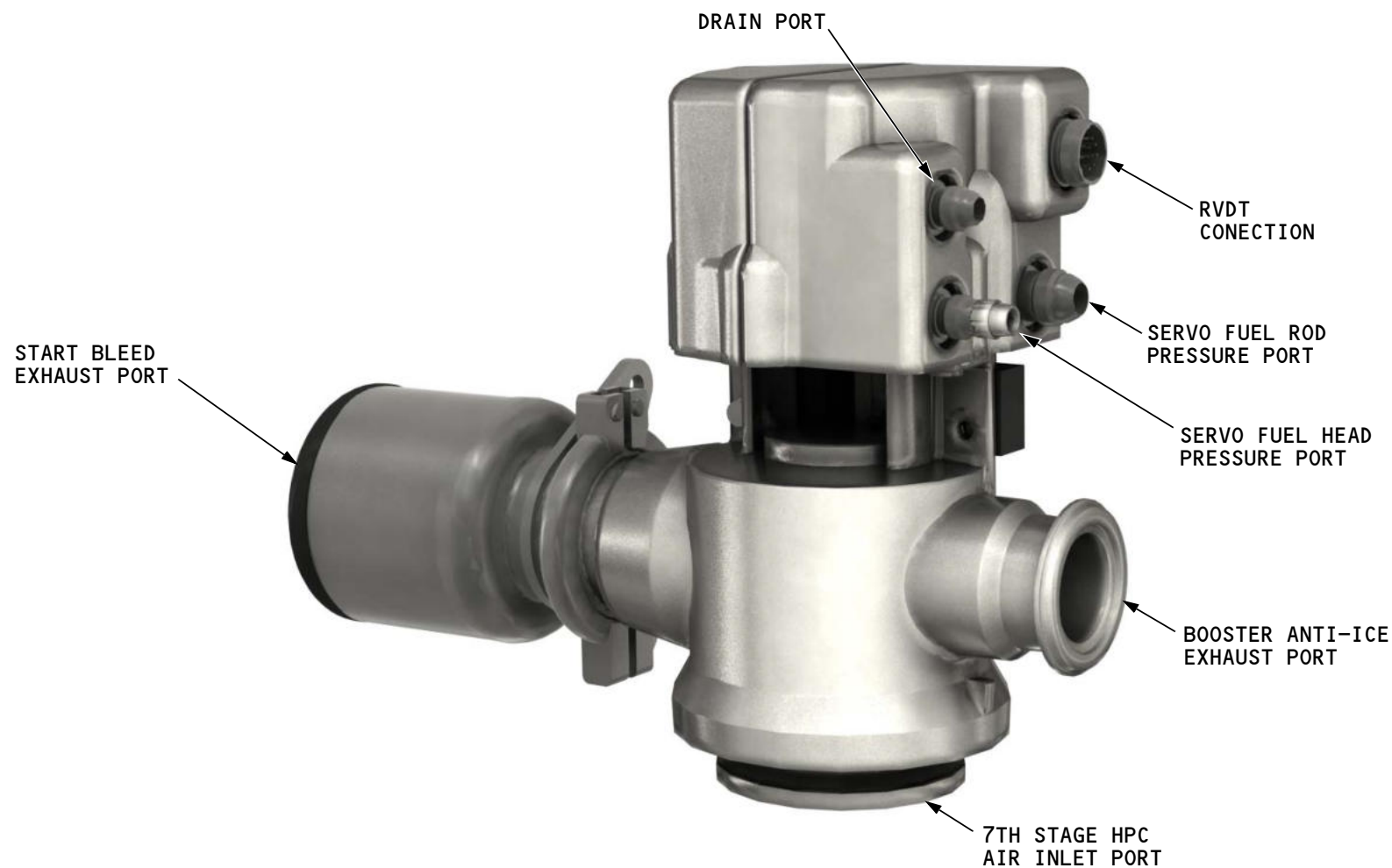
- Start bleed exhaust port
- Booster anti-ice port

The EEC channel in control sends commands to the split control unit (SCU), which then sends servo fuel pressure to the SB/BAI valve. The servo fuel pressure moves the valve actuator to provide engine start bleed or booster anti-ice. The valve has one RVDT that gives position feedback signals to both of the EEC channels. There is only one valve in the component that gives the two functions. Thus the component mixes the bleed function and the booster anti-ice function in some positions.

DMC-B737T-A-75-33-00-03A-034B-T_002-00

02-Dec-2020

75-33-00



ENGINE AIR -- START BLEED/BOOSTER ANTI-ICE VALVE - VALVE ASSEMBLY

75-33-00

DMC-B737T-A-75-33-00-03A-034B-T_002-00
ICN-B737T-A-753300-A-81205-00006-A-002-01

02-Dec-2020

ENGINE AIR -- START BLEED/BOOSTER ANTI-ICE VALVE - FUNCTIONAL DESCRIPTION

General

The start bleed/booster anti-ice (SB/BAI) valve does the start bleed and booster anti-ice functions.

There is one SB/BAI valve per engine. The valve has a single inlet and two outlets. The inlet to the SB/BAI valve air is from the 7th stage of the high pressure compressor. One outlet is for start bleed. The other outlet is for booster anti-ice.

SB/BAI Valve Operation

The start bleed function of the SB/BAI valve operates during engine starts. The booster anti-ice (BAI) function prevents ice at the booster (LPC) inlet.

The SB/BAI valve has an internal ball mechanism. The internal ball mechanism has 3 primary positions:

- No bleed air flow from the valve (closed position)
- Directs bleed air to the booster anti-ice outlet (up to the first 25% of travel)
- Directs bleed air to the booster anti-ice outlet and the start bleed outlet (from 25% to 100% open).

When you start the engine, the EEC channel in control sends a signal to an electro-hydraulic servo valve (EHSV) on the split control unit (SCU). This EHSV then gives servo fuel pressure to turn the SB/BAI valve clockwise. This makes the internal ball mechanism go to the start bleed position. The start bleed position of the valve causes the valve to release the 7th stage bleed air to the ambient under cowl area. Air also flows to the booster anti-ice outlet during start bleed operation because of the valve design.

The EEC activates the BAI function when all of these occur:

- The engine is running
- EEC finds that conditions for ice exist
- $N1 < 2616 \text{ RPM (59.5\%)}$.

The EEC channel in control sends a signal to the EHSV. The EHSV then sends servo fuel pressure to turn the SB/BAI valve counterclockwise. This makes the internal ball mechanism go to the BAI position. The BAI position of the valve causes the valve to release the 7th stage bleed air to the booster splitter (LPC inlet). This increases heat in the area of the splitter to prevent ice.

The SB/BAI valve has one RVDT that has two channels. Each channel of this RVDT gives valve position data to one EEC channel. The SB/BAI valve position shows on the EPCS maintenance data page.

Training Information Point

If the SB/BAI valve fails, these indications will show:

- ENG ANTI-ICE light on wing anti-ice panel comes on
- ENG 1 (2) BAI/SBV VALVE status message
- ENGINE CONTROL light and ENG 1 (2) CONTROL status message.



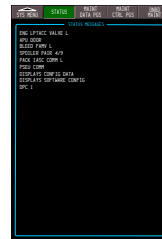
START PANEL (P5)



DPC (2)



MAINT DATA PAGE



STATUS PAGE



WING ANTI-ICE PNL



ENGINE
INTERFACES
AIRPLANE
INTERFACES

EEC A
EEC B

VALVE POSITION

SERVO (TYP)

SCU

SB/BAI
VALVE

START BLEED
OUTLET

BOOSTER
INLET

LPC

BOOSTER
ANTI-ICE
OUTLET

7TH STAGE
COMPRESSOR
BLEED AIR
INLET

HPC

● ● ● SERVO ROTATE CW PRESSURE
○ ○ ○ SERVO ROTATE CCW PRESSURE

ENGINE AIR -- START BLEED/BOOSTER ANTI-ICE VALVE - FUNCTIONAL DESCRIPTION

75-33-00



ENGINE AIR -- FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE AIR -- FUNCTIONAL DESCRIPTION

General

The EECs calculate the position commands for the different engine air components as a function of engine and airplane data. The EEC channel in control sends command signals to the servos in the split control unit (SCU). The actuator or valve provides position sensor feedback to the EECs.

High Pressure Turbine Active Clearance Control (HPTACC)

The HPTACC valve controls the amount of fan air that goes to the HPT air manifold. When the HPTACC valve opens, fan air goes to the high pressure turbine (HPT) case. The EEC channel in control sends a command signal to the SCU. The SCU sends servo fuel pressure to move the actuator in the HPTACC valve. A linear variable differential transducer (LVDT) sends the actuator position data to both of the EECs for closed-loop control and fault monitoring.

Low Pressure Turbine Active Clearance Control (LPTACC)

The LPTACC valve controls the amount of fan air that goes to the LPT cooling air manifold. From the LPT cooling air manifold, the fan air goes to the low pressure turbine (LPT) case. The EEC channel in control sends a command signal to the SCU. The SCU sends servo fuel pressure to move the actuator in the LPTACC valve. A linear variable differential transducer (LVDT) sends the valve position data to both of the EECs for closed-loop control and fault monitoring.

Modulated Turbine Cooling (MTC)

The MTC system controls the amount of cooling air to the first stage of high pressure turbine. This makes the engine use less fuel. The cooling air comes from the engine compressor outlet. The EEC channel in control sends a command signal to the SCU. The SCU

sends servo fuel pressure to move the MTC actuators, which each move an internal MTC valve. Each actuator assembly has an LVDT that sends the valve position data to the respective EEC for closed-loop control and fault monitoring.

Variable Bleed Valves (VBVs)

The VBVs control the amount of LPC discharge air that goes into the fan discharge airflow. The EEC channel in control sends a command signal to the SCU. The SCU sends servo fuel pressure to move the two actuators. The actuators mechanically connect with the bleed valves. An LVDT sends the position of each actuator to the respective EEC for closed-loop control and fault monitoring.

Variable Stator Vanes (VSVs)

The two VSV actuators control the angle of airflow through the forward stages of the HPC. The EEC channel in control sends a command signal to the SCU. The SCU sends servo fuel pressure to move the two actuators. The actuators mechanically connect with the variable inlet guide vanes and variable stator vanes. An LVDT sends the position of each actuator to the respective EEC for closed-loop control and fault monitoring.

Transient Bleed Valve (TBV)

The TBV discharges HPC 10th stage air to the engine exhaust. This improves the stall margin of the compressor. The EEC channel in control sends a command signal to the SCU. The SCU sends servo fuel pressure to move the TBV. An LVDT in the TBV gives valve position to the EECs for closed loop control and fault monitoring.

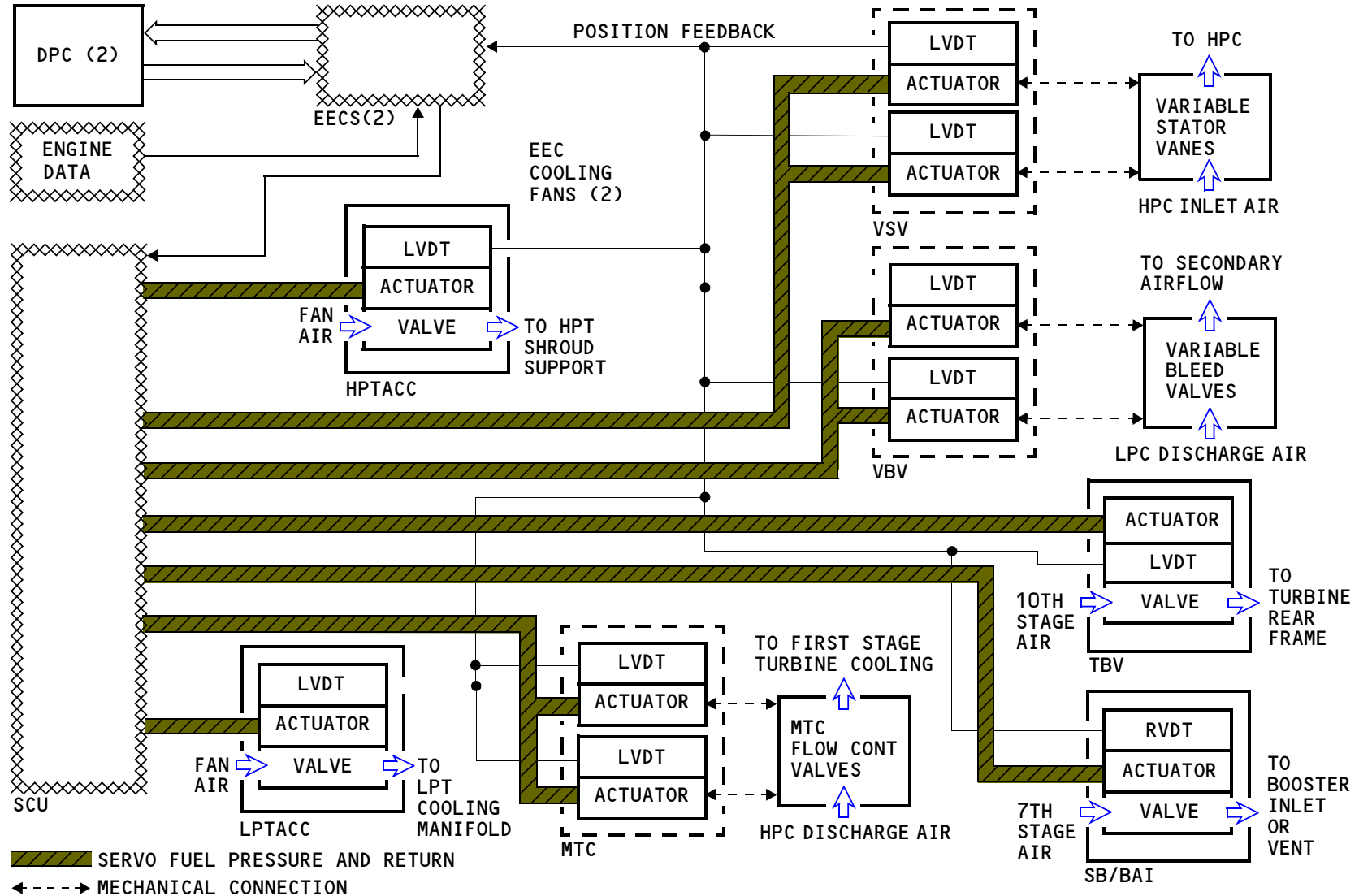
Start Bleed/Booster Anti-Ice (SB/BAI)

The SB/BAI valve gives these two functions:

ENGINE AIR -- FUNCTIONAL DESCRIPTION

- Engine booster anti-ice
- Start bleed.

The EEC channel in control sends a command signal to the SCU. The SCU sends servo fuel pressure to operate the valve. The valve can move to the anti-ice position, start bleed position, or middle closed position. An RVDT gives valve position data to the EECs for closed-loop control and fault monitoring.



DMC-B737T-A-75-00-00-02A-042A-T_006-00
ICN-B737T-A-750000-A-81205-00006-A-005-01

ENGINE AIR -- FUNCTIONAL DESCRIPTION

02-Dec-2020

75-00-00

ENGINE OIL -- INTRODUCTION

Purpose

The engine oil system supplies oil to lubricate, cool, and clean the engine bearings and gears. The engine oil system has these subsystems:

- Storage
- Distribution
- Indicating.

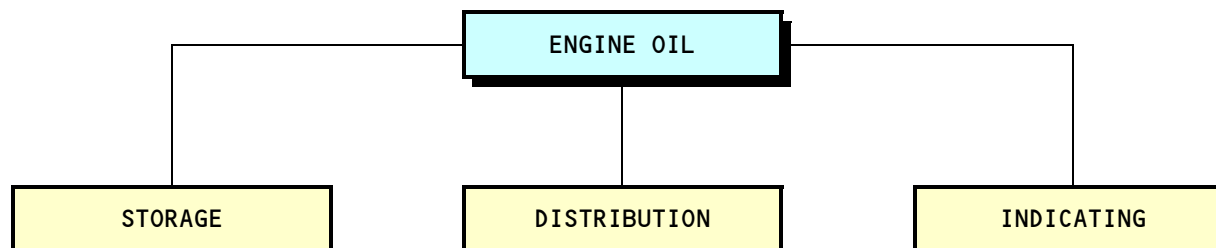
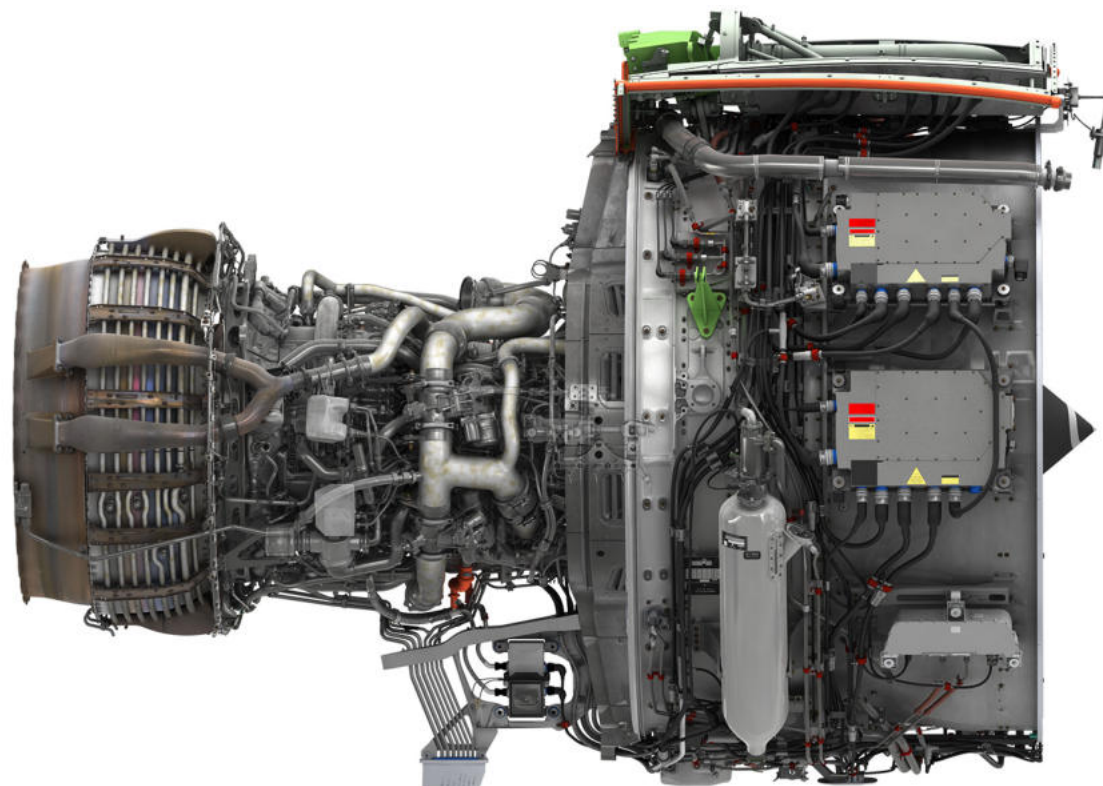
Abbreviations and Acronyms

ACOC	- air cooled oil cooler
AGB	- accessory gear box
AMM	- airplane maintenance manual
DPC	- display processing computer
DU	- display unit
MDS	- MAX display system
MHX	- main fuel/oil heat exchanger
EEC	- electronic engine control
EI	- engine indicator
psid	- pound-per-square-inch differential
TGB	- transfer gear box
TPV	- tank pressurizing valve
ODMS	- oil debris monitoring system
OFDPS	- oil filter differential pressure sensor
OLS	- oil level sensor
OPTS	- oil pressure/temperature sensor
SACOC	- surface air cooling oil cooler

DMC-B737T-A-79-00-00-02A-018A-T_003-00

02-Dec-2020

79-00-00



ENGINE OIL -- INTRODUCTION

79-00-00

DMC-B737T-A-79-00-00-02A-018A-T_003-00
ICN-B737T-A-790000-A-81205-00003-A-002-01

02-Dec-2020

ENGINE OIL -- GENERAL DESCRIPTION

General

The engine oil system has these subsystems:

- Storage
- Distribution
- Indicating.

Storage

The oil tank keeps sufficient oil for a continuous supply to the oil distribution circuit. The oil tank has a gravity fill port and an oil level sight gage for servicing.

Distribution

The oil distribution system has these circuits:

- Supply
- Scavenge
- Vent.

The supply circuit sends oil to lubricate the engine bearings and gears. Oil from the tank goes to the lubrication unit through an internal anti-leak check valve. The lubrication unit pressurizes and filters the oil. The oil then goes to the servo fuel heater, the engine surface air cooling oil coolers, and the main fuel/oil heat exchanger before it goes to the engine.

The scavenge circuit returns oil from the engine to the oil tank. Oil first flows through the lubrication unit. Then it goes through the air/oil/debris separator (Lubriclone). From the air/oil/debris separator, the oil returns to the oil tank.

The vent circuit balances the internal air pressures in the oil system. A vent line connects the engine A sump to the lubriclone. Unwanted air

pressure goes out of the oil tank through the vent line. The engine sumps are vented through the engine center shaft. An oil sump eductor increases vent airflow at low engine speeds to improve oil control.

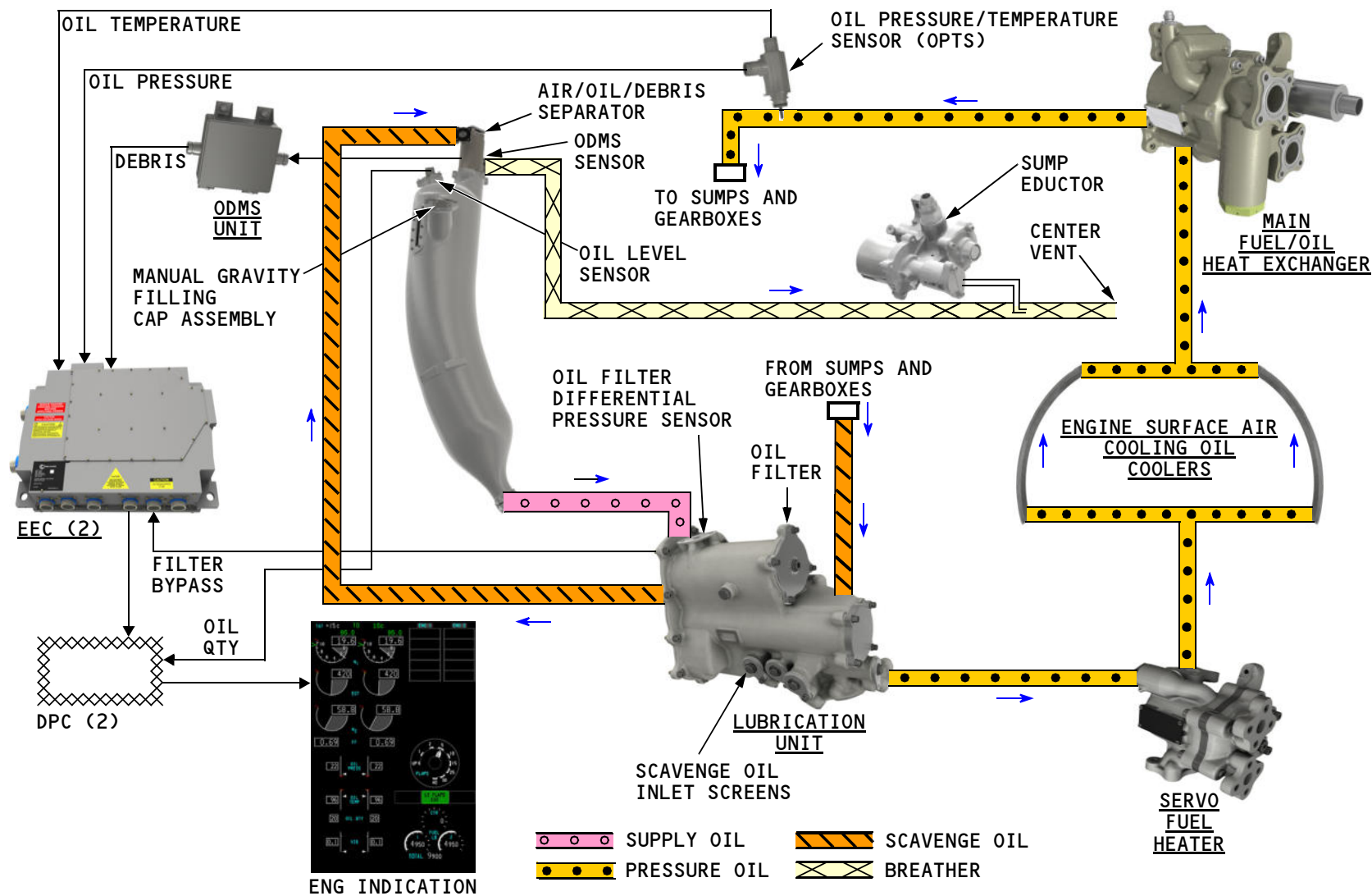
Indication

The oil quantity indicating system sends this data to the engine electronic engine controls (EECs):

- Oil filter differential pressure
- Oil debris monitoring system (ODMS) sensor
- Oil pressure
- Oil temperature.

The oil quantity indicating system sends this data directly to the display processing computers (DPCs):

- Oil quantity.



DMC-B737T-A-79-00-00-02A-040A-T_008-00
ICN-B737T-A-790000-A-81205-00004-A-008-01

ENGINE OIL -- GENERAL DESCRIPTION

02-Dec-2020

79-00-00

ENGINE OIL -- STORAGE - OIL TANK

General

These are the functions of the engine oil tank:

- Stores the engine oil
- Lets you check the oil level
- Lets you service the oil system.

Component location

The engine oil tank is on the fan case, at the 4:00 position. Normally the oil level check and fill of the oil tank is through the oil tank access door. The oil tank access door is on the side of the right fan cowl. Opening the right fan cowl will also give you access to the oil tank.

Physical Description

The oil tank has a oil level sight gage and a gravity fill port.

Use the oil level sight gage on the oil tank to make a visual check of the engine oil quantity. The oil level sight gage is on the front of the oil tank.

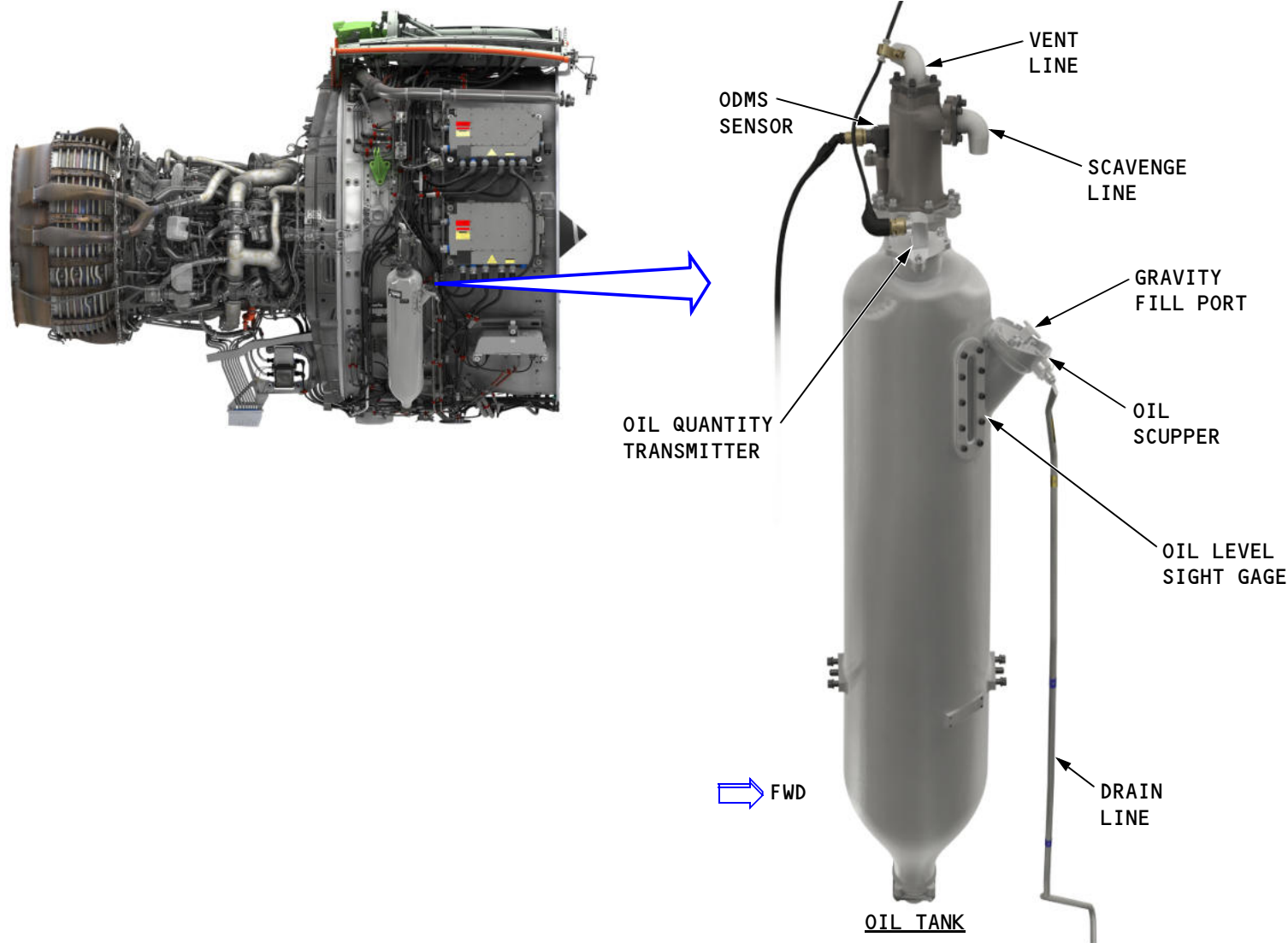
Use the oil tank gravity fill port to fill the oil tank. The gravity fill port is on the right of the oil tank. The oil filler cap has a locking handle. The oil that spills during servicing collects in the oil scupper. The oil scupper connects to a drain line.

To drain the oil tank, you use a drain plug at the low point in the oil supply tube, on the accessory gearbox.

The engine oil tank has a total volume of approximately 27.83 US quarts (26.34 liters). The oil tank for right engine has a total usable oil quantity of 22.61 quarts (21.40 liters) and the oil tank for left engine has a total usable oil quantity of 22.35 quarts (21.15 liters). The

difference in usable oil quantity between the left and right engine is because of the dihedral of the wings.

DMC-B737T-A-79-10-00-02A-034B-T_002-00



ENGINE OIL -- STORAGE - OIL TANK

79-10-00

DMC-B737T-A-79-10-00-02A-034B-T_002-00
ICN-B737T-A-791000-A-81205-00003-A-002-01

02-Dec-2020

ENGINE OIL -- STORAGE - SERVICING

General

Check the engine oil sight gage and fill the engine oil tank for any of these conditions:

- Normal servicing
- After replacement of an oil system component
- Engine oil change.

Engine Oil Level

Open the oil tank access door to access and see the sight gage.

The engine oil level also shows in the flight compartment, on the MAX display system (MDS) engine indication display.

Engine Oil Servicing (Gravity Fill)

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING: DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING: DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR

PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING: DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION: DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

Check the oil level and service the tank not less than five minutes and not more than 60 minutes after engine shutdown, while the oil in the tank is still warm. If the oil in the tank is cold, the oil density will increase. This will let the volume decrease and the oil tank can be over-serviced. If the oil tank is over-serviced, the extra oil will be blown overboard through the vent system. This may also cause an incorrect calculation of the oil consumption rate.

To service the engine oil, open the locking oil filler cap. Do a visual and a smell inspection for the presence of contamination in the oil. Add the engine oil into the gravity fill port until the oil level gets to the full indication on the sight gage. If the engine has been shutdown for longer than 60 minutes the oil level may not be reliable, run the engine at ground idle for a minimum of 5 minutes, and then service as necessary.



OIL TANK ACCESS



OIL TANK

ENGINE OIL -- STORAGE - SERVICING

79-10-00

DMC-B737T-A-79-10-00-02A-200A-T_002-00
ICN-B737T-A-791000-A-81205-00004-A-002-01

02-Dec-2020

ENGINE OIL -- DISTRIBUTION - GENERAL DESCRIPTION

General

The engine oil distribution system supplies oil to cool, clean, and lubricate the engine bearings and gears. The engine oil distribution system also removes oil from sumps and gearboxes. It sends the oil to the storage system. Engine oil distribution has these systems:

- Supply
- Scavenge
- Vent.

Supply System

From the oil tank, the oil flows to the lubrication unit through the anti-leakage valve. In the lubrication unit, the supply pump pressurizes the oil. The oil goes from the supply pump to the oil filter. The oil filter is inside the lubrication unit. The oil flows out of the lubrication unit and through the servo fuel heater. The oil flows out of the servo fuel heater and through the engine surface air cooling oil coolers. The oil flows out of the engine surface air cooling oil coolers and through the main oil/fuel heat exchanger before it goes into the engine to these areas:

- Forward (A) sump and transfer gearbox (TGB)
- Rear (B and C) sumps
- Accessory gearbox (AGB).

Scavenge System

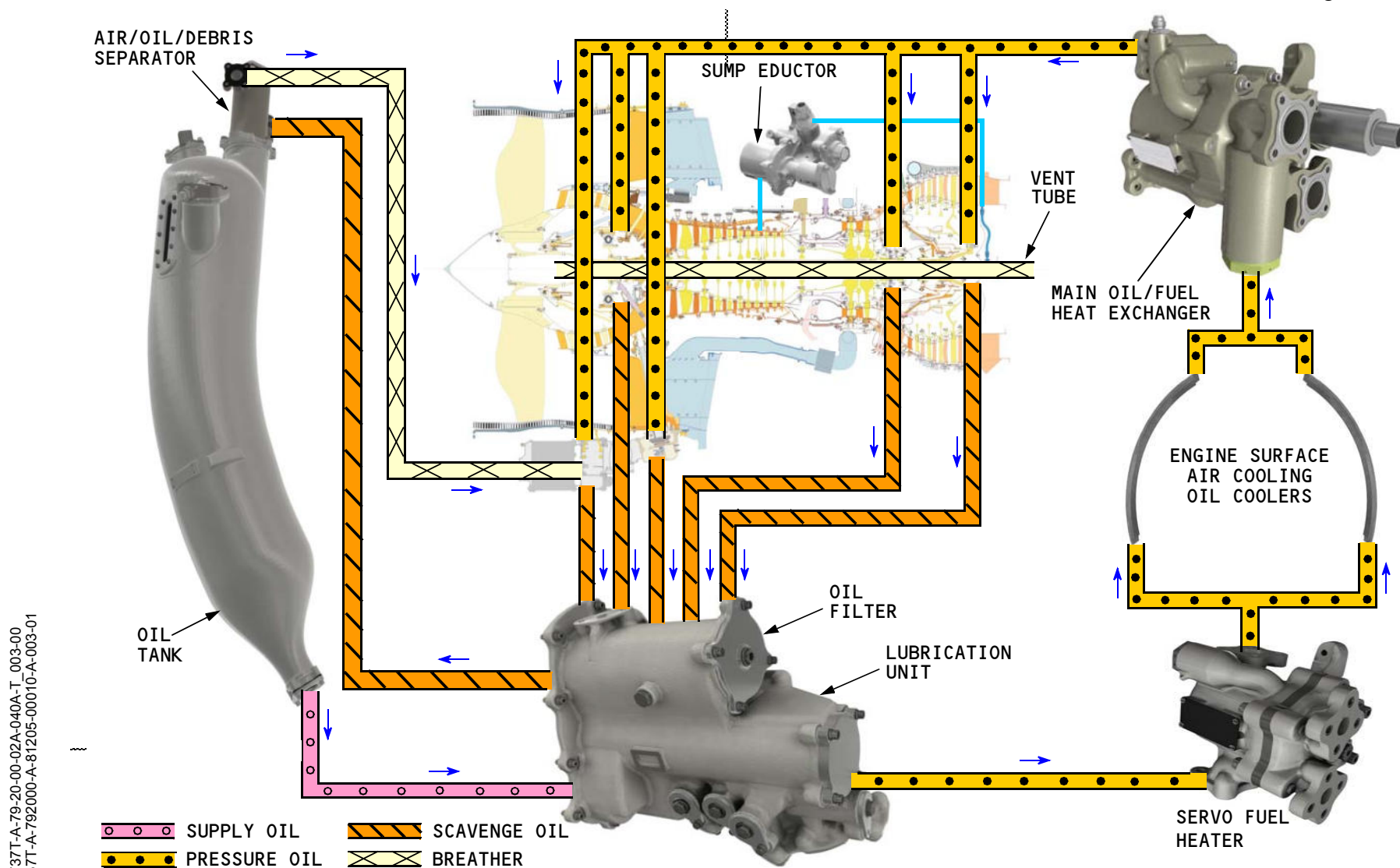
The scavenge system removes the oil that collects at the lowest point of these three areas:

- Forward (A) sump
- Rear (B and C) sump
- AGB and TGB (2 places).

From these areas, the oil flows through six lines to four scavenge screen plugs with magnetic bars. The oil from each scavenge line goes to a dedicated scavenge pump before it goes to the air/oil/debris separator on the top of the oil tank. Air goes to the vent system, and oil returns to the oil tank.

Vent System

A vent line connects the oil tank mounted air/oil/debris separator with the accessory gearbox (AGB). There are also internal connections between the engine sumps and gearboxes. The vent system bleeds air out of the vent tube which goes through the exhaust plug at the rear of the engine. The sump eductor sends 7th stage high pressure compressor (HPC) air through the vent at low engine speeds to improve airflow through the A sump.



DMC-B737T-A-79-20-00-02A-040A-T_003-00
ICN-B737T-A-792000-A-81205-00010-A-003-01

ENGINE OIL -- DISTRIBUTION - GENERAL DESCRIPTION

02-Dec-2020

79-20-00

ENGINE OIL -- DISTRIBUTION - COMPONENT LOCATION

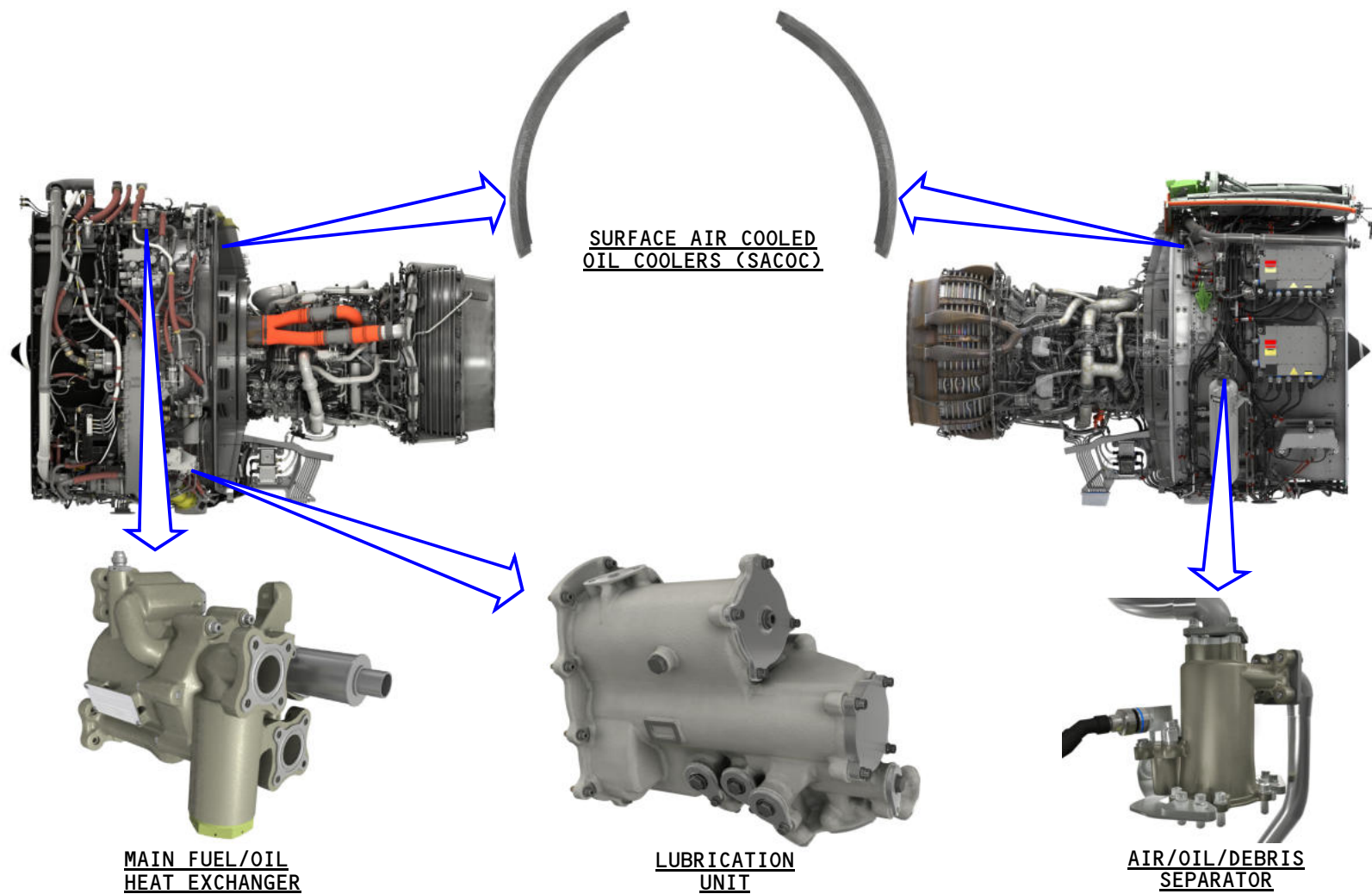
Component Locations

These components of the engine oil distribution system are on the left side, and at the bottom of the fan case:

- Lubrication unit (7:00 position)
- Main oil/fuel heat exchanger (MHX) (10:00 position)
- Left Surface air cooled oil cooler (SACOC) (8:00 to 11:00 position)
- Right Surface air cooled oil cooler (SACOC) (1:00 to 4:00 position)
- Air/oil/debris separator (3:00 position).

The oil supply filter and the two single and two double scavenge inlet screen plugs with magnetic bars are located in the lubrication unit.

Open the left and right fan cowls to get access to the components of the engine oil distribution system.



DMC-B737T-A-79-20-00-02A-055A-T_002-00
ICN-B737T-A-792000-A-81205-00011-A-003-01

ENGINE OIL -- DISTRIBUTION - COMPONENT LOCATION

79-20-00

02-Dec-2020

ENGINE OIL -- DISTRIBUTION - LUBRICATION UNIT

Purpose

The lubrication unit supplies pressurized oil to lubricate the engine bearings and gears. It also takes the oil that collects in the sumps and the gearboxes and sends it back to the oil tank.

The oil supply pump does not control the output pressure. When the engine speed changes, the oil pressure changes.

Location

The lubrication unit is on the rear face of the accessory gearbox at the 7:00 position. Open the left fan cowl to get access to the lubrication unit.

Physical description

The lubrication unit contains these parts:

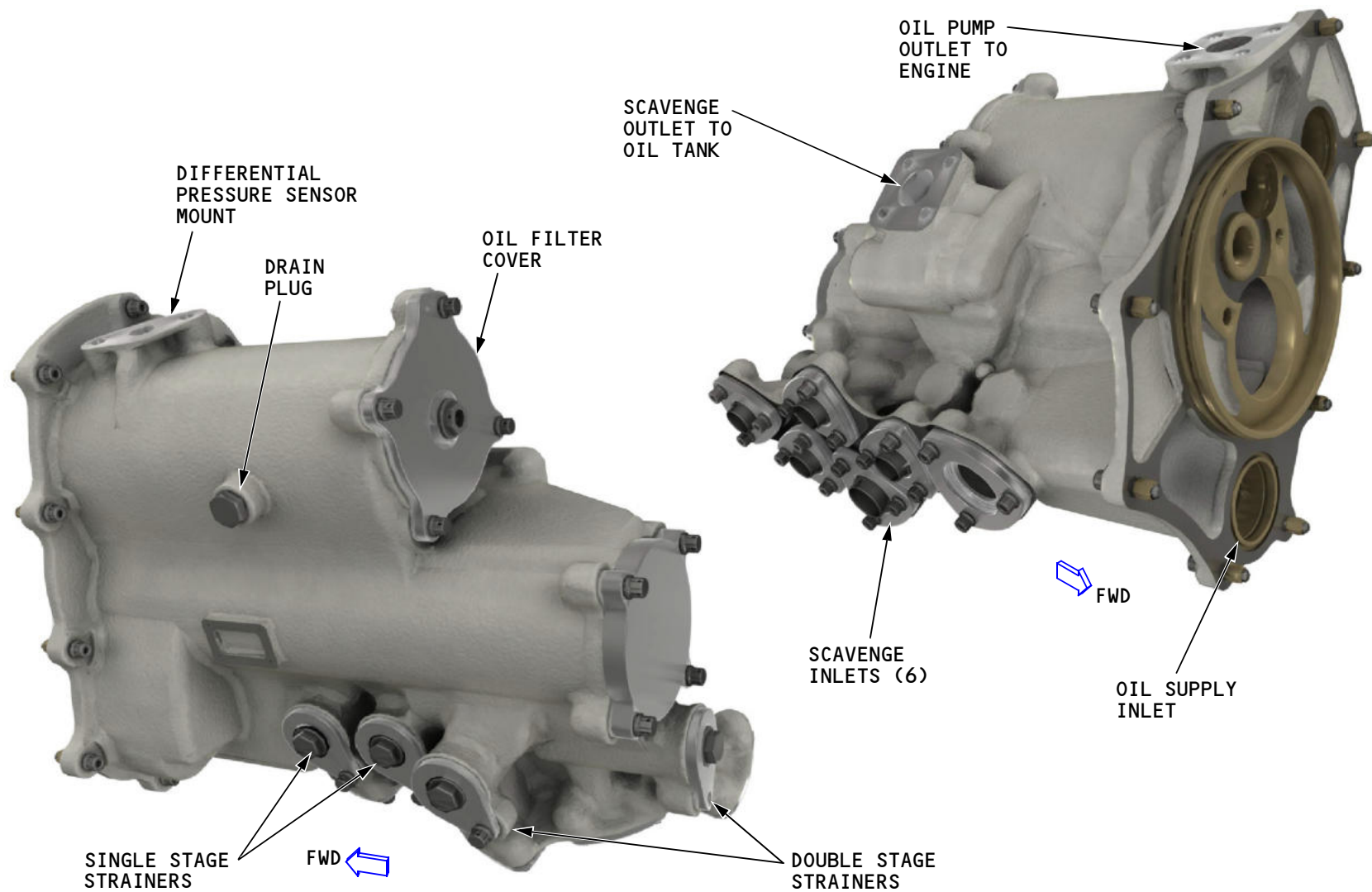
- Oil supply pump (not shown)
- Supply oil filter
- Supply oil filter bypass valve (not shown)
- Supply oil filter differential pressure sensor
- Pressure relief valve (not shown)
- Oil scavenge pumps (6) (not shown)
- Two single stage strainers with magnetic bars.
- Two double stage strainers with magnetic bars.
- Anti-leak valve (not shown)
- Non return valve (not shown)

The lubrication unit is attached to the accessory gearbox by nine (9) screws.

Functional Description

The accessory gearbox turns the oil supply pump and the six scavenge pumps in the lubrication unit. The pumps are on a common shaft.

DMC-B737T-A-79-20-00-07A-034B-T_003-00



DMC-B737T-A-79-20-00-07A-034B-T_003-00
ICN-B737T-A-792000-A-81205-00012-A-003-01

ENGINE OIL -- DISTRIBUTION - LUBRICATION UNIT

79-20-00

02-Dec-2020

ENGINE OIL -- DISTRIBUTION - OIL SUPPLY FILTER

Purpose

The supply oil filter removes and holds unwanted material from the supply oil. The supply oil filter prevents the contamination of the downstream oil circuit.

Location

The supply oil filter is in the lubrication unit. The lubrication unit is on the rear face of the accessory gearbox at the 7:00 position. Open the left fan cowl to get access to the supply oil filter.

Physical Description

The supply oil filter is a disposable paper filter cartridge.

The lubrication unit housing holds the supply oil filter. The cover keeps the supply oil filter in the housing. The oil filter housing cover has a drain plug.

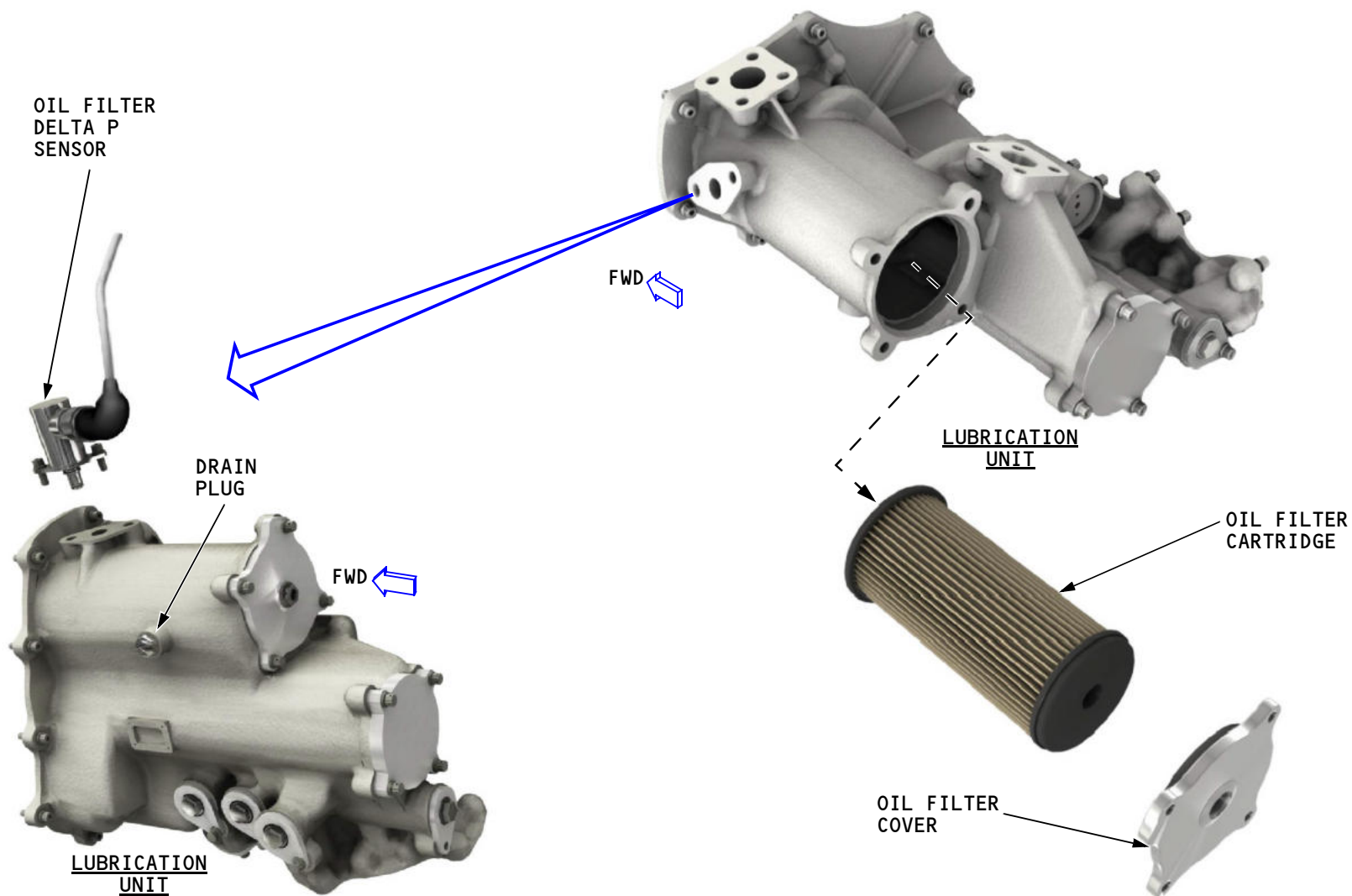
Training Information Point

When the filter clogs, the supply oil filter bypass valve opens. The bypass valve is in the lubrication unit housing. A differential pressure sensor provides indication before the bypass valve opens.

Training Information Point

Remove the drain plug, and drain the oil filter housing before removing the cover and the filter.

DMC-B737T-A-79-20-00-08A-034B-T_002-00



DMC-B737T-A-79-20-00-08A-034B-T_002-00
ICN-B737T-A-792000-A-81205-00013-A-003-01

ENGINE OIL -- DISTRIBUTION - OIL SUPPLY FILTER

79-20-00

ENGINE OIL -- DISTRIBUTION - ENGINE SURFACE AIR COOLING OIL COOLERS

Purpose

The engine surface air cooling oil coolers give additional cooling to the oil.

Location

There are two engine surface air cooling oil coolers. One is on the left side inner wall of the fan frame, at the 08:00 to 11:00 o'clock position, and the other is on the right side at the 01:00 to 04:00 o'clock position. Each engine surface air cooling oil cooler has these features:

- Oil inlet/outlet manifold
- Plain fin core (Matrix)
- Pressure relief valve
- Thermal valve.

Functional Description

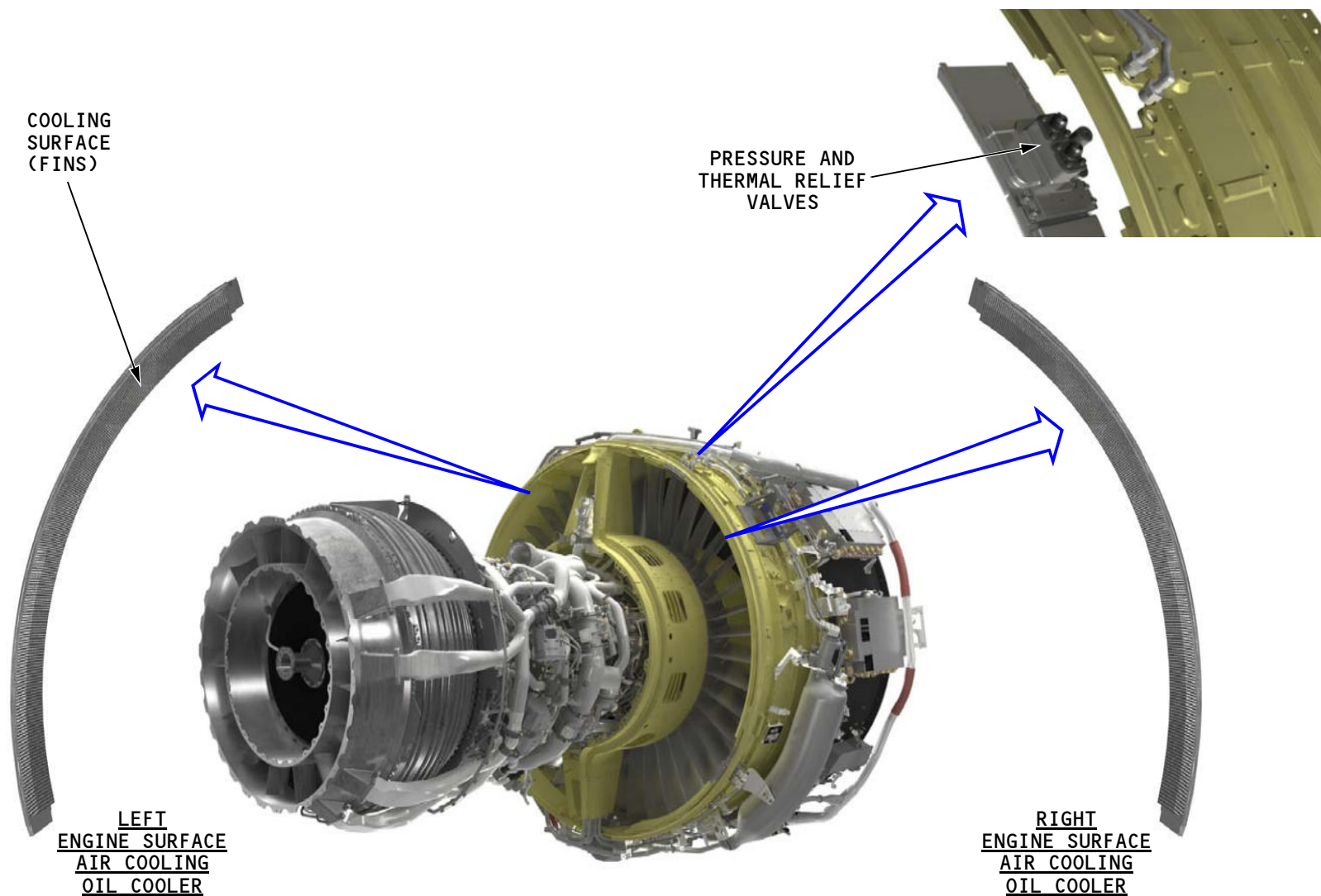
The engine surface air cooling oil coolers are fixed to the inner surface of the fan frame by hard mounts and a sliding feature to relieve thermal strain. The inlet/outlet manifold has a pressure relief valve to allow the oil to bypass if there is a blockage in the engine surface air cooling oil cooler.

The inlet/outlet manifold also has a thermal valve, which regulates the amount of oil cooled by the engine surface air cooling oil cooler. When the oil is cold, the thermal valve is open. This lets oil bypass the engine surface air cooling oil cooler. When the oil is warm, the thermal valve is closes. Warm oil then passes through the engine surface air cooling oil cooler plain fin core.

DMC-B737T-A-79-20-00-20A-034B-T_003-00

02-Dec-2020

79-20-00



DMC-B737T-A-79-20-00-20A-034B-T_003-00
ICN-B737T-A-792000-A-81205-00035-A-002-01

ENGINE OIL -- DISTRIBUTION - ENGINE SURFACE AIR COOLING OIL COOLERS

79-20-00

02-Dec-2020

ENGINE OIL -- DISTRIBUTION - MAIN OIL/FUEL HEAT EXCHANGER

Purpose

The main oil/fuel heat exchanger uses fuel from the fuel pump to decrease the temperature of the supply oil.

See the fuel distribution system section for more information on the fuel system flow (Chapter 73-11).

Location

The main oil/fuel heat exchanger (MHX) is on the fan case at the 10:00 position. You open the left fan cowl to get access to the main oil/fuel heat exchanger.

Physical Description

The main oil/fuel heat exchanger has these parts:

- Housing
- Fuel and oil connections
- Exchanger core (inside the housing)
- Oil Bypass valve
- Oil Pressurizing valve (OPV)
- Fuel delta P sensor.

The exchanger core is where the oil temperature decreases as the oil heats the fuel.

The main oil/fuel heat exchanger has an oil bypass valve. The oil bypass valve opens when the exchanger core is clogged. This lets the oil flow around the exchanger core. The oil temperature increases when the oil bypass valve opens, but the oil continues to flow.

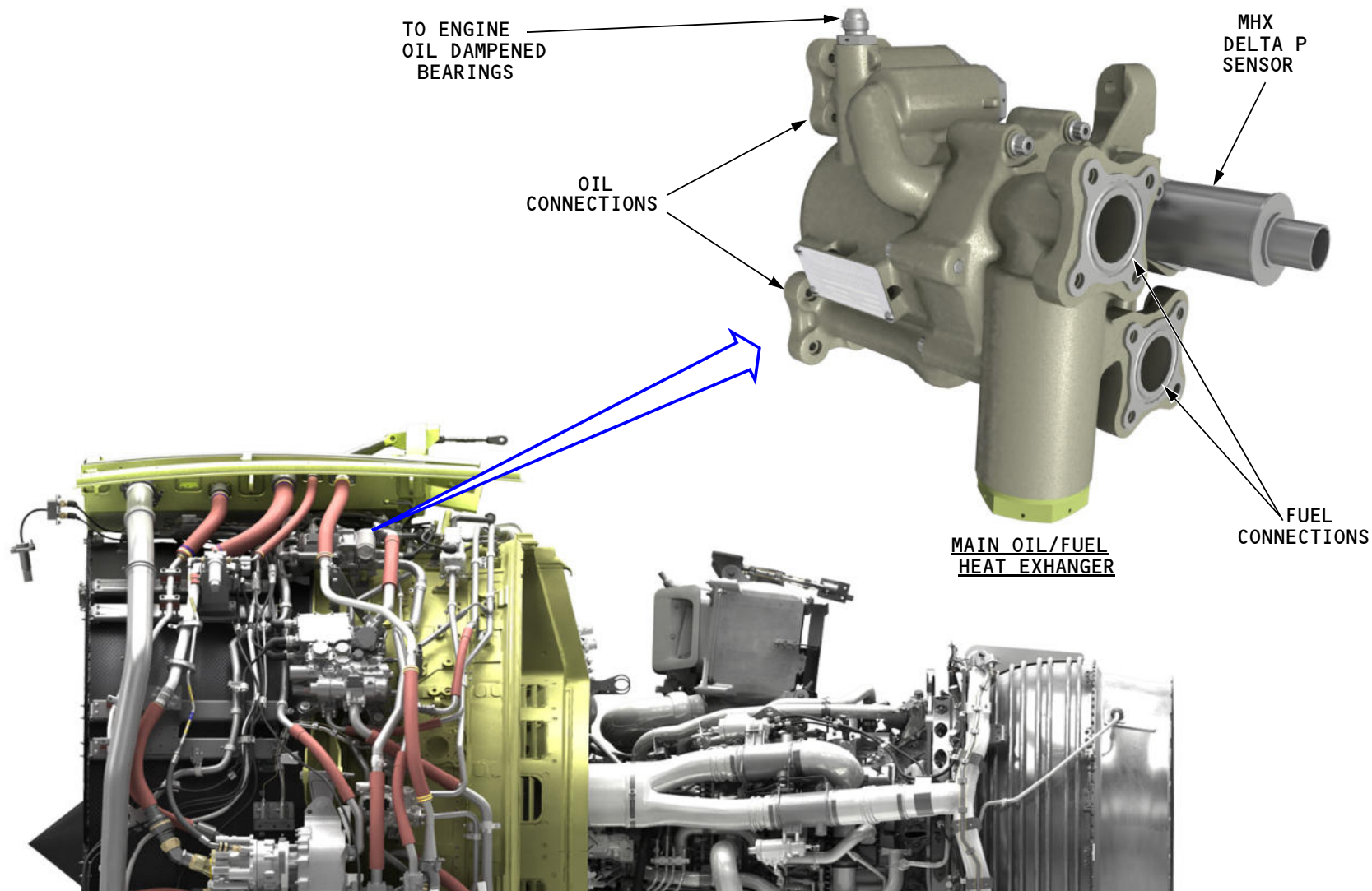
The main oil/fuel heat exchanger has an oil pressurizing valve (OPV). The OPV makes sure that there is a constant supply of pressurised oil to feed the engine oil dampened bearings.

The fuel delta P sensor indicates a blockage in the fuel flow through the heat exchanger.

Training Information Point

You replace the main oil/fuel heat exchanger as an assembly.

If the main oil/fuel heat exchanger has an internal leak, fuel goes into the oil circuit. This occurs because the fuel pressure is higher than the oil pressure. If you smell fuel in the oil tank or if the oil quantity increases, refer to the AMM for more information.



DMC-B737T-A-79-20-00-10A-034B-T_002-00
ICN-B737T-A-792000-A-81205-00015-A-001-01

ENGINE OIL -- DISTRIBUTION - MAIN OIL/FUEL HEAT EXCHANGER

79-20-00

02-Dec-2020

ENGINE OIL -- DISTRIBUTION - SCAVENGE SCREEN PLUGS

Purpose

The scavenge screen plugs collect and keep unwanted materials from the scavenge oil. Collected material can indicate if there is mechanical failure of an engine bearing or gear. The scavenge screen plugs have magnetic bars that keep a sample of any ferrous material and a screen for non-ferrous material pieces larger than 0.71 mm.

There are four scavenge screen plugs with magnetic bars. The plugs for the A and B sump are single stage strainers. The plug for the transfer gear box is a double stage strainer. The plug for the combined accessory gear box and C sump is also double stage strainer.

Location

The lubrication unit housing holds the scavenge screen plugs. They are at the inlet of the scavenge pumps. The lubrication unit is on the rear of the accessory gearbox at the 7:00 position.

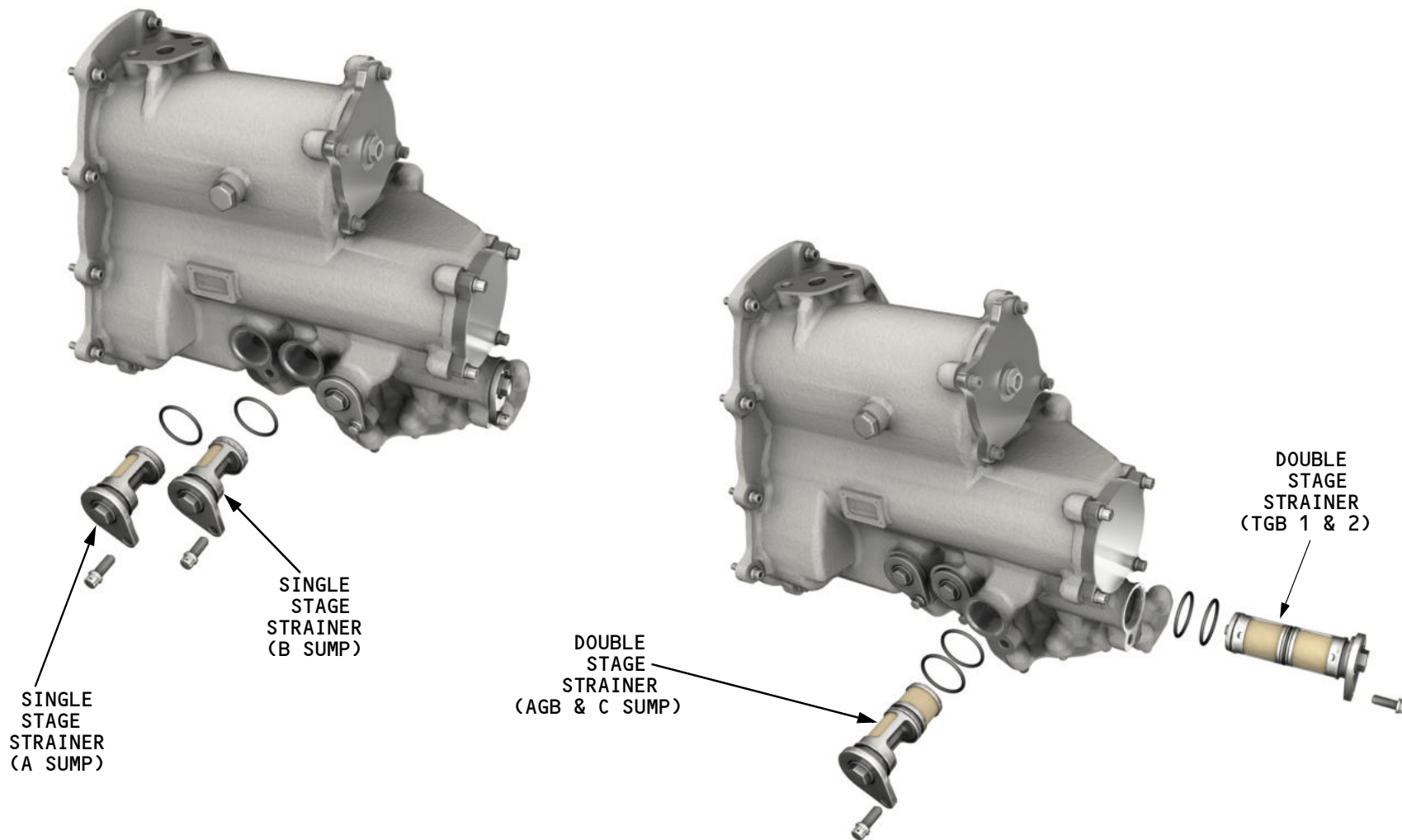
Physical Description

A scavenge screen plug has a magnetic bar and a metallic-mesh screen. Each scavenge screen plug attaches to the lubrication unit housing with a single attach bolt. There are two types of scavenge screen plugs; scavenge screen plugs with single stage strainers, and scavenge screen plugs with double stage strainers.

Training Information Point

The engine also detects ferrous metal in the oil with the oil debris monitoring system (ODMS). Once alerted to the presence of metal in the oil by the ODMS, the scavenge screen plugs would be inspected for the presence of metal particles. This can then identify the area of the failure.

DMC-B737T-A-79-20-00-09A-034B-T_001-00



DMC-B737T-A-79-20-00-09A-034B-T_001-00
ICN-B737T-A-792000-A-81205-00014-A-001-01

ENGINE OIL -- DISTRIBUTION - SCAVENGE SCREEN PLUGS

79-20-00

02-Dec-2020

ENGINE OIL -- DISTRIBUTION - AIR/OIL/DEBRIS SEPARATOR

Purpose

The engine air/oil/debris separator, separates air and debris from the scavenge oil before it goes to the oil tank. The engine air/oil/debris separator is a Lubriclone type separator that uses centrifugal force.

The air/oil/debris separator performs these functions:

- Separates particles from the oil and directs them to the oil debris monitoring system (ODMS) sensor
- Provides head pressure on the oil tank with the tank pressurizing valve (TPV)
- Removes air from the lubrication scavenge system oil.

Location

The air/oil/debris separator is on the top of the oil tank. Open the right fan cowl to get access to the air/oil/debris separator.

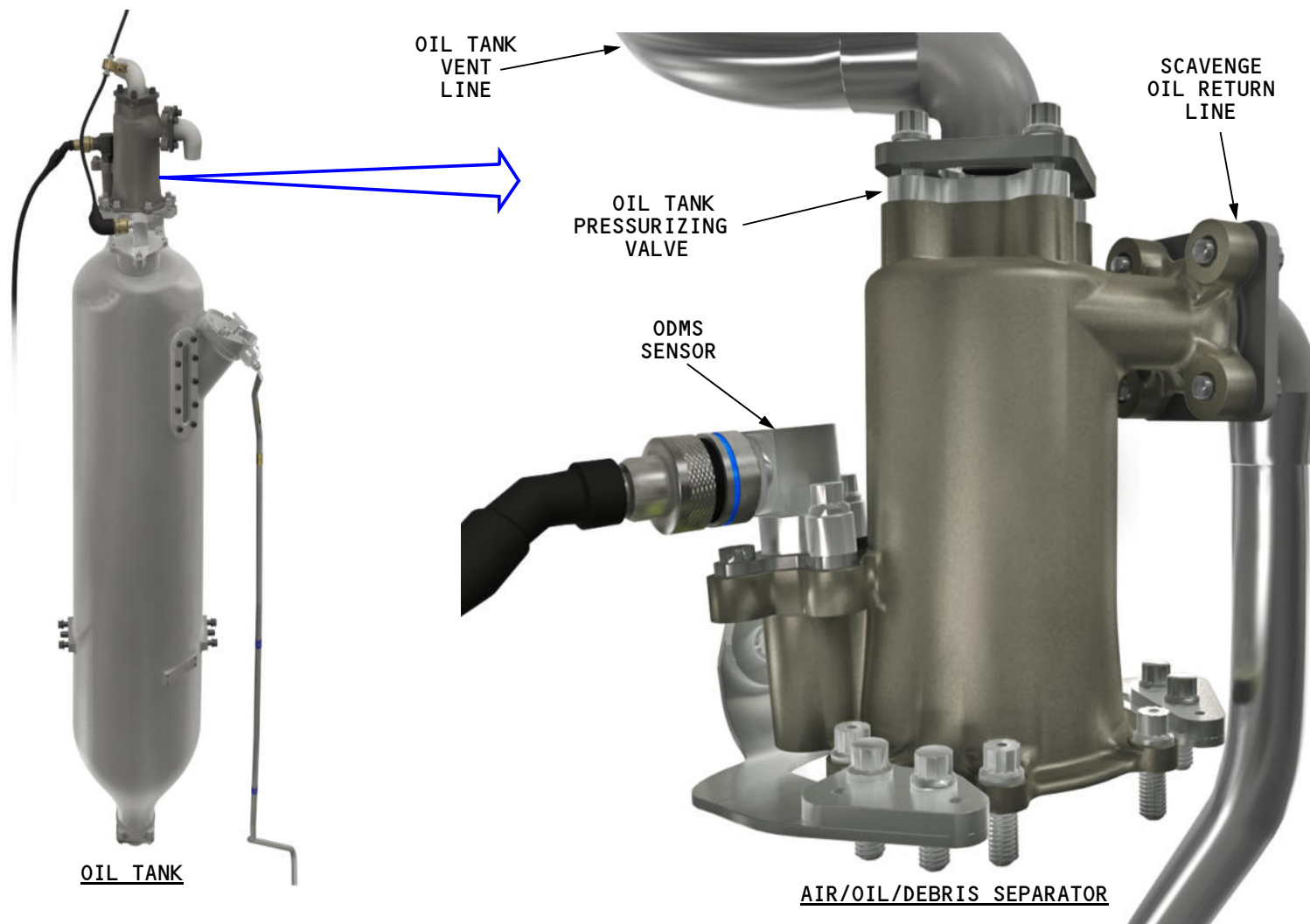
Functional Description

Air and oil in the scavenge oil line separate in the air/oil/debris separator. The air vents through the tank pressurizing valve (TPV) to the accessory gearbox (AGB). The oil goes to the oil tank through the oil outlet on the bottom of the air/oil/debris separator.

DMC-B737T-A-79-20-00-11A-034B-T_001-00

02-Dec-2020

79-20-00



DMC-B737T-A-79-20-00-11A-034B-T_001-00
ICN-B737T-A-792000-A-81205-00016-A-001-01

ENGINE OIL -- DISTRIBUTION - AIR/OIL/DEBRIS SEPARATOR

79-20-00

02-Dec-2020

ENGINE OIL -- DISTRIBUTION -ODMS SENSOR

Purpose

The oil debris monitoring system (ODMS) sensor captures ferromagnetic particles that are present in the scavenge oil.

When the sensor captures ferromagnetic particles, it generates an output pulse proportional to the particle mass.

The sensor also retains the particles for future inspection, and so that they will not continue to circulate in the oil system.

Location

The ODMS sensor is on the air/oil/debris separator, which is on the top of the oil tank. It is on the right side of the engine at the 03:00 o'clock position.

Functional Description

The ODMS sensor is a passive, magnetic, inductive sensor that collects and retains the ferromagnetic particles that are present in the scavenge oil returning to the oil tank. When the sensor captures a ferromagnetic particle, its magnetic field has a fluctuation. This fluctuation causes an output pulse from the sensor. The larger the particle, the larger the pulse is from the sensor. The output of the sensor goes to ODMS unit for signal conditioning.

The ODMS sensor is on the air/oil/debris separator, and is connected to the ODMS unit through a single wire harness.

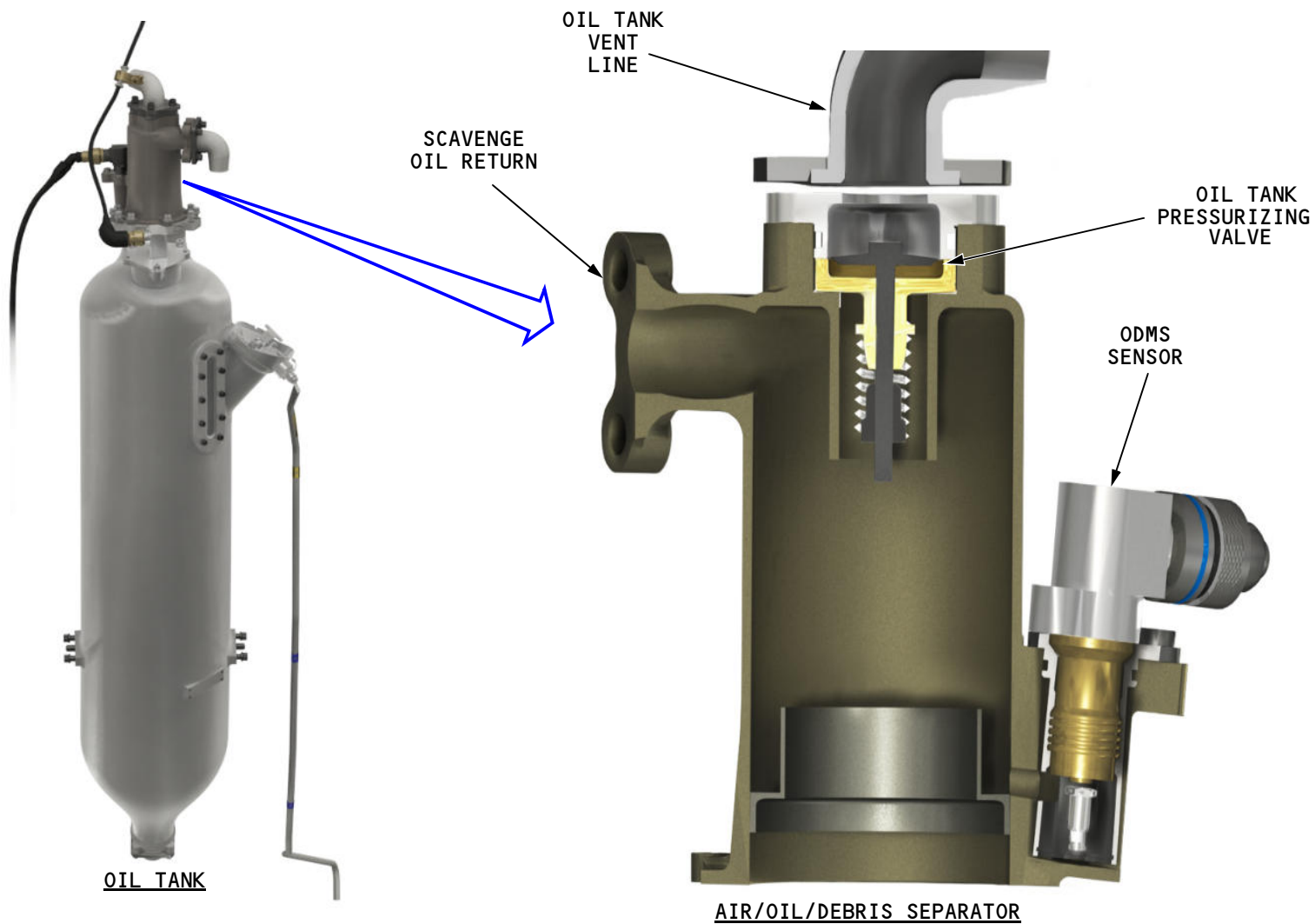
Training Information Point

The ODMS sensor is equipped with an O-ring seal to prevent oil leakage.

DMC-B737T-A-79-20-00-12A-034B-T_001-00

02-Dec-2020

79-20-00



ENGINE OIL -- DISTRIBUTION -ODMS SENSOR

79-20-00

DMC-B737T-A-79-20-00-12A-034B-T_001-00
ICN-B737T-A-792000-A-81205-00017-A-001-01

02-Dec-2020

ENGINE OIL -- DISTRIBUTION - ODMS UNIT

Purpose

The oil debris monitoring system (ODMS) unit filters and conditions the signal from the ODMS sensor, and provides a signal to EEC channel B when it detects ferrous particles in the scavenge oil.

Location

The ODMS unit is under the oil tank, in the 04:00 o'clock position, aft looking forward.

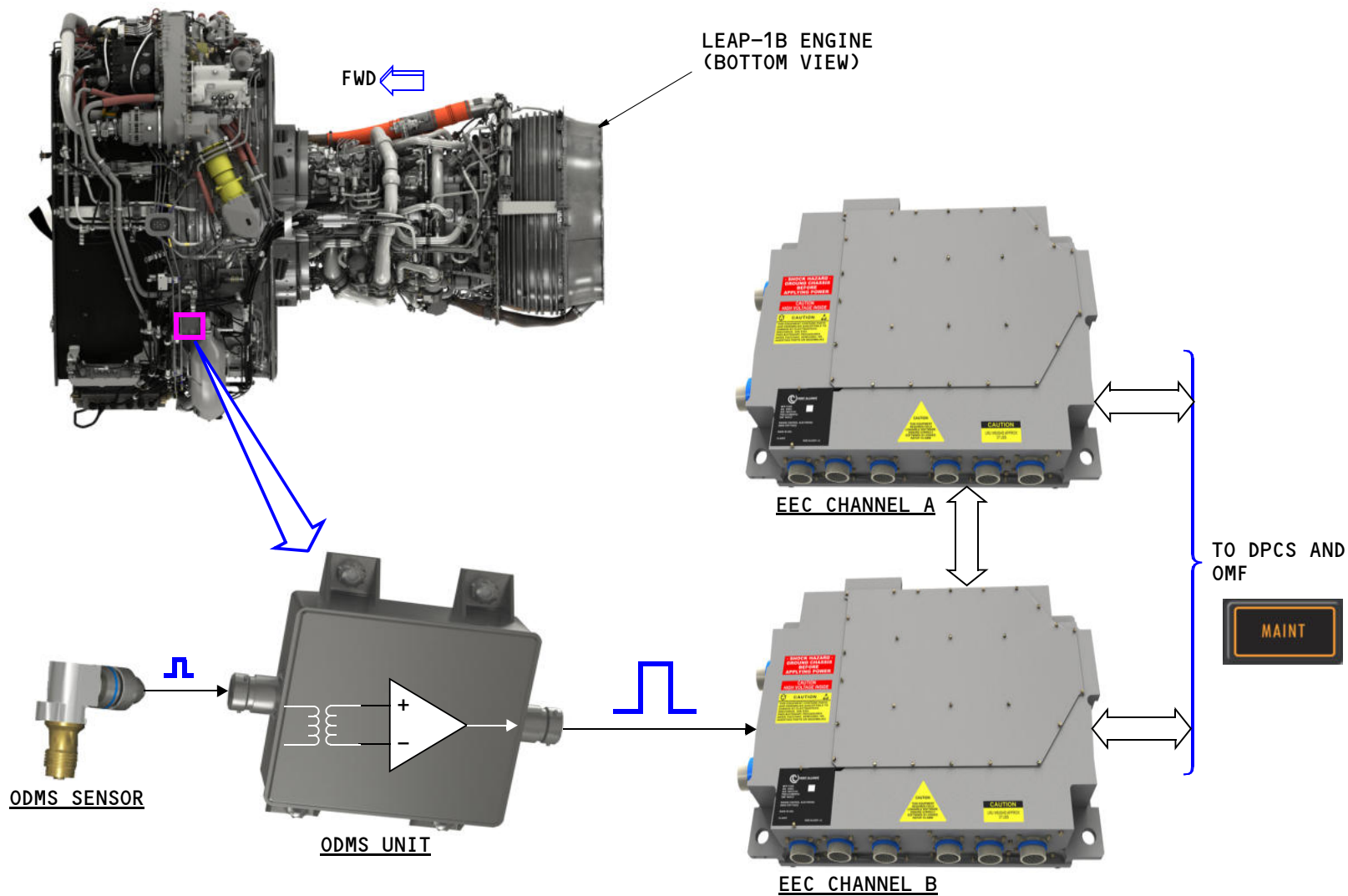
Functional Description

The ODMS unit receives a pulse from the ODMS sensor when it captures a ferromagnetic particle. The ODMS unit amplifies and filters the particle pulse, and compares the pulse magnitude to a threshold. If the signal is larger than the threshold, the signal is conditioned and sent to EEC channel B. The EECs store the number of chips that are counted in NVM memory. If the number of pulses reaches a threshold, the EECs send a maintenance message to the onboard maintenance function (OMF). At another threshold, the EECs send a signal to the display processing computers (DPCs), which then show a status message and turn on the maintenance light. There is an OMF special function to erase the chip count in the EECs.

DMC-B737T-A-79-20-00-22A-034B-T_003-00

02-Dec-2020

79-20-00



ENGINE OIL -- DISTRIBUTION - ODMS UNIT

79-20-00

DMC-B737T-A-79-20-00-22A-034B-T_003-00
ICN-B737T-A-792000-A-81205-00037-A-003-01

02-Dec-2020

ENGINE OIL -- DISTRIBUTION - OIL EDUCTOR VALVE

Purpose

The oil eductor valve increases engine forward sump pressurization by providing primary air to an eductor tube in the center vent tube. This increases the delta pressure across the engine forward (A) sump seals at low engine speeds, which improves oil control.

Location

The oil eductor valve is at the 08:00 O'clock position (aft looking forward), on a bracket fixed to the combustion case and the high pressure turbine case.

Physical Description

The eductor valve is an inlet pressure controlled, altitude compensated poppet style shutoff valve. It is composed of these :

- Housing with 3 mounting lugs and inlet and outlet fittings
- Control piston with evacuated bellows and poppet valve
- Visual position indication.

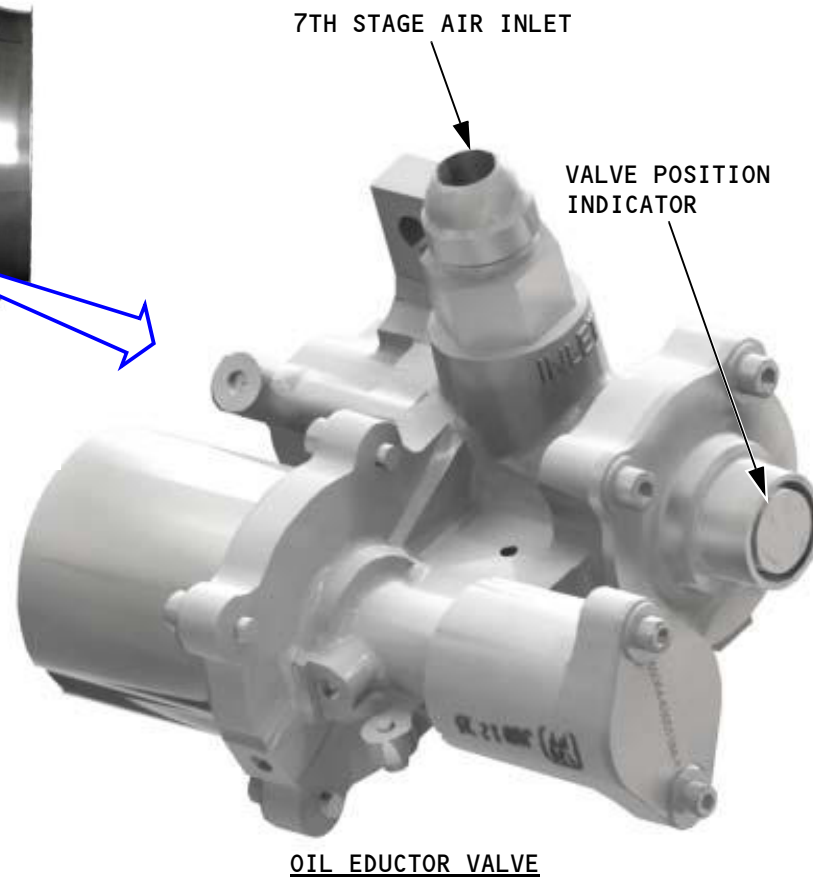
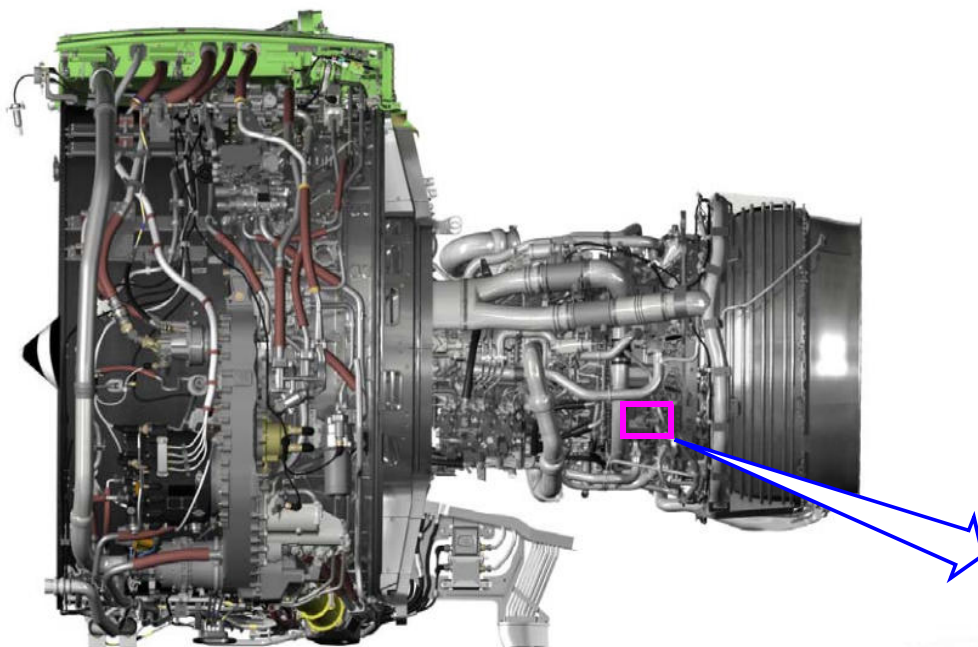
The eductor valve increases the delta pressure across the A sump number 1 and number 3 carbon seals (the pressure outside the A sump minus the pressure inside the A sump). This is necessary to maintain high enough delta pressure across the seals to avoid oil leakage from the A sump into the center vent tube. When the delta pressure across the A sump carbon seals is too low, the seals in the bearing sump may leak. To prevent this, the eductor valve opens to force ventilation of the sump. This decreases pressure inside the sump, which increases the delta pressure and efficiency across the carbon seals in the A sump.

To ventilate the sump, the eductor valve inlet is connected to 7th stage high pressure compressor (HPC) air, and the outlet is connected to the

center vent tube. At low 7th stage HPC inlet pressure, the valve is open. This lets 7th stage HPC air go to the center vent tube, creating a venturi effect in the A sump. At high 7th stage HPC inlet pressure, the control piston bellows closes the valve and shuts off air flow to the center vent tube.

The oil eductor valve has a valve position indication on the housing. A pin extends to show the open position of the valve. The valve should be in the open position when the engine is shutdown on the ground.

DMC-B737T-A-79-20-00-21A-034B-T_002-00



DMC-B737T-A-79-20-00-21A-034B-T_002-00
ICN-B737T-A-792000-A-81205-00036-A-002-01

ENGINE OIL -- DISTRIBUTION - OIL EDUCTOR VALVE

79-20-00

02-Dec-2020



ENGINE OIL -- DISTRIBUTION - FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK

ENGINE OIL -- DISTRIBUTION - FUNCTIONAL DESCRIPTION

Supply Circuit

Oil from the oil tank goes to the lubrication unit through the anti-leakage valve that is internal to the unit.

The anti-leakage valve keeps the oil in the oil tank when the engine is shutdown.

Two oil supply pump gives oil pressure that is not regulated. Oil pressure increases and decreases with engine speed.

The pressure relief valve that is also internal to the lubrication unit sends oil to the inlet of the A sump scavenge pump when the supply pressure is too high.

The supply pump output goes through the internal oil filter. The oil filter has a bypass valve that monitors the pressure difference across the oil filter. If the filter clogs, the valve opens. A differential pressure sensor gives indication of impending bypass of the oil filter.

The filtered oil flows through the servo fuel heater, the surface air cooled oil coolers, and then the main fuel/oil heat exchanger. If the surface air cooled oil coolers or the main fuel/oil heat exchanger clogs, a bypass valve opens to allow the continued flow of oil.

The oil flows through these supply lines:

- Rear (B and C) sump
- Accessory gearbox (AGB)
- Transfer gearbox (TGB) and forward (A) sump.

An oil line from the main fuel/oil heat exchanger sends pressure to the oil dampened bearings 1R and 5R..

Scavenge Circuit

Scavenge pumps move oil from engine sumps and gearboxes through scavenge screens with magnetic bars. There is one scavenge pump and one scavenge inlet screen element for each of these five scavenge lines:

- Rear (C) sump
- Rear (B) sump
- AGB
- TGB (Forward scavenge line)
- TGB (Aft scavenge line).

The forward (A) sump has two scavenge pumps connected to one scavenge inlet screen element. Thus there are a total of seven scavenge pumps in the lubrication unit.

Check valves prevent leakage if a scavenge screen is removed.

The oil from the scavenge pumps contains air. The air/oil mixture from the scavenge pumps goes to the air/oil/debris separator. The air/oil/debris separator separates the air from the oil. The air/oil/debris separator does these with the air from the scavenge oil:

- Uses it to keep a head pressure on the oil in the tank
- Gives it to the accessory gearbox (AGB) through a vent line.

Ferrous debris is captured and indicated by the oil debris monitoring system (ODMS).

Vent Circuit

The air that goes into the AGB from the air/oil/debris separator then goes through the transfer gearbox (TGB) to the A sump (forward sump). Air in the A sump then exits through the engine center vent

ENGINE OIL -- DISTRIBUTION - FUNCTIONAL DESCRIPTION

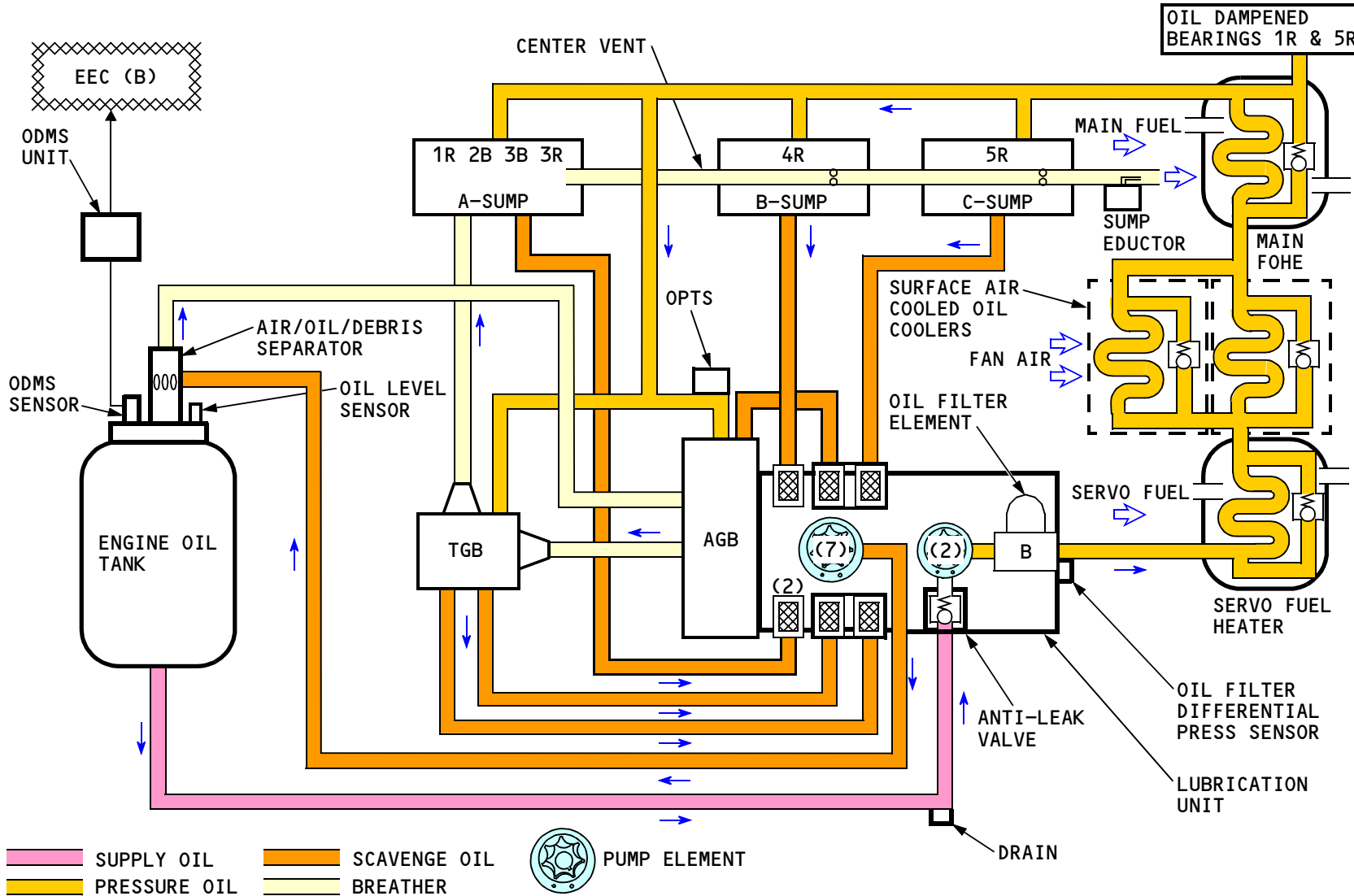
tube through the exhaust plug. There are also internal connections between the engine sumps and gearboxes.

An oil eductor system gives seventh stage compressor bleed air to the center vent tube at low thrust settings. This helps the seals to keep the oil in the sumps.

DMC-B737T-A-79-20-00-02A-042A-T_006-00

02-Dec-2020

79-20-00



ENGINE OIL -- DISTRIBUTION - FUNCTIONAL DESCRIPTION

79-20-00

ENGINE OIL -- INDICATING - GENERAL DESCRIPTION

General

The engine oil indicating system supplies oil system data to the display processing computers (DPCs). The engine indication display shows this data:

- Oil quantity
- Oil pressure
- Oil temperature
- Oil filter condition.

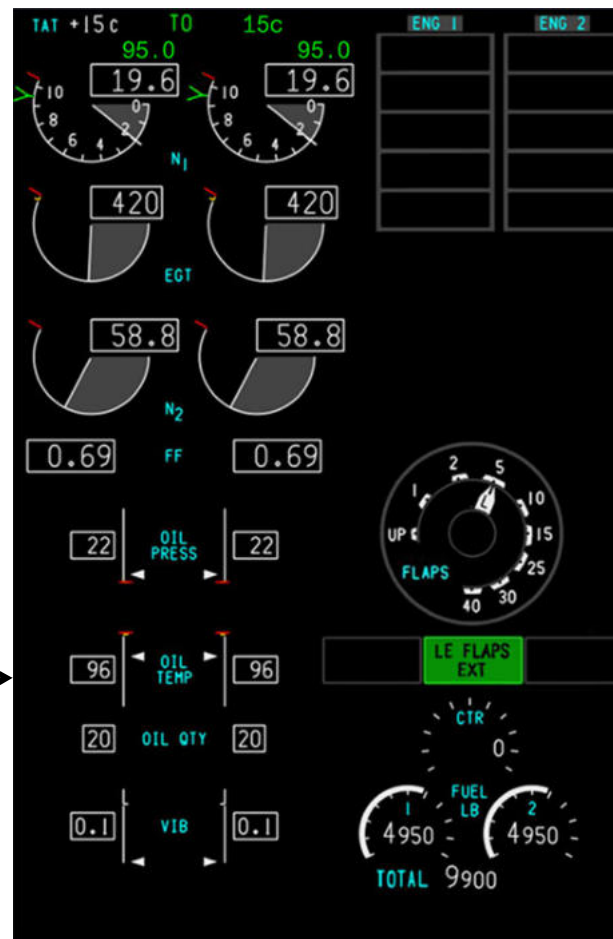
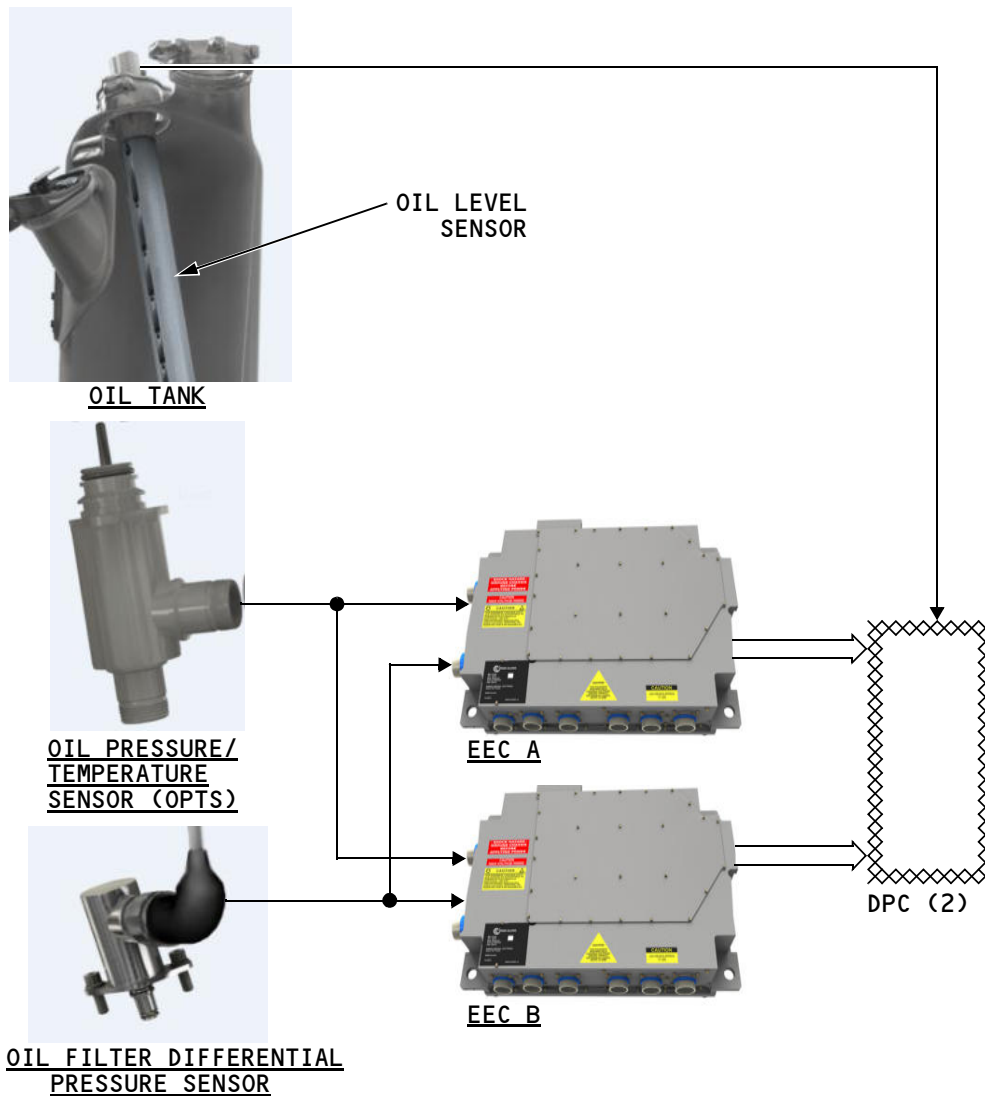
These components monitor the oil system:

- Oil level sensor
- Oil pressure/temperature sensor (OPTS)
- Oil filter differential pressure sensor (OFDPS).

The oil level sensor sends the oil quantity data directly to the DPCs.

The three other components send data to the EECs. The EECs then send the data to the DPCs.

The oil pressure/temperature sensor (OPTS) has dual oil pressure and oil temperature sensors.



ENGINE INDICATION

ENGINE OIL -- INDICATING - GENERAL DESCRIPTION

79-30-00

DMC-B737T-A-79-30-00-02A-040A-T_001-00
ICN-B737T-A-793000-A-81205-00007-A-001-01

02-Dec-2020

ENGINE OIL -- INDICATING - COMPONENT LOCATION

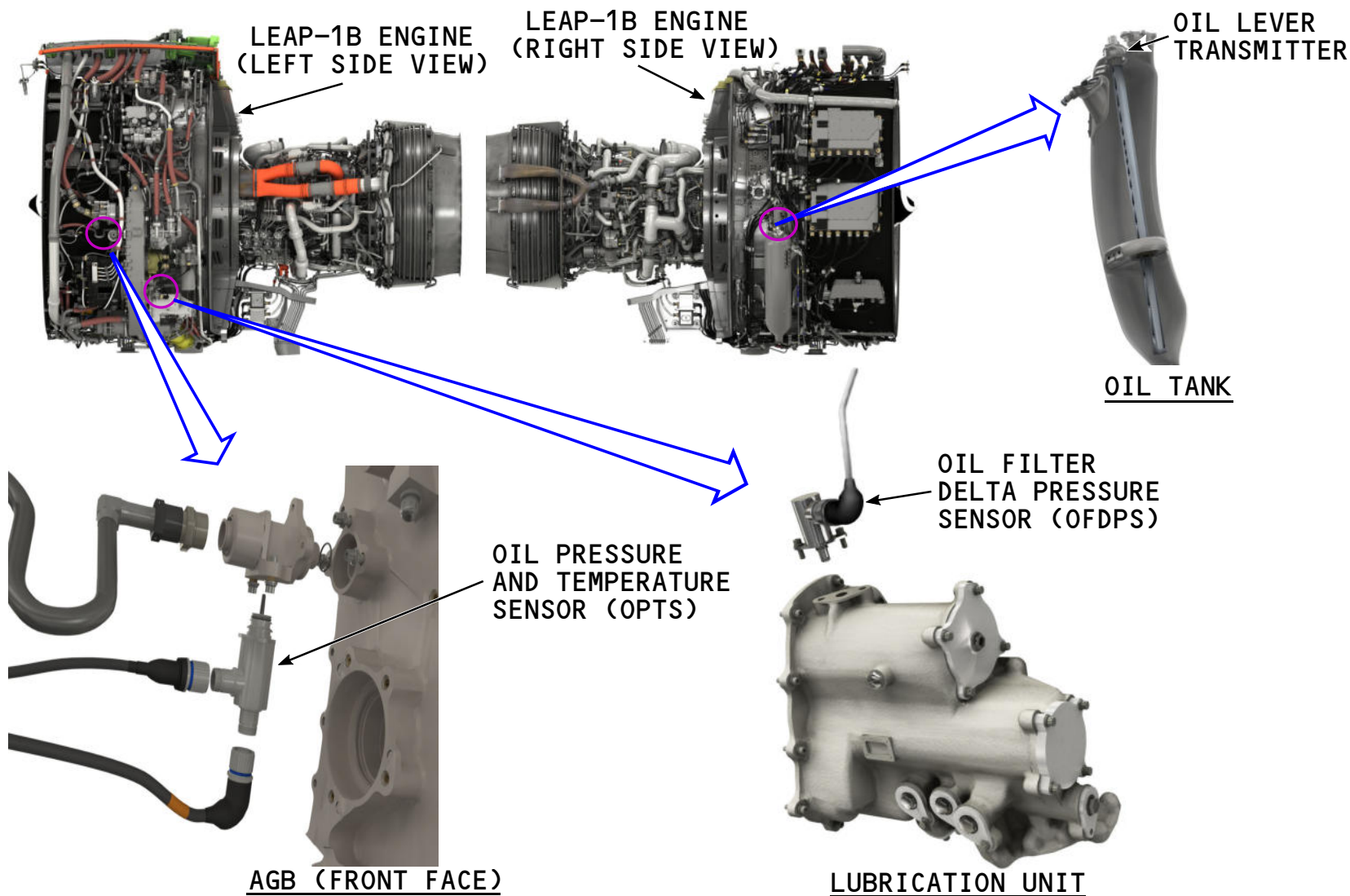
Component Locations

These components of the engine oil indicating system are on the left side of the fan case:

- Oil pressure and temperature sensor (OPTS) (8:00 position), on the AGB (accessory gearbox)
- Oil filter differential pressure sensor (OFDPS) (7:00 position), on the lubrication unit.

The oil level sensor is on the oil tank (4:00 position), on the right side of the fan case.

Open the fan cowls to get access to these components.



DMC-B737T-A-79-30-00-02A-055A-T_005-00
ICN-B737T-A-793000-A-81205-00008-A-005-01

ENGINE OIL -- INDICATING - COMPONENT LOCATION

79-30-00

ENGINE OIL -- INDICATING - OIL QUANTITY INDICATING SYSTEM

General

The oil quantity indicating system shows the engine oil quantity data on the engine indication display. The oil quantity indicating system uses an oil quantity transmitter to measure the oil quantity in the oil tank. The oil quantity transmitter sends the oil quantity data directly to the display processing computers (DPCs).

If the oil quantity data is not valid, the display is blank, but the highlight box continues to show.

Physical Description

The oil quantity transmitter is an electrical resistance sensor. It uses a magnet that floats and reed switches to give the oil level data. The oil quantity transmitter has 1 connector to transmit data to the DPCs.

Functional Description

The DPCs supply an excitation signal to the sensing circuit of the oil quantity transmitter. As the magnet moves up or down with the oil level, the reed switches open or close different resistor circuits. A sensor output signal in proportion to the oil level goes to the DPCs.

Indication

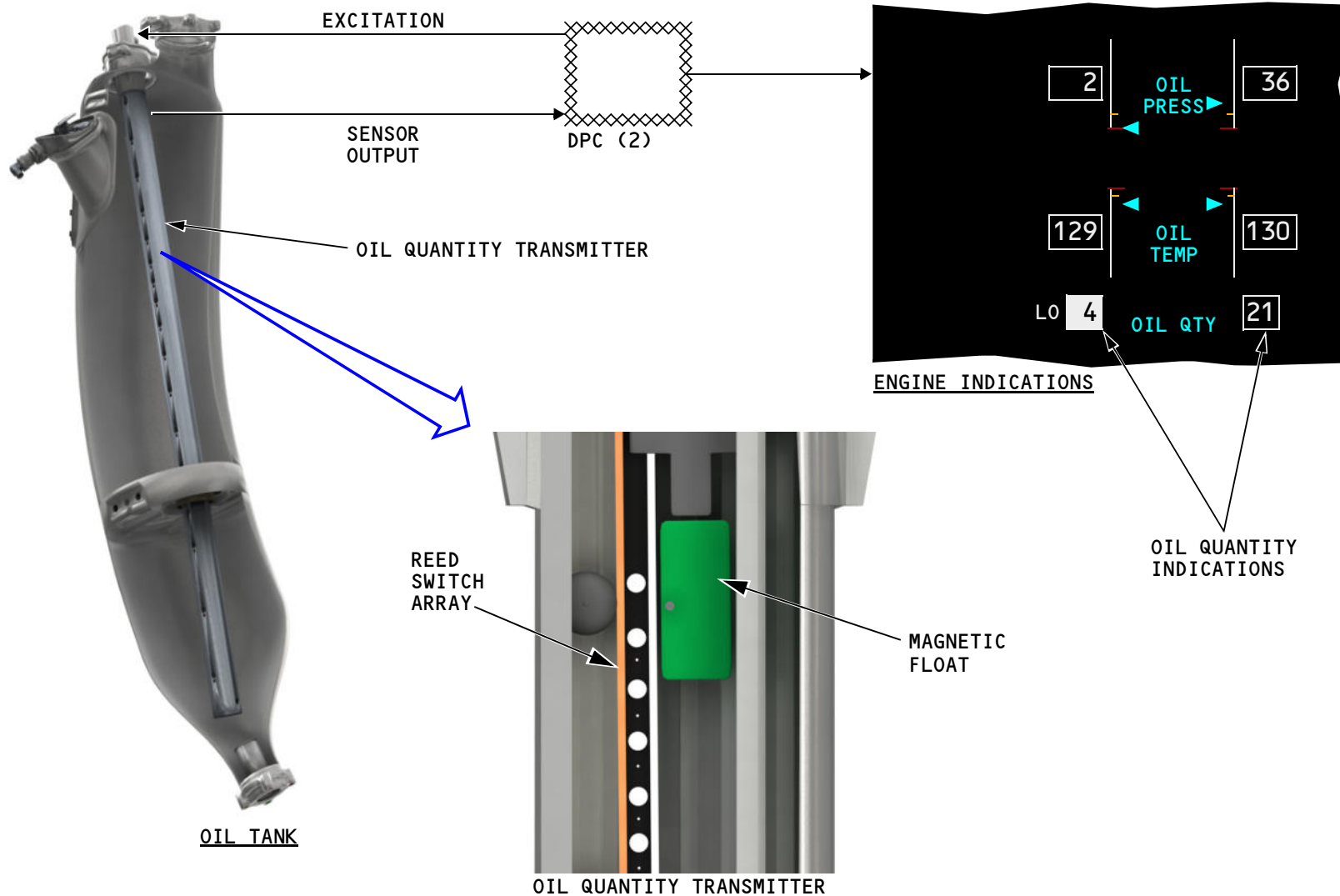
The engine indication display shows usable oil quantities in quarts. The display range is 0 to approximately 23 quarts. The wing dihedral causes the oil tank for engine 2 to hold slightly more oil than engine 1.

The readout changes to a black on white reverse video indication with a LO message adjacent to the readout if these 2 conditions are true:

- Oil quantity goes below 6.6 quarts for more than 35 seconds
- Low oil quantity inhibit is not active.

The low oil quantity inhibit is active for 20 seconds after engine thrust reverser operation and 90 seconds after an airplane landing.

DMC-B737T-A-79-30-00-05A-034B-T_002-00



ENGINE OIL -- INDICATING - OIL QUANTITY INDICATING SYSTEM

79-30-00

ENGINE OIL -- INDICATING - OIL PRESSURE INDICATING SYSTEM

General

The oil pressure indicating system shows the engine oil pressure data on the engine indication display. A combined oil pressure/temperature sensor (OPTS) measures the oil pressure at the inlet to the accessory gearbox (AGB). The LOW OIL PRESSURE message shows when oil pressure goes below the red line limit.

Physical Description

The OPTS has two sensor elements for oil pressure. Each element connects with one of the EEC channels through a separate connector. The OPTS is in an adapter that is on the front face of the AGB. The OPTS also contains the oil temperature sensors.

Functional Description

The oil pressure sensors measure the oil supply pump outlet pressure at the input to the AGB. The oil pressure sensor section of the OPTS is a dual, strain gauge type pressure sensor. Each element in the sensor gets an excitation input, and sends an output to the related EEC channel. The EECs change this signal into an ARINC 429 signal and send it to the DPCs. The DPCs show the oil pressure on the engine indication display.

Indication

Oil pressure shows on two vertical indicators and two digital displays. A pointer shows the oil pressure in psi differential (psid) on each vertical indicator. The indicator has two index markers. The amber index marker shows minimum oil pressure. The red index marker shows the oil pressure red line limit. If the oil pressure shows between the amber limit and the red line limit, the digital display and the box around the digital display are amber. This is the caution range. If the oil

pressure shows at or below the red line limit, the digital display and box around the digital display are red. This is the out-of-limit range.

Minimum oil pressure at ground idle speed is 17.4 PSI. Minimum oil pressure at high N2 speed is 29 psi.

The low oil pressure red line limit at idle is 17.4 psi. This limit increases with N2 speed when N2 is more than 74 percent. The maximum low oil pressure red line limit is 29 psi at 116.5 percent N2 speed.

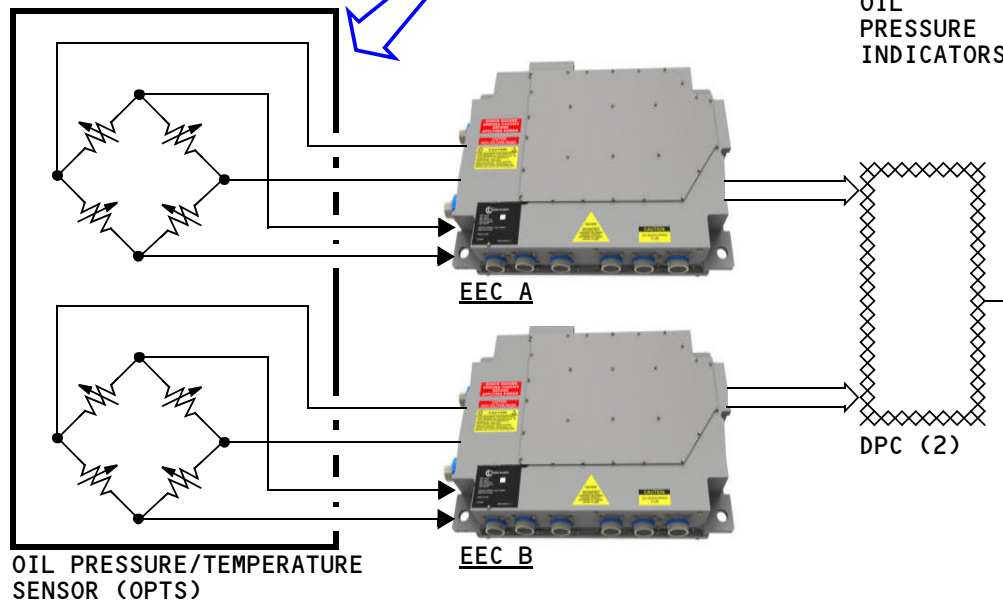
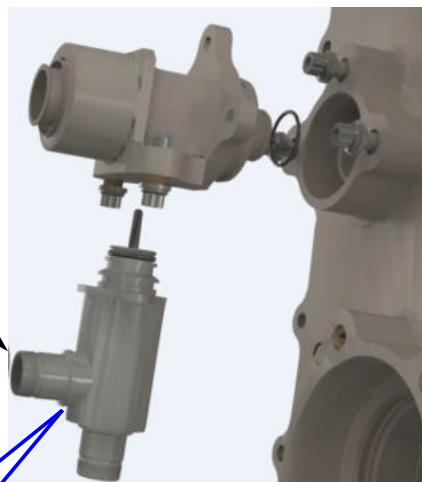
LOW OIL PRESSURE Message

When the oil pressure is less than the red line limit, the EECs send a signal to the DPCs. This causes the engine indication display to show the amber LOW OIL PRESSURE message. There is one LOW OIL PRESSURE message for each engine.

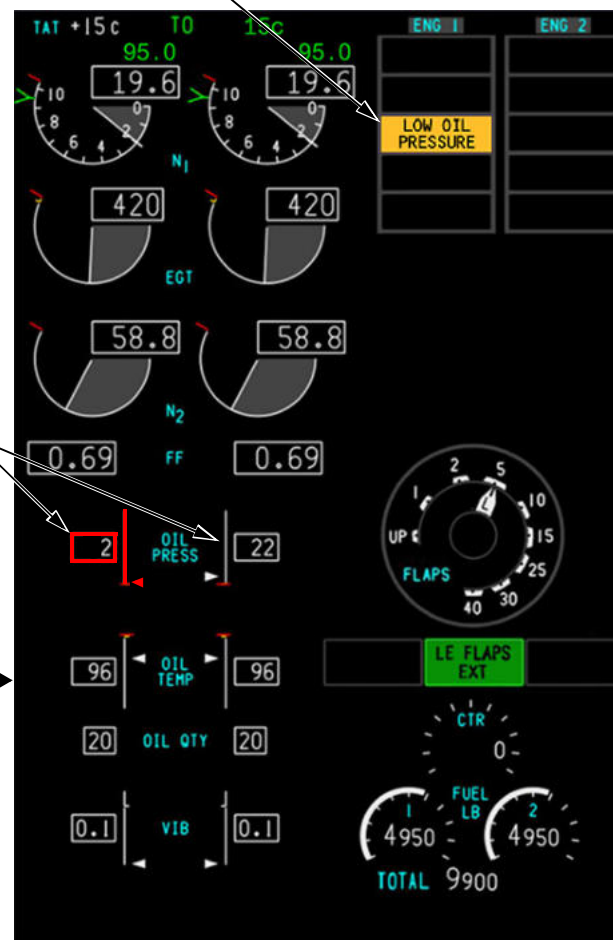
The amber LOW OIL PRESSURE message flashes for 10 seconds then shows continuously when the oil pressure is less than the red line limit. The DPCs inhibit the flash mode for takeoff and landing.

During start, the EEC prevents the indicator and pointer from changing to amber or red.

OIL PRESSURE/TEMPERATURE
SENSOR (OPTS)



LOW OIL
PRESSURE
MESSAGE



ENGINE INDICATION

ENGINE OIL -- INDICATING - OIL PRESSURE INDICATING SYSTEM

79-30-00

DMC-B737T-A-79-30-00-06A-034B-T_003-00
ICN-B737T-A-793000-A-81205-00010-A-002-01

02-Dec-2020

ENGINE OIL -- INDICATING - OIL TEMPERATURE INDICATING SYSTEM

General

The oil temperature indicating system shows the engine oil temperature data on the engine indication display. A combination oil pressure/temperature sensor (OPTS) measures the oil temperature at the inlet to the accessory gearbox (AGB).

Physical Description

The OPTS has two sensor elements for oil temperature. Each element connects with one of the EEC channels through a separate connector. The OPTS is in an adapter that is on the front face of the AGB. The OPTS also contains the oil pressure sensors.

Functional Description

The oil temperature sensors measure oil temperature at the inlet to the AGB. The oil temperature sensor section of the OPTS is a dual, resistive type temperature sensor. Each element in the sensor gets an excitation input, and sends an output to the related EEC channel. The EECs change this signal to an ARINC 429 signal and sends it to the DPCs. The DPCs show the oil temperature on the engine indication display.

Indication

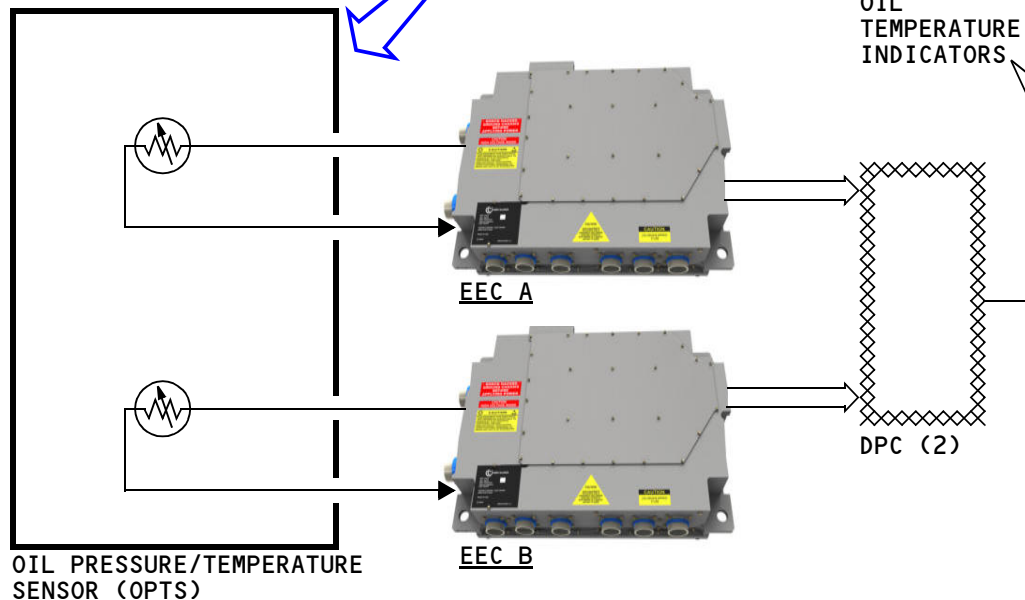
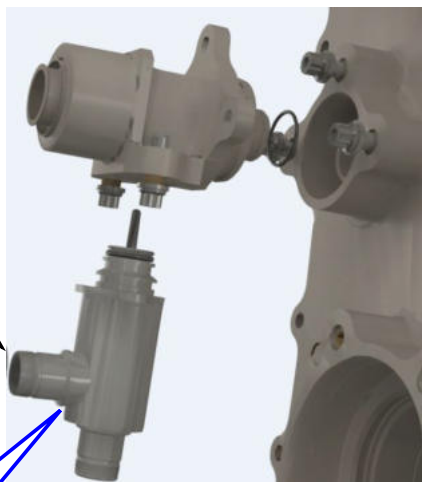
The oil temperature shows on two vertical indicators and two digital displays. A pointer shows the oil temperature in degrees Celsius on each indicator. The vertical indicator has two index markers. The amber index marker shows the oil temperature amber limit. The red index marker shows the oil temperature red line limit.

If the oil temperature is between the amber limit and the red line limit, the digital display and the box around the digital display are amber. This is the caution range. If the oil temperature is more than the red

line limit, the digital display and the box around the digital display are red. This is the out-of-limit range.

The maximum oil temperature limit for continuous operation, (or amber limit), is 320 degrees F (160 degrees C). The red line limit for oil temperature is 350 degrees F (177 degrees C). The usual oil temperature range is between 170 degrees F (77 degrees C) and 320 degrees F (160 degrees C).

OIL PRESSURE/TEMPERATURE
SENSOR (OPTS)



ENGINE INDICATION

ENGINE OIL -- INDICATING - OIL TEMPERATURE INDICATING SYSTEM

79-30-00

DMC-B737T-A-79-30-00-07A-034B-T_001-00
ICN-B737T-A-793000-A-81205-00011-A-001-01

02-Dec-2020

ENGINE OIL -- INDICATING - OIL FILTER BYPASS WARNING SYSTEM

General

The oil filter bypass warning system shows a message when the oil filter is near a bypass condition. The message shows on the engine indication display. The oil filter bypass indicating system uses a single, dual element differential pressure sensor. The EECs send a signal to the display processing computers (DPCs) just before the oil filter bypass valve opens.

Physical Description

The oil filter differential pressure sensor connects with both of the EECs through a single connector. The transmitter is on the lubrication unit.

Functional Description

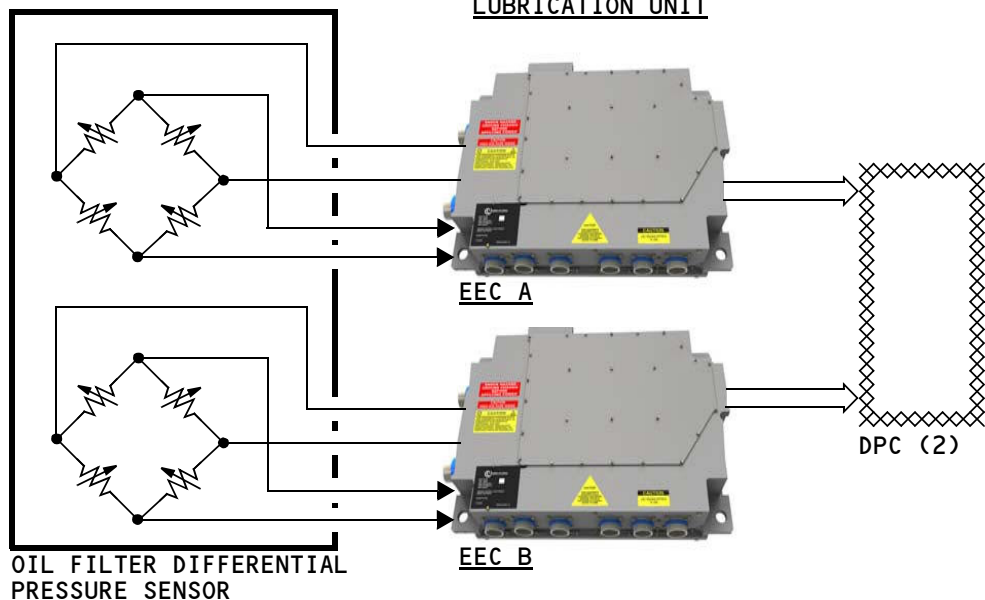
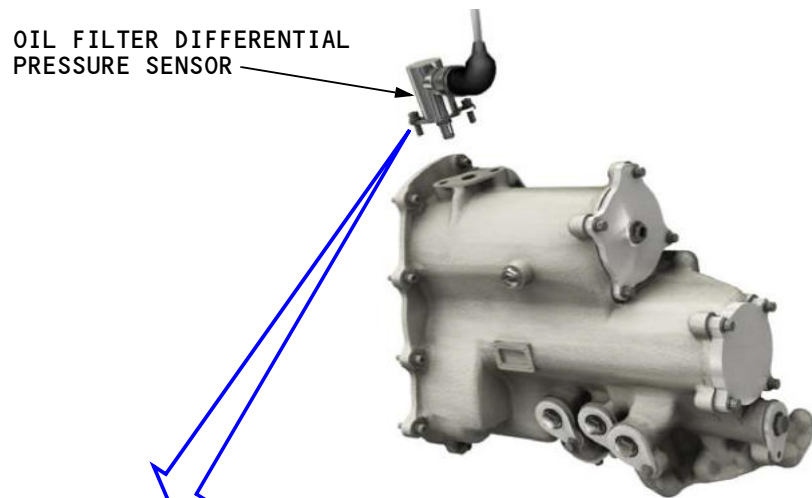
The oil filter differential pressure sensor monitors the oil pressure difference between the inlet and the outlet of the oil filter. The oil filter differential pressure sensor is a dual, strain gauge type pressure sensor. Each element in the sensor gets an excitation input, and sends an output to the related EEC channel. The EECs change this signal to an ARINC 429 signal and send it to the DPCs.

Oil Filter Bypass Message

Just before the oil filter bypass valve opens, the EEC sends a signal to the DPCs. This causes the engine indication display to show the amber OIL FILTER BYPASS message. There is one OIL FILTER BYPASS message for each engine.

The amber OIL FILTER BYPASS message flashes initially for 10 seconds, and then shows continuously. The DPCs inhibit the flash mode during takeoff and landing.

DMC-B737T-A-79-30-00-08A-034B-T_001-00



ENGINE OIL -- INDICATING - OIL FILTER BYPASS WARNING SYSTEM

79-30-00

DMC-B737T-A-79-30-00-08A-034B-T_001-00
ICN-B737T-A-793000-A-81205-00012-A-002-01

02-Dec-2020



IGNITION -- INTRODUCTION

General

The ignition system supplies electrical sparks to the combustion chamber. Each engine has two ignition systems that operate independently. The ignition system has manual and automatic modes of operation. The electronic engine controls (EECs) select which igniter(s) operate.

Ignition operates during:

- Starting
- Takeoff (RTO)
- In-flight (Continuous)
- Flameout or rundown.

Abbreviations and Acronyms

AUTO	- automatic
DU	- display unit
CONT	- continuous
DPC	- display processing computer
EEC	- electronic engine control
FLT	- flight
GRD	- ground
IGN	- ignition
L	- left
MDS	- MAX display system
R	- right

DMC-B737T-A-74-00-00-02A-018A-T_005-00

02-Dec-2020

74-00-00



POWER PLANT
(71)

ENGINE
(72)

ENGINE
INDICATING
(77)

ENGINE
CONTROLS
(76)

ENGINE FUEL
AND CONTROL
(73)



TAKEOFF (RT0)



GROUND STARTING



AIR STARTING
CONTINUOUS / AUTOMATIC

ENGINE AIR
(75)

ENGINE OIL
(79)

IGNITION
(74)

STARTING
(80)

EXHAUST
(78)

IGNITION -- INTRODUCTION

74-00-00

DMC-B737T-A-74-00-00-02A-018A-T_005-00
ICN-B737T-A-740000-A-81205-00009-A-002-01

02-Dec-2020

IGNITION -- GENERAL DESCRIPTION

General

These components control ignition:

- Ignition selector switch (P5)
- Engine start switch (P5)
- Start lever (P8)
- Display processing computers (DPC)
- Electronic engine controls (EEC).

The start lever controls ignition system power to the EECs. The start switch and the ignition selector switch supply inputs to the DPCs. The DPCs send the start switch and the ignition switch position to the EECs. The EECs give power to the selected ignition exciter(s) during starting or automatic ignition operation. The ignition exciters supply power to the engine igniters.

Start Lever Switch Module

The engine start lever in the control stand has internal switches. Some of those switches control electrical power to the EECs for engine ignition. The EECs get 115v ac power for ignition only when the start lever is in the IDLE position.

Engine Start Switch

There are four engine start switch positions:

- GRD (ground start)
- OFF or AUTO
- CONT (continuous ignition)
- FLT (flight).

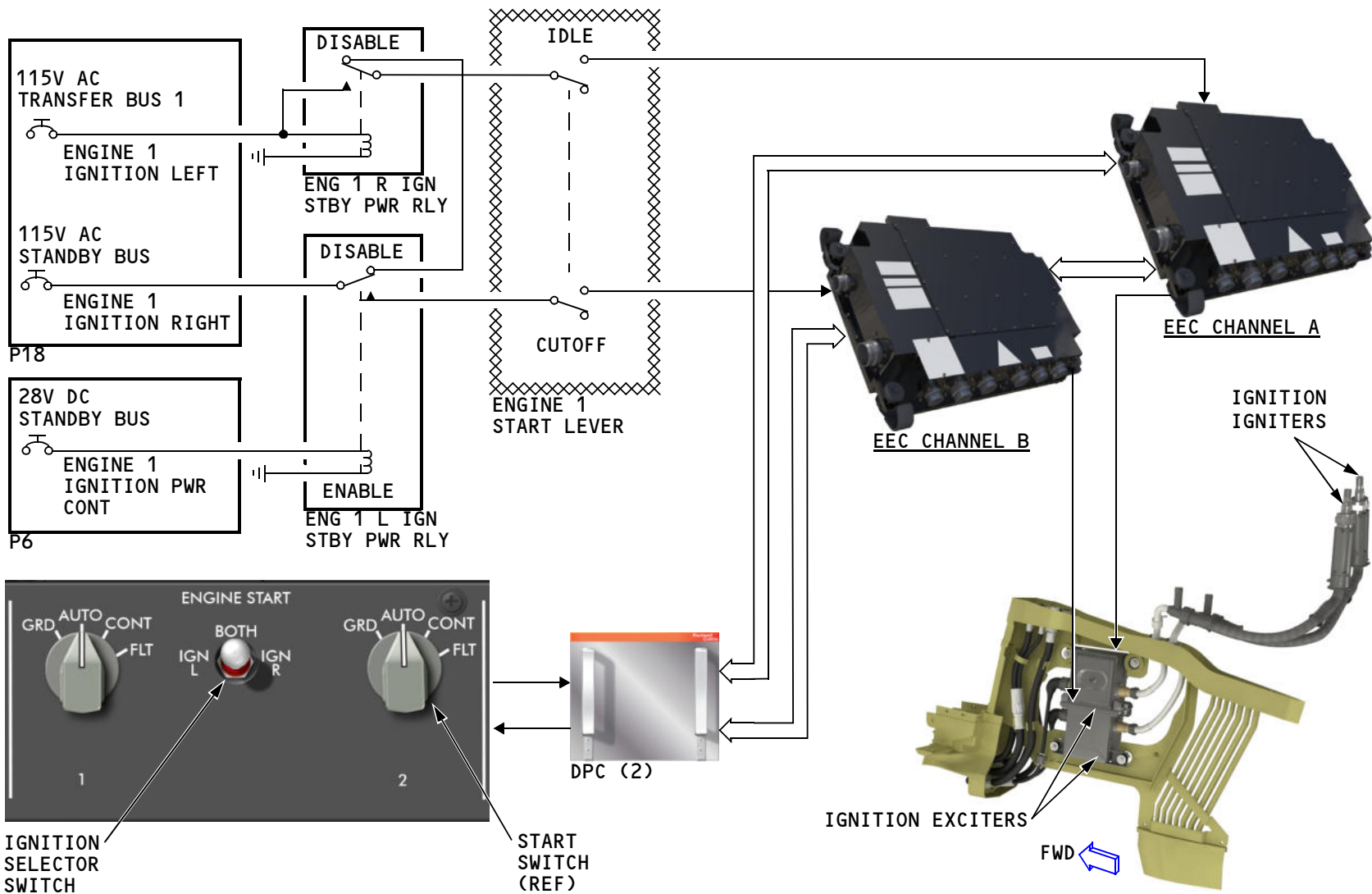
Ignition Selector Switch

There are three ignition selector switch positions:

- IGN L (left spark igniter)
- BOTH (both spark igniters)
- IGN R (right spark igniter).

Ignition Power

EEC channel A normally gets electrical power for the upper ignition exciter from the ac transfer bus. EEC channel B gets electrical power for the lower ignition exciter from the ac standby bus. EEC channel A can get electrical power for the upper ignition exciter from the ac standby bus if you pull the ignition power control circuit breaker. The EECs have logic that controls when 115v ac electrical power goes to the appropriate ignition exciter. The ignition exciter converts the 115v ac into high energy pulses that fire the ignition igniter.



DMC-B737T-A-74-00-00-02A-040A-T_002-00
ICN-B737T-A-740000-A-81205-00010-A-002-01

IGNITION -- GENERAL DESCRIPTION

02-Dec-2020

74-00-00

IGNITION -- FLIGHT COMPARTMENT - COMPONENT LOCATION

General

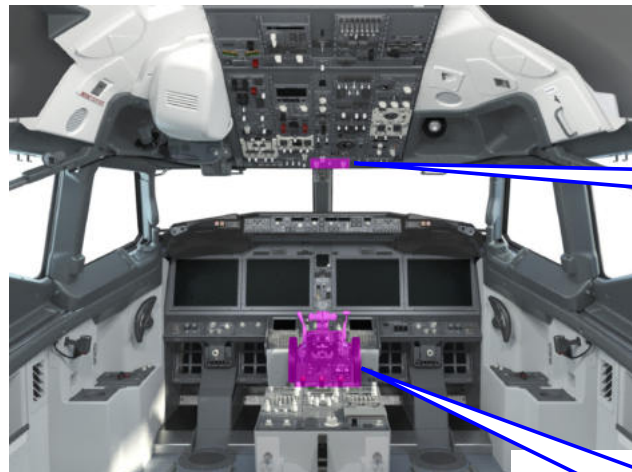
These components that control the ignition system are in the flight compartment:

- Engine start switches
- Ignition selector switch
- Start levers.

Component Locations

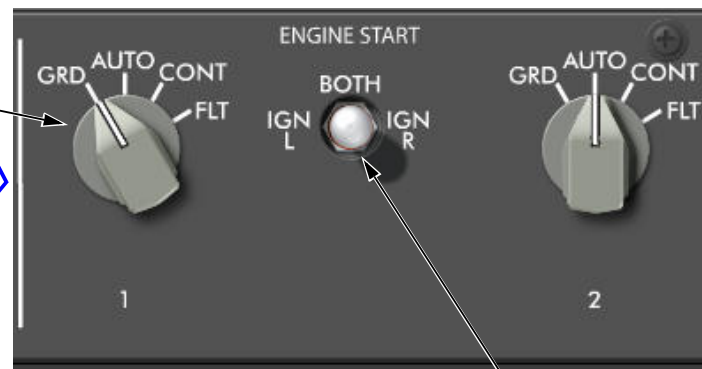
The ignition selector switch and engine start switches are on the forward overhead panel (P5).

The start levers are on the control stand.



FLIGHT COMPARTMENT

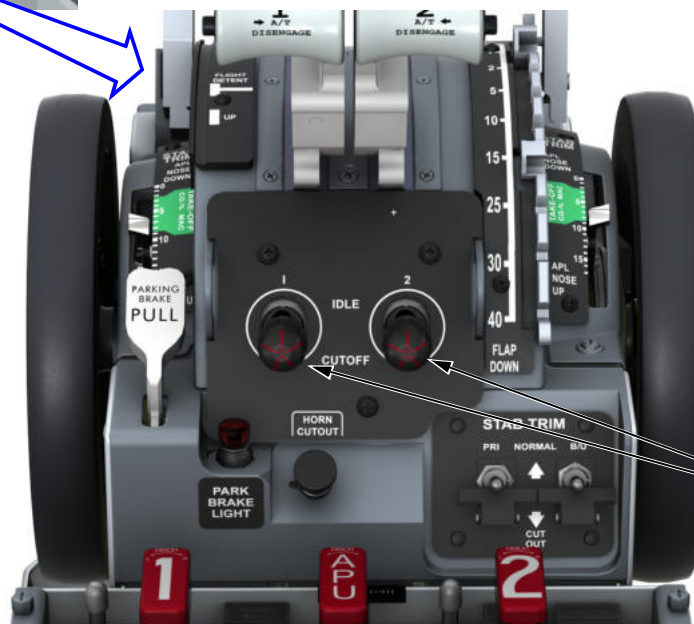
ENGINE START SWITCH (2)



P5 PANEL

IGNITION SELECTOR SWITCH

FWD



CONTROL STAND

START LEVERS

IGNITION -- FLIGHT COMPARTMENT - COMPONENT LOCATION

74-00-00

IGNITION -- ENGINE - COMPONENT LOCATION

General

Each engine has a right and a left ignition system. These are the components of each ignition system:

- Ignition exciter
- Ignition lead
- Air manifold
- Ignition igniter.

Component Locations

There are two ignition exciters on each engine. They are on the bottom of the aft fan case.

There are two igniters on each engine. The left ignition igniter is on the left side of the combustor near the 6:00 position. The right ignition igniter is on the right side of the combustor at the 5:00 position.

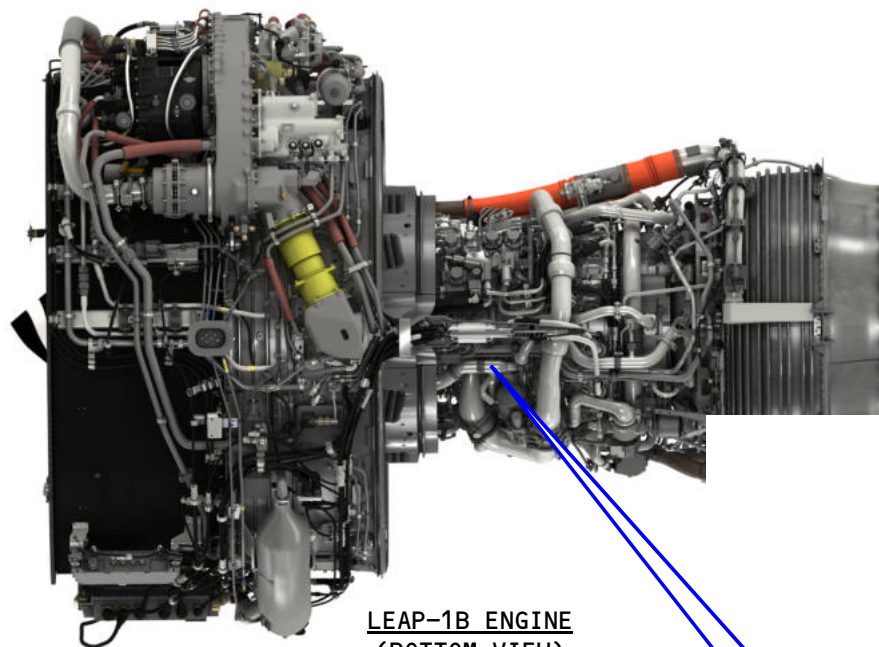
There are two ignition leads on each engine. They go from the ignition exciters to the ignition igniters.

Air from the core compartment cooling air manifold cools the ignition leads and igniters. This cooling air enters the ignition leads near the ignition exciters and exits at the igniter cooling shroud.

DMC-B737T-A-74-00-00-03A-055A-T_003-00

02-Dec-2020

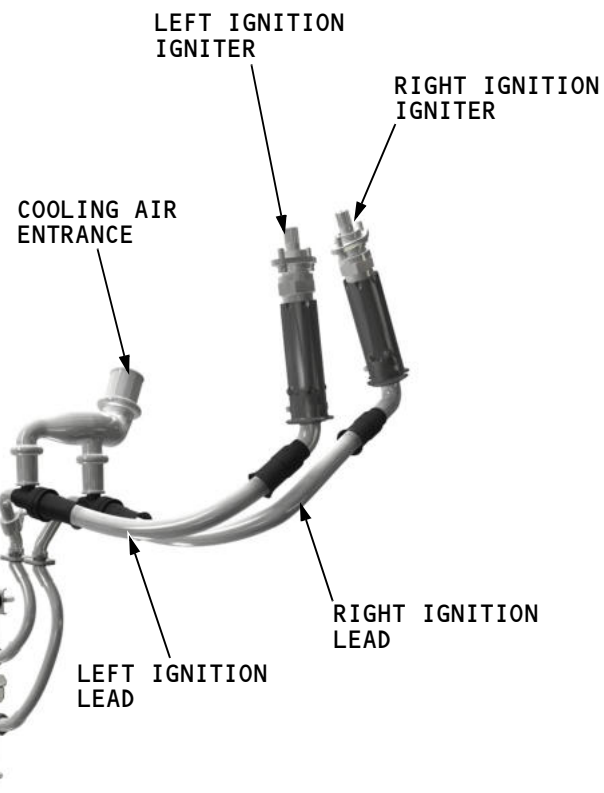
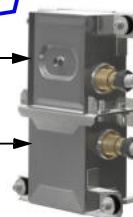
74-00-00



LEAP-1B ENGINE
(BOTTOM VIEW)

LEFT IGNITION EXCITER
(EEC A)

RIGHT IGNITION EXCITER
(EEC B)



IGNITION -- ENGINE - COMPONENT LOCATION

74-00-00

IGNITION -- IGNITION EXCITER

Purpose

The ignition exciters supply power to the ignition igniters.

General

The ignition exciters change 115v ac to 15,000v dc - 20,000v dc. The output of the ignition exciter is 14.5 to 16 joules. The ignition igniter uses this power to ignite the fuel/air mixture in the combustion chamber.

Physical Description

A 115v ac electrical connector attaches to the forward face of the ignition exciter. The ignition lead attaches to the aft face of the ignition exciter.

The ignition exciters are in the lower bifurcation section of the bottom of the aft fan case. The two exciters connect together with a bracket. They attach to the fan case with four shock mounts. Two shock mounts are on the top of the left (upper) ignition exciter. Two more shock mounts are on the bottom of the right (lower) ignition exciter.

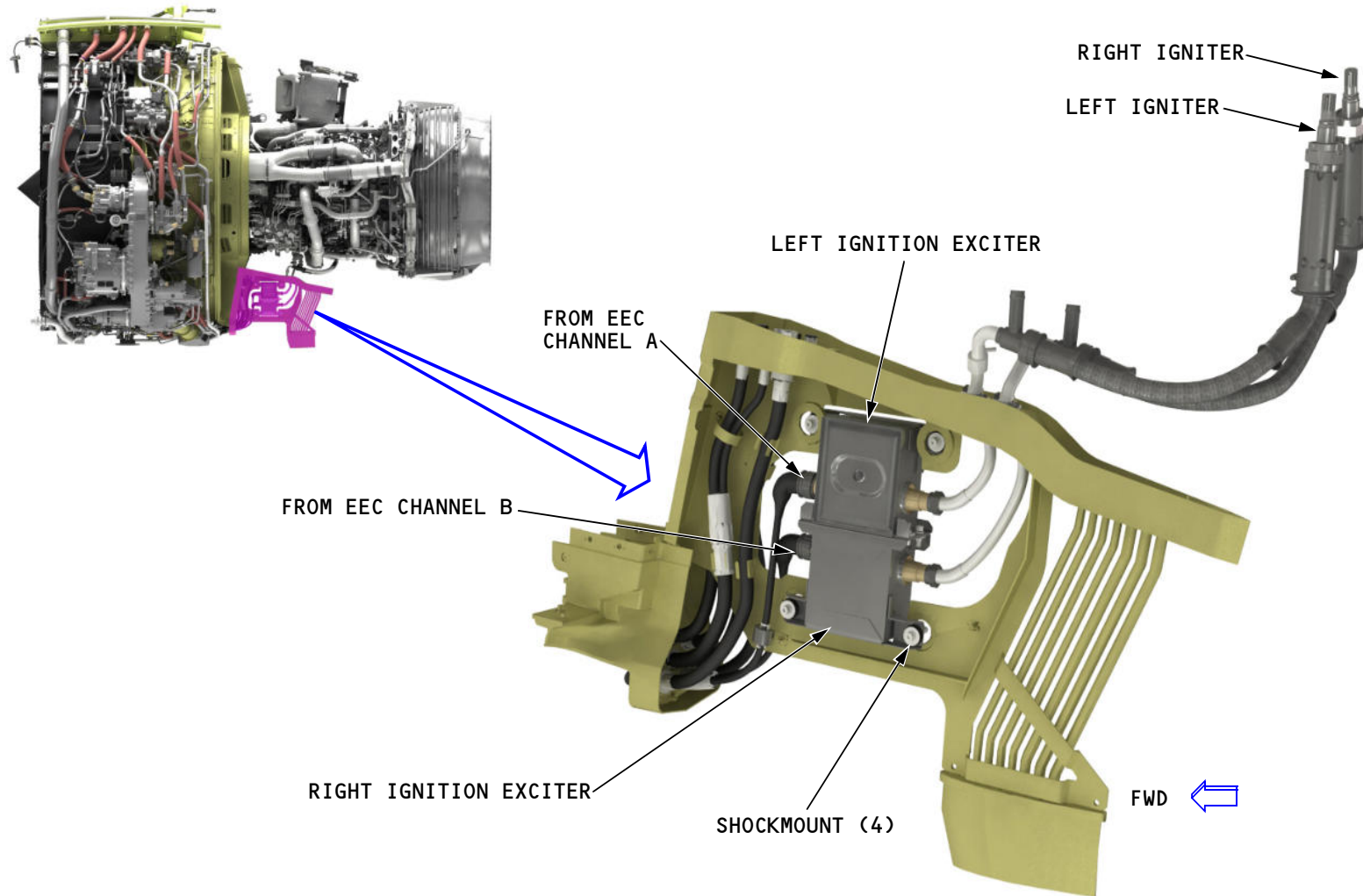
WARNING: MAKE SURE THE IGNITION EXCITERS ARE DE-ENERGIZED BEFORE YOU START WORK ON THE IGNITION SYSTEM. IF YOU DO NOT OBEY THIS WARNING, INJURY OR DEATH CAN OCCUR.

Training Information Point

An MEL procedure lets you change the source of electrical power to the left ignition system of an engine. This procedure connects ac standby bus power to the left exciter of the left or right engine. This

occurs when you open one of these circuit breakers for the associated engine :

- Engine 1 ignition power control left
- Engine 2 ignition power control left.



IGNITION -- IGNITION EXCITER

74-00-00

IGNITION -- DISTRIBUTION - IGNITION LEADS AND SPARK IGNITERS

General Description

Each ignition igniter has its own ignition lead. Cooling air flows around each ignition lead and igniter plug to decrease the temperature of the lead and the plug.

Ignition Lead Cooling

Each ignition lead goes into an air manifold at the 6:00 position at the aft end of the fan case. From the air manifold to the igniter plug, the ignition lead is inside of the igniter lead shroud.

Low pressure compressor discharge air is used for cooling the ignition leads and igniters. The air goes into the ignition lead shroud and then exits at the igniter plug through the igniter plug shroud.

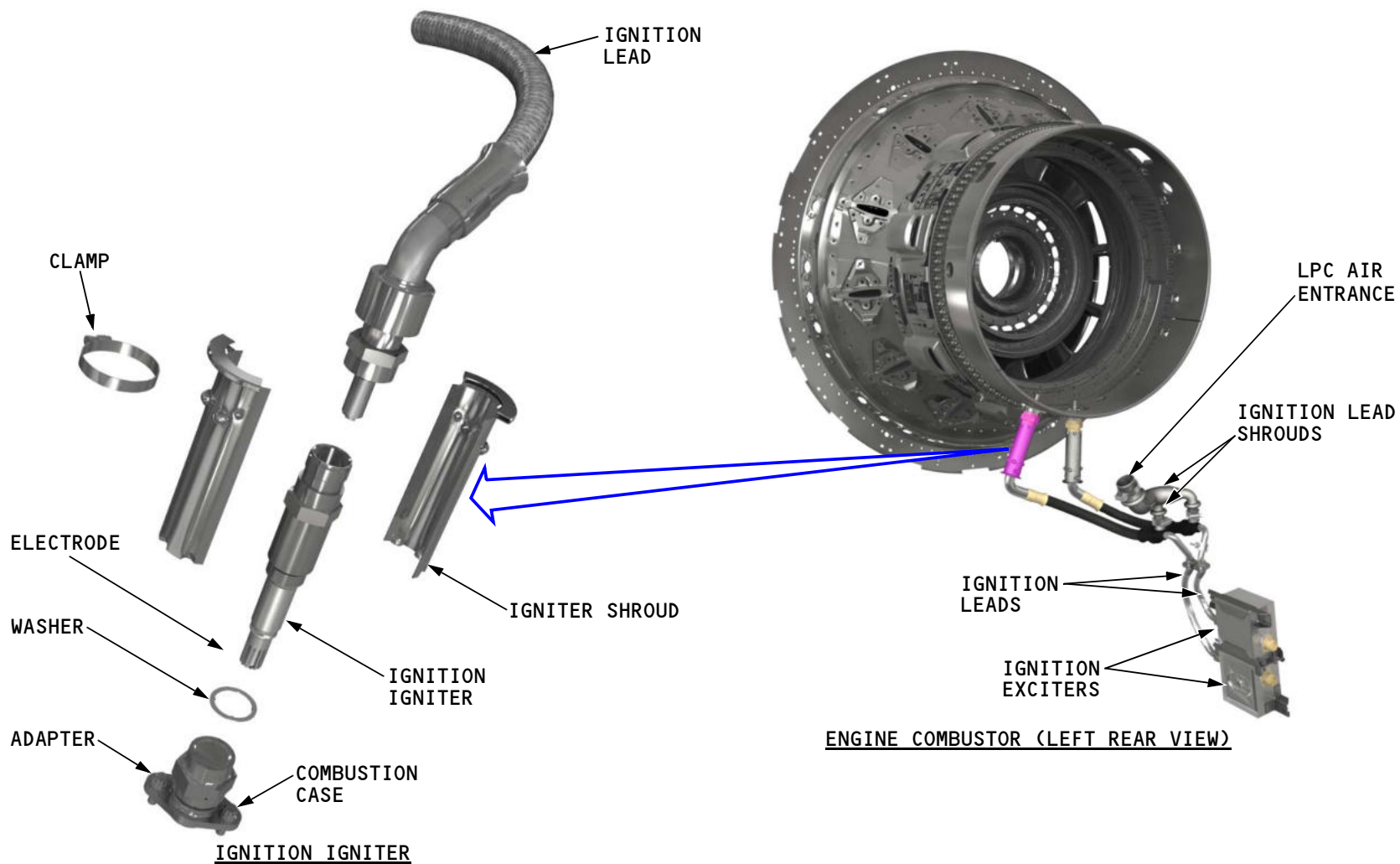
Spark Igniters

Each ignition igniter installs into an adapter at the 5:00 and 6:00 positions on the engine. The spark igniter electrode extends into the combustion chamber.

DMC-B737T-A-74-00-00-04A-034B-T_003-00

02-Dec-2020

74-00-00



DMC-B737T-A-74-00-00-04A-034B-T_003-00
ICN-B737T-A-740000-A-81205-00014-A-003-01

IGNITION -- DISTRIBUTION - IGNITION LEADS AND SPARK IGNITERS

74-00-00

IGNITION -- FUNCTIONAL DESCRIPTION

General

The electronic engine controls (EECs) turn the two igniters on or off. Input from the flight compartment switches give manual control. The EECs have internal logic for automatic ignition system control.

Ignition Inputs

The EEC and the display processing computers (DPCs) monitor the position of these flight compartment components:

- Ignition selector switch
- Engine start switch
- Start lever.

Each DPC sends digital data signals to the EECs with the start and ignition selector switch position information. When you move a start lever to IDLE, 115v ac ignition power goes to each EEC channel.

The EECs control internal EEC ignition on/off switches. This controls the 115v ac power that goes to the ignition exciters. Each EEC channel controls one ignition exciter. The EEC channel in control can control the igniter in the other channel through the cross channel data bus.

The ignition exciters get 115v ac power and then supply pulses of 15,000 to 20,000v dc electrical power to the ignition igniters. This high voltage causes sparks across the electrodes of the igniter.

GND Position

The EECs make the selected igniter(s) operate when you select the GND position of the start switch and these conditions exist :

- Engine below idle
- Start logic enables ignition operation (bowed rotor start).

Automatic Ignition (Start Switch in the AUTO or OFF Position)

The EECs make both the igniters operate if one or more of these conditions exist :

- Rejected takeoff
- Flameout or a rollback
- Quick windmill relight engine start try - (start lever cycled from idle - cutoff - idle)
- Rotating stall.

Continuous Ignition

The EECs make the selected igniter(s) operate when you select the CONT position of the start switch and these conditions exist :

- Engine at or above idle
- Ps3 is below the threshold for igniter sparking.

FLT Position

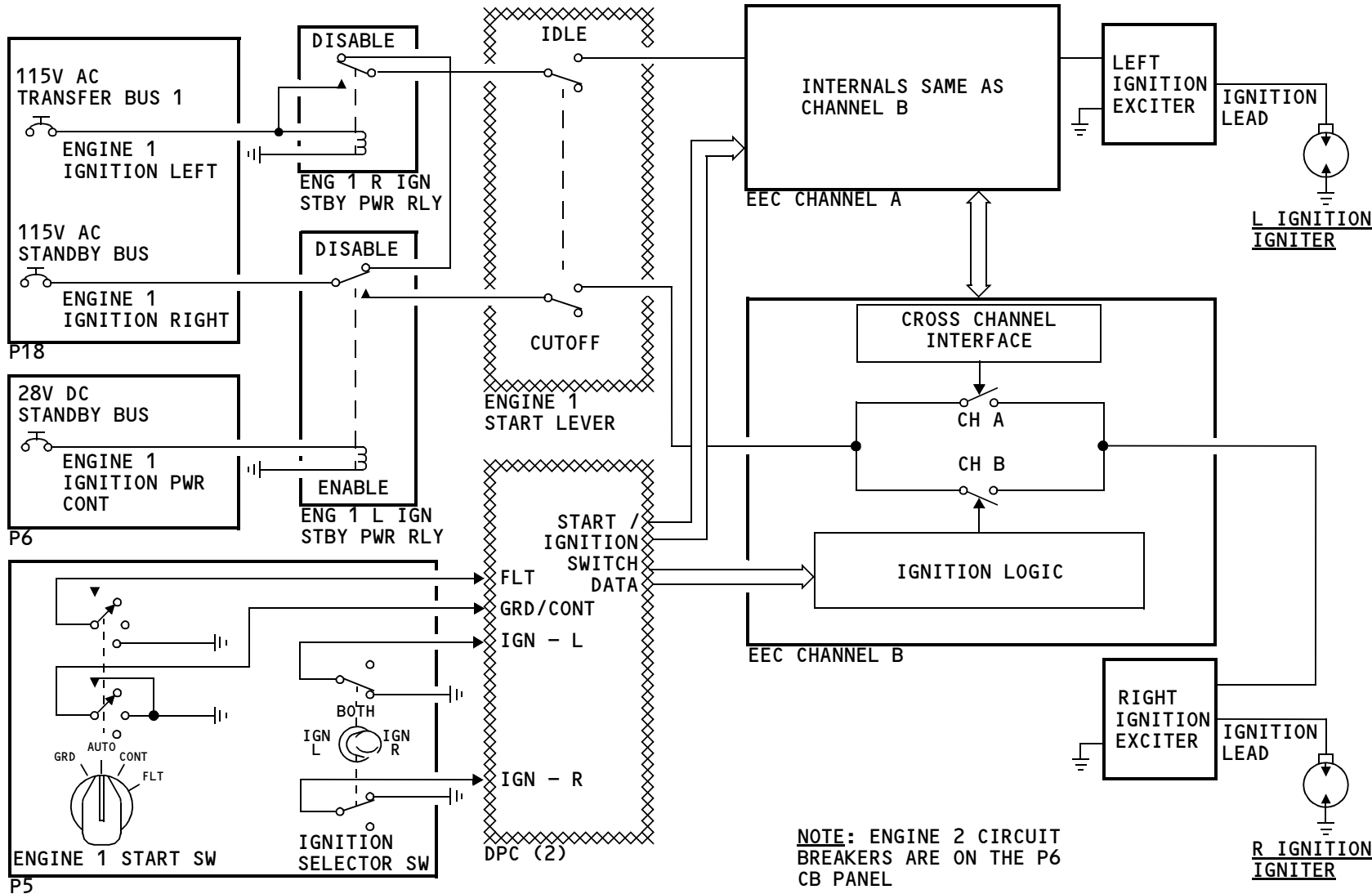
The EECs makes both igniters operate when you select the FLT position of the start switch and these conditions exist :

- Engine below idle
- Airplane in the air.

Damage Protection

The EEC can also stop the operation of the igniters if it senses one or more other these conditions :

- Hot start on the ground
- No-lightoff condition
- Start stall
- Ground start rollback overtemperature.



IGNITION -- FUNCTIONAL DESCRIPTION

74-00-00

IGNITION -- OPERATION

Engine Start Switch Position (GRD)

The engine starter engages and turns the engine when you move the engine start switch to the GRD position. The electronic engine controls (EECs) turn on ignition and fuel to the engine combustor after the appropriate delay time when you then move the start lever to the idle position.

The ignition selector switch tells the EECs which, or both igniters to use for ground start operation. Normal operation is for you to alternate igniters every ground start.

You use the GRD position to start the engine on the ground, or in the air if starter assist is necessary.

Engine Start Switch Position (OFF or AUTO)

Usually the igniters do not operate when the start switch is in the OFF or AUTO position. The EECs sense conditions where both igniters operate continuously for the engine to continue operation. An example of this is flameout is protection. Ignition system operation is the same if the panel position is either OFF or AUTO (customer option).

Engine Start Switch Position (CONT)

The flight crew may turn the engine start switch to the CONT position at these times:

- Takeoff
- Approach
- Landing
- Bad Weather.

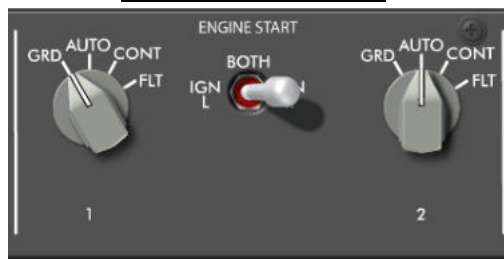
The ignition igniter(s), as selected by the ignition selector switch, turn on and operate continuously.

Engine Start Switch Position (FLT)

The crew puts the engine start switch to the FLT position for an inflight windmill start. A windmill start does not use the starter.

Both ignition igniters operate continuously when the engine start switch is in the FLT position.

IGNITION SELECTION



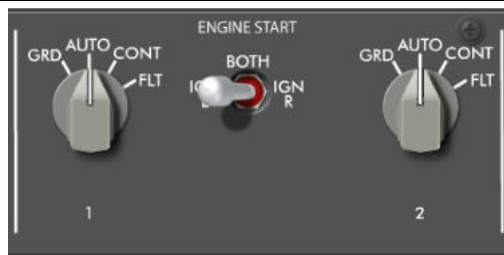
AIRPLANE CONDITION



ENGINE 1 RIGHT IGNITER SELECTION

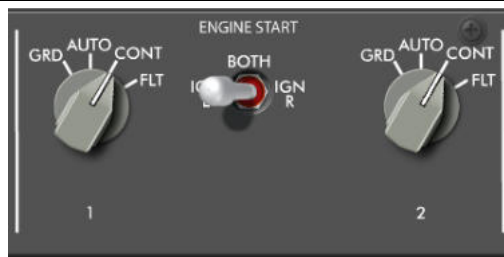
IGNITER OPERATION

LEFT IGNITER RIGHT IGNITER



ENGINE 1 & 2 CRUISE, NO IGNITION (UNLESS FLAMEOUT)

LEFT IGNITER RIGHT IGNITER



ENGINE 1 & 2, LEFT IGNITER SELECTION

LEFT IGNITER RIGHT IGNITER



ENGINE 1 OR 2 FLIGHT, BOTH IGNITERS OPERATE

LEFT IGNITER RIGHT IGNITER



IGNITION -- OPERATION

74-00-00

ENGINE STARTING -- INTRODUCTION

General

SB/BAI - start bleed/booster anti-ice

The engine starting system uses bleed air to turn the engine N2 rotor during a start or motor procedure. Bleed air comes from one of these sources:

- APU
- Ground pneumatic equipment
- Opposite engine.

These components control the engine start system:

- Flight compartment switches
- Display processing computers (DPCs)
- Electronic engine controls (EECs).

The engine starting system operates on the ground and in flight.

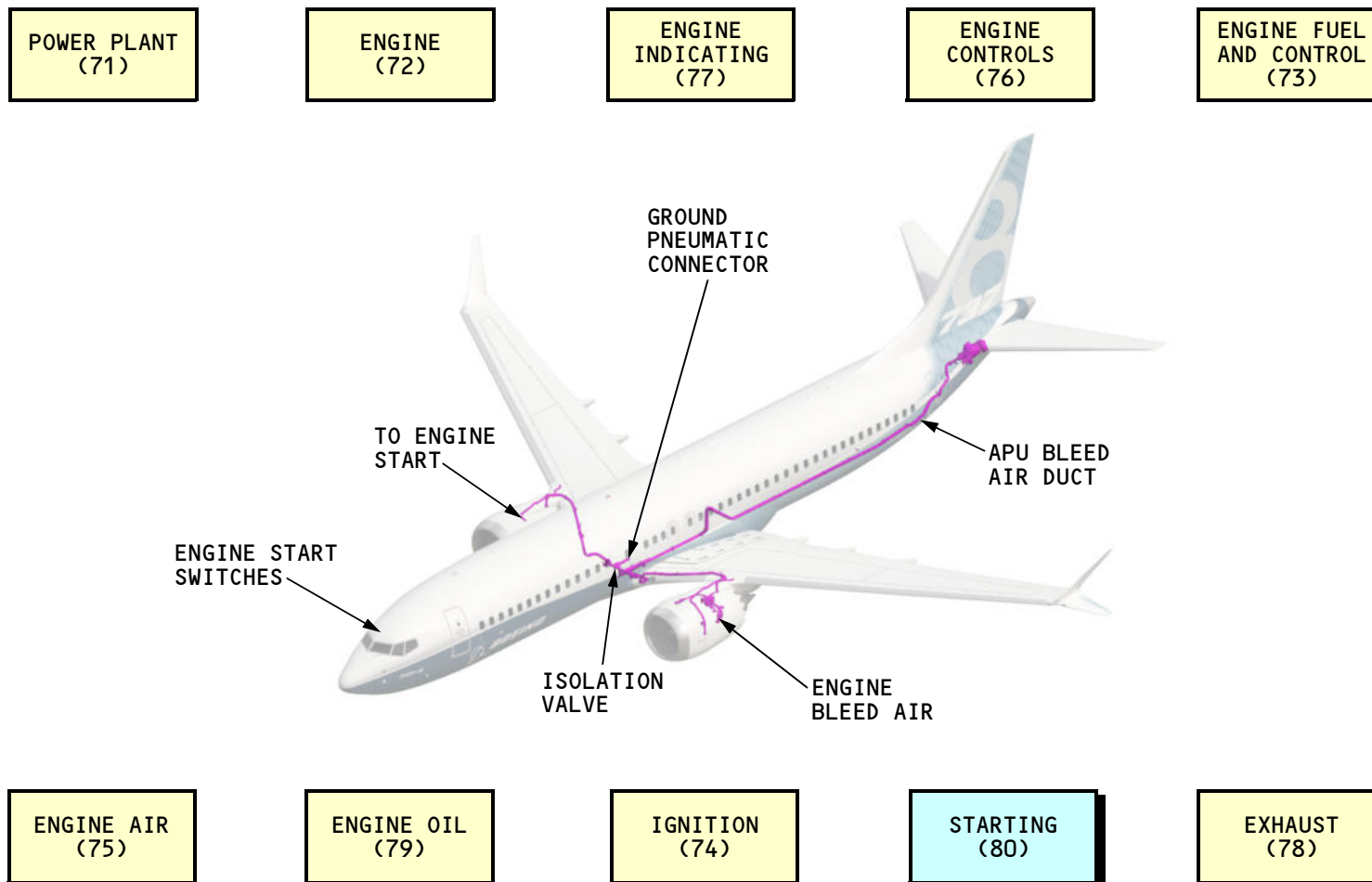
Abbreviations & Acronyms

AGB	- engine accessory gearbox
ALF	- aft looking forward
BRS	- bowed rotor start
DPC	- display processing computer
DU	- display unit
EEC	- electronic engine control
FLT	- flight
FOD	- foreign object debris/foreign object damage
FMCS	- flight management computer system
GRD	- ground
IGV	- inlet guide vanes
IGVA	- inlet guide vane actuator
MDS	- max display system
MES	- main engine start
SAV	- starter air valve

DMC-B737T-A-80-00-00-02A-018A-T_004-00

02-Dec-2020

80-00-00



ENGINE STARTING -- INTRODUCTION

80-00-00

DMC-B737T-A-80-00-00-02A-018A-T_004-00
ICN-B737T-A-800000-A-81205-00010-A-002-01

02-Dec-2020

ENGINE STARTING -- GENERAL DESCRIPTION

General

The engine starting system uses the engine start switch and these systems:

- Electrical power
- Bleed air
- Air turbine starter (ATS)
- Starter air valve (SAV)
- Electronic engine controls (EECs)
- APU electronic control unit (APU ECU)
- Display Processing Computers (DPCs).

Electrical Power

The airplane normally has 115v ac power established before engine starting. If necessary, you can operate the start system with airplane battery power only.

Bleed Air

These systems can supply bleed air for engine start:

- APU
- Ground pneumatic equipment
- Opposite engine.

Engine Start Switch

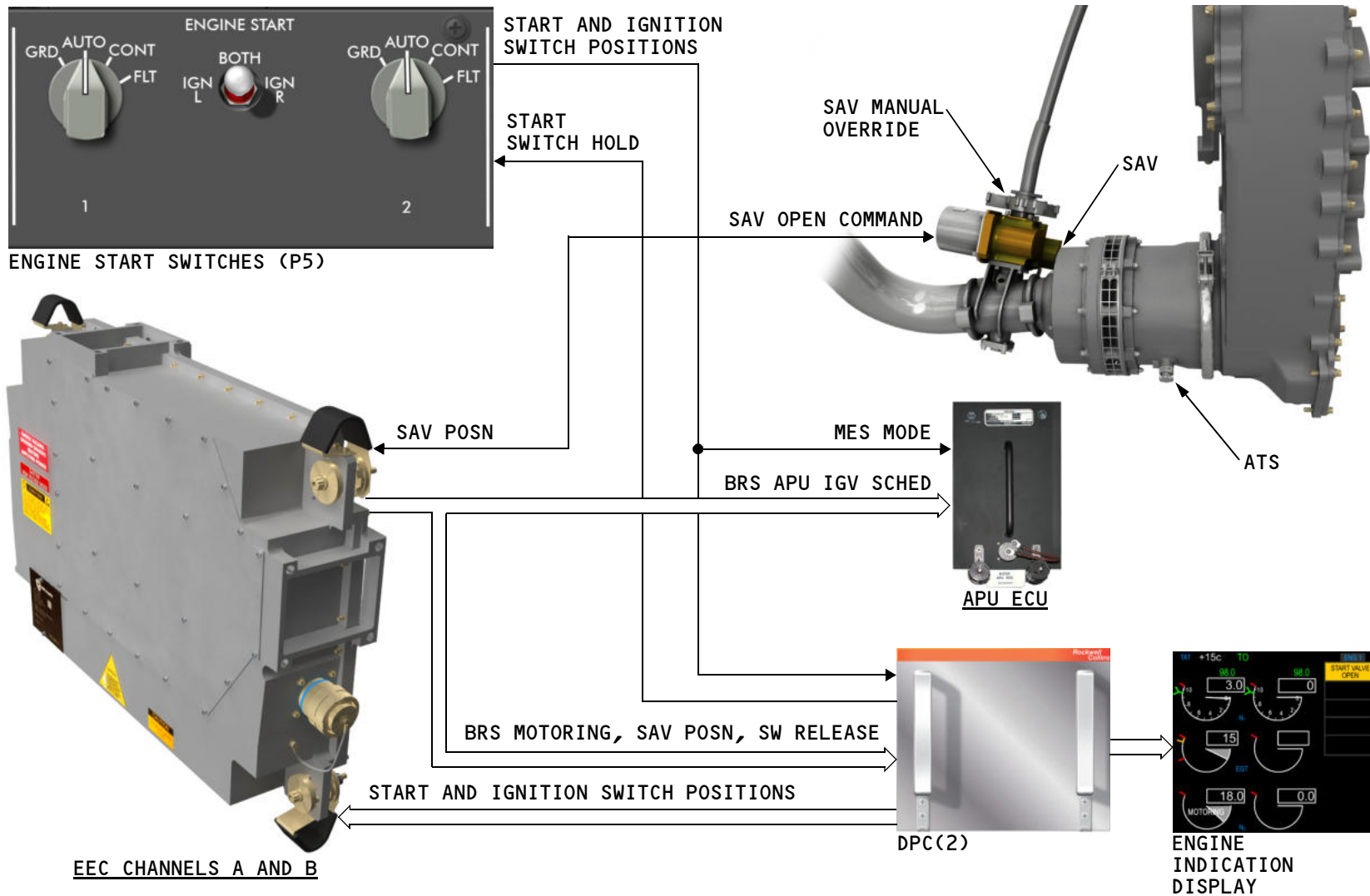
You move the engine start switch to the GRD position to turn the engine with the ATS. The ATS can turn the N2 rotor on the ground and in-flight. The switch automatically moves to the OFF position when starting is complete.

When electrical power and bleed air are available, this happens when you move the switch to the GRD position:

- APU ECU receives a main engine start (MES) signal
- DPCs receive the start and ignition switch positions
- DPCs send a start switch hold signal to the start switch
- DPCs send the start and ignition switch positions to the EECs
- EECs send an open command to the SAV, and the ATS begins to turn the engine
- The SAV sends an open position signal to the EECs. The EECs send this signal to the DPCs, which then show the start valve open message
- The EECs send a signal to the DPCs and APU ECU during bowed rotor start (BRS) motoring. The APU IGVs move to a more closed position, and the DPCs show the MOTORING message
- The EECs modulate the start valve signal to limit N2 speed during BRS motoring
- After BRS motoring, the start valve and the APU IGVs move to the full open position
- When the engine reaches idle speed, the EECs send a start switch release signal to the DPCs. The DPCs then remove the switch hold signal, and the start switch moves to the OFF position.

The crew uses the FLT position to start the engine in-flight when the ATS is not necessary.

See the ignition chapter 74 for more information on the ignition system.



DMC-B737T-A-80-00-00-02A-040A-T_003-00
ICN-B737T-A-800000-A-81205-00011-A-003-01

ENGINE STARTING -- GENERAL DESCRIPTION

80-00-00

02-Dec-2020

ENGINE STARTING -- COMPONENT LOCATIONS - ENGINE

General

The engine starting system has these components on the left side of the engine:

- Pneumatic starter duct assembly
- Starter air valve (SAV)
- Air turbine starter (ATS).

Open the left fan cowl to get access to these components.

Pneumatic Starter Duct Assembly

There is one pneumatic starter duct assembly on the fan case. The assembly has a top tube and a bottom tube. The top end of the top tube of the assembly attaches to the pylon duct at the strut. Flexible joints join pieces of the top tube.

The bottom tube has an elbow shape. The forward end of the bottom tube attaches to the top tube. The aft end of the bottom tube attaches to the SAV.

Starter Air Valve (SAV)

The forward end of the SAV connects to the aft end of the starter duct assembly. The aft end of the SAV connects to the forward end of the ATS.

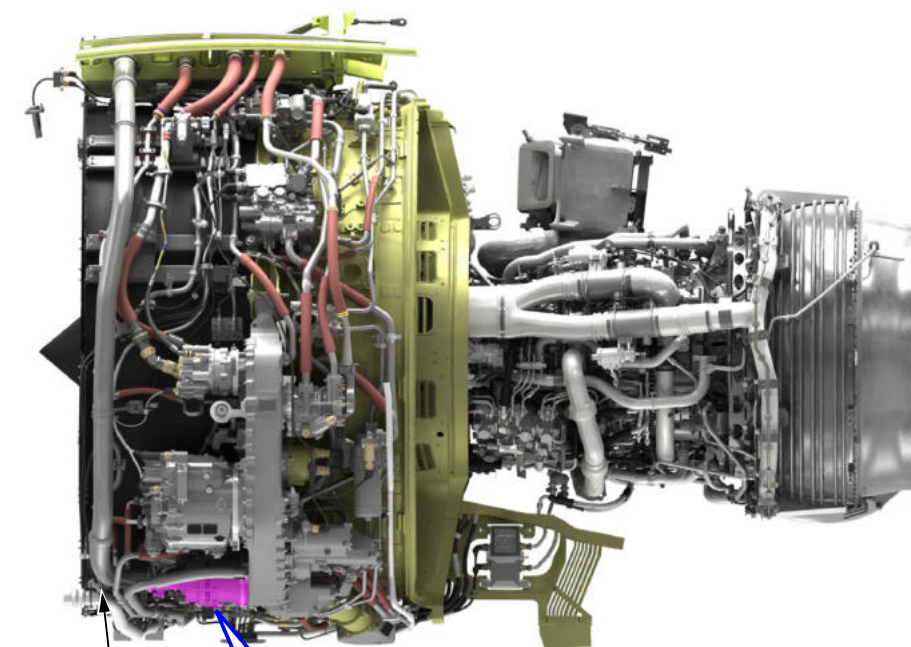
Air Turbine Starter (ATS)

The ATS is on the forward face of the engine accessory gearbox (AGB) at the 7:00 position.

DMC-B737T-A-80-00-00-03A-055A-T_002-00

02-Dec-2020

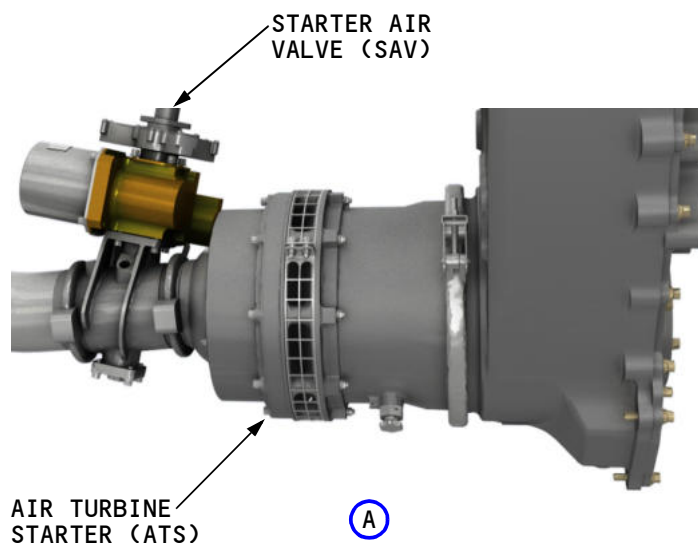
80-00-00



LEAP-1B ENGINE
(LEFT VIEW)

SEE (A)

PNEUMATIC STARTER
DUCT ASSEMBLY



STARTER AIR
VALVE (SAV)

AIR TURBINE
STARTER (ATS)

(A)

ENGINE STARTING -- COMPONENT LOCATIONS - ENGINE

80-00-00

DMC-B737T-A-80-00-00-03A-055A-T_002-00
ICN-B737T-A-800000-A-81205-00012-A-002-01

02-Dec-2020

ENGINE STARTING -- COMPONENT LOCATIONS - FLIGHT COMPARTMENT AND EE COMPARTMENT

Flight Compartment

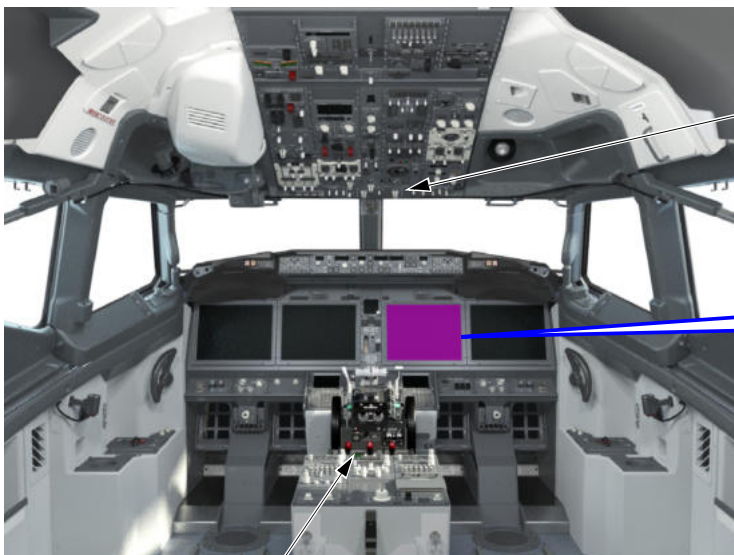
The engine start switches and the ignition selector switch are on the P5 forward overhead panel.

The engine parameters show on the engine indication display. You can see the engine indication displays on the display units.

The engine start levers are on the control stand, aft of the thrust levers.

Electronic Equipment Compartment

The display processing computers (DPC)s are on the E3 rack.

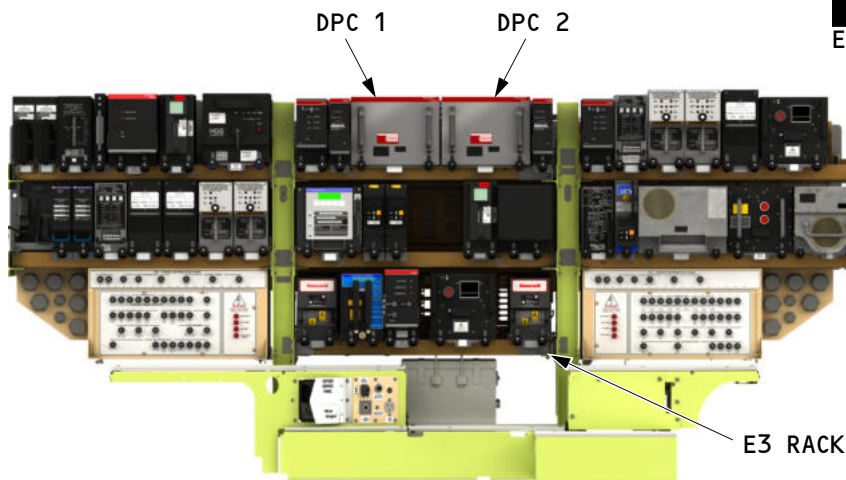


P5 FORWARD OVERHEAD PANEL
- ENGINE START SWITCHES
- IGNITION SELECTOR SWITCH



ENGINE INDICATION DISPLAY

CONTROL STAND
- ENGINE START LEVERS



ELECTRONIC EQUIPMENT COMPARTMENT

ENGINE STARTING -- COMPONENT LOCATIONS - FLIGHT COMPARTMENT AND EE COMPARTMENT

80-00-00

DMC-B737T-A-80-00-00-04A-055A-T_003-00
ICN-B737T-A-800000-A-81205-00013-A-002-01

02-Dec-2020

ENGINE STARTING -- STARTER AIR VALVE (SAV)

Purpose

The starter air valve (SAV) delivers bleed air to the starter. The valve opens for these engine operations:

- Motoring
- Ground start
- In-flight starts which require starter assist.

Functional Description

The primary components are the valve body assembly and the actuator assembly.

The valve is normally electrically controlled and pneumatically operated. The valve is spring-loaded closed and opened with bleed air. Strain gauge type pressure sensors send the start valve position to the electronic engine controls (EECs). A visual indicator on the valve body also shows the valve position.

The valve can be manually operated using a 3/8 inch square drive tool. A small hole in the left fan cowl gives access to operate the manual override. A flexible drive shaft connects from the fan cowl opening down to the start valve. A placard is near the hole.

WARNING: REMOVE PNEUMATIC POWER FROM THE AIRPLANE BEFORE YOU REMOVE THE START VALVE. AIR PRESSURE IN THE PNEUMATIC STARTER DUCT COULD CAUSE INJURY.

CAUTION: DO NOT OPERATE THE MANUAL DRIVE WITHOUT PNEUMATIC POWER AVAILABLE. YOU MUST SUPPLY PNEUMATIC POWER TO THE VALVE TO PREVENT DAMAGE TO THE ACTUATOR DIAPHRAGMS. IF YOU

DO NOT OBEY PRECAUTIONS, DAMAGE TO THE VALVE CAN OCCUR.

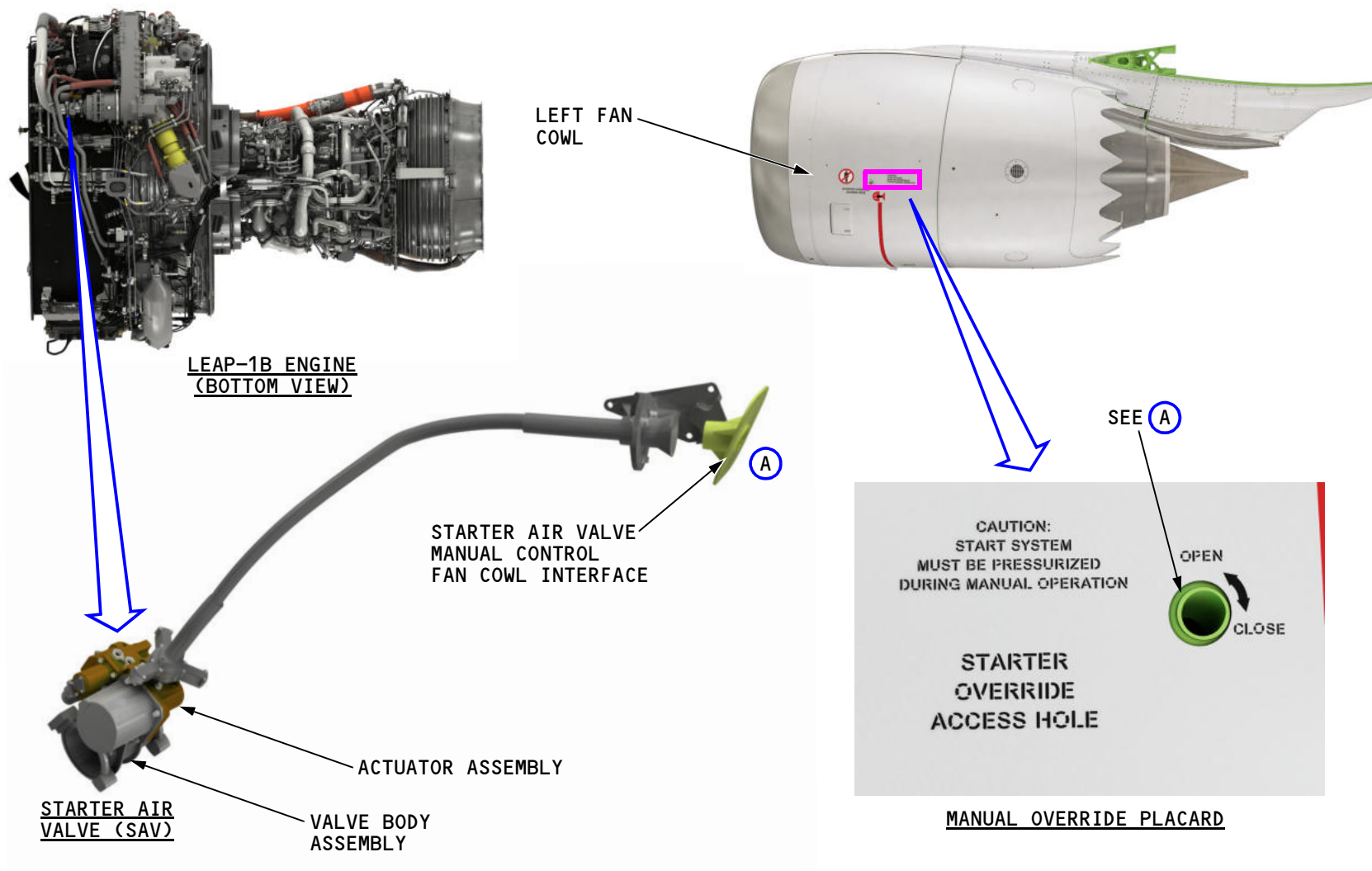
Training Information Point

Many SAV removals that occur are not necessary. Many valve assemblies sent out for maintenance have no defects. You must not temporarily bend the valve body while you install the valve assembly. This could occur if you apply too much torque on the clamps. A bent valve body can prevent the correct movement of the component's internal parts.

DMC-B737T-A-80-00-00-03A-034B-T_004-00

02-Dec-2020

80-00-00



ENGINE STARTING -- STARTER AIR VALVE (SAV)

80-00-00

DMC-B737T-A-80-00-00-03A-034B-T_004-00
ICN-B737T-A-800000-A-81205-00014-A-005-01

02-Dec-2020

ENGINE STARTING -- STARTER

Purpose

The air turbine starter (ATS) turns the engine for engine starting or motoring procedures. The ATS changes air pressure into mechanical energy.

Physical Description

The ATS has a single-stage, axial-flow, turbine air motor. The starter uses reduction gears and a clutch to turn the output shaft to the engine accessory gearbox (AGB).

The ATS has one port for oil drain and fill. This port also has a magnetic chip detector. There is no sight glass to find the oil level when you add oil.

A coupling at the aft end of the ATS attaches the ATS to the AGB. One more coupling at the forward end of the ATS attaches the ATS to the starter air valve (SAV).

Functional Description

Bleed air goes through the open SAV to turn the turbine air motor in the ATS. The ATS output shaft turns the AGB. The AGB then gives torque to the transfer gearbox. The transfer gearbox gives that torque to the inlet gearbox, which makes the engine core turn.

At approximately 60 percent N2, the SAV closes and removes bleed air from the ATS. The turbine and reduction gears decrease speed and the clutch disengages. The ATS output shaft then turns with the AGB and engine. The turbine and reduction gears continue to decrease speed until they stop.

The ATS receives a continuous flow of oil from the AGB when the engine turns. This maintains the correct amount of oil in the starter.

The oil reduces the temperature of and lubricates these components of the ATS:

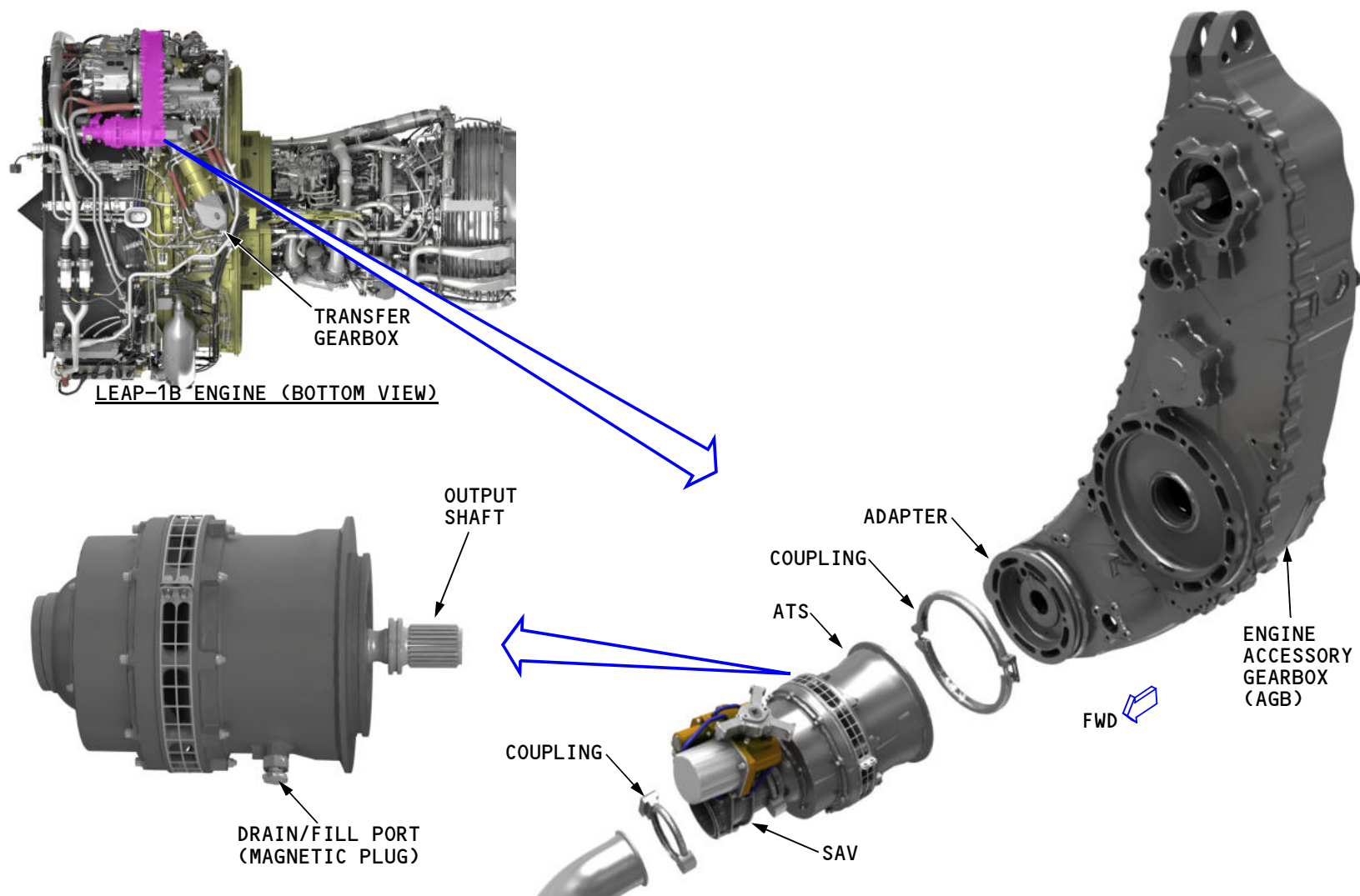
- Clutch
- Gears
- Bearings.

Oil goes back to the AGB. A small quantity of oil stays in the ATS.

Training Information Point

The maintenance manual tells you to give a new ATS oil only when you install it. After you give a new ATS oil, there is no scheduled oil servicing requirement.

DMC-B737T-A-80-00-00-04A-034B-T_003-00



DMC-B737T-A-80-00-00-04A-034B-T_003-00
ICN-B737T-A-800000-A-81205-00015-A-002-01

ENGINE STARTING -- STARTER

80-00-00



ENGINE STARTING -- FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK



ENGINE STARTING -- FUNCTIONAL DESCRIPTION

Start Control

Electrical power for the starter air valve (SAV) control comes from the battery bus. The engine electronic controls (EECs) control the SAV during engine start.

The EECs use inputs from these components to control the engine start system:

- Display processing computers (DPCs) provide the engine start switch and ignition select switch positions
- Start valve position sensor (strain gauge pressure sensors)
- Start levers
- Start bleed/booster anti-ice valve (SB/BAI)
- Engine sensors.

Indication

The DPCs use input data from the EECs to show the start valve position on the display units (DUs).

These are the 2 modes of operation for the amber START VALVE OPEN message:

- Steady
- Flash (crew alert).

The START VALVE OPEN message is on steady when the start valve is open, and the start switch is in the ground (GND) position.

The START VALVE OPEN message flashes for 10 seconds and then is on steady when the start valve is open, and the start switch is not in the GND position.

Start Valve

Bleed air and the start valve solenoids control opening of the start valve. Normally bleed air is on both sides of the valve diaphragm. This pressure equalizes across the valve diaphragm and the spring holds the start valve closed. During start, the EECs give power to the SAV solenoids. This causes the internal vent valve to open. The spring side of the valve diaphragm is then vented. The other side of the diaphragm with bleed pressure over-comes the spring pressure and the valve opens. The EECs modulate the start valve during bowed rotor start (BRS) motoring to keep the engine below fuel-on speed. This is done by intermittently removing the voltage to the solenoids. After BRS motoring, the EECs then command the start valve to the full open position for the rest of the start.

The EECs command the SAV to close when one of these occurs:

- You move the start switch to OFF
- The engine start cycle concludes
- The EECs find that the engine start switch latch/release mechanism did not release the switch correctly.

When the start switch is in the OFF position the start valve solenoids are de-energized, and the internal vent valve moves to the closed position. The bleed air pressure then equalizes across the valve diaphragm and spring pressure closes the start valve.

Start Switch

When bleed air and electrical power are supplied, and the start switch is placed in the GRD position, the following occur:

- EECs receive an engine start signal
- APU electronic control unit (ECU) receives a signal to increase bleed air supply, main engine start (MES). The APU ECU also

ENGINE STARTING -- FUNCTIONAL DESCRIPTION

gets alternate inlet guide vane schedule (IGV) data to help reduce motoring speed during bowed rotor start (BRS) motoring

- Start valve solenoid energizes, and the valve opens
- Air turbine starter (ATS) clutch engages, and the ATS turns the N2 rotor
- Solenoid in the P5 panel energizes to hold the switch in the GRD position.

Starter Cutout

When the engine reaches approximately 60 percent N2, the EECs send a switch-release signal to the DPCs. These conditions then occur:

- DPCs disconnect the electrical ground for the start switch solenoid
- Engine start switch goes to the OFF position
- Start valve solenoid de-energizes, and the valve closes.

The start switch can be moved manually to the OFF position at any time during the start, or after a start, if the start switch solenoid does not de-energize.

Starter Duty Cycles

The start has two duty cycles. The usual starter duty cycle is observed during engine starts. The extended duty cycle is observed during maintenance motoring.

The usual duty cycle limits are as follows:

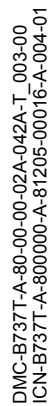
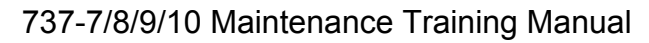
- No limit to the number of engine starts
- Maximum operating time is 2 minutes
- Wait 10 seconds between each start.

The EECs receive temperature and vibration data from engine sensors. It uses that data and internal bowed rotor start logic to calculate if it is safe to let the engine start. The EECs can prevent engine fuel flow and ignition automatically until conditions are safe. While the EEC is preventing the engine start, the ATS can continue to be in operation. If you do not obey the ATS duty cycle limits, damage to the ATS can occur. You must be sure to obey the usual duty cycle limits. See the bowed rotor start section of this chapter for more information.

The extended duty cycle limits are as follows:

- 2 extended starter operations - maximum time for each is less than 15 minutes
- Wait 2 minutes between each operation
- After the 2 extended operations the maximum time for subsequent starter operations is 5 minutes
- Wait a minimum of 10 minutes between each subsequent motor of the engine.

Exceeding any of the extended duty cycle motoring limits will require starter replacement.



80-00-00

ENGINE STARTING -- BOWED ROTOR

General

Bowed rotor is a temporary condition that occurs in the LEAP-1B engine because of heat soak after shutdown. The difference in heat between the higher and lower parts of the N2 rotor after engine shutdown causes a small bend in the rotor. This bend is also called a bow. The bow starts to form in less than 30 minutes after you stop the engine. The bow can stay for a more than 8 hours after you stop the engine.

When the N2 rotor has a bow, it can cause engine damage or malfunctions during start such as:

- Rubbing of compressor blades and compressor discharge pressure (CDP) seal
- Start stalls due to more open clearances on one side of the rotor
- High vibration.

To correct this condition and prevent engine damage, the temperature of the different parts of the N2 rotor must become equal before fuel and ignition come on. Engine motoring reduces the difference in heat, and thus reduces the bow and subsequent vibration.

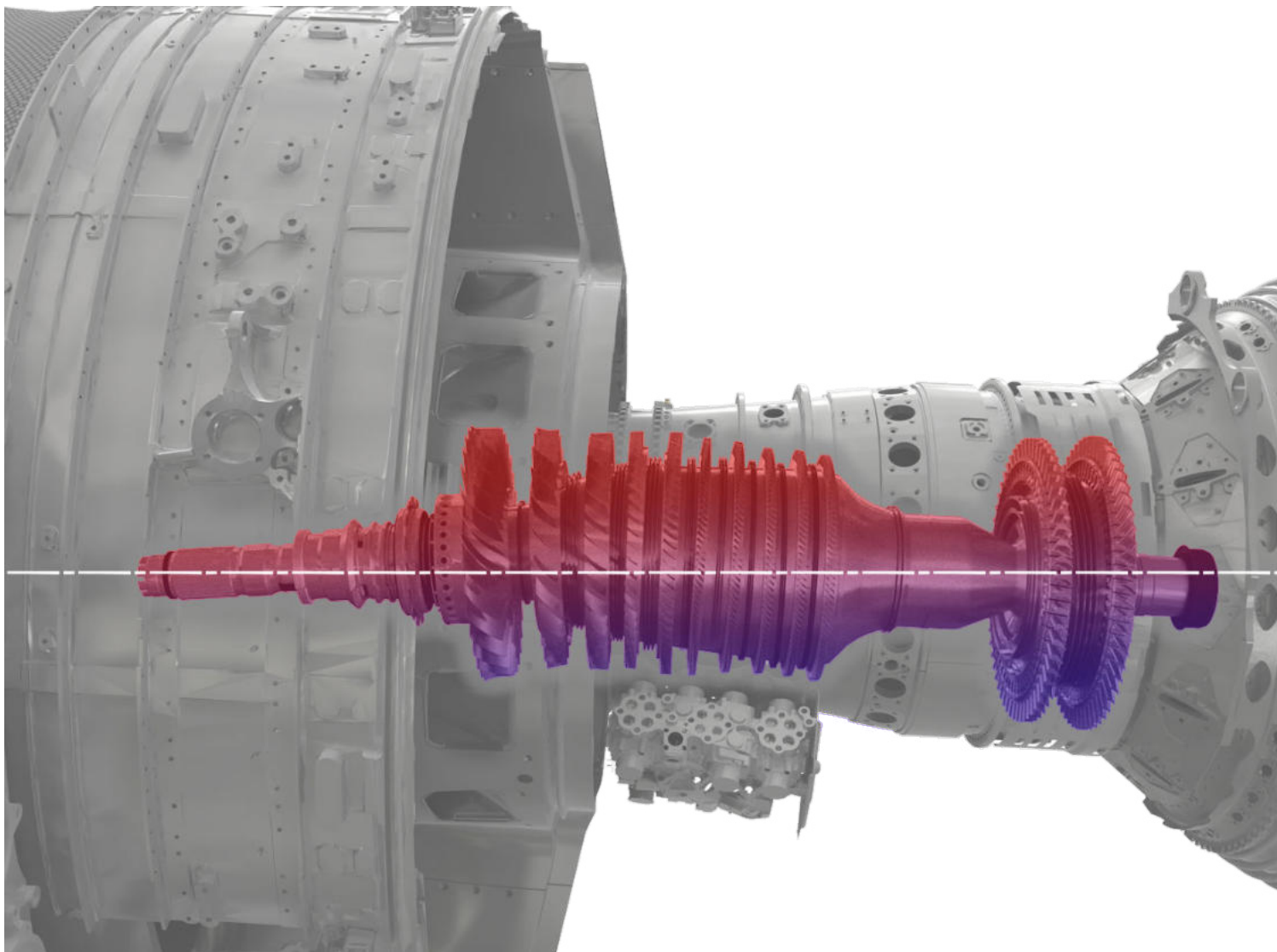
The electronic engine controls (EECs) protect the engine against damage due to the bowed rotor effect during start by doing the following:

- Limit the engine motoring RPM
- Calculate the motoring dwell time before fuel/ignition on.

DMC-B737T-A-80-00-00-06A-040A-T_002-00

02-Dec-2020

80-00-00



ENGINE STARTING -- BOWED ROTOR

80-00-00

DMC-B737T-A-80-00-00-06A-040A-T_002-00
ICN-B737T-A-800000-A-81205-00072-A-002-01

02-Dec-2020

ENGINE STARTING -- BOWED ROTOR PROTECTION

Bowed Rotor Start (BRS) Logic

The EECs contain bowed rotor start (BRS) logic. BRS logic is active for all engine motoring and ground starts. BRS logic extends the engine motoring time during the start sequence. This extended motoring time is called the motoring dwell time. The EEC calculates the minimum motoring dwell time.

The EECs use the following to calculate the motoring dwell time:

- Engine off time
- Residual compressor discharge temperature (T3-TAT)
- Turbine center frame (TCF) core vibration level.

The usual minimum motoring dwell time is between 6 and 85 seconds. The EECs extend the motoring dwell time if:

- EGT is too high after the initial bowed rotor start motoring dwell time
- Engine vibration is too high after the initial bowed rotor start motoring dwell time.

The maximum bowed rotor start motoring dwell time is 86 seconds. At this time the start valve goes to full open. When the APU gives the bleed air for the engine start, the APU bleed air pressure goes to maximum. The engine start can then continue.

The EECs do the following during the motoring dwell time:

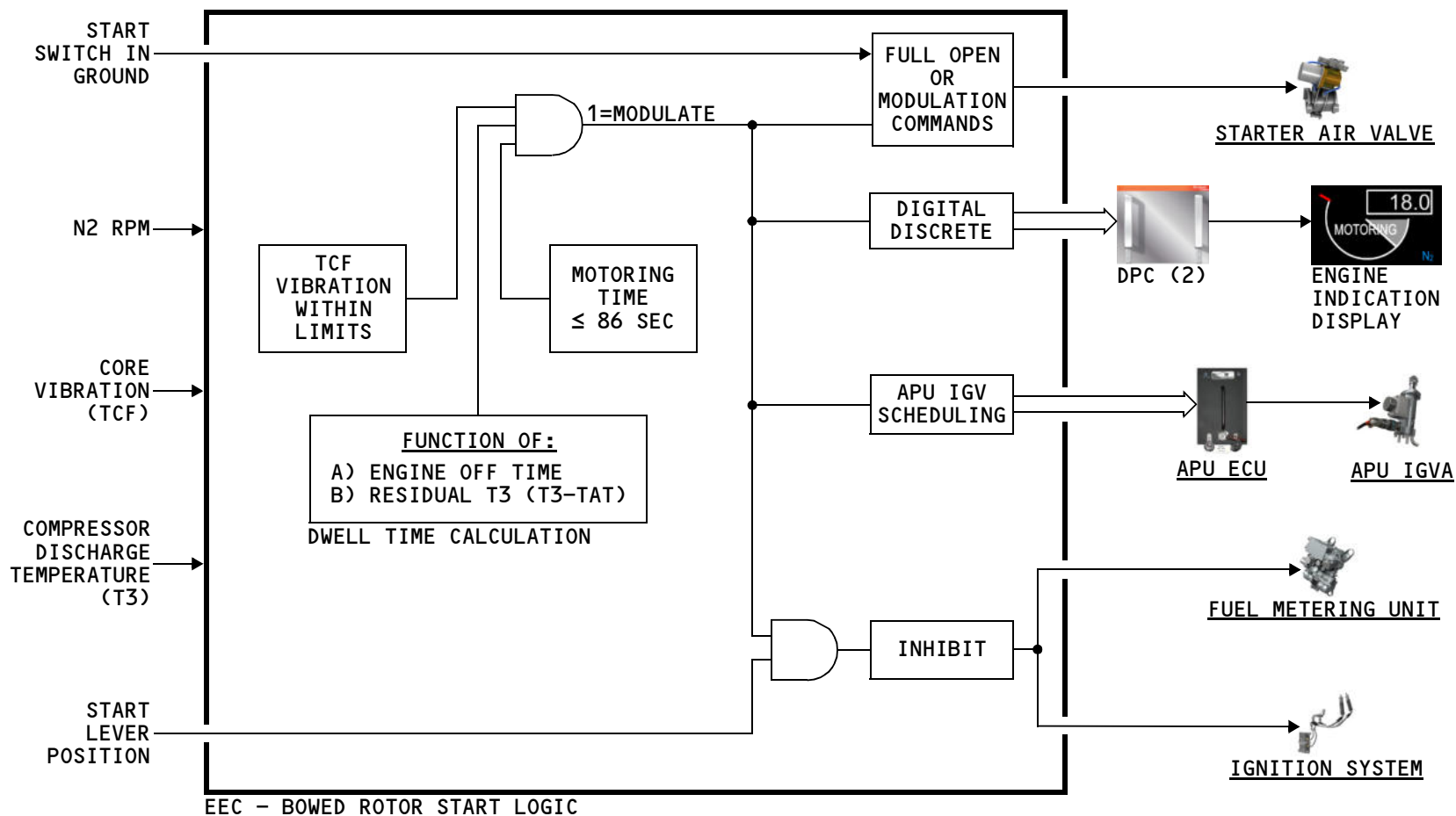
- Send a discrete signal to the digital processing computers (DPCs). The DPCs then show the white MOTORING message on the engine's N2 indication at approximately 18 percent
- Send modulation commands to the starter air valve to keep N2 RPM at approximately 23 percent
- Signal the APU electronic control unit to use a different bleed schedule. This modulates the APU inlet guide vanes toward a more closed position

- Prevent fuel flow
- Prevent ignition.

Do not move the engine start lever to IDLE until both of these occur:

- The MOTORING message goes away
- The engine goes to maximum motoring speed.

DMC-B737T-A-80-00-00-01A-028A-T_009-00



DMC-B737T-A-80-00-00-01A-028A-T_009-00
ICN-B737T-A-800000-A-81205-00019-A-007-01

ENGINE STARTING -- BOWED ROTOR PROTECTION

80-00-00

02-Dec-2020

ENGINE STARTING -- FUNCTIONAL DESCRIPTION - START PROFILE

Starting Control

The engine starting system uses an enhanced-manual control mode for all engine starts. This mode includes automatic abnormal start protection features. Inputs from these two flight compartment controls tell the EECs to do an engine start:

- Engine start switch
- Engine start lever.

Start Switch Latching and Release

The start switch is spring-loaded to return to the OFF/AUTO position from the GRD position. A latching solenoid holds it in the GRD position until starter cutout speed is reached. The MAX display system (MDS) uses a digital discrete input from the EECs to control the latching solenoid. The EEC sends the discrete to the MDS when:

- Engine core speed exceeds starter cutout speed
- EECs stop an engine start due to bowed rotor speed exceedance.

The EECs initiate a ground start sequence when these occur:

- Engine start switch is moved to GRD
- Engine start lever is moved to IDLE
- Primary air/ground signal shows airplane on-ground.

Ground Start

These three valves control engine fuel flow:

- Engine fuel spar valve
- High-pressure shutoff valve (HPSOV)
- Fuel metering valve (FMV).

The EECs activate one or both igniters based on inputs from these flight deck controls:

- Ignition select switch
- Engine start switch
- Engine start lever.

The EECs control the starter air valve (SAV) using inputs from these:

- Engine start switch
- N2 speed sensor.

When you move the engine start switch to GRD, these occur:

- EECs send a voltage to open the starter air valve
- N2 speed increases.

At 18% N2 RPM, the EECs tell the DPCs to show MOTORING on the N2 indicator for the applicable engine. At the same time, the EECs also start the bowed rotor motoring timer. The EECs then use bowed rotor start (BRS) logic to keep N2 RPM at or below approximately 23%. This is below the usual fuel-on speed for the engine. The low N2 speed helps prevent engine damage during start. After the BRS motoring, you move the start lever to IDLE. With the start lever in IDLE, the EECs do the engine fuel manifold prefill function.

8 to 10 seconds can be necessary for the engine fuel manifold prefill function. After the engine fuel manifold prefill, the EECs do a test of the electronic over-speed and thrust control malfunction accommodation (EOS/TCMA) system. If the EOS/TCMA system test shows no faults, the EECs let fuel and ignition go to the engine combustor for engine start.



ENGINE STARTING -- FUNCTIONAL DESCRIPTION - START PROFILE

Fuel Manifold Prefill

On ground starts, the EEC opens the fuel metering valve (FMV) to fill the fuel manifold. This procedure gives these functions:

- Prevents irregular fuel flow when the engine is cold-soaked
- Improves engine light-off time
- Allows a test of the (EOS/TCMA) system.

Engine Start Abort

There are two types of procedures to stop an engine start:

- Manual procedure by pilot or technician
- Automatic procedure through EEC failed start detection logic (ground starts only).

The pilot or technician can stop the engine start by moving the start lever to CUTOFF. When this occurs, fuel and ignition will be commanded off. If an EEC finds that a ground engine start did not complete correctly, it can stop the engine start. This is done by closing the FMV. When this occurs, fuel flow to the combustor is stopped.

When an EEC stops the engine start, a flammable fuel and air mixture will be in the engine. You must remove this mixture before you try the next engine start. When you keep the engine start switch in GND with the engine start lever in CUTOFF, you dry-motor the engine. When you dry-motor the engine, you make the engine turn but do not give it fuel or ignition. This removes the fuel and air mixture from the engine. After a period of time, you move the engine start switch to OFF/AUTO to stop the dry-motor procedure. You can then try to start the engine again.

Unusual Engine Start Conditions

The EECs can find start parameters that are not correct. These conditions are unusual, but can occur during engine start:

- Starter air valve failed closed
- No indicated core rotation
- Low oil pressure
- Low starter air pressure
- Locked fan rotor (ground starts only)
- No light-off
- Overtemperature start
- Hot start
- Hung start
- Start stall
- Starter air valve failed open
- Dual N2 sensor failure (ground only)
- Starter duty cycle exceedance
- Bowed rotor speed exceedance (ground starts only).

Not all of the unusual engine start conditions in this list cause an EEC to stop the engine start.

EEC Automatic Start Abort

The EECs automatically stop the engine start if it finds one of these unusual start conditions:

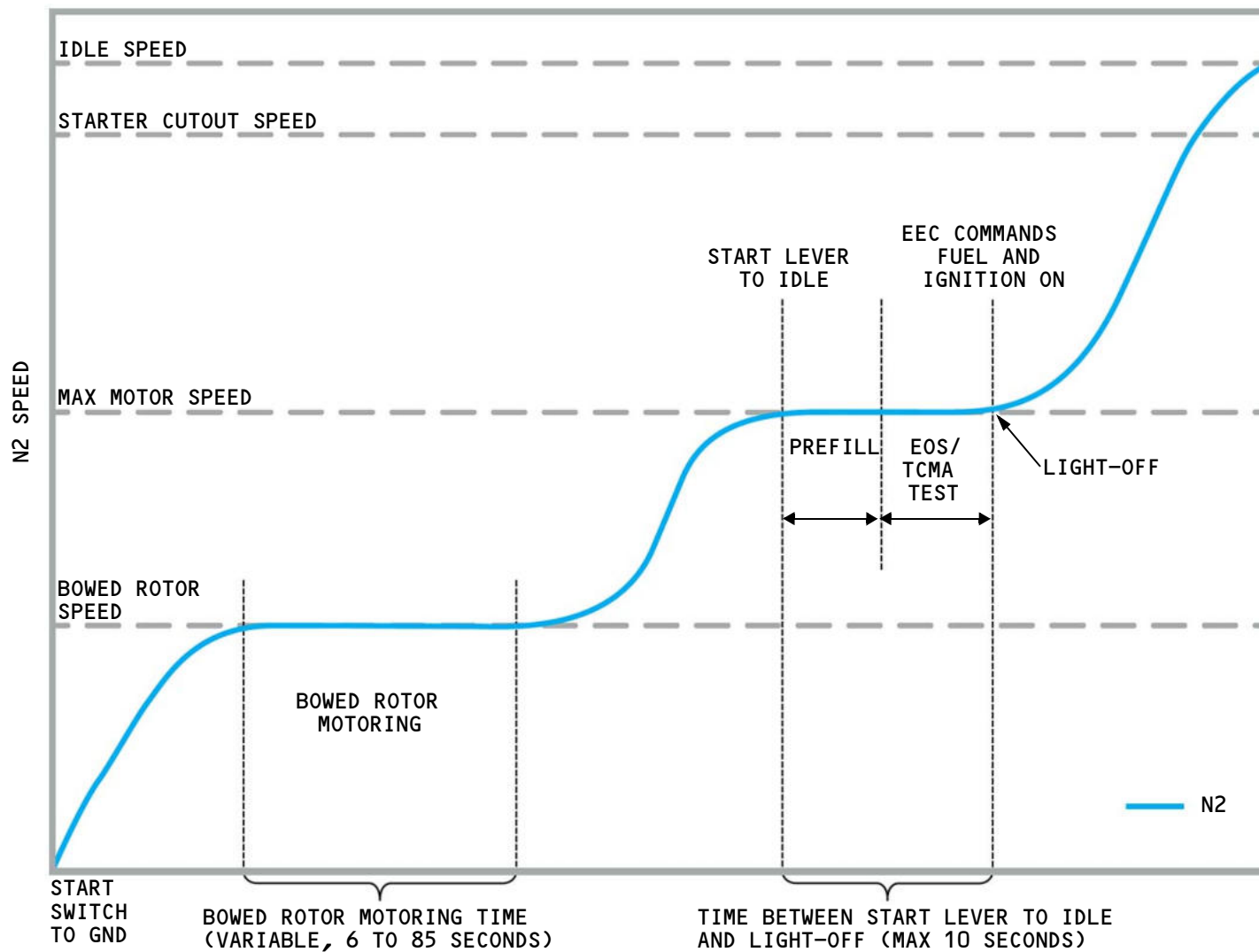
- No indicated core rotation
- Bowed rotor speed exceedance (ground starts only)
- Locked fan rotor (ground starts only)
- No light-off
- Hot start
- Overtemperature start
- Start stall

DMC-B737T-A-80-00-00-03A-042A-T_004-00

ENGINE STARTING -- FUNCTIONAL DESCRIPTION - START PROFILE

- Dual N2 sensor failure (ground only).

After an EEC stops an engine start, you can try to start the engine again. When you do, the EECs will automatically try to correct for the problem that made it stop the engine start. To try an engine start again, move the start lever to CUTOFF, then back to IDLE. The EECs provide protection for all subsequent start attempts. EEC start abort logic is different for in-flight operation.



ENGINE STARTING -- FUNCTIONAL DESCRIPTION - START PROFILE

80-00-00

ENGINE STARTING -- OPERATION

General

These are the flight compartment controls you use to start the engine:

- Engine start switches
- Ignition select switch
- Engine start levers.

Pneumatic power must be available to start the engine on the ground.

You must monitor these engine parameters during engine starting:

- N2
- Oil pressure
- N1
- Fuel flow
- EGT.

Engine Start

CAUTION: PERFORM FOREIGN OBJECT DEBRIS (FOD) WALK IN FRONT OF AND AROUND ENGINE INGESTION AREA PRIOR TO ENGINE START.

This is a summary of the engine start procedure:

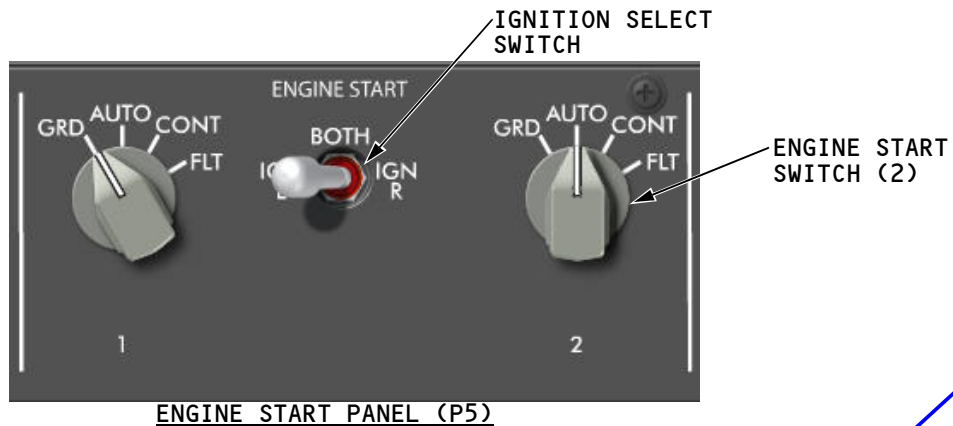
- Follow safety procedures and airplane and engine limitations
- Set the ignition select switch to IGN L or IGN R
- Put the engine start switch to the GRD position
- Look for start valve open indication
- Monitor N2
- Make sure oil pressure increases
- Speak with ground personnel to make sure the N1 fan begins to turn counterclockwise

- Move the engine start lever to the idle position (forward) at a minimum of 25 percent N2 (or max motoring if 25 percent RPM is not possible)
- Make sure fuel flow is in limits
- Make sure EGT increases
- Monitor EGT and N2 increase until starter cutout at 60 percent N2
- Make sure engine start switch goes back to the OFF position at 60 percent N2
- Monitor all engine parameters as engine speed increases to idle.

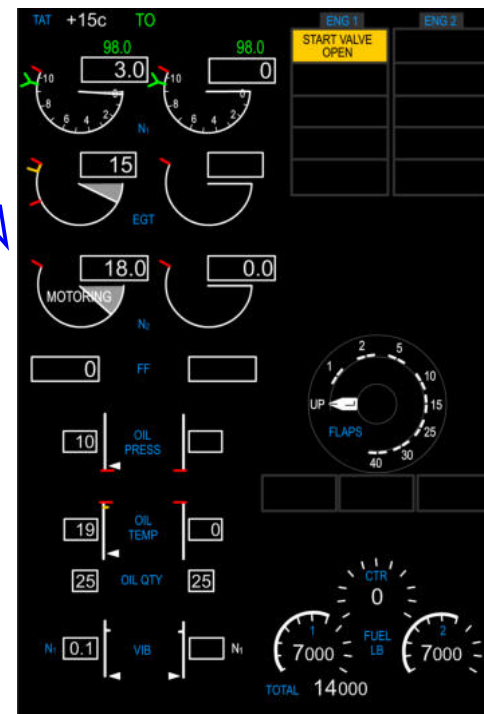
Training Information Point

Usually, only one igniter is necessary for starting. You should use a different igniter selection for each start.

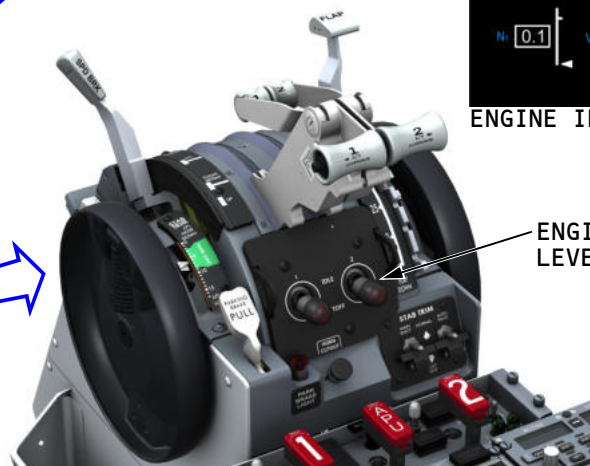
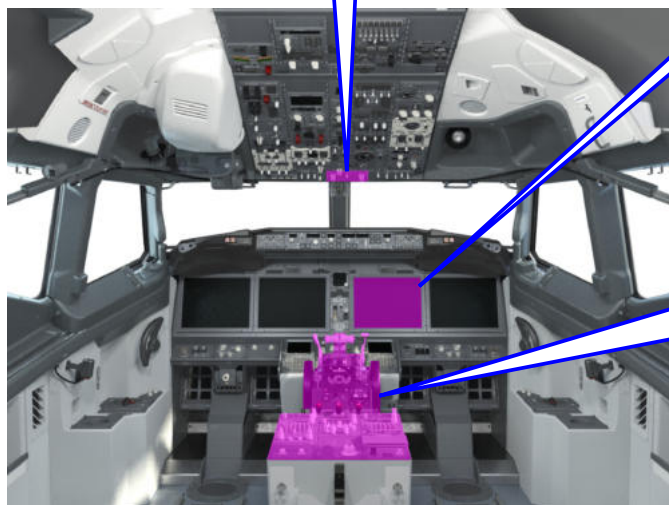
DMC-B737T-A-80-00-00-02A-100A-T_003-00



ENGINE START PANEL (P5)



ENGINE INDICATION DISPLAY



ENGINE STARTING -- OPERATION

80-00-00



ENGINE EXHAUST SYSTEM -- INTRODUCTION

Purpose

The engine exhaust system controls the direction of the engine exhaust gases.

The engine exhaust system has these sub-systems:

- Turbine exhaust
- Thrust reverser (T/R).

SL	- sync lock
stby	- standby
SW	- switch
sync	- synchronizing
sys	- system
T/R	- thrust reverser
TRAS	- thrust reverser actuation system

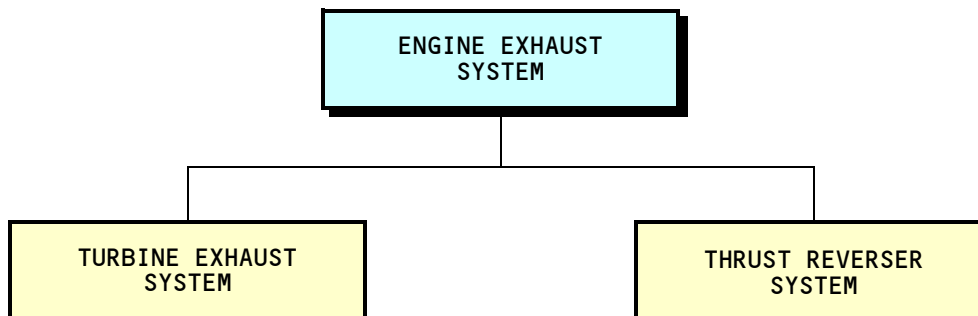
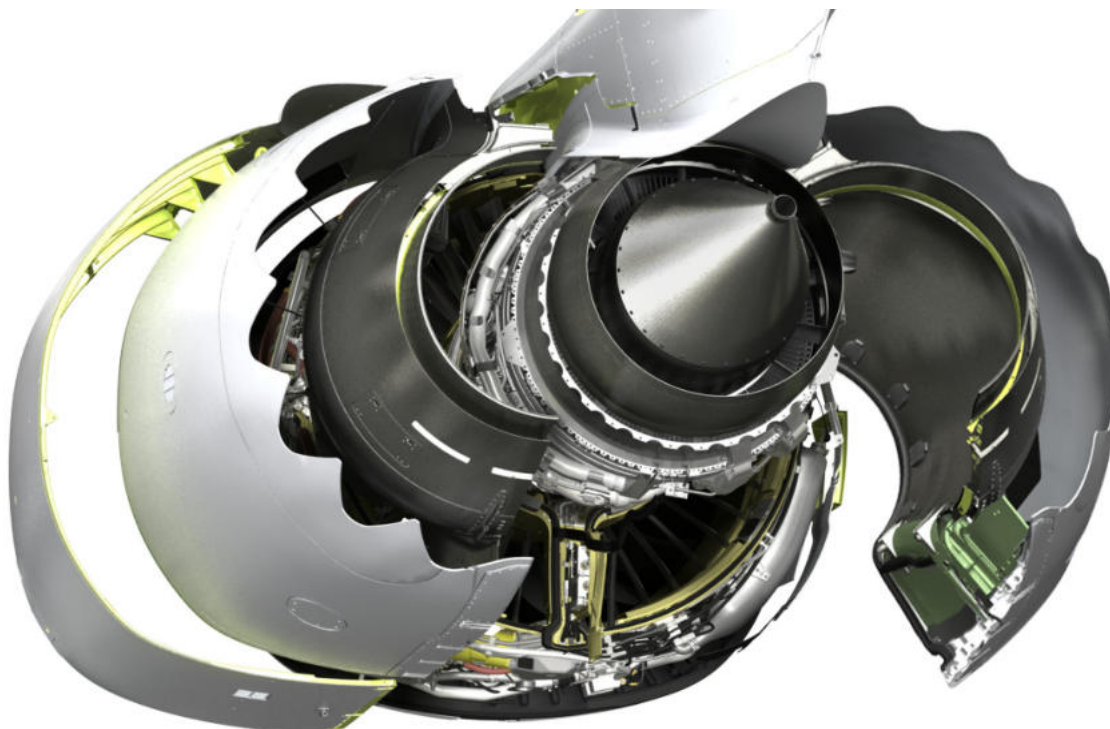
Abbreviations and Acronyms

A/T	- autothrottle
CDU	- control display unit
DCV	- directional control valve
DPC	- display processing computer
EEC	- electronic engine control
ELEC	- electrical
ENG	- engine
FCC	- flight control computer
GND	- ground
HIV	- hydraulic isolation valve
INBD	- inboard
ind	- indication
ISV	- isolation valve
LVDT	- linear variable differential transformer
MDS	- Max display system
MDOS	- manual door opening system
P	- panel
prox	- proximity
REF	- reference
RLY	- relay
RTO	- rejected take off
S	- seconds
seq	- sequence

DMC-B737T-A-78-00-00-02A-018A-T_001-00

02-Dec-2020

78-00-00



ENGINE EXHAUST SYSTEM -- INTRODUCTION

78-00-00

DMC-B737T-A-78-00-00-02A-018A-T_001-00
ICN-B737T-A-780000-A-81205-00003-A-001-01

02-Dec-2020

ENGINE EXHAUST SYSTEM -- GENERAL DESCRIPTION

General

The engine exhaust system controls the direction of the turbine exhaust gases and the fan air exhaust gases.

Turbine Exhaust System

The turbine exhaust system supplies an exit for the engine exhaust gases. This exit increases the velocity of the exhaust gases. This increases engine thrust.

The primary components of the turbine exhaust system are the exhaust nozzle and the exhaust plug.

Thrust Reverser System

The thrust reverser (T/R) system changes the direction of the fan air exhaust to give reverse thrust. The flight crew uses reverse thrust to decrease the airplane's ground speed after landing or during a rejected takeoff (RTO). The turbine exhaust airflow direction does not change during reverse thrust. The T/R system has an electrohydraulic control system and an indicating system.

The T/R system has two thrust reversers. T/R 1 is the thrust reverser for engine 1 (left). T/R 2 is the thrust reverser for engine 2 (right).

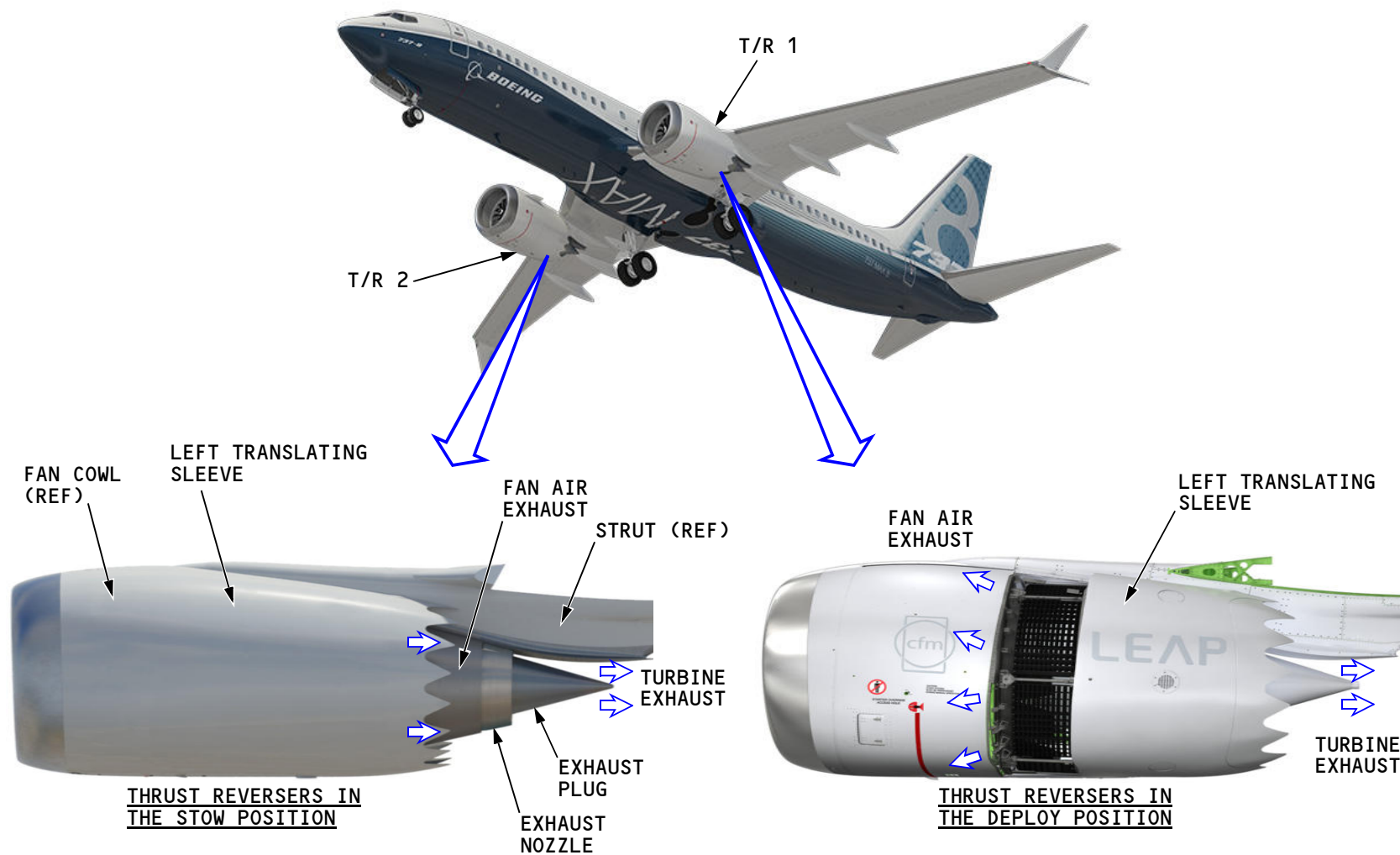
Each T/R has a left and right half. Each half has a translating sleeve which moves aft (deploy position) for reverse thrust. The two sleeves work independently from each other. Fan air exhaust goes out radially and forward when the translating sleeves are in the deploy position.

Hinges attach each T/R half to the strut. To prevent injury and damage the thrust reverser must be deactivated before opening a T/R half. Latches are at the bottom of the two halves. The latches keep the two halves together.

DMC-B737T-A-78-00-00-02A-040A-T_002-00

02-Dec-2020

78-00-00



DMC-B737T-A-78-00-00-02A-040A-T_002-00
ICN-B737T-A-780000-A-81205-00004-A-002-01

ENGINE EXHAUST SYSTEM -- GENERAL DESCRIPTION

78-00-00

02-Dec-2020



TURBINE EXHAUST SYSTEM -- INTRODUCTION

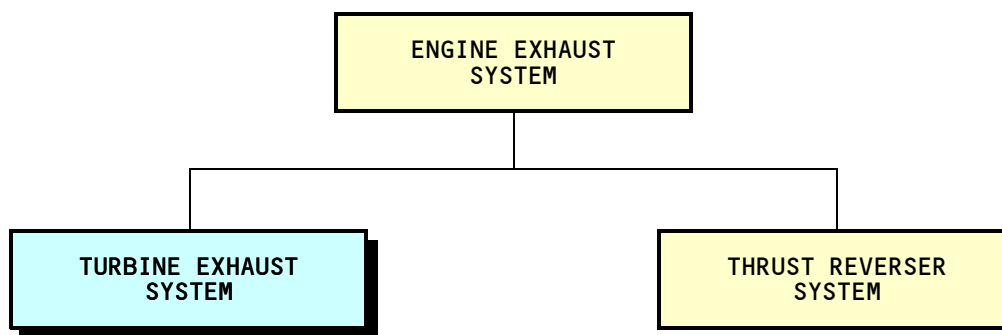
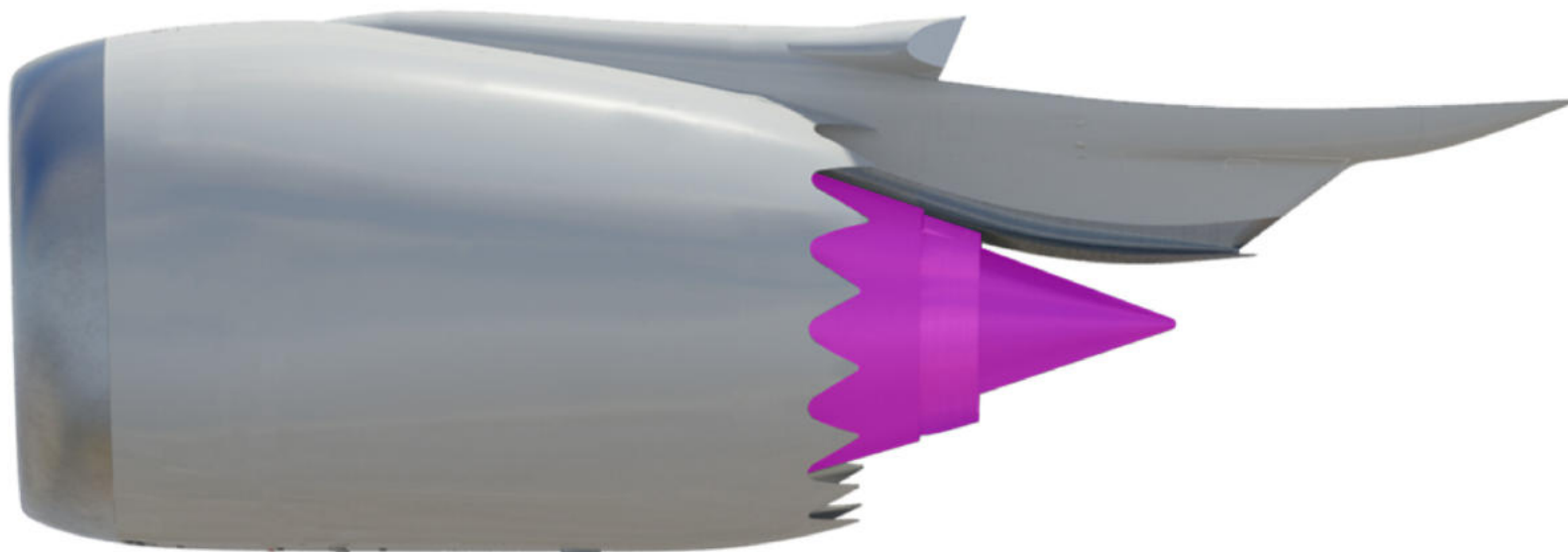
Purpose

The turbine exhaust system supplies an exit for the turbine exhaust gases. The system increases the turbine exhaust gas velocity to increase engine thrust.

DMC-B737T-A-78-11-00-02A-018A-T_001-00

02-Dec-2020

78-11-00



TURBINE EXHAUST SYSTEM -- INTRODUCTION

78-11-00

DMC-B737T-A-78-11-00-02A-018A-T_001-00
ICN-B737T-A-781100-A-81205-00005-A-001-01

02-Dec-2020

TURBINE EXHAUST SYSTEM -- GENERAL DESCRIPTION

General

The turbine exhaust system uses a nozzle and a plug to control the direction of the turbine exhaust gases.

Components

These are the turbine exhaust system components:

- Plug
- Nozzle.

Physical Description

The exhaust nozzle controls the outer edge of the turbine exhaust flow.
The nozzle attaches to the engine turbine case.

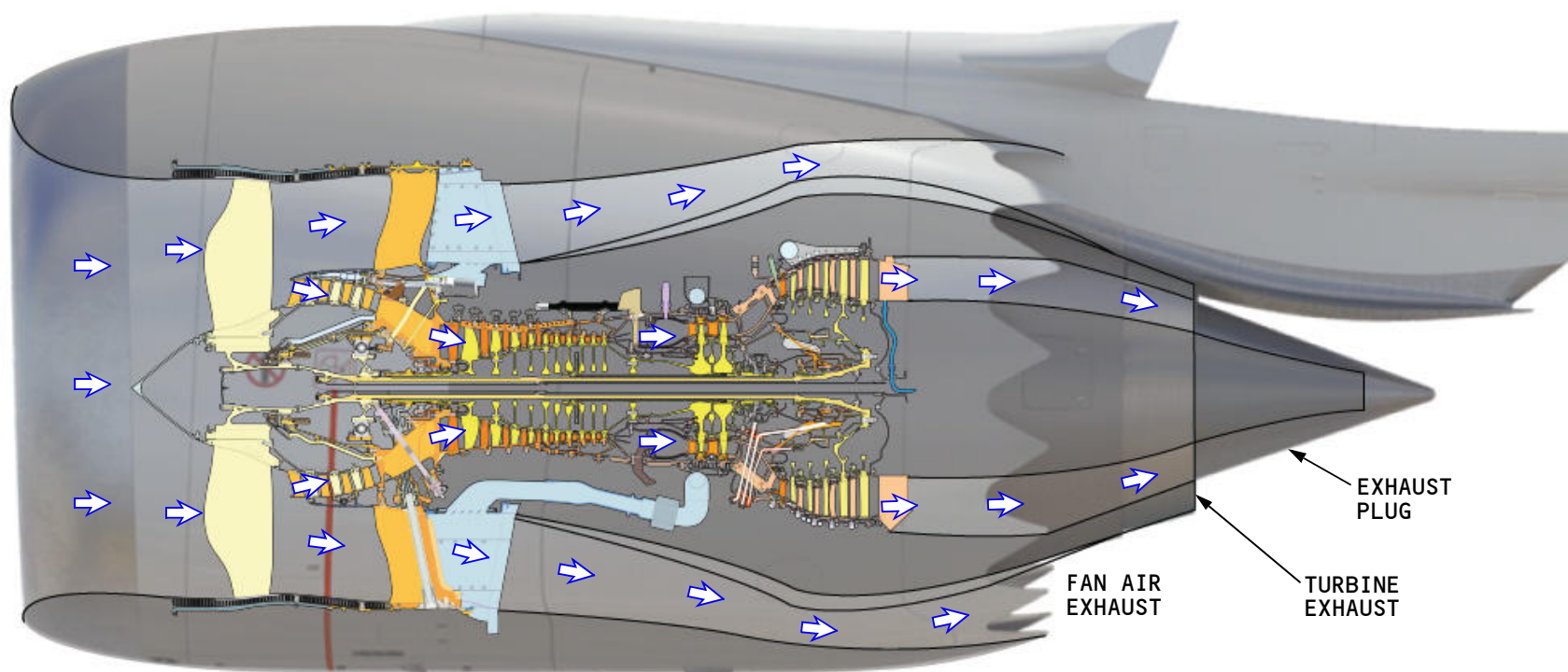
The exhaust plug controls the inner edge of the turbine exhaust flow.
The plug attaches to the engine turbine case.

The plug and the nozzle are made of nickel alloy.

DMC-B737T-A-78-11-00-02A-040A-T_001-00

02-Dec-2020

78-11-00



ENGINE AIRFLOW

TURBINE EXHAUST SYSTEM -- GENERAL DESCRIPTION

78-11-00

DMC-B737T-A-78-11-00-02A-040A-T_001-00
ICN-B737T-A-781100-A-81205-00006-A-001-01

02-Dec-2020

TURBINE EXHAUST SYSTEM -- EXHAUST NOZZLE

Purpose

The exhaust nozzle assembly controls the outer edge of the turbine exhaust flow. The aft part also controls the inner edge of the fan air exhaust flow.

General

The exhaust nozzle assembly has these parts:

- Attach flange
- Honeycomb panel
- Fairing
- Aft skin
- Labyrinth seal.

Bolts attach the nozzle to the turbine exhaust case.

The exhaust nozzle uses labyrinth seals to contain fire. Bolts attach the labyrinth seals to the exhaust nozzle assembly.

Location

The exhaust nozzle assembly is on the aft end of the engine. Open the fan cowl and thrust reverser halves to get access.

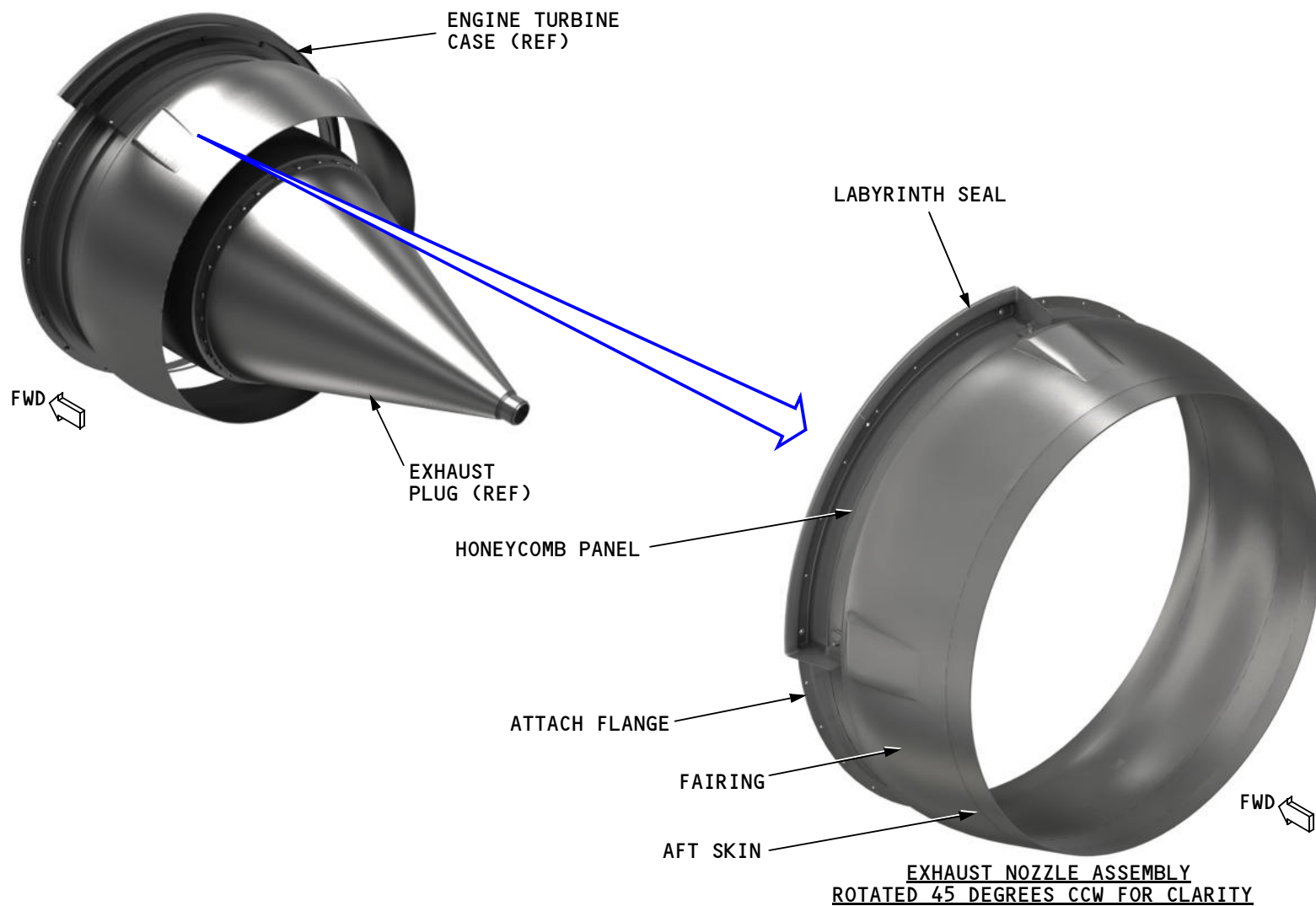
WARNING: DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSERS. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

An alignment pin on the inner sleeve, near the 12:00 position, aligns the nozzle assembly during installation.

The leading edge flaps can extend and damage a T/R half when the half is in the open (maintenance) position. Follow the AMM procedures to prevent leading edge flaps operation.

Deploying the T/R when the T/R halves are open will damage the leading edge flaps.

DMC-B737T-A-78-11-00-03A-034B-T_004-00



DMC-B737T-A-78-11-00-03A-034B-T_004-00
ICN-B737T-A-781100-A-81205-00007-A-003-01

TURBINE EXHAUST SYSTEM -- EXHAUST NOZZLE

78-11-00

02-Dec-2020

TURBINE EXHAUST SYSTEM -- EXHAUST PLUG

Purpose

The exhaust plug assembly controls the inner edge of the turbine exhaust flow.

The engine vent system uses a hole at the aft end of the exhaust plug.

General

The exhaust plug assembly has these major components:

- Forward plug
- Aft plug.

Torx head fasteners attach the aft plug to the forward plug. Bolts attach the forward plug to the engine turbine exhaust case.

The center vent tube goes aft through the exhaust plug assembly. The center vent tube connects to a hole in the center of the aft exhaust plug. The center vent tube releases air aft from the vent circuit of the engine oil system.

Location

The exhaust plug is on the aft end of the engine turbine case. Do these tasks to get access to the exhaust plug bolts:

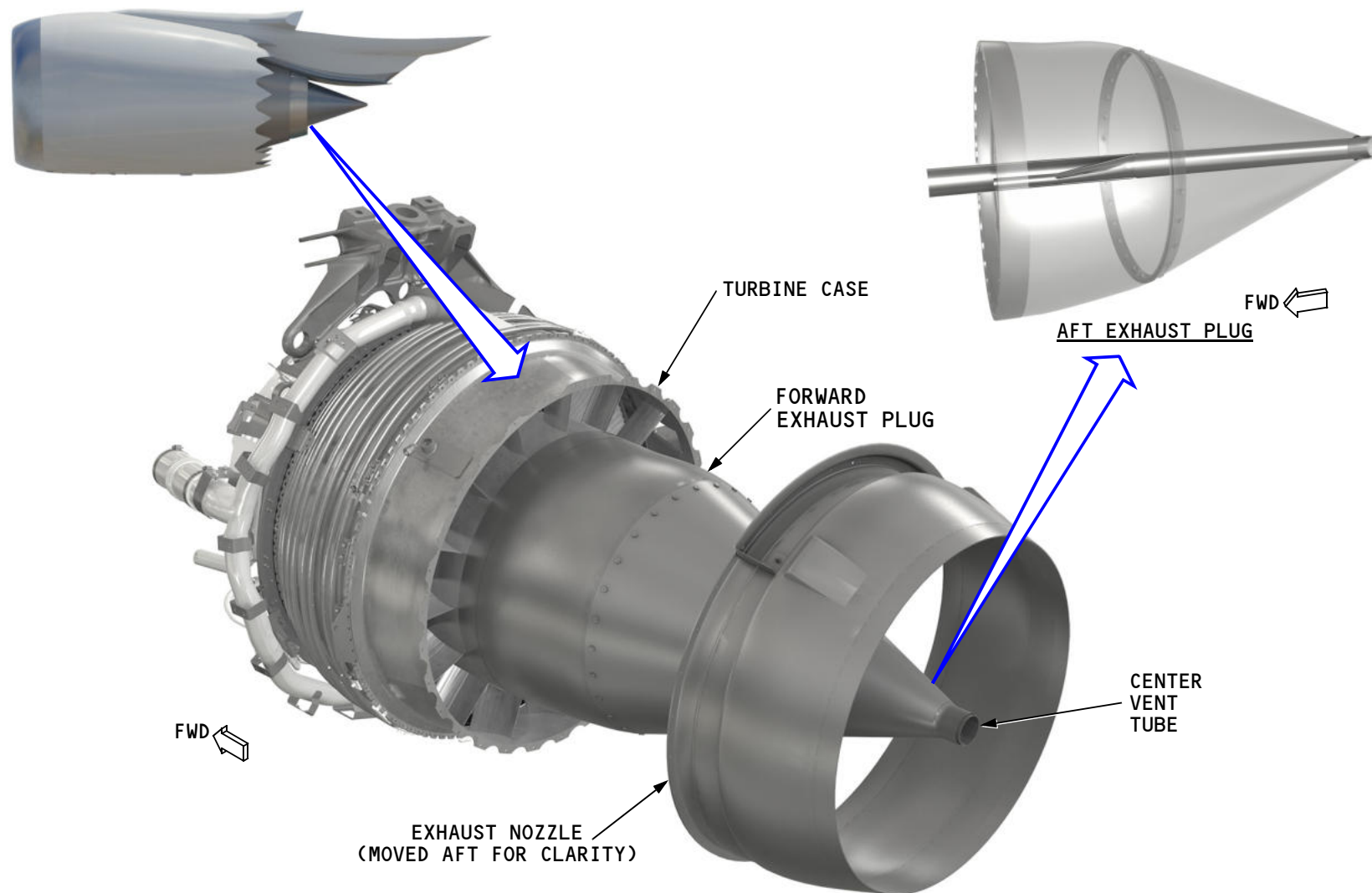
- Open the fan cowls to the full open position
- Open the thrust reverser halves
- Remove the exhaust nozzle.

Training Information Point

Remove the exhaust nozzle and the aft plug before removing the forward plug.

An alignment pin on the forward plug, near the 12:00 position, aligns the forward plug during installation.

The aft plug has a small notch near the 12:00 position. This notch aligns the aft plug during installation.



TURBINE EXHAUST SYSTEM -- EXHAUST PLUG

78-11-00

DMC-B737T-A-78-11-00-04A-034B-T_001-00
ICN-B737T-A-781100-A-81205-00008-A-001-01

02-Dec-2020

THRUST REVERSER SYSTEM -- INTRODUCTION

General

The thrust reverser (T/R) system has these subsystems:

- Thrust reverser
- Control
- Indicating.

Thrust Reverser System

The T/R system controls the direction of engine fan air exhaust for forward and reverse thrust.

Reverse thrust helps decrease the speed of the airplane after landing or during a rejected take off (RTO).

T/R Control System

The T/R control system controls electrical and hydraulic power to the T/R system.

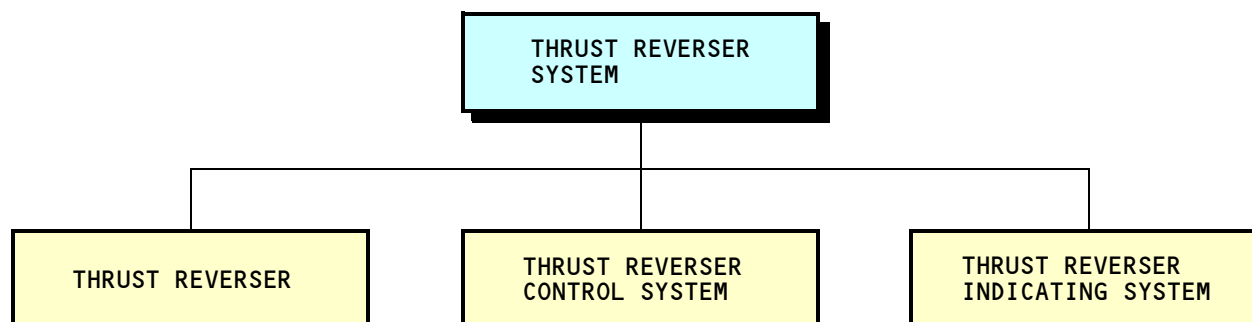
T/R Indicating System

The T/R indicating system supplies T/R system and T/R control system indication in the flight compartment.

DMC-B737T-A-78-30-00-02A-018A-T_002-00

02-Dec-2020

78-30-00



THRUST REVERSER SYSTEM -- INTRODUCTION

78-30-00

DMC-B737T-A-78-30-00-02A-018A-T_002-00
ICN-B737T-A-783000-A-81205-00002-A-003-01

02-Dec-2020



THRUST REVERSER -- INTRODUCTION

Purpose

The thrust reverser (T/R) system controls the direction of engine fan air, or secondary airflow exhaust for forward and reverse thrust.

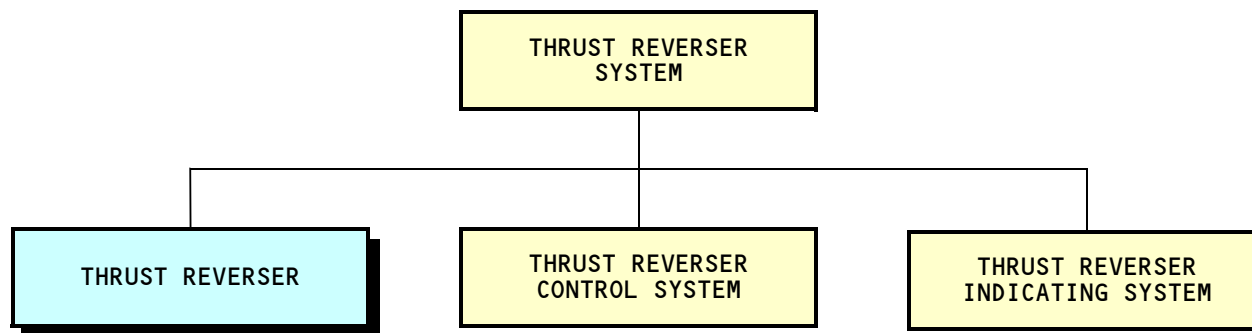
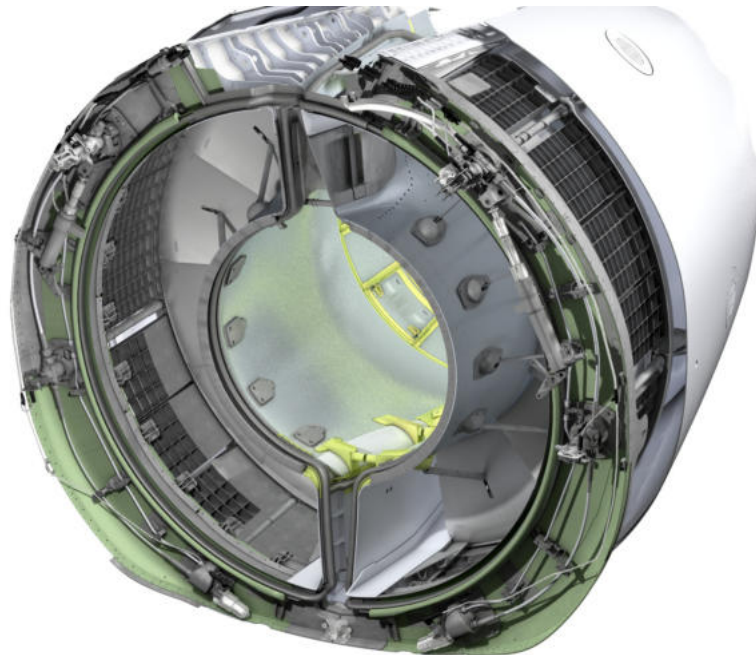
Reverse thrust helps decrease the speed of the airplane after landing or during a rejected take off (RTO).

Each thrust engine thrust reverser system has two translating sleeves. The two sleeves are electrically controlled by a common source. These sleeves are not connected but they must move together.

DMC-B737T-A-78-31-00-02A-018A-T_001-00

02-Dec-2020

78-31-00



THRUST REVERSER -- INTRODUCTION

78-31-00

DMC-B737T-A-78-31-00-02A-018A-T_001-00
ICN-B737T-A-783100-A-81205-00012-A-001-01

02-Dec-2020

THRUST REVERSER -- GENERAL DESCRIPTION

General

The T/R is a translating sleeve and cascade design. Each T/R has a left and right fan duct half with a translating outer wall (sleeve). The translating sleeves are V-shaped with the open end forward. The two sleeves on each T/R operate at the same time. However, the sleeves are independent of each other.

Four hinges attach each T/R half to the strut. Nine tension latches at the bottom of the T/R keep the two halves together at the bottom.

Each T/R half has these components:

- Translating sleeve
- Three Hydraulic actuators and 2 sync shafts
- Five Blocker doors
- 5 Blocker door drag links
- T/R opening actuator
- Torque box
- Three access doors
- Sliders and tracks.

Both the outboard and inboard thrust reverser sleeves have 8 cascade segments each.

The sleeves are in the stow position when they are in the full forward position. The sleeves are in the deploy position when they are in the full aft position. Sliders let the sleeves move forward and aft in tracks.

Each blocker door drag link attaches to a blocker door on the inner fan duct.

Functional Description

This is the configuration of the thrust reverser with the translating sleeves stowed:

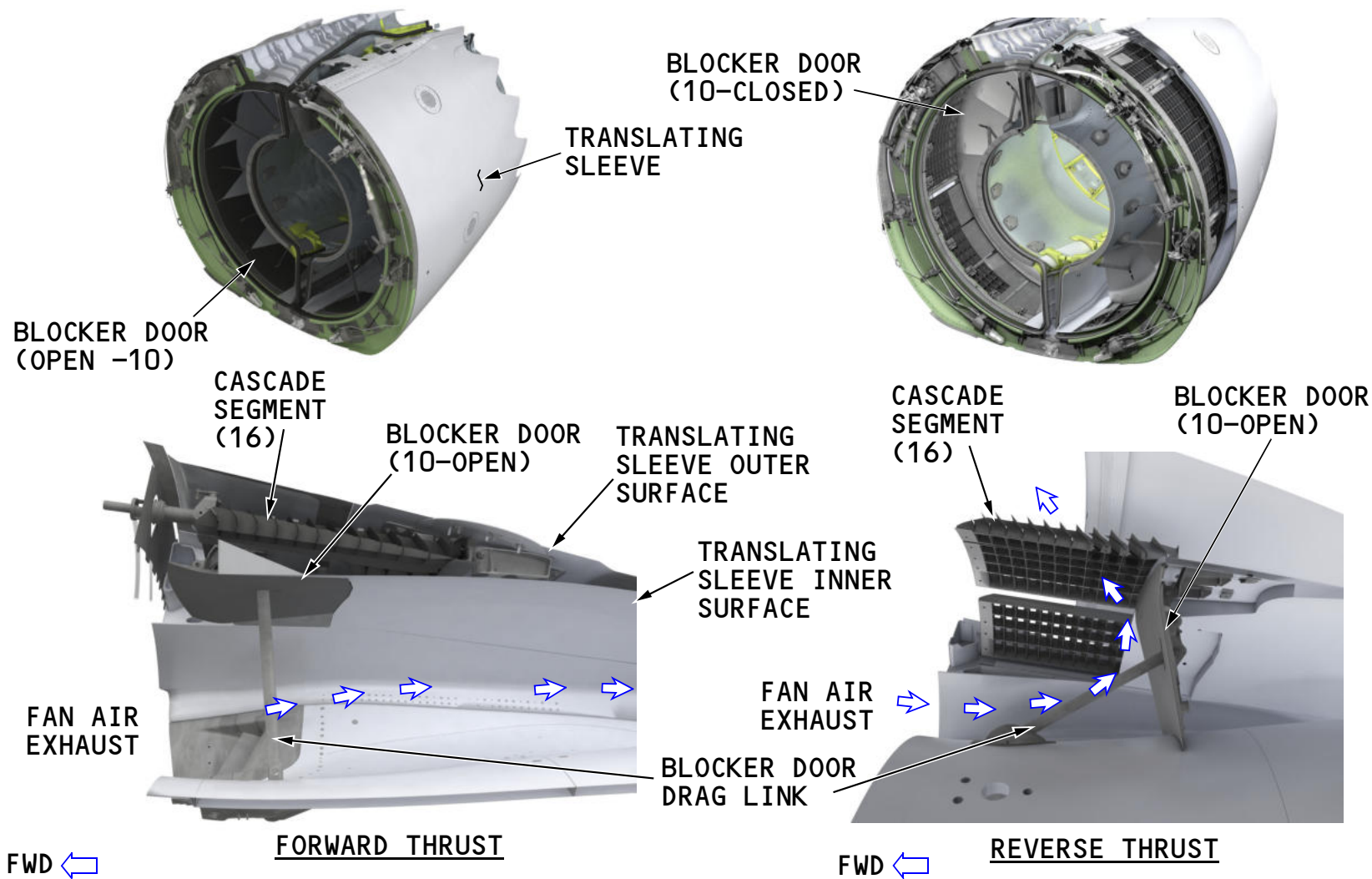
- The cascade segments are between the inner and outer surfaces of the translating sleeves
- The blocker doors are in the open position.

Fan air exhaust can go through the fan duct to give forward thrust to the airplane.

The T/R control system uses the hydraulic actuators to move the translating sleeves. The sleeves move aft of the cascades during a deploy operation. This is the configuration of the thrust reverser with the translating sleeves deployed:

- The translating sleeves do not cover the cascade segments
- The blocker doors are in the closed position.

Each drag link makes a blocker door move into the fan air exhaust flow as the sleeve moves aft. With the sleeves deployed, the blocker doors are closed. Fan air exhaust can not go through the fan duct because the blocker doors stop the air. This air now goes through the cascade segments. The cascade segments change the direction of the fan air. This gives the airplane reverse thrust.



THRUST REVERSER -- GENERAL DESCRIPTION

78-31-00

DMC-B737T-A-78-31-00-04A-040A-T_002-00
ICN-B737T-A-783100-A-81205-00013-A-002-01

02-Dec-2020

THRUST REVERSER -- COMPONENT LOCATION

General

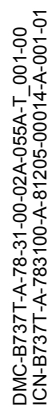
Each thrust reverser has two halves. These are the thrust reverser components on each half:

- Translating sleeve
- Latch beam
- Hinge beam
- Sync shafts
- Cascade segments
- Aft cascade support ring
- Inner wall
- TRAS actuators
- Blocker doors
- Blocker door drag links
- Opening actuator
- Tension latches
- Fire seals
- Torque box
- Access doors
- Krueger flap deflector.

DMC-B737T-A-78-31-00-02A-055A-T_001-00

02-Dec-2020

78-31-00



78-31-00

THRUST REVERSER -- HYDRAULIC ACTUATORS AND SYNC SHAFTS - GENERAL DESCRIPTION

Purpose

The hydraulic actuators move the translating sleeves during T/R deploy and stow operations.

The sync shafts make the hydraulic actuators extend and retract together. The shafts are connected to a sync lock. Each sync lock has a manual drive fitting. The sync locks give these two functions to the T/Rs:

- A secondary lock for the T/R translating sleeves
- Manual deploy and stow the T/R translating sleeves.

General

Each T/R half has three hydraulic actuators.

- One top actuator that can lock closed and has a linear variable differential transformer (LVDT)
- One middle actuator that can lock closed but does not have an LVDT
- One bottom actuator that can not lock closed and has a sync lock, but no LVDT.

The actuators extend during a deploy operation and retract during a stow operation. The actuators with lock mechanisms must unlock for their thrust reverser half to operate.

There are two sync shafts on each T/R half. The top sync shaft connects between the top actuator and middle actuator. The bottom sync shaft connects between the middle actuator and the bottom actuator. The lower side of the bottom actuator has a sync lock with a manual drive fitting. The sync lock does these functions:

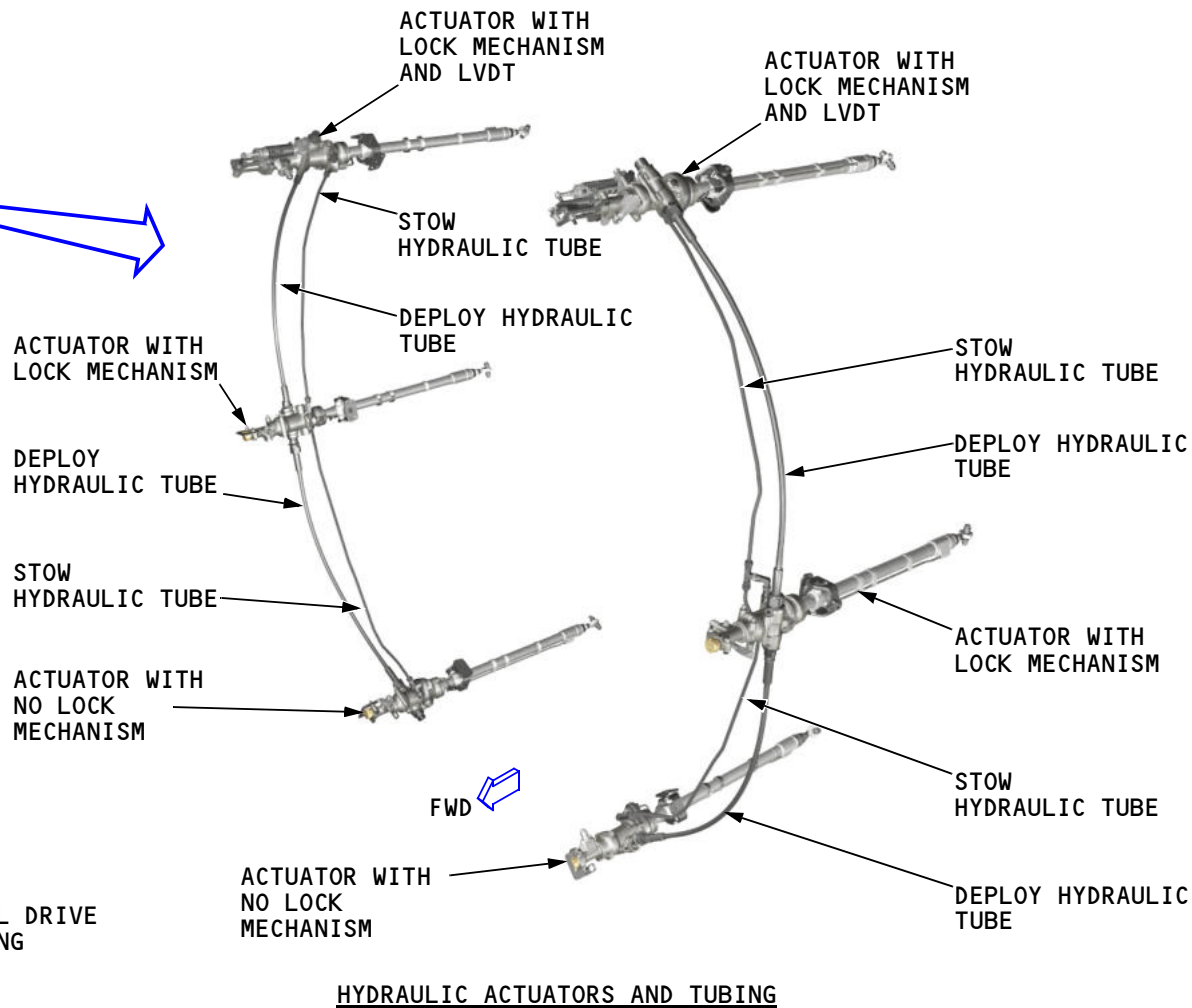
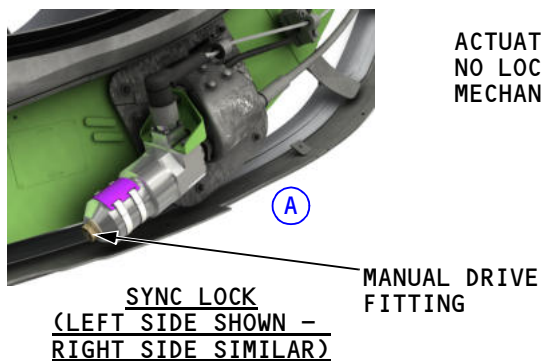
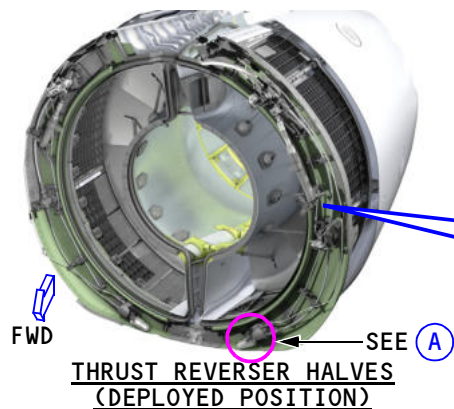
- Does not let the sync shafts turn when locked
- Lets you manually deploy the thrust reverser half.

Location

Open the fan cowl and move the translating sleeve aft to get access to the hydraulic actuators.

The upper sync shaft is inside the deploy hydraulic tube, between the upper and center actuators. The lower sync shaft is inside the deploy hydraulic tube, between the center and lower actuators. The deploy tubes are larger than the stow tubes. You must open the fan cowl to get access to the tubing, actuators, and the sync locks with manual drive fittings on each side.

DMC-B737T-A-78-31-00-05A-040A-T_002-00



THRUST REVERSER -- HYDRAULIC ACTUATORS AND SYNC SHAFTS - GENERAL DESCRIPTION

78-31-00



THRUST REVERSER -- HYDRAULIC ACTUATORS - GENERAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK

THRUST REVERSER -- HYDRAULIC ACTUATORS - GENERAL DESCRIPTION

General

Each engine has two thrust reverser assemblies. There is one thrust reverser assembly on each side of the engine. Each thrust reverser assembly has these components:

- T/R hydraulic actuators (3)
- Manual unlock levers (2)
- LVDT (1)
- Sync shafts (2)
- Sync lock (1)

Thrust Reverser (T/R) Hydraulic Actuators

There are two primary types of thrust reverser (T/R) hydraulic actuators:

- Actuators with lock mechanisms
- Actuators without lock mechanisms.

Each T/R assembly has an upper, center, and lower hydraulic actuator. The forward end of all T/R hydraulic actuators attaches to the torque box. The aft end of the all T/R hydraulic actuators attaches to the T/R translating sleeve. This table gives the actuator configurations:

T/R Hydraulic Actuator	Lock Mechanism installed?	Manual Unlock Lever installed?	Proximity Sensor installed?	LVDT installed?
Upper	Yes	Yes	Yes	Yes
Middle	Yes	Yes	Yes	No
Lower	No	No	No	No

Manual Unlock Levers

If a T/R hydraulic actuator has a lock mechanism, it also has a proximity sensor and a manual unlock lever. You use the manual unlock lever when you deploy the thrust reverser manually for maintenance.

Sleeve Lock Proximity Sensor

The manual unlock lever is also a target for the T/R sleeve lock proximity sensor. The proximity sensor gives the EECs a signal when the actuator is locked in the stow position.

Linear Variable Differential Transformer (LVDT)

The LVDT gives data about thrust reverser movement and position to the EECs. The EECs send T/R position data to the DPCs.

Location

Open the fan cowl and move the translating sleeve aft to get access to the hydraulic actuators.

T/R Actuator Connections

All of the hydraulic actuators have these parts and connections:

- Deploy (extend) pressure port
- Stow (retract) pressure port
- Gimbal assembly
- Sync shaft and tubing connections.

T/R Deploy and Stow Operation

The T/R control valve module controls the hydraulic power to the actuators. Hydraulic fluid goes to the extend and retract pressure ports

THRUST REVERSER -- HYDRAULIC ACTUATORS - GENERAL DESCRIPTION

of each actuator during a deploy operation. Hydraulic power unlocks the locking actuators so they can extend. All actuators extend and move the translating sleeve aft.

The T/R control valve module sends hydraulic fluid to the retract pressure ports of each actuator during a stow operation. With hydraulic pressure at these ports, the actuators retract and move the translating sleeve forward. The hydraulic fluid at the extend port returns through the T/R control valve module to the hydraulic reservoir. The locking actuators lock when they are in the fully retracted position.

Gimbal Assembly

A gimbal assembly attaches the head end of each hydraulic actuator to the aft side of the torque box. The translating sleeves must be deployed to access the gimbal assembly.

Sync Shaft and Tubing Connections

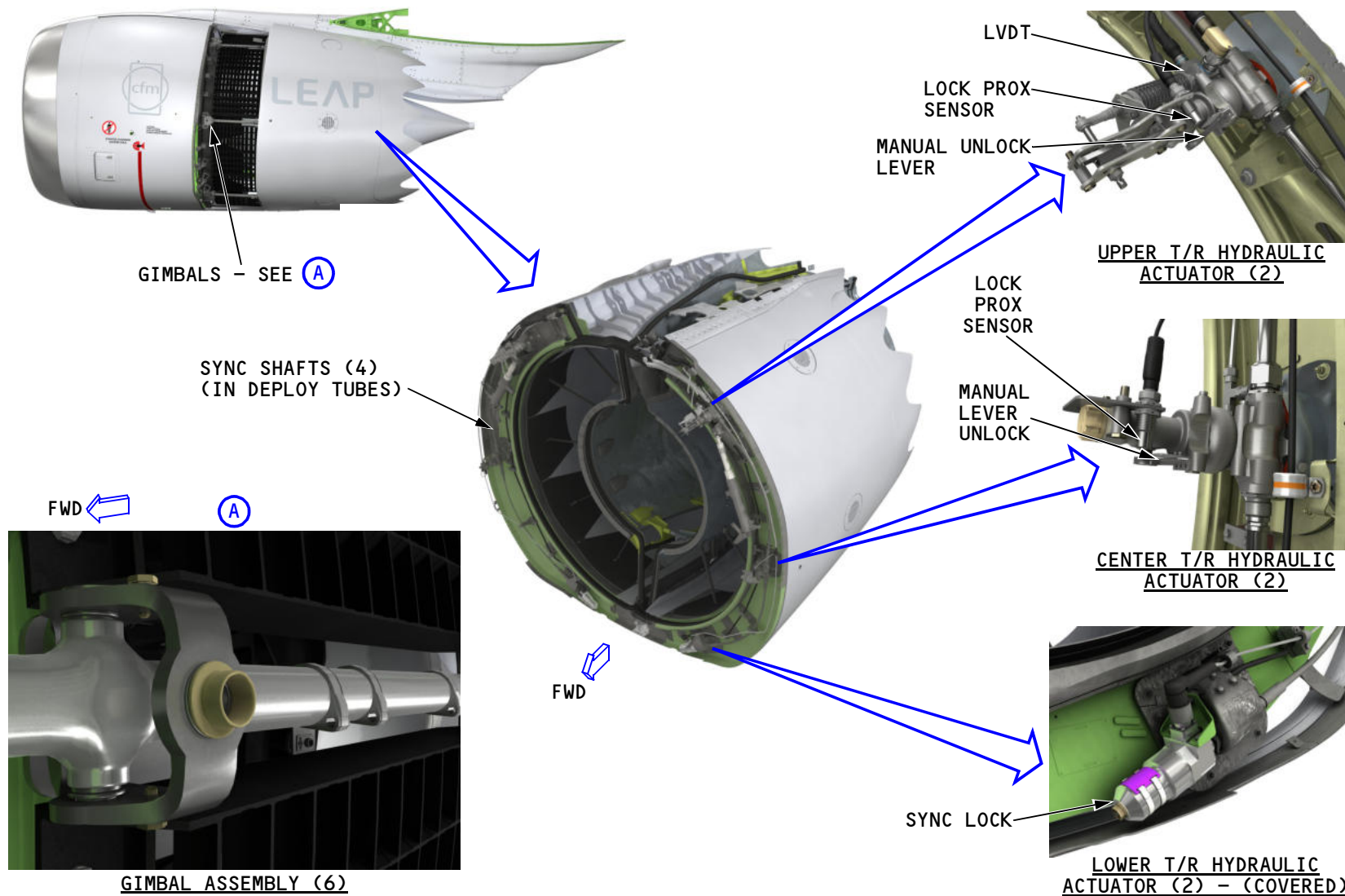
The sync shafts connect the drive mechanisms of the actuators together. The shafts are inside of the deploy hydraulic tubing which connects to the extend pressure port.

CAUTION: DO NOT PERMIT THE UPPER ACTUATOR ROD END TO TURN OR BE TURNED WHEN YOU HANDLE THE ACTUATOR. DAMAGE TO THE INTERNAL FEEDBACK MECHANISM (LVDT) CAN OCCUR.

DMC-B737T-A-78-31-00-06A-040A-T_005-00

02-Dec-2020

78-31-00



DMC-B737T-A-78-31-00-06A-040A-T_005-00
ICN-B737T-A-783100-A-81205-00020-A-007-01

THRUST REVERSER -- HYDRAULIC ACTUATORS - GENERAL DESCRIPTION

78-31-00

02-Dec-2020

THRUST REVERSER -- CASCADE SEGMENTS

Purpose

The cascades control the direction of the fan air exhaust during a T/R deploy operation. This gives reverse thrust. The cascades also give structural strength to the T/R.

General

The thrust reverser of each LEAP-1B engine has 16 cascade segments. There are 8 cascade segments on each thrust reverser half. The left and right engine cascade segment configurations are opposite. Numbers identify the cascade locations. T/R 1 cascade numbers increase clockwise when viewed from behind. T/R 2 cascade numbers increase counterclockwise when viewed from behind.

Location

Bolts attach the cascades to the torque box on the forward edge and to the cascade support ring on the aft edge.

Extend the T/R to get access to the cascades.

Physical Description

The cascades are graphite epoxy.

There are different types of cascade segments. Each type causes the fan exhaust airflow to go out in a different direction. The cascade segments are interchangeable to adjust the thrust reversers for installation on a different engine if necessary. The different types of cascade vanes can be:

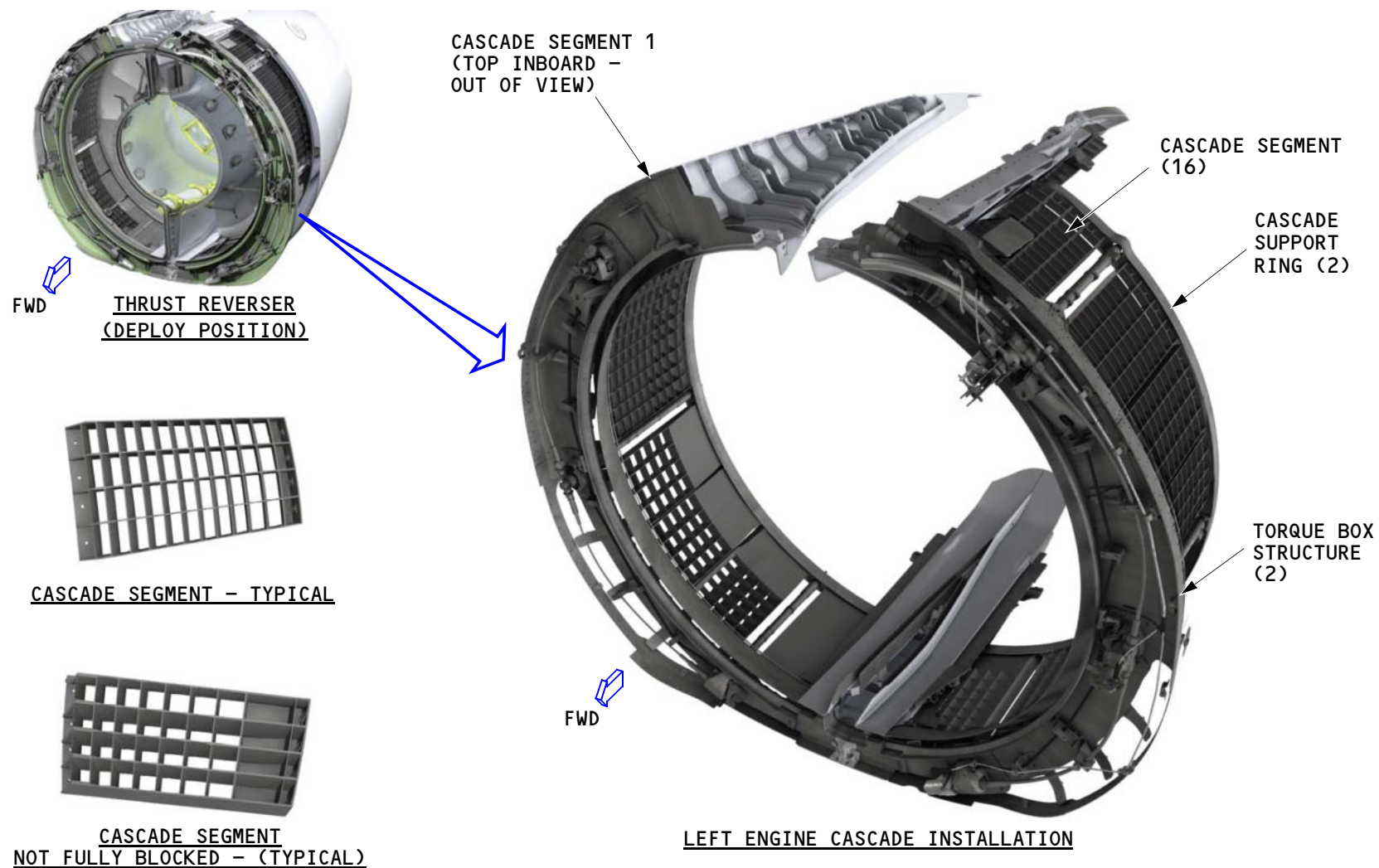
- Fully open
- Fully blocked
- Not fully blocked.

The technician must be sure to obey all effectivity specifications for locations of each cascade segment. Fully blocked or not fully blocked cascade segments do not let air go through the blocked part of their vanes. This does the following during thrust reverser operation:

- Prevents damage to the airplane fuselage
- Prevents change of airflow on the airplane's flight control surfaces
- Decreases cabin noise
- Prevents FOD damage to engine.

Training Information Point

The T/R sleeves must be in the deploy position to examine, remove, or install the cascades.



THRUST REVERSER -- CASCADE SEGMENTS

78-31-00

THRUST REVERSER -- ACCESS PANELS

General

Each T/R cowl (half) has three access panels.

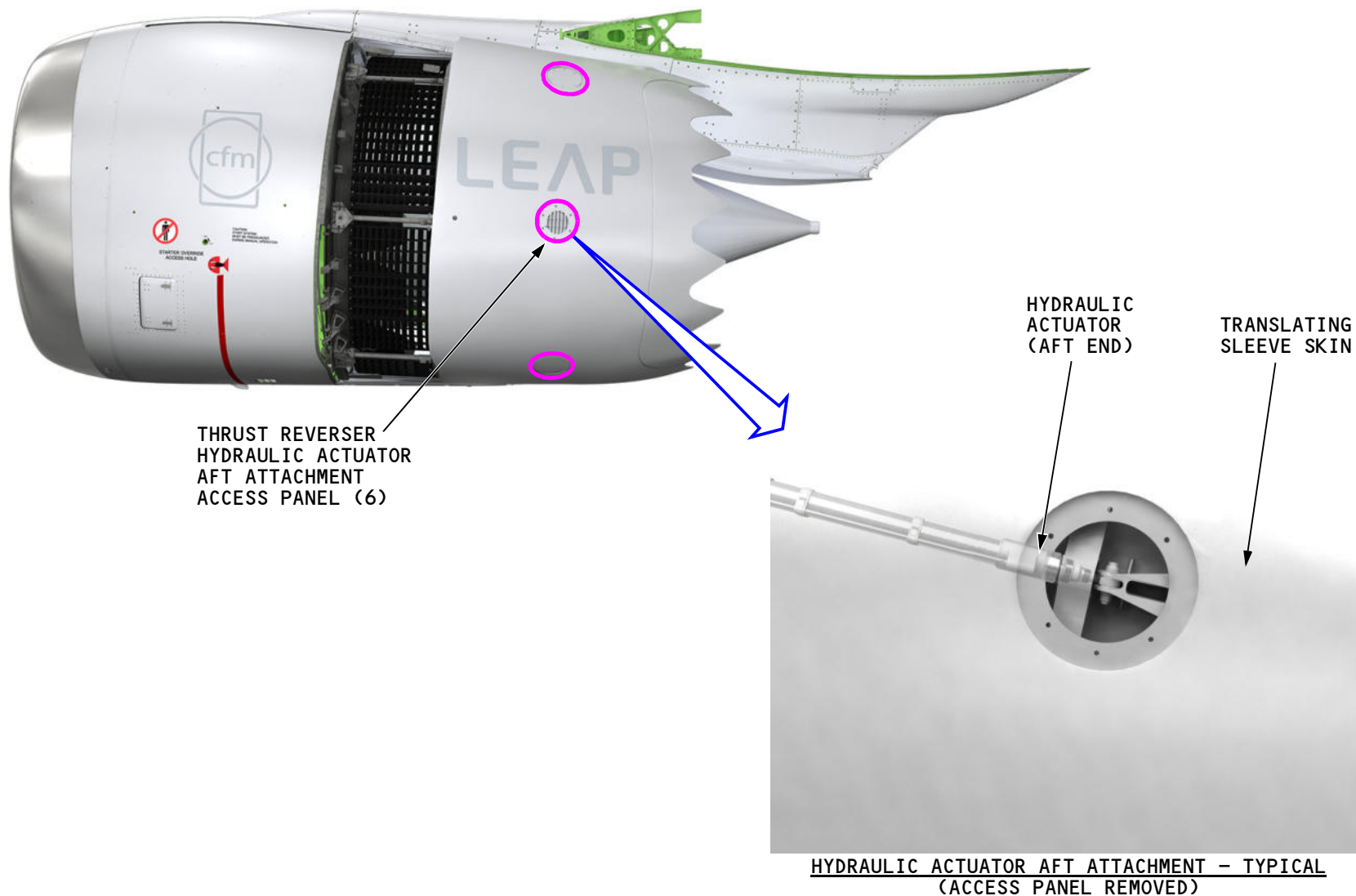
Remove the access panels to get access to the T/R hydraulic actuator's aft attach point.

The center access panel also supplies an exit for fan air exhaust if it goes by the bullnose seal when the T/R is in the stow position.

DMC-B737T-A-78-31-00-08A-034B-T_002-00

02-Dec-2020

78-31-00



DMC-B737T-A-78-31-00-08A-034B-T_002-00
ICN-B737T-A-783100-A-81205-00019-A-002-01

02-Dec-2020

THRUST REVERSER -- ACCESS PANELS

78-31-00

THRUST REVERSER -- DEACTIVATION FOR FLIGHT DISPATCH - TRAINING INFORMATION POINTS

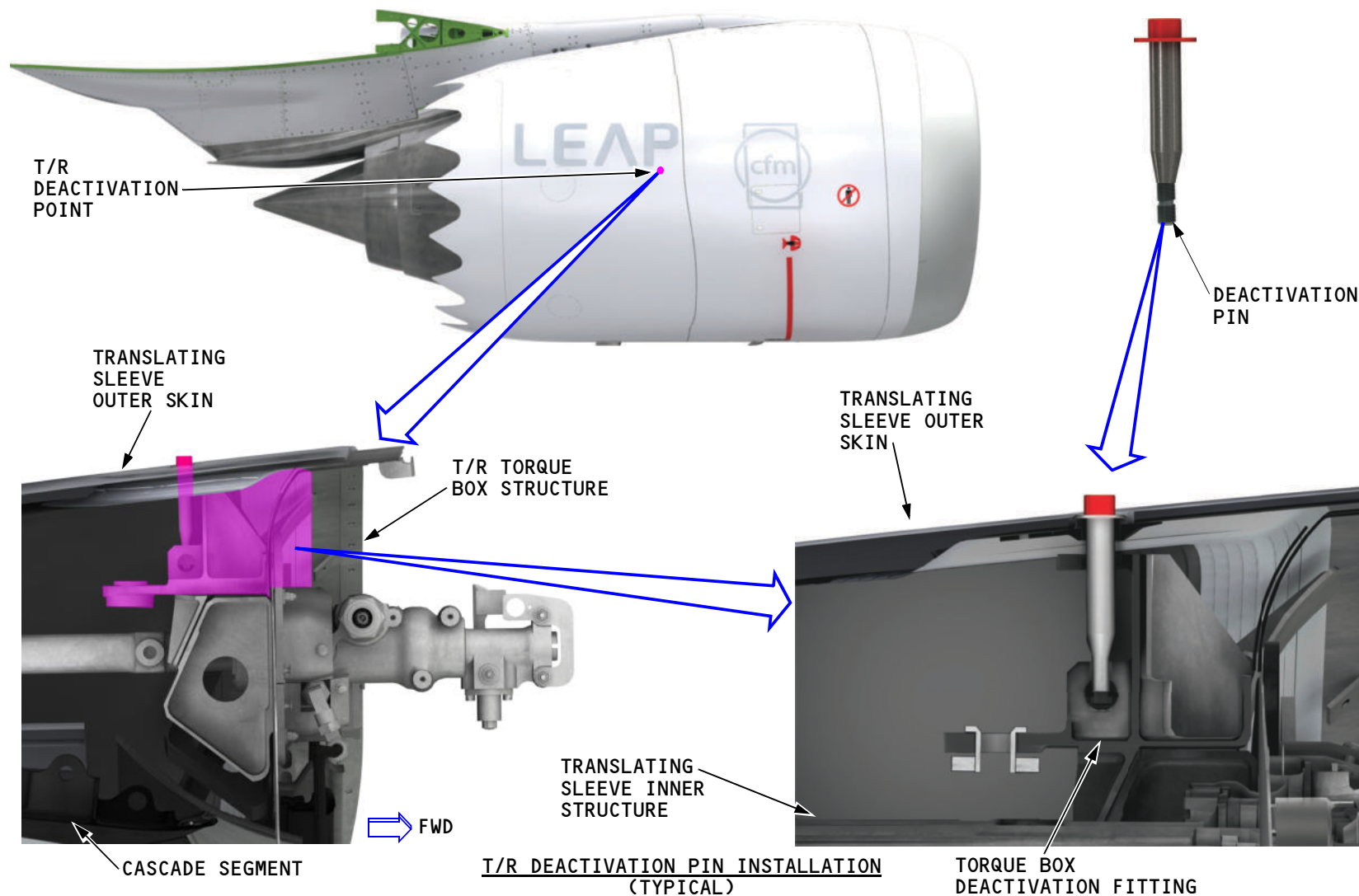
General

Each T/R translating sleeve has one deactivation point. Install one deactivation pin at this point to deactivate the T/R for airplane dispatch. The pins are usually in the fly away kit and are painted red. When you install a deactivation pin, you put Never-Seez NSBT compound, D00006 on the threads. This helps make sure that you can easily remove the pin when necessary.

Each translating sleeve has one hole at the deactivation point. You install one deactivation pin through the hole to deactivate the thrust reverser on that side of the engine.

The pins mechanically connect the translating sleeve structure to a deactivation fitting on the thrust reverser torque box. This prevents the movement of the T/R sleeve.

Follow the thrust reverser deactivation for flight dispatch procedure found in the airplane maintenance manual (AMM).



DMC-B737T-A-78-31-00-02A-300A-T_003-00
ICN-B737T-A-783100-A-81205-00022-A-003-01

THRUST REVERSER -- DEACTIVATION FOR FLIGHT DISPATCH - TRAINING INFORMATION POINTS

02-Dec-2020

78-31-00



THRUST REVERSER -- LOCKING HYDRAULIC ACTUATORS -- FUNCTIONAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK

THRUST REVERSER -- LOCKING HYDRAULIC ACTUATORS -- FUNCTIONAL DESCRIPTION

Physical Description

The actuators with lock mechanisms have these parts and connections:

- Deploy and sync shaft tubing connections
- Stow tubing connections
- Gimbal assembly
- Rod end assembly
- Manual unlock lever
- Position feedback rod (upper actuator only)
- 2-channel LVDT (upper actuator only)
- Acme nut and screw (internal)
- Piston (internal)
- Piston stop (internal)
- Worm wheel and shaft (internal)
- Lock mechanism (internal).

Hydraulic pressure at the deploy port unlocks the actuator internal lock mechanism during usual T/R operation. You can use the manual unlock lever to unlock the actuator to deploy a translating sleeve manually.

T/R Actuators with Internal Lock Mechanisms

The T/R control valve module supplies hydraulic power to the actuators. The actuator contains a lock mechanism at the head end which prevents piston movement. When the actuators are in the stow position and locked, a spring positions a lock sleeve, holding lock keys radially into a cavity in the piston. When a deploy signal is given, pressure to the head end of the actuator shuttles the lock sleeve allowing the lock keys to be ramped outward into an unlocked position as the actuator starts to deploy. The lock sleeve can be manually

operated with the manual unlock latch (lever) for manual deployment of the thrust reversers.

During deployment hydraulic pressure is ported to both the head and rod ends of the actuator. A differential pressure area moves the piston and acme nut towards the deployed position. The nut motion causes rotation of a multi-start acme screw with a worm gear attached to one end. The worm gear drives a worm that is perpendicular to the actuator centerline and attached to a flexible sync shaft cable connecting the actuators. When stowing, the head end pressure drops to return line pressure causing the piston to move towards the stow (head end) position. The actuator is sized to deploy the reverser in a maximum of 3 seconds and stow it in a maximum of 5 seconds. The actuator stroke is 20.55 inches.

When the thrust reverser control valve module is commanded to the stow mode, hydraulic pressure goes to the rod end of the actuators only. The actuators now move the T/R sleeves to the stowed position and the lock keys go into the locked position. The lock sleeve spring then moves the lock sleeve over the lock keys and the actuator is locked.

NOTE: T/R actuators without internal lock mechanisms deploy and stow similarly to T/R actuators with internal lock mechanisms.

Position Feedback Mechanism

Position feedback is by an internally threaded feedback nut connected to the actuator acme screw and an externally threaded feedback rod. Rotary motion of the acme screw is changed to linear motion of the feedback rod through the feedback nut. During actuator extension, the feedback rod moves toward the actuator. During actuator stow, the feedback rod moves away from the actuator. Differences in the pitch of the acme screw/nut threads and the threads of the feedback rod/nut

THRUST REVERSER -- LOCKING HYDRAULIC ACTUATORS -- FUNCTIONAL DESCRIPTION

cause total feedback rod travel to be 1.58 inches and total actuator travel to be 20.55 inches.

Gimbal Assembly

A gimbal assembly attaches the head end of each hydraulic actuator to the aft side of the T/R torque box. You must deploy the translating sleeves aft to get access to the gimbal assembly.

Sync Shaft and Tubing Connections

The sync shafts connect the drive mechanisms of the actuators together. The shafts are inside of the deploy hydraulic tubing which connects to the deploy pressure port. The sync shafts synchronize the motion of the actuators so that all the actuators on a T/R assembly move together. The sync shaft connection is at the deploy pressure port.

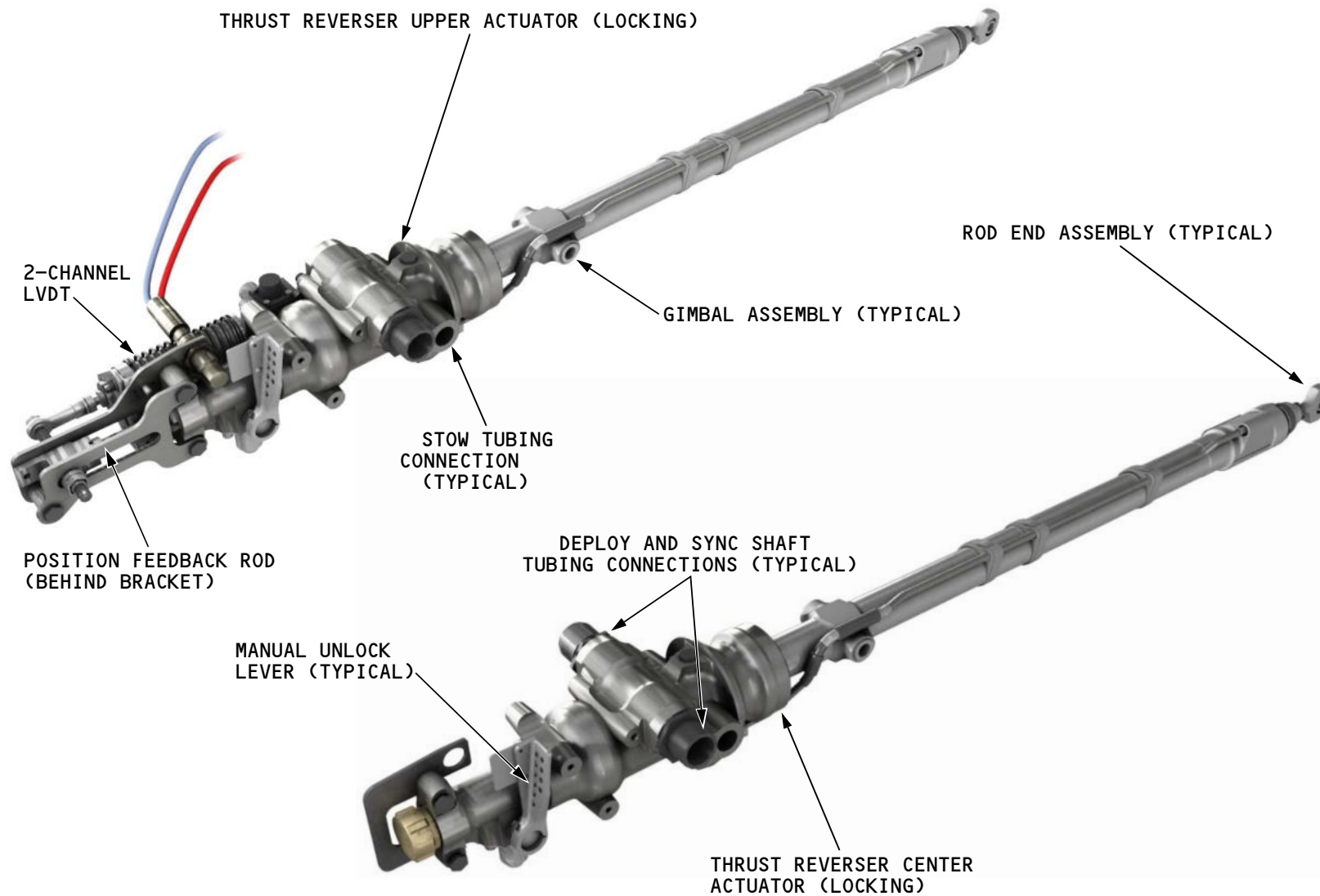
Training Information Point

Do not let the actuator rod end turn when you handle the actuator. Rigging of the feedback rod will be necessary.

DMC-B737T-A-78-31-00-24A-034B-T_002-00

02-Dec-2020

78-31-00



THRUST REVERSER -- LOCKING HYDRAULIC ACTUATORS -- FUNCTIONAL DESCRIPTION

78-31-00



THRUST REVERSER CONTROL SYSTEM -- INTRODUCTION

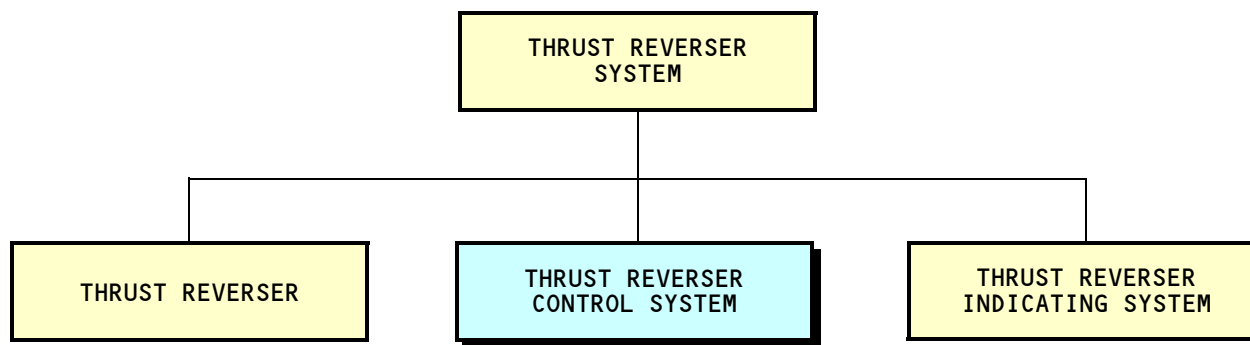
Purpose

The thrust reverser control system controls hydraulic and electrical power to the thrust reverser for stow and deploy operations.

DMC-B737T-A-78-34-00-02A-018A-T_002-00

02-Dec-2020

78-34-00



THRUST REVERSER CONTROL SYSTEM -- INTRODUCTION

78-34-00

DMC-B737T-A-78-34-00-02A-018A-T_002-00
ICN-B737T-A-783400-A-81205-00017-A-002-01

02-Dec-2020



THRUST REVERSER CONTROL SYSTEM -- GENERAL DESCRIPTION

THIS PAGE INTENTIONALLY LEFT BLANK

THRUST REVERSER CONTROL SYSTEM -- GENERAL DESCRIPTION

General

The thrust reverser control system uses hydraulic power and electrical power to deploy and stow the thrust reverser (T/R) translating sleeves. The system uses 28v dc electrical power and reverse thrust lever position for control. Indications show on the max display system (MDS) and engine control panel.

You can deploy the thrust reversers when the airplane is less than 10 feet (3 meters) from the ground. The flight control computers (FCC) and a radio altimeter operated relay supply airplane air/ground information. The proximity sensor electronics unit (PSEU) also supplies air/ground logic. The fire handle must also be down for the control system to receive electrical power.

Each T/R control valve module contains the electrical and hydraulic components necessary to control the flow to the hydraulic actuators. There are two T/R control valve modules, one for each T/R.

Sync shafts on each translating sleeve make sure the sleeve's three actuators operate at the same speed. The actuators can operate only if the shaft is free to turn.

A sync lock connects to the bottom hydraulic actuator on each T/R half. The sync lock must unlock for the sync shafts to turn. During normal T/R operation, the sync locks energize to unlock. The sync lock is also a manual drive mechanism. You use the sync lock to manually move the T/R translating sleeves for maintenance operations.

The reverse thrust lever operates internal switches in the autothrottle switch pack (S4) and in the lever (control switch). These switches send control signals to these components:

- Sync locks
- Thrust reverser control valve module.

Deploy Operation

This happens when you raise the reverse thrust lever to deploy a T/R:

- The S4 switch in the autothrottle switch pack moves to energize the sync locks (unlock)
- EEC channel A uses thrust lever angle (TLA) logic to open the hydraulic isolation valve (ARM)
- The T/R control switch moves, and a deploy signal goes through the T/R control valve module
- The T/R control valve module sends hydraulic fluid to the actuators to move the translating sleeves aft.
- The hydraulic fluid pressure operates the stow hydraulic pressure switch. The switch sends a signal to the EEC. This tells the EEC the T/R system is armed.

The T/R sequence relay adds a 100 msec delay to make sure that the sync locks unlock before hydraulic pressure goes to the actuators.

When the translating sleeves are more than 60% deployed, EEC channel A sends a signal to the interlock solenoid. This lets you move the reverse thrust lever to the full reverse position.

Stow Operation

This happens when you lower the reverse thrust lever back to the stow position:

- The T/R control switch removes the deploy signal to the T/R control valve module
- EEC channel A sends power to the stow solenoid in the T/R control valve module
- Time delays in the sync lock control relay and EEC channel A keep the sync locks energized (unlocked) and the ARM solenoid energized for ten seconds

THRUST REVERSER CONTROL SYSTEM -- GENERAL DESCRIPTION

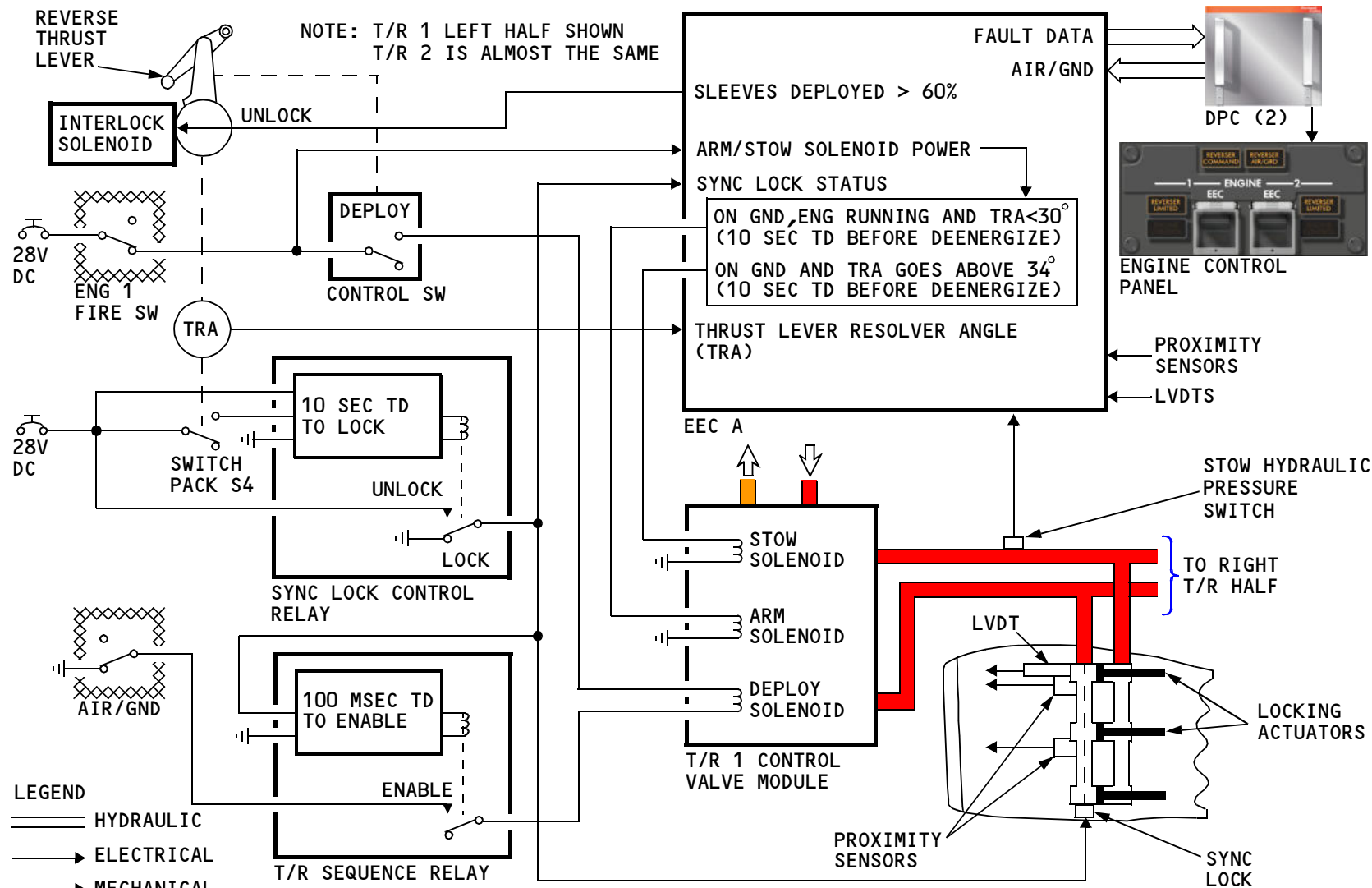
- The T/R control valve module sends hydraulic fluid to the actuators to move the translating sleeves back to the stow position.

EEC channel A removes power from the arm and stow solenoids 10 seconds after the reverse thrust lever goes above 34° TRA, or until a proximity sensor on each sleeve indicates locked, (whichever is longer). EEC channel A also removes power from the interlock solenoid when the translating sleeves are less than 40% deployed.

DMC-B737T-A-78-34-00-02A-040A-T_006-00

02-Dec-2020

78-34-00



THRUST REVERSER CONTROL SYSTEM -- GENERAL DESCRIPTION

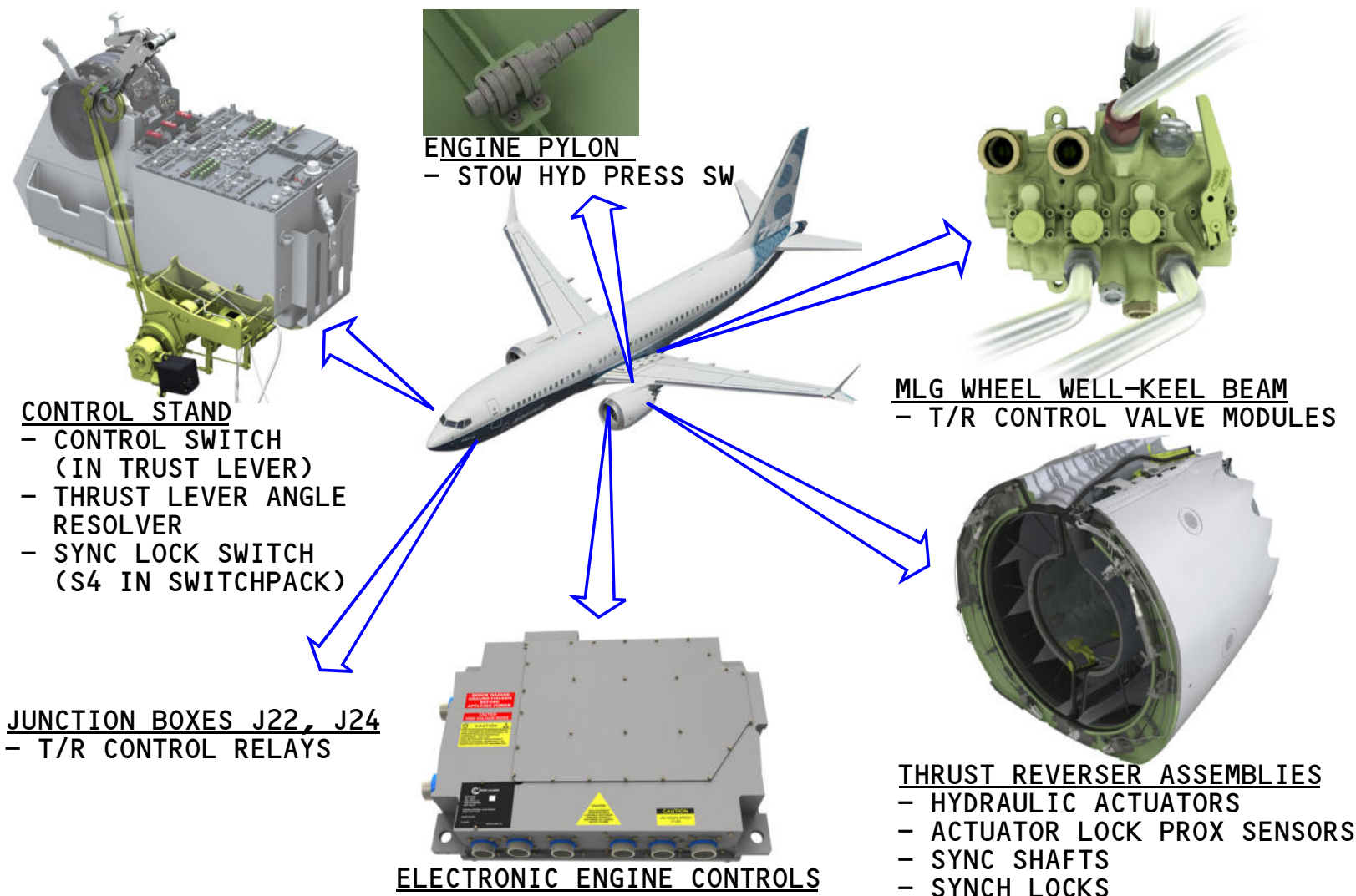
78-34-00

THRUST REVERSER CONTROL SYSTEM -- COMPONENT LOCATIONS

General

The thrust reverser (T/R) control components are at these areas of the airplane:

- Upper and lower control stand
- Engine pylon
- Engine electronic control
- Thrust reverser assemblies
- Main gear wheel well.



THRUST REVERSER CONTROL SYSTEM -- COMPONENT LOCATIONS

78-34-00

DMC-B737T-A-78-34-00-02A-055A-T_003-00
ICN-B737T-A-783400-A-81205-00019-A-003-01

02-Dec-2020

THRUST REVERSER CONTROL SYSTEM -- SYNC LOCK SWITCHES

Purpose

The sync lock switches control power to the sync locks. Power to the sync locks will unlock the flexible drive shafts.

Location

The sync lock switches, (S4) are in the autothrottle (A/T) switch packs, below the flight compartment aisle stand. There are two switch packs, one for each T/R. The switch packs are between the two A/T servo mechanisms.

Access to the A/T switch packs is through the lower nose compartment access door.

General

The sync lock switch is a microswitch that is part of the switch pack assembly. The S4 switch can not be removed individually. You must remove and replace the entire assembly if there is a malfunction of any one of the microswitches.

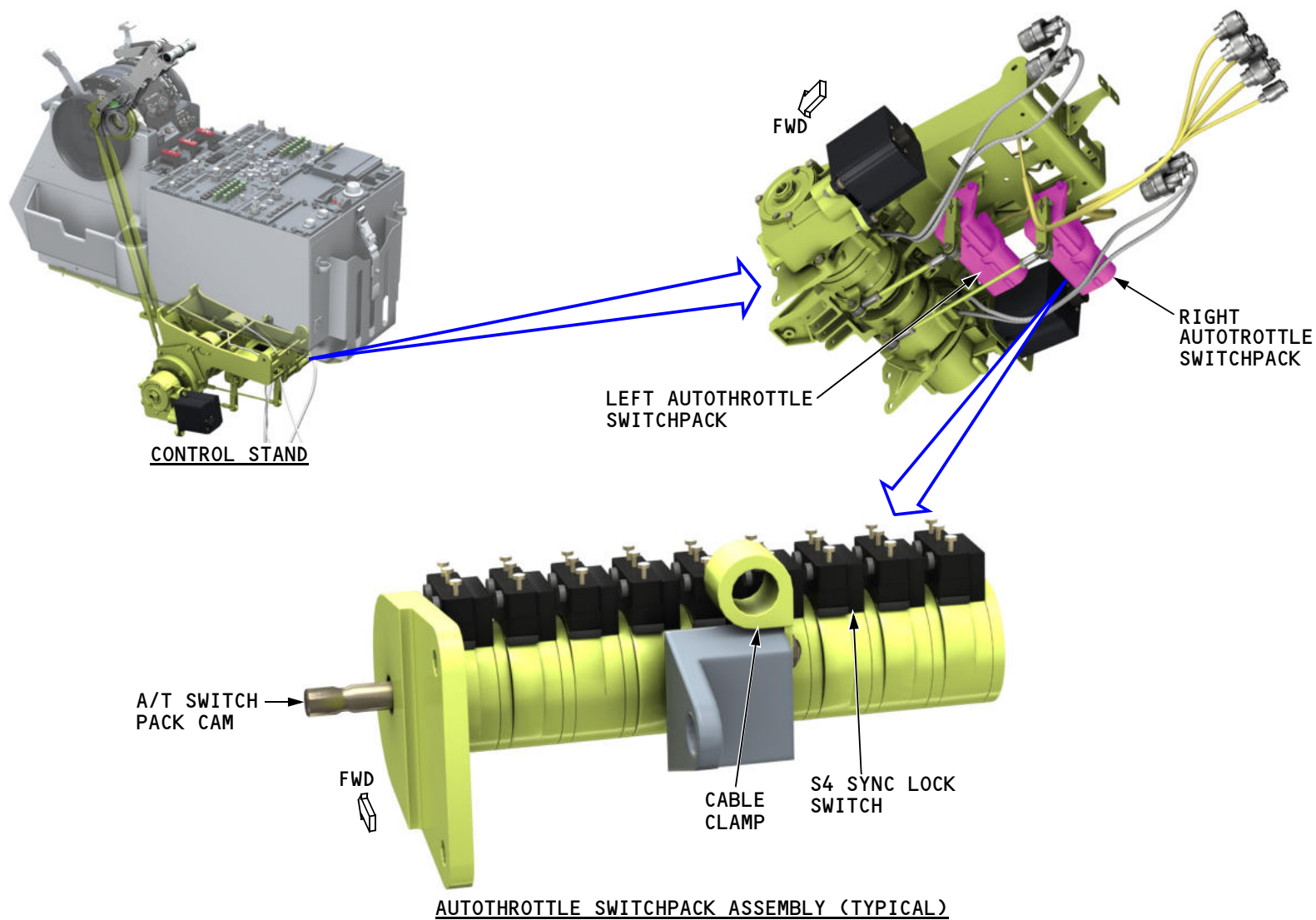
Functional Description

The input crank turns the A/T switch pack cam. The cam operates all of the switches in the switch pack assembly. There is no rigging of the switch pack assembly. The rigging adjustment are the control rods that operate the cams.

DMC-B737T-A-78-34-00-09A-110A-T_001-00

02-Dec-2020

78-34-00



THRUST REVERSER CONTROL SYSTEM -- SYNC LOCK SWITCHES

78-34-00

DMC-B737T-A-78-34-00-09A-110A-T_001-00
ICN-B737T-A-783400-A-81205-00020-A-001-01

02-Dec-2020

THRUST REVERSER CONTROL SYSTEM -- SYNC LOCK

Purpose

The sync lock has two purposes:

- Locks the sync shafts to prevent operation of the hydraulic actuators when there is no T/R deploy signal
- A manual drive for manually operating the hydraulic actuators.

Location

A sync lock is on the lower side of the bottom non-locking hydraulic actuator of each T/R half.

Physical Description

The sync lock is a mechanical lock which uses electrical power to operate (unlock) during normal T/R operation. The sync lock has one electrical connector. The manual drive is on the bottom. Use the manual drive to unlock the lock mechanism. You do not need electrical power to unlock the sync lock when using the manual drive.

Functional Description

Without electrical power, the sync lock is in the locked position, and the sync shafts can not turn. Therefore the T/R hydraulic actuators can not move the T/R translating sleeves.

During a T/R deploy operation, the internal solenoid energizes to unlock the sync lock mechanism. The sync shafts can then turn, which lets the hydraulic actuators move the T/R translating sleeves.

The sync lock solenoid de-energizes 10 seconds after the stow command goes away. This ensures that the translating sleeve has sufficient time to go to the stow position before the sync lock locks the sync shafts again.

The manual drive portion of the sync lock engages internal mechanisms to turn let you turn the sync shafts.

Training Information Point

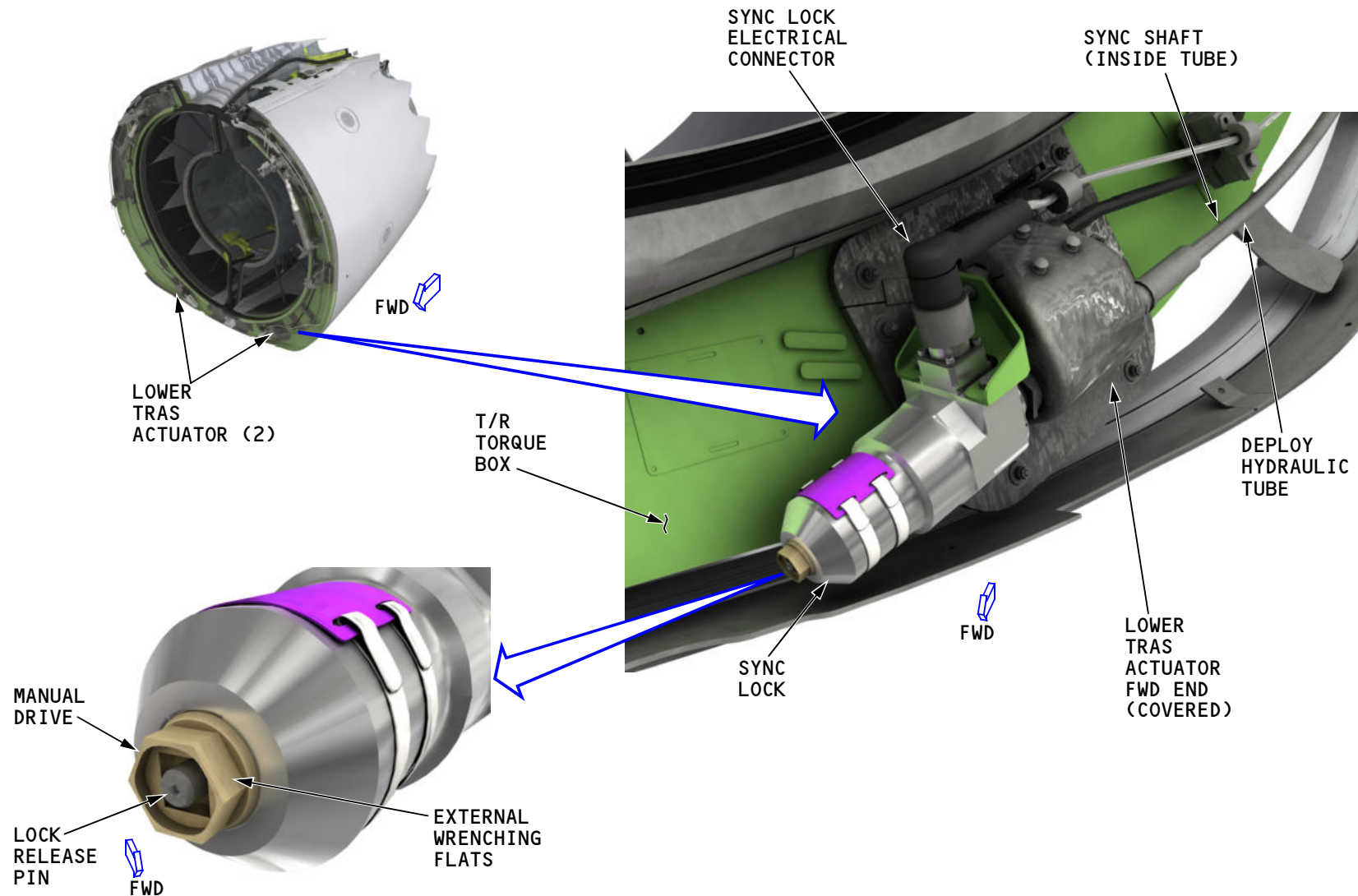
To manually unlock the sync lock, push a 3/8 square drive tool into the manual drive fitting. The square drive tool pushes in the lock release pin inside the drive connector. The sync lock manually unlocks when this pin is pushed in completely.

NOTE: You may need to use a wrench on the external wrenching flats to turn the manual drive fitting. Turn the manual drive fitting to align the internal lock mechanism so that you can push the release pin in completely.

The T/R hydraulic control valve must have the lockout pin installed, and the manual unlock lever on the locking actuators (center and upper) must also be unlocked before you can manually extend and retract the hydraulic actuators.

Follow the procedures in the airplane maintenance manual (AMM) for manual operation of the T/R.

DMC-B737T-A-78-34-00-12A-110A-T_001-00



DMC-B737T-A-78-34-00-12A-110A-T_001-00
ICN-B737T-A-783400-A-81205-00023-A-001-01

THRUST REVERSER CONTROL SYSTEM -- SYNC LOCK

78-34-00

02-Dec-2020

THRUST REVERSER CONTROL SYSTEM -- CONTROL SWITCH

Purpose

The control switch supplies electrical power to the deploy solenoid in the T/R control valve module during a T/R deploy operation. There are two control switches, one for each T/R.

Location

The control switches are in the thrust levers. Each thrust lever has one switch. Remove the cover on the side of the thrust lever to access the switch.

Functional Description

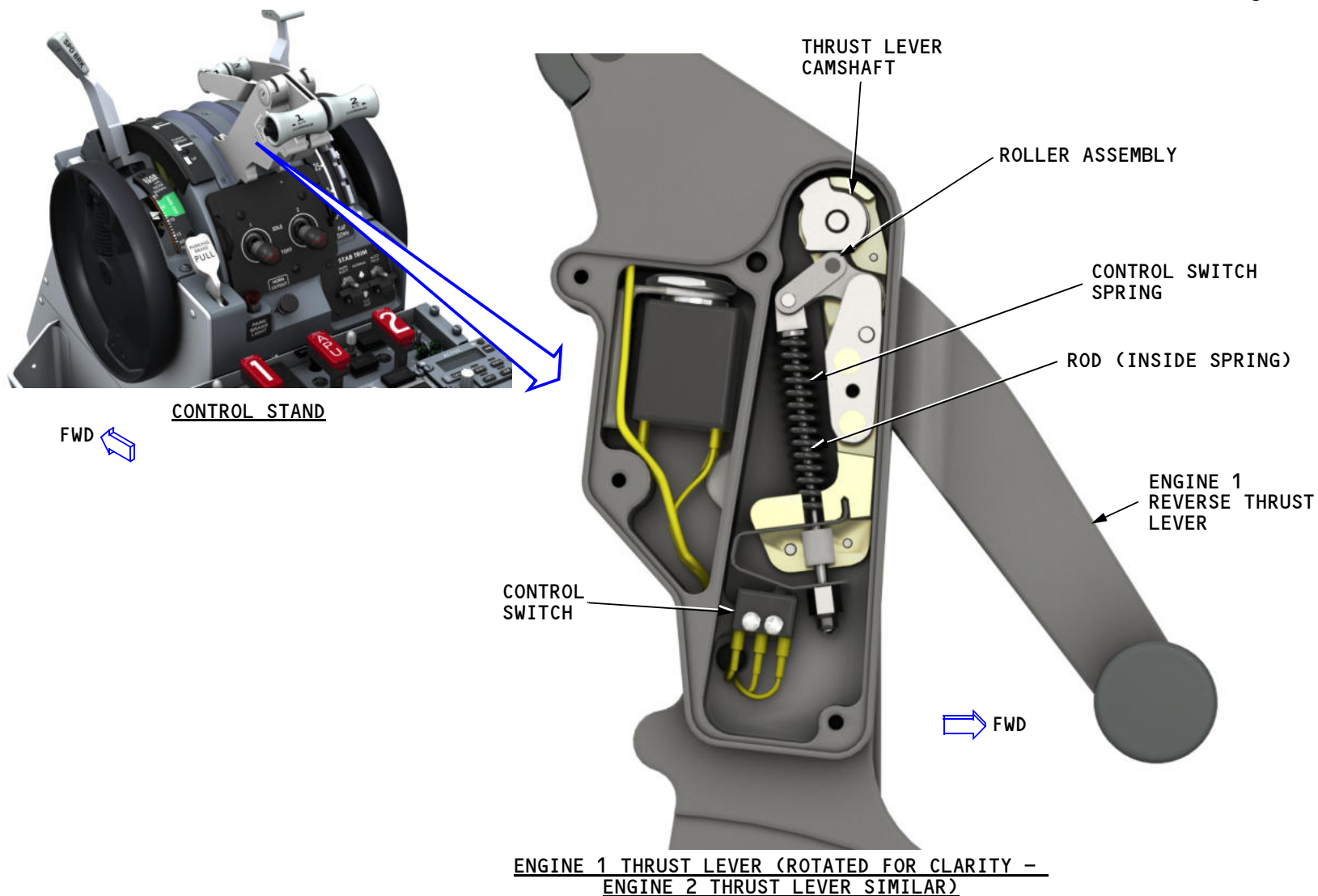
The reverse thrust lever turns the thrust lever camshaft. The camshaft causes the roller assembly and rod to move down when the reverse thrust lever is raised to the deploy position. The control switch spring compresses and operates the control switch.

The spring around the rod keeps the rod and the roller assembly against the cam. The rod and the roller move back to their normal position when the reverse thrust lever is lowered to the stow position. The control switch spring relaxes and the switch moves back to the stow position.

DMC-B737T-A-78-34-00-10A-110A-T_001-00

02-Dec-2020

78-34-00



THRUST REVERSER CONTROL SYSTEM -- CONTROL SWITCH

78-34-00

DMC-B737T-A-78-34-00-10A-110A-T_001-00
ICN-B737T-A-783400-A-81205-00021-A-001-01

02-Dec-2020

THRUST REVERSER CONTROL SYSTEM -- CONTROL VALVE MODULE

Purpose

The T/R control valve module controls hydraulic pressure to the T/R hydraulic actuators. Each module has a handle that deactivates the T/R for maintenance.

Physical Description

Each T/R control valve module has these internal components:

- Isolation valve
- Directional control valve
- Arm solenoid valve
- Stow solenoid valve
- Deploy solenoid valve
- Manual isolation (shutoff) valve.

NOTE: The three solenoid valves are not line replaceable units (LRUs).

The manual isolation valve handle has a hole for the T/R maintenance deactivation pin.

Location

The T/R control valve modules are in the main landing gear wheel well, on the keel beam. The T/R 1 control valve module is on the left side of the keel beam. The T/R 2 control valve module is on the right side of the keel beam.

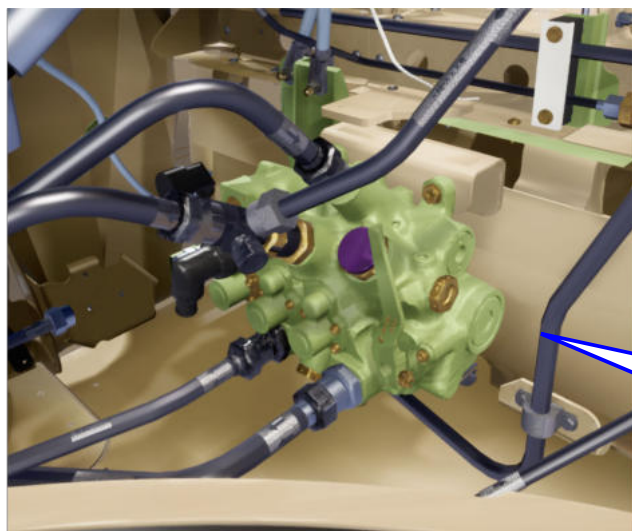
Training Information Point

Use the manual isolation valve handle to prevent accidental operation of T/Rs when persons or equipment are near the T/R. Also use the

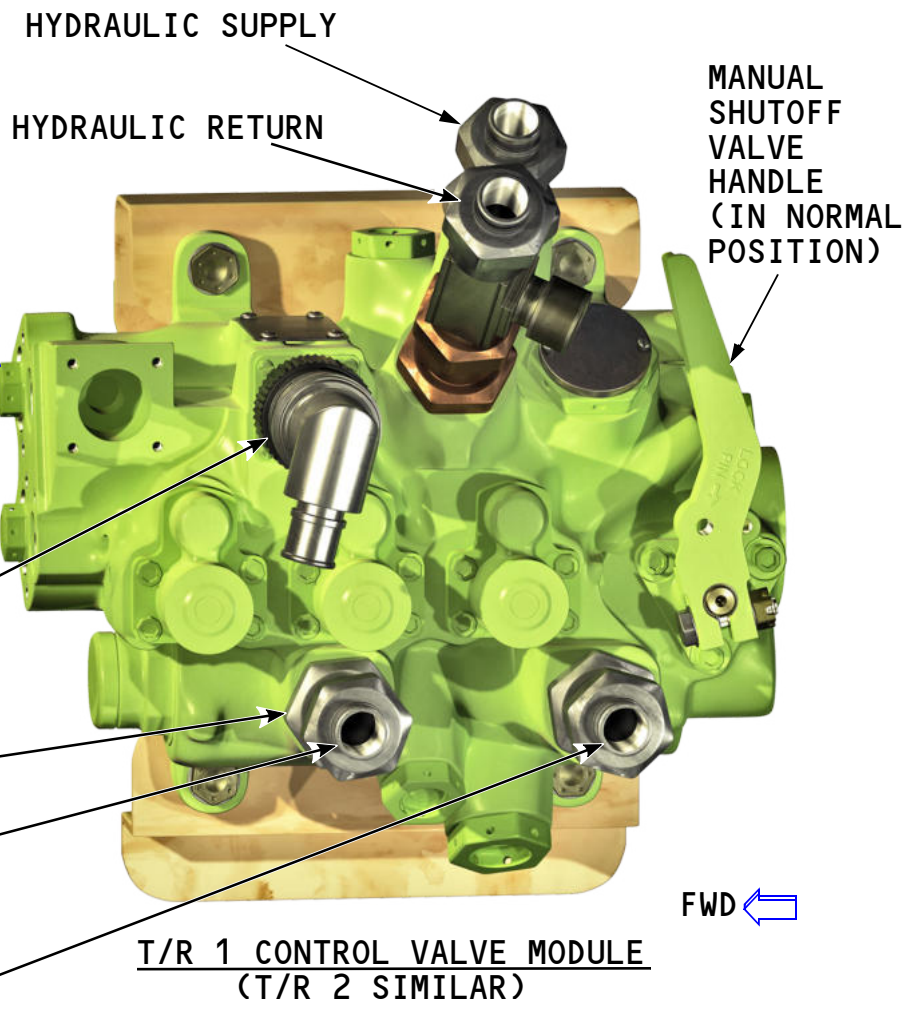
manual isolation valve handle when the T/R cowls are open. Install a lock pin to keep the handle in the shutoff position.

The T/R deactivation for maintenance is different than the T/R deactivation for airplane dispatch. Be sure to follow the airplane maintenance manual (AMM) procedures.

WARNING: BEFORE YOU REMOVE THE LOCK PIN FROM THE T/R CONTROL VALVE MODULE FOR THE THRUST REVERSER, MAKE SURE THE THRUST REVERSER SLEEVES ARE IN THE CORRECT POSITION. IF THE THRUST REVERSER SLEEVES ARE IN THE STOWED POSITION, THE REVERSE THRUST LEVER MUST BE FULLY FORWARD. IF THE THRUST REVERSER SLEEVES ARE IN THE DEPLOYED POSITION, THE REVERSE THRUST LEVER MUST NOT BE IN THE STOWED POSITION. IF THE POSITION OF THE REVERSE THRUST LEVER DOES NOT AGREE WITH THE POSITION OF THE THRUST REVERSER SLEEVES, THE THRUST REVERSER SLEEVES COULD MOVE WHEN THE HYDRAULIC SYSTEM IS ACTIVATED. THIS COULD CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.



MLG WHEEL WELL KEEL BEAM



THRUST REVERSER CONTROL SYSTEM -- CONTROL VALVE MODULE

78-34-00

DMC-B737T-A-78-34-00-14A-110A-T-004-00
ICN-B737T-A-783400-A-81205-00025-A-004-01

THRUST REVERSER CONTROL SYSTEM -- THRUST REVERSER ACTUATOR LOCK PROXIMITY SENSORS

Purpose

The thrust reverser sleeve lock proximity sensors supply a hydraulic actuator lock/unlock signal to the engine electronic controls (EECs).

The EECs use this signal for these functions:

- T/R stow control
- Fault logic and fault indication
- Fault isolation.

Location

The actuator lock proximity sensors are on the forward end of the T/R hydraulic actuators with internal lock mechanisms (top and center).

Open the fan cowl to get access to the actuators and sensors. Each T/R locking hydraulic actuator with an internal lock mechanism has a lock proximity sensor. Therefore, there are two lock sensors for each T/R translating sleeve.

General

Each sensor has two output levels. One level for the lock position (far), and the other level for the unlock position, (near).

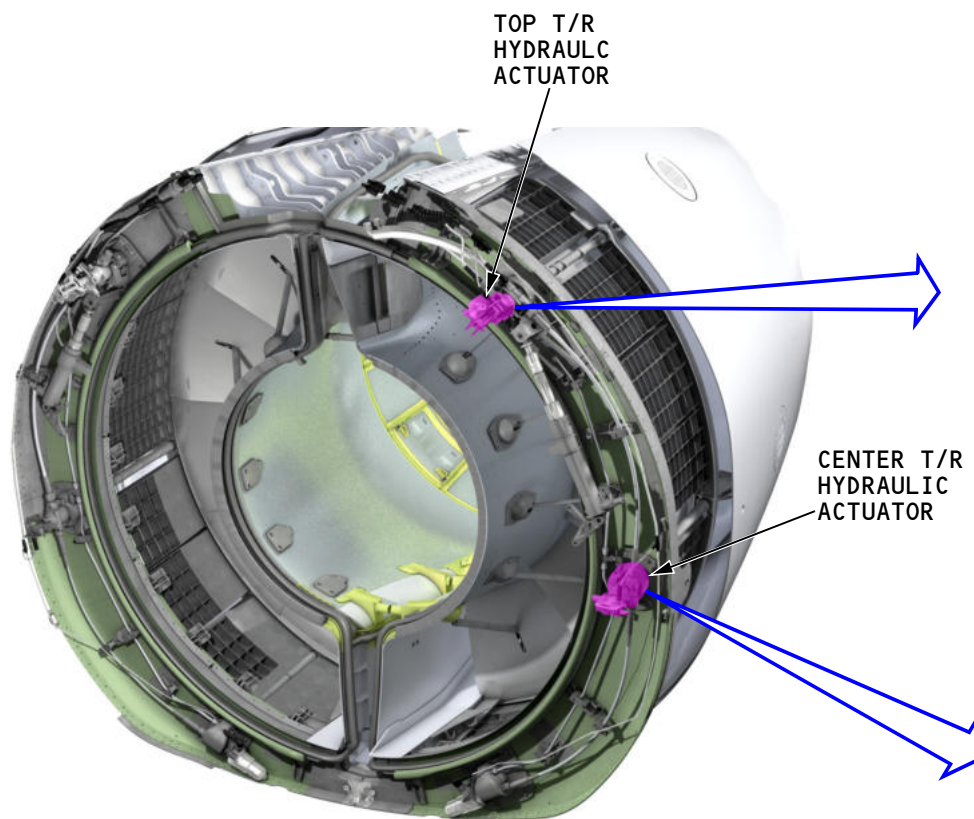
The sensor is a proximity type sensor. The actuator's manual unlock lever is the target for the sensor. The sensor's output changes when the target is close (near).

Functional Description

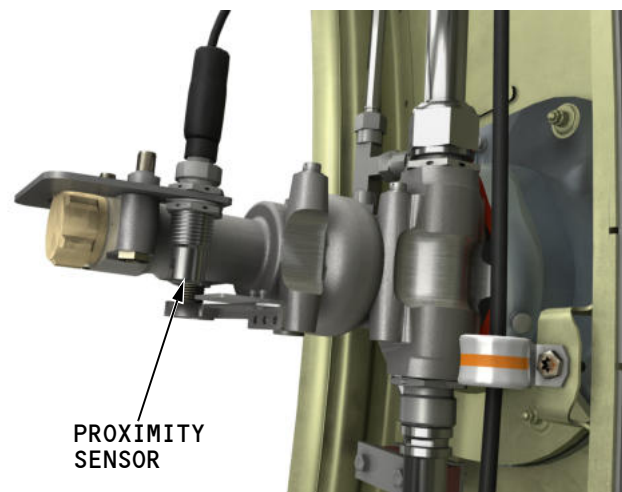
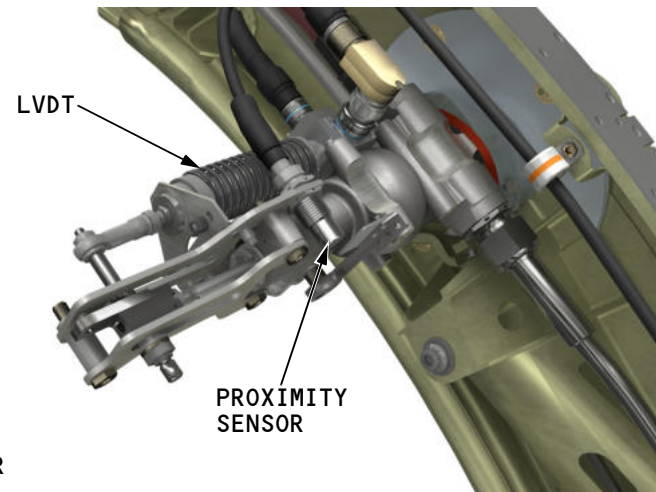
The EECs supply excitation to the sensors. The EECs see a large change in the impedance of the sensor's output when the target is near. The EECs use the target near the sensor input in the T/R stow control circuit.

The manual unlock lever moves the target near to the sensor when the hydraulic actuator unlocks during a T/R deploy operation. The lever stays in this position until the sleeve comes back to the stow position and the hydraulic actuator locks, which moves the target far to the sensor. The EECs use the sensor inputs for fault detection and to reset the T/R stow control circuit.

DMC-B737T-A-78-34-00-11A-110A-T_003-00



ENGINE THRUST REVERSER ASSEMBLIES



THRUST REVERSER CONTROL SYSTEM -- THRUST REVERSER ACTUATOR LOCK PROXIMITY SENSORS

78-34-00

DMC-B737T-A-78-34-00-11A-110A-T_003-00
ICN-B737T-A-783400-A-81205-00022-A-001-01

02-Dec-2020

THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - DEPLOY AND STOW CONTROL

Thrust Reverser Control System - General

The EECs contain logic for thrust reverser system control and monitoring. Only Channel A sends control signals to the T/R control valve module. There are also thrust reverser control system switches and relays that use 28v dc standby bus power.

Thrust Reverser Deploy - Sequence

To select reverse thrust, the forward thrust lever must be in the idle position, and then you firmly lift the reverse thrust lever to the reverse idle position. This starts the thrust reverser deploy sequence.

The thrust reverser deploy sequence has these four steps:

- Sync lock
- Arm solenoid
- Deploy solenoid
- Interlock solenoid.

Thrust Reverser Deploy - Sync Locks

The ENGINE 1 T/R SYNC LOCK circuit breaker sends power through switch S4 in the left autothrottle switch pack. From switch S4, this power energizes the coil of the sync lock control relay solenoid. When this solenoid energizes, the power goes to these:

- T/R sequence relay
- T/R 1 L sync lock solenoid (unlocks)
- T/R 1 R sync lock solenoid (unlocks)
- T/R 1 LH sync lock indication relay.

The T/R sequence relay has a time 100-millisecond time delay. This delay lets the sync locks unlock before the thrust reverser hydraulic actuators get pressure.

Thrust Reverser Deploy - Arm Solenoid

EEC channel A has this logic for the arm solenoid power:

- The airplane must be on the ground
- The engine must be running
- The forward thrust lever must be at the reverse idle position (TRA less than 30 degrees)
- The engine fan speed must be at or near idle.

The arm solenoid opens the hydraulic isolation valve in the T/R control valve module.

Thrust Reverser Deploy - Deploy Solenoid

After the 100-millisecond time delay, the T/R sequence relay energizes. This gives a path to ground on the deploy solenoid in the T/R control valve module. 28v dc power for the deploy solenoid comes from the ENGINE 1 T/R CONTROL circuit breaker. From this circuit breaker, power goes through the lower contacts of the ENG 1 fire switch. Power then goes through the deploy contacts of the control switch in the thrust lever. Power then goes to the deploy solenoid, which energizes. The T/R control valve module gives hydraulic pressure to the thrust reverser deploy and stow side of the actuators at this time.

The air/ground signal for the deploy solenoid comes from one or more of these:

- Flight control computer (FCC) A
- FCC B
- Air sensing relay (PSEU).

THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - DEPLOY AND STOW CONTROL

Thrust Reverser Deploy - Interlock Solenoid

The interlock solenoid does not let the reverse thrust lever go past reverse idle position until the thrust reverser sleeves are almost fully deployed. When energized, the solenoid unlocks to let you select full reverse thrust. This prevents damage to the engine and airplane.

28v dc power to unlock the Interlock solenoid comes from the ENGINE 1 T/R INTLK circuit breaker. The solenoid gets a ground from internal EEC logic when all of these occur:

- Thrust resolver angle is less than 30 degrees (at or below reverse idle)
- Both thrust reverser sleeves on the left engine are greater than 60% deployed.

Thrust Reverser Stow - Conditions

Reverse thrust slows the airplane on the ground. When continued reverse thrust is not necessary, you firmly lower the reverse thrust lever to the stow position. This starts the thrust reverser stow operation.

Thrust Reverser Stow - Sequence

When you lower the reverse thrust lever to stow the thrust reverser sleeves, these occur:

- Switch S4 of the left autothrottle switch pack opens. The sync lock control relay starts a 10 second timer to de-energize
- The EEC continues to send power to the arm solenoid for ten seconds
- The control switch in the thrust lever opens, and the deploy solenoid de-energizes
- The stow solenoid gets power when the TRA goes back to forward idle position (TRA greater than 34 degrees). The T/R

control valve module sends only stow pressure to the actuators, and the sleeves retract. The stow solenoid continues to get power for ten second or until at least one T/R proximity sensor indicates locked.

Thrust Reverser Stow - T/R Sync Lock Solenoids

The sync locks must stay unlocked while the thrust reverser sleeves move to the stow position. To stay unlocked, the L and R sync lock solenoids must stay energized. The solenoids stay energized for ten seconds after the start of a stow operation. This is because of the timer in the sync lock control relay.

After ten seconds, the sync lock control relay de-energizes. This removes power from the T/R 1 L and R sync lock solenoids. It also removes power from the T/R 1 LH sync lock indication relay.

Thrust Reverser Stow - Arm Solenoid

The EEC internal logic for the stow operation has a 10 second time delay. This delay keeps the arm solenoid in the T/R control valve module energized for 10 seconds. The arm solenoid must stay energized for the stow operation to occur. The delay starts after you move the T/R lever to the stow position (TRA greater than 30 degrees).

Thrust Reverser Stow - Stow Solenoid

EEC internal logic energizes the stow solenoid when all these occur:

- Related engine is running
- Thrust lever moved to the stow position (TRA greater than 34 degrees)
- All T/R hydraulic actuators are unlocked.

THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - DEPLOY AND STOW CONTROL

The T/R control valve module now gives hydraulic pressure to the stow side of the thrust reverser actuators. The top and middle T/R hydraulic actuators have internal lock mechanisms. These internal lock mechanisms lock when their related T/R sleeve is in the stow position. One T/R top or middle hydraulic actuator on each T/R sleeve must lock for a stow operation to finish. If this does not occur, stow hydraulic pressure will continue to each T/R hydraulic actuator on the related engine.

Thrust Reverser Stow - Interlock Solenoid

When the T/R sleeves are almost stowed (<40%), internal EEC logic de-energizes the interlock solenoid. The interlock then returns to the blocking position.

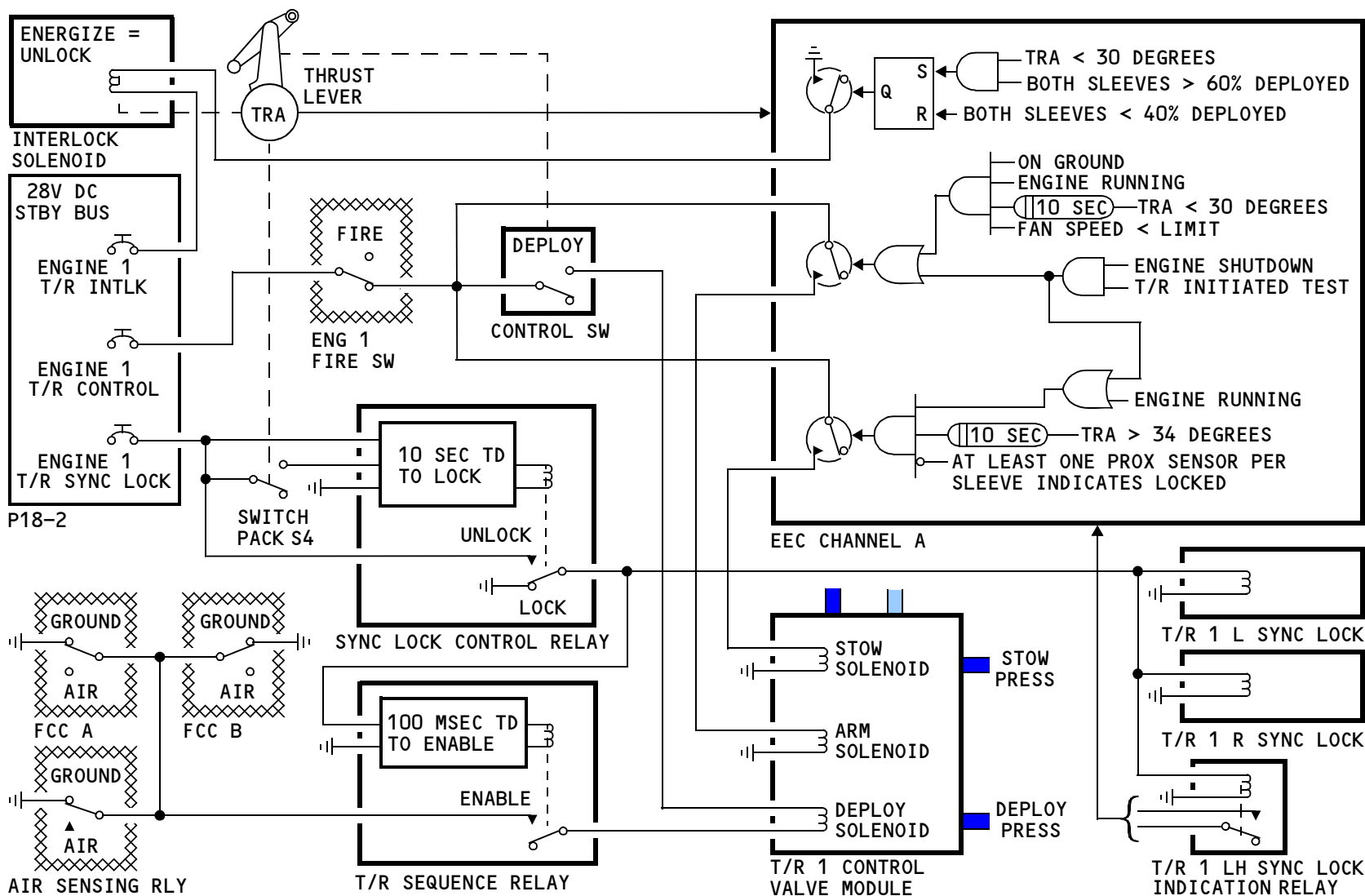
Thrust Reverser Initiated Test Function

You can operate the T/R sleeves similarly to usual operation mode when the engine is not in operation. To do this however, you must use the thrust reverser initiated test special function. This function is done using the onboard maintenance function (OMF). In this function, the OMF communicates with EEC channel A only. When you do this function, internal EEC logic lets the T/R sleeves operate. See the AMM for further information to perform this test.

DMC-B737T-A-78-34-00-05A-042A-T_002-00

02-Dec-2020

78-34-00



THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - DEPLOY AND STOW CONTROL

02-Dec-2020

78-34-00



THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - DEPLOY HYDRAULIC FLOW

THIS PAGE INTENTIONALLY LEFT BLANK

THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - DEPLOY HYDRAULIC FLOW

General

The T/R control valve modules control hydraulic power to the hydraulic actuators for T/R deploy and stow.

Hydraulic system A supplies hydraulic power for T/R 1. System B supplies hydraulic power for T/R 2. The standby hydraulic system supplies backup hydraulic power through shuttle valves if the normal system source fails.

The graphic shows T/R 1 operation. T/R 2 operation is almost the same.

Deploy

The arm and deploy solenoid valves energize when the reverse thrust lever moves to the deploy position. These valves are normally spring-loaded closed.

The hydraulic isolation valve (HIV) is spring loaded to the close position. The HIV needs hydraulic pressure to move to the arm position.

This happens when the arm solenoid energizes after the sync lock solenoids get electrical power to unlock:

- The solenoid valve opens, and hydraulic fluid goes through the valve to the HIV
- The HIV moves to the arm position
- Hydraulic pressure goes through open HIV to the directional control valve (DCV)
- Hydraulic pressure also goes through the manual shutoff valve to the rod side of the actuator pistons.
- Hydraulic pressure causes the stow hydraulic pressure switch to send a signal to the EEC so the EEC knows the HIV is armed.

The DCV needs hydraulic pressure to move to either the deploy or the stow position. A spring-loaded mechanical detent keeps the DCV in the last commanded position.

This happens when the deploy solenoid energizes:

- The valve opens, and hydraulic fluid flows through the valve to the DCV
- The DCV moves to the deploy position
- Hydraulic fluid from the HIV flows through the DCV, and then through the manual shutoff valve to the head side and the rod side of the T/R actuator pistons
- The deploy hydraulic pressure causes the locking actuators to unlock.

The hydraulic pressure on both sides of each actuator piston are equal. The surface area of the head side is larger than the rod side of the actuator pistons. Since area multiplied by pressure equals force, the larger force on the head side causes the actuator to extend. As the actuators extend, the fluid on the rod side of the actuators goes to the manual shutoff valve and mixes with the fluid going to the head side. The rod side pressure helps to control the speed at which the actuators move the T/R translating sleeves to deploy.

Manual Shutoff Valve

Each control valve module has a manual shutoff valve. The manual shutoff valve is normally spring-loaded open. Close the manual shutoff valve whenever you are doing maintenance on or around the T/Rs. Also, make sure to open the manual shutoff valve when maintenance is no longer being done on or around the T/Rs.

You use a lock pin to keep the manual shutoff valve closed against the spring pressure.

THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - DEPLOY HYDRAULIC FLOW

One Way Flow Restrictors

The one-way restrictors restrict the mass flow rate of hydraulic fluid to the T/R hydraulic actuators. They permit free hydraulic flow from the T/R hydraulic actuators. This prevents a possible hydraulic pressure build up across the head of the actuators when the T/Rs are not operating.

Proximity Position Sensors

The EECs uses proximity type position sensors on the locking actuators (upper and center) and the stow pressure sensor for fault detection.

NOTE: See the thrust reverser indicating system (ATA 78-36) for more information on the T/R proximity sensors.

Training Information Point

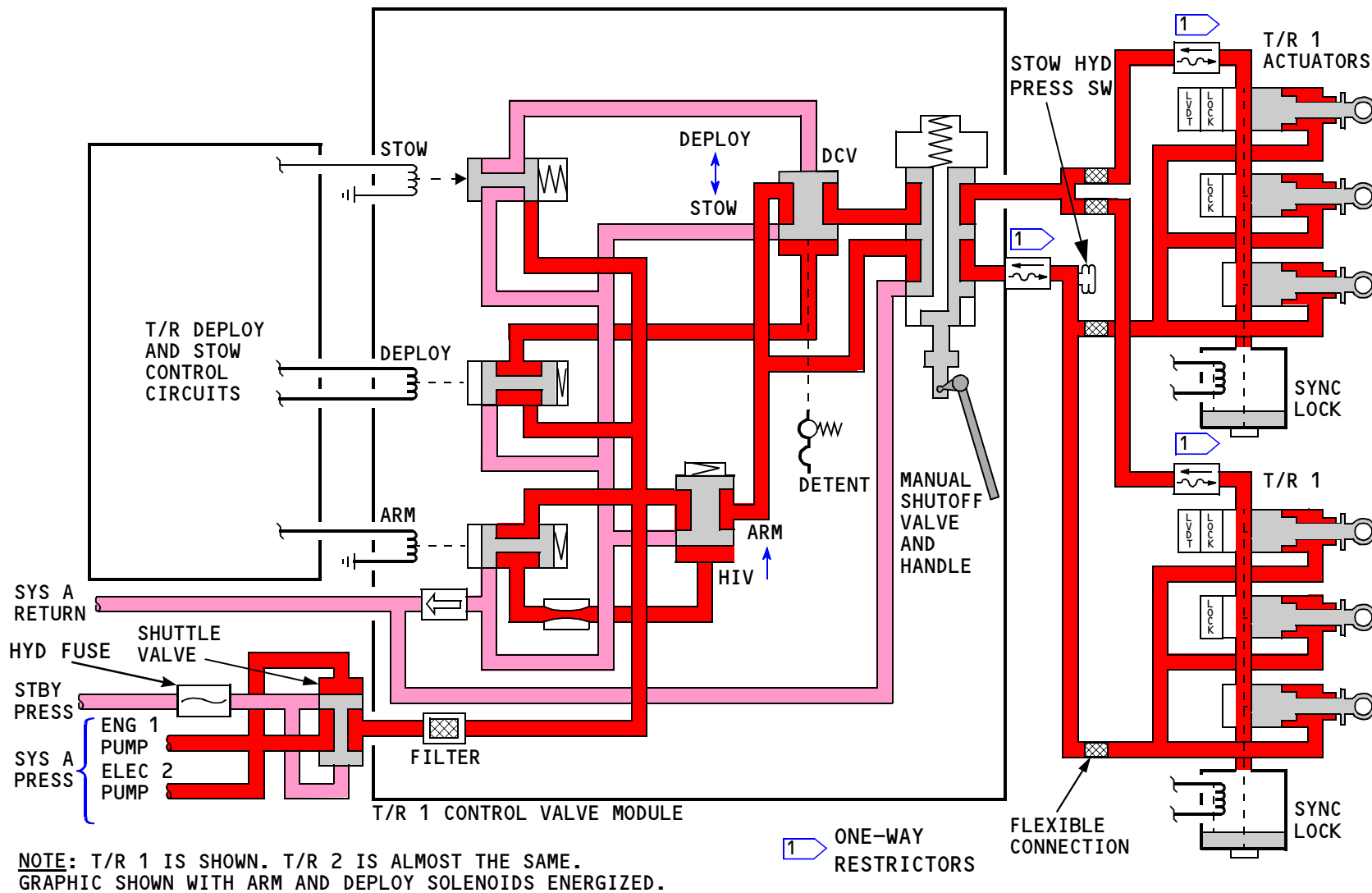
You can hydraulically or manually deploy the T/R for maintenance. To deploy the T/Rs manually you first turn off the hydraulic pressure, then pin the T/R control valve module. Push the locking levers on the locking actuators and then push the pin in to hold them in the unlocked position. You can then insert a tool into the sync lock to unlock it and manually drive the sync shaft, deploying the sleeve. You can also operate the T/R hydraulically on the ground without the engine running. To operate the T/Rs hydraulically you must first do these steps:

- Energize the EEC for the T/R to be operated
- Put the ground test switch in the TEST position
- Activate the OMF thrust reverser special function
- Operate the reverse thrust lever.

There is no hydraulic return to the standby system for T/R 1. There is a transfer of hydraulic fluid from the standby system to system A during stow if the standby system deploys the T/R.

There is not a similar transfer of hydraulic fluid for T/R 2 operation. The standby reservoir and the system B reservoir are connected by the fill and balance line, and share hydraulic fluid.

The standby hydraulic pressure to T/R 1 and T/R 2 goes through a Volume (Hydraulic) Fuse. The fuse closes when the hydraulic flow increases to 175 cubic inches. This is to prevent the loss of hydraulic fluid when a leak occurs in the T/R system.



THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - DEPLOY HYDRAULIC FLOW

02-Dec-2020

78-34-00



THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - STOW HYDRAULIC FLOW

THIS PAGE INTENTIONALLY LEFT BLANK

THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - STOW HYDRAULIC FLOW

General

The thrust reverser (T/R) control valve modules control hydraulic power to the hydraulic actuators for T/R sleeve deploy and stow operations.

Hydraulic system A supplies hydraulic power for T/R 1. System B supplies hydraulic power for T/R 2. The standby hydraulic system supplies backup hydraulic power through the shuttle valves if system A and/or system B fails.

The graphic shows T/R 1 operation. T/R 2 operation is almost the same.

Stow

The logic in electronic engine control (EEC) channel A keeps the ARM solenoid energized for ten seconds after the reverse thrust lever moves to the stowed position. The thrust lever angle (TLA) gives the EEC the reverse thrust lever position.

When the reverse thrust lever TLA moves from the deploy to the stow position, EEC channel A energizes the stow solenoid. The stow solenoid remains energized until at least one lock proximity sensor indicates locked.

These are the effects when the arm solenoid remains energized with hydraulic pressure available to the T/R control valve module:

- The hydraulic isolation valve (HIV) remains in the arm position
- Hydraulic pressure causes the stow hydraulic pressure switch to send a signal to the EEC so the EEC knows the HIV is armed.
- Active hydraulic pressure remains on the rod side of the T/R actuator pistons

- Hydraulic pressure remains at the directional control valve (DCV).

This happens when the stow solenoid valve energizes:

- The valve moves against its spring, and hydraulic fluid flows through the valve to the DCV
- The DCV moves against the mechanical detent from the deploy position to the stow position
- Hydraulic pressure on the actuator pistons head side becomes static return pressure
- Hydraulic pressure at the actuator pistons rod side cause the actuators to retract and stow the T/R
- The fluid returns to the A hydraulic system through the manual shutoff valve and the DCV.

Ten seconds after the reverse thrust lever gets moved to the stow position EEC de-energizes the ARM solenoid. These are the effects with the sleeves in the stow and locked positions:

- The stow solenoid valve de-energizes and closes
- The arm solenoid valve de-energizes and closes
- The HIV moves away from the arm position
- The DCV remains held in the stow position by the mechanical detent
- All hydraulic fluid in the T/R actuators becomes static return pressure.

NOTE: Both T/R sleeves must remain stowed and mechanically locked for the stow control circuit to reset. A 10 second time delay gets activated when the reverse thrust lever goes to the stow position. The 10 second time delay can keep stow hydraulic pressure on the T/R actuator pistons after the T/R sleeves stow. This helps to make sure the mechanical locks in the upper and center actuators go back to the locked position.

THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - STOW HYDRAULIC FLOW

Manual Shutoff Valve

The manual shutoff valve is normally open. Close manual shutoff valve whenever doing maintenance on or around the T/Rs. Also make sure you re-open the manual shutoff valve before hydraulic pressure is on the airplane, and there is no maintenance being done on the T/Rs.

The standby hydraulic pressure to T/R 1 and T/R 2 goes through a Volume (Hydraulic) Fuse. The fuse closes when the hydraulic flow increases to 175 cubic inches. This is to prevent the loss of hydraulic fluid when a leak occurs in the T/R system.

One Way Flow Restrictors

The one-way restrictors restrict the mass flow rate of hydraulic fluid to the T/R hydraulic actuators. They permit free hydraulic flow from the T/R hydraulic actuators. This prevents a possible hydraulic pressure build up across the head of the actuators when the T/Rs are not operating.

Proximity Position Sensors

The EECs use the position sensors on the T/R upper and center actuators and the stow pressure sensor for fault detection and stow control enabling.

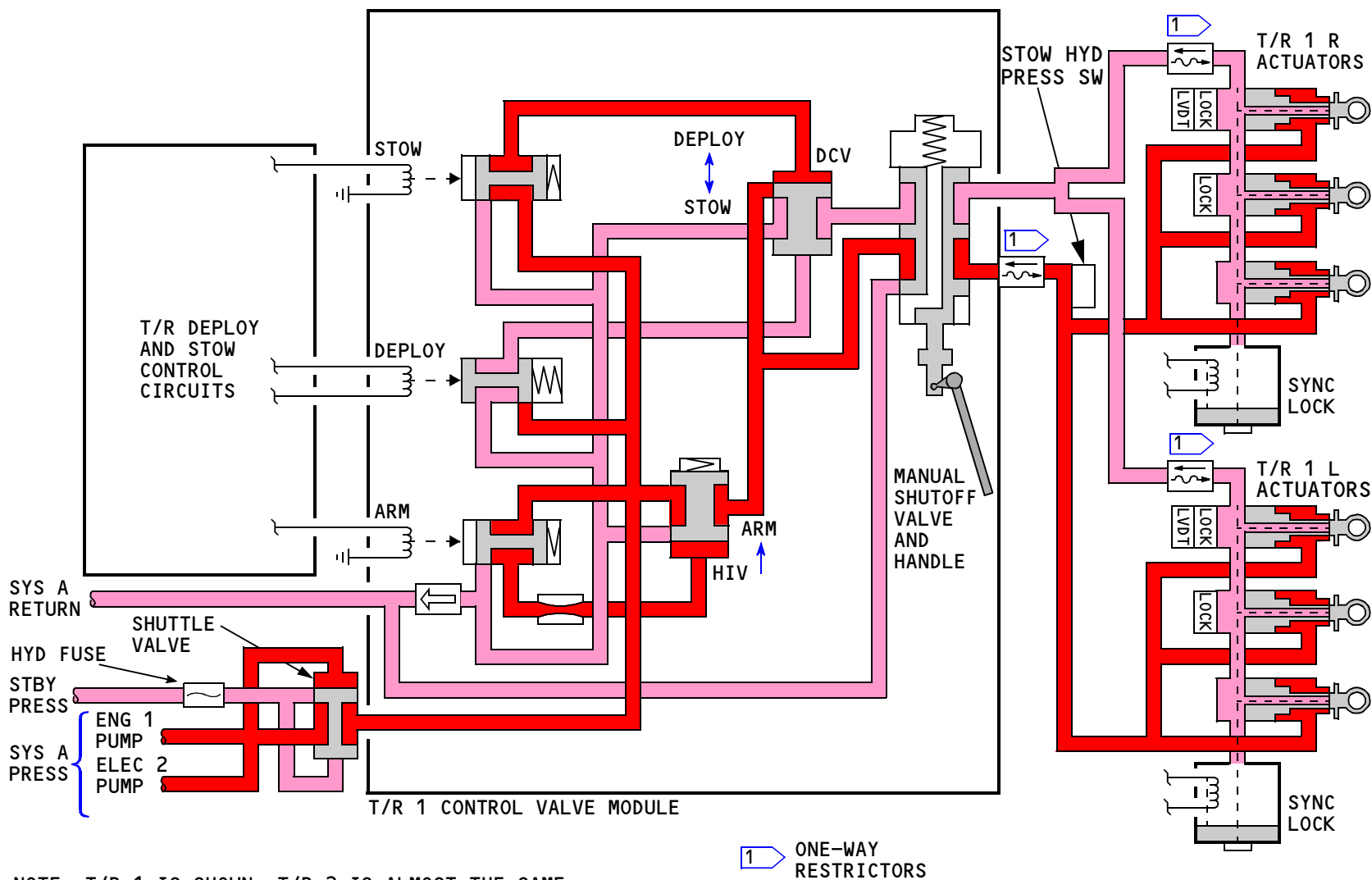
NOTE: See 78-36-00 THRUST REVERSER INDICATING SYSTEM for more information on these proximity position sensors.

Training Information Point

There is no hydraulic return to the standby system for T/R 1. There is a transfer of hydraulic fluid from the standby system to system A during stow if the standby system deploys the T/R.

There is not a similar transfer of hydraulic fluid for T/R 2 operation, because the standby reservoir and the system B reservoir are connected by the fill and balance line, and share hydraulic fluid.

DMC-B737T-A-78-34-00-08A-042A-T_003-00



THRUST REVERSER CONTROL SYSTEM -- FUNCTIONAL DESCRIPTION - STOW HYDRAULIC FLOW

78-34-00

THRUST REVERSER INDICATING SYSTEM -- INTRODUCTION

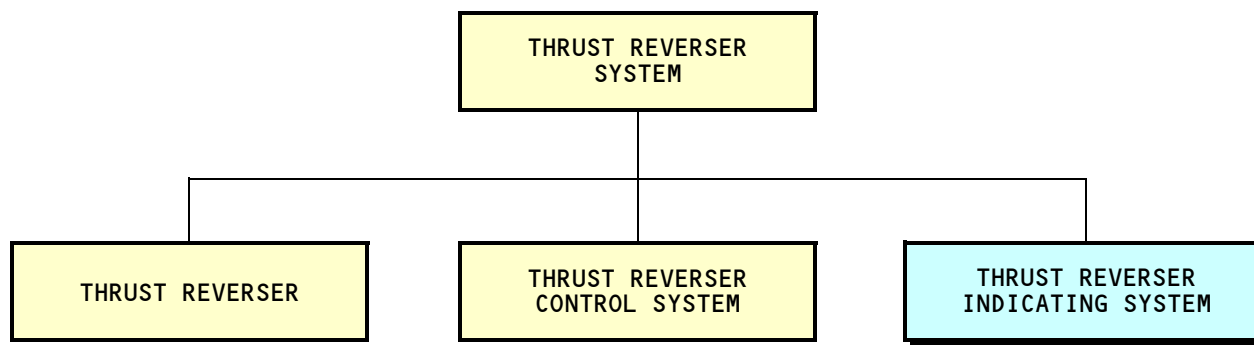
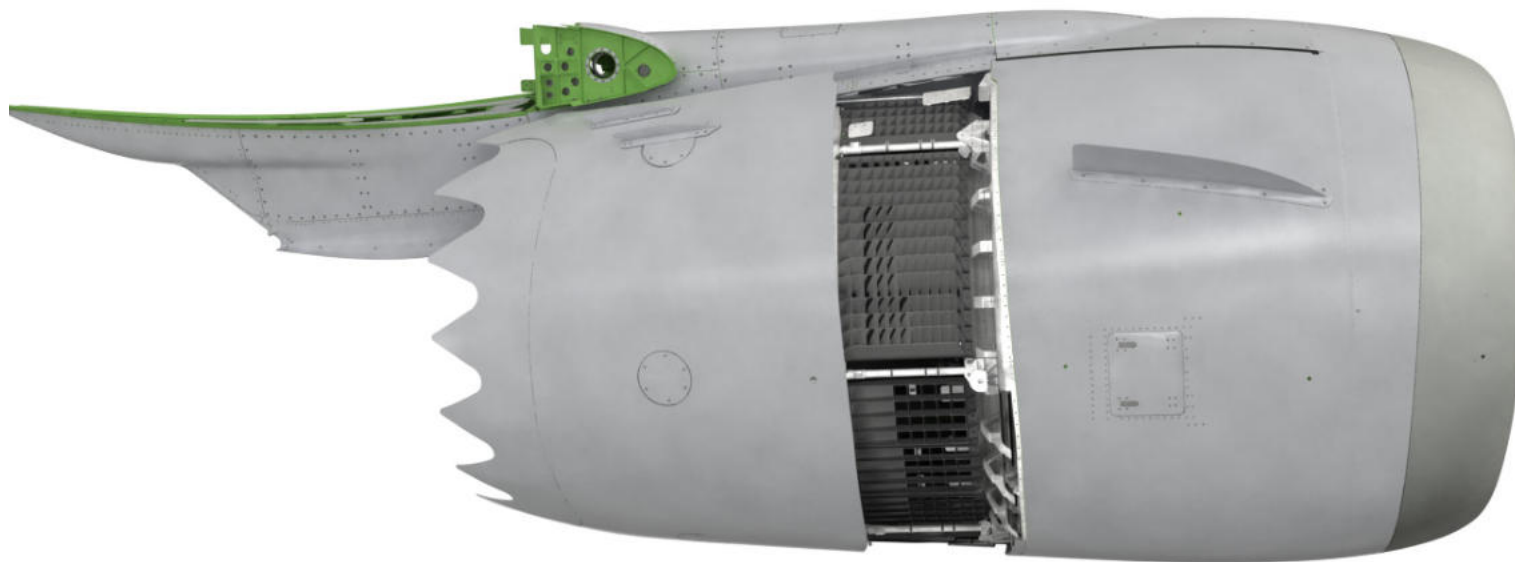
Purpose

The thrust reverser (T/R) indicating system supplies these on the Max display system (MDS) in the flight compartment:

- REV message
- Linear variable differential transformer (LVDT) real time data and failure data.

The thrust reverser (T/R) indicating system also supplies these indications on the P-5 aft overhead panel in the flight compartment:

- REVERSER LIMITED fault light (aft P5)
- REVERSER COMMAND fault light (aft P5)
- REVERSER AIR/GRD fault light (aft P5)
- MAINT light.



THRUST REVERSER INDICATING SYSTEM -- INTRODUCTION

78-36-00

DMC-B737T-A-78-36-00-02A-018A-T_001-00
ICN-B737T-A-783600-A-81205-00007-A-001-01

02-Dec-2020

THRUST REVERSER INDICATING SYSTEM -- GENERAL DESCRIPTION

General

The thrust reverser (T/R) indicating system supplies T/R translating sleeve position data to the Max display system (MDS). The REV message comes on as an indication of translating sleeve position.

Annunciator Lights

The T/R indicating system uses lights on the engine overhead panel to show system malfunctions. When one of these lights comes on, the MASTER CAUTION light will also come on. These are the T/R lights on the engine overhead panel:

- REVERSER LIMITED
- REVERSER COMMAND
- REVERSER AIR/GRD.

The REVERSER LIMITED light shows that the related thrust reverser assembly is not fully functional. The related T/R assembly will not deploy when necessary, or it will deploy but not give full reverse thrust. This light means that there is a reduced level of safety or loss of position indication. Defects that cause this light to come on must be either corrected or deferred. If you defer the defect, you must be sure to follow all applicable MEL specifications for the deferral.

The REVERSER COMMAND light comes on in flight. It shows that a thrust resolver gives reverse thrust signals to the EEC while the airplane is airborne.

The REVERSER AIR/GRD light shows that there is no thrust reverser air/ground protection for one or two engines. A difference between the thrust reverser air/ground input and the EEC air/ground input causes this defect.

Use the OMF to see T/R indicating system fault data.

REV Messages

The REV messages show just above the engine N1 indicators on the engine indication display. One message shows for each T/R.

When REV shows in amber, the T/Rs for the related engine are between the stowed and fully deployed positions. When REV shows in green, the T/Rs for the related engine are fully deployed. When REV does not show, the T/Rs for the related engine are stowed and locked.

The EEC and the display processing computers (DPCs) contain the logic necessary to operate the REV message.

Stow Hydraulic Pressure Switch

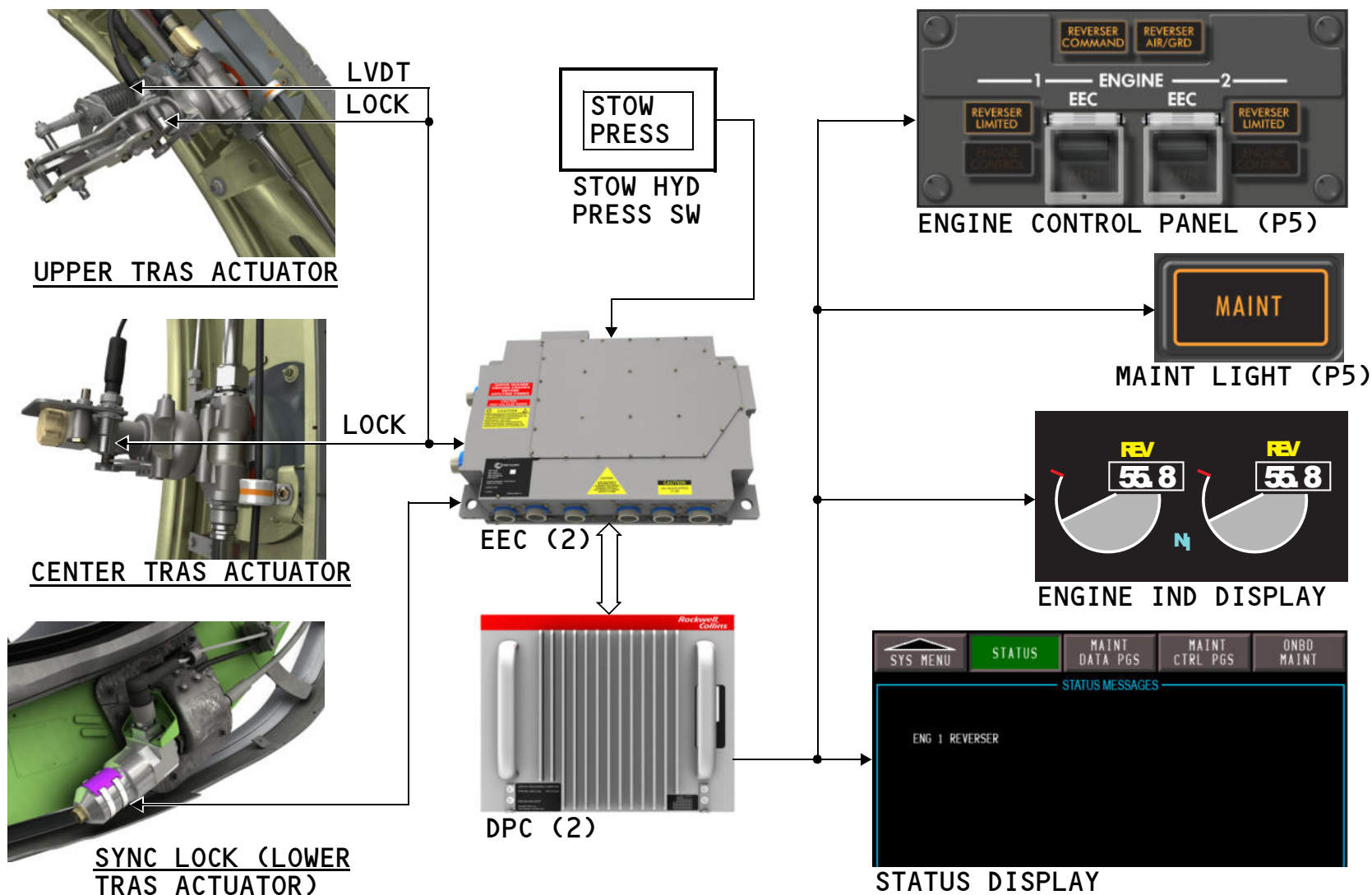
There is one stow hydraulic pressure switch in each pylon. The switch gives pressure information about the applicable engine's thrust reverser system to the EEC. The switch closes when hydraulic pressure is more than 750 PSI +/- 50 PSI (hydraulic isolation valve open). The EECs give a signal to the DPC when the switch gives a signal that is incorrect for the conditions.

Status Level Message

When the EEC finds a non-dispatchable defect in the thrust reverser system, it sends a signal to the DPCs. The DPCs then do these:

- Turn on the MAINT light
- Turn on the MASTER CAUTION light
- Show the ENG 1,2) REVERSER message on the MDS system status display page.

Most T/R actuation system faults that cause the REVERSER LIMITED light to come also cause the ENG (1,2) REVERSER status message to show.



DMC-B737T-A-78-36-00-02A-040A-T_004-00
ICN-B737T-A-783600-A-81205-00008-A-004-01

THRUST REVERSER INDICATING SYSTEM -- GENERAL DESCRIPTION

02-Dec-2020

78-36-00

THRUST REVERSER INDICATING SYSTEM -- COMPONENT LOCATION - FLIGHT COMPARTMENT

Annunciator Lights

These amber annunciator lights are on the engine overhead panel.
This panel is on the P5 aft overhead panel in the flight compartment.

- REVERSER LIMITED
- REVERSER AIR/GRD
- REVERSER COMMAND.

MAINT Light

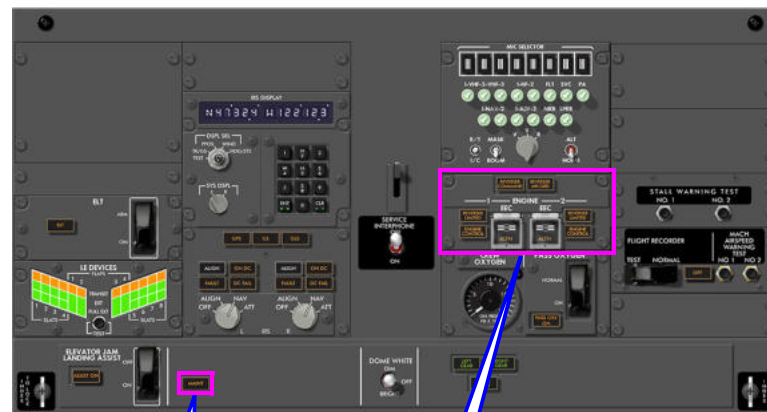
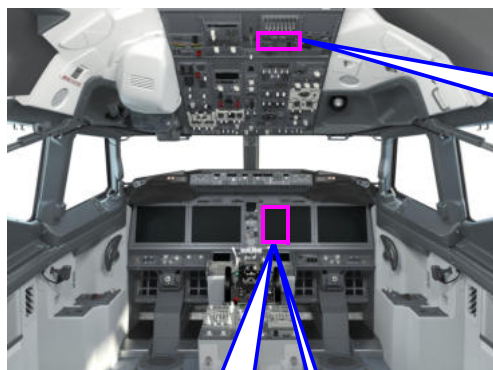
The MAINT light is on the left side of the P5 aft overhead panel.

REV Messages

The REV messages show on the engine indication display. The message shows just above the engine N1 indicator.

Status Messages

The ENG (1,2) REVERSER status message shows on the system status display page of the MDS.



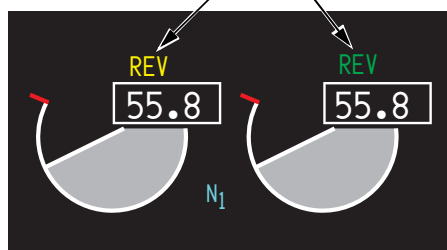
P5 AFT

SYS MENU	STATUS	MAINT DATA PGS	MAINT CTRL PGS	ONBD MAINT
PG 1 OF 4				
ENGINE 1		ENGINE 2		
A	B			
XXX.X	XXX.X	N1	XXX.X	XXX.X
XXX.X	XXX.X	N2	XXX.X	XXX.X
XX.X	XX.X	TRA	XX.X	XX.X
XXX.X	XXX.X	T/R SLEEVE L	XXX.X	XXX.X
XXX.X	XXX.X	T/R SLEEVE R	XXX.X	XXX.X
XX.X	XX.X	P0	XX.X	XX.X
XXX.X	XXX.X	PS3	XXX.X	XXX.X
XXX.X	XXX.X	P3B	XXX.X	XXX.X
XXX.X	XXX.X	BLEED FLOW	XXX.X	XXX.X
XXX.X	XXX.X	T12	XXX.X	XXX.X
XXXX	XXXX	T25	XXXX	XXXX
XXXX	XXXX	T3	XXXX	XXXX
XXX.X	XXX.X	VBV	XXX.X	XXX.X
XXX.X	XXX.X	VBV CMD	XXX.X	XXX.X
XXX.X	XXX.X	N1K12	XXX.X	XXX.X
XXX.X	XXX.X	VSV	XXX.X	XXX.X
XXX.X	XXX.X	VSV CMD	XXX.X	XXX.X
XXX.X	XXX.X	N2K25	XXX.X	XXX.X
XXX.X	XXX.X	FUEL STR DP	XXX.X	XXX.X
XXX.X	XXX.X	FCOC DP	XXX.X	XXX.X
XXX.X	XXX.X	FUEL FLT DP	XXX.X	XXX.X
XXXX	XXXX	FMV	XXX.X	XXX.X
XXXX	XXXX	FF (LBS/HR)	XXXX	XXXX
XXXX	XXXX	PMV	XXX.X	XXX.X
XXXX	XXXX	PMV CMD	XXX.X	XXX.X
XXX	XXX	PFMN	XXX	XXX
DATE XX XX XX UTC XX:XX:XX				
PREV MENU	PRINT	SEND	RECORD	PREV PAGE
		NEXT PAGE		

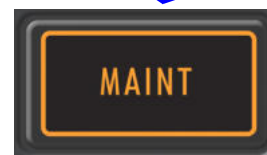
ELECTRONIC PROPULSION CONTROL SYSTEM PAGES

ENGINE 1 CONTROL LIGHT

REV MESSAGES

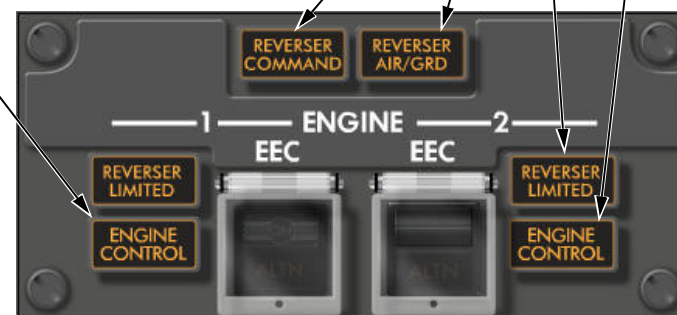


PRIMARY ENGINE DISPLAY (P2)



REVERSER ANNUNCIATOR LIGHTS

ENGINE 2 CONTROL LIGHT



ENGINE CONTROL PANEL (P5)

DMC-B737T-A-78-36-00-02A-055A-T_001-00
ICN-B737T-A-783600-A-81205-00009-A-002-01

THRUST REVERSER INDICATING SYSTEM -- COMPONENT LOCATION - FLIGHT COMPARTMENT

02-Dec-2020

78-36-00

THRUST REVERSER INDICATING SYSTEM -- LINEAR VARIABLE DIFFERENTIAL TRANSFORMER

Purpose

The linear variable differential transformer (LVDT) supplies T/R sleeve position data to the electronic engine control (EEC).

The EEC uses the LVDT signal for these functions:

- Control the REV message on the Max display system (MDS)
- Control the REVERSER LIMITED light (aft P5)
- LVDT failure isolation through the OMF(MDS).
- Reverse thrust interlock control
- Engine forward and reverse thrust control.

The EEC supplies excitation to the LVDT. The LVDT's output voltage is proportional to its armature position. The LVDT's output is proportional to the sleeve position. Each sleeve position shows on the max display system (MDS) EPCS maintenance data page.

Physical Description

The LVDT is a two-channel transducer with separate armatures. The LVDT has one electrical connector each for EEC channel A and channel B wiring.

The upper T/R actuator has a feedback rod. This rod moves when the actuator extends and retracts. The rod connects to the LVDT's armature assembly. The movement of the feedback rod changes the signal that the LVDT gives the EEC. The EEC uses this signal to calculate the position of the related thrust reverser sleeve.

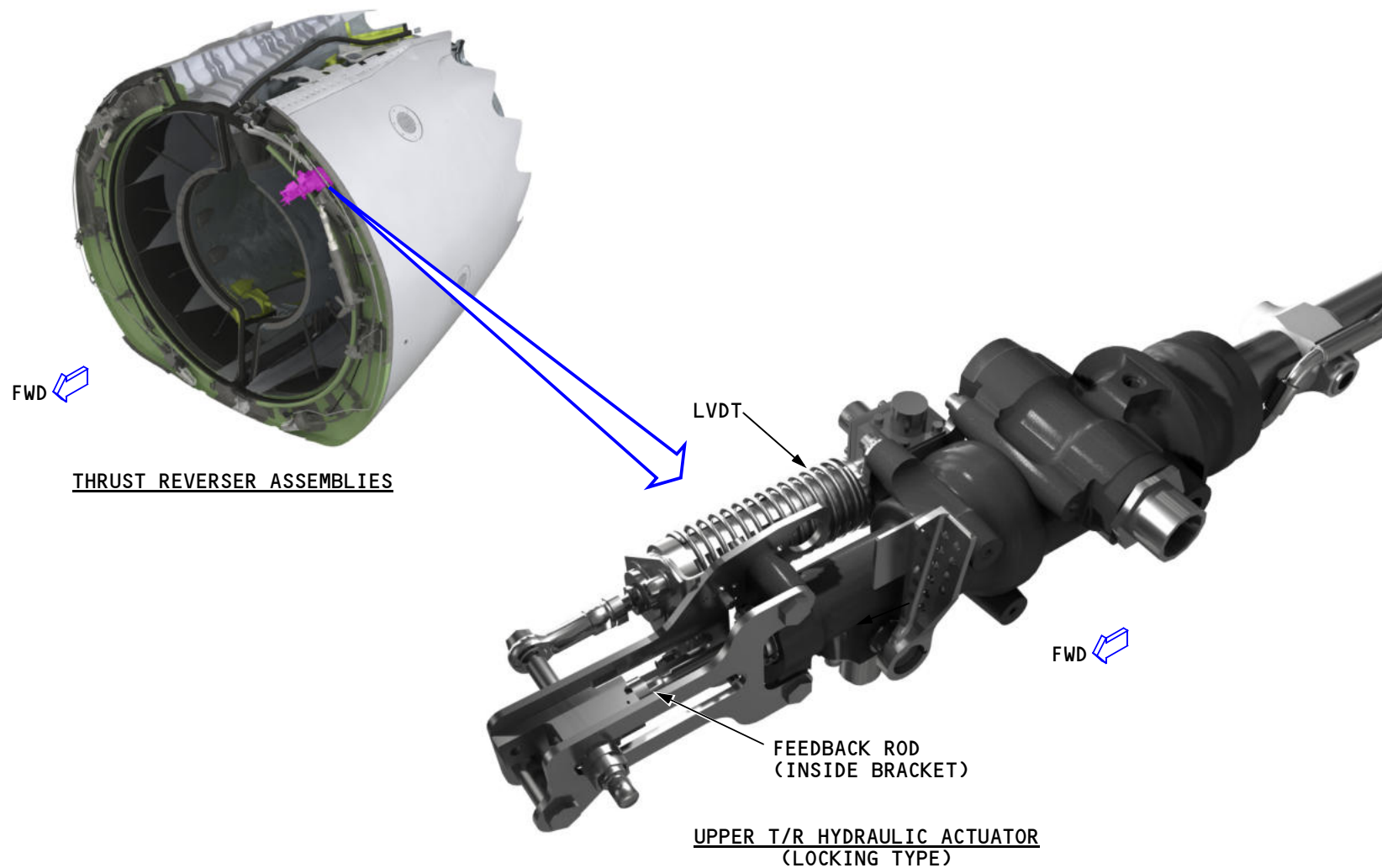
Location

The LVDT is on the head end of the upper locking actuator. Open the fan cowl to get access to the head of the locking actuator. Both of the upper locking actuators on the thrust reverser assemblies have an LVDT sleeve position sensor.

Functional Description

The feedback rod of the locking actuators moves the LVDT's armature assembly as the T/R sleeve deploys or stows.

DMC-B737T-A-78-36-00-02A-034B-T_002-00



THRUST REVERSER INDICATING SYSTEM -- LINEAR VARIABLE DIFFERENTIAL TRANSFORMER

78-36-00

DMC-B737T-A-78-36-00-02A-034B-T_002-00
ICN-B737T-A-783600-A-81205-00010-A-001-01

02-Dec-2020

THRUST REVERSER INDICATING SYSTEM -- REVERSER AIR/GRD LIGHT - FUNCTIONAL DESCRIPTION

General

The REVERSER AIR/GRD light comes on when the thrust reverser air/ground logic is failed. The light tells the flight crew that the thrust reverser will deploy if commanded in-flight.

Thrust Reverser Air/Ground Selection Logic

The EECs uses the inputs from these sources to give an air/ground logic protection for TR deploy:

- Proximity Sensing Electronics Unit (PSEU)
- Low Range Radio Altimeters (LRRAs)
- N2 sensor
- EEC Corrected Air Speed.

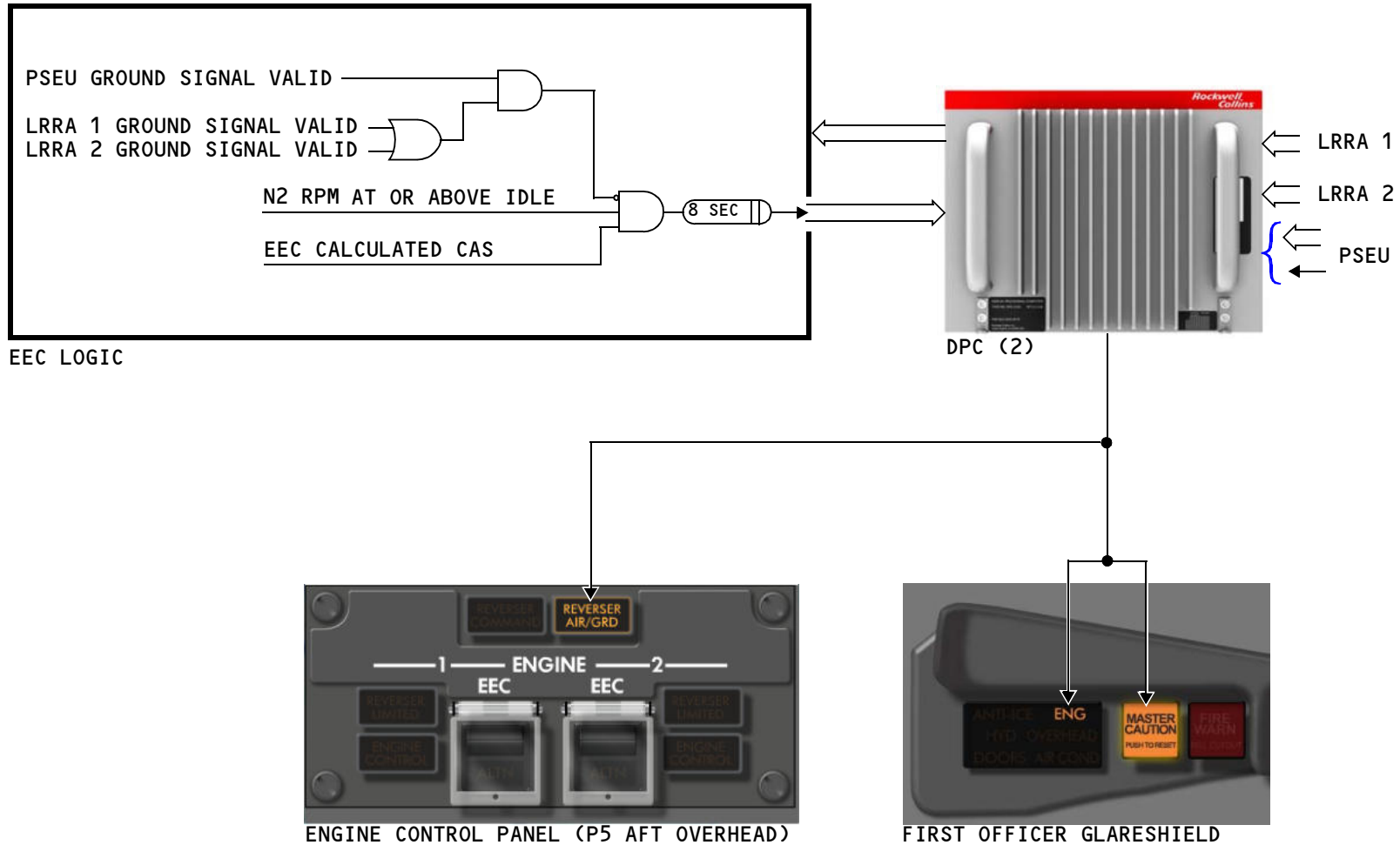
The REVERSER AIR/GR amber light comes on when these logic conditions are present:

- Losses of PSEU sensor signal with any one of the two LRRAs signals
- Engine is operating
- EEC calculated CAS (corrected air speed) differ with the TR ground signals.

DMC-B737T-A-78-36-00-01A-110A-T_002-01

02-Dec-2020

78-36-00



THRUST REVERSER INDICATING SYSTEM -- REVERSER AIR/GRD LIGHT - FUNCTIONAL DESCRIPTION

02-Dec-2020

78-36-00

THRUST REVERSER INDICATING SYSTEM -- REVERSER COMMAND LIGHT - FUNCTIONAL DESCRIPTION

General

The amber REVERSER COMMAND annunciator light is on the engine control panel (P5). This light comes on if reverse thrust is selected with the airplane in the air mode, and the related engine is running. When the REVERSER COMMAND light comes on, the master caution and the ENG annunciator also come on.

The electronic engine controls (EECs) contain the logic for this indication. The EECs send a signal to the display processing computers (DPCs) when all of these conditions apply:

- Airplane is in the air mode for at least ten seconds
- Reverse thrust is selected
- Engine is running.

The DPCs then turn on the ENG annunciator light and the master caution lights.

Flight Deck Light Logic

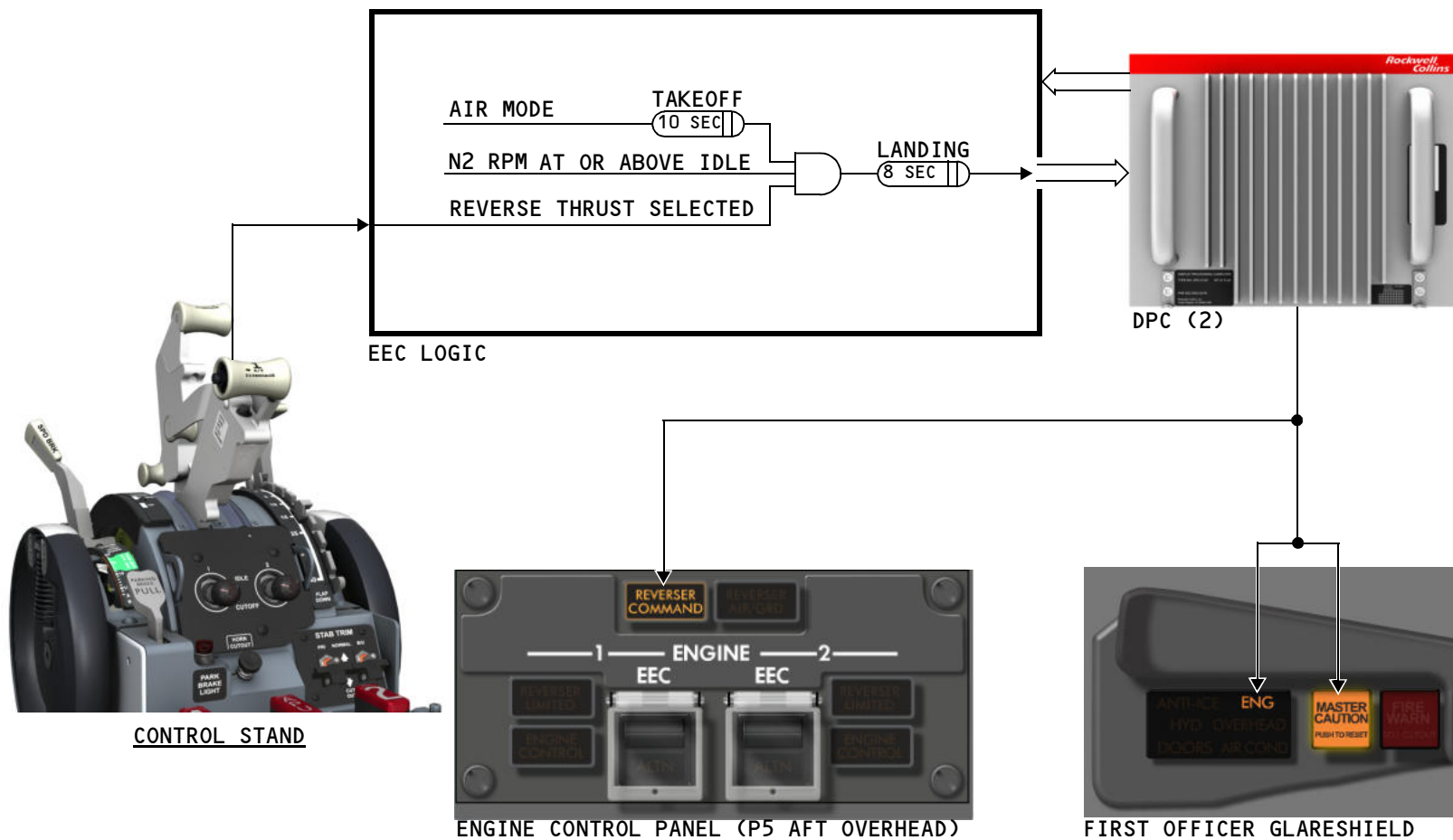
The EEC compares the air/ground condition with the reverse thrust selection. The reverse thrust selection signal comes from thrust lever angle (TLA) resolvers.

A 10-second time period starts when the airplane changes from the ground mode to the air mode on takeoff. During that time, nuisance REVERSER COMMAND indications are prevented.

The REVERSER COMMAND light also has an 8-second time delay on landing. This prevents the indication when one of these occurs:

- Pilot selects reverse thrust just before the airplane has changed from the air mode to the ground mode, or
- The airplane momentarily changes between the air mode and the ground mode on landing.

DMC-B737T-A-78-36-00-04A-042A-T_002-00



THRUST REVERSER INDICATING SYSTEM -- REVERSER COMMAND LIGHT - FUNCTIONAL DESCRIPTION

02-Dec-2020

78-36-00

THRUST REVERSER INDICATING SYSTEM -- REVERSER LIMITED LIGHT - FUNCTIONAL DESCRIPTION

General

A REVERSER LIMITED light comes on for any of these conditions:

- Reverser is inhibited due to detectable faults
- System has reduced level of safety
- System has loss of position indication.

The engine electronic control (EEC) channels A and B contain the logic necessary to identify T/R control system component faults. The EECs send a signal to the display processing computers (DPCs) when it finds a defect. The DPCs control the REVERSER LIMITED lights.

Logic to make the REVERSER LIMITED light come on in flight is different than on the ground.

REVERSER LIMITED Light (Aircraft In Flight)

The REVERSER LIMITED light can come on in the air mode. This is when the airplane is airborne or when it is at or more than 80 knots on the ground. The light comes on to indicate:

- Thrust reversers will not deploy when commanded on the ground
- Thrust reversers will deploy on the ground but thrust will be limited to idle.

The REVERSER LIMITED light comes on in the air mode when any of these occur:

- Dual-channel thrust resolver failure
- Thrust lever interlock system failure
- Dual-channel LVDT failure
- Thrust resolver channel A/B disagree
- Air/Ground logic defaults to "AIR"
- Hydraulic isolation valve (HIV) failure (stow hydraulic pressure switch)

- Left and right translating sleeves not in commanded position.

REVERSER LIMITED Light (Aircraft on Ground)

The REVERSER LIMITED light can also come on when the aircraft is not in flight mode. This is when the aircraft is below 80 knots. The light tells the flight crew that there is one or more system faults that could decrease safety. The faults must be addressed before the next flight with passengers. If you defer the defects by the MEL, you must deactivate the thrust reverser system on the applicable engine.

A normal stow of the T/R is complete in approximately 4 seconds.

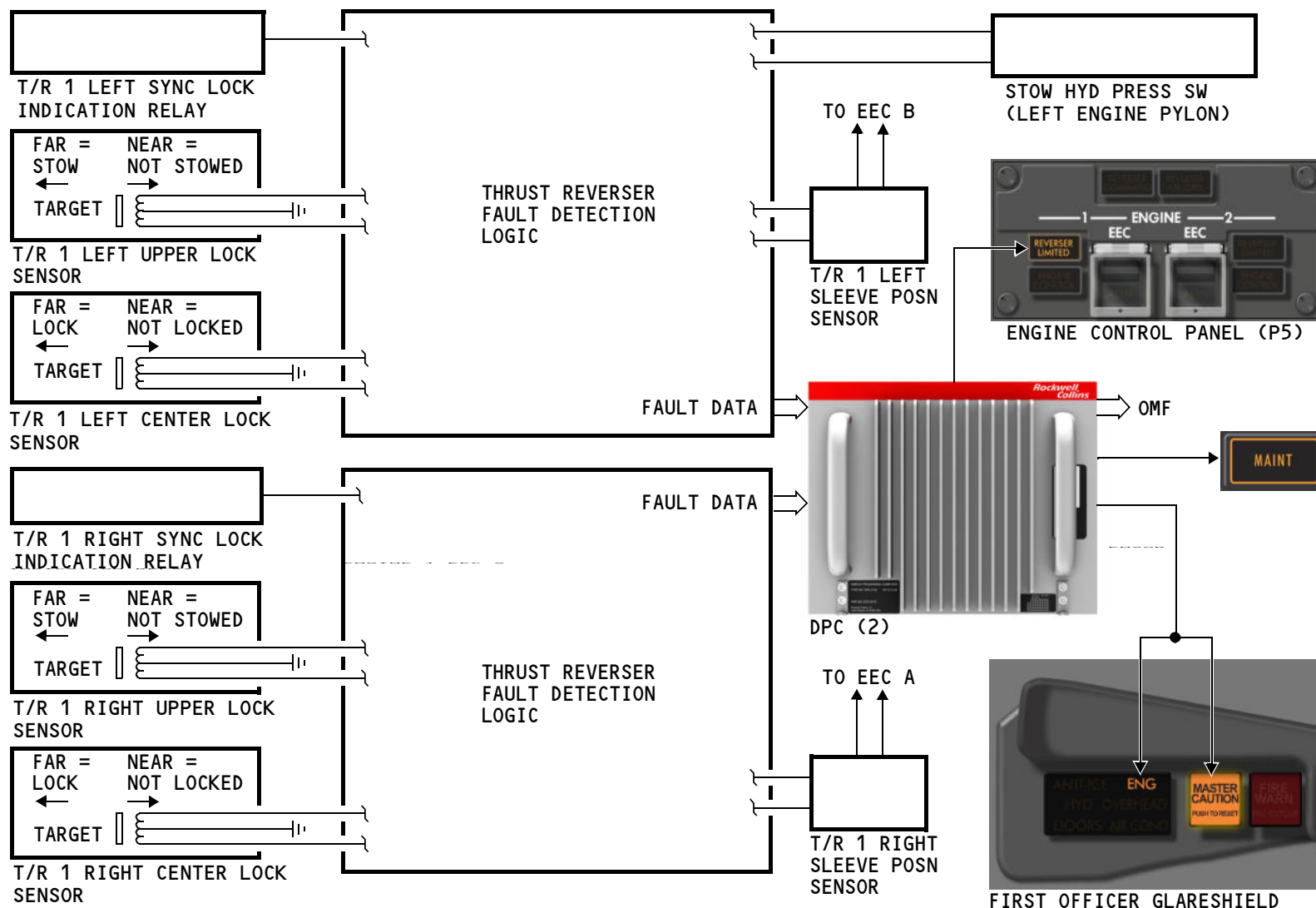
Fault Logic

The REVERSER LIMITED light can also come on in the ground mode. It comes on in the ground mode when there is one of the same defects specified for the air mode. It can also come on when one of these other defects occur:

- Loss of cross-channel data link
- Loss of an EEC channel
- Sync lock failure
- Locking actuator prox sensor failure
- Locking actuator LVDT indication failure.

When the REVERSER LIMITED light comes on, The DPCs also turn on the master caution and the ENG annunciator.

DMC-B737T-A-78-36-00-03A-042A-T_002-00



DMC-B737T-A-78-36-00-03A-042A-T_002-00
ICN-B737T-A-783600-A-81205-00011-A-002-01

02-Dec-2020

78-36-00



737-7/8/9/10 Maintenance Training Manual

Boeing Training and Professional Services
FS328

Boeing Proprietary

The information contained herein is proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company.

Copyright © 2020 The Boeing Company. All rights reserved.

EXPORT CONTROLLED - This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN 9E991

ENGINE MANUFACTURER PROPRIETARY INFORMATION

A portion of the information in this training material is the property of Rolls-Royce plc, General Electric or CFM International, Inc. (the "Engine Manufacturer") and has been included with the Engine Manufacturer's permission. It may not be copied, or communicated to a third party, or used for any purpose other than that for which it is supplied without the express written consent of the applicable Engine Manufacturer. In this regard, the Engine Manufacturer training material herein is not an official publication of the Engine Manufacturer and must not be used for operating or maintaining the equipment described. Official Engine Manufacturer publications and manuals must be used for those purposes.

To the extent the Engine Manufacturer has an agreement covering the control and disclosure of information with the company, organization or individual receiving this training material then the Engine Manufacturer portion of the training material must be held in confidence and controlled in accordance with such agreement's terms. While this information is given in good faith based upon the latest information available to the Engine Manufacturer, no warranty or representation is given concerning such information, which must not be taken as establishing any contractual or other commitment binding upon the Engine Manufacturer or any of its subsidiary or associated companies.